Eclipses by Appointment - Supplement

Volume 3 of 3

R410-R539 and remaining reviewed evidence

Original continuous pages 1944-2691 of 2691

Navigation: each volume adds this one-page guide.

A page in this volume = original page minus 1943 plus 1.

The public supplement-review-pages.csv maps every R key to volume

and exact page. The scientific-review-register.csv maps status to report.

Volume 1: Original A catalogue and R001-R216 (original pages 1-1182).

Volume 2: Fixed magnitude comparison and R217-R409 (original pages 1183-1943).

Volume 3: R410-R539 and remaining reviewed evidence (original pages 1944-2691).

Original illustrated A entries and all R full reports remain in the

continuous source HTML and the public reports/evidence archive.

Cross-volume references use the page map above.

Spectroscopic-orbit period folds: brightest remaining screened candidates

The next 130 screened candidates in actual Gaia DR3 G order (12.0036-12.3371) have 103 physically confirmed, 24 probable and three plausible outcomes. Two confirmed known recoveries are excluded, leaving 101 confirmed and 24 probable unique additions. The same bounded physical, catalogue and source-attribution criteria apply; the fixed balanced 120-target comparison remains unchanged. The 136 unreviewed screened candidates are 56 at G 12.25-12.5 and 80 at G 12.5-12.75.

Periods and epochs below retain their per-object eclipse-kind and error meanings. Same-sector QLP and TESScut share photons; distinct sectors provide separate observing occasions on the same instrument. Specific morphology, alias and localization limits remain in each full report.

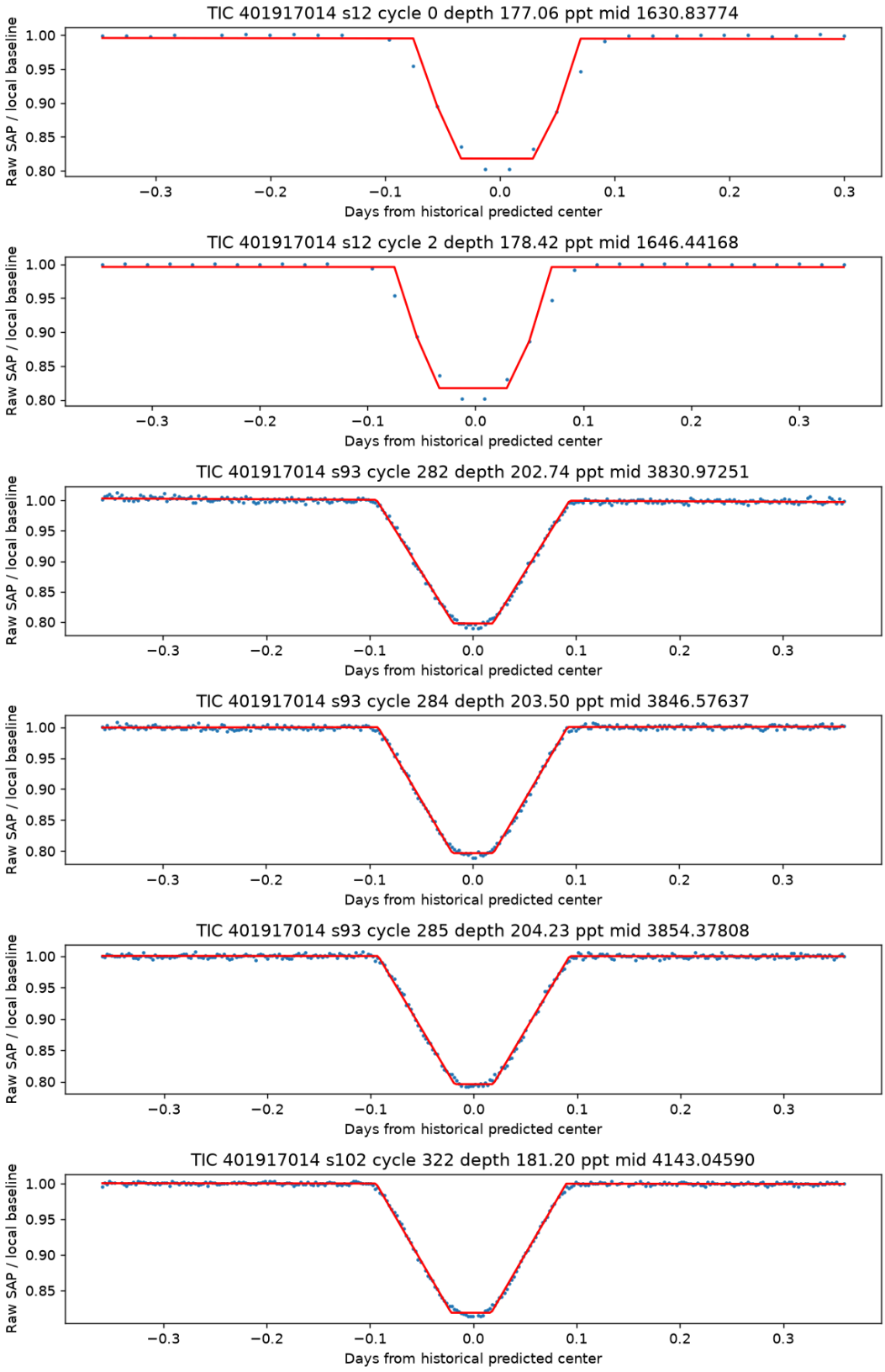
R410. TIC 401917014

physical_verdict: CONFIRMED

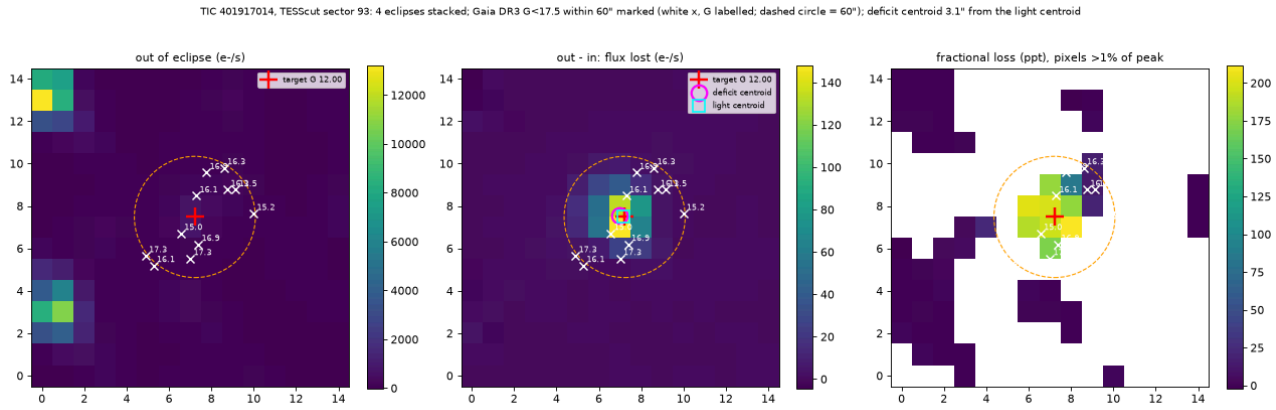
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R410; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R410; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 401917014: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured eclipses in representative sectors [12, 93, 102]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 8 actual midpoints yield P = 7.8018659062 ± 0.0000181348 d and reference epoch BTJD 1630.8460077 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 93 cutout puts the deficit 4.91 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.577; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 5796293562665379456: separation 46.11 arcsec, G 13.487, approximate intrinsic eclipse requirement 0.923. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 196.20/192.24/161.00 ppt in 4 stacked events; median selected QLP raw fit depth 191.97 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5796293459586158976 has orbital P = 7.8019703401 d, e = 0.02120; closest edge-on conjunction offset at the measured epoch is -0.02604 d (-0.00334 phase), with 0.13110 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.66 h versus observed median 4.41 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Eight deep raw eclipses across S12/93/102; capable G-band neighbor 46.1 arcsec away requires an extreme intrinsic loss, while the deficit core is target-centered. Pixel residual limits subpixel claims.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	0	1630.8377438	0.010417	177.06	used
12	2	1646.4416770	0.010417	178.42	used
93	282	3830.9725084	0.001157	202.74	used
93	283	3838.7746987	0.001157	205.41	used
93	284	3846.5763723	0.001157	203.50	used
93	285	3854.3780807	0.001157	204.23	used
102	321	4135.2443979	0.001157	180.80	used
102	322	4143.0459048	0.001157	181.20	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s93.json, historical/diff_s93.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R410. Public evidence files: diff_s93.json, diff_s93.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

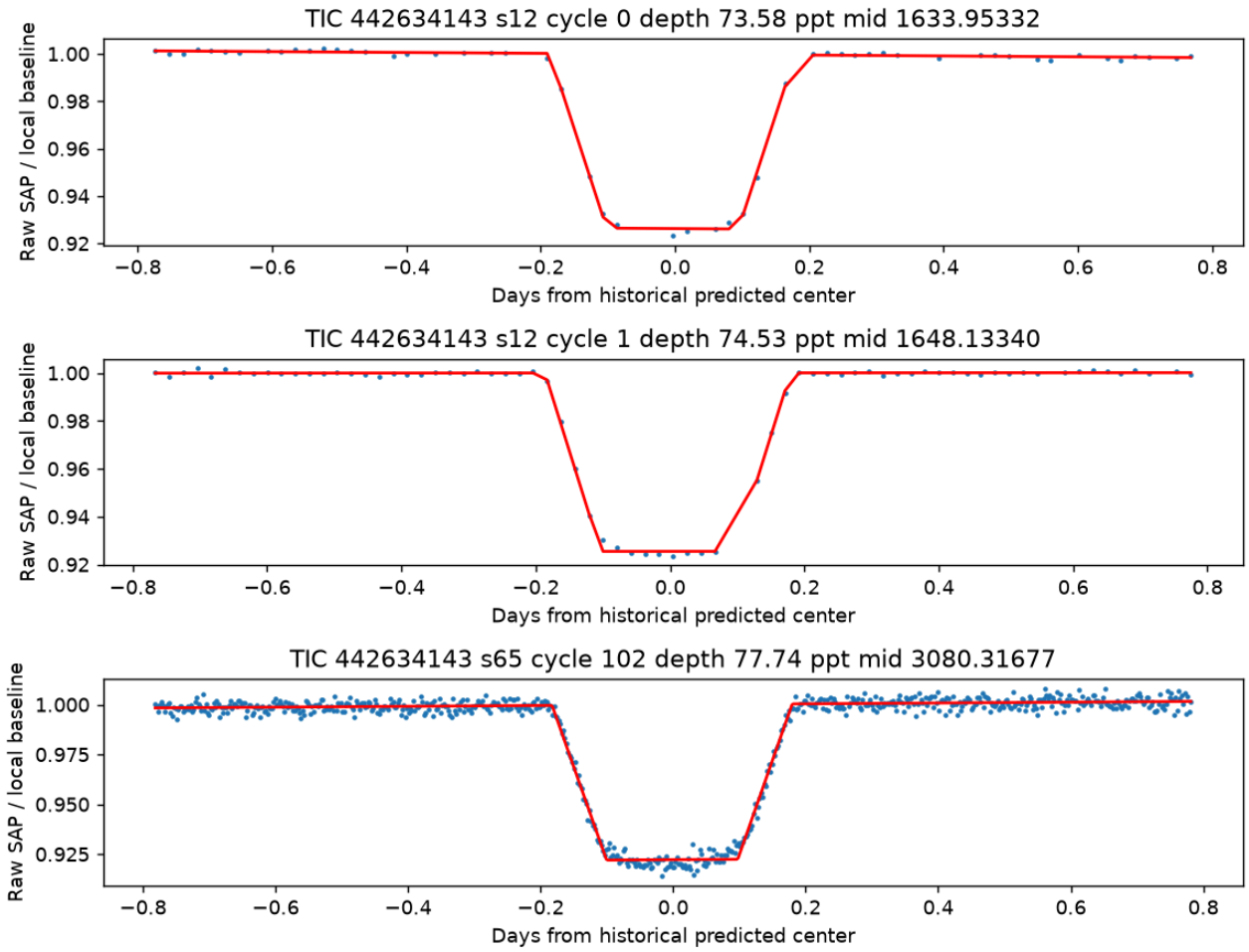
R411. TIC 442634143

physical_verdict: CONFIRMED

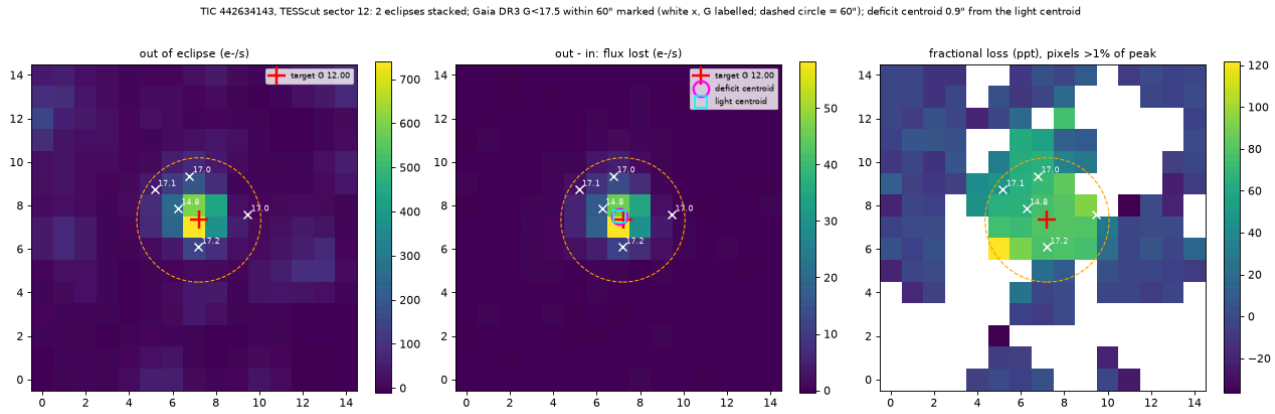
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R411; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R411; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 442634143: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 3 timing-usable measured eclipses in representative sectors [12, 65]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 3 actual midpoints yield P = 14.1800335439 ± 0.0000734560 d and reference epoch BTJD 1633.9533458 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 12 cutout puts the deficit 4.58 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.123; this is not a calibrated localization error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 78.72/77.08/75.71 ppt in 2 stacked events; median selected QLP raw fit depth 74.53 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 6040478190203632640 has orbital P = 14.1830337728 d, e = 0.14754; closest edge-on conjunction offset at the measured epoch is -0.20084 d (-0.01416 phase), with 0.38892 d linearly propagated full-SB1-covariance timing error. The rough central duration is 6.68 h versus observed median 8.78 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Three deep raw eclipses in S12/65. Observed duration exceeds a rough central radius/minimum-mass estimate; the estimate is model dependent and does not resolve detailed binary geometry.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	0	1633.9533229	0.010416	73.58	used
12	1	1648.1334025	0.010416	74.53	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
65	102	3080.3167673	0.001157	77.74	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s12.json, historical/diff_s12.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R411. Public evidence files: diff_s12.json, diff_s12.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

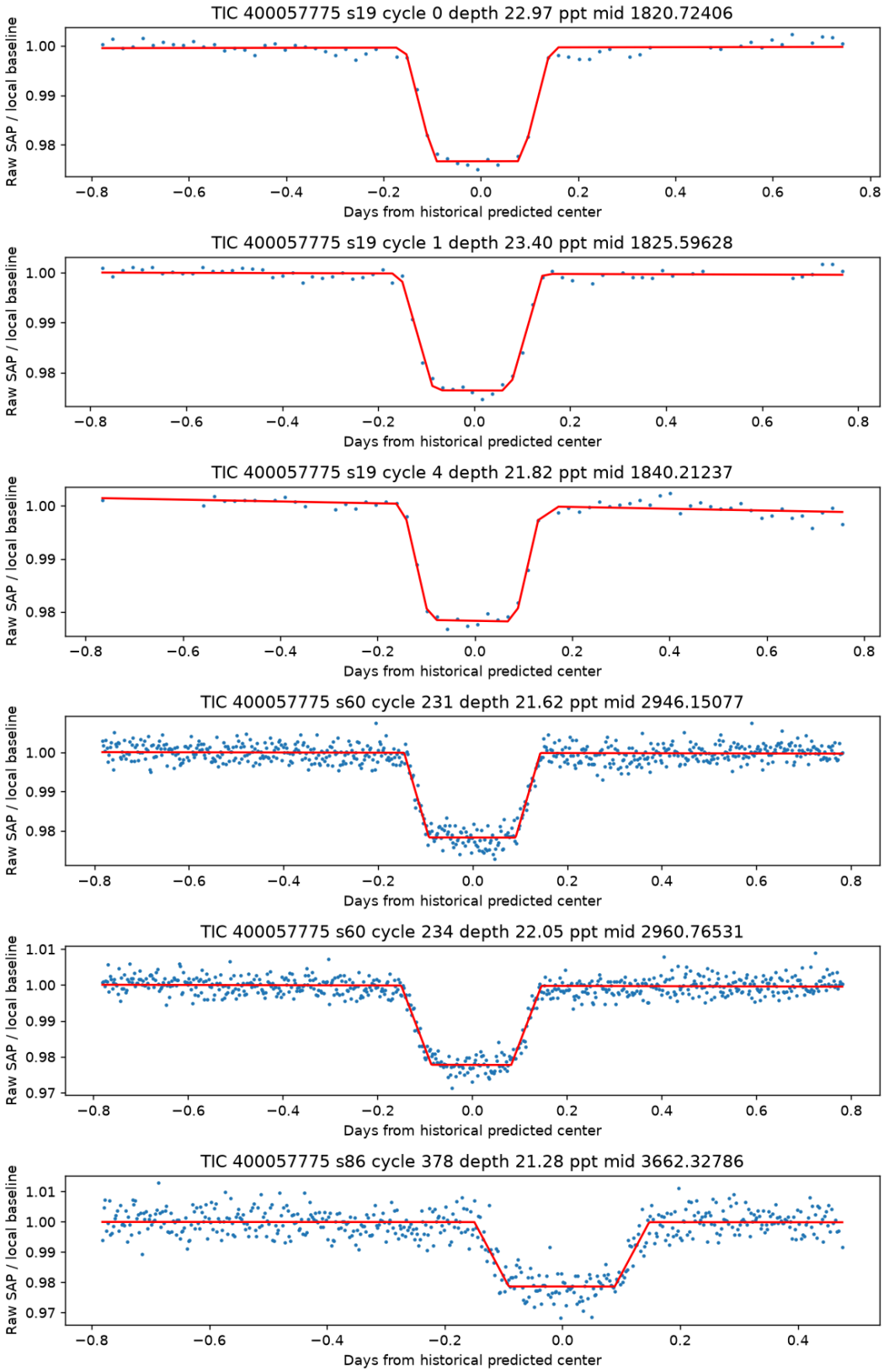
R412. TIC 400057775

physical_verdict: CONFIRMED

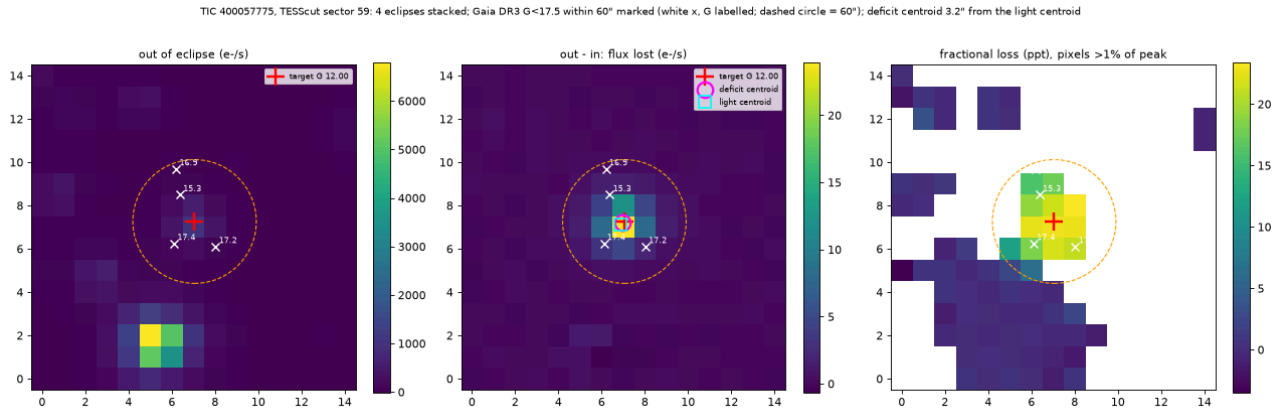
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R412; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R412; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 400057775: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 10 timing-usable measured eclipses in representative sectors [19, 60, 86]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 10 actual midpoints yield P = 4.8719600113 ± 0.0000070752 d and reference epoch BTJD 1820.7274949 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 59 cutout puts the deficit 1.73 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.822; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 480689479890567936: separation 29.38 arcsec, G 15.294, approximate intrinsic eclipse requirement 0.466. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 22.99/22.01/20.61 ppt in 4 stacked events; median selected QLP raw fit depth 21.93 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 480689509953048960 has orbital P = 4.8719149116 d, e = 0.08656; closest edge-on conjunction offset at the measured epoch is -0.01917 d (-0.00393 phase), with 0.12594 d linearly propagated full-SB1-covariance timing error. The rough central duration is 8.83 h versus observed median 7.09 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Ten repeated raw eclipses S19/60/86. A capable neighbor 29.4 arcsec away is distinct from the target-centered deficit; high image-model residual limits precision.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
19	0	1820.7240571	0.010417	22.97	used
19	1	1825.5962845	0.010417	23.40	used
19	3	1835.3436536	0.010417	23.15	used
19	4	1840.2123729	0.010417	21.82	used
60	230	2941.2790493	0.001157	21.44	used
60	231	2946.1507725	0.001157	21.62	used
60	233	2955.8940093	0.001157	22.42	used
60	234	2960.7653054	0.001157	22.05	used
86	375	3647.7126339	0.001157	21.72	used
86	378	3662.3278567	0.001562	21.28	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s59.json, historical/diff_s59.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R412. Public evidence files: diff_s59.json, diff_s59.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

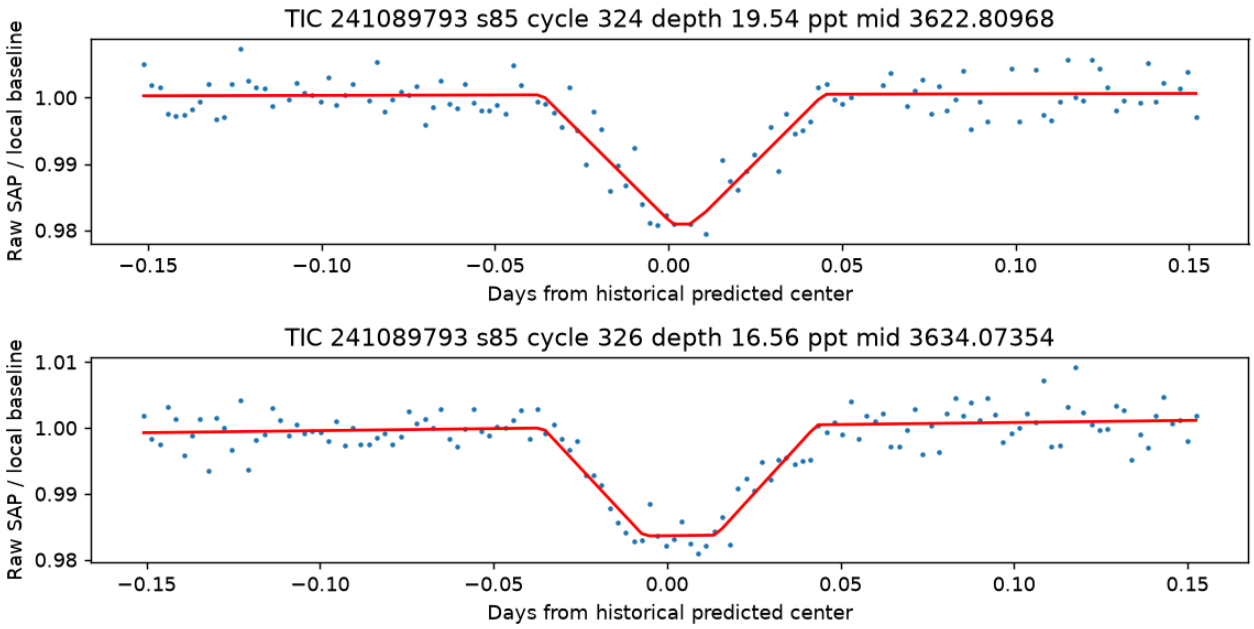
R413. TIC 241089793

physical_verdict: CONFIRMED

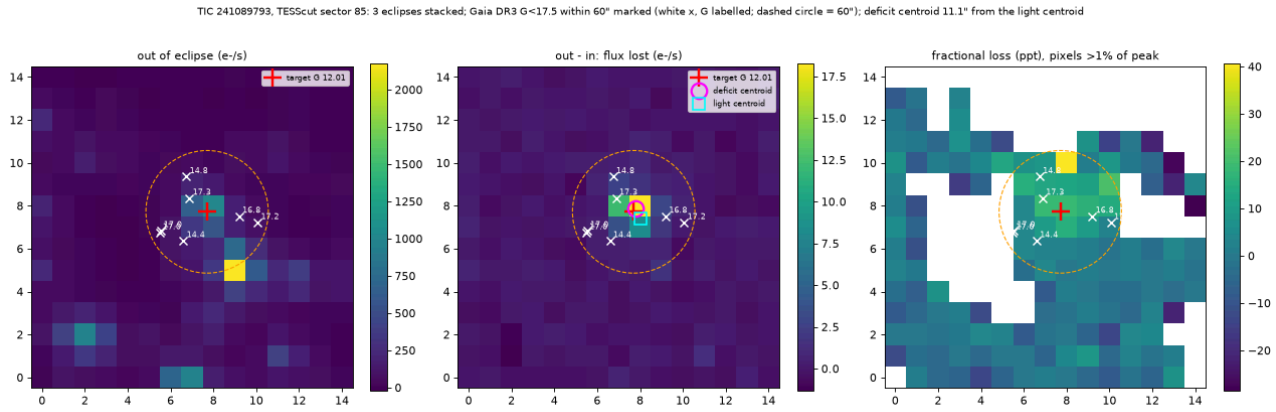
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R413; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R413; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 241089793: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured eclipses in representative sectors [18, 85]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 5 actual midpoints yield P = 5.6321610433 ± 0.0000186954 d and reference epoch BTJD 1792.3571061 (repeating eclipse; primary by convention). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear fit to independently measured raw S18 and S85 midpoints, with max(formal center covariance, half cadence) per event and scatter inflation; formal fit error, not total calibrated uncertainty. Historical period/cycle predictions guided extraction only. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 85 cutout puts the deficit 3.60 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.847; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 408289628502534144: separation 36.59 arcsec, G 14.375, approximate intrinsic eclipse requirement 0.166; Gaia 408289727280756608: separation 39.76 arcsec, G 14.819, approximate intrinsic eclipse requirement 0.242. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 18.09/15.54/10.73 ppt in 3 stacked events; median selected QLP raw fit depth 17.95 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 408289727280589312 has orbital P = 5.6321186642 d, e = 0.02967; closest edge-on conjunction offset at the measured epoch is -0.03436 d (-0.00610 phase), with 0.12821 d linearly propagated full-SB1-covariance timing error. The rough central duration is 4.30 h versus observed median 1.88 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: S18 clean raw eclipses at successive cycles -1 and 0 plus cycle 2 test the 5.632-day recurrence; measured S85 cycles 324/326 extend the timing baseline. S85 cycle 325 and S18 cycle 1 are flagged, not clean nulls. S18 partial cycle 3 is excluded from timing. The formal period error is not total calibrated uncertainty.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
18	-1	1792.3580780	0.010420	16.04	used
18	0	1797.9886680	0.010420	17.95	used
18	2	1809.2530956	0.010420	18.69	used
18	3	1814.8824888	0.010420	19.46	excluded: Partial coverage; excluded from joint timing fit.
85	324	3622.8096759	0.001157	19.54	used
85	326	3634.0735380	0.001157	16.56	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s85.json, historical/diff_s85.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). This target also has ../241089793-refined-timing.json, ../241089793-alias-coverage.json and s18-cycle-check.png. Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R413. Public evidence files: diff_s85.json, diff_s85.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv, s18-cycle-check.png. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

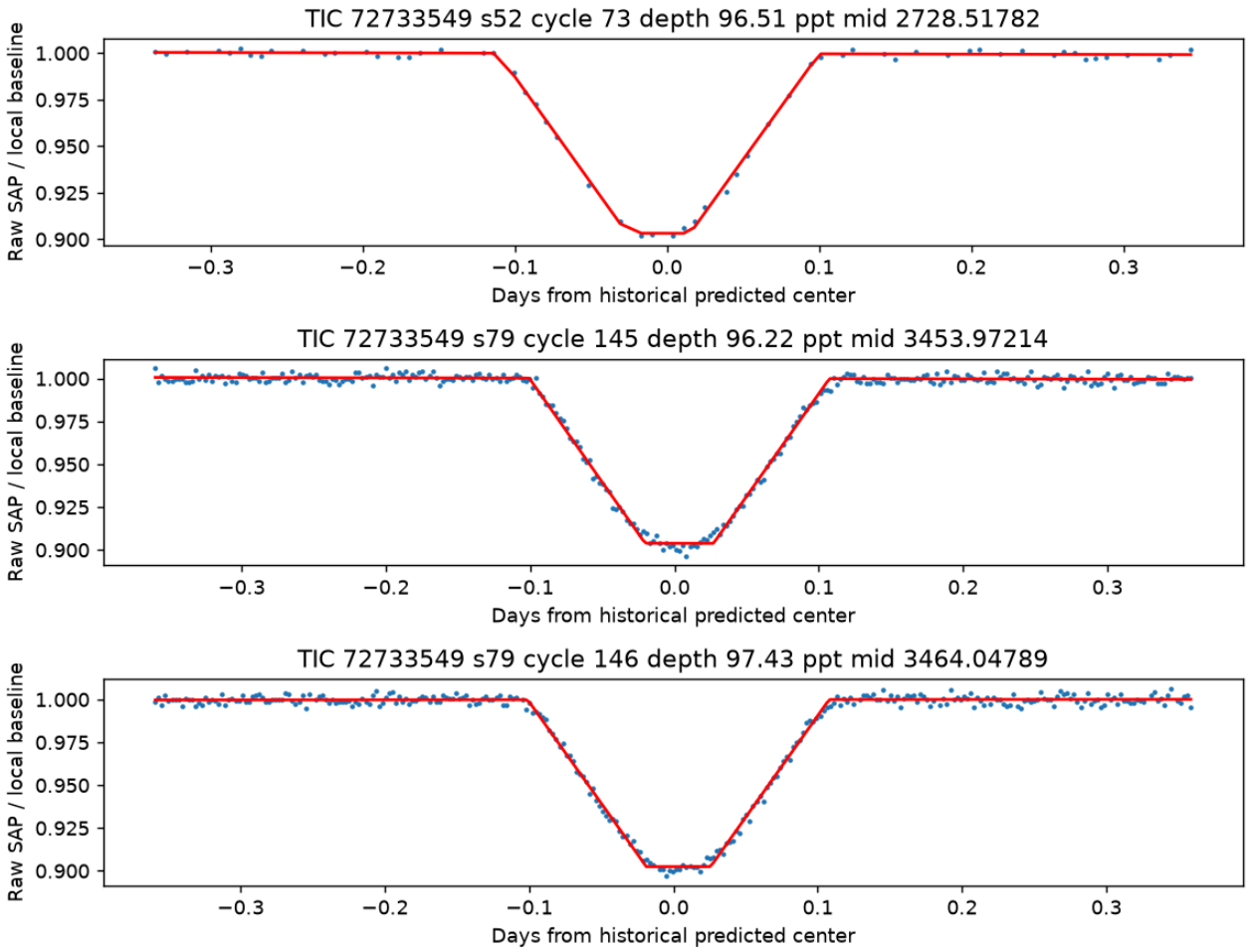
R414. TIC 72733549

physical_verdict: CONFIRMED

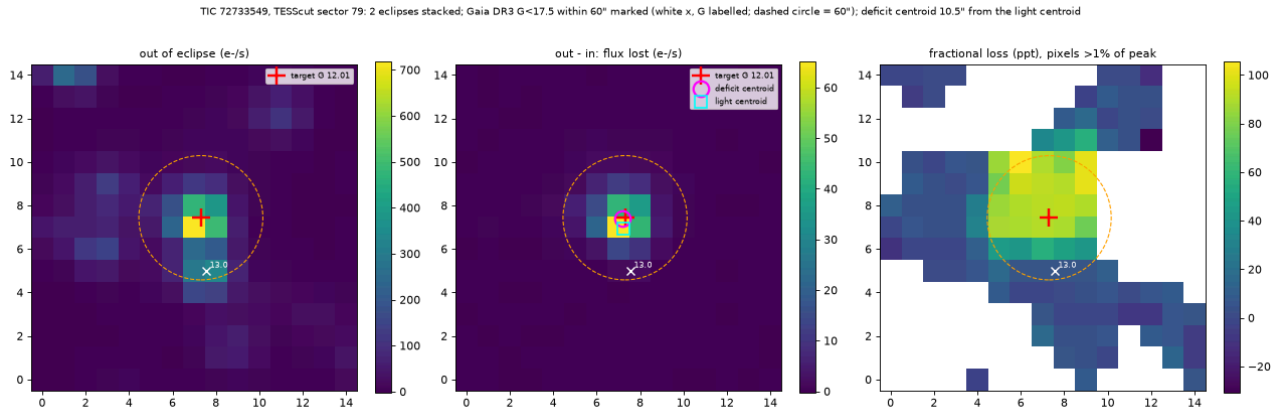
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R414; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R414; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 72733549: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 3 timing-usable measured eclipses in representative sectors [52, 79]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 3 actual midpoints yield P = 10.0757543600 ± 0.0000962802 d and reference epoch BTJD 2728.5178240 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 79 cutout puts the deficit 2.70 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.430; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 1351135967600649984: separation 48.12 arcsec, G 12.966, approximate intrinsic eclipse requirement 0.312. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 90.72/83.62/69.70 ppt in 2 stacked events; median selected QLP raw fit depth 96.51 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1351133729921222400 has orbital P = 10.0743628417 d, e = 0.01363; closest edge-on conjunction offset at the measured epoch is 0.28926 d (0.02871 phase), with 0.46418 d linearly propagated full-SB1-covariance timing error. The rough central duration is 7.25 h versus observed median 5.01 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Three deep raw eclipses S52/79, including successive S79 cycles. A capable neighbor 48.1 arcsec away is distinct from the target-centered deficit.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
52	73	2728.5178238	0.006944	96.51	used
79	145	3453.9721385	0.001157	96.22	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
79	146	3464.0478918	0.001157	97.43	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s79.json, historical/diff_s79.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R414. Public evidence files: diff_s79.json, diff_s79.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

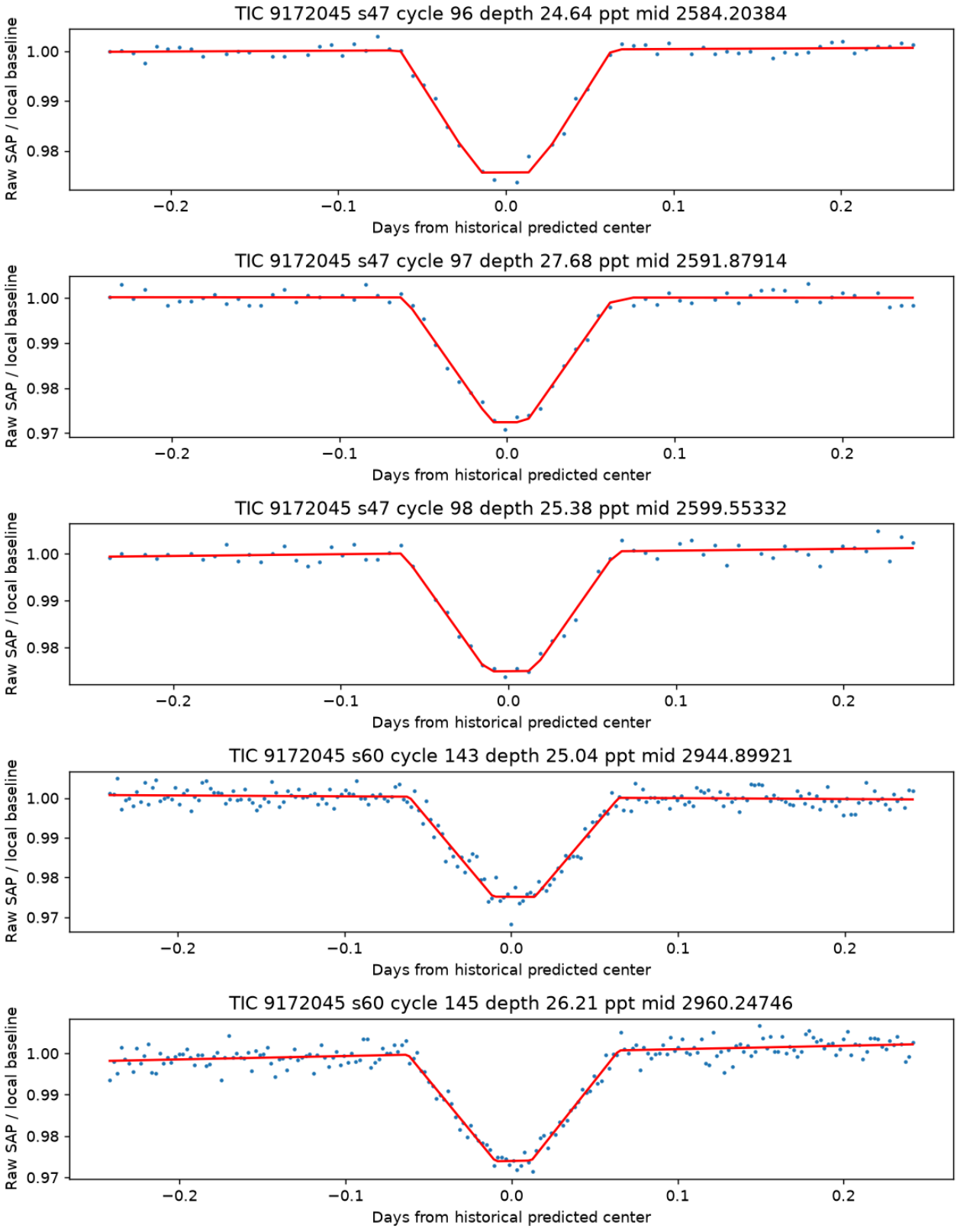
R415. TIC 9172045

physical_verdict: CONFIRMED

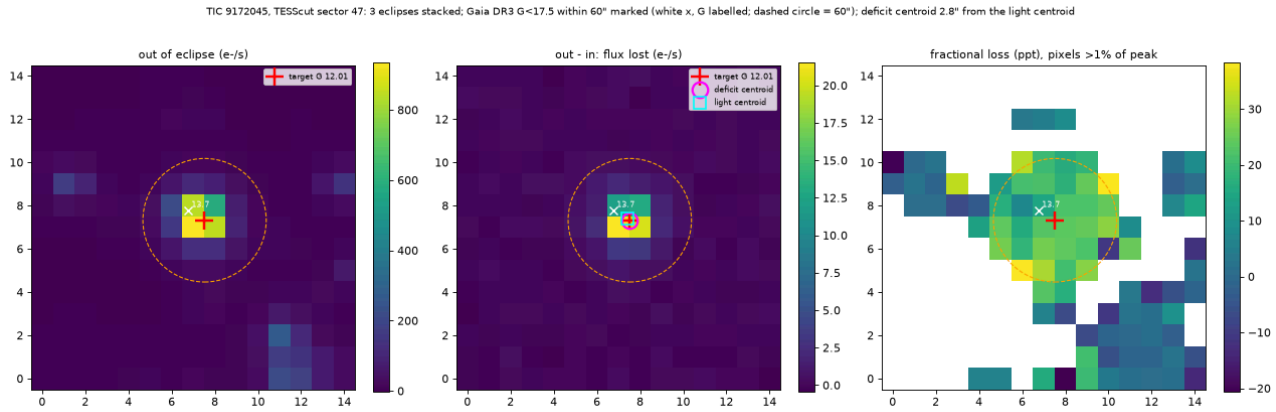
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R415; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R415; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 9172045: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured eclipses in representative sectors [47, 60]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 5 actual midpoints yield P = 7.6743519360 ± 0.0000459896 d and reference epoch BTJD 2584.2044385 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 47 cutout puts the deficit 0.24 arcsec from the target coordinate; peak [7, 7] versus target pixel [8, 7]. Image-model fractional residual 0.213; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 919146748219604736: separation 18.80 arcsec, G 13.721, approximate intrinsic eclipse requirement 0.139. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 23.77/21.23/20.42 ppt in 3 stacked events; median selected QLP raw fit depth 25.38 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 919146679500127616 has orbital P = 7.6744160954 d, e = 0.24606; closest edge-on conjunction offset at the measured epoch is -0.07616 d (-0.00992 phase), with 0.15053 d linearly propagated full-SB1-covariance timing error. The rough central duration is 6.62 h versus observed median 3.04 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Five repeated raw eclipses S47/60. A capable neighbor 18.8 arcsec away can in principle produce the blended loss, but the inspected deficit core falls on the target and shows no distinct neighbor peak; attribution is favored at TESS resolution, not proven subpixel.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
47	96	2584.2038405	0.003472	24.64	used
47	97	2591.8791377	0.003472	27.68	used
47	98	2599.5533210	0.003472	25.38	used
60	143	2944.8992147	0.001157	25.04	used
60	145	2960.2474561	0.001157	26.21	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s47.json, historical/diff_s47.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R415. Public evidence files: diff_s47.json, diff_s47.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

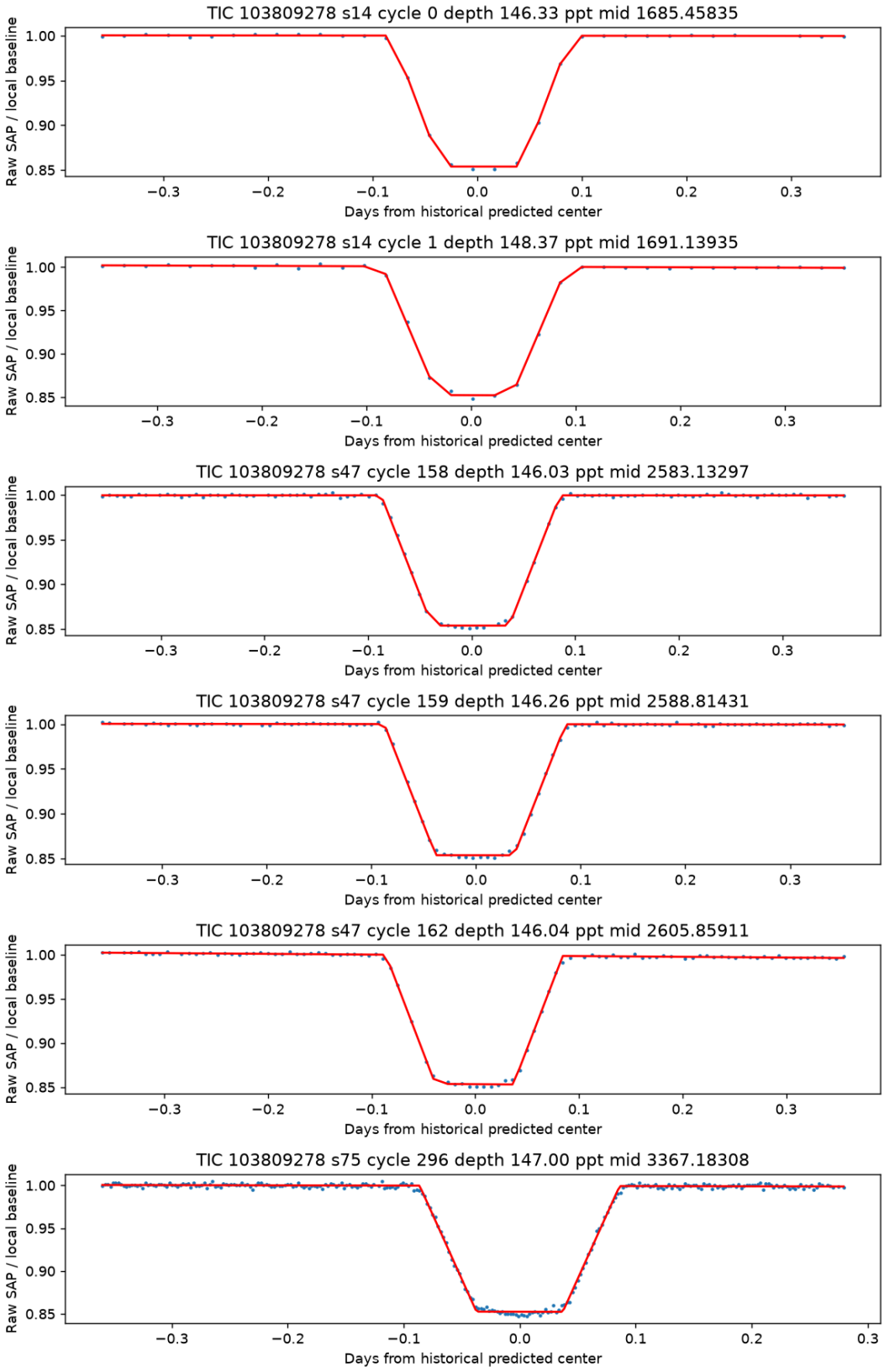
R416. TIC 103809278

physical_verdict: CONFIRMED

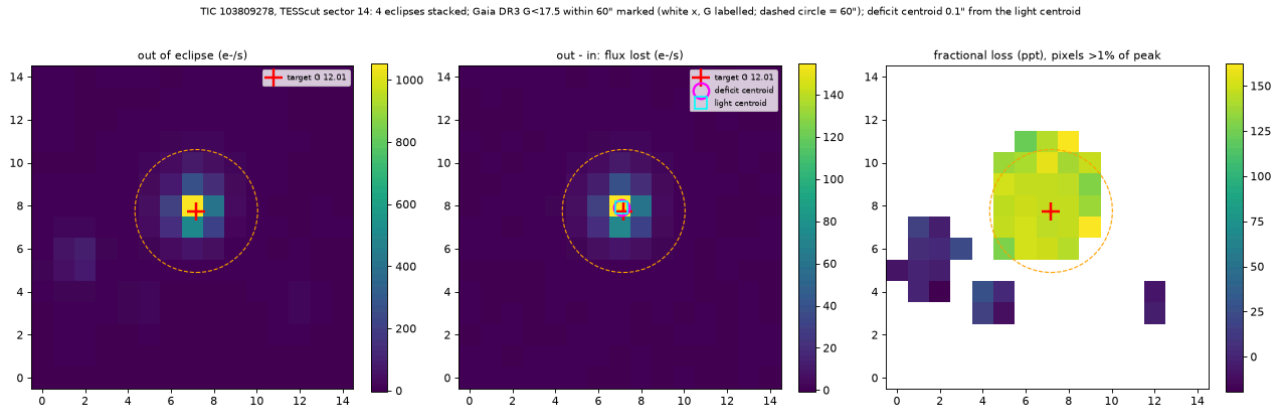
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R416; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R416; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 103809278: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 9 timing-usable measured eclipses in representative sectors [14, 47, 75]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 9 actual midpoints yield P = 5.6815171354 ± 0.0000120548 d and reference epoch BTJD 1685.4540122 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 14 cutout puts the deficit 3.79 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.023; this is not a calibrated localization error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 147.02/147.02/146.82 ppt in 4 stacked events; median selected QLP raw fit depth 146.26 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1059801184791364480 has orbital P = 5.6814017817 d, e = 0.08033; closest edge-on conjunction offset at the measured epoch is 0.13967 d (0.02458 phase), with 0.10317 d linearly propagated full-SB1-covariance timing error. The rough central duration is 4.38 h versus observed median 4.14 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Nine deep raw eclipses across S14/47/75; target-centered difference core and no capable Gaia neighbor under the brightness screen.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
14	0	1685.4583462	0.010416	146.33	used
14	1	1691.1393536	0.010416	148.37	used
14	3	1702.5021041	0.010416	141.24	used
47	158	2583.1329651	0.003472	146.03	used
47	159	2588.8143143	0.003472	146.26	used
47	161	2600.1777778	0.003472	146.26	used
47	162	2605.8591093	0.003472	146.04	used
75	293	3350.1387125	0.001157	147.35	used
75	296	3367.1830767	0.001157	147.00	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s14.json, historical/diff_s14.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R416. Public evidence files: diff_s14.json, diff_s14.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

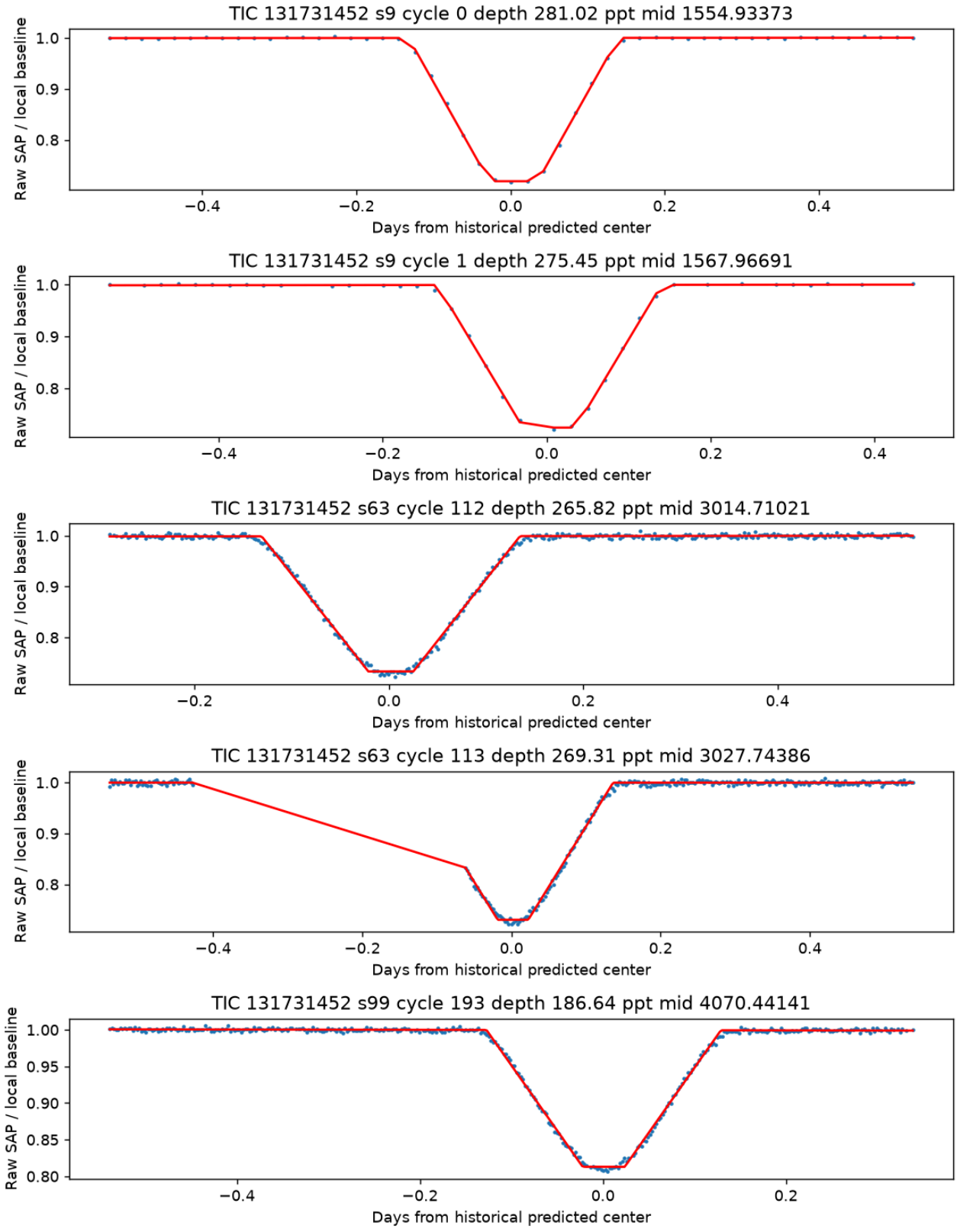
R417. TIC 131731452

physical_verdict: CONFIRMED

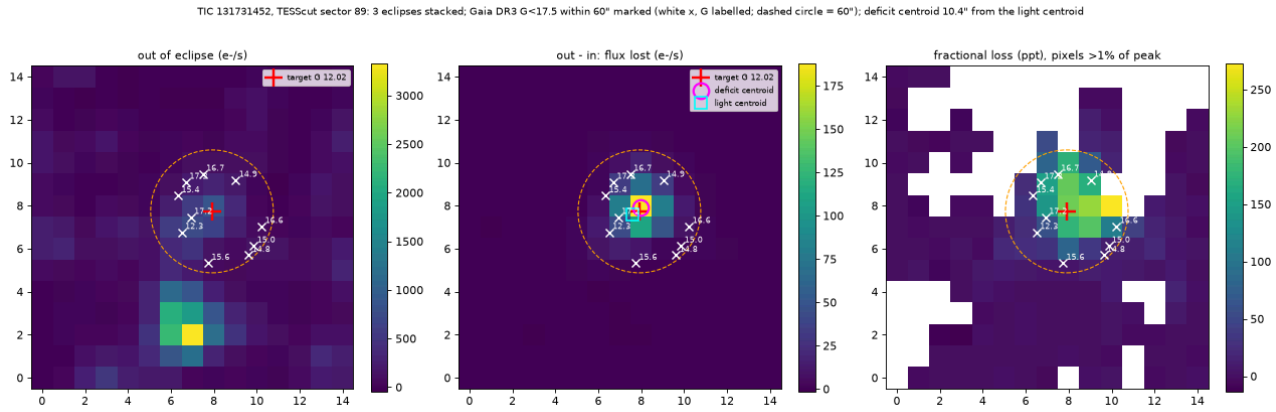
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R417; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R417; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 131731452: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured eclipses in representative sectors [9, 63, 99]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 5 actual midpoints yield P = 13.0337190439 ± 0.0000167190 d and reference epoch BTJD 1554.9336392 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 89 cutout puts the deficit 3.67 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.764; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 5406501911631243520: separation 32.41 arcsec, G 12.282, approximate intrinsic eclipse requirement 0.553. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 210.62/164.58/119.31 ppt in 3 stacked events; median selected QLP raw fit depth 269.31 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5406501915941740928 has orbital P = 13.0317563802 d, e = 0.15286; closest edge-on conjunction offset at the measured epoch is -0.06192 d (-0.00475 phase), with 0.15776 d linearly propagated full-SB1-covariance timing error. The rough central duration is 7.22 h versus observed median 6.44 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Five deep raw eclipses S9/63/99. A capable neighbor 32.4 arcsec away is distinct from the target-centered core; high pixel-model residual and changing aperture dilution limit intrinsic depth interpretation.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
9	0	1554.9337346	0.010417	281.02	used
9	1	1567.9669072	0.010417	275.45	used
63	112	3014.7102093	0.001157	265.82	used
63	113	3027.7438640	0.001157	269.31	used
99	193	4070.4414090	0.001157	186.64	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s89.json, historical/diff_s89.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R417. Public evidence files: diff_s89.json, diff_s89.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

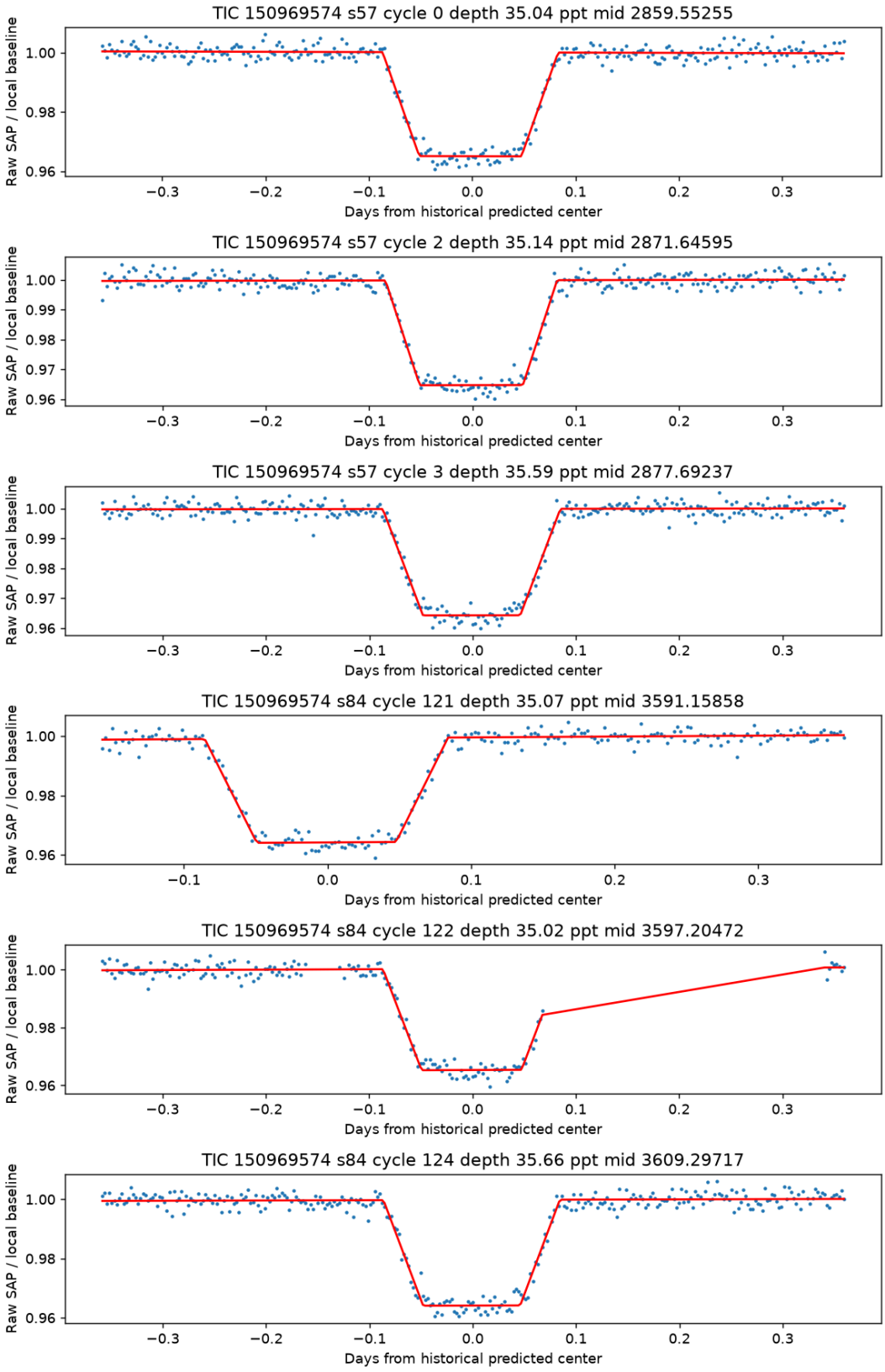
R418. TIC 150969574

physical_verdict: CONFIRMED

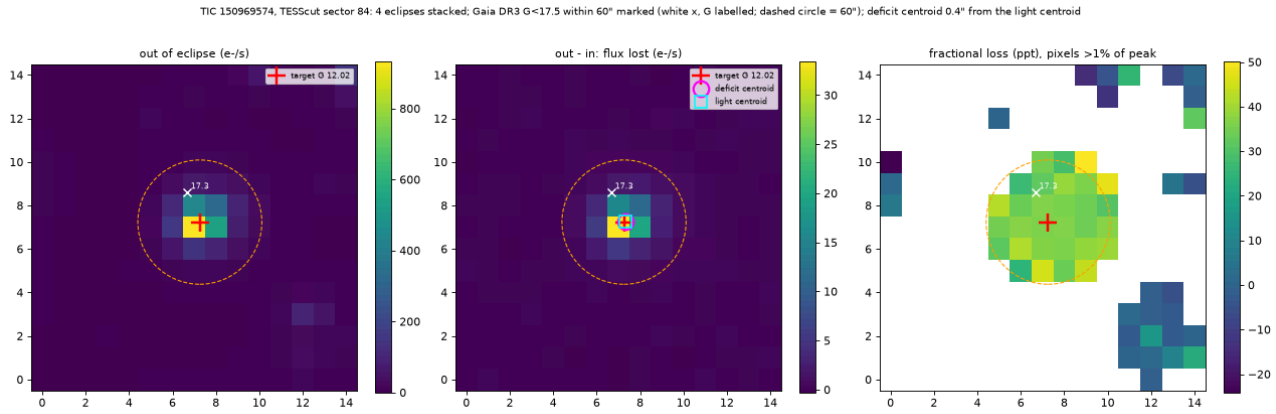
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R418; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R418; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 150969574: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured eclipses in representative sectors [57, 84]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 8 actual midpoints yield P = 6.0463235428 ± 0.0000070222 d and reference epoch BTJD 2859.5530800 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 84 cutout puts the deficit 1.35 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.031; this is not a calibrated localization error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 35.85/36.10/35.95 ppt in 4 stacked events; median selected QLP raw fit depth 35.32 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 2799480161937494912 has orbital P = 6.0462607078 d, e = 0.08689; closest edge-on conjunction offset at the measured epoch is 0.12220 d (0.02021 phase), with 0.45776 d linearly propagated full-SB1-covariance timing error. The rough central duration is 4.48 h versus observed median 4.08 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Eight repeated raw eclipses S57/84; target-centered difference core and no capable Gaia neighbor under the brightness screen.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
57	0	2859.5525475	0.001157	35.04	used
57	2	2871.6459482	0.001157	35.14	used
57	3	2877.6923742	0.001157	35.59	used
84	120	3585.1115748	0.001157	35.53	used
84	121	3591.1585797	0.001157	35.07	used
84	122	3597.2047197	0.001157	35.02	used
84	123	3603.2507075	0.001157	35.51	used
84	124	3609.2971671	0.001157	35.66	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s84.json, historical/diff_s84.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R418. Public evidence files: diff_s84.json, diff_s84.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

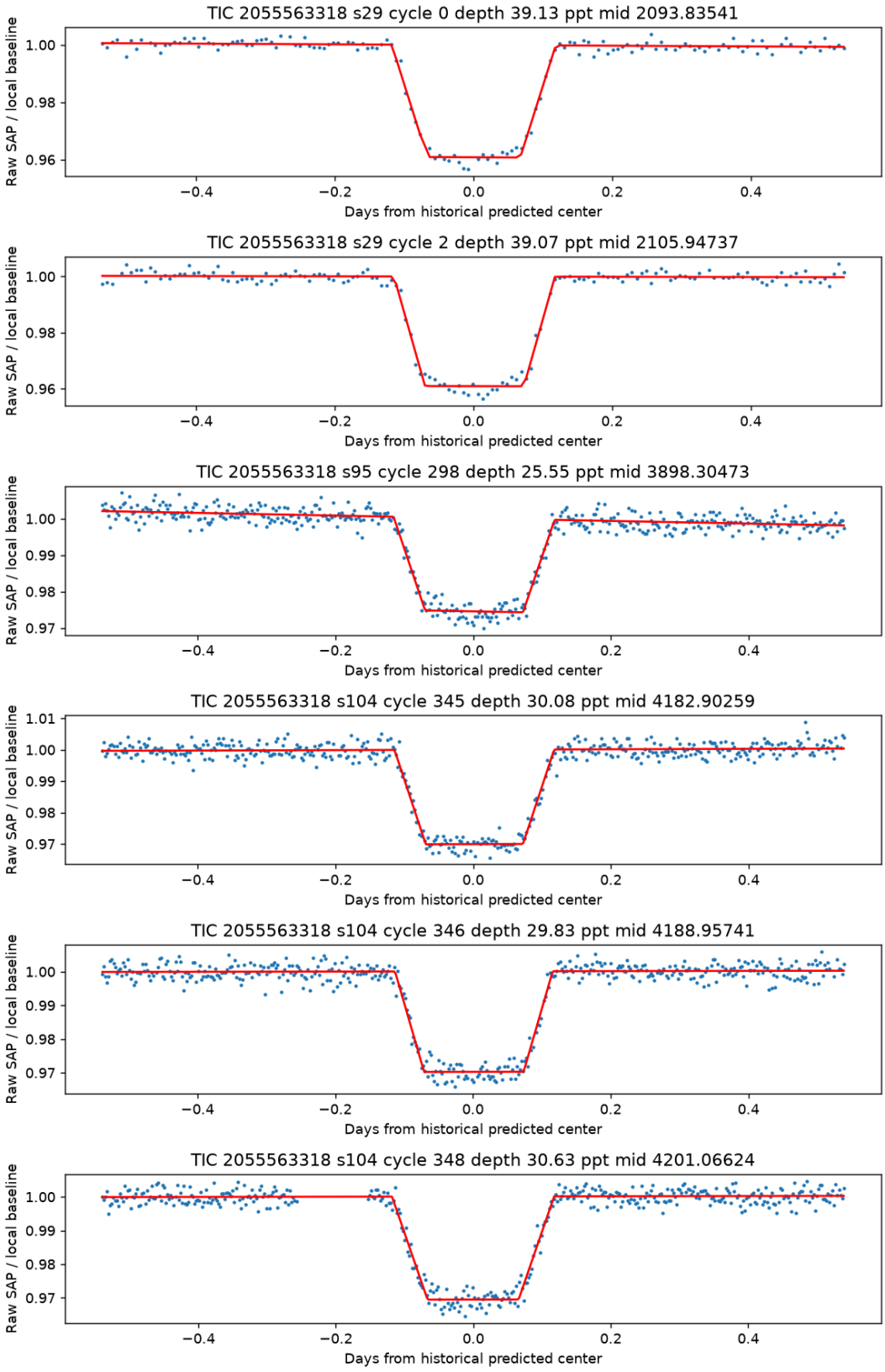
R419. TIC 2055563318

physical_verdict: PROBABLE

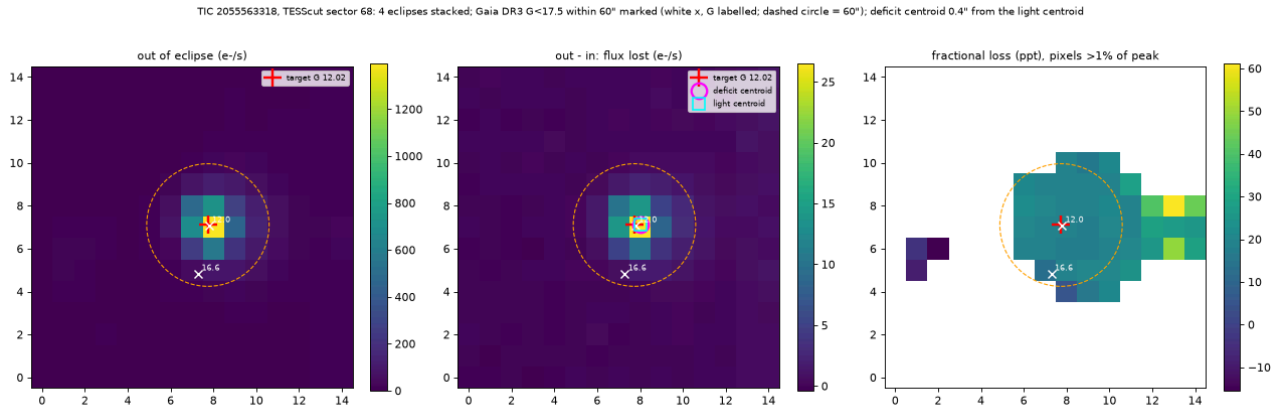
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry / multi-source PRF localization



R419; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R419; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 2055563318: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured eclipses in representative sectors [29, 95, 104]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 8 actual midpoints yield P = 6.0552614725 ± 0.0000071310 d and reference epoch BTJD 2093.8367290 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: unresolved after completed check.** Historical sector 68 cutout puts the deficit 5.79 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.044; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 6524911983920449664: separation 1.11 arcsec, G 11.971, approximate intrinsic eclipse requirement 0.059. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 19.01/19.01/19.12 ppt in 4 stacked events; median selected QLP raw fit depth 30.23 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 6524911983916150016 has orbital P = 6.0544220136 d, e = 0.14861; closest edge-on conjunction offset at the measured epoch is 0.12018 d (0.01985 phase), with 0.29407 d linearly propagated full-SB1-covariance timing error. The rough central duration is 3.87 h versus observed median 5.59 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Eight repeated raw eclipses S29/95/104 show a real recurring signal, but a capable Gaia neighbor only 1.1 arcsec away shares the central TESS pixel. The image cannot uniquely identify the eclipsing star; geometry agrees with the selected Gaia orbit but does not cure this attribution limit.

Next check: Resolved eclipse photometry or multi-source PRF localization of the 1.1-arcsec pair.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
29	0	2093.8354098	0.003472	39.13	used
29	2	2105.9473696	0.003472	39.07	used
95	298	3898.3047275	0.001157	25.55	used
95	299	3904.3607308	0.001157	25.95	used
104	345	4182.9025932	0.001157	30.08	used
104	346	4188.9574118	0.001157	29.83	used
104	347	4195.0123085	0.001157	30.38	used
104	348	4201.0662356	0.001157	30.63	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s68.json, historical/diff_s68.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R419. Public evidence files: diff_s68.json, diff_s68.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

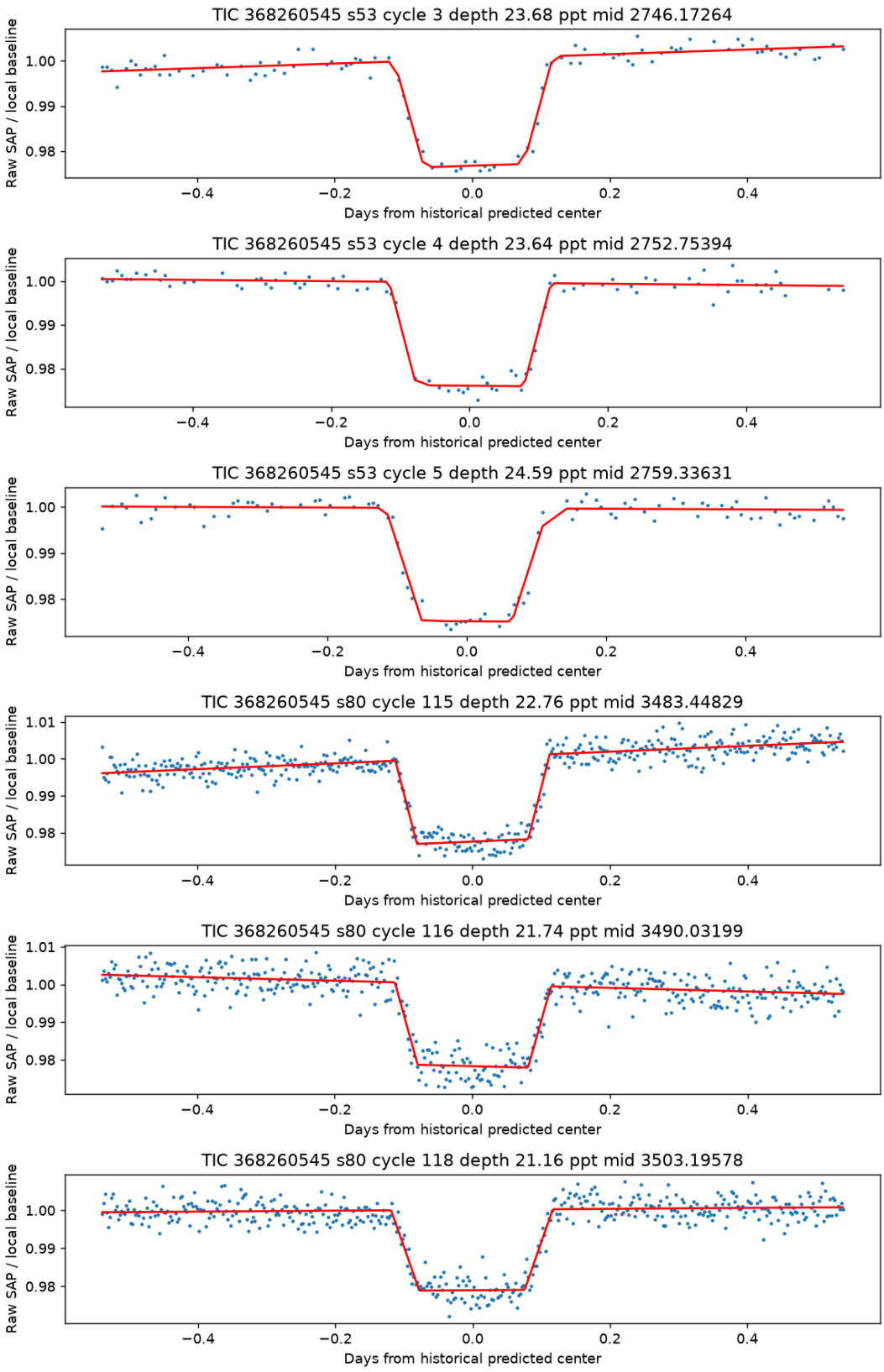
R420. TIC 368260545

physical_verdict: CONFIRMED

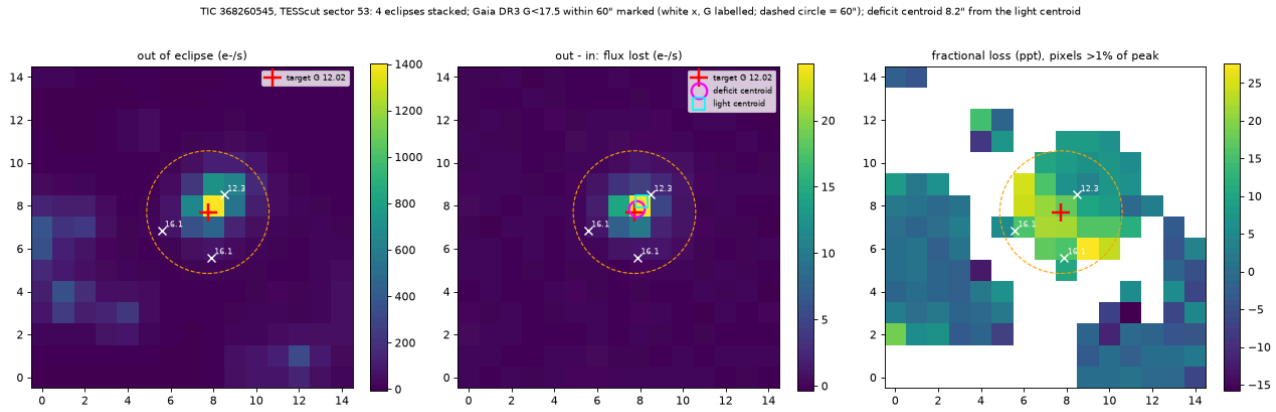
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional pixel orientations or spatially resolved photometry can strengthen attribution and refine the ephemeris.



R420; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R420; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 368260545: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 11; Gaia DR3 source 4556083135603641856; actual DR3 G=12.021002. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 8 usable measured eclipse centers across sectors 53, 80; 8 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 23.20 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=6.582836723 ± 0.000016330 d and reference center BTJD 2746.170982. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=6.583413245 d; difference=-0.000577 d.
- Pixel attribution and alternative reduction: historical TESScut sector 53 stacked 4 eclipse windows; the flux-deficit centroid is 0.206 pixel from the Gaia target and the peak deficit pixel is [8, 8] versus target pixel [8, 8]. Custom 1×1/3×3/5×5 pixel apertures give 17.3/14.4/14.0 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 18 Gaia neighbors within 60 arcsec were tested for flux capability; 1 is capable. Nearest capable: G=12.30, 22.87 arcsec, requiring 5.4% intrinsic eclipse, 0.92 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset -0.02615 in orbital phase, linear covariance scale 0.220 d. Approximate central duration 5.80 h versus observed median 5.57 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

A similarly bright G=12.30 Gaia neighbor 22.87 arcsec away is flux-capable, but the deficit is 0.206 pixel from the target versus 0.92 pixel from that neighbor and peaks on the target pixel. A targeted recomputation of all four stacked windows puts every event deficit closer to the target (0.13–0.47 pixel) than the neighbor (0.75–1.07 pixel). Positional uncertainty remains descriptive rather than calibrated.

Additional pixel orientations or spatially resolved photometry can strengthen attribution and refine the ephemeris.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/368260545/diff_s53.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet368260545/tesscut_s053.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`. Targeted four-window centroid audit: `pixel-event-audit.json`, reproduced by `../audit_368_pixels.py`.

Stable review key: R420. Public evidence files: diff_s53.json, diff_s53.png, gaia.json, measurements.json, pixel-event-audit.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

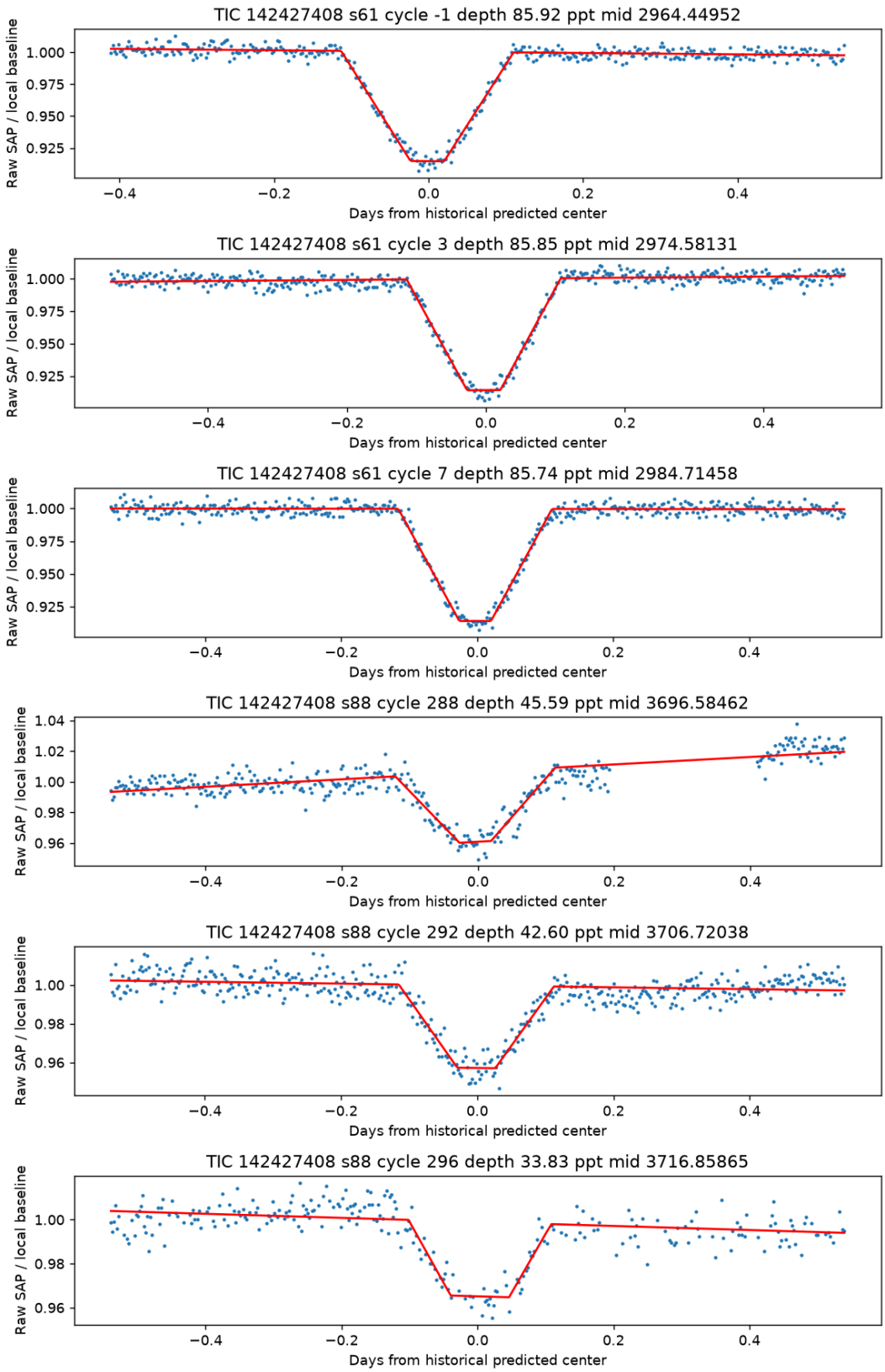
R421. TIC 142427408

physical_verdict: CONFIRMED

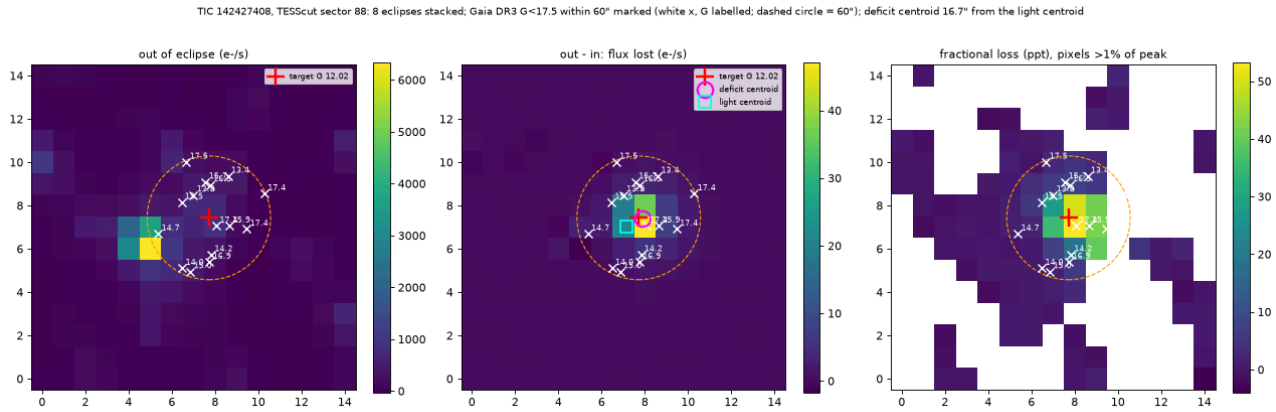
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R421; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R421; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 142427408: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 12; Gaia DR3 source 5711508945433894656; actual DR3 G=12.021337. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 19 usable measured eclipse centers across sectors 61, 88; 9 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 85.74 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=5.066684108 ± 0.000005802 d and reference center BTJD 2964.448321. Epoch kind: Selected deeper repeating eclipse; odd historical half-period cycles. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=5.066415582 d; difference=+0.000269 d.
- Pixel attribution and alternative reduction: historical TESScut sector 88 stacked 8 eclipse windows; the flux-deficit centroid is 0.172 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 53.3/37.1/18.8 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 71 Gaia neighbors within 60 arcsec were tested for flux capability; 7 are capable. Nearest capable: G=13.77, 27.00 arcsec, requiring 23.3% intrinsic eclipse, 1.42 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset +0.03100 in orbital phase, linear covariance scale 0.240 d. Approximate central duration 8.26 h versus observed median 5.38 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage. Alternating half-period events were separated by eclipse kind before orbital timing. Crowded pixel PSF fit is poor, but all flux-capable neighbors are at least 27 arcsec from the target and farther from the deficit.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/142427408/diff_s88.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet142427408/tesscut_s088.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R421. Public evidence files: diff_s88.json, diff_s88.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

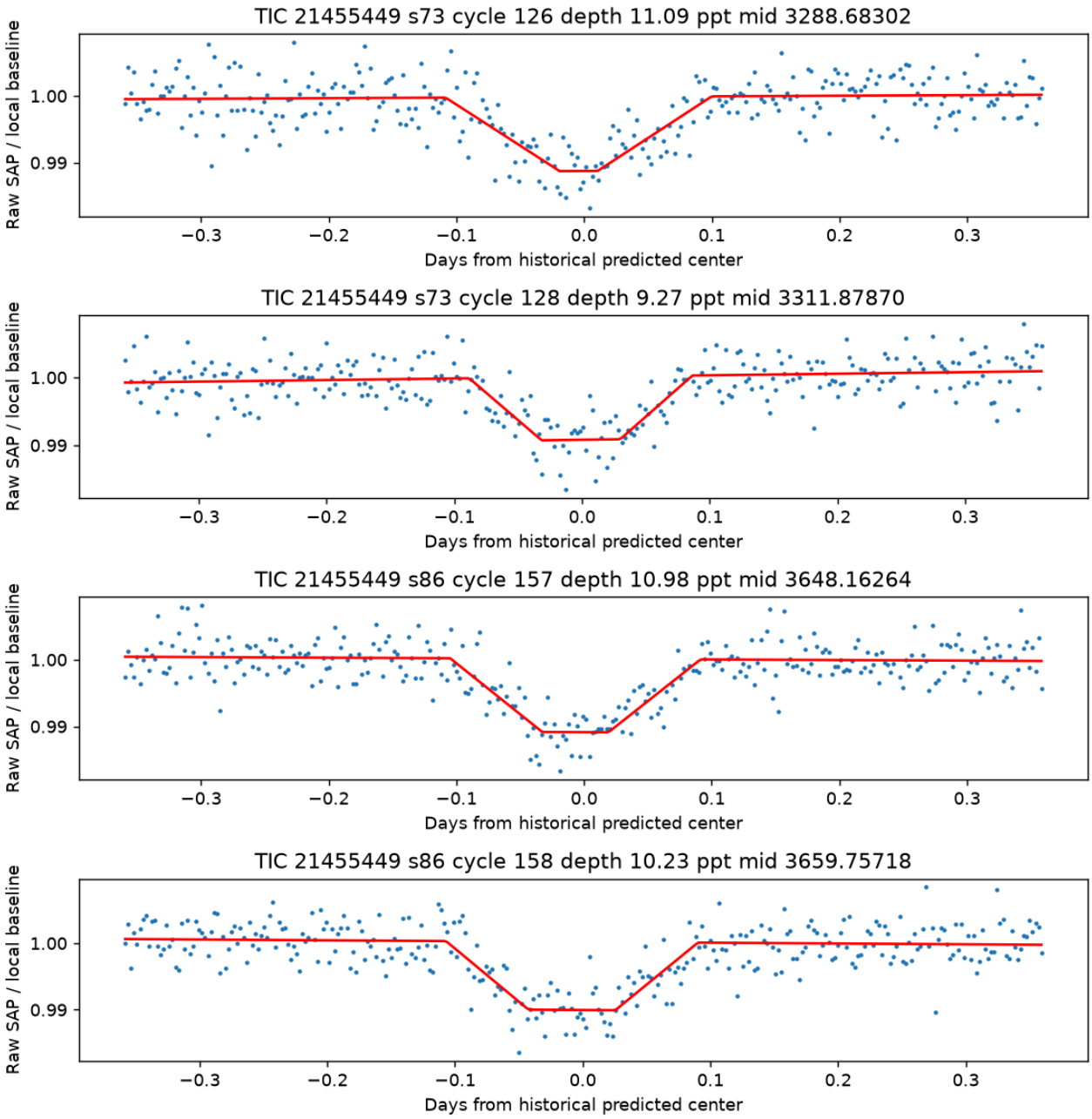
R422. TIC 21455449

physical_verdict: CONFIRMED

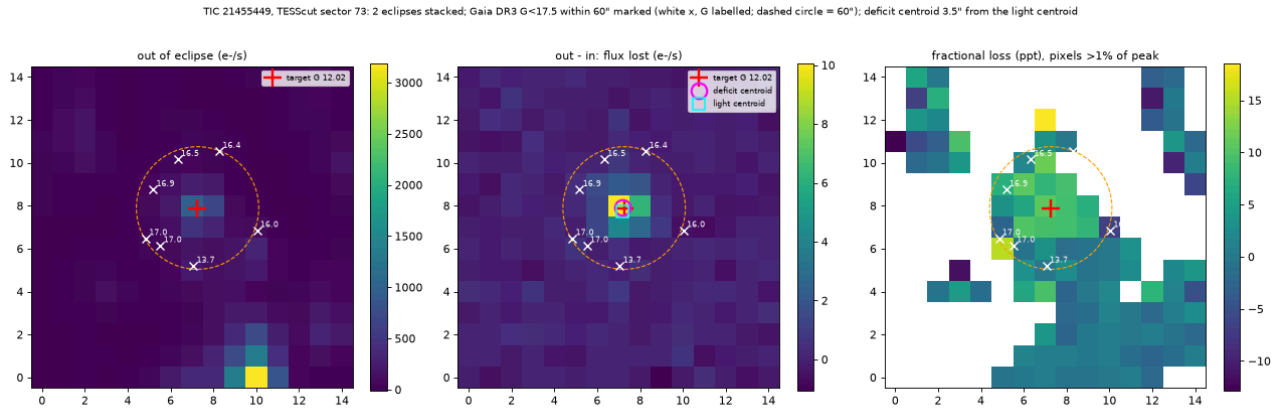
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R422; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R422; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 21455449: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 13; Gaia DR3 source 254615934869825920; actual DR3 G=12.021999. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 4 usable measured eclipse centers across sectors 73, 86; 4 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 10.61 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=11.596027763 ± 0.000084033 d and reference center BTJD 3288.685159. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=11.595182256 d; difference=+0.000846 d.
- Pixel attribution and alternative reduction: historical TESScut sector 73 stacked 2 eclipse windows; the flux-deficit centroid is 0.080 pixel from the Gaia target and the peak deficit pixel is [7, 8] versus target pixel [7, 8]. Custom 1×1/3×3/5×5 pixel apertures give 10.6/9.4/8.0 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 43 Gaia neighbors within 60 arcsec were tested for flux capability; 7 are capable. Nearest capable: G=16.92, 44.98 arcsec, requiring 87.2% intrinsic eclipse, 2.17 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset +0.01425 in orbital phase, linear covariance scale 0.610 d. Approximate central duration 14.66 h versus observed median 4.72 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage. Gaia duration estimate is crude and exceeds the observed narrow event; geometry phase and period remain consistent.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/21455449/diff_s73.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet21455449/tesscut_s073.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R422. Public evidence files: diff_s73.json, diff_s73.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

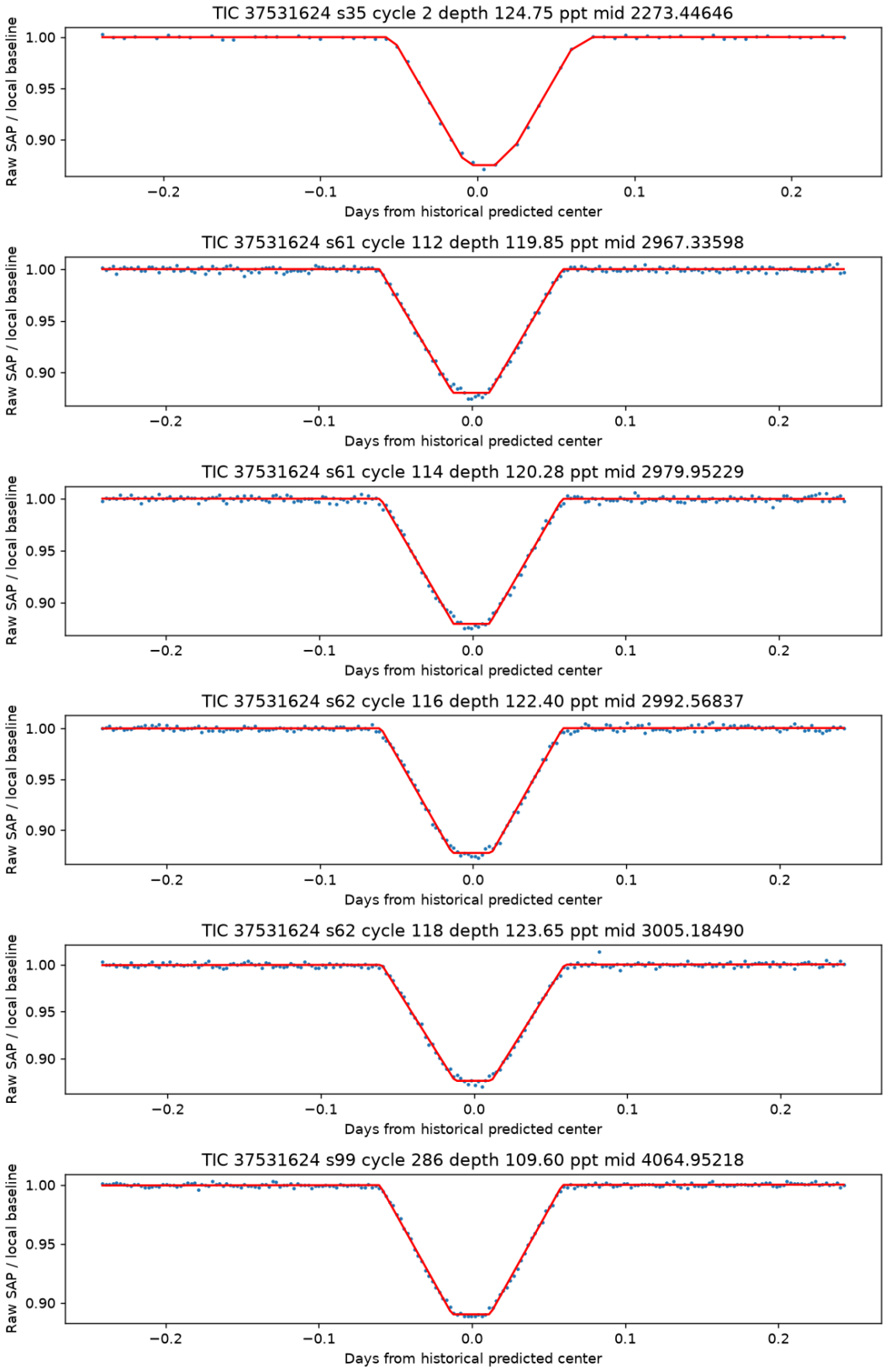
R423. TIC 37531624

physical_verdict: CONFIRMED

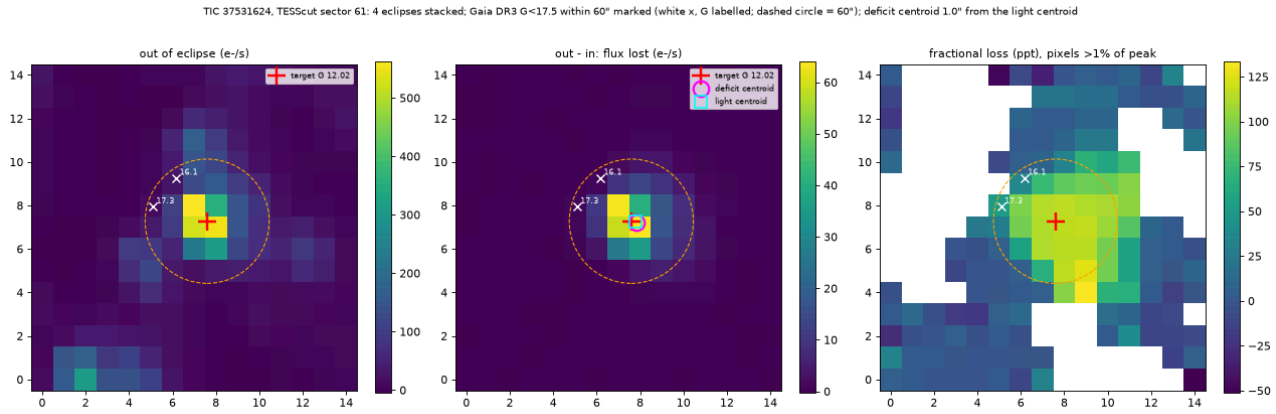
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R423; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R423; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 37531624: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 14; Gaia DR3 source 5654299985347511552; actual DR3 G=12.022275. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 12 usable measured eclipse centers across sectors 35, 61, 62, 99; 12 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 121.06 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=6.308135631 ± 0.000005140 d and reference center BTJD 2273.441351. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=6.308348116 d; difference=-0.000212 d.
- Pixel attribution and alternative reduction: historical TESScut sector 61 stacked 4 eclipse windows; the flux-deficit centroid is 0.224 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 116.4/114.2/107.2 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 13 Gaia neighbors within 60 arcsec were tested for flux capability; 0 are capable. No Gaia neighbor is flux-capable.
- Gaia geometry: selected edge-on conjunction offset -0.00871 in orbital phase, linear covariance scale 0.079 d. Approximate central duration 4.47 h versus observed median 2.85 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/37531624/diff_s61.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet37531624/tesscut_s061.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R423. Public evidence files: diff_s61.json, diff_s61.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

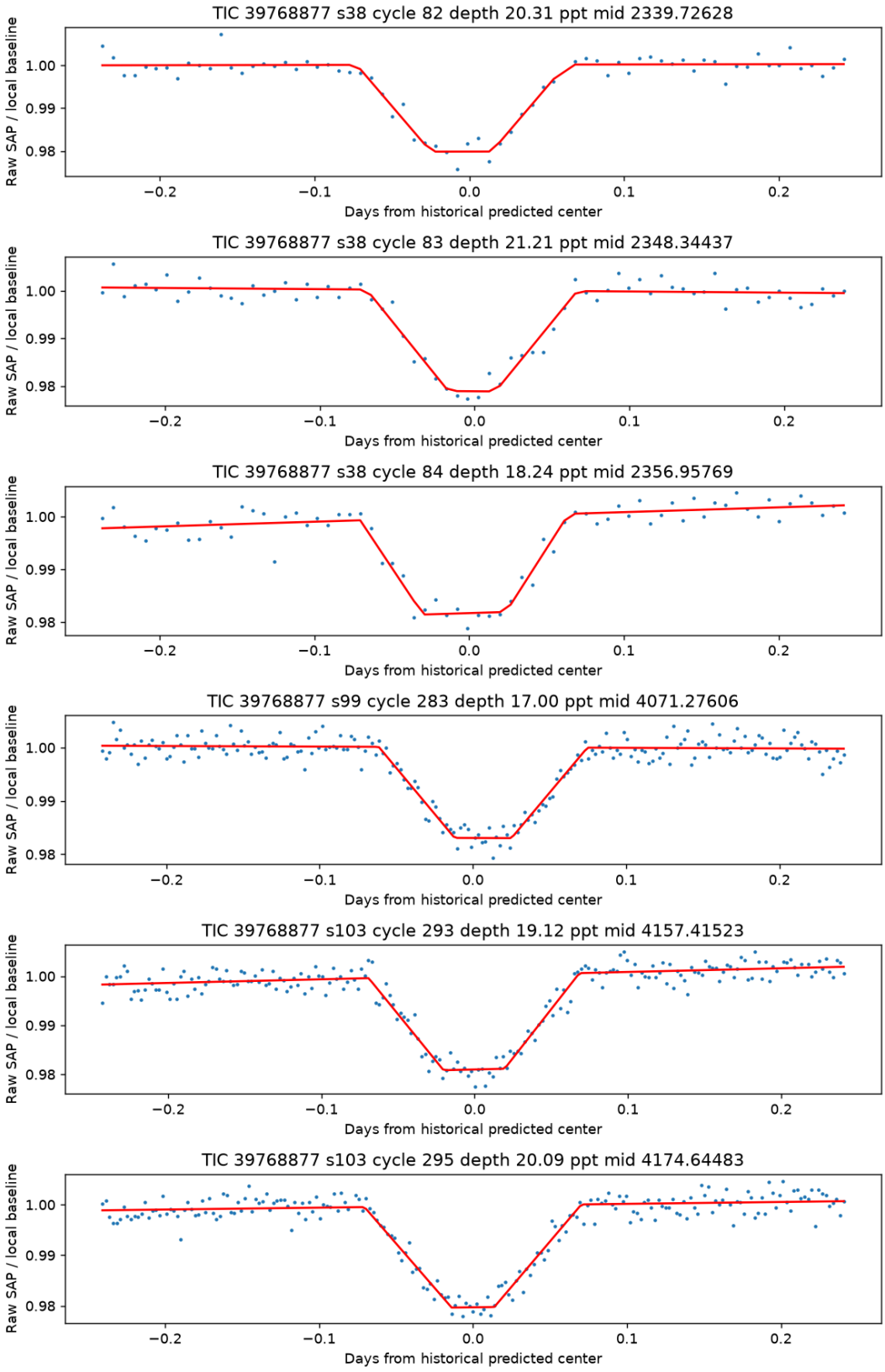
R424. TIC 39768877

physical_verdict: CONFIRMED

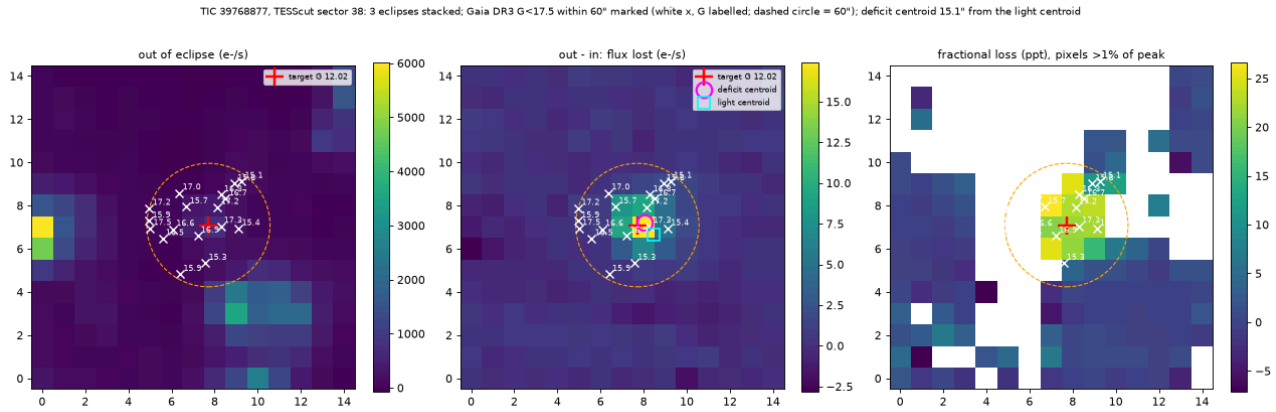
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R424; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R424; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 39768877: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 15; Gaia DR3 source 5887384768167350784; actual DR3 G=12.022689. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 8 usable measured eclipse centers across sectors 38, 99, 103; 8 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 18.97 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=8.614638525 ± 0.000026357 d and reference center BTJD 2339.729538. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=8.614908563 d; difference=-0.000270 d.
- Pixel attribution and alternative reduction: historical TESScut sector 38 stacked 3 eclipse windows; the flux-deficit centroid is 0.342 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 21.8/22.0/16.5 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 186 Gaia neighbors within 60 arcsec were tested for flux capability; 8 are capable. Nearest capable: G=15.69, 27.54 arcsec, requiring 57.4% intrinsic eclipse, 1.53 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset -0.00312 in orbital phase, linear covariance scale 0.159 d. Approximate central duration 6.91 h versus observed median 3.28 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/39768877/diff_s38.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet39768877/tesscut_s038.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R424. Public evidence files: diff_s38.json, diff_s38.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

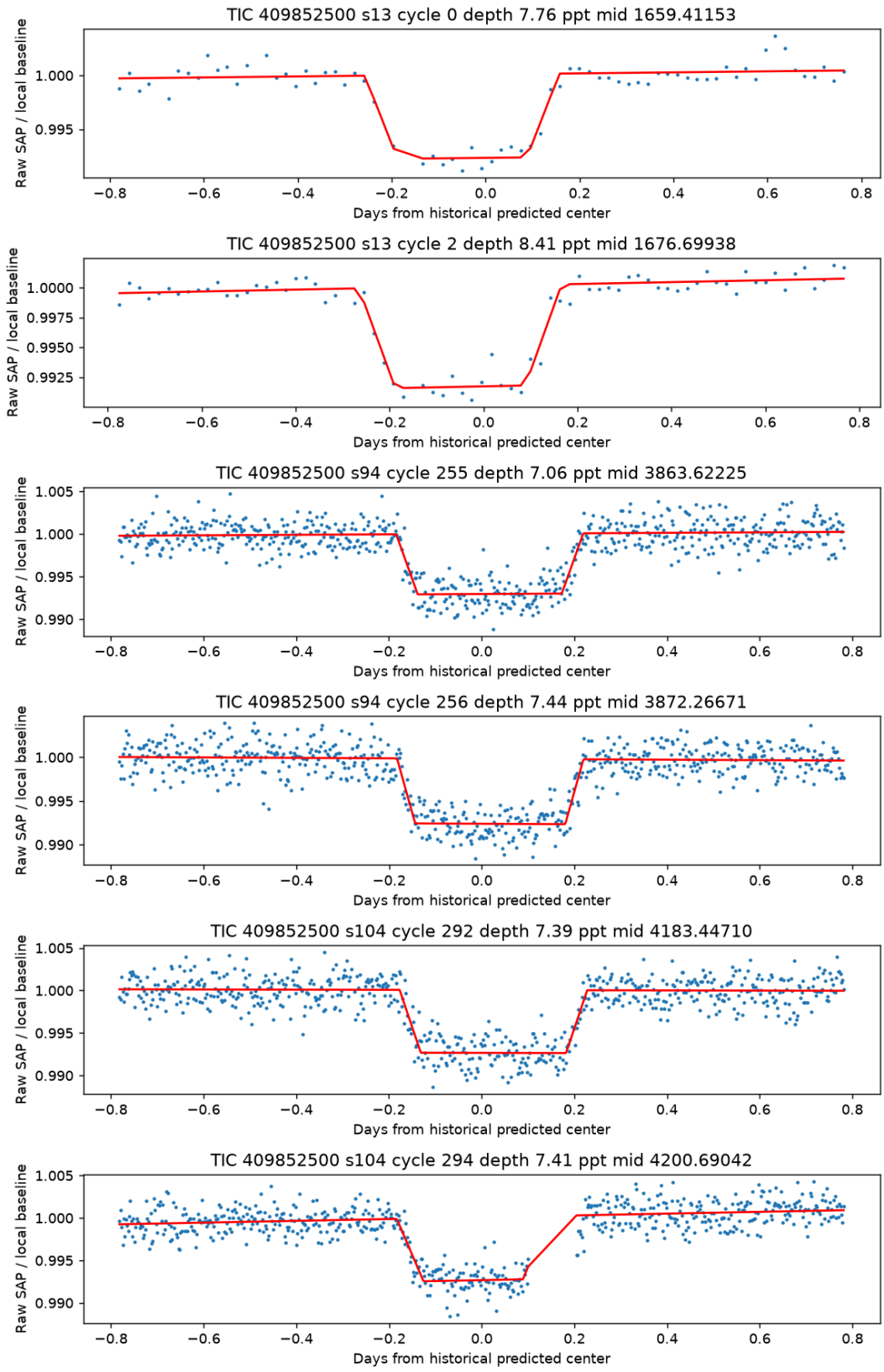
R425. TIC 409852500

physical_verdict: PROBABLE

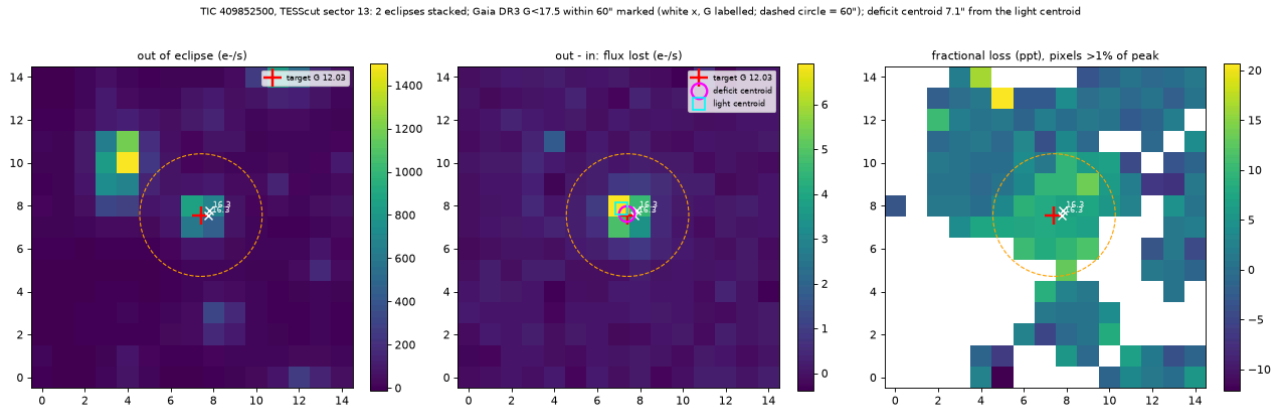
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Obtain spatially resolved eclipse photometry, or a stronger difference image in a suitable sector.



R425; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R425; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 409852500: batch 02 review

Physical verdict: probable. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 16; Gaia DR3 source 6445307372129779072; actual DR3 G=12.026384. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 5 usable measured eclipse centers across sectors 13, 94, 104; 5 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 7.44 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=8.643951250 ± 0.000025003 d and reference center BTJD 1659.414449. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=8.642794775 d; difference=+0.001156 d.
- Pixel attribution and alternative reduction: historical TESScut sector 13 stacked 2 eclipse windows; the flux-deficit centroid is 0.107 pixel from the Gaia target and the peak deficit pixel is [7, 8] versus target pixel [7, 8]. Custom 1×1/3×3/5×5 pixel apertures give 7.9/7.6/6.9 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 11 Gaia neighbors within 60 arcsec were tested for flux capability; 2 are capable. Nearest capable: G=16.33, 7.31 arcsec, requiring 41.0% intrinsic eclipse, 0.43 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset -0.00905 in orbital phase, linear covariance scale 0.421 d. Approximate central duration 11.27 h versus observed median 9.70 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Two G~16.3 neighbors at 7.3 and 9.5 arcsec are capable in flux and share a TESS pixel; the deficit centroid favors the target but does not securely resolve them. The sector-104 cycle-294 egress gap excludes that center from timing.

Obtain spatially resolved eclipse photometry, or a stronger difference image in a suitable sector.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/409852500/diff_s13.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet409852500/tesscut_s013.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R425. Public evidence files: diff_s13.json, diff_s13.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

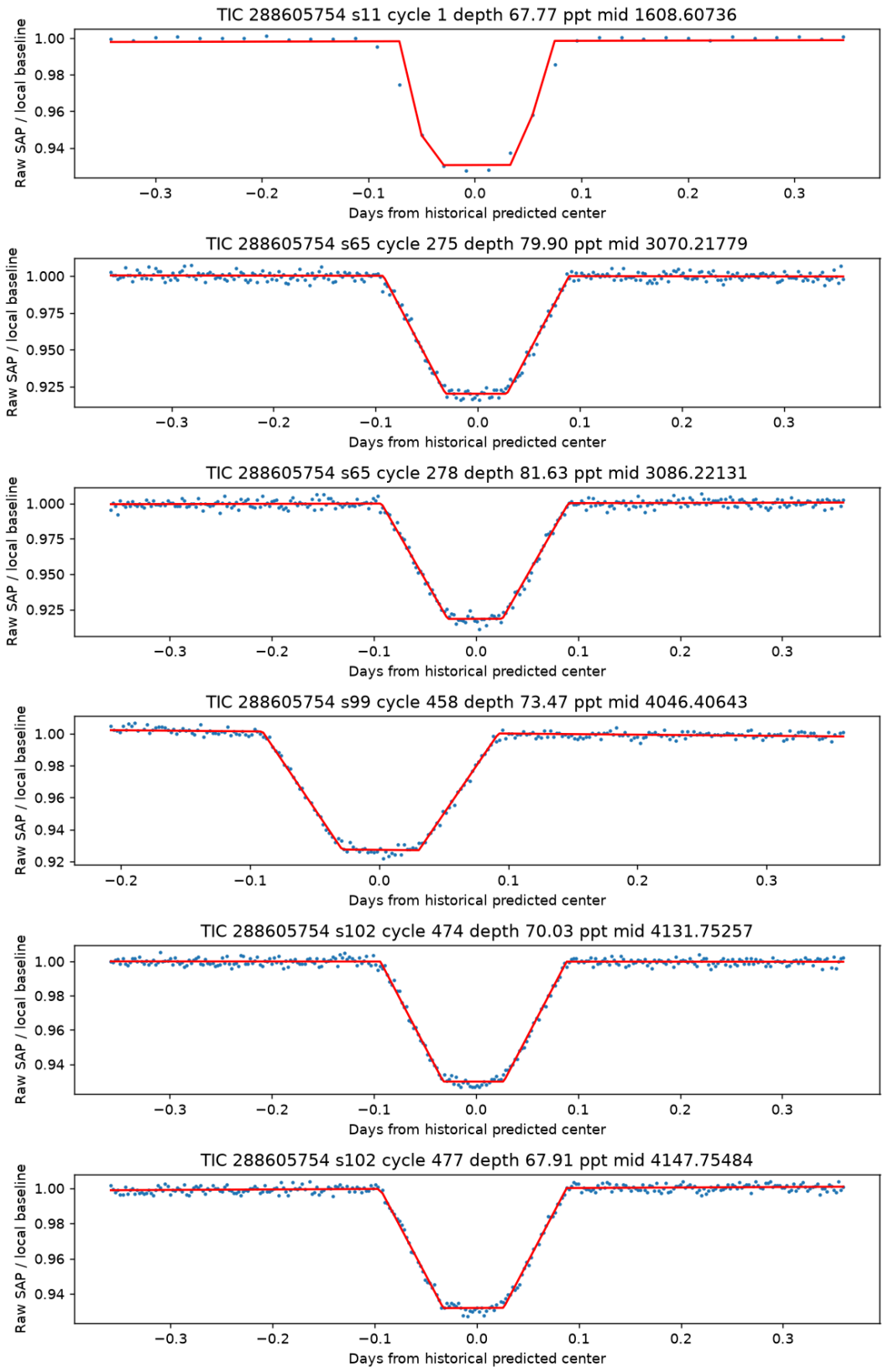
R426. TIC 288605754

physical_verdict: CONFIRMED

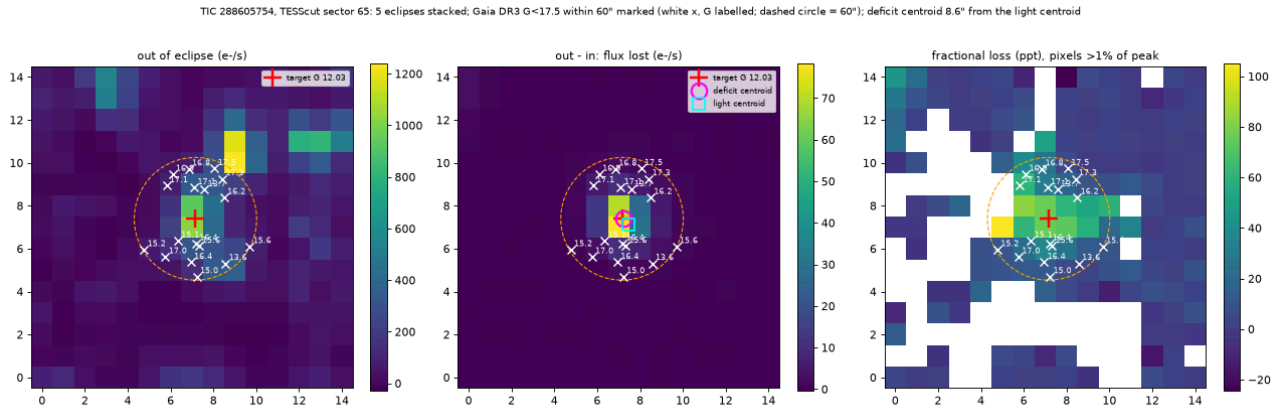
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R426; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R426; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 288605754: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 17; Gaia DR3 source 5894485208091053056; actual DR3 G=12.027896. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 14 usable measured eclipse centers across sectors 11, 65, 99, 102; 14 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 72.32 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=5.334347769 ± 0.000004228 d and reference center BTJD 1608.607034. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=5.334169975 d; difference=+0.000178 d.
- Pixel attribution and alternative reduction: historical TESScut sector 65 stacked 5 eclipse windows; the flux-deficit centroid is 0.059 pixel from the Gaia target and the peak deficit pixel is [7, 7] versus target pixel [7, 7]. Custom 1×1/3×3/5×5 pixel apertures give 76.6/64.3/47.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 164 Gaia neighbors within 60 arcsec were tested for flux capability; 2 are capable. Nearest capable: G=13.72, 29.39 arcsec, requiring 41.2% intrinsic eclipse, 1.38 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset +0.01007 in orbital phase, linear covariance scale 0.130 d. Approximate central duration 6.51 h versus observed median 4.38 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/288605754/diff_s65.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet288605754/tesscut_s065.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R426. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

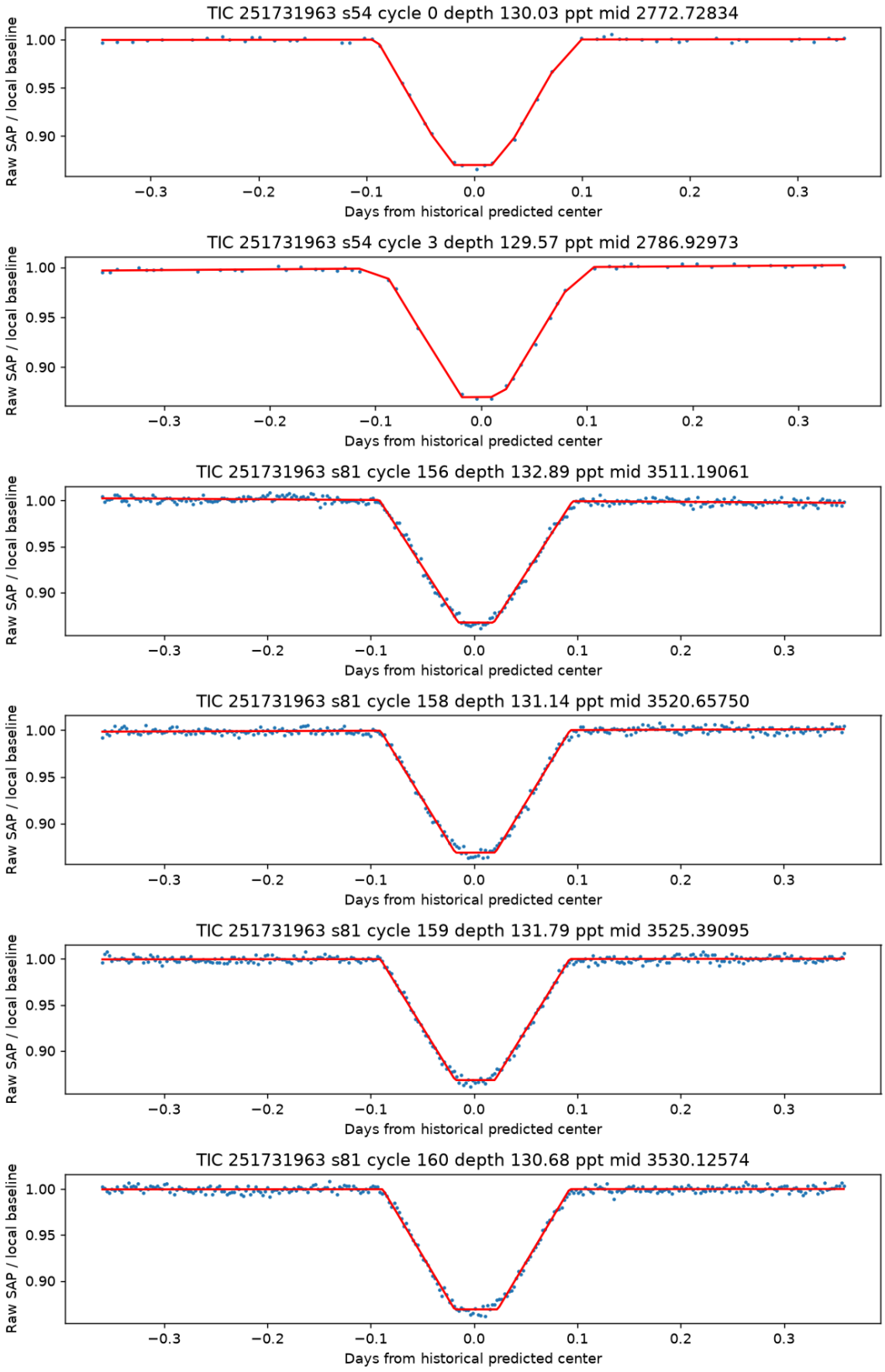
R427. TIC 251731963

physical_verdict: PROBABLE

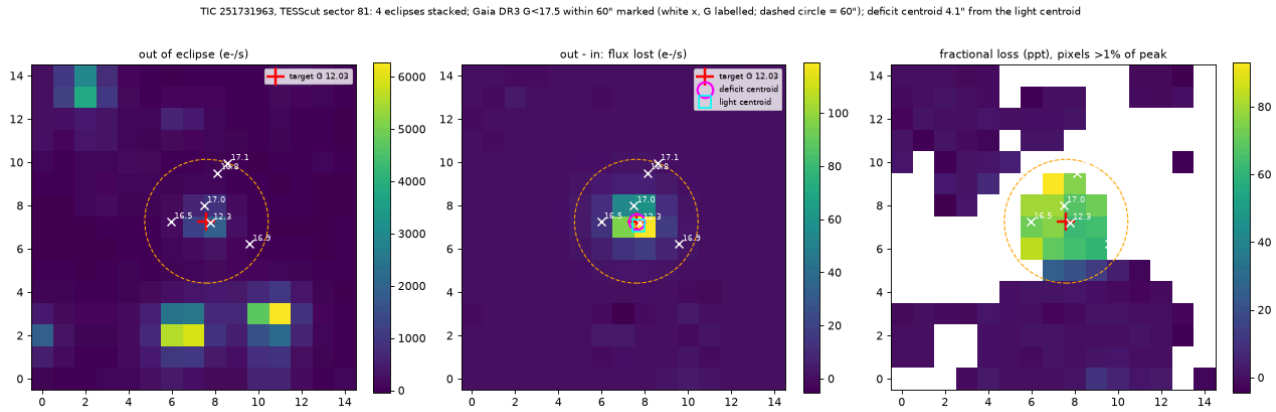
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolve the two stars in higher-spatial-resolution time-series photometry at predicted eclipse times.



R427; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R427; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 251731963: batch 02 review

Physical verdict: probable. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 18; Gaia DR3 source 4238560567095727744; actual DR3 G=12.032766. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 6 usable measured eclipse centers across sectors 54, 81; 6 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 130.91 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=4.733729823 ± 0.000031440 d and reference center BTJD 2772.728456. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=4.733742635 d; difference=-0.000013 d.
- Pixel attribution and alternative reduction: historical TESScut sector 81 stacked 4 eclipse windows; the flux-deficit centroid is 0.024 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 63.0/69.0/66.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 73 Gaia neighbors within 60 arcsec were tested for flux capability; 1 is capable. Nearest capable: G=12.27, 3.69 arcsec, requiring 27.5% intrinsic eclipse, 0.18 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset +0.01300 in orbital phase, linear covariance scale 0.108 d. Approximate central duration 6.03 h versus observed median 4.40 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

A G=12.27 Gaia star 3.69 arcsec away can produce the dip with a ~27% intrinsic eclipse; the TESS pixel centroid cannot resolve the pair.

Resolve the two stars in higher-spatial-resolution time-series photometry at predicted eclipse times.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/251731963/diff_s81.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet251731963/tesscut_s081.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R427. Public evidence files: diff_s81.json, diff_s81.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

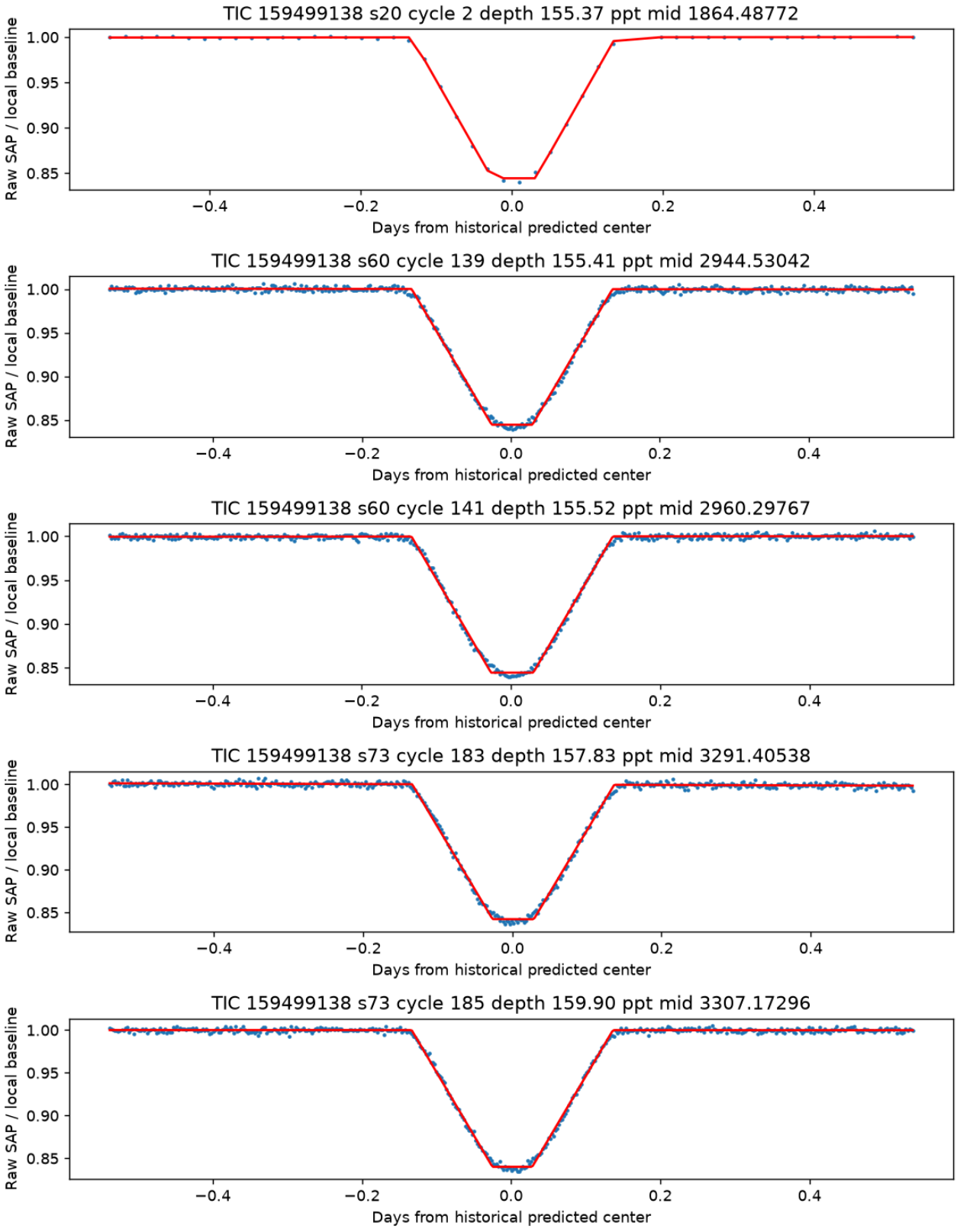
R428. TIC 159499138

physical_verdict: CONFIRMED

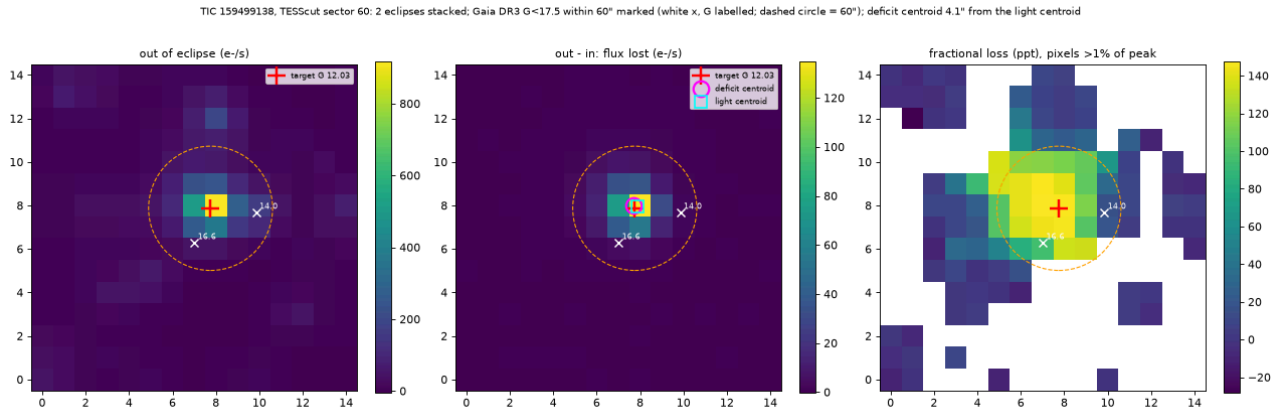
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R428; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R428; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 159499138: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 19; Gaia DR3 source 967967877695470848; actual DR3 G=12.033017. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 5 usable measured eclipse centers across sectors 20, 60, 73; 5 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 155.52 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=7.883525722 ± 0.000024373 d and reference center BTJD 1864.487494. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=7.883934523 d; difference=-0.000409 d.
- Pixel attribution and alternative reduction: historical TESScut sector 60 stacked 2 eclipse windows; the flux-deficit centroid is 0.135 pixel from the Gaia target and the peak deficit pixel is [8, 8] versus target pixel [8, 8]. Custom 1×1/3×3/5×5 pixel apertures give 147.0/139.6/127.9 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 10 Gaia neighbors within 60 arcsec were tested for flux capability; 0 are capable. No Gaia neighbor is flux-capable.
- Gaia geometry: selected edge-on conjunction offset -0.01428 in orbital phase, linear covariance scale 0.133 d. Approximate central duration 9.62 h versus observed median 6.48 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/159499138/diff_s60.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet159499138/tesscut_s060.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R428. Public evidence files: diff_s60.json, diff_s60.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

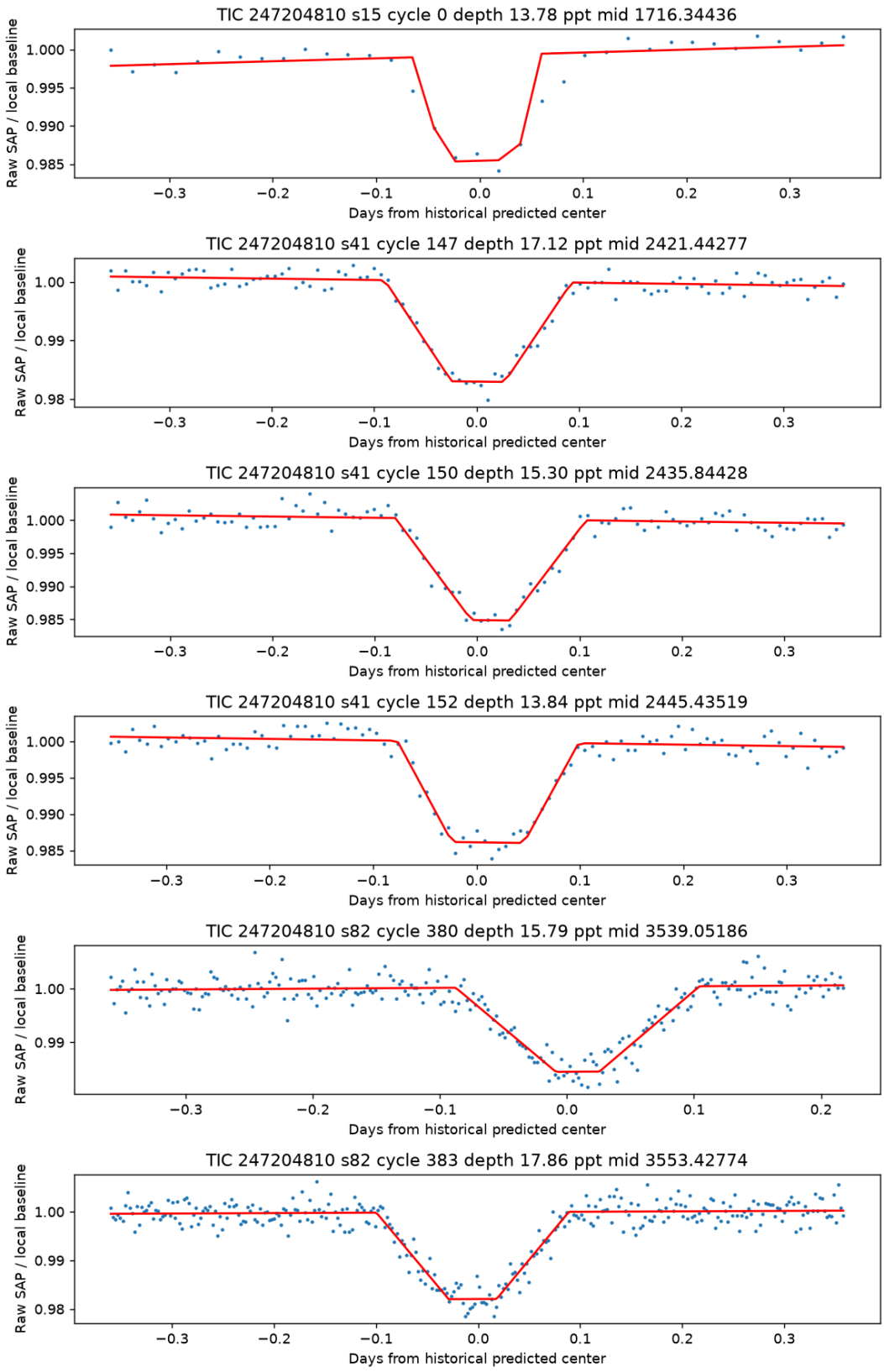
R429. TIC 247204810

physical_verdict: CONFIRMED

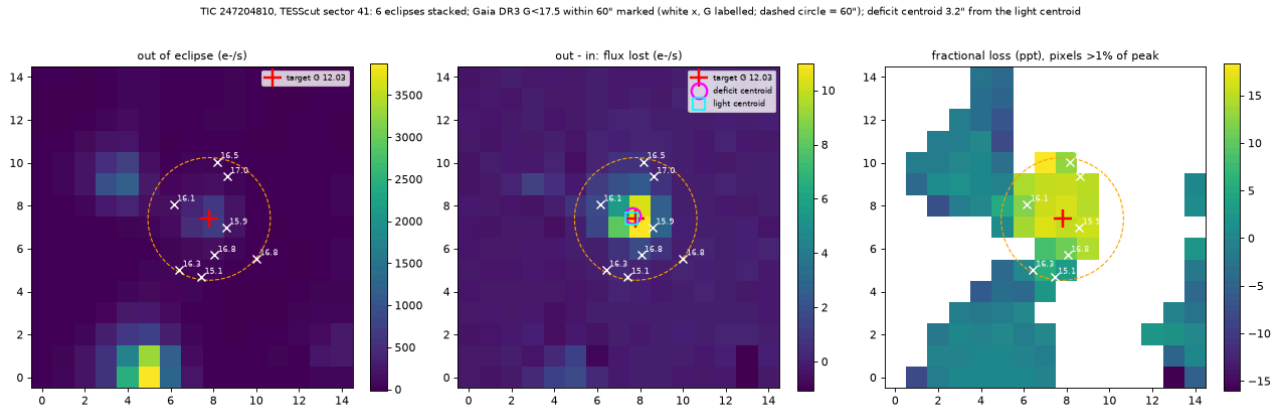
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R429; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R429; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 247204810: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 20; Gaia DR3 source 1842700536609008000; actual DR3 G=12.034502. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 12 usable measured eclipse centers across sectors 15, 41, 82; 7 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 17.16 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=9.593093220 ± 0.000017436 d and reference center BTJD 1730.740727. Epoch kind: Selected deeper repeating eclipse; odd historical half-period cycles. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=9.594030212 d; difference=-0.000937 d.
- Pixel attribution and alternative reduction: historical TESScut sector 41 stacked 6 eclipse windows; the flux-deficit centroid is 0.192 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 15.4/15.3/14.4 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 42 Gaia neighbors within 60 arcsec were tested for flux capability; 5 are capable. Nearest capable: G=15.90, 17.29 arcsec, requiring 57.7% intrinsic eclipse, 1.07 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset -0.01414 in orbital phase, linear covariance scale 0.111 d. Approximate central duration 10.53 h versus observed median 4.53 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage. Alternating half-period events were separated by eclipse kind before orbital timing.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/247204810/diff_s41.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet247204810/tesscut_s041.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R429. Public evidence files: diff_s41.json, diff_s41.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

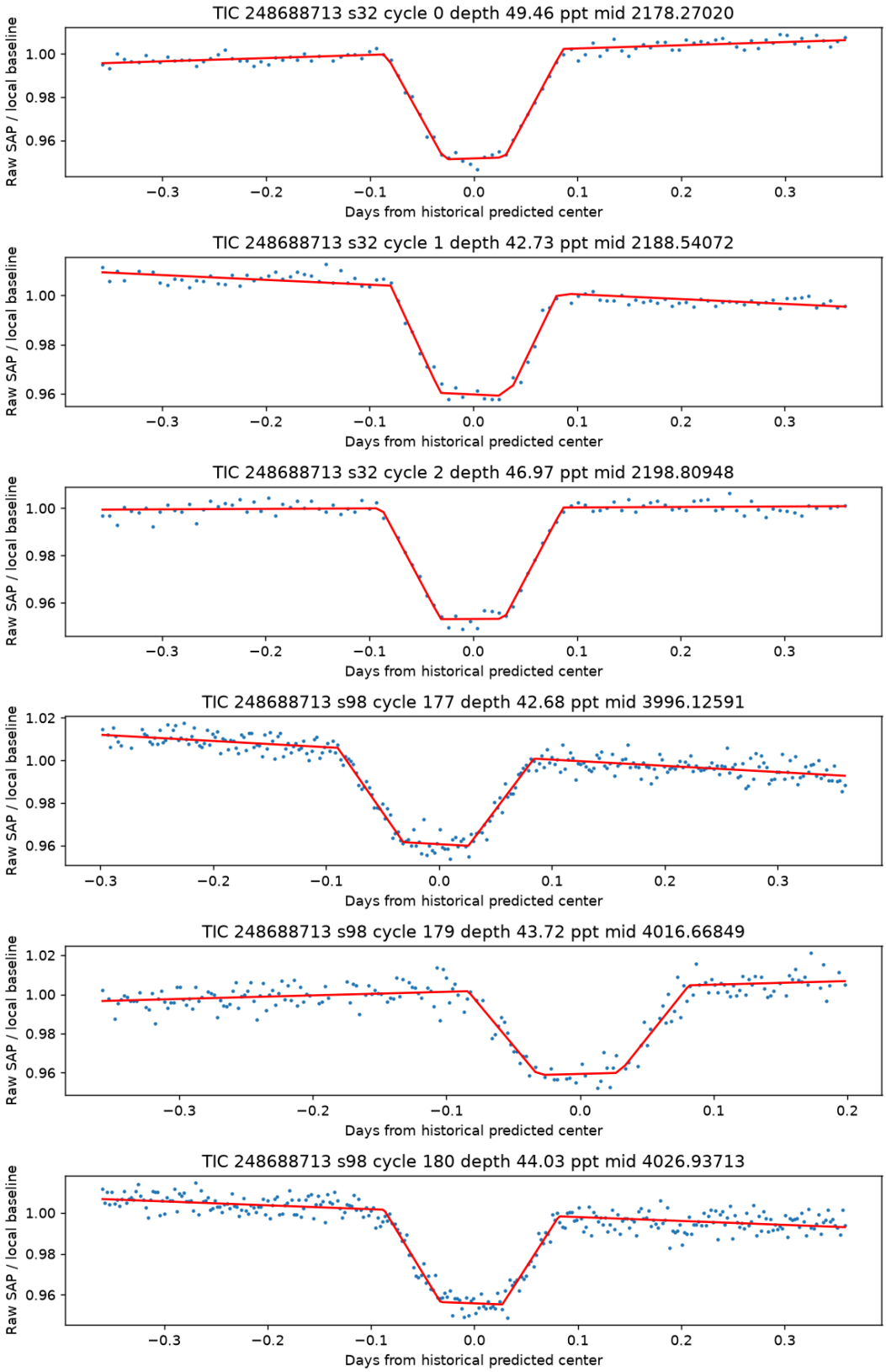
R430. TIC 248688713

physical_verdict: CONFIRMED

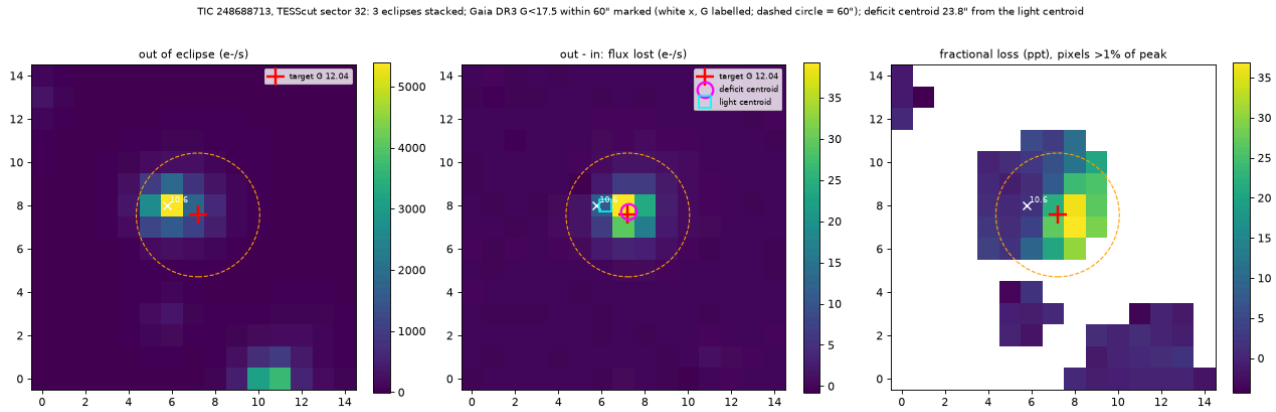
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R430; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R430; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 248688713: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured centers in sectors [32, 98]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 10.2703783974 ± 0.0000121037 d and reference epoch BTJD 2178.2697573 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 3 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 32 TESScut difference image puts the deficit 3.00 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.867 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 3205176248663357440 (31.57 arcsec, G 10.569, intrinsic loss approx. 0.054). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 22.05/10.14/8.07 ppt in 3 stacked historical events; representative QLP raw median fit depth is 43.88 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 3205176283023095424 has SB1 P = 10.2702471905 d and e = 0.09526. Closest edge-on conjunction offset is -0.01040 d (-0.00101 phase), with 0.82596 d linearly propagated full source-specific SB1 covariance. Rough central duration 5.62 h versus observed median fit 4.08 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Adjacent S32 cycles and S98 repeats establish recurrence. The brighter capable neighbor at 31.6 arcsec is distinct from the target-centered deficit; high image-model residual limits precision.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
32	0	2178.2702044	0.003472	49.46	used
32	1	2188.5407239	0.003472	42.73	used
32	2	2198.8094783	0.003472	46.97	used
98	177	3996.1259138	0.001736	42.68	used
98	179	4016.6684904	0.001197	43.72	used
98	180	4026.9371326	0.001314	44.03	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s32.json, historical/diff_s32.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R430. Public evidence files: diff_s32.json, diff_s32.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

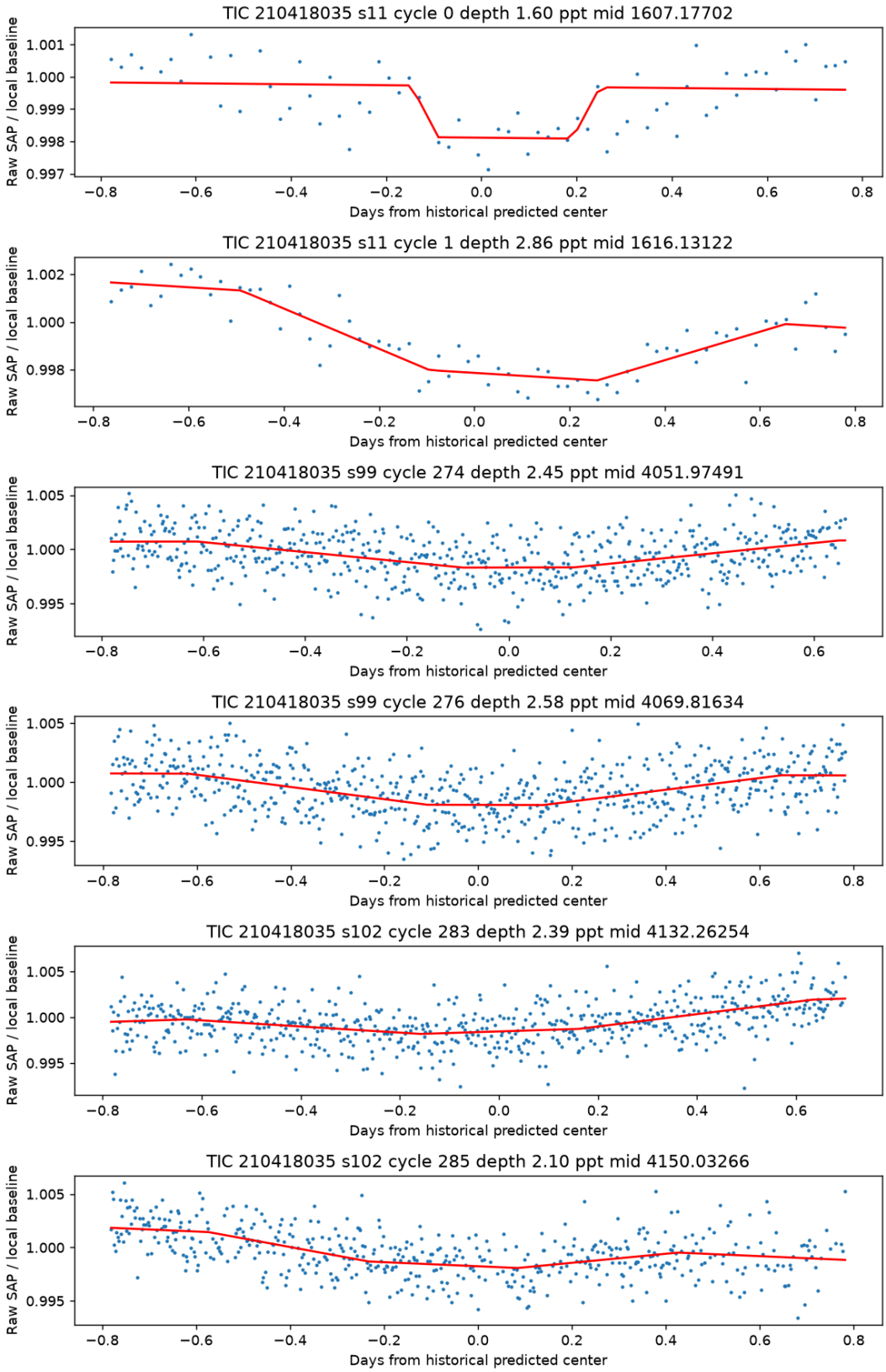
R431. TIC 210418035

physical_verdict: PLAUSIBLE

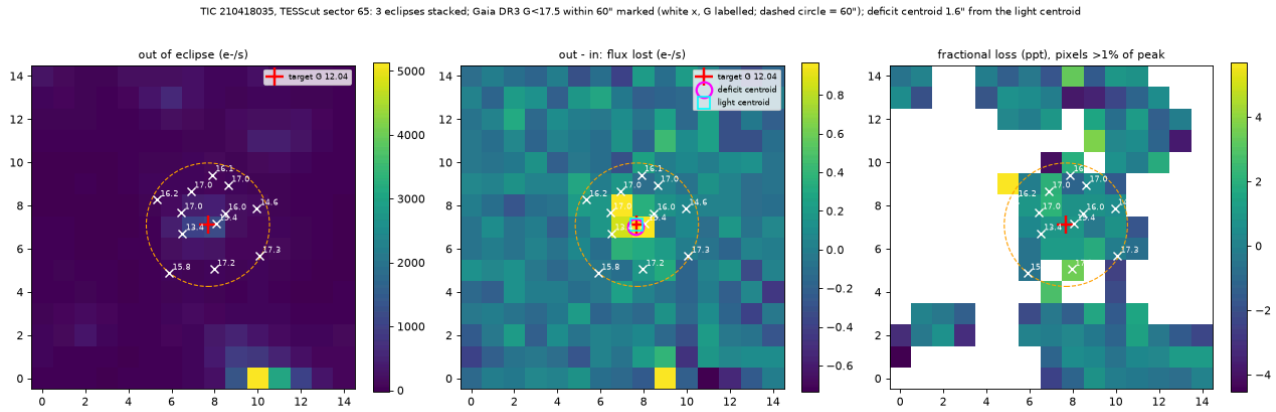
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved time-series photometry and a phase-resolved orbital model to distinguish broad variability from eclipses; repeat difference imaging in a cleaner sector.



R431; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R431; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 210418035: PLAUSIBLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 7 timing-usable measured centers in sectors [11, 99, 102]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. The morphology is eclipse-shaped but inconsistent across sectors.
2. **Timing and aliases.** Measured centers yield P = 8.9225385101 ± 0.0000978228 d and reference epoch BTJD 1607.1807945 (selected repeating phase; eclipse kind unestablished). There are 3 within-sector adjacent selected-phase pairs (../alias-audit.json). These are selected-phase adjacent pairs; the event kind is uncertain. The formal fit describes phase recurrence of a variable signal, not a secure eclipse ephemeris. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 65 TESScut difference image puts the deficit 2.12 arcsec from the target coordinate; peak [9, 0] versus target pixel [8, 7]. Image-model fractional residual 0.709 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 5896574623770523776 (7.68 arcsec, G 15.438, intrinsic loss approx. 0.021); Gaia 5896574619459900800 (9.18 arcsec, G 18.511, intrinsic loss approx. 0.346); Gaia 5896574623759948800 (10.27 arcsec, G 19.663, intrinsic loss approx. 0.999); Gaia 5896574623759957760 (15.97 arcsec, G 18.988, intrinsic loss approx. 0.537); Gaia 5896574619457099008 (18.58 arcsec, G 18.799, intrinsic loss approx. 0.451); Gaia 5896574623759957888 (19.52 arcsec, G 18.876, intrinsic loss approx. 0.485); Gaia 5896574619455713664 (19.90 arcsec, G 15.984, intrinsic loss approx. 0.035); Gaia 5896574619457093248 (21.97 arcsec, G 18.749, intrinsic loss approx. 0.431); Gaia 5896574619457101952 (22.44 arcsec, G 18.796, intrinsic loss approx. 0.450); Gaia 5896574619457093376 (23.21 arcsec, G 19.170, intrinsic loss approx. 0.635); Gaia 5896574623759952384 (23.80 arcsec, G 19.118, intrinsic loss approx. 0.605); Gaia 5896574623759938432 (24.48 arcsec, G 19.572, intrinsic loss approx. 0.920); Gaia 5896574619457102464 (24.49 arcsec, G 19.161, intrinsic loss approx. 0.630); Gaia

5896574619457548288 (25.42 arcsec, G 19.630, intrinsic loss approx. 0.969); Gaia
5896574623759979136 (25.77 arcsec, G 19.080, intrinsic loss approx. 0.585); Gaia
5896574413297283200 (25.91 arcsec, G 13.377, intrinsic loss approx. 0.004); Gaia
5896574623759954304 (26.82 arcsec, G 19.487, intrinsic loss approx. 0.850); Gaia
5896574413297281408 (28.11 arcsec, G 17.001, intrinsic loss approx. 0.087); Gaia
5896574619457547776 (28.82 arcsec, G 19.221, intrinsic loss approx. 0.665); Gaia
5896574658119698304 (29.26 arcsec, G 19.457, intrinsic loss approx. 0.827); Gaia
5896574623759928192 (32.64 arcsec, G 19.418, intrinsic loss approx. 0.798); Gaia
5896574619457104768 (33.21 arcsec, G 18.480, intrinsic loss approx. 0.337); Gaia
5896574653816841728 (33.51 arcsec, G 19.571, intrinsic loss approx. 0.918); Gaia
5896574619457105152 (33.74 arcsec, G 19.092, intrinsic loss approx. 0.591); Gaia
5896574417612080000 (34.55 arcsec, G 18.078, intrinsic loss approx. 0.233); Gaia
5896574516376495872 (34.90 arcsec, G 17.875, intrinsic loss approx. 0.193); Gaia
5896574692490007680 (35.11 arcsec, G 19.516, intrinsic loss approx. 0.873); Gaia
5896574417612079488 (35.91 arcsec, G 16.963, intrinsic loss approx. 0.084); Gaia
5896574688176582528 (37.61 arcsec, G 17.814, intrinsic loss approx. 0.183); Gaia
5896574653816838016 (37.91 arcsec, G 18.545, intrinsic loss approx. 0.358); Gaia
5896574516377875328 (37.91 arcsec, G 19.489, intrinsic loss approx. 0.852); Gaia
5896574417612078592 (38.25 arcsec, G 17.853, intrinsic loss approx. 0.189); Gaia
5896574658119723648 (38.42 arcsec, G 19.538, intrinsic loss approx. 0.891); Gaia
5896574585097358208 (38.53 arcsec, G 18.301, intrinsic loss approx. 0.286); Gaia
5896574413297282304 (38.64 arcsec, G 17.838, intrinsic loss approx. 0.187); Gaia
5896574692479462528 (38.71 arcsec, G 19.535, intrinsic loss approx. 0.888); Gaia
5896574688176583296 (40.38 arcsec, G 19.346, intrinsic loss approx. 0.747); Gaia
5896574413297279616 (40.88 arcsec, G 17.527, intrinsic loss approx. 0.141); Gaia
5896574653824506368 (42.90 arcsec, G 17.239, intrinsic loss approx. 0.108); Gaia
5896574520691300352 (43.04 arcsec, G 17.019, intrinsic loss approx. 0.088); Gaia
5896574688176583936 (45.47 arcsec, G 19.544, intrinsic loss approx. 0.896); Gaia
5896574413298657280 (45.69 arcsec, G 19.247, intrinsic loss approx. 0.681); Gaia
5896574589400232192 (46.92 arcsec, G 18.166, intrinsic loss approx. 0.252); Gaia
5896574447658407040 (47.01 arcsec, G 19.215, intrinsic loss approx. 0.662); Gaia
5896574688182055552 (47.10 arcsec, G 19.232, intrinsic loss approx. 0.672); Gaia
5896574413298659712 (47.19 arcsec, G 18.123, intrinsic loss approx. 0.243); Gaia
5896574413298662528 (47.26 arcsec, G 18.059, intrinsic loss approx. 0.229); Gaia
5896574653816845056 (47.66 arcsec, G 17.986, intrinsic loss approx. 0.214); Gaia
5896574520691294336 (48.26 arcsec, G 16.112, intrinsic loss approx. 0.039); Gaia
5896574722536323456 (48.27 arcsec, G 18.348, intrinsic loss approx. 0.298); Gaia
5896574417601476096 (48.55 arcsec, G 19.477, intrinsic loss approx. 0.842); Gaia
5896574585095976960 (49.70 arcsec, G 14.577, intrinsic loss approx. 0.010); Gaia
5896574314522261248 (50.06 arcsec, G 19.479, intrinsic loss approx. 0.844); Gaia
5896574692493101056 (50.92 arcsec, G 17.866, intrinsic loss approx. 0.192); Gaia
5896574688176585216 (51.45 arcsec, G 19.374, intrinsic loss approx. 0.766); Gaia
5896574451961279616 (52.23 arcsec, G 19.453, intrinsic loss approx. 0.824); Gaia
5896574589400233472 (52.75 arcsec, G 18.864, intrinsic loss approx. 0.479); Gaia
5896574447658408064 (52.75 arcsec, G 19.371, intrinsic loss approx. 0.764); Gaia
5896574447658406528 (53.28 arcsec, G 17.713, intrinsic loss approx. 0.167); Gaia
5896574413297279232 (54.11 arcsec, G 16.211, intrinsic loss approx. 0.042); Gaia
5896574722536324480 (55.11 arcsec, G 18.205, intrinsic loss approx. 0.262); Gaia

5896574585097810816 (55.12 arcsec, G 19.395, intrinsic loss approx. 0.781); Gaia 5896574413298655104 (55.31 arcsec, G 19.128, intrinsic loss approx. 0.611); Gaia 5896574692479480192 (55.72 arcsec, G 19.163, intrinsic loss approx. 0.631); Gaia 5896574310219439616 (55.76 arcsec, G 18.804, intrinsic loss approx. 0.453); Gaia 5896574413298655488 (56.70 arcsec, G 18.386, intrinsic loss approx. 0.309); Gaia 5896574692490013440 (57.19 arcsec, G 17.313, intrinsic loss approx. 0.116); Gaia 5896574417612073856 (59.74 arcsec, G 18.966, intrinsic loss approx. 0.527); Gaia 5896574451971834240 (59.95 arcsec, G 15.806, intrinsic loss approx. 0.029). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.

4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 0.83/0.97/0.78 ppt in 3 stacked historical events; representative QLP raw median fit depth is 2.39 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 5896574619455481216 has SB1 P = 8.9213571665 d and e = 0.27356. Closest edge-on conjunction offset is -0.15304 d (-0.01715 phase), with 0.15937 d linearly propagated full source-specific SB1 covariance. Rough central duration 5.84 h versus observed median fit 27.45 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Broad, changing 0.96–2.86 ppt depressions repeat near the selected phase, but fitted duration is 27.45 h versus a rough 5.84 h central eclipse duration. S11 includes a narrow-looking dip while S99/S102 are broad. A phase-dependent orbital variable, blend or extraction structure remains possible. The timing fit does not establish stable eclipses; crowded S65 pixels are noisy.

Next check: Resolved time-series photometry and a phase-resolved orbital model to distinguish broad variability from eclipses; repeat difference imaging in a cleaner sector.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
11	0	1607.1770181	0.017043	1.60	used
11	1	1616.1312173	0.048568	2.86	used
99	274	4051.9749060	0.047515	2.45	used
99	276	4069.8163427	0.027080	2.58	used
102	283	4132.2625396	0.031026	2.39	used
102	284	4141.2054083	0.018705	0.96	used
102	285	4150.0326581	0.026959	2.10	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s65.json, historical/diff_s65.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R431. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

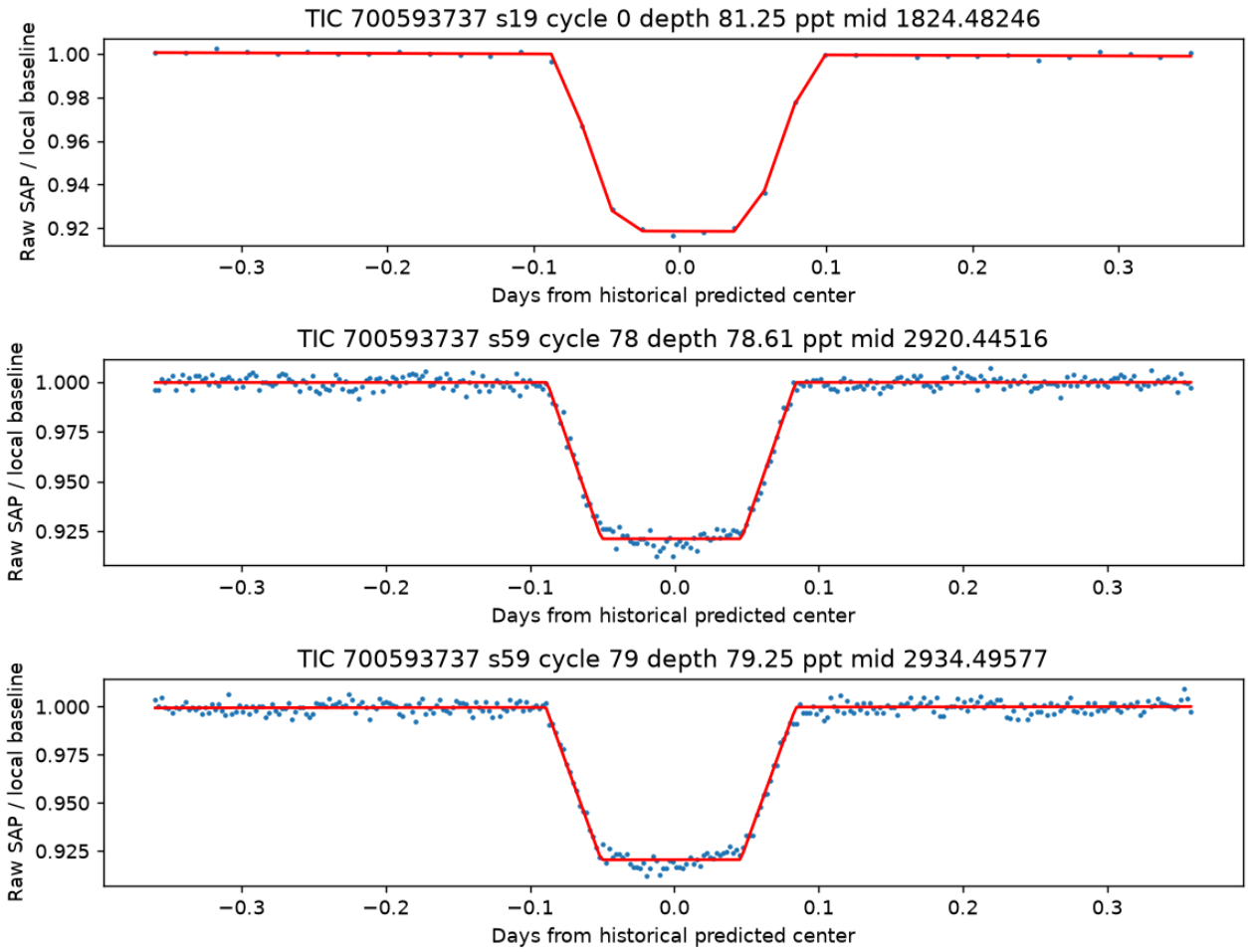
R432. TIC 700593737

physical_verdict: PROBABLE

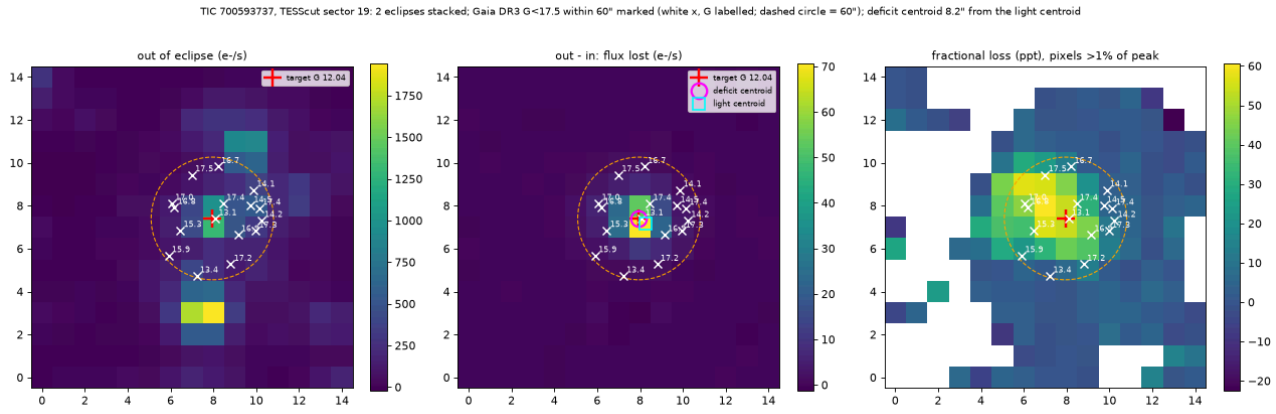
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PSF localization of the 2.8 arcsec pair.



R432; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R432; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 700593737: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 3 timing-usable measured centers in sectors [19, 59]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 14.0508013034 ± 0.0001326694 d and reference epoch BTJD 1824.4825619 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 19 TESScut difference image puts the deficit 0.69 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.866 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 190955384147342592 (2.80 arcsec, G 13.139, intrinsic loss approx. 0.301); Gaia 190955384151069696 (38.88 arcsec, G 14.451, intrinsic loss approx. 0.819); Gaia 190956857322461184 (47.10 arcsec, G 14.240, intrinsic loss approx. 0.688); Gaia 190956861618883840 (48.54 arcsec, G 14.129, intrinsic loss approx. 0.629); Gaia 190955349790083200 (58.08 arcsec, G 13.440, intrinsic loss approx. 0.371). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 54.10/49.36/34.54 ppt in 2 stacked historical events; representative QLP raw median fit depth is 79.25 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 190955379851574912 has SB1 P = 14.0495286778 d and e = 0.40569. Closest edge-on conjunction offset is 0.01921 d (0.00137 phase), with 0.42354 d linearly propagated full source-specific SB1 covariance. Rough central duration 3.87 h versus observed median fit 4.16 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: A raw S19 adjacent-cycle pair confirms repetition. A capable 2.8 arcsec Gaia companion shares the central TESS core; source attribution is unresolved.

Next check: Resolved eclipse photometry or multi-source PSF localization of the 2.8 arcsec pair.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
19	0	1824.4824625	0.010417	81.25	used
59	78	2920.4451605	0.001157	78.61	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
59	79	2934.4957691	0.001157	79.25	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s19.json, historical/diff_s19.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R432. Public evidence files: diff_s19.json, diff_s19.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

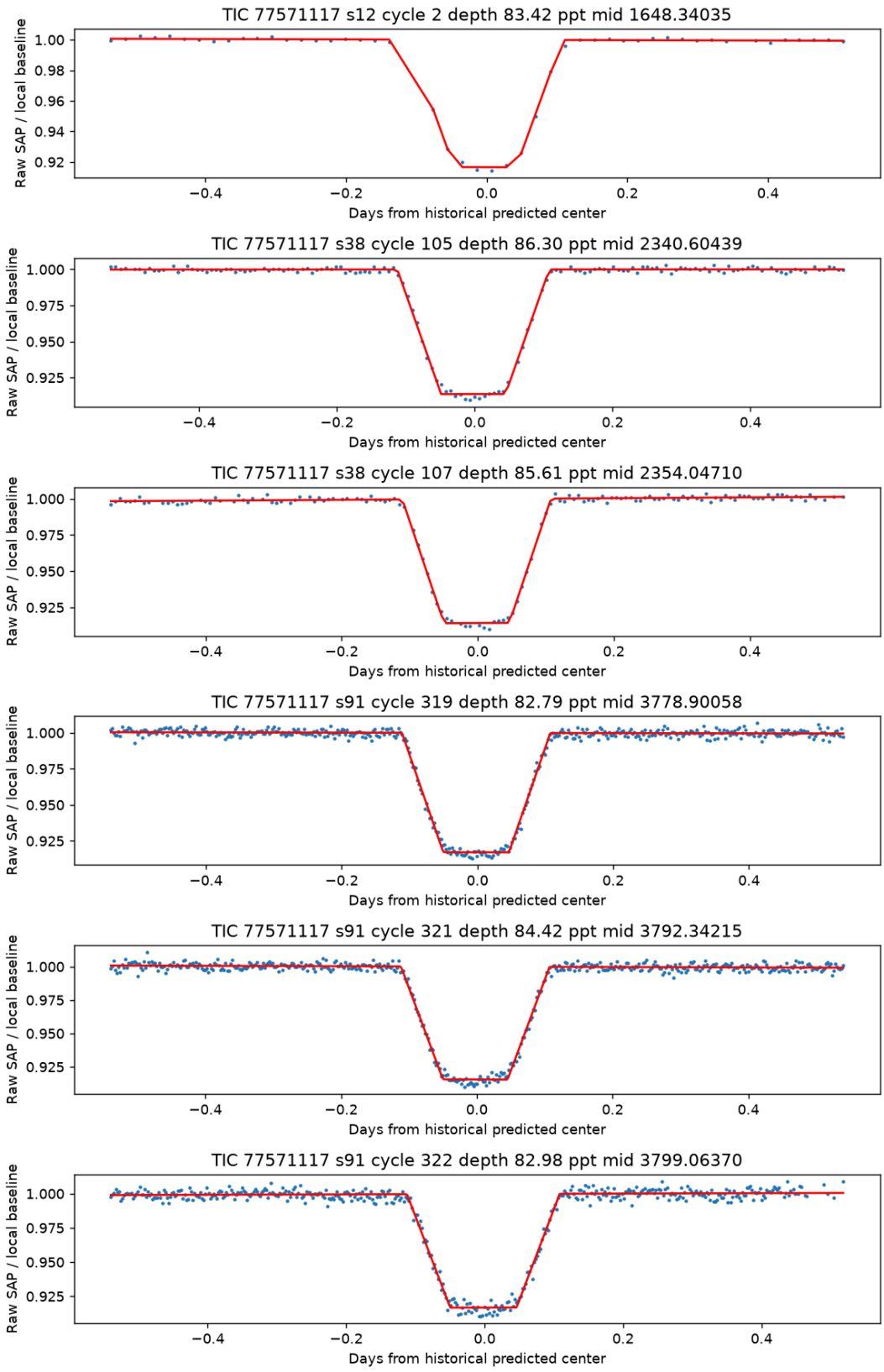
R433. TIC 77571117

physical_verdict: CONFIRMED

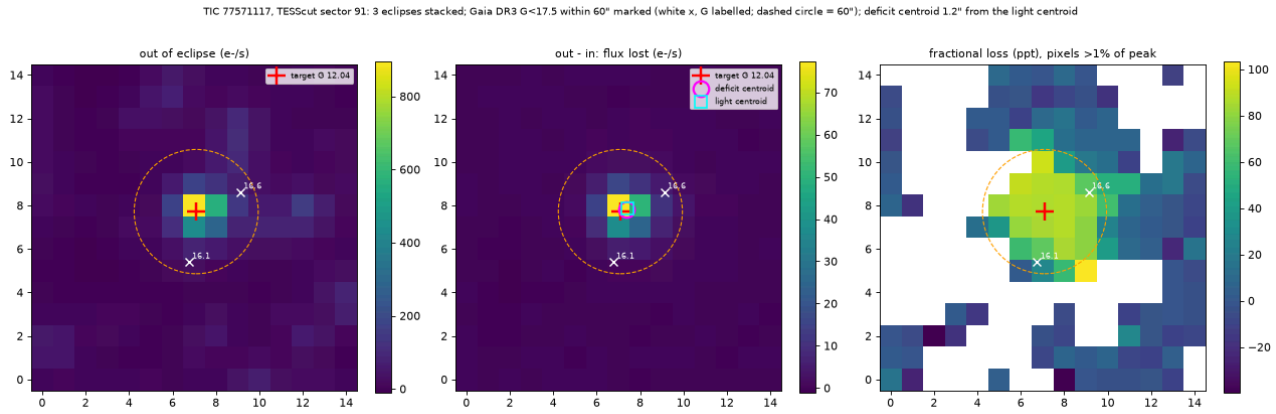
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R433; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R433; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 77571117: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured centers in sectors [12, 38, 91]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 6.7210082588 ± 0.0000112025 d and reference epoch BTJD 1648.3408444 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 91 TESScut difference image puts the deficit 5.93 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.176 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under approximate G-band flux budget. Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 86.36/86.68/82.50 ppt in 3 stacked historical events; representative QLP raw median fit depth is 83.92 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 6041559422449793792 has SB1 P = 6.7218718431 d and e = 0.00991. Closest edge-on conjunction offset is -0.13771 d (-0.02049 phase), with 0.25422 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.62 h versus observed median fit 5.26 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Six raw repeats span three sectors. The deficit is target-centered and no Gaia neighbor is capable under the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	2	1648.3403506	0.010416	83.42	used
38	105	2340.6043885	0.003472	86.30	used
38	107	2354.0471022	0.003472	85.61	used
91	319	3778.9005767	0.001157	82.79	used
91	321	3792.3421496	0.001157	84.42	used
91	322	3799.0636992	0.001157	82.98	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s91.json, historical/diff_s91.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R433. Public evidence files: diff_s91.json, diff_s91.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

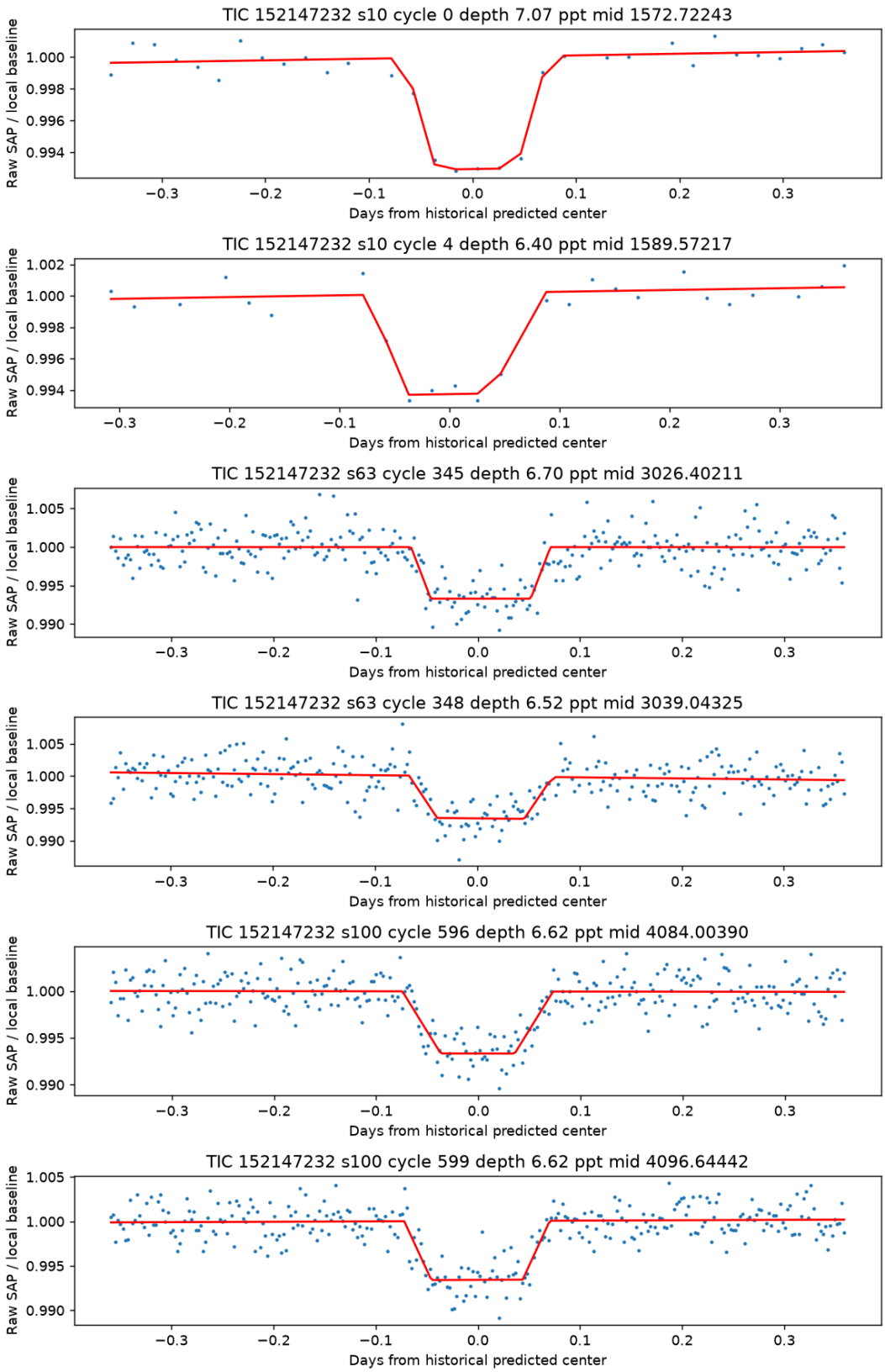
R434. TIC 152147232

physical_verdict: PROBABLE

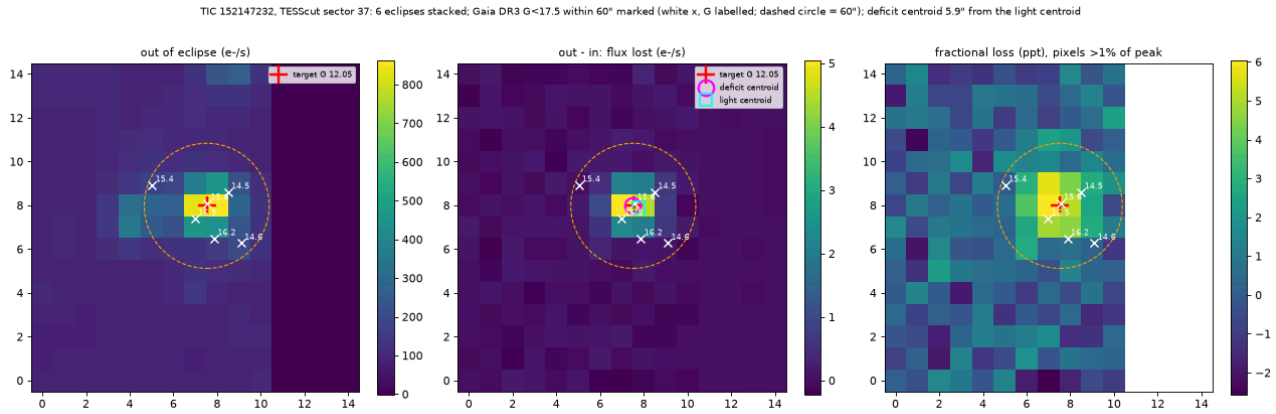
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PSF localization of the 2.7 arcsec pair.



R434; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R434; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 152147232: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 15 timing-usable measured centers in sectors [10, 63, 100]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 4.2135536598 ± 0.0000034278 d and reference epoch BTJD 1572.7251990 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 10 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 37 TESScut difference image puts the deficit 0.69 arcsec from the target coordinate; peak [7, 8] versus target pixel [8, 8]. Image-model fractional residual 0.531 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 5395959695358663040 (2.69 arcsec, G 15.822, intrinsic loss approx. 0.216); Gaia 5395959695356170624 (15.98 arcsec, G 16.524, intrinsic loss approx. 0.407); Gaia 5395959386121017472 (23.16 arcsec, G 14.509, intrinsic loss approx. 0.069); Gaia 5395959695358664832 (31.29 arcsec, G 16.245, intrinsic loss approx. 0.316); Gaia 5395959764078141952 (46.20 arcsec, G 14.642, intrinsic loss approx. 0.077); Gaia 5395959489200231936 (52.26 arcsec, G 15.438, intrinsic loss approx. 0.154). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 5.19/4.77/3.58 ppt in 6 stacked historical events; representative QLP raw median fit depth is 6.70 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 5395959695358663296 has SB1 P = 4.2132911086 d and e = 0.09094. Closest edge-on conjunction offset is -0.03876 d (-0.00920 phase), with 0.37454 d linearly

propagated full source-specific SB1 covariance. Rough central duration 4.84 h versus observed median fit 3.35 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Fifteen raw repeats include many within-sector pairs. A capable 2.7 arcsec Gaia neighbor shares the TESS core and cannot be resolved.

Next check: Resolved eclipse photometry or multi-source PSF localization of the 2.7 arcsec pair.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
10	0	1572.7224256	0.010417	7.07	used
10	2	1581.1488039	0.010417	7.22	used
10	4	1589.5721707	0.010417	6.40	used
63	343	3017.9727299	0.001818	7.87	used
63	344	3022.1868246	0.002043	7.09	used
63	345	3026.4021096	0.001357	6.70	used
63	346	3030.6159883	0.001675	6.81	used
63	347	3034.8274841	0.001542	7.44	used
63	348	3039.0432524	0.002049	6.52	used
100	594	4075.5762282	0.001268	6.27	used
100	595	4079.7915930	0.001251	6.81	used
100	596	4084.0038984	0.001917	6.62	used
100	597	4088.2146792	0.001852	6.67	used
100	598	4092.4283531	0.001173	6.28	used
100	599	4096.6444202	0.001402	6.62	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s37.json, historical/diff_s37.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R434. Public evidence files: diff_s37.json, diff_s37.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

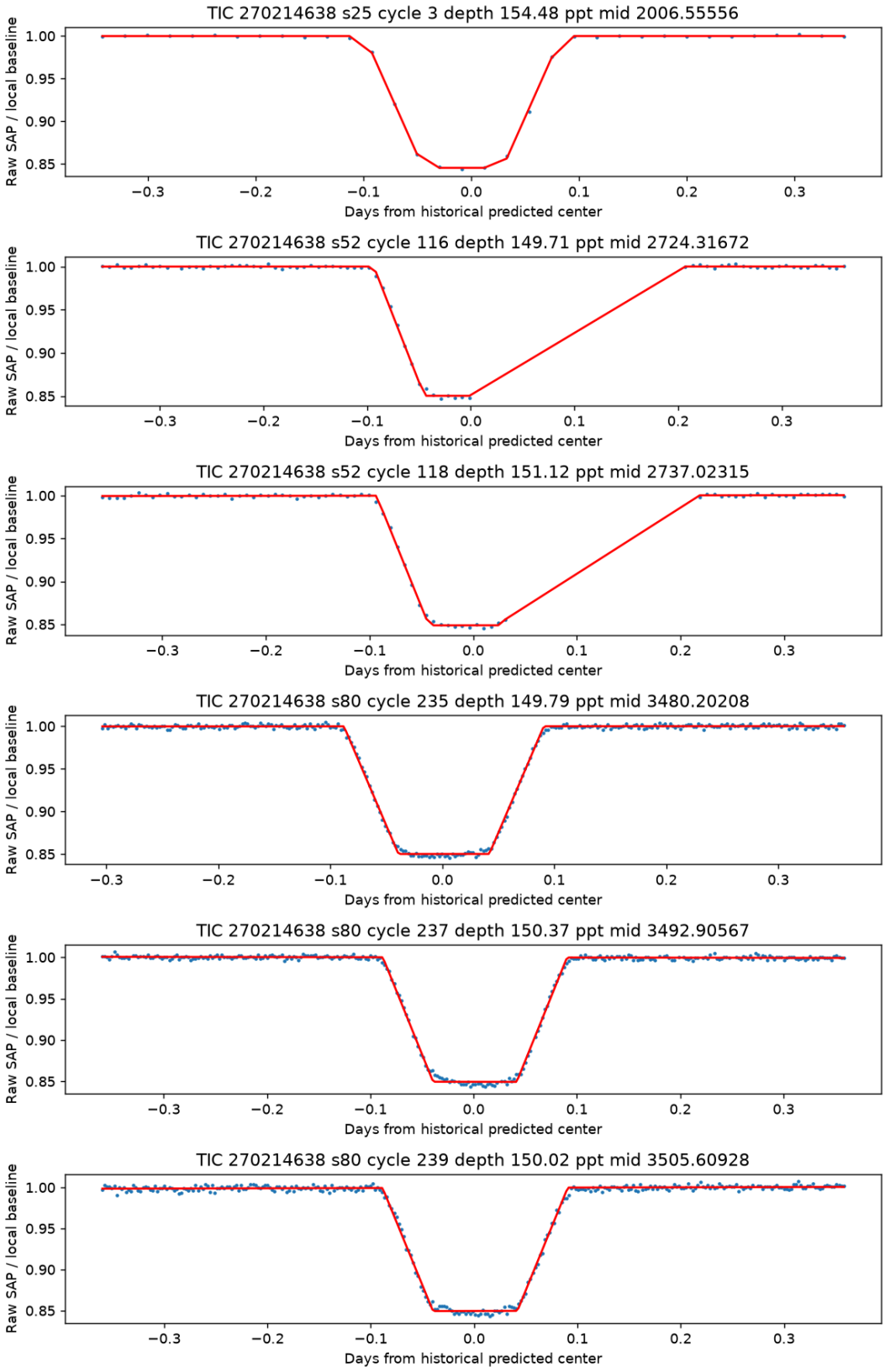
R435. TIC 270214638

physical_verdict: CONFIRMED

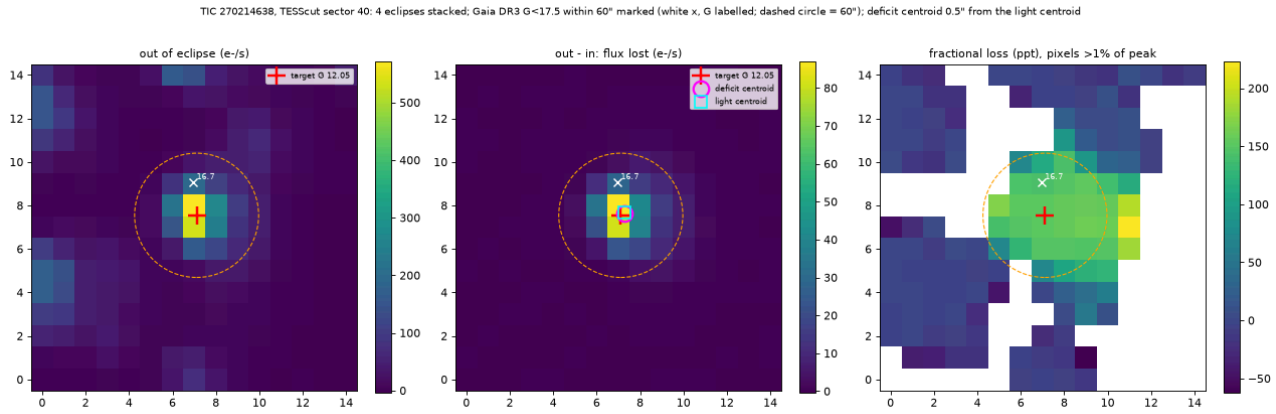
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R435; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R435; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 270214638: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured centers in sectors [25, 52, 80]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 6.3519439147 ± 0.0000246588 d and reference epoch BTJD 2006.5507448 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 40 TESScut difference image puts the deficit 4.28 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.143 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under approximate G-band flux budget. Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 152.27/151.57/149.77 ppt in 4 stacked historical events; representative QLP raw median fit depth is 150.54 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 1360991195262446720 has SB1 P = 6.3527030171 d and e = 0.00643. Closest edge-on conjunction offset is -0.19216 d (-0.03025 phase), with 0.19511 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.95 h versus observed median fit 4.28 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Six deep raw repeats span three sectors. The deficit is target-centered and no Gaia neighbor is capable under the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
25	3	2006.5555644	0.010417	154.48	used
52	116	2724.3167245	166474.921768	149.71	excluded: Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
52	118	2737.0231476	0.003472	151.12	used
80	235	3480.2020761	0.001157	149.79	used
80	237	3492.9056726	0.001157	150.37	used
80	238	3499.2574710	0.001157	150.70	used
80	239	3505.6092753	0.001157	150.02	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s40.json, historical/diff_s40.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R435. Public evidence files: diff_s40.json, diff_s40.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

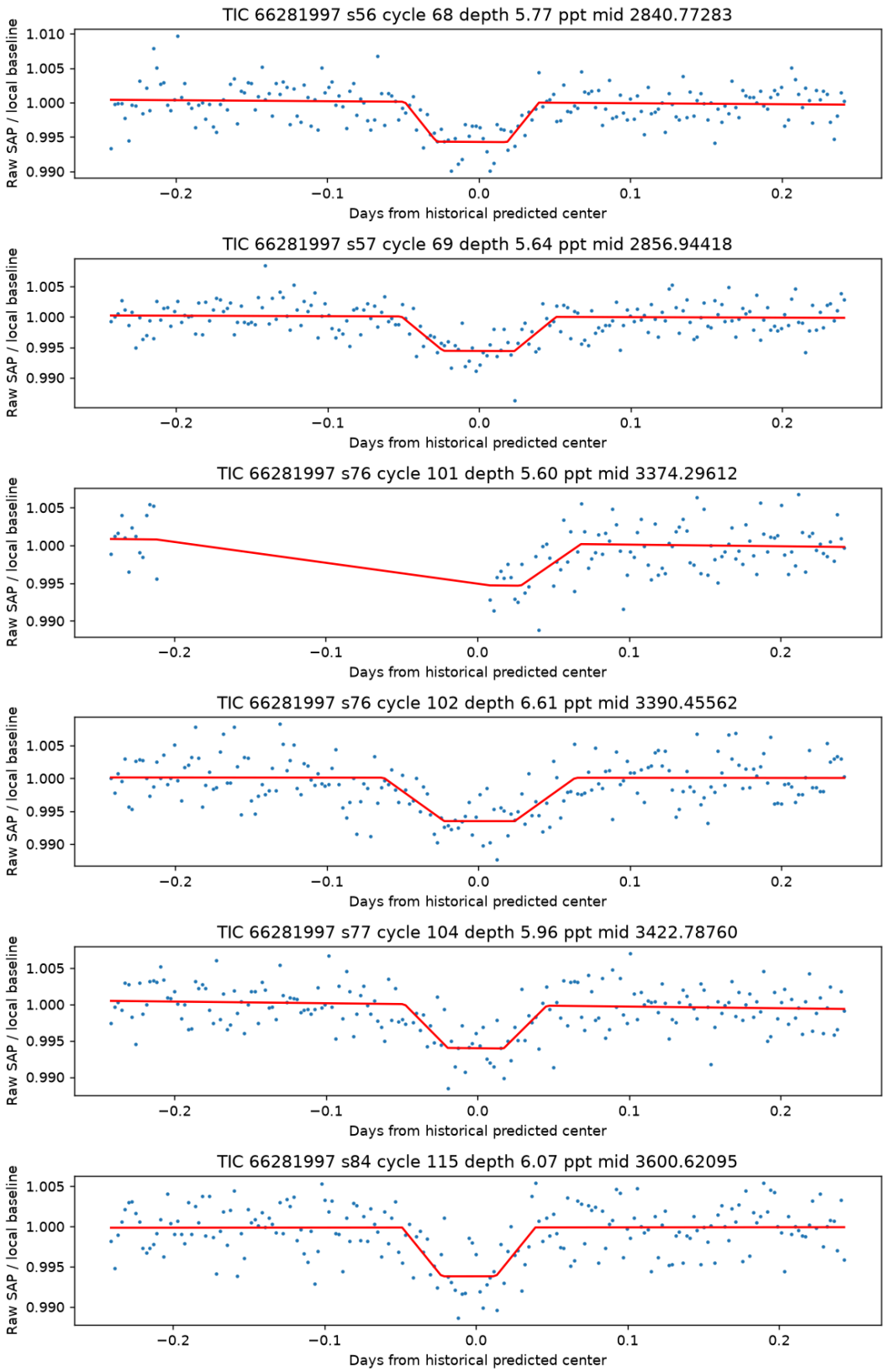
R436. TIC 66281997

physical_verdict: PROBABLE

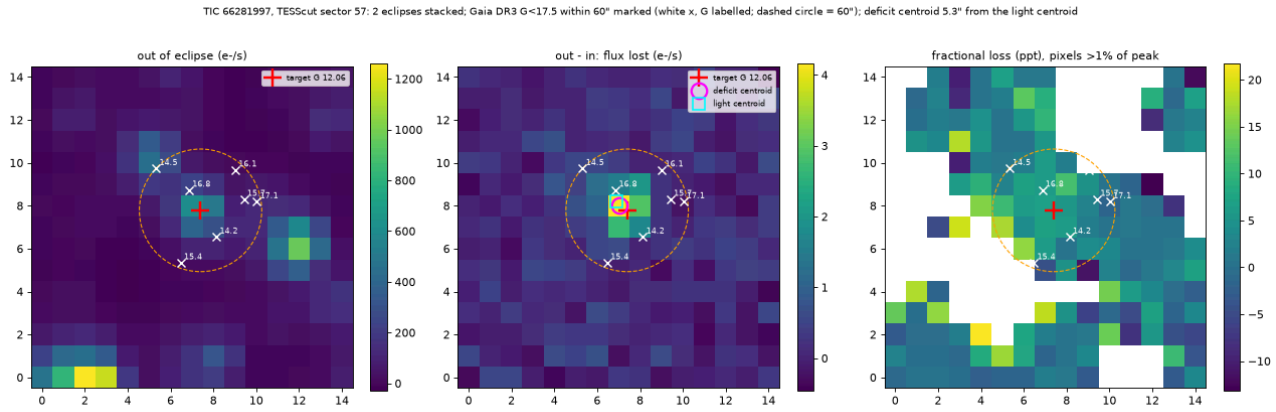
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Independent high-resolution eclipse photometry or multi-source PSF localization of the crowded deficit core.



R436; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R436; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 66281997: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured centers in sectors [56, 57, 76, 77, 83, 84]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 16.1669963889 ± 0.0000470947 d and reference epoch BTJD 2840.7742919 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 57 TESScut difference image puts the deficit 9.07 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.659 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 1988346111733826048 (21.78 arcsec, G 16.847, intrinsic loss approx. 0.477); Gaia 1988346180453300224 (28.72 arcsec, G 14.216, intrinsic loss approx. 0.047); Gaia 1988344634265076480 (42.81 arcsec, G 15.709, intrinsic loss approx. 0.171); Gaia 1988346111733827840 (51.33 arcsec, G 16.147, intrinsic loss approx. 0.253); Gaia 1988346936367541632 (53.51 arcsec, G 15.384, intrinsic loss approx. 0.128); Gaia 1988344634265076608 (54.24 arcsec, G 17.139, intrinsic loss approx. 0.622); Gaia 1988346146093565824 (56.84 arcsec, G 14.496, intrinsic loss approx. 0.060). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 6.83/6.71/5.38 ppt in 2 stacked historical events; representative QLP raw median fit depth is 5.95 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.

5. **Gaia geometry.** Matched DR3 source 1988346180453301632 has SB1 P = 16.1641569409 d and e = 0.40209. Closest edge-on conjunction offset is 0.43325 d (0.02680 phase), with 0.51578 d linearly propagated full source-specific SB1 covariance. Rough central duration 7.71 h versus observed median fit 2.48 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Eight usable raw centers lie in six sectors after all eight archived QLP sectors were examined; they include adjacent S56/S57 and S76/S77 cycles. The historical S57 deficit lies 9.1 arcsec from the target, toward a capable G16.8 star 21.8 arcsec away. Noisy pixels and other capable neighbors prevent unique attribution.

Next check: Independent high-resolution eclipse photometry or multi-source PSF localization of the crowded deficit core.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
56	68	2840.7728335	0.002230	5.77	used
57	69	2856.9441832	0.002354	5.64	used
57	70	2873.1056462	0.002773	5.92	used
76	101	3374.2961244	392421.524476	5.60	excluded: Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
76	102	3390.4556240	0.003173	6.61	used
77	103	3406.6183094	0.002544	6.55	used
77	104	3422.7876011	0.002693	5.96	used
83	113	3568.2897722	0.003193	5.94	used
84	115	3600.6209460	0.002463	6.07	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s57.json, historical/diff_s57.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R436. Public evidence files: diff_s57.json, diff_s57.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

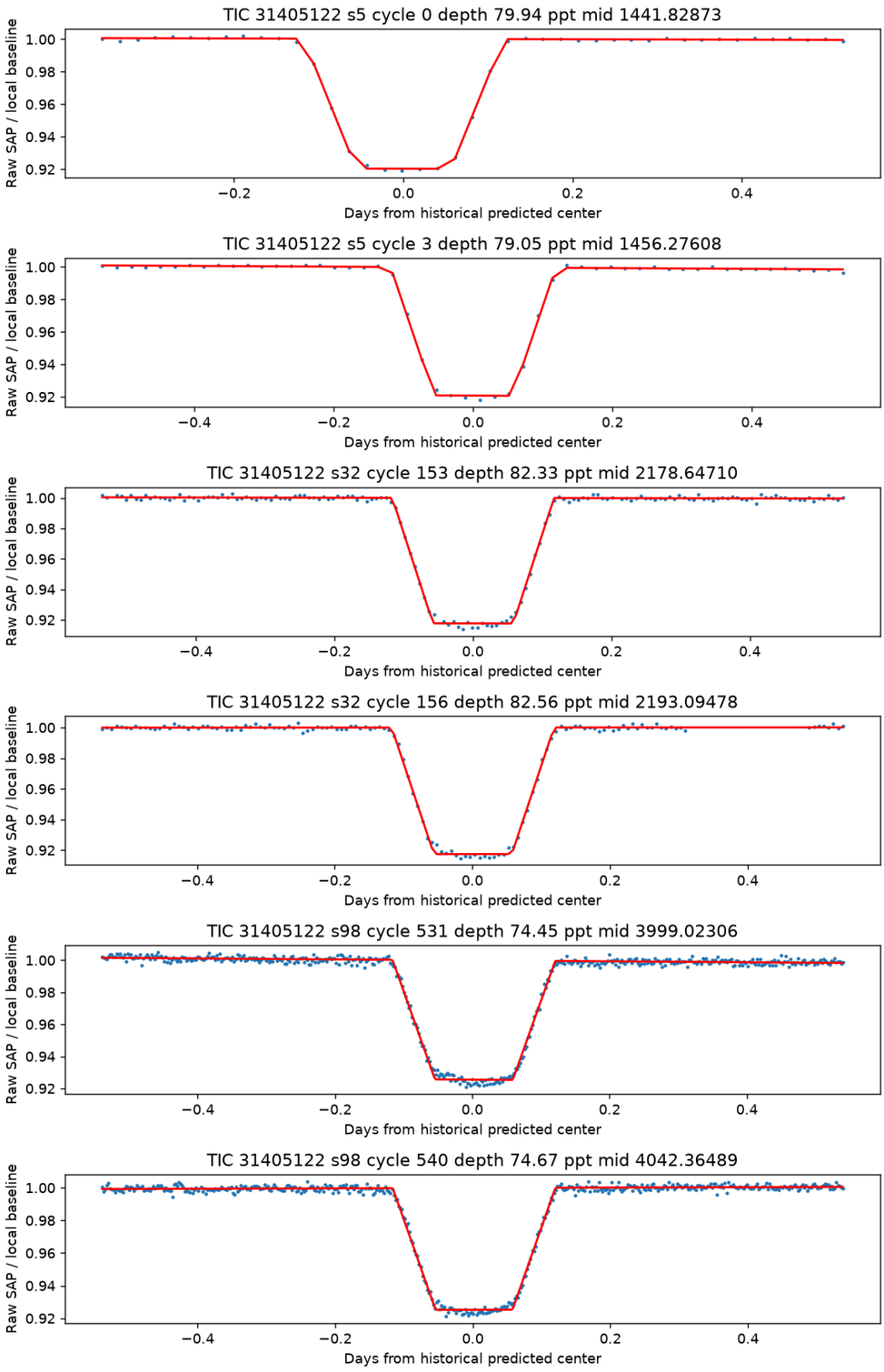
R437. TIC 31405122

physical_verdict: CONFIRMED

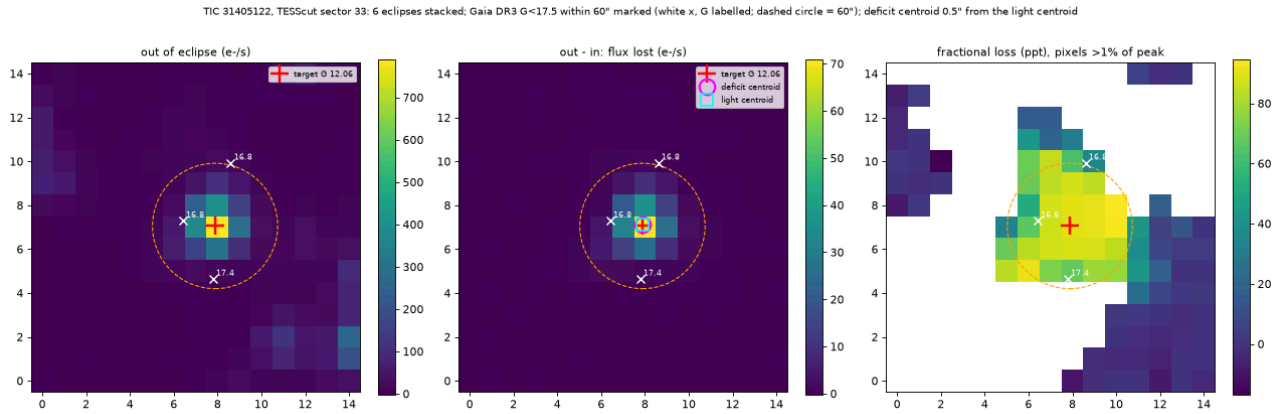
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R437; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R437; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 31405122: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 13 timing-usable measured centers in sectors [5, 32, 98]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 4.8158084158 ± 0.0000040354 d and reference epoch BTJD 1441.8286729 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 6 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 33 TESScut difference image puts the deficit 1.77 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.070 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under approximate G-band flux budget. Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 89.80/88.84/87.53 ppt in 6 stacked historical events; representative QLP raw median fit depth is 80.36 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 2909098108269909376 has SB1 P = 4.8156658314 d and e = 0.04267. Closest edge-on conjunction offset is 0.07549 d (0.01568 phase), with 0.06737 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.26 h versus observed median fit 5.67 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Thirteen deep raw repeats include six adjacent-cycle pairs. The deficit is target-centered and no Gaia neighbor is capable under the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
5	0	1441.8287271	0.010417	79.94	used
5	1	1446.6437397	0.010417	80.36	used
5	3	1456.2760786	0.010417	79.05	used
5	4	1461.0923054	0.010417	80.75	used
32	153	2178.6470952	0.003472	82.33	used
32	154	2183.4632085	0.003472	82.12	used
32	155	2188.2792504	0.003472	82.21	used
32	156	2193.0947793	0.003472	82.56	used
32	157	2197.9105536	0.003472	82.79	used
98	531	3999.0230649	0.001157	74.45	used
98	534	4013.4704924	0.001157	73.93	used
98	537	4027.9178740	0.001157	74.82	used
98	540	4042.3648909	0.001157	74.67	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s33.json, historical/diff_s33.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R437. Public evidence files: diff_s33.json, diff_s33.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

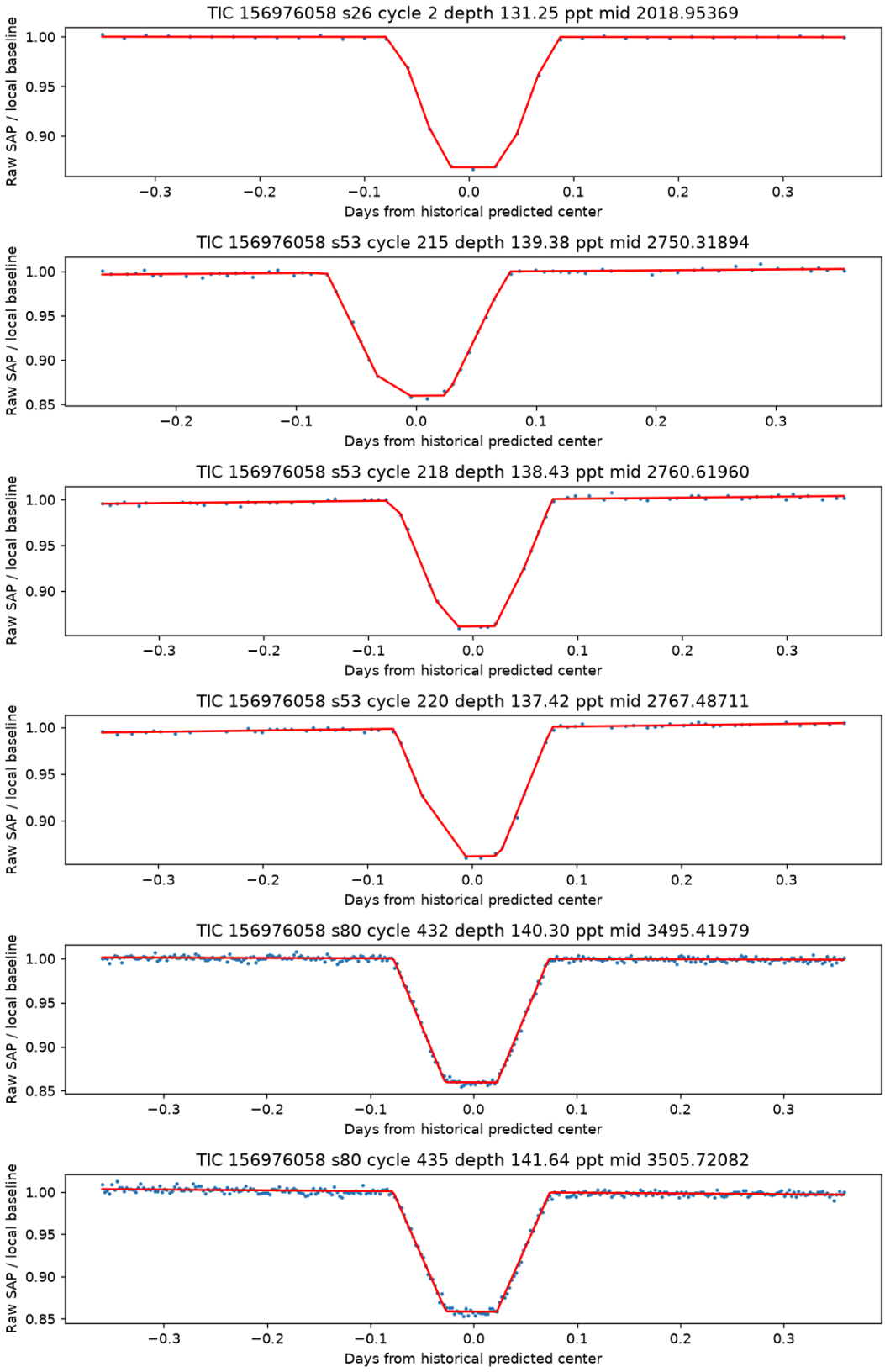
R438. TIC 156976058

physical_verdict: CONFIRMED

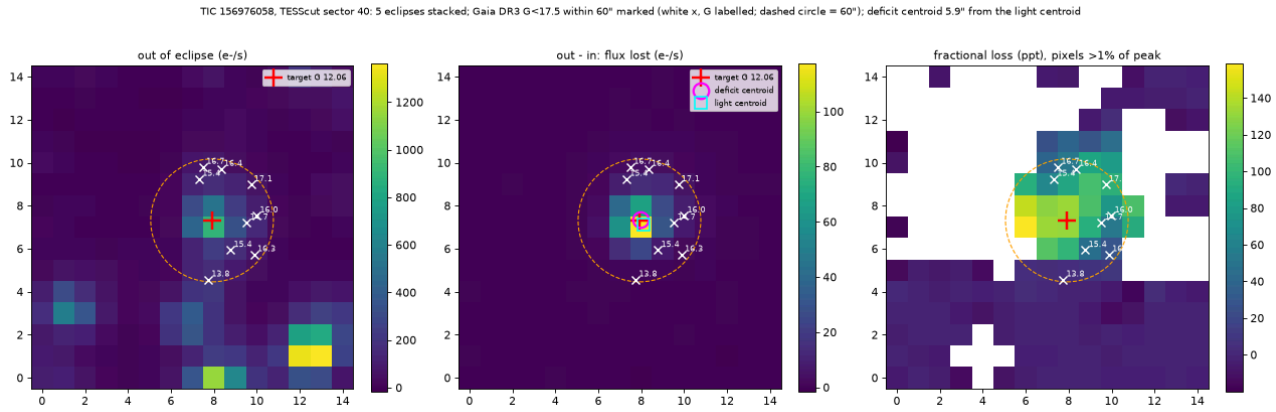
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R438; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R438; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 156976058: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 14 timing-usable measured centers in sectors [26, 53, 80]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 3.4336457438 ± 0.0000062829 d and reference epoch BTJD 2018.9522980 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 10 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 40 TESScut difference image puts the deficit 1.65 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.465 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 4523316211621183872 (55.81 arcsec, G 13.768, intrinsic loss approx. 0.794). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 132.09/118.89/100.11 ppt in 5 stacked historical events; representative QLP raw median fit depth is 138.82 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 4523316314700411136 has SB1 P = 3.4333917898 d and e = 0.03177. Closest edge-on conjunction offset is 0.07574 d (0.02206 phase), with 0.08230 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.49 h versus observed median fit 3.61 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Fourteen deep raw repeats include adjacent cycles in S53 and S80. The only nominally capable neighbor is 55.8 arcsec away, distinct from the target-centered deficit.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
26	2	2018.9536896	0.010417	131.25	used
53	214	2746.8856511	0.003472	136.41	used
53	215	2750.3189356	0.003472	139.38	used
53	216	2753.7522898	0.003472	138.79	used
53	217	2757.1857718	0.003472	137.72	used
53	218	2760.6195976	0.003472	138.43	used
53	219	2764.0532691	0.003472	137.64	used
53	220	2767.4871060	0.003472	137.42	used
80	428	3481.6852727	0.001157	138.85	used
80	431	3491.9865294	0.001157	140.49	used
80	432	3495.4197879	0.001157	140.30	used
80	433	3498.8537895	0.001157	139.67	used
80	434	3502.2872671	0.001157	139.86	used
80	435	3505.7208234	0.001157	141.64	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s40.json, historical/diff_s40.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R438. Public evidence files: diff_s40.json, diff_s40.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

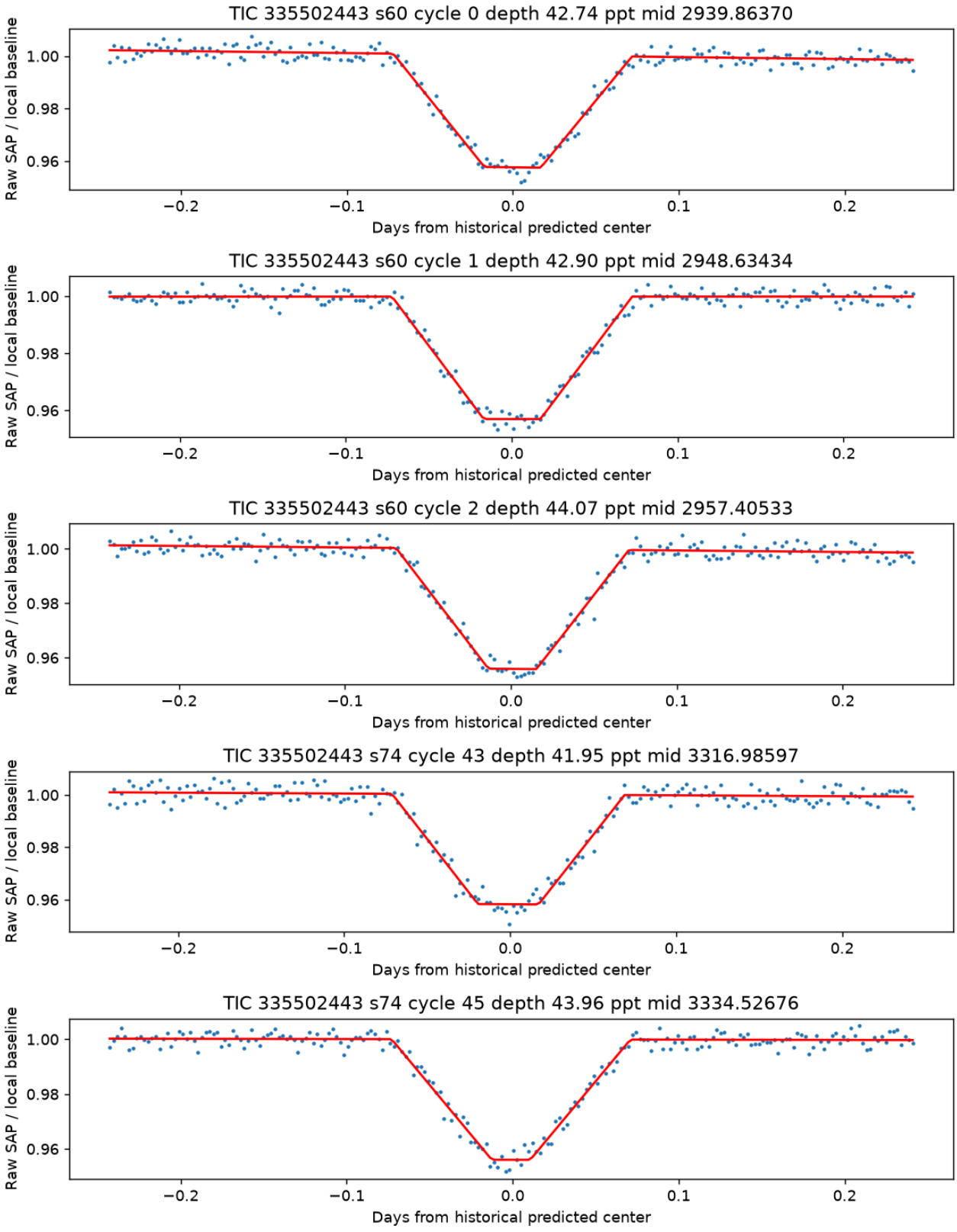
R439. TIC 335502443

physical_verdict: CONFIRMED

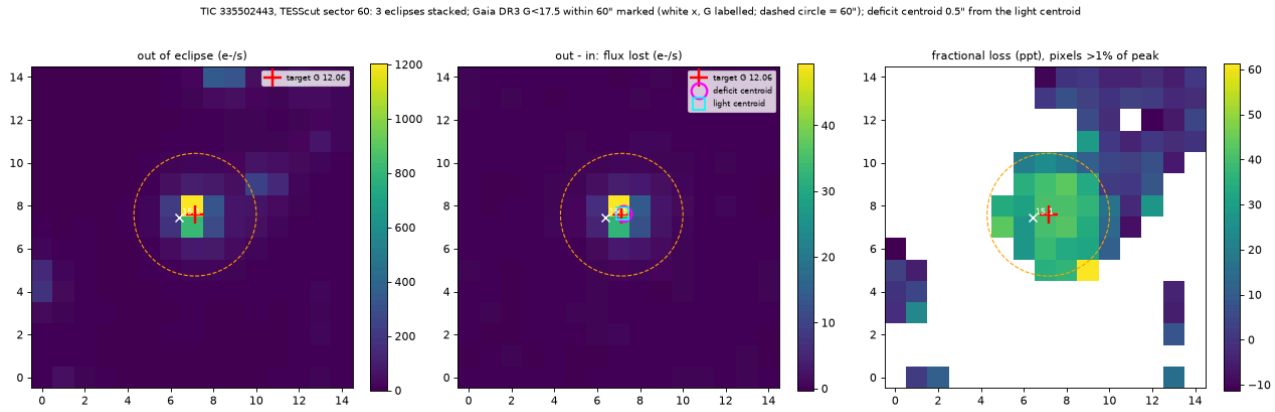
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R439; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R439; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 335502443: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured centers in sectors [60, 74]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 8.7702776381 ± 0.0000245490 d and reference epoch BTJD 2939.8641685 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 60 TESScut difference image puts the deficit 1.55 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.193 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 1088529297616817152 (16.22 arcsec, G 15.075, intrinsic loss approx. 0.704). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 41.06/39.39/38.17 ppt in 3 stacked historical events; representative QLP raw median fit depth is 42.90 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 1088529293321259648 has SB1 P = 8.7704426450 d and e = 0.03074. Closest edge-on conjunction offset is -0.01668 d (-0.00190 phase), with 0.24762 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.49 h versus observed median fit 3.45 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Five raw repeats include adjacent S60 cycles. A capable neighbor 16.2 arcsec away is distinct from the target-centered deficit core at TESS resolution.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
60	0	2939.8637048	0.001157	42.74	used
60	1	2948.6343381	0.001157	42.90	used
60	2	2957.4053258	0.001157	44.07	used
74	43	3316.9859743	0.001157	41.95	used
74	45	3334.5267647	0.001157	43.96	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s60.json, historical/diff_s60.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R439. Public evidence files: diff_s60.json, diff_s60.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

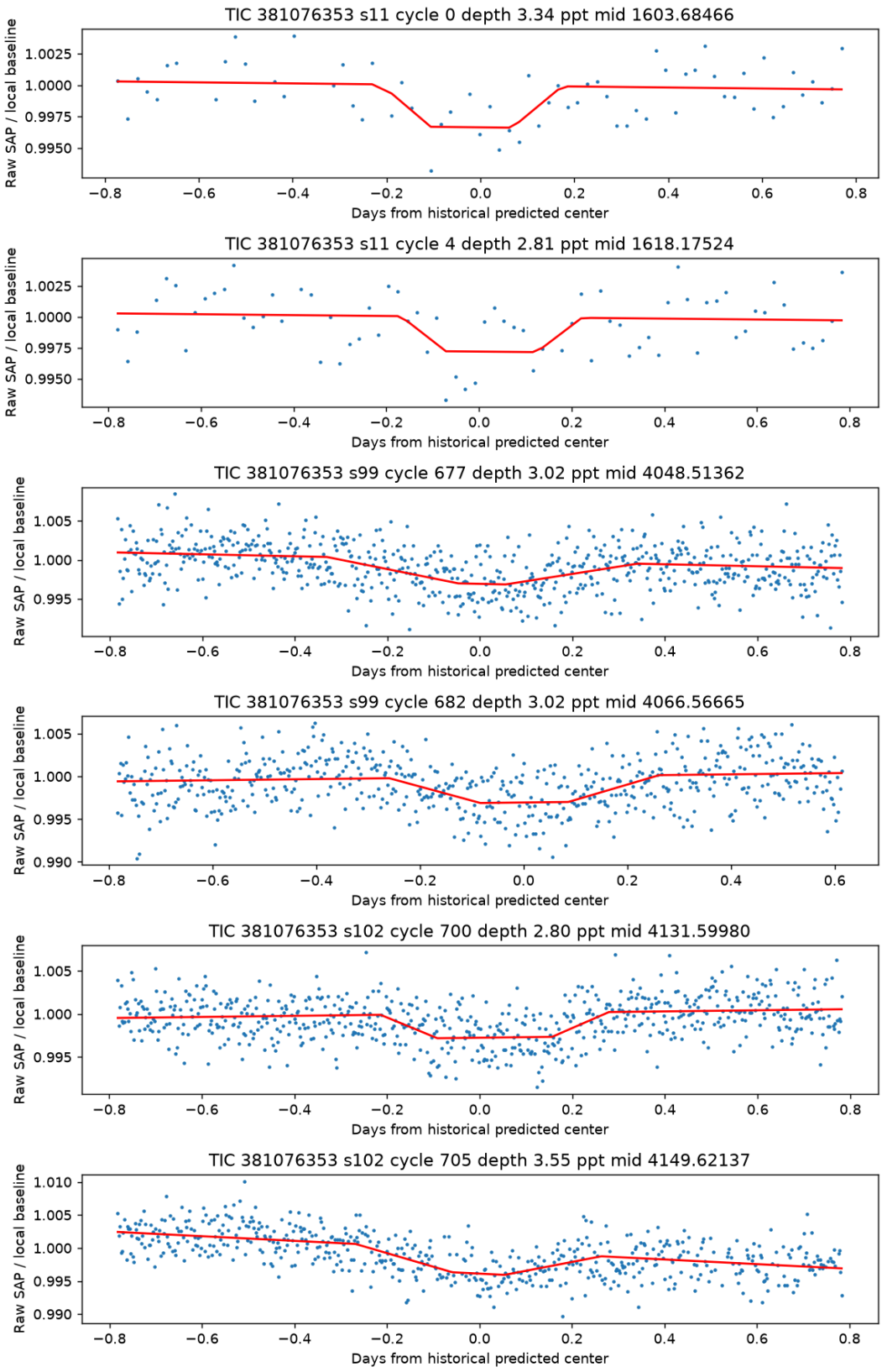
R440. TIC 381076353

physical_verdict: PLAUSIBLE

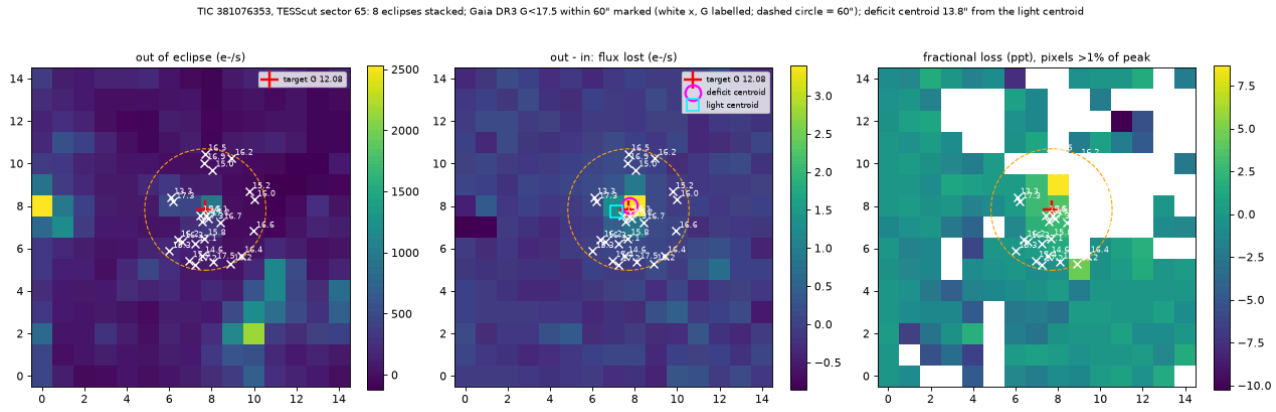
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

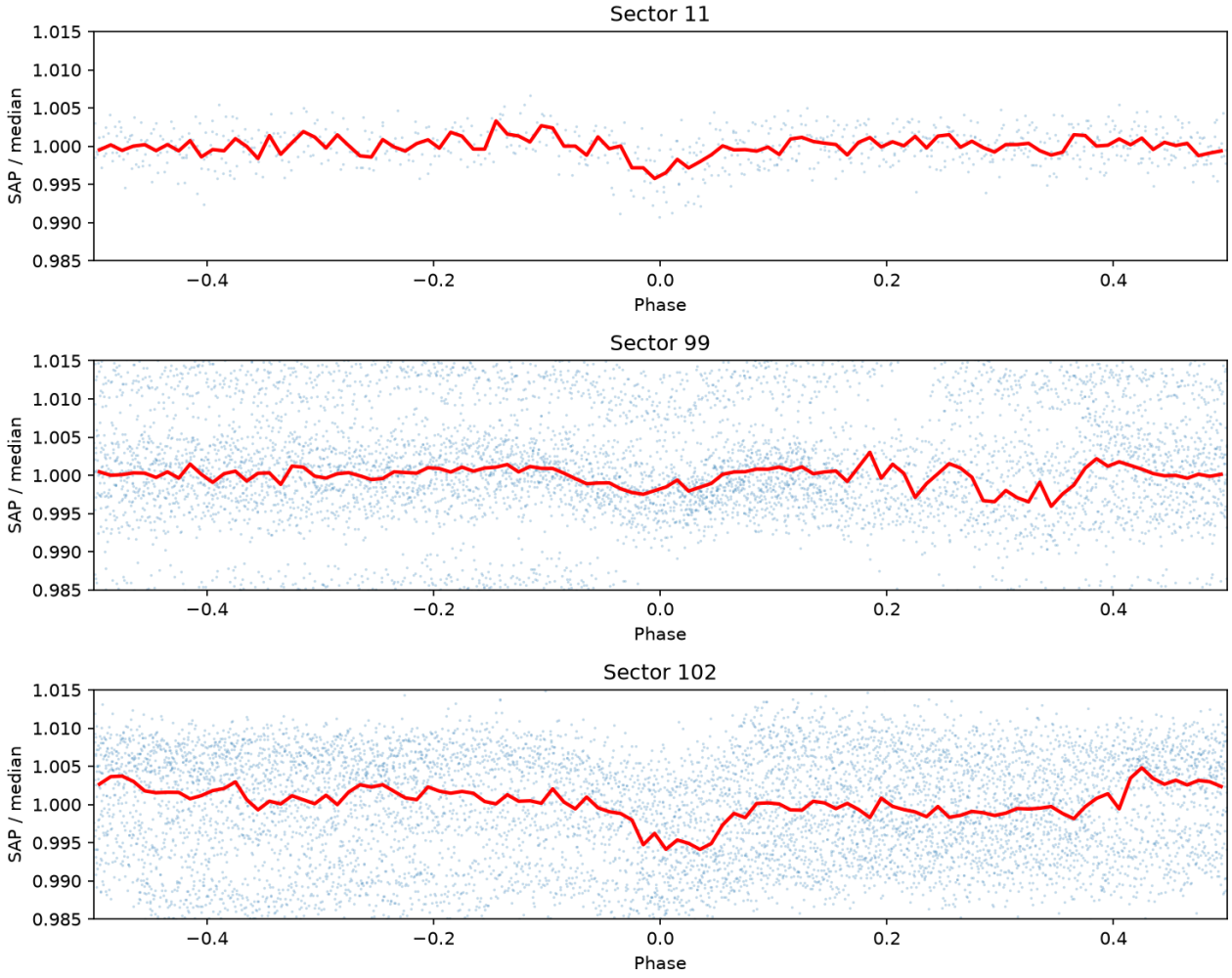
next_check: Test morphology and resolve source attribution



R440; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R440; Representative eclipse-deficit image; source attribution retains the report limits.



R440; Independent sector folds expose the variable eclipse morphology; this signal remains plausible and excluded.

Archived source: sanitized report text.

TIC 381076353: PLAUSIBLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 12 timing-selected candidate depressions in sectors [11, 99, 102]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured repeating depression centers give P=3.6112653345 d, fit error=0.0000376302 d, epoch BTJD 1603.6849541 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 consecutive within-sector same-kind cycle pairs. The inventory period was 3.61123 d and the historical SAP search branch was 3.6112336247 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: unresolved at TESS resolution.** Historical S65 TESScut deficit core is 3.71 arcsec from the target coordinate; peak pixel [8, 8] and target pixel [8, 8]. Fractional PSF model residual is 0.909, not a localization uncertainty. Capable Gaia neighbors: 1.9 arcsec, G=17.55, intrinsic-loss requirement ~0.48; 5.8 arcsec, G=16.08, intrinsic-loss requirement ~0.13; 6.8 arcsec, G=18.17, intrinsic-loss requirement ~0.85; 7.8 arcsec, G=15.47, intrinsic-loss requirement ~0.07; 9.2 arcsec, G=17.17, intrinsic-loss requirement ~0.34; 45 further capable sources in the 60 arcsec list. The full list and flux-budget calculation are in historical/diff_s65.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 3.60/3.63/2.38 ppt from 8 selected windows. Median selected QLP raw fit depth is 3.05 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 5864386386590001024 has P=3.6112336247 d and e=0.38173. Closest edge-on conjunction differs from the measured epoch by 0.00267 in phase (0.00964 d); linear propagation of the source-specific full SB1 covariance gives 0.07820 d as a descriptive timing scale. The rough central-duration estimate is 3.02 h versus trapezoid median 12.16 h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Recurrent ~3 ppt depressions are comparable to local scatter. Folded S11 and S102 show localized troughs and recovery, but S99 is rounded and broad variability is not decisively excluded; eclipse morphology remains uncertain. Fifty capable Gaia sources within 60 arcsec include several within 8 arcsec; the target-centered core has 0.909 image-model fractional residual, leaving attribution unresolved. Trapezoid durations and timing are shape-sensitive (up to 113 min timing residual).

Next check. Higher-resolution time-series photometry to test eclipse morphology and source attribution.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
11	0	1603.6846601	0.023909	3.34	used
11	1	1607.2888763	0.013657	3.93	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
11	4	1618.1752398	0.033312	2.81	used
11	5	1621.7413685	0.022295	2.20	used
99	677	4048.5136194	0.016278	3.02	used
99	678	4052.0649452	0.015813	3.08	used
99	682	4066.5666518	0.013352	3.02	used
99	683	4070.2579081	0.018299	5.10	used
102	700	4131.5997979	0.009527	2.80	used
102	701	4135.1379910	0.012265	4.25	used
102	703	4142.4034336	0.010024	2.88	used
102	705	4149.6213680	0.015593	3.55	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s65.json, historical/diff_s65.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R440. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

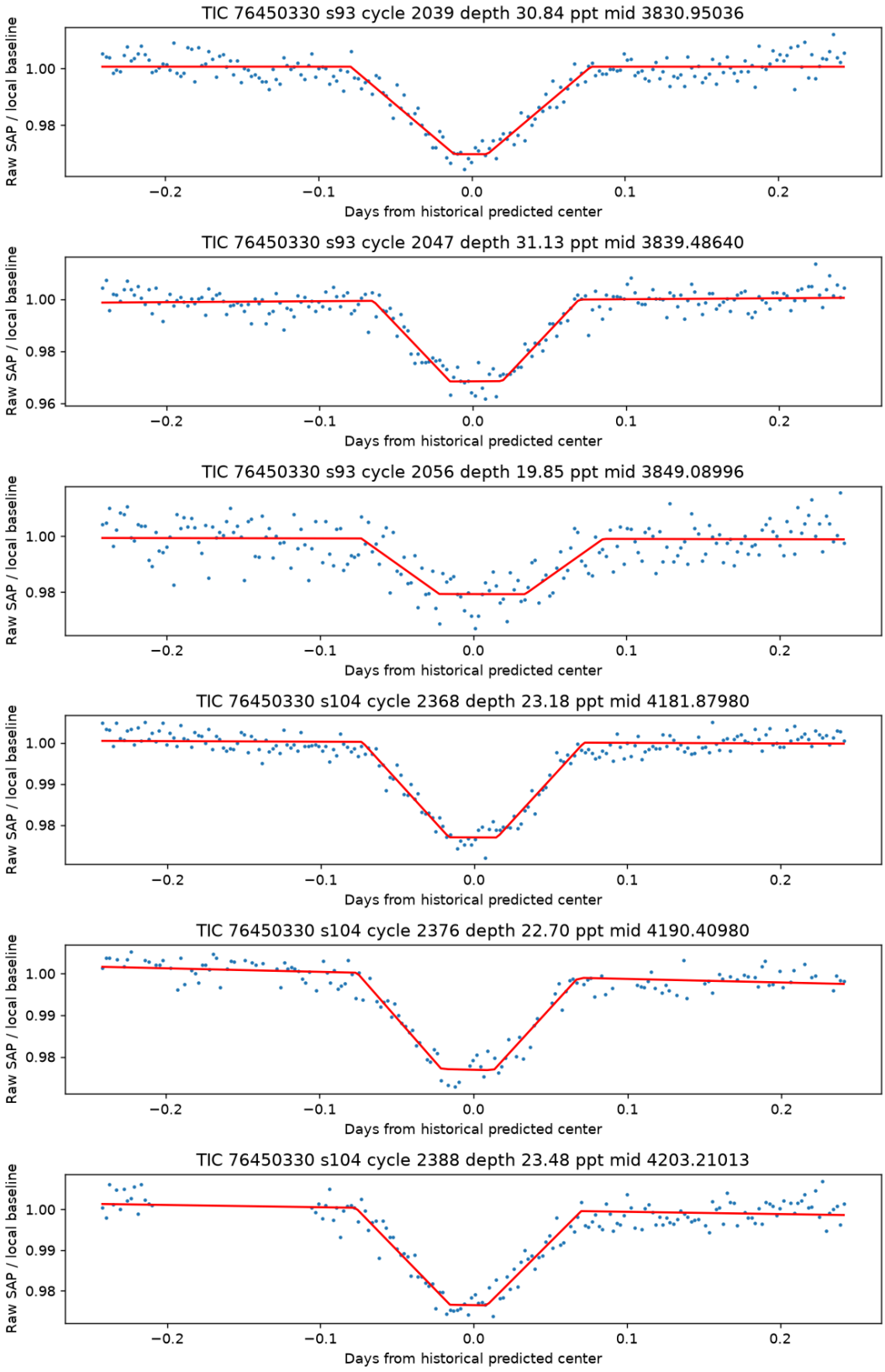
R441. TIC 76450330

physical_verdict: CONFIRMED

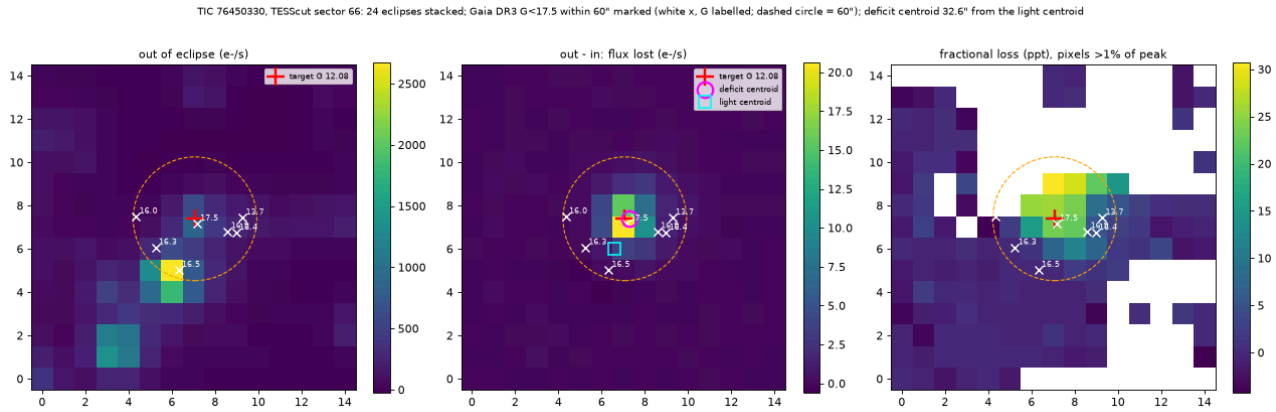
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R441; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R441; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 76450330: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 17 timing-selected measured eclipses in sectors [93, 104]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=2.1332813704 d, fit error=0.0000038241 d, epoch BTJD 3830.9530096 (Deeper repeating eclipse; historical half-period parity 1). There are 10 consecutive within-sector same-kind cycle pairs. The inventory period was 2.13331 d and the historical SAP search branch was 1.0666548545 d. Deeper parity 1 of historical half-period cycles; consecutive same-kind events have orbital-cycle count one. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S66 TESScut deficit core is 4.40 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Fractional PSF model residual is 0.952, not a localization uncertainty. Capable Gaia neighbors: 45.6 arcsec, G=13.73, intrinsic-loss requirement ~0.13; 56.6 arcsec, G=15.96, intrinsic-loss requirement ~0.83. The full list and flux-budget calculation are in historical/diff_s66.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 23.47/17.15/7.81 ppt from 24 selected windows. Median selected QLP raw fit depth is 31.53 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 5918304478211801728 has P=2.1333285880 d and e=0.04973. Closest edge-on conjunction differs from the measured epoch by -0.03250 in phase (-0.06933 d); linear propagation of the source-specific full SB1 covariance gives 0.07043 d as a descriptive timing scale. The rough central-duration estimate is 6.24 h versus trapezoid median 3.47

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Strong alternating depths establish the 2.1333 d orbital cycle; the historical SAP 1.06665 d branch is the half-period. The deeper parity is fitted for the reported epoch.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
93	2039	3830.9503554	0.001244	30.84	used
93	2040	3832.0188593	0.001403	23.02	other eclipse kind
93	2041	3833.0874435	0.001157	38.33	used
93	2042	3834.1488187	0.001425	25.77	other eclipse kind
93	2043	3835.2217774	0.001351	32.71	used
93	2045	3837.3508688	0.001157	34.70	used
93	2046	3838.4181594	0.001330	25.12	other eclipse kind
93	2047	3839.4864046	0.001157	31.13	used
93	2048	3840.5553168	0.001157	22.95	other eclipse kind
93	2051	3843.7553337	0.001602	29.37	used
93	2052	3844.8200618	0.001520	24.12	other eclipse kind
93	2053	3845.8862230	0.001157	33.14	used
93	2054	3846.9531815	0.001265	23.20	other eclipse kind
93	2055	3848.0204345	0.001252	30.82	used
93	2056	3849.0899565	0.002583	19.85	other eclipse kind
93	2058	3851.2177570	0.001737	23.20	other eclipse kind
93	2059	3852.2846859	0.001157	31.88	used
93	2060	3853.3545826	0.001509	23.26	other eclipse kind
93	2061	3854.4187693	0.001342	32.23	used
104	2366	4179.7433388	0.002615	24.03	other eclipse kind
104	2367	4180.8106391	0.001157	31.56	used
104	2368	4181.8797986	0.001157	23.18	other eclipse kind
104	2369	4182.9451359	0.001157	31.53	used
104	2370	4184.0114345	0.001157	22.49	other eclipse kind
104	2372	4186.1448417	0.001169	22.25	other eclipse kind
104	2373	4187.2111476	0.001157	31.41	used
104	2374	4188.2772217	0.001360	23.53	other eclipse kind
104	2375	4189.3448112	0.001157	30.98	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
104	2376	4190.4097993	0.001374	22.70	other eclipse kind
104	2380	4194.6770309	0.001425	24.09	other eclipse kind
104	2381	4195.7440795	0.001157	32.82	used
104	2382	4196.8101891	0.001319	25.07	other eclipse kind
104	2384	4198.9436754	0.001157	23.39	other eclipse kind
104	2385	4200.0099345	0.001157	31.36	used
104	2386	4201.0779941	0.001157	22.29	other eclipse kind
104	2387	4202.1439058	0.001157	30.93	used
104	2388	4203.2101329	0.001157	23.48	other eclipse kind

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s66.json, historical/diff_s66.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R441. Public evidence files: diff_s66.json, diff_s66.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

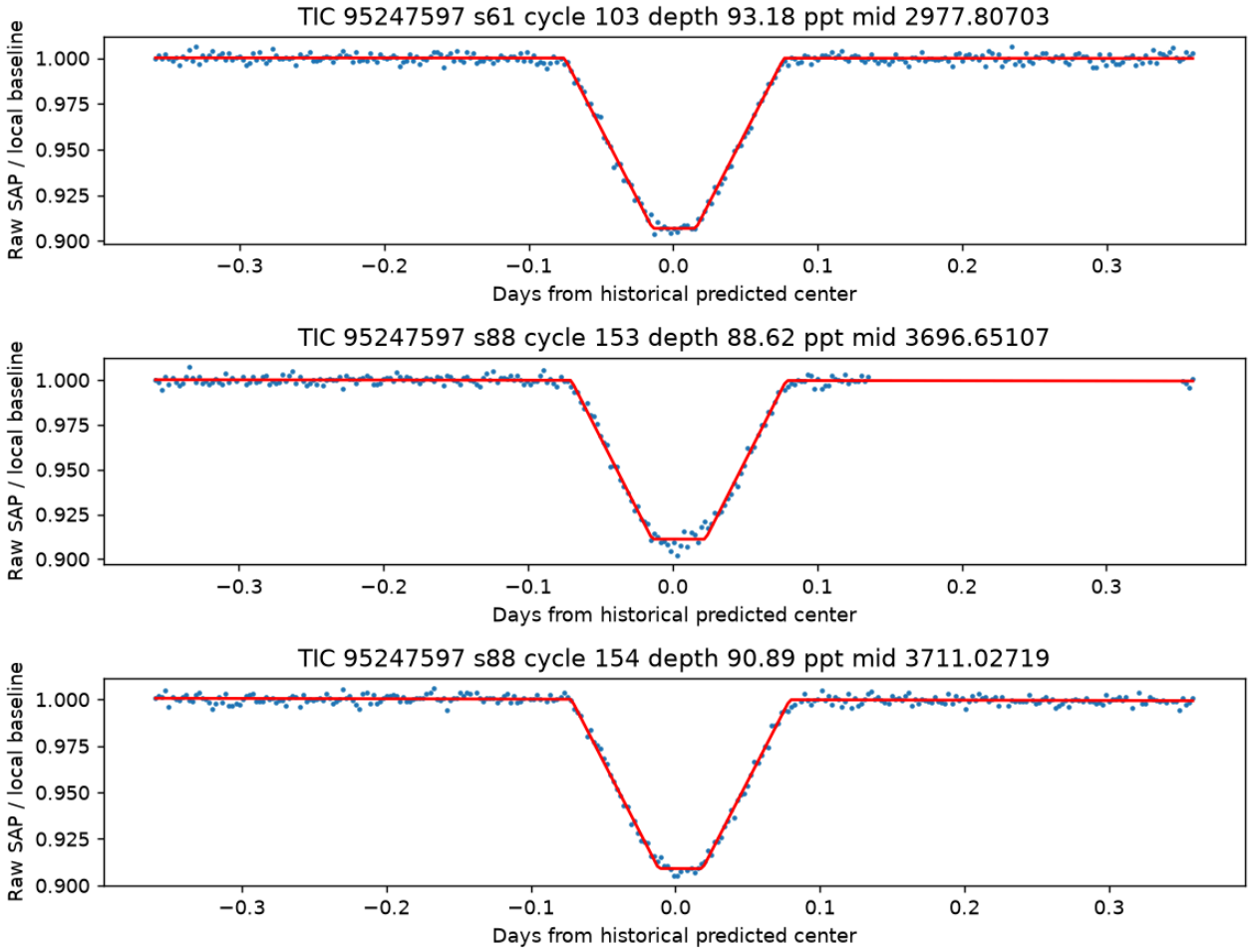
R442. TIC 95247597

physical_verdict: CONFIRMED

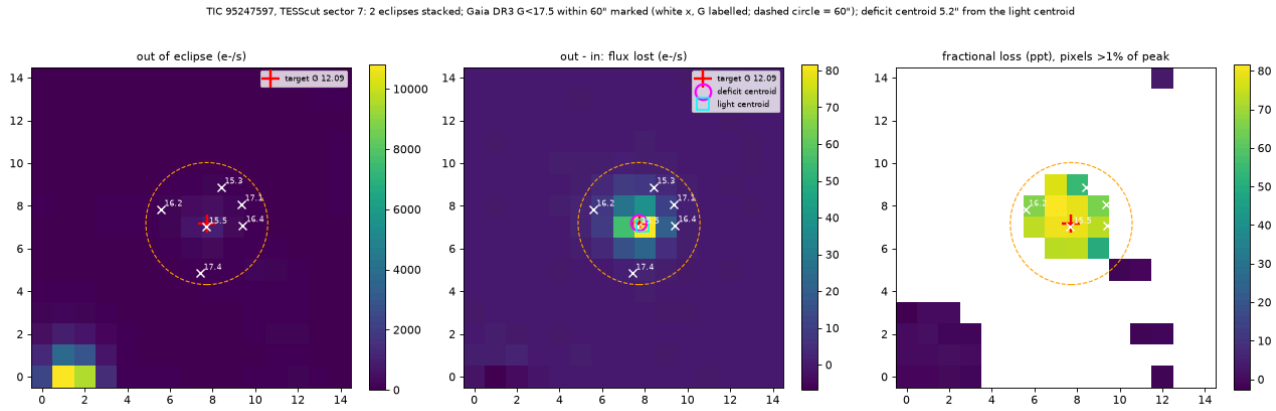
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R442; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R442; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 95247597: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 3 timing-selected measured eclipses in sectors [61, 88]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=14.3768729995 d, fit error=0.0000280654 d, epoch BTJD 2977.8070390 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 consecutive within-sector same-kind cycle pairs. The inventory period was 14.37681 d and the historical SAP search branch was 14.3768112970 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S7 TESScut deficit core is 1.48 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Fractional PSF model residual is 0.250, not a localization uncertainty. Capable Gaia neighbors: none under the approximate G-band flux budget. The full list and flux-budget calculation are in historical/diff_s07.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 79.56/76.84/68.07 ppt from 2 selected windows. Median selected QLP raw fit depth is 90.89 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 3042854243248092928 has P=14.3765198811 d and e=0.11457. Closest edge-on conjunction differs from the measured epoch by -0.02690 in phase (-0.38667 d); linear propagation of the source-specific full SB1 covariance gives 0.41907 d as a descriptive timing scale. The rough central-duration estimate is 6.15 h versus trapezoid median 3.63 h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. S88 has a consecutive 14.376 d same-kind recurrence and S61 adds a long baseline. Earlier S7 windows have only five quality-zero points each, so they are not clean nulls. Current MAST QLP availability includes S7/34/61/88; availability alone is not an eclipse detection.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
61	103	2977.8070316	0.001157	93.18	used
88	153	3696.6510671	0.001157	88.62	used
88	154	3711.0271913	0.001157	90.89	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s07.json, historical/diff_s07.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R442. Public evidence files: diff_s07.json, diff_s07.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

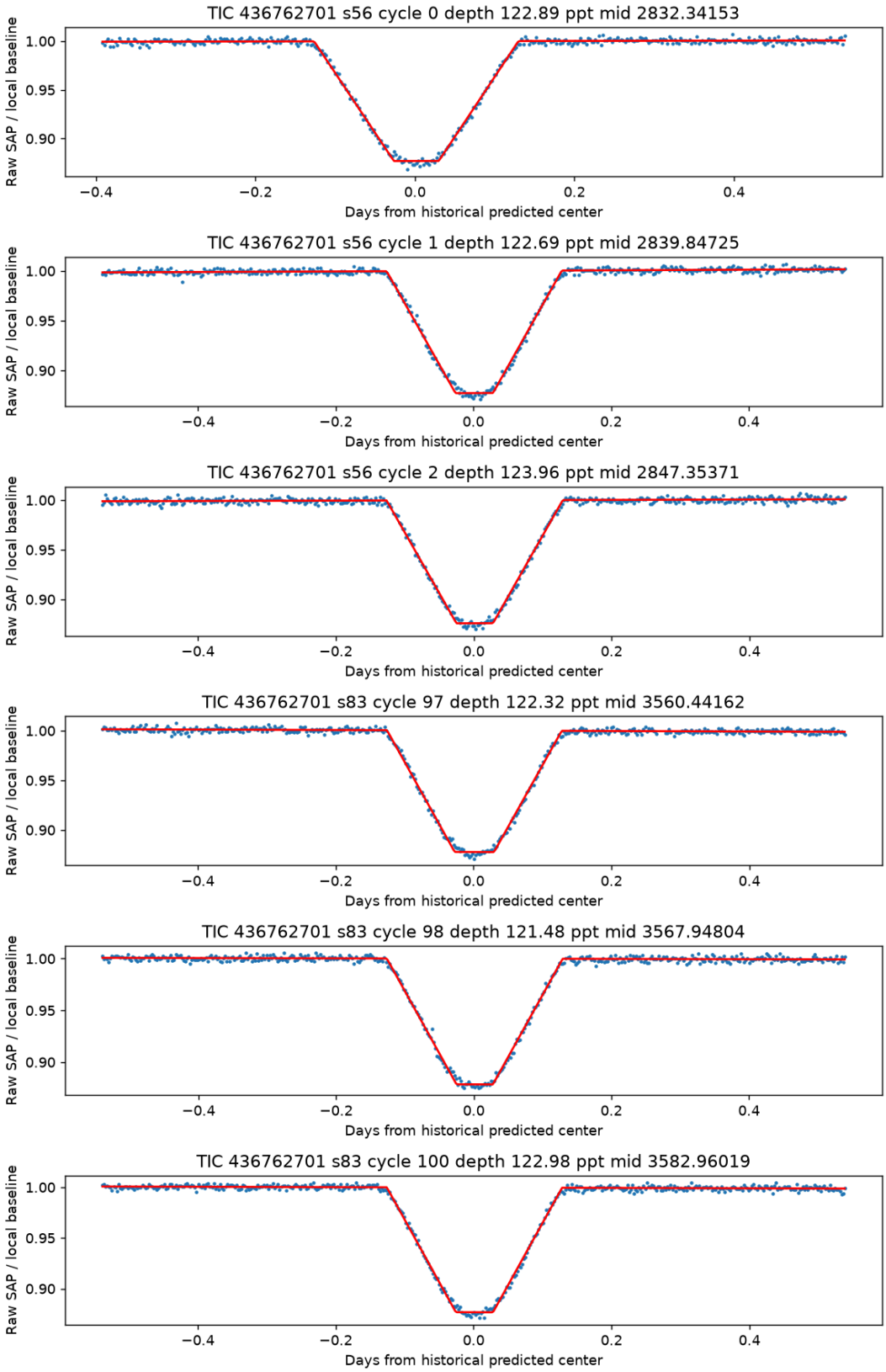
R443. TIC 436762701

physical_verdict: CONFIRMED

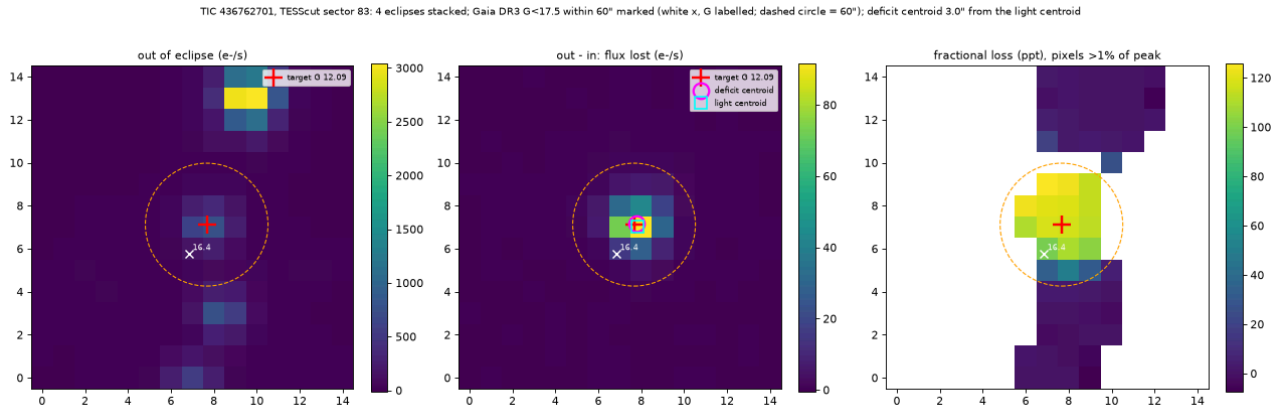
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R443; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R443; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 436762701: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 7 timing-selected measured eclipses in sectors [56, 83]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=7.5061902904 d, fit error=0.0000090645 d, epoch BTJD 2832.3413064 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 consecutive within-sector same-kind cycle pairs. The inventory period was 7.50619 d and the historical SAP search branch was 7.5061923952 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S83 TESScut deficit core is 3.16 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Fractional PSF model residual is 0.667, not a localization uncertainty. Capable Gaia neighbors: none under the approximate G-band flux budget. The full list and flux-budget calculation are in historical/diff_s83.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 117.56/115.42/108.68 ppt from 4 selected windows. Median selected QLP raw fit depth is 122.69 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 1885079155014227584 has P=7.5069767062 d and e=0.18531. Closest edge-on conjunction differs from the measured epoch by -0.02604 in phase (-0.19548 d); linear propagation of the source-specific full SB1 covariance gives 0.13174 d as a descriptive timing scale. The rough central-duration estimate is 9.36 h versus trapezoid median 6.11

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Seven deep raw events across S56 and S83 include consecutive cycles, with consistent depths and a target-centered deficit.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
56	0	2832.3415322	0.001157	122.89	used
56	1	2839.8472492	0.001157	122.69	used
56	2	2847.3537074	0.001157	123.96	used
83	97	3560.4416157	0.001157	122.32	used
83	98	3567.9480430	0.001157	121.48	used
83	99	3575.4543530	0.001157	121.25	used
83	100	3582.9601899	0.001157	122.98	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s83.json, historical/diff_s83.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R443. Public evidence files: diff_s83.json, diff_s83.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

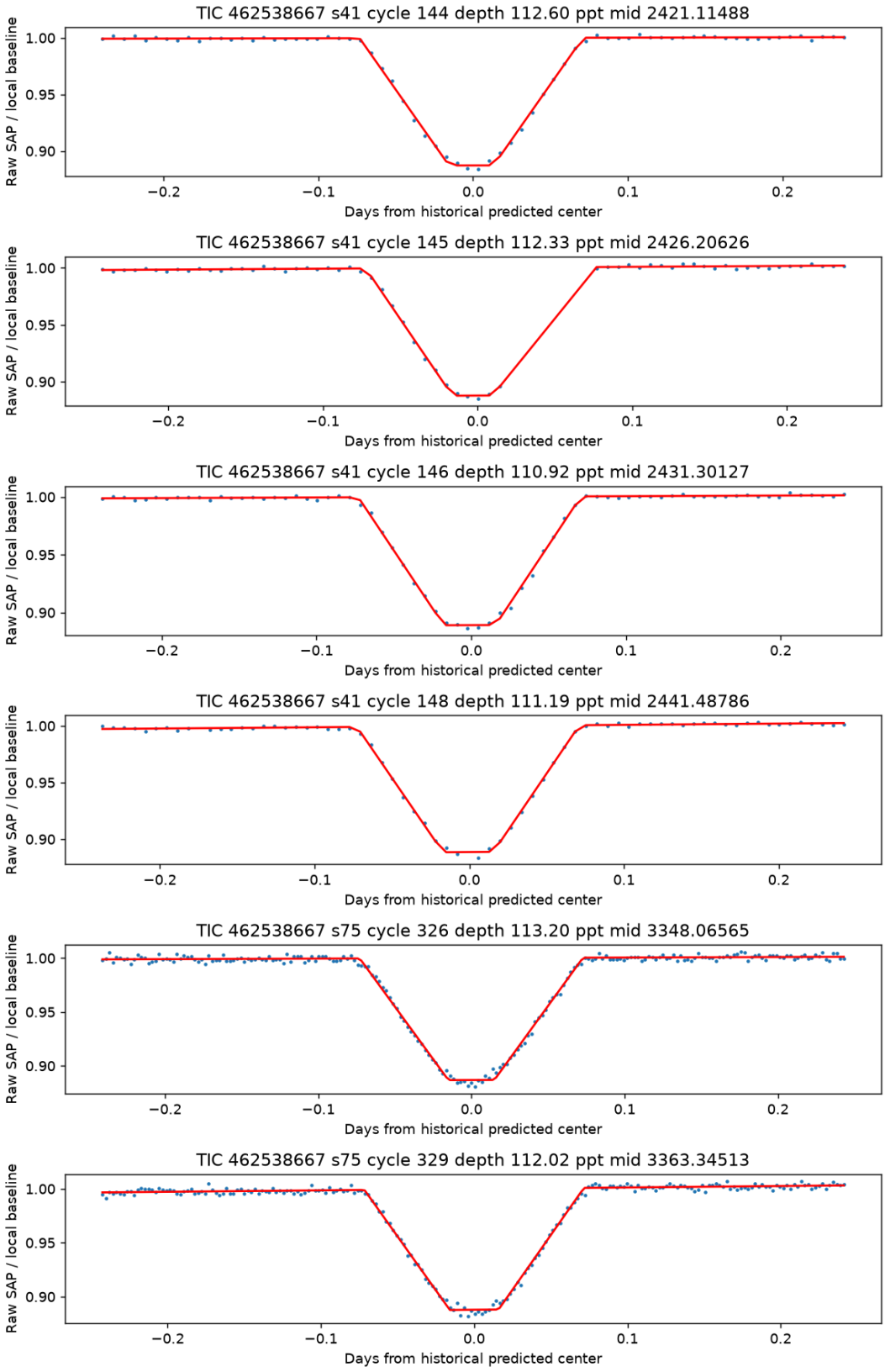
R444. TIC 462538667

physical_verdict: CONFIRMED

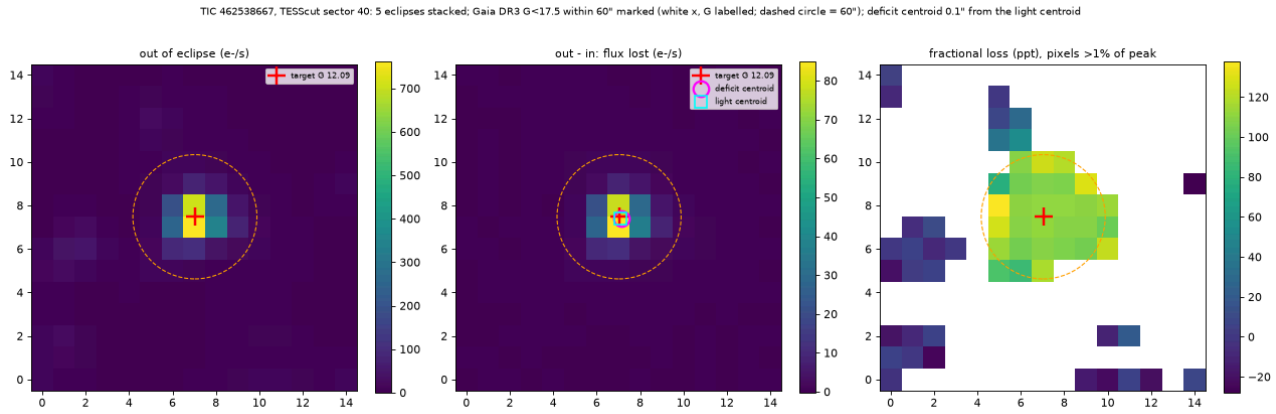
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R444; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R444; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 462538667: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 8 timing-selected measured eclipses in sectors [41, 75]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=5.0931372216 d, fit error=0.0000093209 d, epoch BTJD 2421.1146206 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 consecutive within-sector same-kind cycle pairs. The inventory period was 5.09313 d and the historical SAP search branch was 5.0931300597 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S40 TESScut deficit core is 2.62 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Fractional PSF model residual is 0.021, not a localization uncertainty. Capable Gaia neighbors: none under the approximate G-band flux budget. The full list and flux-budget calculation are in historical/diff_s40.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 111.52/110.09/109.81 ppt from 5 selected windows. Median selected QLP raw fit depth is 112.46 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 1688501177639000320 has P=5.0931642420 d and e=0.03904. Closest edge-on conjunction differs from the measured epoch by 0.02129 in phase (0.10842 d); linear propagation of the source-specific full SB1 covariance gives 0.13996 d as a descriptive timing scale. The rough central-duration estimate is 4.85 h versus trapezoid median 3.46

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Eight deep raw events across S41 and S75 include consecutive cycles, with a well-centered pixel deficit and stable depths.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
41	144	2421.1148783	0.003472	112.60	used
41	145	2426.2062636	0.003472	112.33	used
41	146	2431.3012696	0.003472	110.92	used
41	147	2436.3942317	0.003472	113.62	used
41	148	2441.4878561	0.003472	111.19	used
75	326	3348.0656501	0.001157	113.20	used
75	327	3353.1585480	0.001157	113.04	used
75	329	3363.3451330	0.001157	112.02	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s40.json, historical/diff_s40.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R444. Public evidence files: diff_s40.json, diff_s40.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

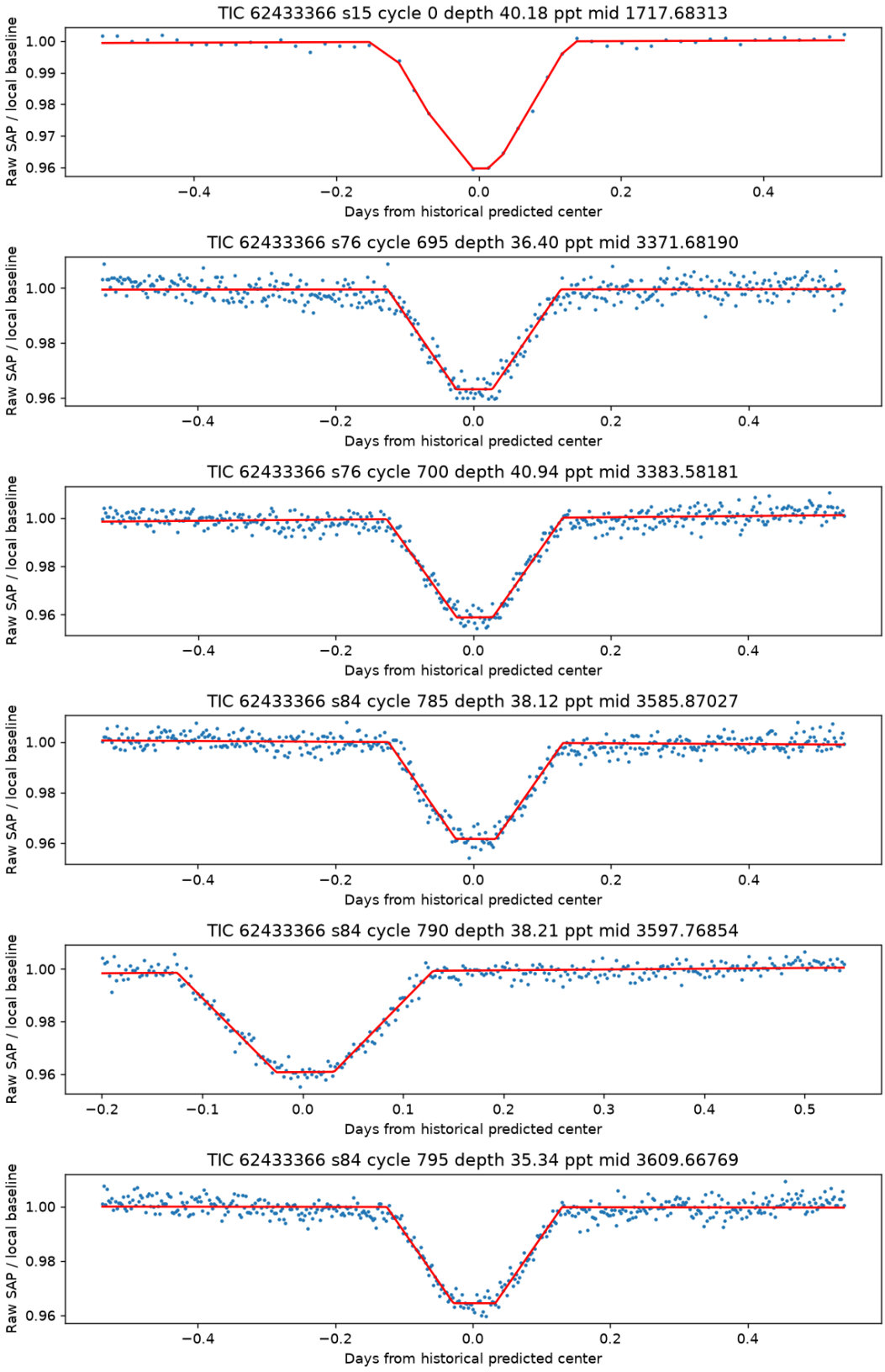
R445. TIC 62433366

physical_verdict: CONFIRMED

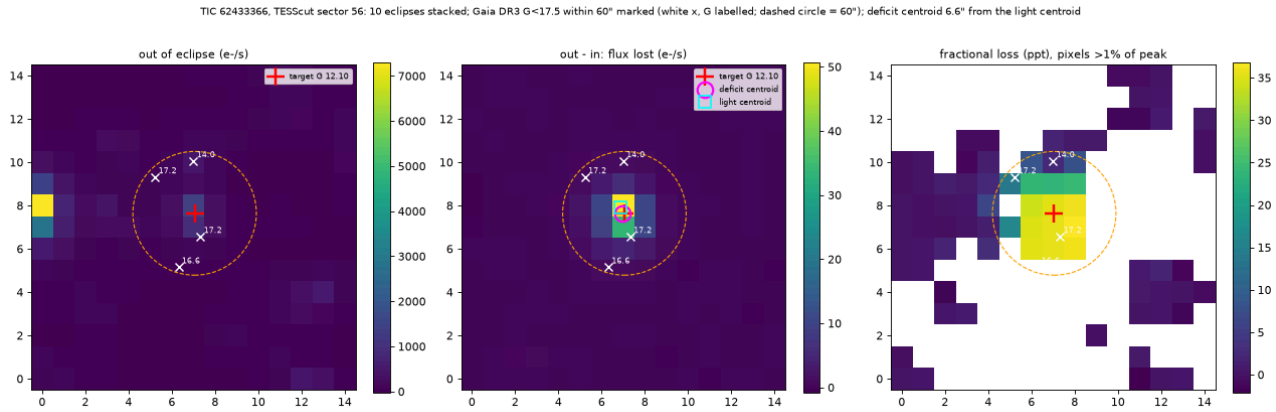
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R445; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R445; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 62433366: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 12 timing-selected measured eclipses in sectors [15, 76, 84]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=4.7597184350 d, fit error=0.0000117166 d, epoch BTJD 1717.6797337 (Deeper repeating eclipse; historical half-period parity 0). There are 6 consecutive within-sector same-kind cycle pairs. The inventory period was 4.75970 d and the historical SAP search branch was 2.3798505904 d. Deeper parity 0 of historical half-period cycles; consecutive same-kind events have orbital-cycle count one. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S56 TESScut deficit core is 1.91 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Fractional PSF model residual is 0.405, not a localization uncertainty. Capable Gaia neighbors: 51.0 arcsec, G=14.00, intrinsic-loss requirement ~0.23. The full list and flux-budget calculation are in historical/diff_s56.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 35.29/34.01/29.65 ppt from 10 selected windows. Median selected QLP raw fit depth is 40.07 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 2175713785143071360 has P=4.7596067442 d and e=0.01266. Closest edge-on conjunction differs from the measured epoch by 0.01077 in phase (0.05124 d); linear propagation of the source-specific full SB1 covariance gives 0.05299 d as a descriptive timing scale. The rough central-duration estimate is 13.51 h versus trapezoid median 6.17

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Even historical half-period cycles are modestly but persistently deeper than odd cycles in S76 and S84, supporting the 4.7597 d orbital period. The duration is only a rough trapezoid shape measure.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
15	0	1717.6831262	0.010417	40.18	used
15	4	1727.1993403	0.010417	38.05	used
15	5	1729.5828742	0.010417	35.04	other eclipse kind
15	6	1731.9623506	0.010417	38.58	used
76	694	3369.3007811	0.001157	39.46	used
76	695	3371.6818953	0.001157	36.40	other eclipse kind
76	696	3374.0525045	69105.867428	39.51	excluded: unidentifiable center covariance
76	697	3376.4414970	0.001157	36.42	other eclipse kind
76	698	3378.8210860	0.001157	41.23	used
76	699	3381.2015616	0.001183	36.31	other eclipse kind
76	700	3383.5818120	0.001157	40.94	used
76	701	3385.9613345	0.001157	37.25	other eclipse kind
76	702	3388.3405430	74615.274606	41.44	excluded: unidentifiable center covariance
76	703	3390.7214454	0.001157	37.46	other eclipse kind
76	704	3393.1009750	0.001157	41.70	used
84	785	3585.8702706	0.001157	38.12	other eclipse kind
84	786	3588.2496231	0.001157	40.09	used
84	787	3590.6295307	0.001157	36.54	other eclipse kind
84	788	3593.0082983	0.001157	40.05	used
84	789	3595.3870047	0.001157	35.52	other eclipse kind
84	790	3597.7685395	0.001157	38.21	used
84	791	3600.1486071	0.001157	35.99	other eclipse kind
84	792	3602.5287240	0.001157	39.26	used
84	793	3604.9075849	0.001157	37.18	other eclipse kind
84	794	3607.2879584	0.001157	41.04	used
84	795	3609.6676856	0.001157	35.34	other eclipse kind

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R445. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

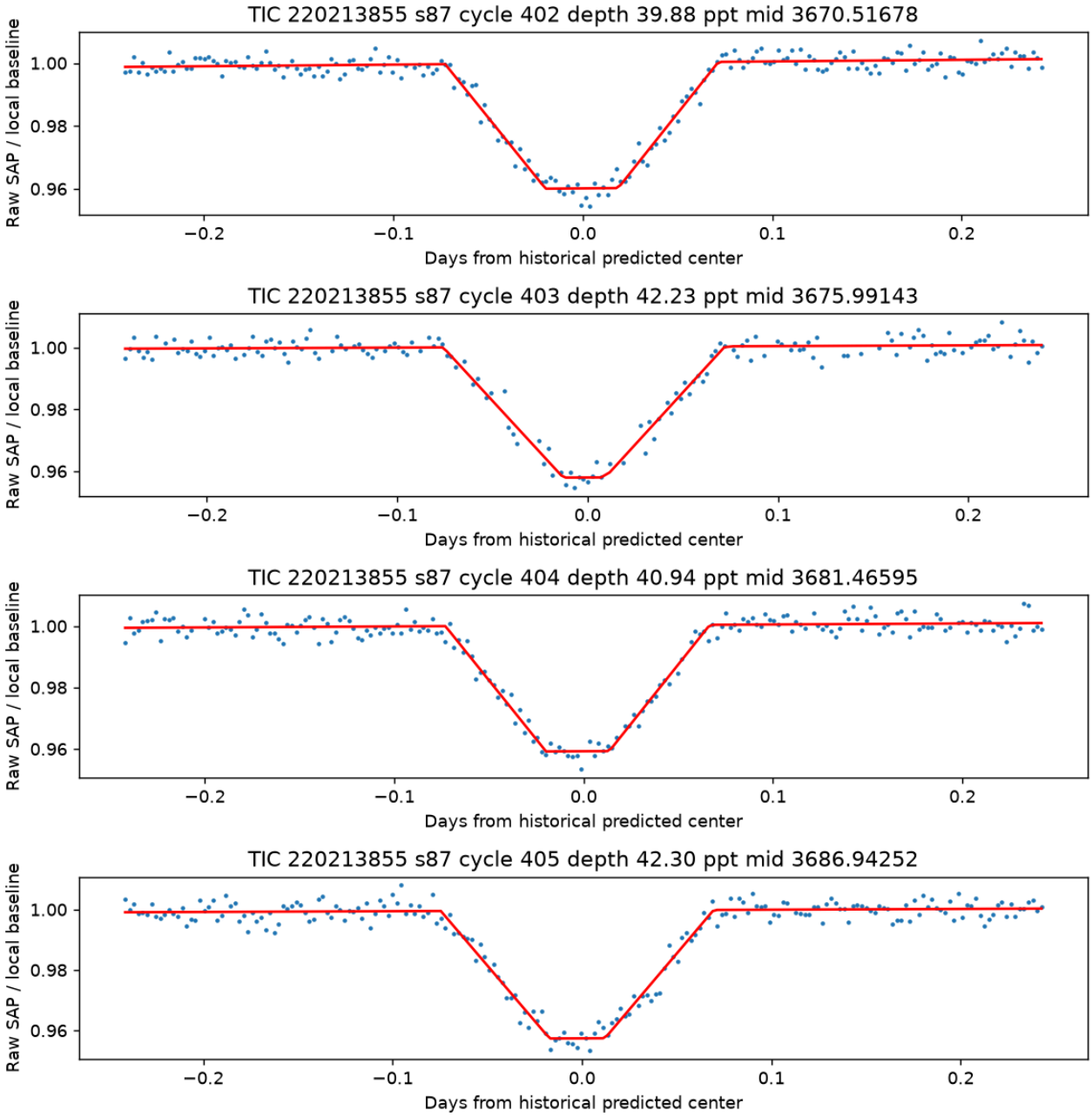
R446. TIC 220213855

physical_verdict: CONFIRMED

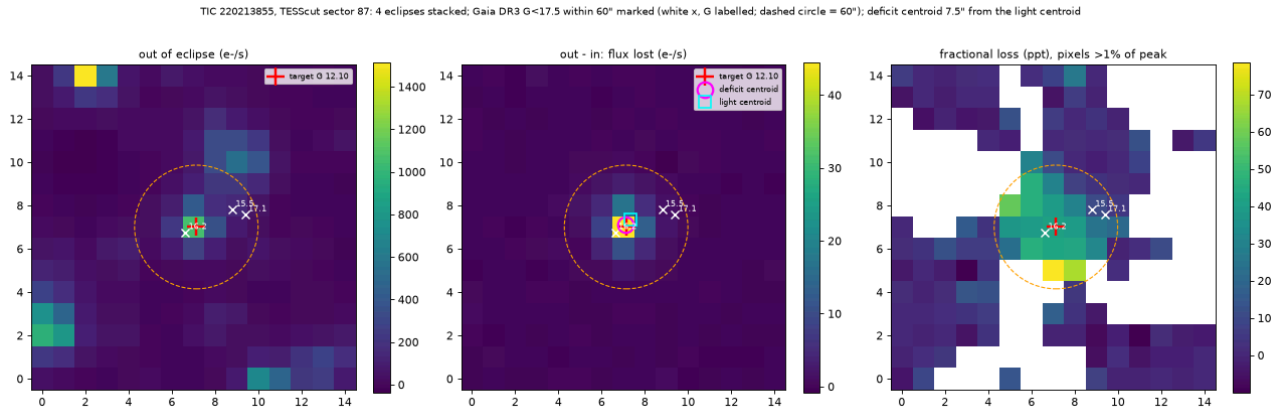
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R446; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R446; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 220213855: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 4 timing-selected measured eclipses in sectors [87]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=5.4751727504 d, fit error=0.0005176080 d, epoch BTJD 3670.5164102 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 3 consecutive within-sector same-kind cycle pairs. The inventory period was 5.47591 d and the historical SAP search branch was 5.4759094123 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S87 TESScut deficit core is 2.58 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Fractional PSF model residual is 0.608, not a localization uncertainty. Capable Gaia neighbors: 38.8 arcsec, G=15.53, intrinsic-loss requirement ~0.99. The full list and flux-budget calculation are in historical/diff_s87.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 41.78/40.53/35.53 ppt from 4 selected windows. Median selected QLP raw fit depth is 41.58 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 3326997734678765696 has P=5.4759465374 d and e=0.03177. Closest edge-on conjunction differs from the measured epoch by 0.01870 in phase (0.10240 d); linear propagation of the source-specific full SB1 covariance gives 0.34526 d as a descriptive timing scale. The rough central-duration estimate is 5.81 h versus trapezoid median 3.46

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Four consecutive raw eclipses in S87 establish recurrence across 16.43 d. S6 is available but its candidate windows have at most four quality-zero cadences inside half the search duration; no clean null is claimed. The fitted period error is limited by one sector and is not total ephemeris uncertainty.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
87	402	3670.5167841	0.001157	39.88	used
87	403	3675.9914259	0.001157	42.23	used
87	404	3681.4659477	0.001157	40.94	used
87	405	3686.9425194	0.001157	42.30	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s87.json, historical/diff_s87.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R446. Public evidence files: diff_s87.json, diff_s87.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

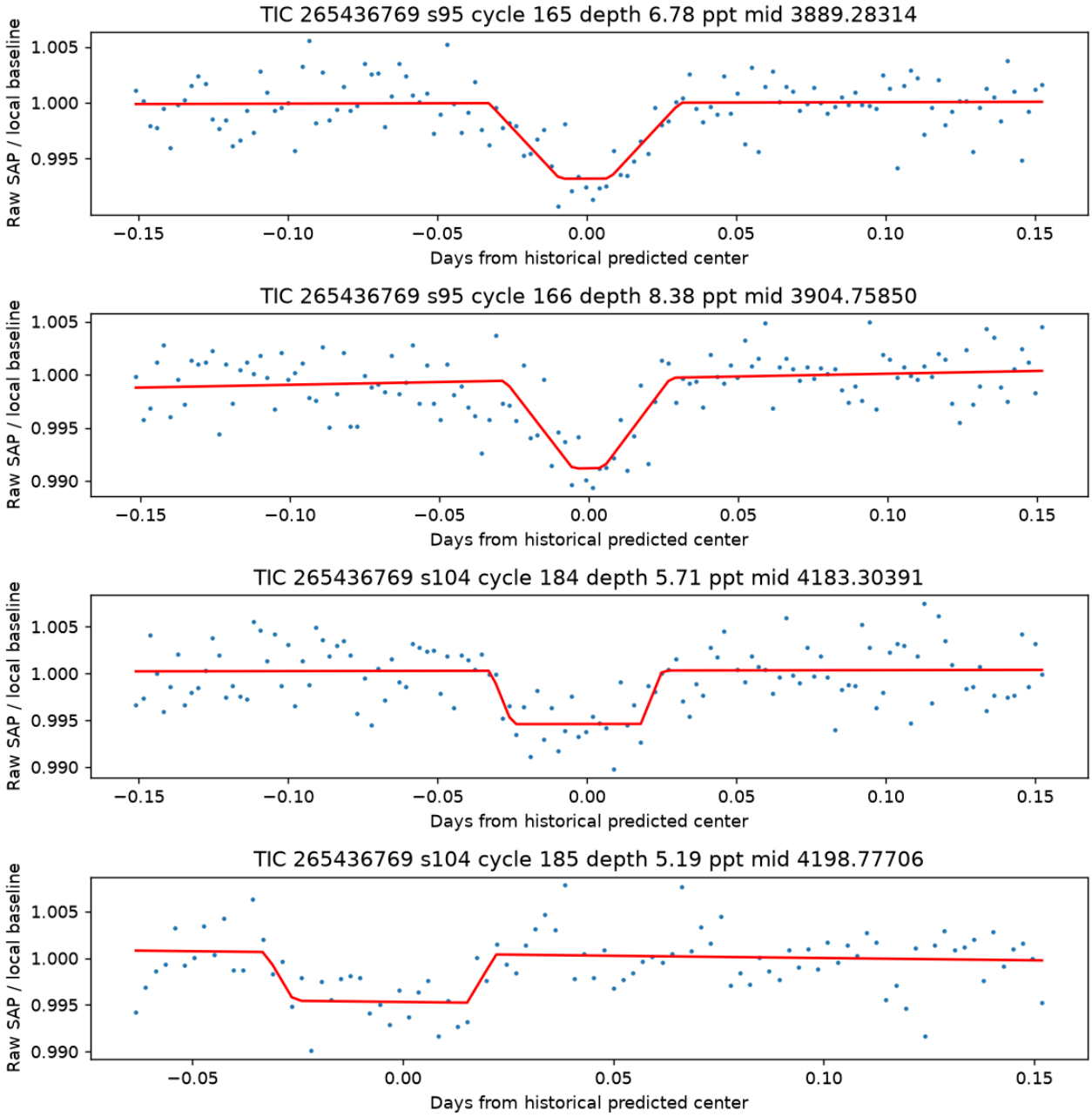
R447. TIC 265436769

physical_verdict: PROBABLE

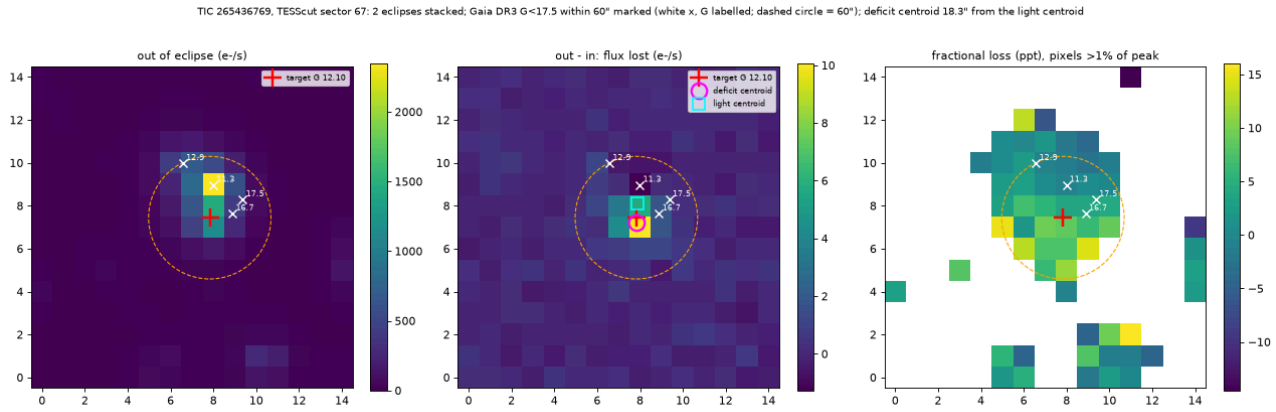
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved source localization



R447; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R447; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 265436769: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 4 timing-selected measured eclipses in sectors [95, 104]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=15.4747037950 d, fit error=0.0000777982 d, epoch BTJD 3889.2835333 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 consecutive within-sector same-kind cycle pairs. The inventory period was 15.47493 d and the historical SAP search branch was 15.4749291199 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: unresolved at TESS resolution.** Historical S67 TESScut deficit core is 4.64 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Fractional PSF model residual is 0.870, not a localization uncertainty. Capable Gaia neighbors: 22.6 arcsec, G=16.71, intrinsic-loss requirement ~0.42; 32.1 arcsec, G=11.32, intrinsic-loss requirement ~0.01; 36.8 arcsec, G=17.48, intrinsic-loss requirement ~0.84; 59.2 arcsec, G=12.94, intrinsic-loss requirement ~0.02. The full list and flux-budget calculation are in historical/diff_s67.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 8.63/6.38/3.66 ppt from 2 selected windows. Median selected QLP raw fit depth is 6.24 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 6403358774444776960 has P=15.4729899462 d and e=0.21828. Closest edge-on conjunction differs from the measured epoch by 0.02964 in phase

(0.45867 d); linear propagation of the source-specific full SB1 covariance gives 0.48556 d as a descriptive timing scale. The rough central-duration estimate is 4.96 h versus trapezoid median 1.36 h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Four shallow raw events in S95 and S104 include same-sector consecutive cycles. The S67 difference core favors the target, yet a capable source 22.6 arcsec away and a high image-model residual (0.87) leave attribution unresolved. Early S1 windows are sparse or flagged, not clean nulls.

Next check. Resolved eclipse photometry or multi-source PRF localization for source attribution.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
95	165	3889.2831391	0.001679	6.78	used
95	166	3904.7584970	0.001422	8.38	used
104	184	4183.3039144	0.001691	5.71	used
104	185	4198.7770617	0.001266	5.19	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s67.json, historical/diff_s67.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R447. Public evidence files: diff_s67.json, diff_s67.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

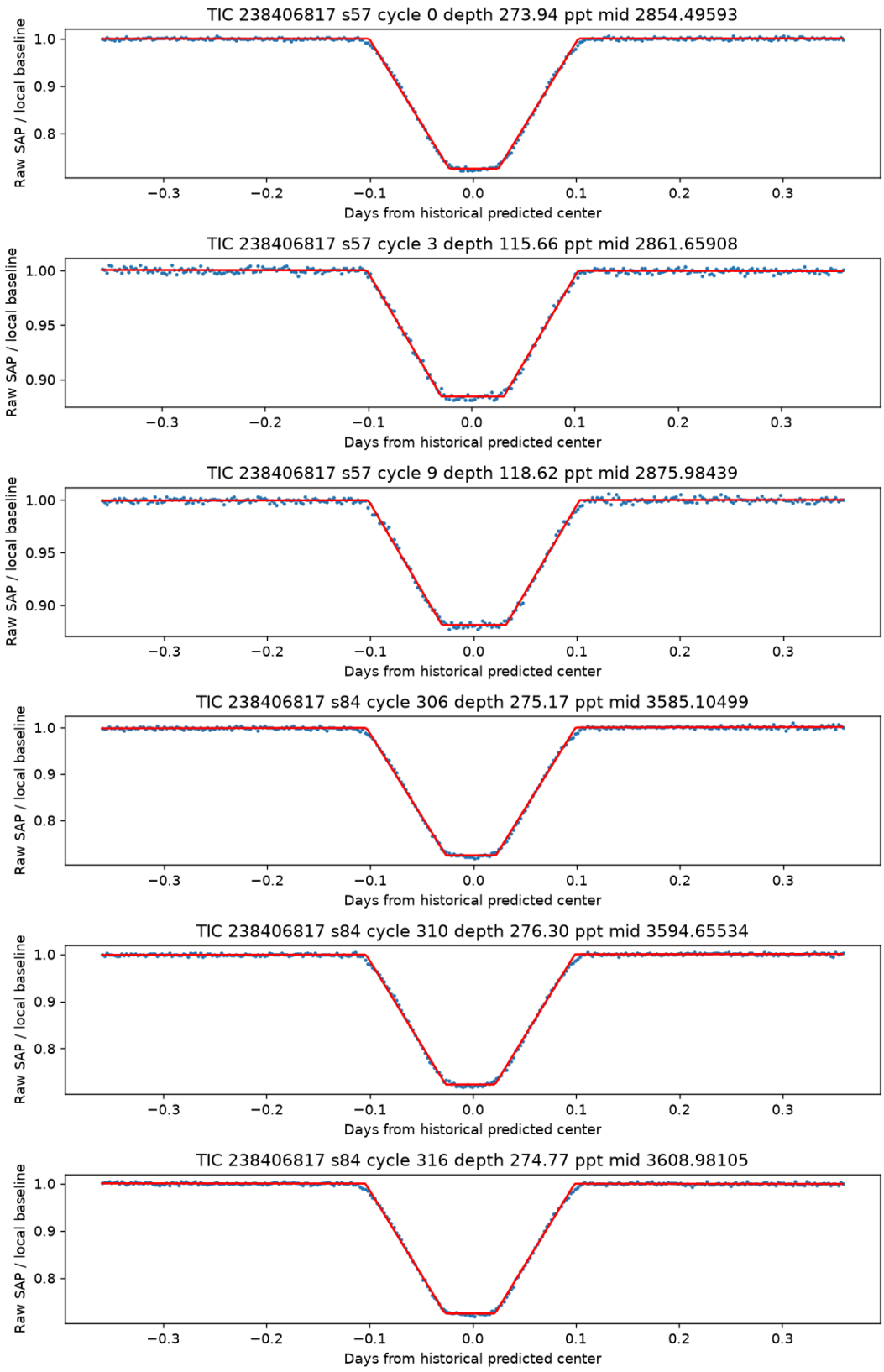
R448. TIC 238406817

physical_verdict: CONFIRMED

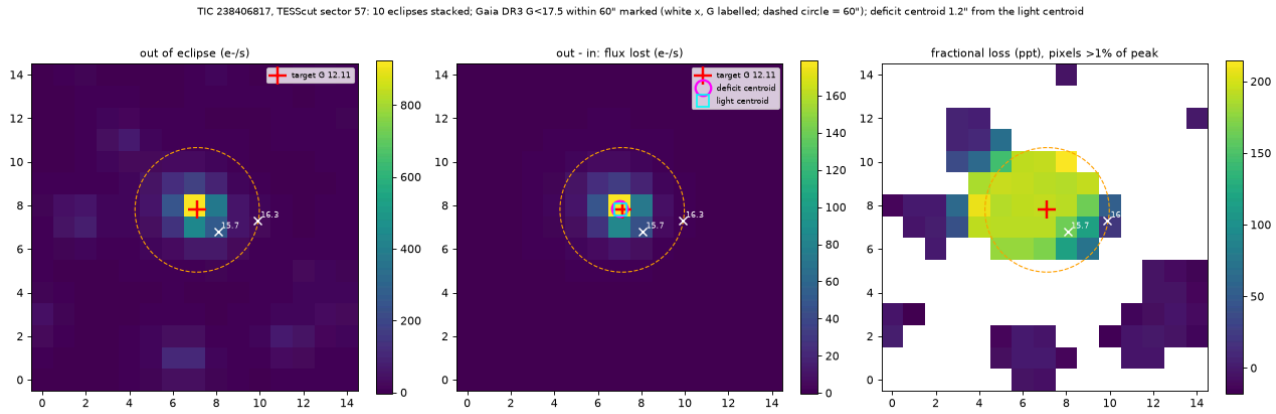
novelty_status: known prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R448; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R448; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 238406817: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 9 timing-selected measured eclipses in sectors [57, 84]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=4.7752225643 d, fit error=0.0000050930 d, epoch BTJD 2854.4959258 (Deeper repeating eclipse; historical half-period parity 0). There are 5 consecutive within-sector same-kind cycle pairs. The inventory period was 4.77524 d and the historical SAP search branch was 2.3876213686 d. Deeper parity 0 of historical half-period cycles; consecutive same-kind events have orbital-cycle count one. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S57 TESScut deficit core is 2.54 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Fractional PSF model residual is 0.089, not a localization uncertainty. Capable Gaia neighbors: none under the approximate G-band flux budget. The full list and flux-budget calculation are in historical/diff_s57.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 193.58/189.24/185.20 ppt from 10 selected windows. Median selected QLP raw fit depth is 274.77 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 2854017381783700352 has P=4.7751949852 d and e=0.01333. Closest edge-on conjunction differs from the measured epoch by 0.00195 in phase (0.00933 d); linear propagation of the source-specific full SB1 covariance gives 0.12817 d as a descriptive timing scale. The rough central-duration estimate is 5.98 h versus trapezoid median 4.87

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Alternating ~275 and ~116 ppt eclipses identify the 4.7752 d orbital period; the historical 2.38762 d SAP branch is the half-period. The deeper parity defines the epoch.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
57	0	2854.4959299	0.001157	273.94	used
57	1	2856.8836580	0.001157	114.76	other eclipse kind
57	2	2859.2710510	0.001157	273.97	used
57	3	2861.6590785	0.001157	115.66	other eclipse kind
57	6	2868.8215384	0.001157	273.49	used
57	7	2871.2094824	0.001157	117.85	other eclipse kind
57	8	2873.5968356	0.001157	274.92	used
57	9	2875.9843939	0.001157	118.62	other eclipse kind
57	10	2878.3721685	0.001157	274.42	used
57	11	2880.7598585	0.001157	118.44	other eclipse kind
84	306	3585.1049933	0.001157	275.17	used
84	307	3587.4930233	0.001157	114.11	other eclipse kind
84	308	3589.8803134	0.001157	276.62	used
84	309	3592.2681450	0.001157	116.02	other eclipse kind
84	310	3594.6553399	0.001157	276.30	used
84	313	3601.8182245	0.001157	115.69	other eclipse kind
84	314	3604.2136516	53785.751904	275.92	excluded: unidentifiable center covariance
84	315	3606.5929309	0.001157	114.09	other eclipse kind
84	316	3608.9810451	0.001157	274.77	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s57.json, historical/diff_s57.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R448. Public evidence files: diff_s57.json, diff_s57.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

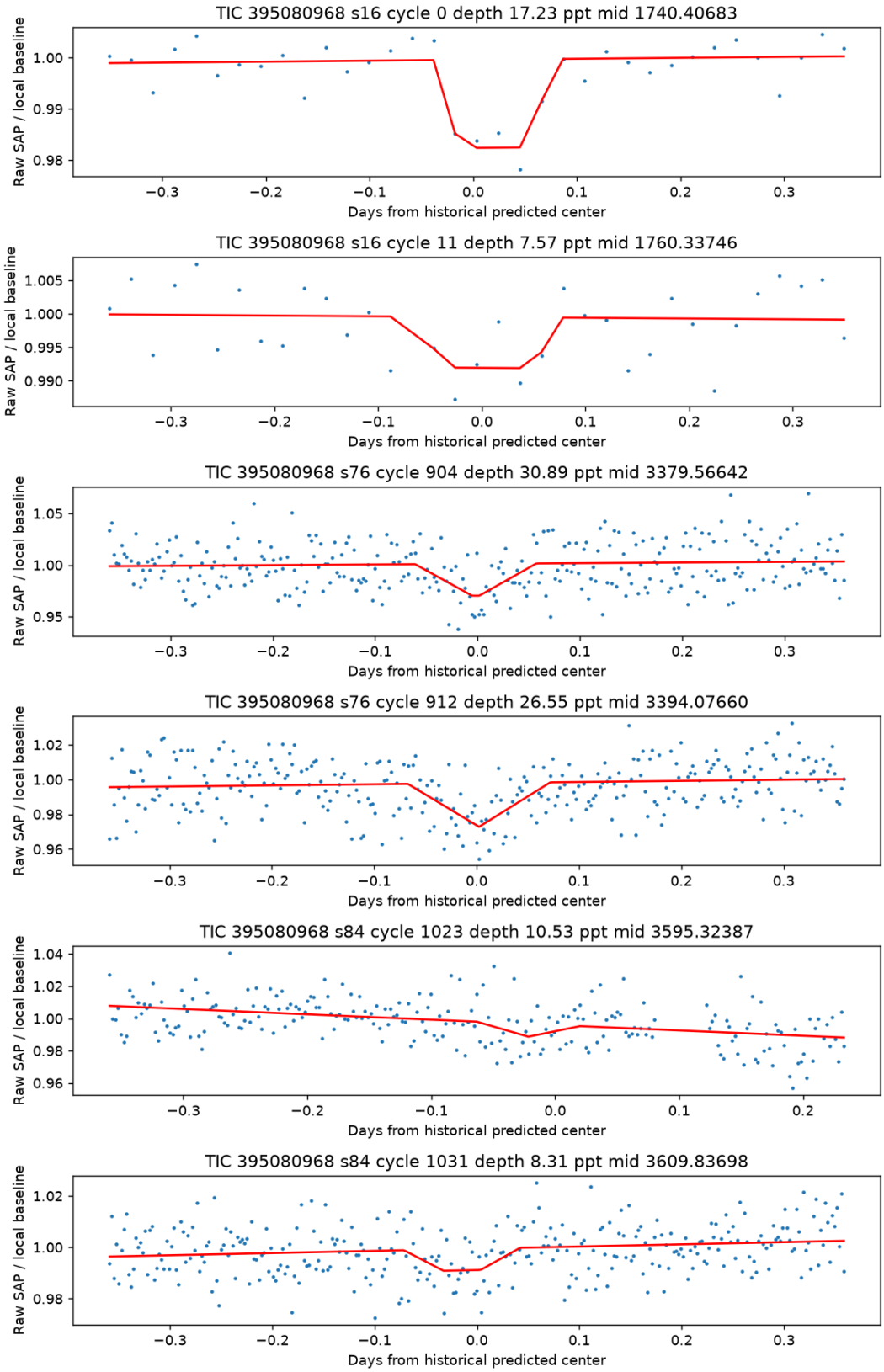
R449. TIC 395080968

physical_verdict: PROBABLE

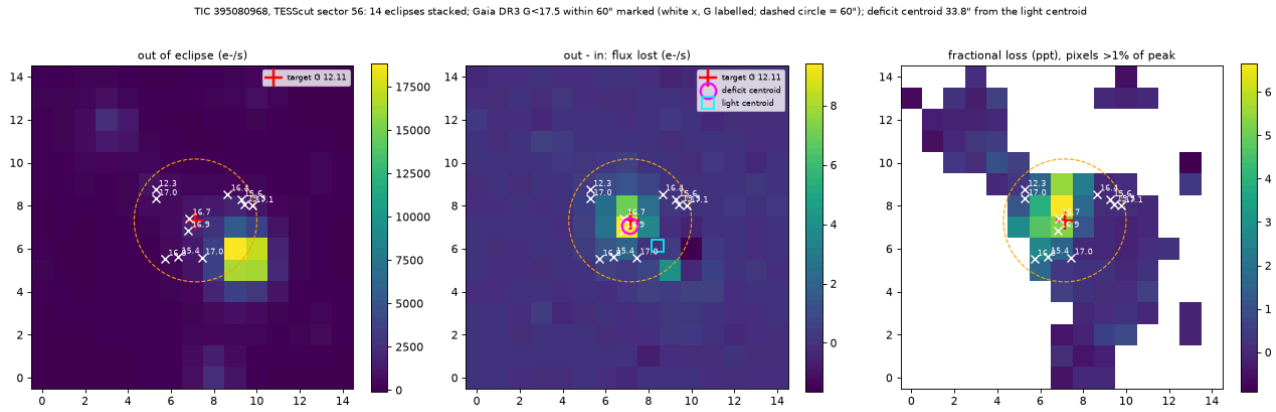
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved source localization



R449; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R449; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 395080968: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 17 timing-selected measured eclipses in sectors [16, 76, 84]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=3.6265035299 d, fit error=0.0000164528 d, epoch BTJD 1740.3942050 (Deeper repeating eclipse; historical half-period parity 0). There are 12 consecutive within-sector same-kind cycle pairs. The inventory period was 3.62651 d and the historical SAP search branch was 1.8132553250 d. Deeper parity 0 of historical half-period cycles; consecutive same-kind events have orbital-cycle count one. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: unresolved at TESS resolution.** Historical S56 TESScut deficit core is 4.65 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Fractional PSF model residual is 0.982, not a localization uncertainty. Capable Gaia neighbors: 6.3 arcsec, G=16.71, intrinsic-loss requirement ~0.75; 11.9 arcsec, G=16.91, intrinsic-loss requirement ~0.91; 35.7 arcsec, G=16.98, intrinsic-loss requirement ~0.96; 38.5 arcsec, G=15.37, intrinsic-loss requirement ~0.23; 40.1 arcsec, G=16.38, intrinsic-loss requirement ~0.56; 5 further capable sources in the 60 arcsec list. The full list and flux-budget calculation are in historical/diff_s56.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 5.29/2.22/0.65 ppt from 14 selected windows. Median selected QLP raw fit depth is 18.54 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 2171489770715391616 has P=3.6264741663 d and e=0.06738. Closest edge-on conjunction differs from the measured epoch by -0.00711 in phase

(-0.02580 d); linear propagation of the source-specific full SB1 covariance gives 0.07572 d as a descriptive timing scale. The rough central-duration estimate is 8.38 h versus trapezoid median 3.11 h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Alternating depths support the 3.6265 d orbital period, but raw fits scatter strongly and residuals reach 75 min. The S56 alternative aperture stack mixes eclipse kinds; its depth is not an intrinsic primary depth. Capable neighbors at 6.3 and 11.9 arcsec share the TESS image core and the image-model residual is 0.982, so source attribution and precision ephemeris remain limited.

Next check. Resolved eclipse photometry or multi-source PRF localization for source attribution.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
16	0	1740.4068340	0.010417	17.23	used
16	1	1742.1753092	0.010417	15.29	other eclipse kind
16	2	1744.0135012	0.010417	21.36	used
16	7	1753.0596536	0.019758	5.08	other eclipse kind
16	8	1754.8942611	0.010417	21.08	used
16	9	1756.6989109	0.010417	9.32	other eclipse kind
16	11	1760.3374561	0.016615	7.57	other eclipse kind
76	898	3368.7464702	0.014812	16.95	used
76	899	3370.4767960	0.003860	7.00	other eclipse kind
76	900	3372.3198869	0.010680	18.60	used
76	902	3375.9517373	0.005173	22.92	used
76	903	3377.7228961	0.008759	16.70	other eclipse kind
76	904	3379.5664159	0.005774	30.89	used
76	906	3383.2096794	0.005946	20.55	used
76	907	3385.0113163	0.007454	8.18	other eclipse kind
76	908	3386.8231146	0.004738	17.53	used
76	909	3388.6222925	0.003181	54.45	other eclipse kind
76	910	3390.4504345	0.004672	14.95	used
76	911	3392.2404815	0.004192	46.66	other eclipse kind
76	912	3394.0765956	0.005304	26.55	used
84	1018	3586.2803727	0.004857	20.02	used
84	1019	3588.0793575	0.007339	11.74	other eclipse kind
84	1020	3589.9091733	0.004168	18.54	used
84	1021	3591.7124273	0.008708	11.84	other eclipse kind
84	1022	3593.5427983	0.004557	16.65	used
84	1023	3595.3238697	0.010836	10.53	other eclipse kind

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
84	1025	3598.9485479	0.005482	15.84	other eclipse kind
84	1026	3600.7852493	0.003297	15.68	used
84	1027	3602.5784254	0.006899	9.21	other eclipse kind
84	1028	3604.4200775	0.003981	18.16	used
84	1029	3606.2246957	0.008835	8.32	other eclipse kind
84	1030	3608.0478675	0.003847	16.65	used
84	1031	3609.8369787	0.007819	8.31	other eclipse kind

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R449. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

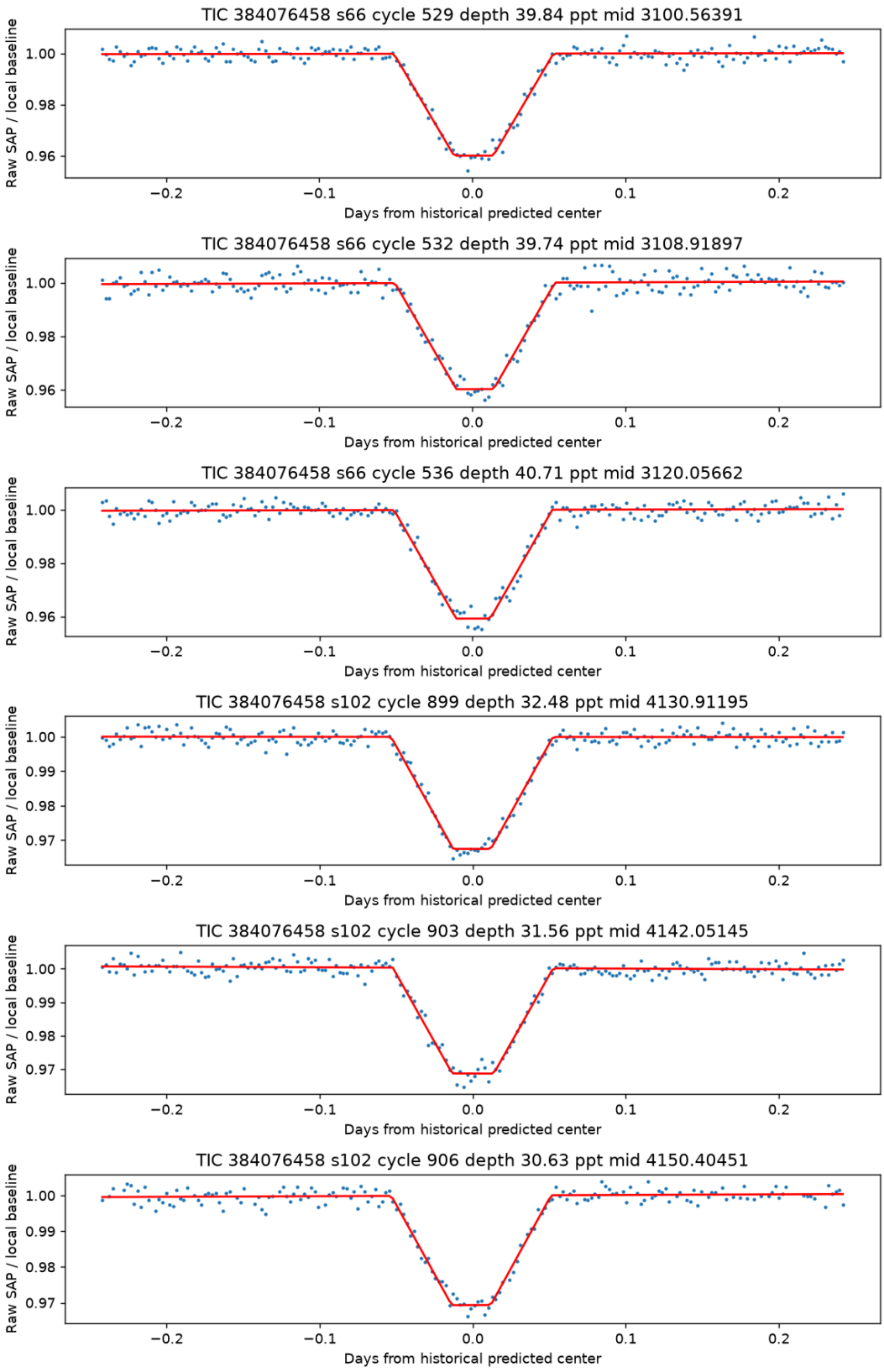
R450. TIC 384076458

physical_verdict: CONFIRMED

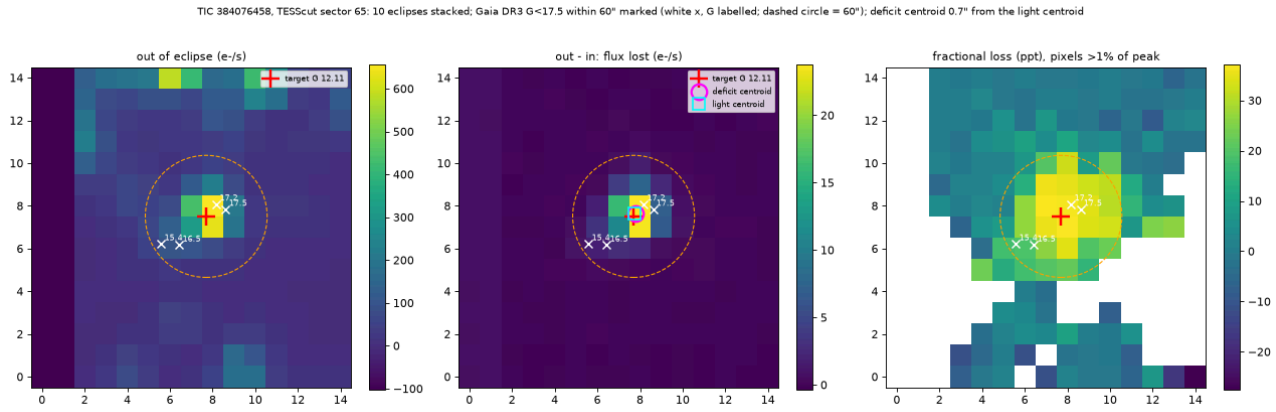
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R450; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R450; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 384076458: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 16 timing-selected measured centers in sectors [66, 102]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 2.7847243427 ± 0.0000015681 d and reference epoch BTJD 3100.5639723 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 12 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 2.78472813 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 65 TESScut difference image puts the deficit 3.54 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.204 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 5778272945037909632 (46.63 arcsec, G 15.372, intrinsic required loss about 0.728). Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 36.20/35.87/33.61 ppt in 10 stacked historical events; representative QLP raw median selected depth is 35.73 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 5778272841958695040 has SB1 P = 2.7847986876 d and e = 0.02887. Closest edge-on conjunction offset is -0.05097 d (-0.01830 phase), with 0.08625 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.37 h versus observed median fit 2.49 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Sixteen measured raw eclipses span sectors 66 and 102 with twelve adjacent-cycle pairs. The only approximately capable Gaia neighbor is 46.6 arcsec away from the target-centered deficit.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
66	529	3100.5639140	0.001157	39.84	used
66	530	3103.3478225	0.001157	38.49	used
66	531	3106.1333249	0.001157	39.70	used
66	532	3108.9189679	0.001157	39.74	used
66	534	3114.4877752	0.001157	39.69	used
66	535	3117.2728831	0.001157	39.11	used
66	536	3120.0566188	0.001157	40.71	used
66	537	3122.8416413	0.001157	38.09	used
102	898	4128.1276997	0.001157	32.10	used
102	899	4130.9119539	0.001157	32.48	used
102	900	4133.6967499	0.001157	32.44	used
102	901	4136.4814578	0.001157	32.36	used
102	903	4142.0514537	0.001157	31.56	used
102	904	4144.8357621	0.001157	33.36	used
102	905	4147.6196430	0.001157	33.10	used
102	906	4150.4045059	0.001157	30.63	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s65.json, historical/diff_s65.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R450. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

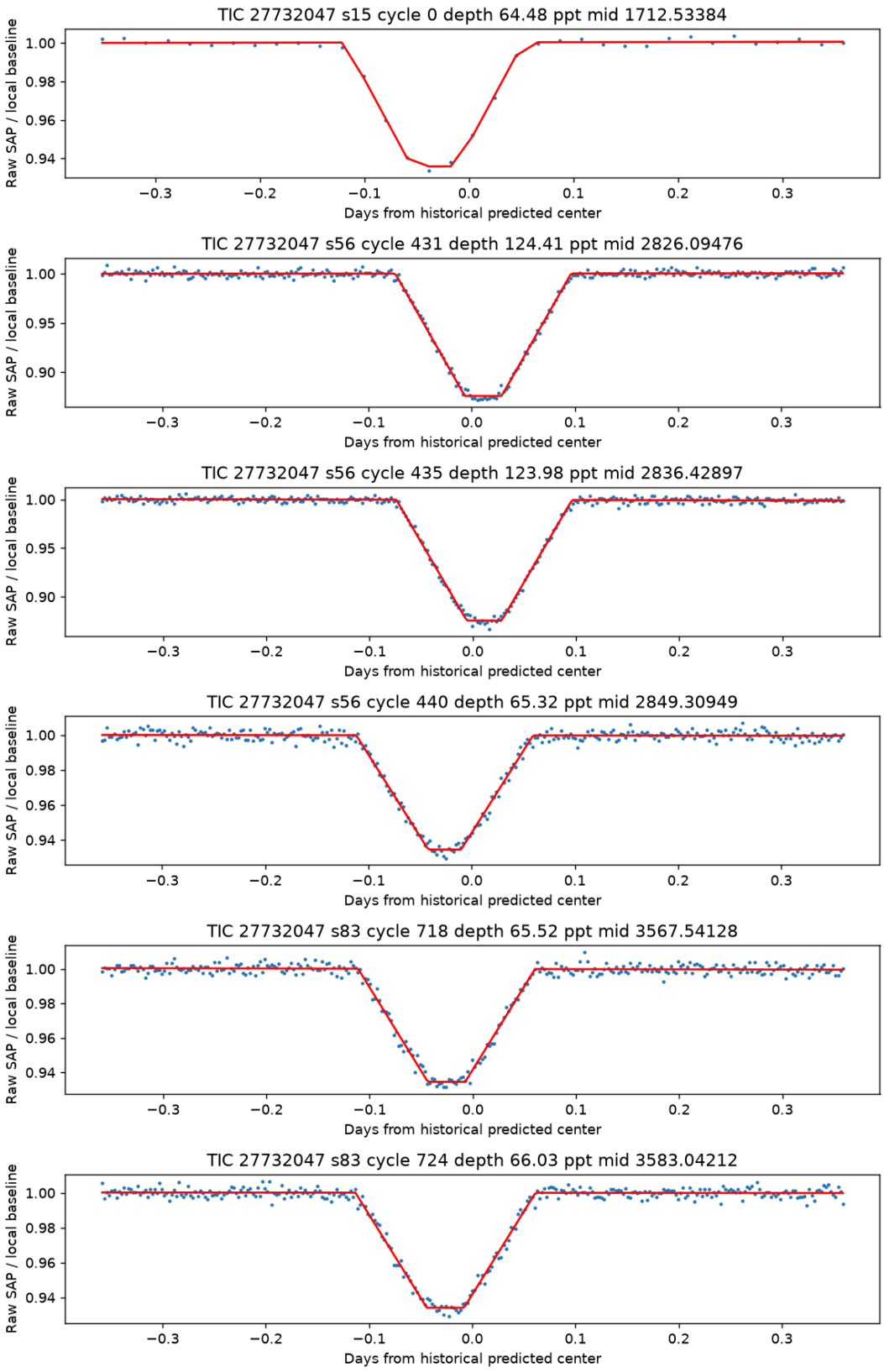
R451. TIC 27732047

physical_verdict: CONFIRMED

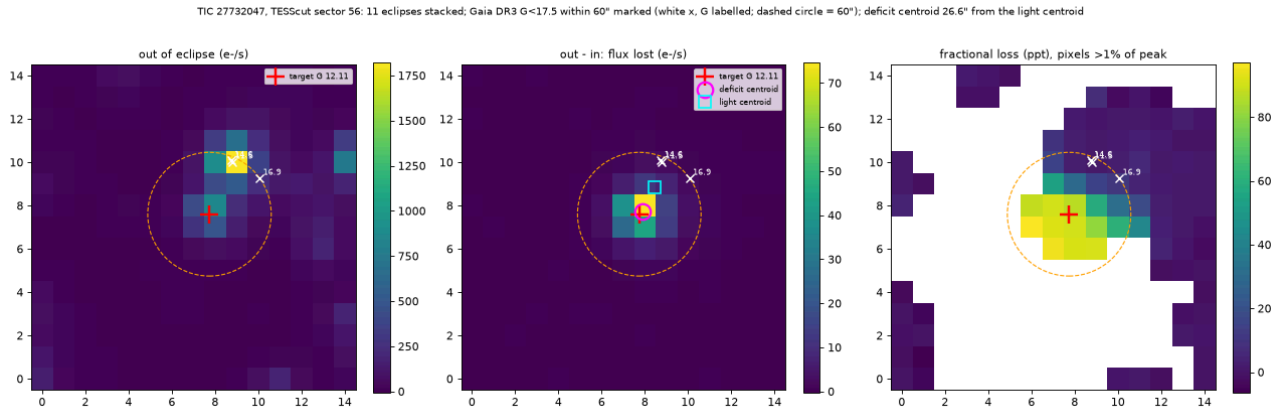
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R451; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R451; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 27732047: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 23 timing-selected measured centers in sectors [15, 56, 83]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 5.1671309470 ± 0.0000052202 d and reference epoch BTJD 1730.6628793 (Deeper repeating eclipse; historical half-period parity 1). There are 7 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 2.58356477 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 56 TESScut difference image puts the deficit 3.97 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.864 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 1893685410481942016 (54.40 arcsec, G 11.479, intrinsic required loss about 0.093); Gaia 1893686166394478208 (55.48 arcsec, G 14.610, intrinsic required loss about 0.655). Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 86.89/67.21/37.77 ppt in 11 stacked historical events; representative QLP raw median selected depth is 123.94 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 1893685406185047936 has SB1 P = 5.1675861098 d and e = 0.01880. Closest edge-on conjunction offset is -0.09725 d (-0.01882 phase), with 0.07040 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.54 h versus observed median fit 4.09 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Deep and shallow eclipses alternate at the 2.58356-d photometric search period. The 5.16713-d orbital ephemeris uses the eleven deeper odd-parity centers; shallow even-parity eclipses independently support the double-period interpretation. Two nominally capable Gaia neighbors lie beyond 54 arcsec; the source image is structured, but its strongest deficit is on the target core.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
15	0	1712.5338395	0.010417	64.48	used
15	2	1717.7013600	0.010417	65.30	used
15	6	1728.0365599	0.010417	65.75	used
15	7	1730.6575933	0.010417	124.34	used
56	431	2826.0947577	0.001157	124.41	used
56	432	2828.6404641	0.001157	65.54	used
56	433	2831.2618576	0.001157	124.37	used
56	434	2833.8087977	0.001157	65.96	used
56	435	2836.4289748	0.001157	123.98	used
56	436	2838.9748702	0.001157	64.57	used
56	437	2841.5959721	0.001157	125.20	used
56	438	2844.1422710	0.001157	64.04	used
56	439	2846.7629037	0.001157	123.38	used
56	440	2849.3094867	0.001157	65.32	used
56	441	2851.9304934	0.001157	123.68	used
83	715	3559.8273915	0.001157	123.43	used
83	716	3562.3739161	0.001157	64.78	used
83	718	3567.5412835	0.001157	65.52	used
83	719	3570.1615750	0.001157	123.76	used
83	720	3572.7081264	0.001157	65.19	used
83	721	3575.3285814	0.001157	123.94	used
83	723	3580.4954790	0.001157	122.52	used
83	724	3583.0421157	0.001157	66.03	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R451. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

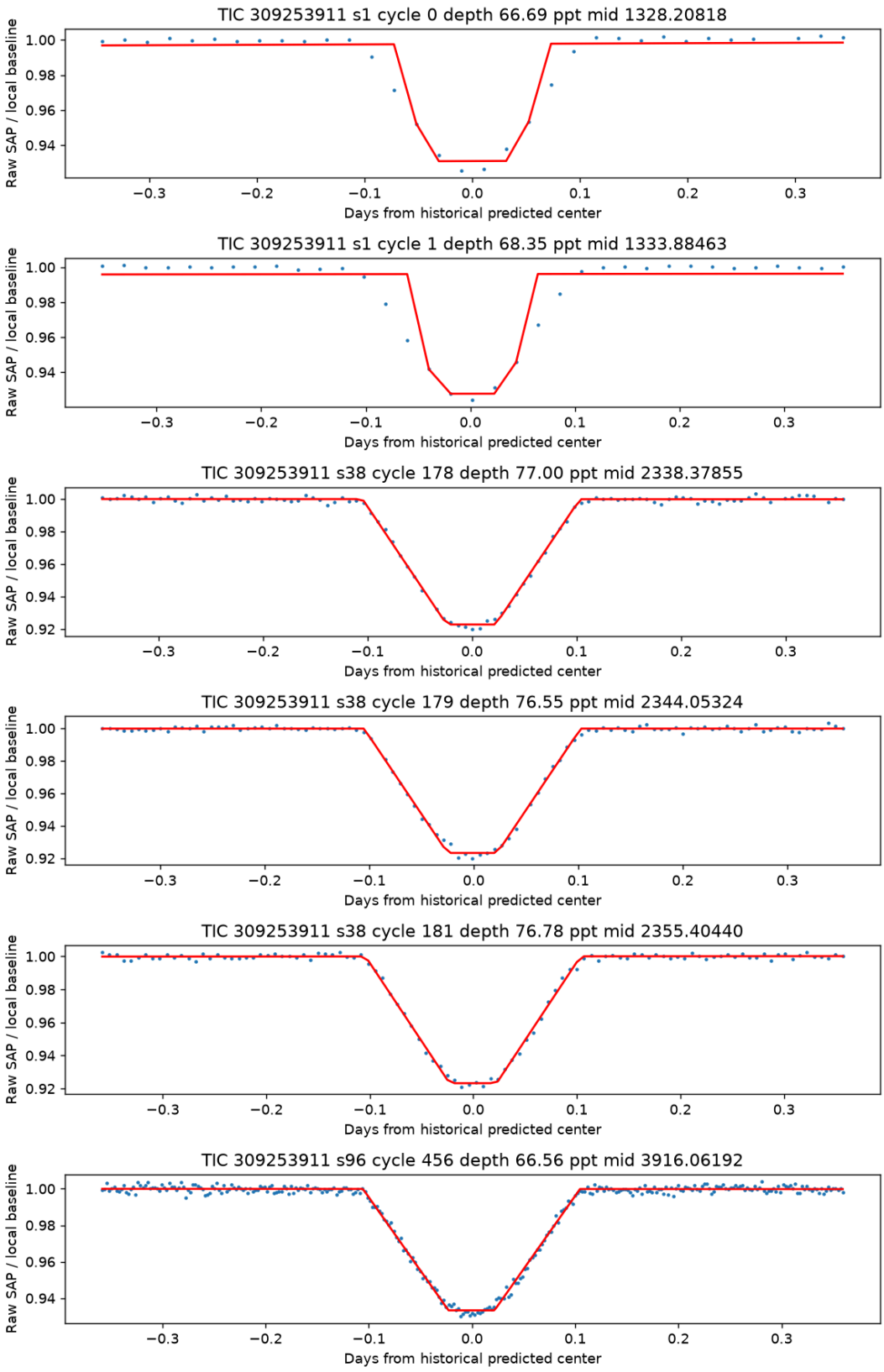
R452. TIC 309253911

physical_verdict: CONFIRMED

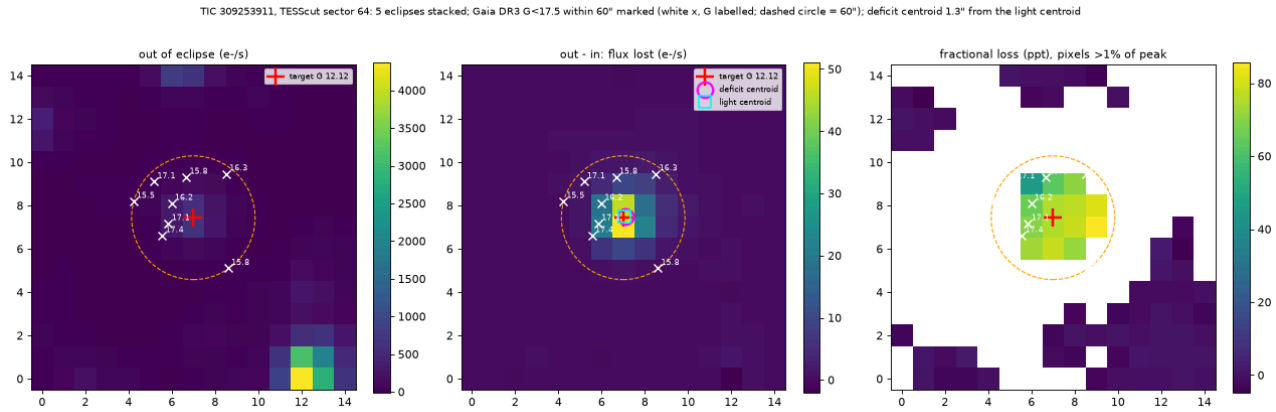
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R452; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R452; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 309253911: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 9 timing-selected measured centers in sectors [1, 38, 96]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 5.6751203957 ± 0.0000063088 d and reference epoch BTJD 1328.2070579 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 5.67511914 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 64 TESScut difference image puts the deficit 2.46 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.179 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 75.40/74.07/69.71 ppt in 5 stacked historical events; representative QLP raw median selected depth is 68.35 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 5276097362970850432 has SB1 P = 5.6754892774 d and e = 0.03670. Closest edge-on conjunction offset is -0.05998 d (-0.01057 phase), with 0.04432 d linearly propagated full source-specific SB1 covariance. Rough central duration 7.71 h versus observed median fit 4.98 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Nine measured eclipses in sectors 1, 38 and 96 include five adjacent-cycle pairs. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
1	0	1328.2081849	0.010416	66.69	used
1	1	1333.8846253	0.010416	68.35	used
1	3	1345.2296355	0.010416	66.37	used
38	178	2338.3785526	0.003472	77.00	used
38	179	2344.0532360	0.003472	76.55	used
38	180	2349.7283396	0.003472	77.20	used
38	181	2355.4044016	0.003472	76.78	used
96	455	3910.3868805	0.001157	65.77	used
96	456	3916.0619223	0.001157	66.56	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s64.json, historical/diff_s64.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R452. Public evidence files: diff_s64.json, diff_s64.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

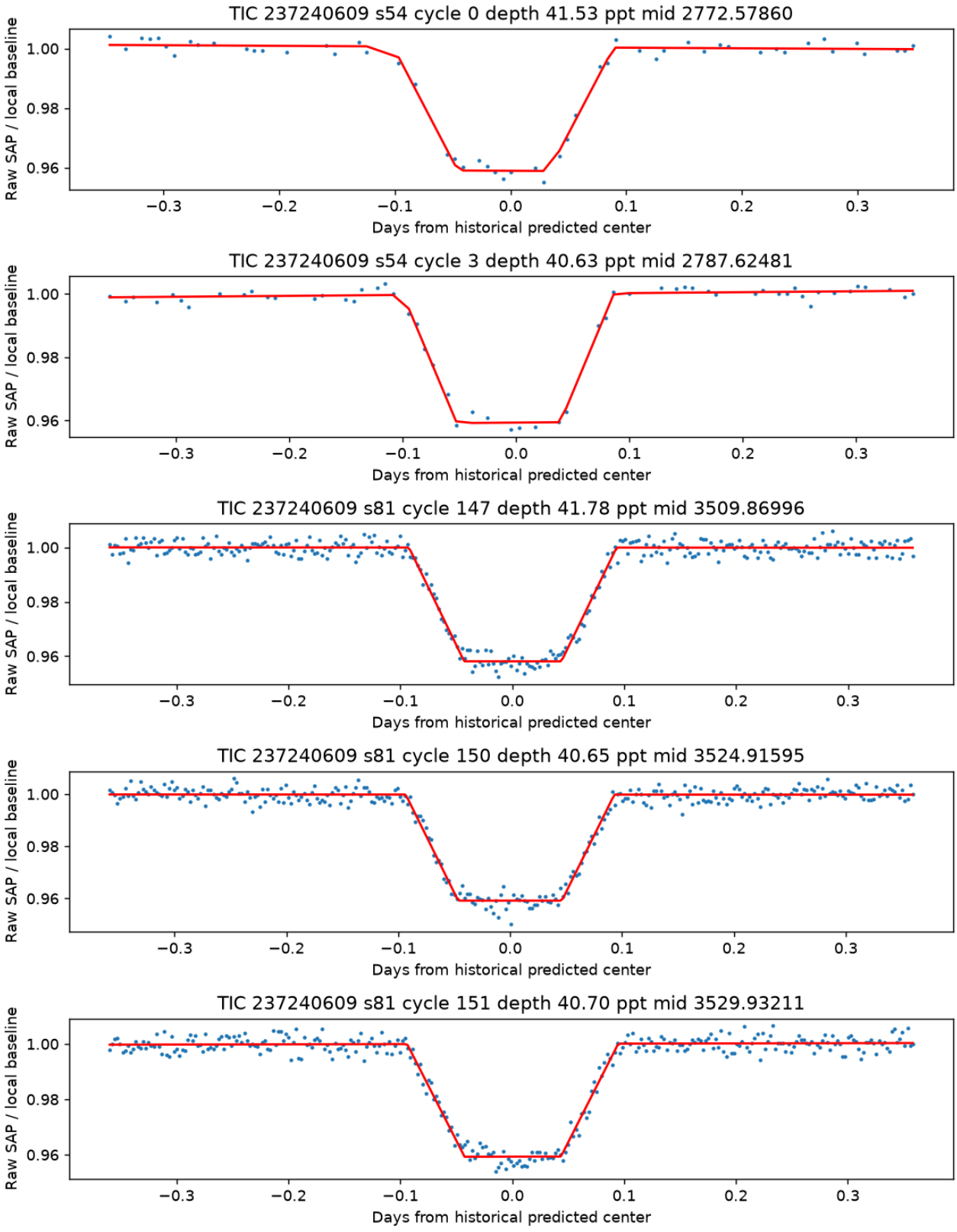
R453. TIC 237240609

physical_verdict: CONFIRMED

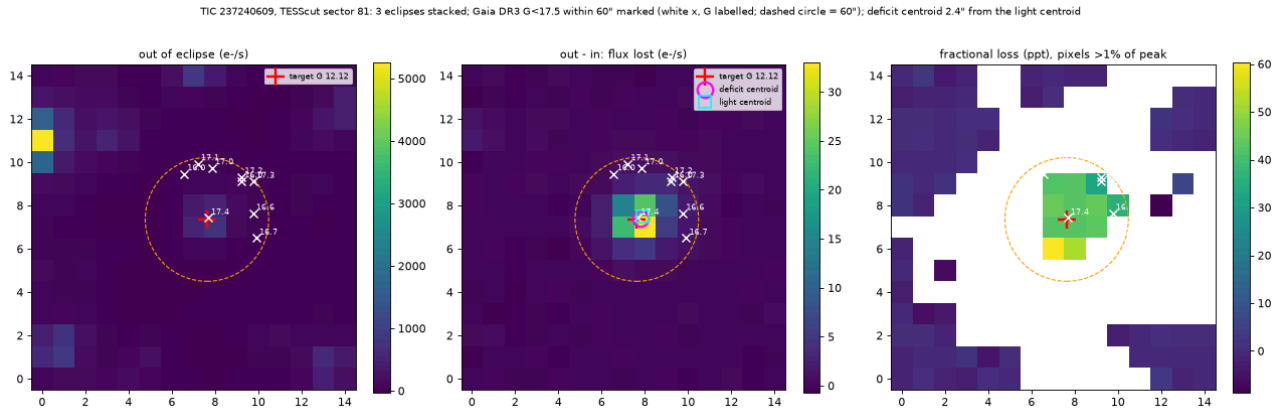
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R453; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R453; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 237240609: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 5 timing-selected measured centers in sectors [54, 81]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 5.0155864471 ± 0.0000333988 d and reference epoch BTJD 2772.5784287 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 5.01554178 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 81 TESScut difference image puts the deficit 3.87 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.262 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 42.53/43.97/44.53 ppt in 3 stacked historical events; representative QLP raw median selected depth is 40.70 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 4196687071198155520 has SB1 P = 5.0156410953 d and e = 0.01696. Closest edge-on conjunction offset is -0.03987 d (-0.00795 phase), with 0.31438 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.29 h versus observed median fit 4.47 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Five measured eclipses in sectors 54 and 81 include a sector-81 adjacent pair. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
54	0	2772.5785995	0.006944	41.53	used
54	3	2787.6248058	0.006944	40.63	used
81	147	3509.8699618	0.001157	41.78	used
81	150	3524.9159492	0.001157	40.65	used
81	151	3529.9321092	0.001157	40.70	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s81.json, historical/diff_s81.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R453. Public evidence files: diff_s81.json, diff_s81.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

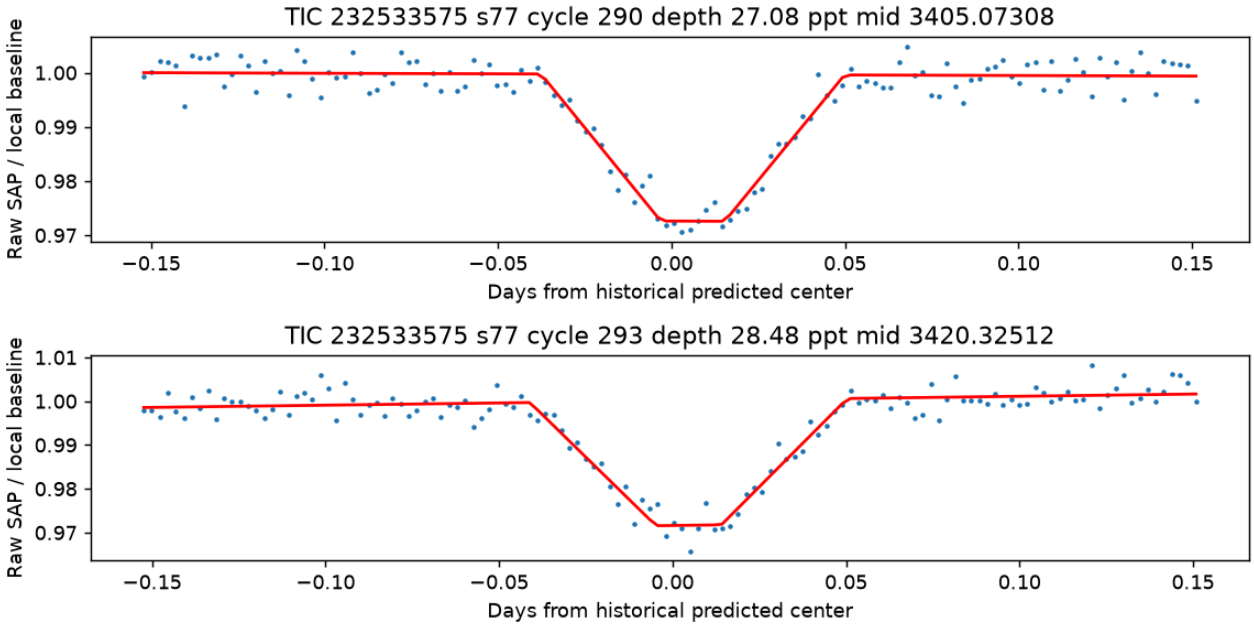
R454. TIC 232533575

physical_verdict: PLAUSIBLE

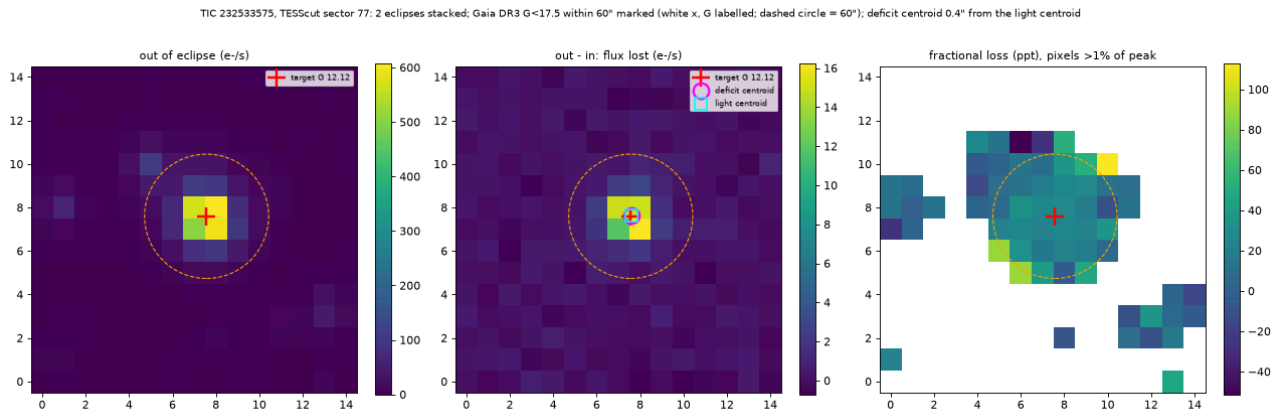
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: A third measured eclipse in future coverage to distinguish 15.252 d divided by three from other integer-cycle aliases.



R454; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R454; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 232533575: PLAUSIBLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 2 timing-selected measured centers in sectors [77]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Two observed eclipse-shaped events do not test the integer-cycle period.
2. **Timing and aliases.** Selected centers yield P = 5.0840126999 ± 0.0005456124 d and reference epoch BTJD 3405.0730806 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 0 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 5.08450258 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search. Live MAST timeseries availability query is retained in ../232533575-availability.json; availability is not an eclipse detection. The two intervening predicted cycles have no quality-zero or flagged QLP cadences within 0.25 d, so neither is a clean null.
3. **Source attribution.** Historical sector 77 TESScut difference image puts the deficit 0.98 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 8]. Image-model fractional residual 0.154 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 24.67/25.12/24.73 ppt in 2 stacked historical events; representative QLP raw median selected depth is 27.78 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 1285502678268839680 has SB1 P = 5.0852473937 d and e = 0.04997. Closest edge-on conjunction offset is -0.35556 d (-0.06992 phase), with 0.18332 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.50 h versus observed median fit 2.15 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Two target-centered eclipses in sector 77 are separated by 15.252 d. The Gaia-like 5.084-d period assigns three cycles, but cycles 291 and 292 fall in an observation gap. Sector 23 has no usable repeated center and current MAST lists only sectors 23 and 77. Thus the integer-cycle alias is untested by a third eclipse; the reported period is conditional on the Gaia-supported three-cycle assignment.

Next check: A third measured eclipse in future coverage to distinguish 15.252 d divided by three from other integer-cycle aliases.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
77	290	3405.0730806	0.001157	27.08	used
77	293	3420.3251187	0.001157	28.48	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s77.json, historical/diff_s77.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json).

Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R454. Public evidence files: diff_s77.json, diff_s77.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

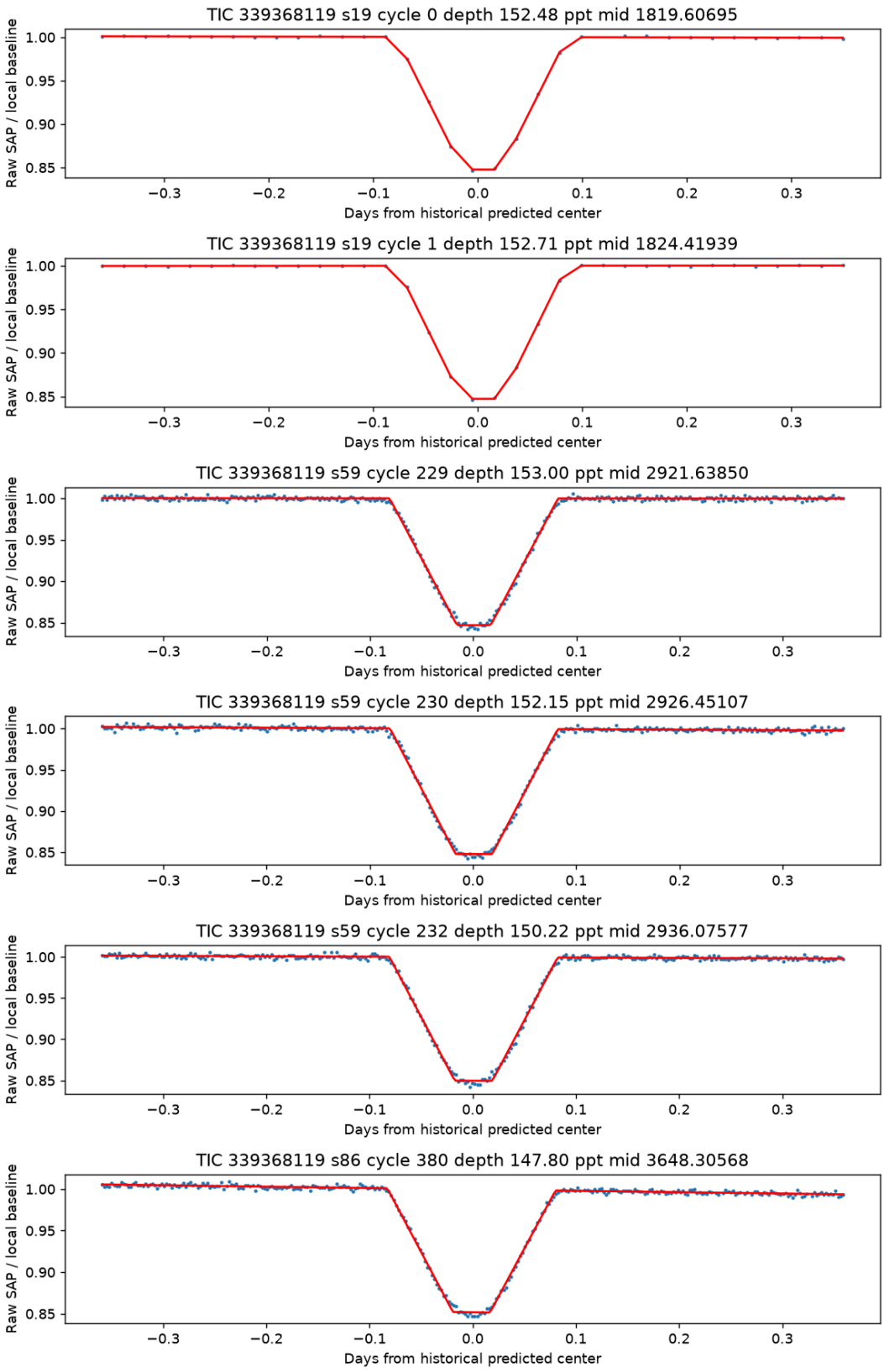
R455. TIC 339368119

physical_verdict: CONFIRMED

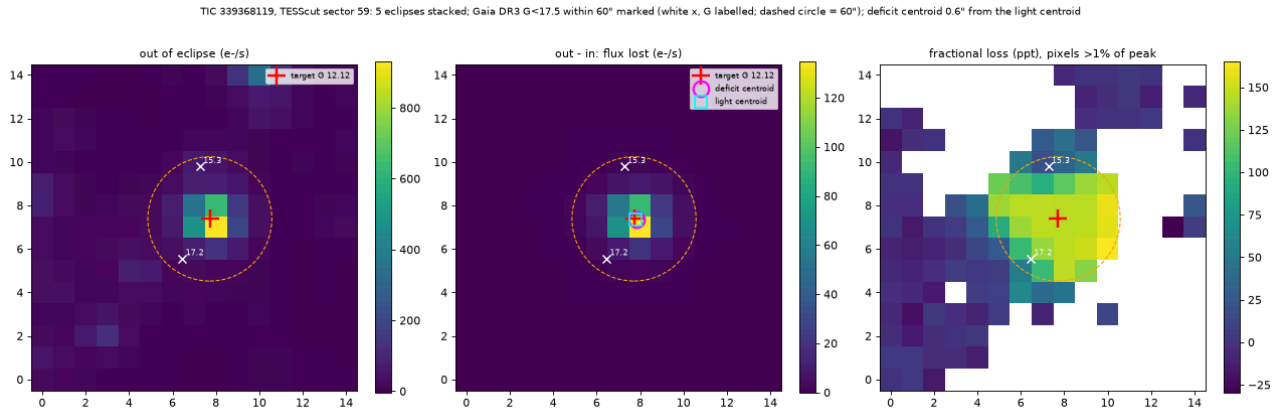
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R455; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R455; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 339368119: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 9 timing-selected measured centers in sectors [19, 59, 86]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 4.8123646667 ± 0.0000062908 d and reference epoch BTJD 1819.6071844 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 4.81237757 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 59 TESScut difference image puts the deficit 2.41 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.110 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 144.77/145.89/142.74 ppt in 5 stacked historical events; representative QLP raw median selected depth is 152.15 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 287971617259251968 has SB1 P = 4.8121222282 d and e = 0.04395. Closest edge-on conjunction offset is 0.04761 d (0.00989 phase), with 0.06365 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.47 h versus observed median fit 3.92 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Nine deep measured eclipses in sectors 19, 59 and 86 include five adjacent-cycle pairs. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
19	0	1819.6069525	0.010417	152.48	used
19	1	1824.4193945	0.010417	152.71	used
59	227	2912.0141496	0.001157	152.95	used
59	229	2921.6385015	0.001157	153.00	used
59	230	2926.4510671	0.001157	152.15	used
59	231	2931.2634482	0.001157	151.22	used
59	232	2936.0757667	0.001157	150.22	used
86	379	3643.4934662	0.001157	148.42	used
86	380	3648.3056804	0.001157	147.80	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s59.json, historical/diff_s59.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R455. Public evidence files: diff_s59.json, diff_s59.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

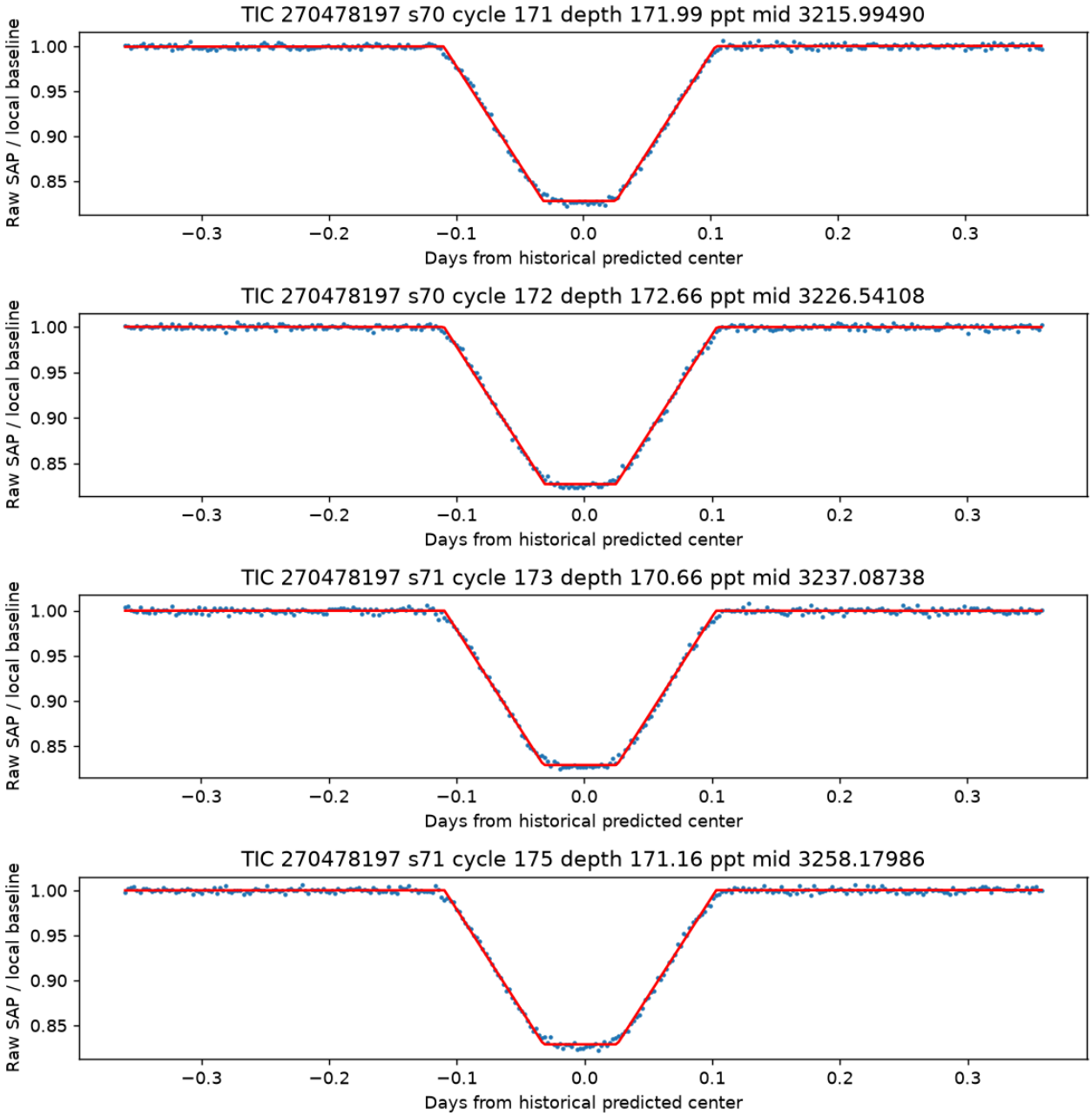
R456. TIC 270478197

physical_verdict: CONFIRMED

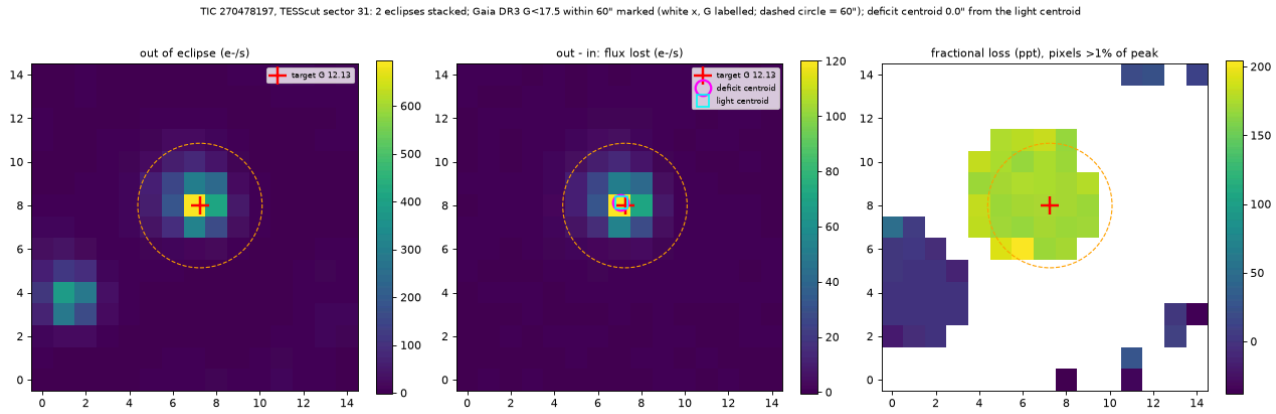
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R456; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R456; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 270478197: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 4 timing-selected measured centers in sectors [70, 71]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 10.5462440932 ± 0.0003912788 d and reference epoch BTJD 3215.9948789 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 10.54632873 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 31 TESScut difference image puts the deficit 5.02 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.036 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 172.94/172.35/172.38 ppt in 2 stacked historical events; representative QLP raw median selected depth is 171.57 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 2513059066528401664 has SB1 P = 10.5462213303 d and e = 0.09909. Closest edge-on conjunction offset is 0.19326 d (0.01833 phase), with 0.54444 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.32 h versus observed median fit 5.12 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Four deep measured eclipses include an adjacent sector-70 pair and a sector-71 repeat. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
70	171	3215.9949044	0.001157	171.99	used
70	172	3226.5410800	0.001157	172.66	used
71	173	3237.0873804	0.001157	170.66	used
71	175	3258.1798593	0.001157	171.16	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s31.json, historical/diff_s31.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R456. Public evidence files: diff_s31.json, diff_s31.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

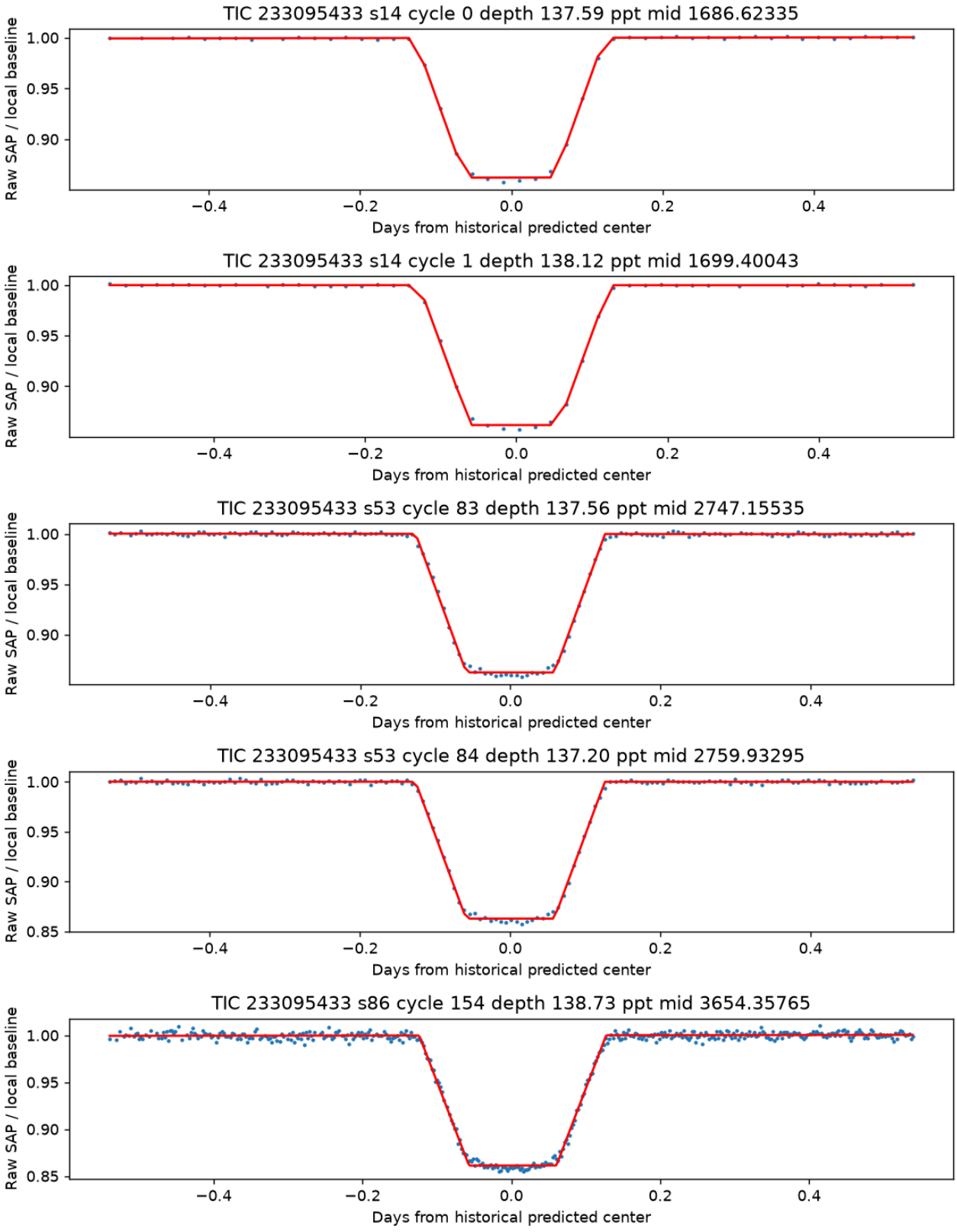
R457. TIC 233095433

physical_verdict: CONFIRMED

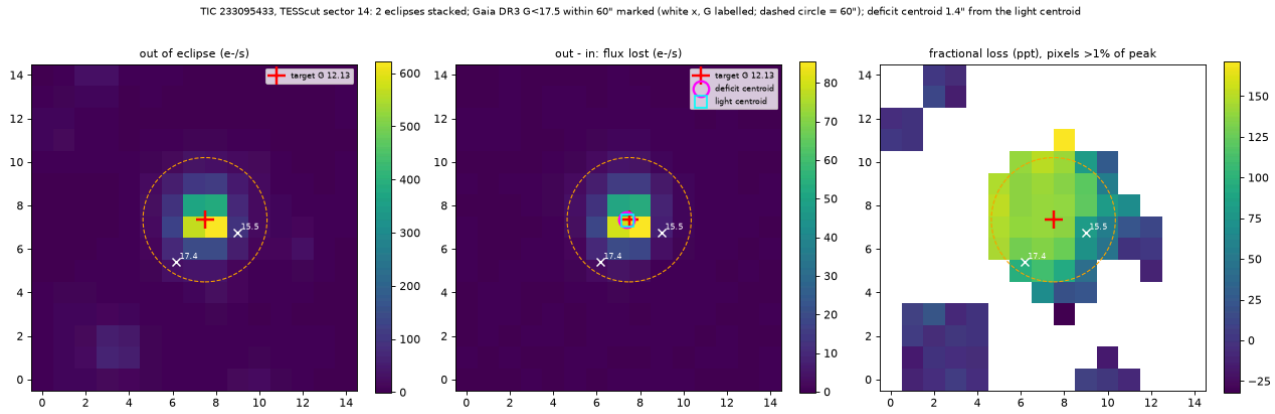
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R457; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R457; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 233095433: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 5 timing-selected measured centers in sectors [14, 53, 86]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 12.7774965312 ± 0.0000311255 d and reference epoch BTJD 1686.6231819 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 12.77746923 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 14 TESScut difference image puts the deficit 2.97 arcsec from the target coordinate; peak [8, 7] versus target pixel [7, 7]. Image-model fractional residual 0.067 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 139.98/137.52/132.99 ppt in 2 stacked historical events; representative QLP raw median selected depth is 137.59 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 2160869549233604608 has SB1 P = 12.7774692281 d and e = 0.04193. Closest edge-on conjunction offset is 0.24719 d (0.01935 phase), with 0.26052 d linearly propagated full source-specific SB1 covariance. Rough central duration 7.34 h versus observed median fit 6.05 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Five deep measured eclipses include adjacent pairs in sectors 14 and 53 and a sector-86 repeat. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
14	0	1686.6233507	0.010417	137.59	used
14	1	1699.4004320	0.010417	138.12	used
53	83	2747.1553542	0.003472	137.56	used
53	84	2759.9329495	0.003472	137.20	used
86	154	3654.3576465	0.001157	138.73	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s14.json, historical/diff_s14.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R457. Public evidence files: diff_s14.json, diff_s14.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

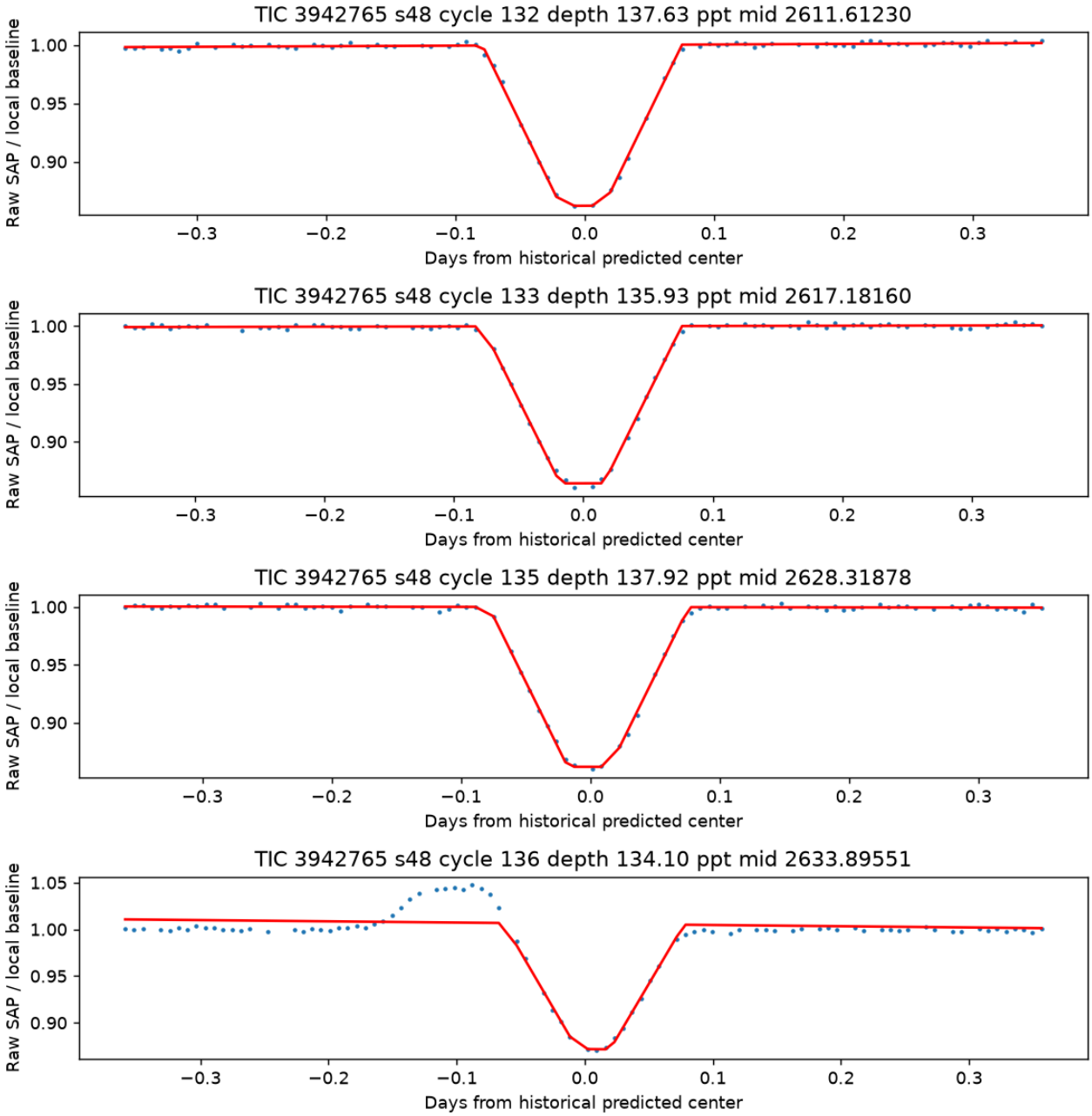
R458. TIC 3942765

physical_verdict: CONFIRMED

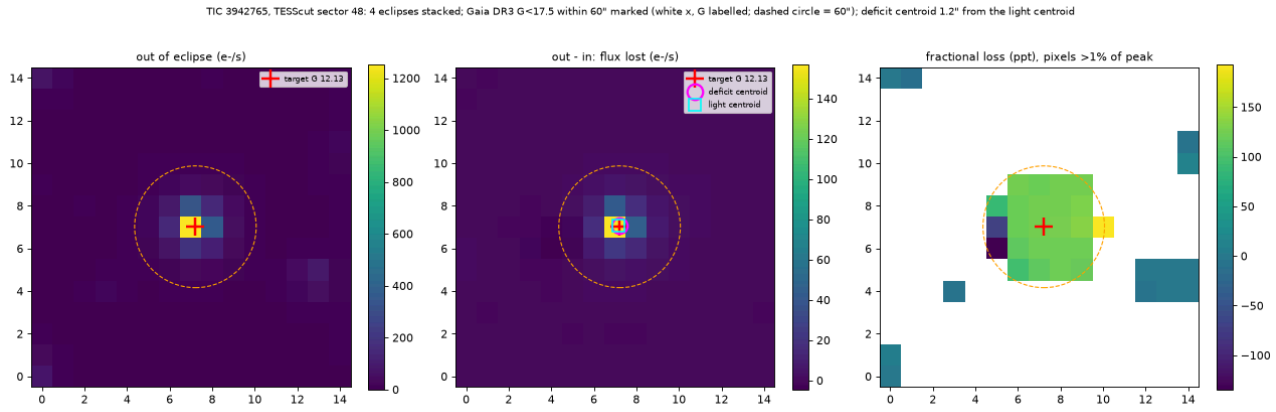
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R458; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R458; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 3942765: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 3 timing-selected measured centers in sectors [48]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 5.5687926067 ± 0.0016073409 d and reference epoch BTJD 2611.6125011 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 5.56862471 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 48 TESScut difference image puts the deficit 1.56 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.063 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 125.29/124.28/120.53 ppt in 4 stacked historical events; representative QLP raw median selected depth is 137.63 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 740708994716293120 has SB1 P = 5.5682939164 d and e = 0.01875. Closest edge-on conjunction offset is 0.12378 d (0.02223 phase), with 0.32903 d linearly propagated full source-specific SB1 covariance. Rough central duration 5.06 h versus observed median fit 3.70 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Four eclipse-shaped events occur in sector 48; three centers give the ephemeris. Cycle 136 has a positive pre-ingress excursion that biases its fitted baseline and midpoint, so it is retained as morphology and excluded from timing. The remaining adjacent pair tests recurrence; no capable Gaia neighbor lies within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
48	132	2611.6122989	0.003472	137.63	used
48	133	2617.1815969	0.003472	135.93	used
48	135	2628.3187778	0.003472	137.92	used
48	136	2633.8955128	0.003472	134.10	excluded: Pre-ingress positive flux excursion biases the local linear baseline and center; retained as eclipse morphology, excluded from timing fit.

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s48.json, historical/diff_s48.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R458. Public evidence files: diff_s48.json, diff_s48.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

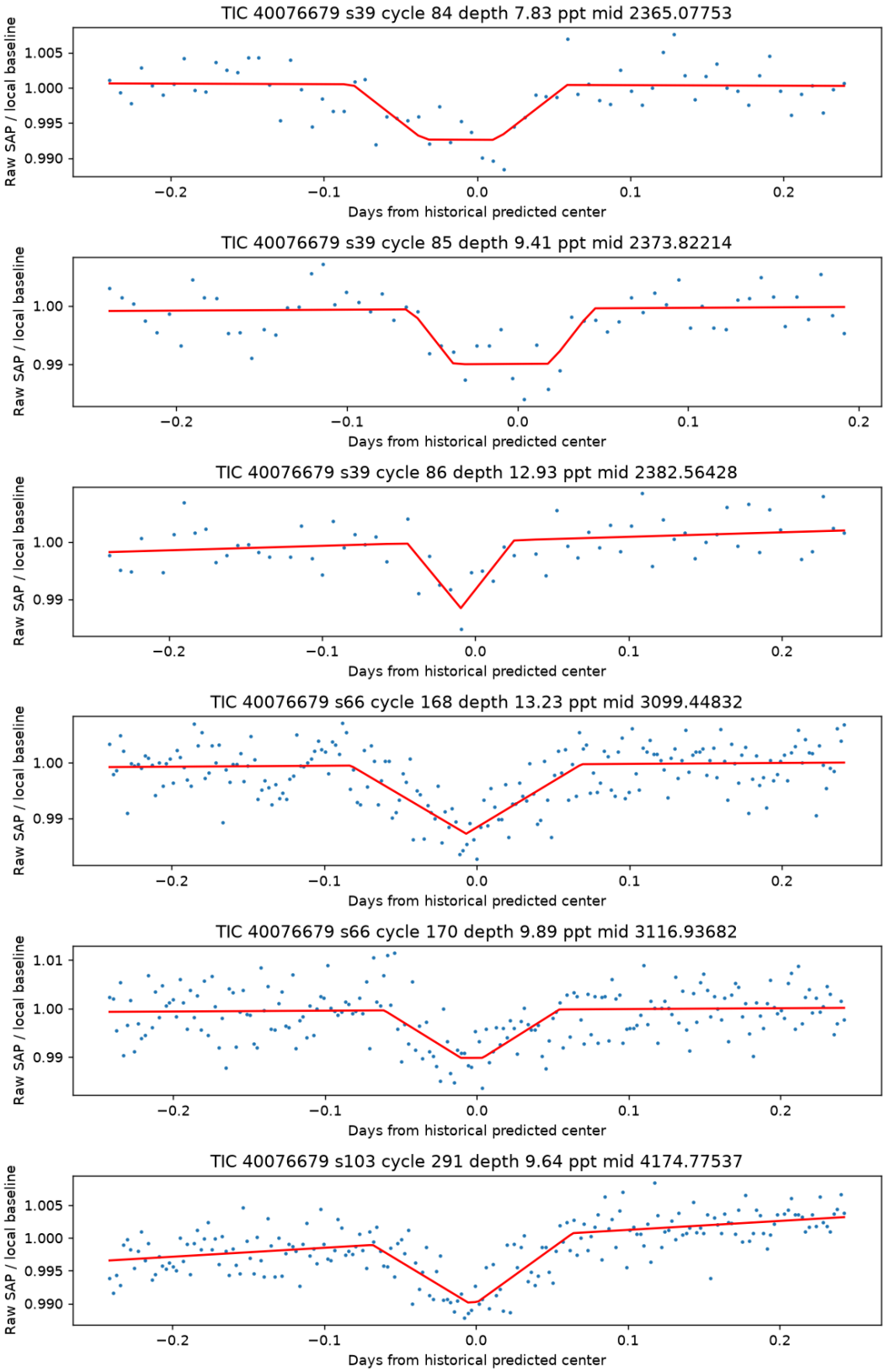
R459. TIC 40076679

physical_verdict: PROBABLE

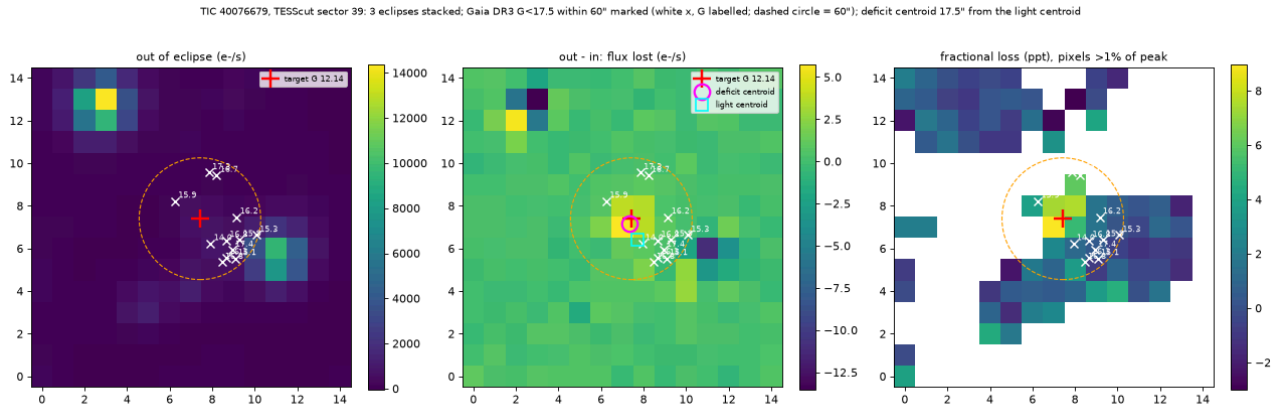
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved time-series photometry or multi-source pixel modeling during an eclipse to identify the variable source.



R459; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R459; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 40076679: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 7 timing-selected measured centers in sectors [39, 66, 103]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 8.7425051500 ± 0.0000139891 d and reference epoch BTJD 2365.0791284 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 8.74245731 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 39 TESScut difference image puts the deficit 5.05 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.856 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 5937789026883670528 (25.82 arcsec, G 14.049, intrinsic required loss about 0.062); Gaia 5937789026883674624 (29.68 arcsec, G 15.869, intrinsic required loss about 0.291); Gaia 5937788270969425664 (32.98 arcsec, G 16.761, intrinsic required loss about 0.649); Gaia 5937794146484859520 (35.77 arcsec, G 16.205, intrinsic required loss about 0.393); Gaia 5937788270969425408 (43.72 arcsec, G 15.946, intrinsic required loss about 0.311); Gaia 5937794902399104128 (43.96 arcsec, G 16.674, intrinsic required loss about 0.600); Gaia 5937788270969424256 (44.85 arcsec, G 16.253, intrinsic required loss about 0.410); Gaia 5937788236609685632 (46.54 arcsec, G 16.804, intrinsic required loss about 0.675); Gaia 5937788270969423744 (51.00 arcsec, G 15.052, intrinsic required loss about 0.142); Gaia 5937788270969470208 (57.24 arcsec, G 15.251, intrinsic required loss about 0.168). Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss. A fresh sector-66 TESScut stack of two actual measured windows gives a 7.48-arcsec target offset, target-core peak, 1x1/3x3/5x5 depths 8.10/6.43/3.15 ppt and image-model residual 0.999; see historical/diff_s66.json and .png. These residuals prevent unique source attribution.

4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 8.97/5.31/3.45 ppt in 3 stacked historical events; representative QLP raw median selected depth is 9.64 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 5937789022547559296 has SB1 P = 8.7431794619 d and e = 0.06793. Closest edge-on conjunction offset is -0.13259 d (-0.01516 phase), with 0.31187 d linearly propagated full source-specific SB1 covariance. Rough central duration 8.16 h versus observed median fit 3.01 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Seven shallow raw eclipses recur in sectors 39, 66 and 103; three adjacent sector-39 cycles directly test the period. Both sector-39 and newly measured sector-66 difference images have strong structured residuals in a crowded field. Ten Gaia neighbors within 60 arcsec can in principle produce the blended depth. The sector-66 deficit core is near the target, yet the pixel data cannot uniquely assign the signal.

Next check: Resolved time-series photometry or multi-source pixel modeling during an eclipse to identify the variable source.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
39	84	2365.0775254	0.004904	7.83	used
39	85	2373.8221439	0.003974	9.41	used
39	86	2382.5642846	0.003740	12.93	used
66	168	3099.4483213	0.003055	13.23	used
66	170	3116.9368236	0.003608	9.89	used
103	289	4157.2935634	0.002077	7.92	used
103	291	4174.7753719	0.003248	9.64	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s39.json, historical/diff_s39.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R459. Public evidence files: diff_s39.json, diff_s39.png, diff_s66.json, diff_s66.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

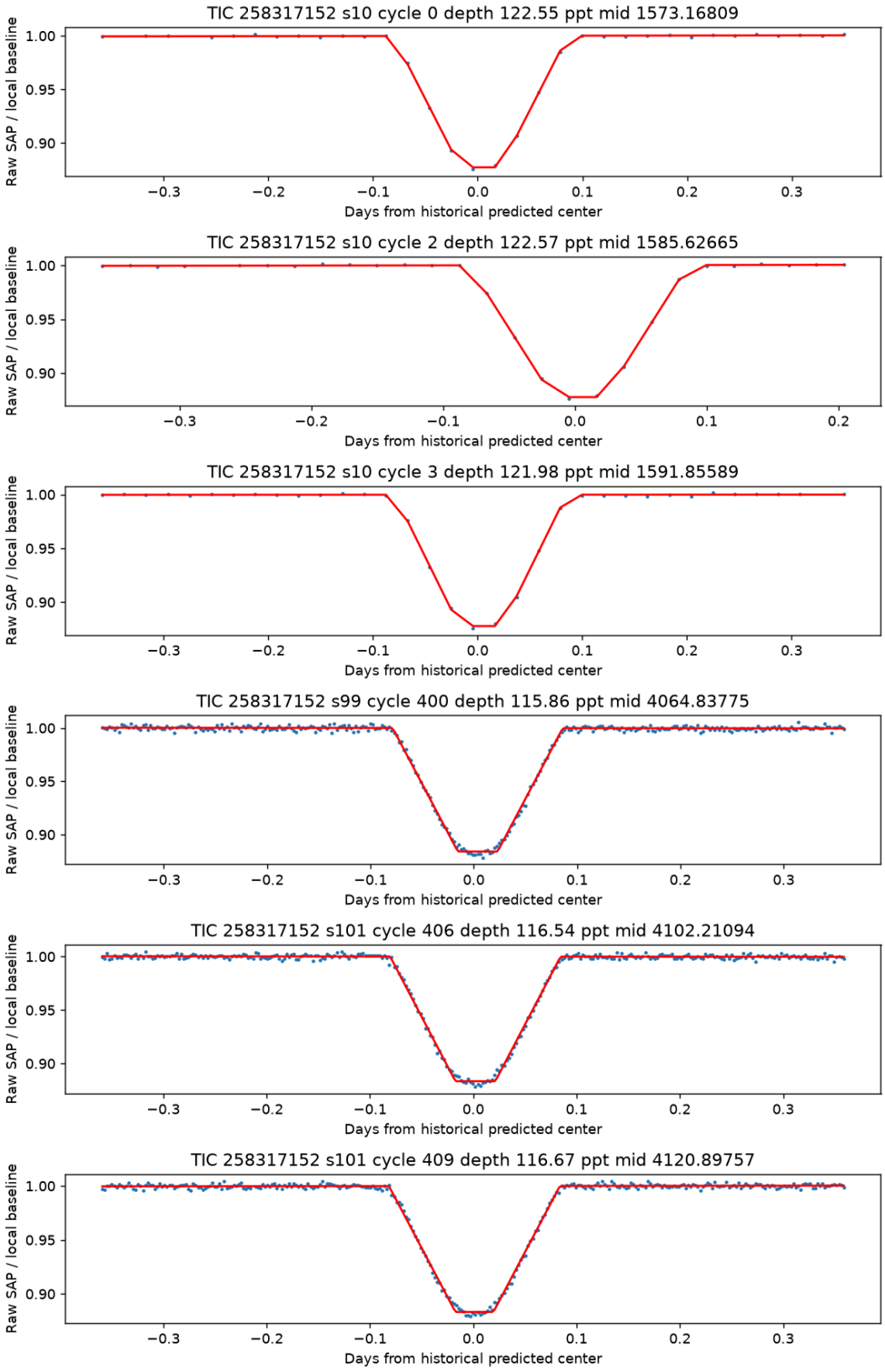
R460. TIC 258317152

physical_verdict: CONFIRMED

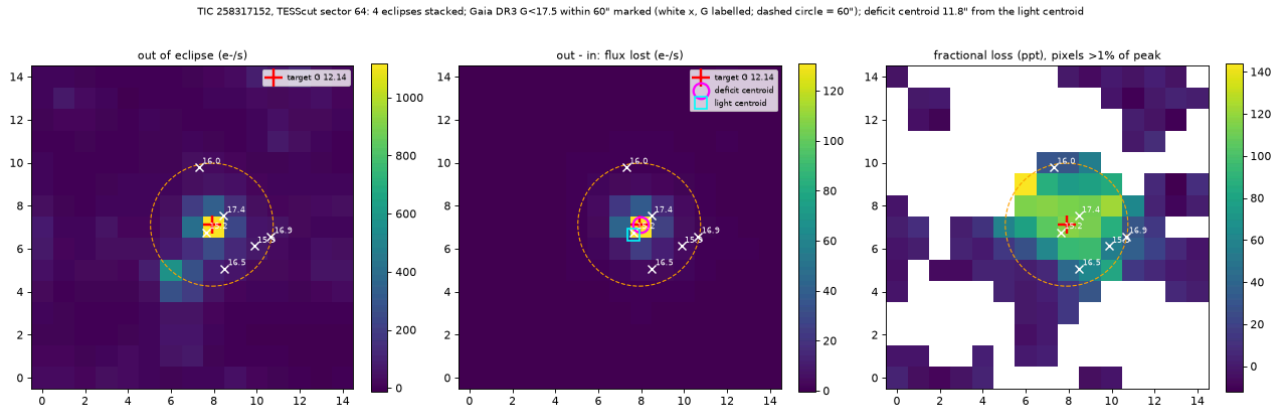
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R460; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R460; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 258317152: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 8 measured same-kind eclipses in representative QLP SAP_FLUX sectors [10, 99, 101]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 6.2291664080 ± 0.0000148911 d and reference BTJD 1573.1699570 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S64 TESScut deficit lies 1.37 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Image-model fractional residual 0.484 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 117.22/108.87/83.39 ppt in 4 stacked events; selected QLP raw depth is 116.70 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 6143372790427445888 has SB1 period 6.2285564049 d, e = 0.07165. The closest edge-on conjunction differs from the measured eclipse epoch by 0.08404 d (0.01349 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.15193 d. Rough central duration 5.14 h versus measured median 3.97 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Eight strong raw eclipses across S10/99/101. The historical S64 deficit peaks on the target, with no capable Gaia neighbor under the approximate flux budget; the image-model residual is large, so its centroid is only descriptive.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
10	0	1573.1680904	0.010417	122.55	used
10	2	1585.6266461	0.010417	122.57	used
10	3	1591.8558908	0.010417	121.98	used
99	398	4052.3797987	0.001157	115.32	used
99	400	4064.8377476	0.001157	115.86	used
101	406	4102.2109375	0.001157	116.54	used
101	407	4108.4399395	0.001157	116.73	used
101	409	4120.8975692	0.001157	116.67	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s64.json, historical/diff_s64.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R460. Public evidence files: diff_s64.json, diff_s64.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

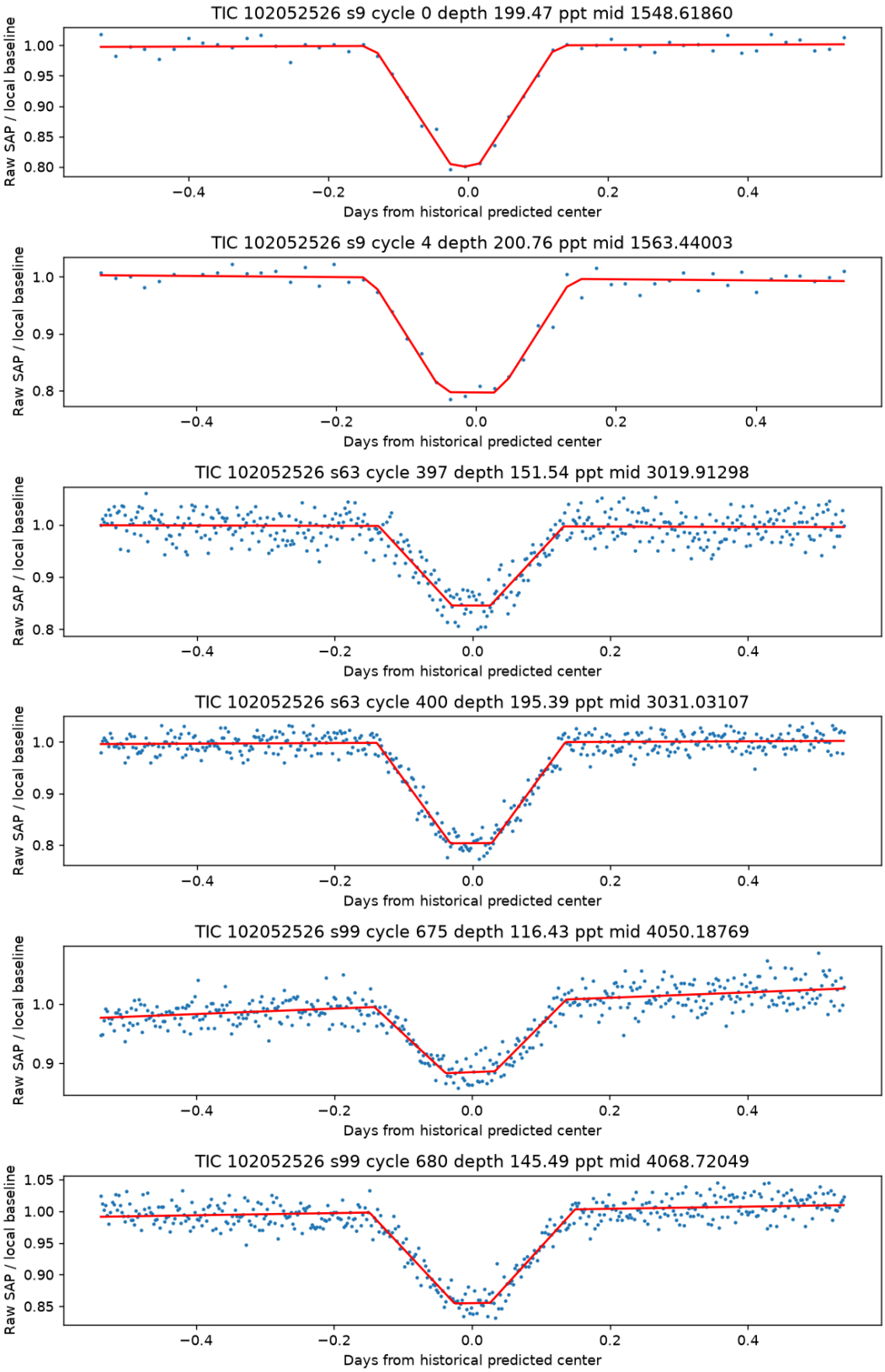
R461. TIC 102052526

physical_verdict: CONFIRMED

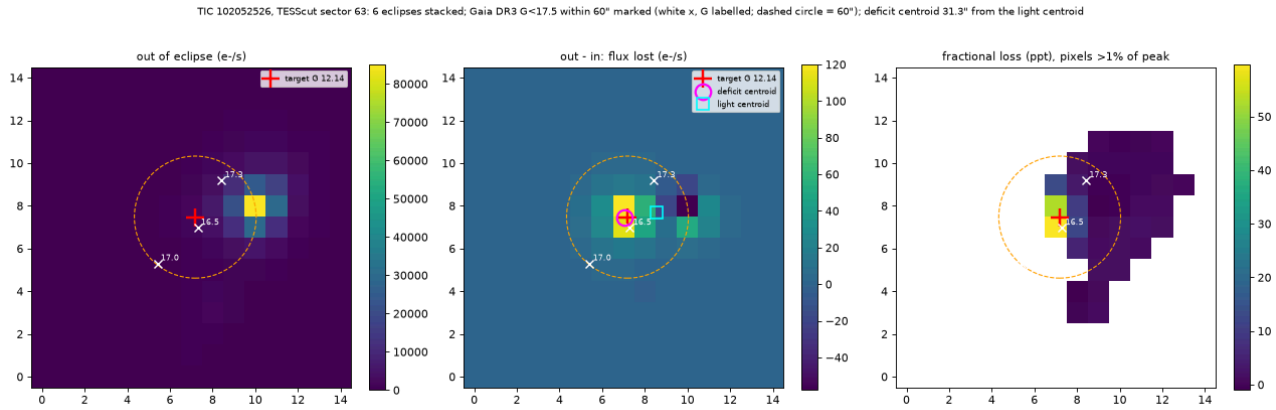
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R461; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R461; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 102052526: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 8 measured same-kind eclipses in representative QLP SAP_FLUX sectors [9, 63, 99]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 7.4120614461 ± 0.0000108580 d and reference BTJD 1548.6197006 (Deeper repeating eclipse; even half-period cycles). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S63 TESScut deficit lies 2.66 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 7]. Image-model fractional residual 0.997 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 59.76/24.22/6.56 ppt in 6 stacked events; selected QLP raw depth is 195.82 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 5416772041726772864 has SB1 period 7.4112548348 d, e = 0.01800. The closest edge-on conjunction differs from the measured eclipse epoch by 0.14870 d (0.02006 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.05727 d. Rough central duration 8.59 h versus measured median 6.72 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Deep and shallow events alternate in S9/63/99. The 3.706-day photometric half-period is not the orbital period: the deeper even parity recurs at 7.412 days. The historical S63 aperture depths are strongly diluted and its image-model residual is high, but the

deficit core is on the target and no capable Gaia neighbor is found.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
9	0	1548.6186004	0.010417	199.47	used
9	3	1559.7334572	0.010417	147.91	excluded from primary timing
9	4	1563.4400284	0.010417	200.76	used
9	5	1567.1466206	0.010417	122.32	excluded from primary timing
63	396	3016.2076632	0.001257	196.25	used
63	397	3019.9129756	0.002086	151.54	excluded from primary timing
63	398	3023.6202250	0.001414	207.16	used
63	399	3027.3144753	0.007049	118.26	excluded from primary timing
63	400	3031.0310660	0.001157	195.39	used
63	401	3034.7348220	0.002138	155.44	excluded from primary timing
63	402	3038.4460476	0.001594	186.63	used
99	675	4050.1876910	0.002411	116.43	excluded from primary timing
99	676	4053.8959706	0.003968	150.78	used
99	679	4065.0094469	0.001528	158.59	excluded from primary timing
99	680	4068.7204943	0.001582	145.49	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s63.json, historical/diff_s63.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R461. Public evidence files: diff_s63.json, diff_s63.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

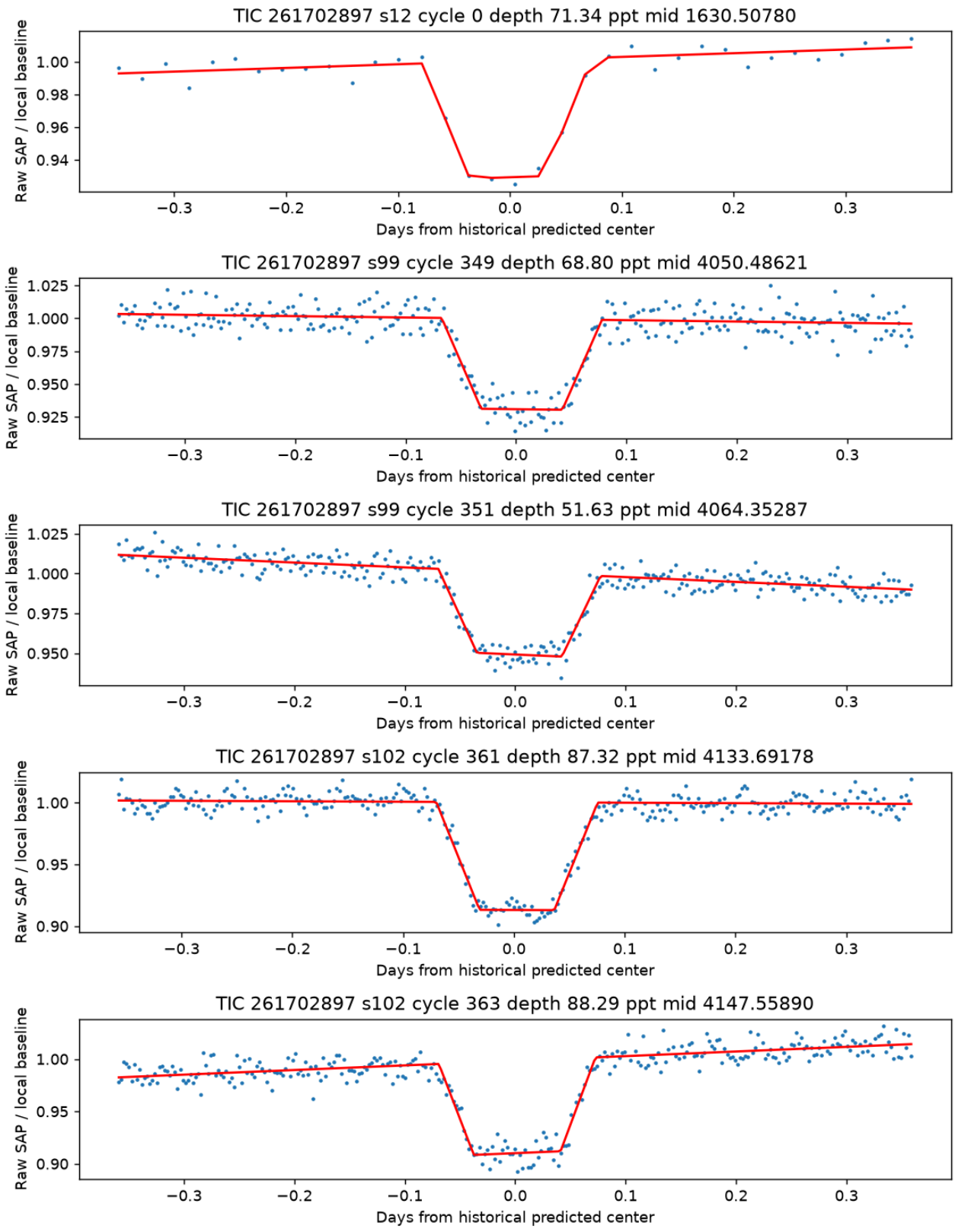
R462. TIC 261702897

physical_verdict: CONFIRMED

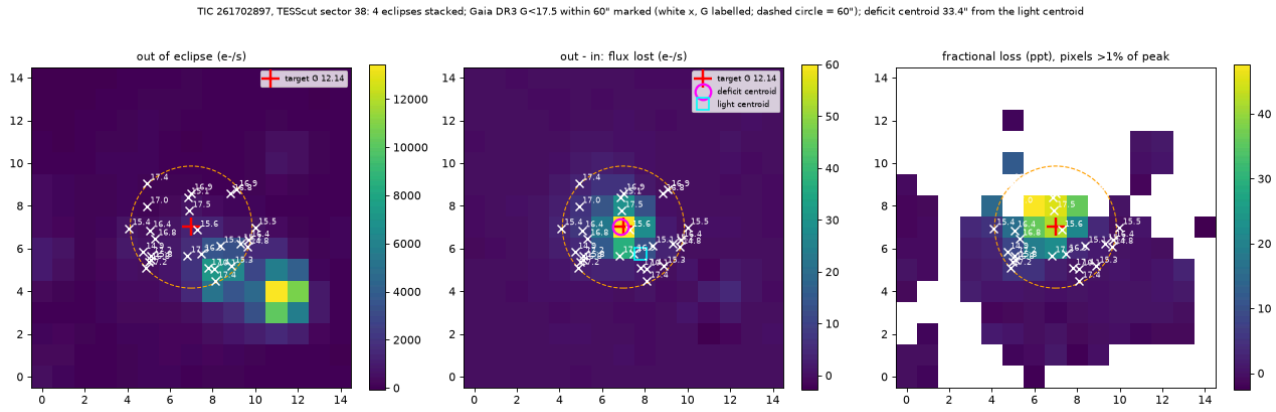
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R462; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R462; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 261702897: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 5 measured same-kind eclipses in representative QLP SAP_FLUX sectors [12, 99, 102]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 6.9340108453 ± 0.0000376792 d and reference BTJD 1630.5145301 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S38 TESScut deficit lies 3.09 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Image-model fractional residual 0.985 is not a calibrated position error. Capable Gaia neighbors: Gaia 5824727895463607808, 44.9 arcsec, G 10.06, approximate intrinsic requirement 0.09; Gaia 5824727517506486016, 55.3 arcsec, G 14.78, approximate intrinsic requirement 0.95. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 41.85/21.50/8.76 ppt in 4 stacked events; selected QLP raw depth is 71.34 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 5824728342140218880 has SB1 period 6.9344260455 d, e = 0.04238. The closest edge-on conjunction differs from the measured eclipse epoch by -0.06187 d (-0.00892 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.09078 d. Rough central duration 4.51 h versus measured median 3.50 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Five raw eclipses across S12/99/102. Two capable Gaia neighbors at about 45 and 55 arcsec require separate evaluation; the S38 deficit peak is at the target pixel, distinct from the bright southeast neighbor. The poor image-model fit limits centroid precision.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	0	1630.5077983	0.010417	71.34	used
99	349	4050.4862110	0.001157	68.80	used
99	351	4064.3528726	0.001319	51.63	used
102	361	4133.6917844	0.001157	87.32	used
102	363	4147.5589027	0.001157	88.29	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s38.json, historical/diff_s38.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R462. Public evidence files: diff_s38.json, diff_s38.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

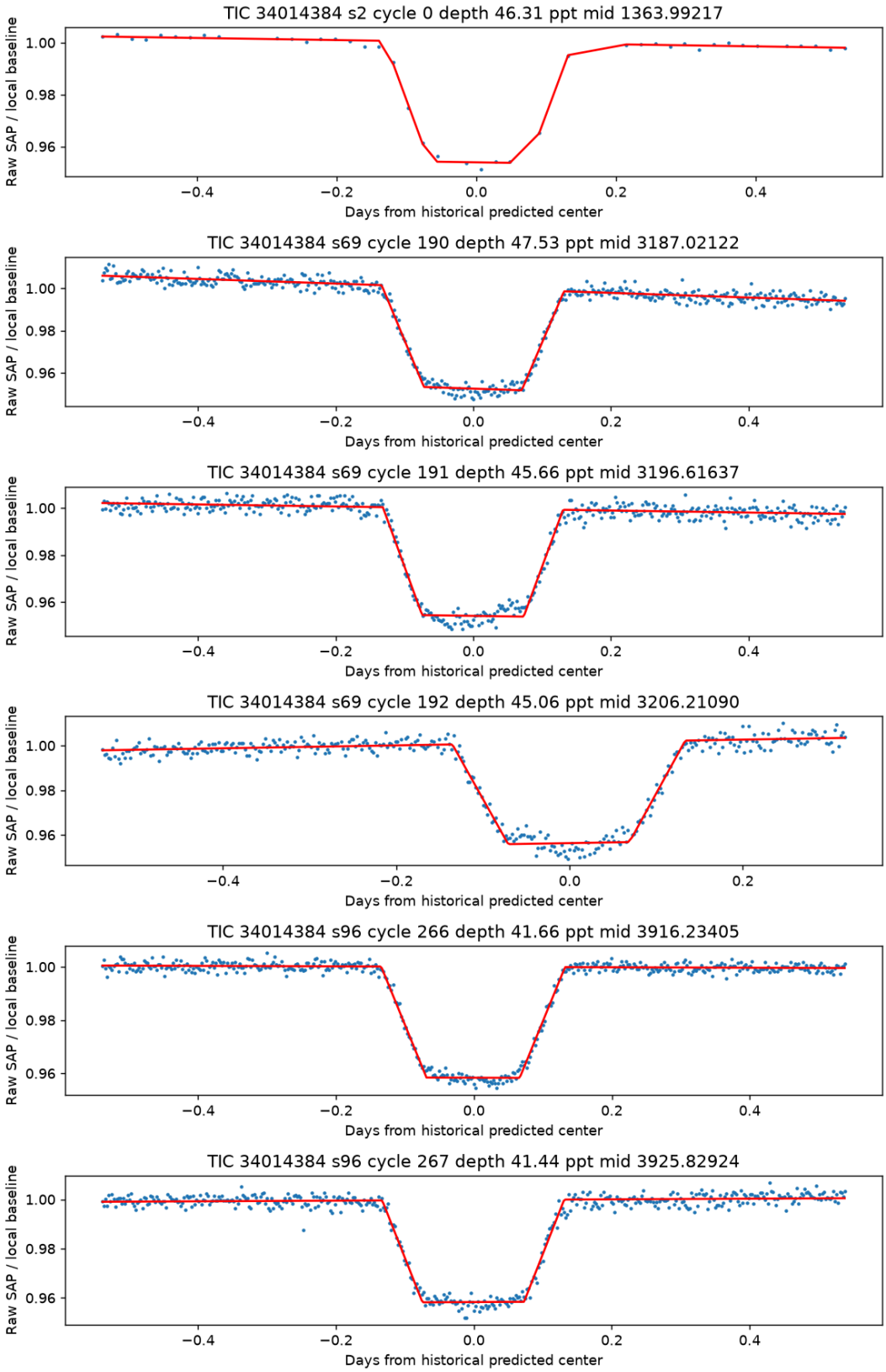
R463. TIC 34014384

physical_verdict: CONFIRMED

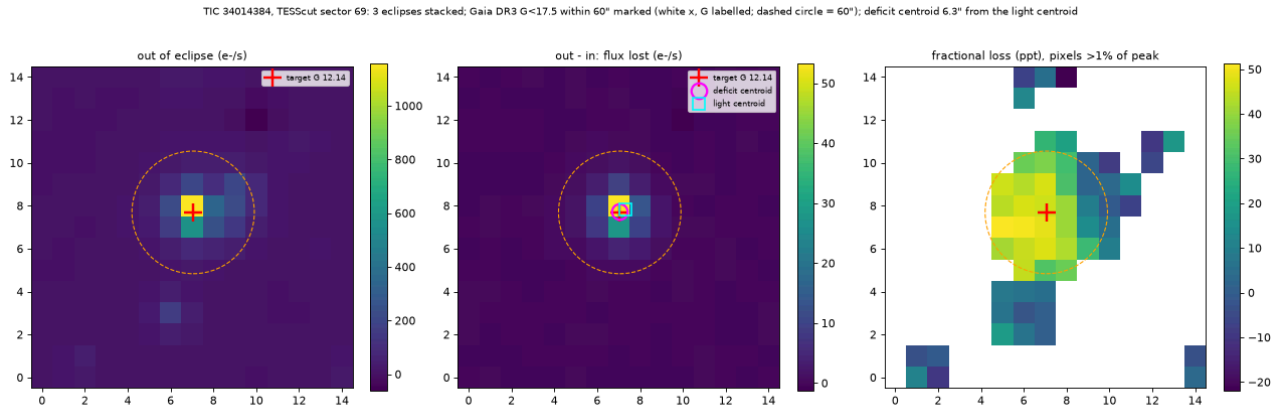
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R463; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R463; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 34014384: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 6 measured same-kind eclipses in representative QLP SAP_FLUX sectors [2, 69, 96]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 9.5949058113 ± 0.0000134136 d and reference BTJD 1363.9891996 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S69 TESScut deficit lies 0.80 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Image-model fractional residual 0.224 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 46.04/45.25/39.29 ppt in 3 stacked events; selected QLP raw depth is 45.36 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 2328077646884230912 has SB1 period 9.5946733453 d, e = 0.07996. The closest edge-on conjunction differs from the measured eclipse epoch by 0.19039 d (0.01984 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.52291 d. Rough central duration 7.88 h versus measured median 6.38 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Six raw eclipses across S2/69/96, including three consecutive S69 cycles. The S69 deficit peaks at the target and the aperture reductions retain the eclipse.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
2	0	1363.9921725	0.010417	46.31	used
69	190	3187.0212237	0.001157	47.53	used
69	191	3196.6163746	0.001157	45.66	used
69	192	3206.2109008	0.001157	45.06	used
96	266	3916.2340458	0.001157	41.66	used
96	267	3925.8292435	0.001157	41.44	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s69.json, historical/diff_s69.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R463. Public evidence files: diff_s69.json, diff_s69.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

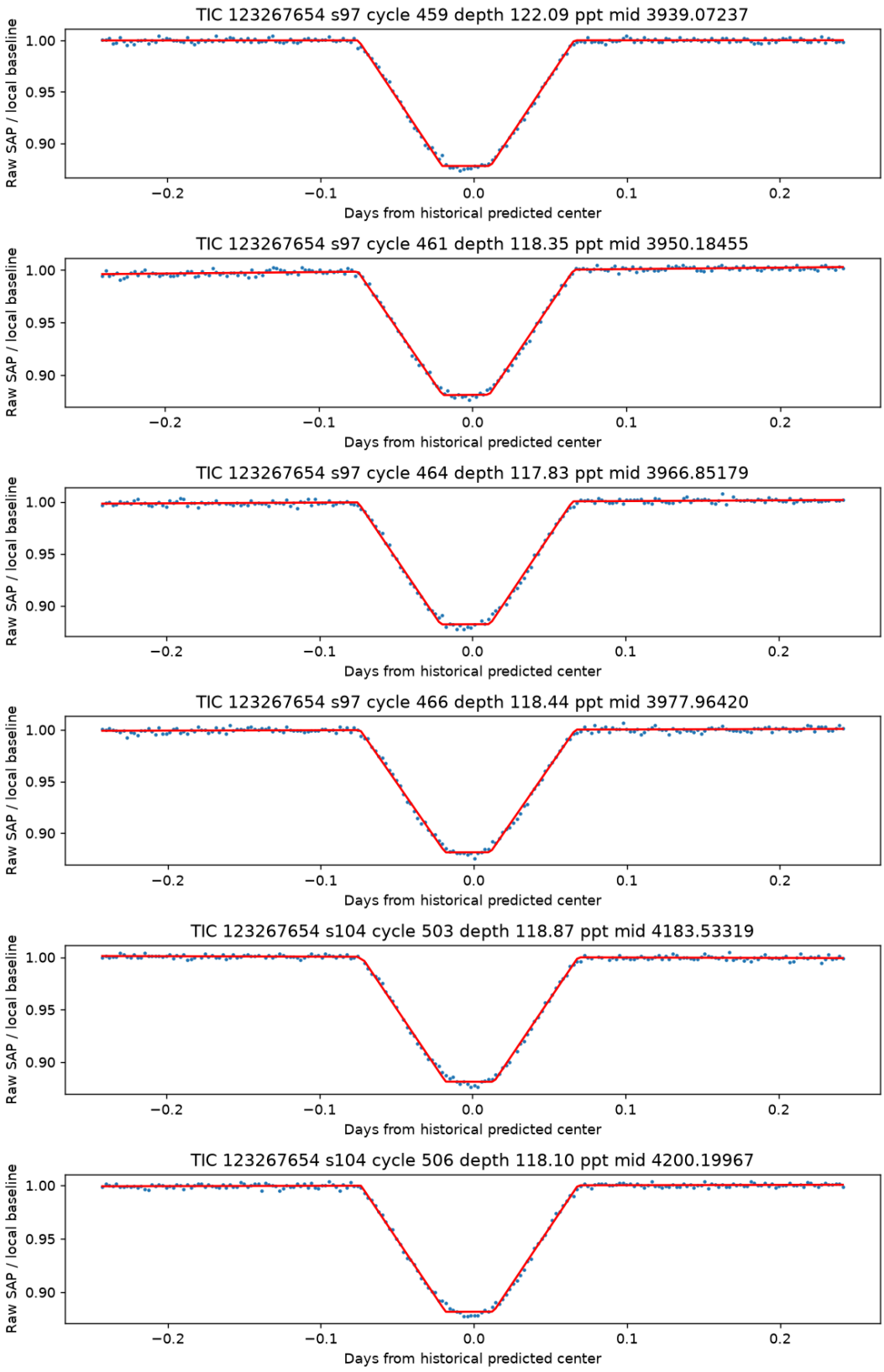
R464. TIC 123267654

physical_verdict: CONFIRMED

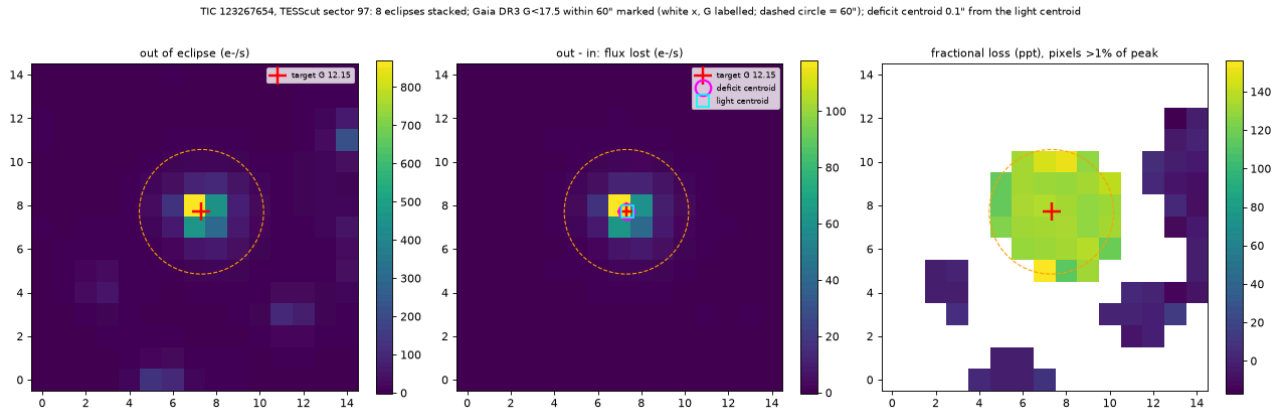
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R464; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R464; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 123267654: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 12 measured same-kind eclipses in representative QLP SAP_FLUX sectors [97, 104]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 5.5559075318 ± 0.0000169590 d and reference BTJD 3939.0725724 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S97 TESScut deficit lies 0.46 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Image-model fractional residual 0.060 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 135.68/133.72/133.25 ppt in 8 stacked events; selected QLP raw depth is 118.29 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 5018762885599910016 has SB1 period 5.5561122262 d, e = 0.07698. The closest edge-on conjunction differs from the measured eclipse epoch by -0.17143 d (-0.03085 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.25667 d. Rough central duration 4.65 h versus measured median 3.38 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Twelve raw eclipses across S97/104, with consecutive cycles and consistent depths. The S97 cutout has a clean target-centered deficit. The selected earlier S3 QLP file supplied no usable midpoint; it is not treated as a null eclipse.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
97	459	3939.0723654	0.001157	122.09	used
97	460	3944.6289202	0.001157	120.31	used
97	461	3950.1845485	0.001157	118.35	used
97	462	3955.7403975	0.001157	118.23	used
97	464	3966.8517920	0.001157	117.83	used
97	465	3972.4078429	0.001157	114.57	used
97	466	3977.9642012	0.001157	118.44	used
97	467	3983.5194413	0.001157	116.32	used
104	503	4183.5331930	0.001157	118.87	used
104	504	4189.0888142	0.001157	118.09	used
104	505	4194.6438920	0.001157	118.65	used
104	506	4200.1996719	0.001157	118.10	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R464. Public evidence files: diff_s97.json, diff_s97.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

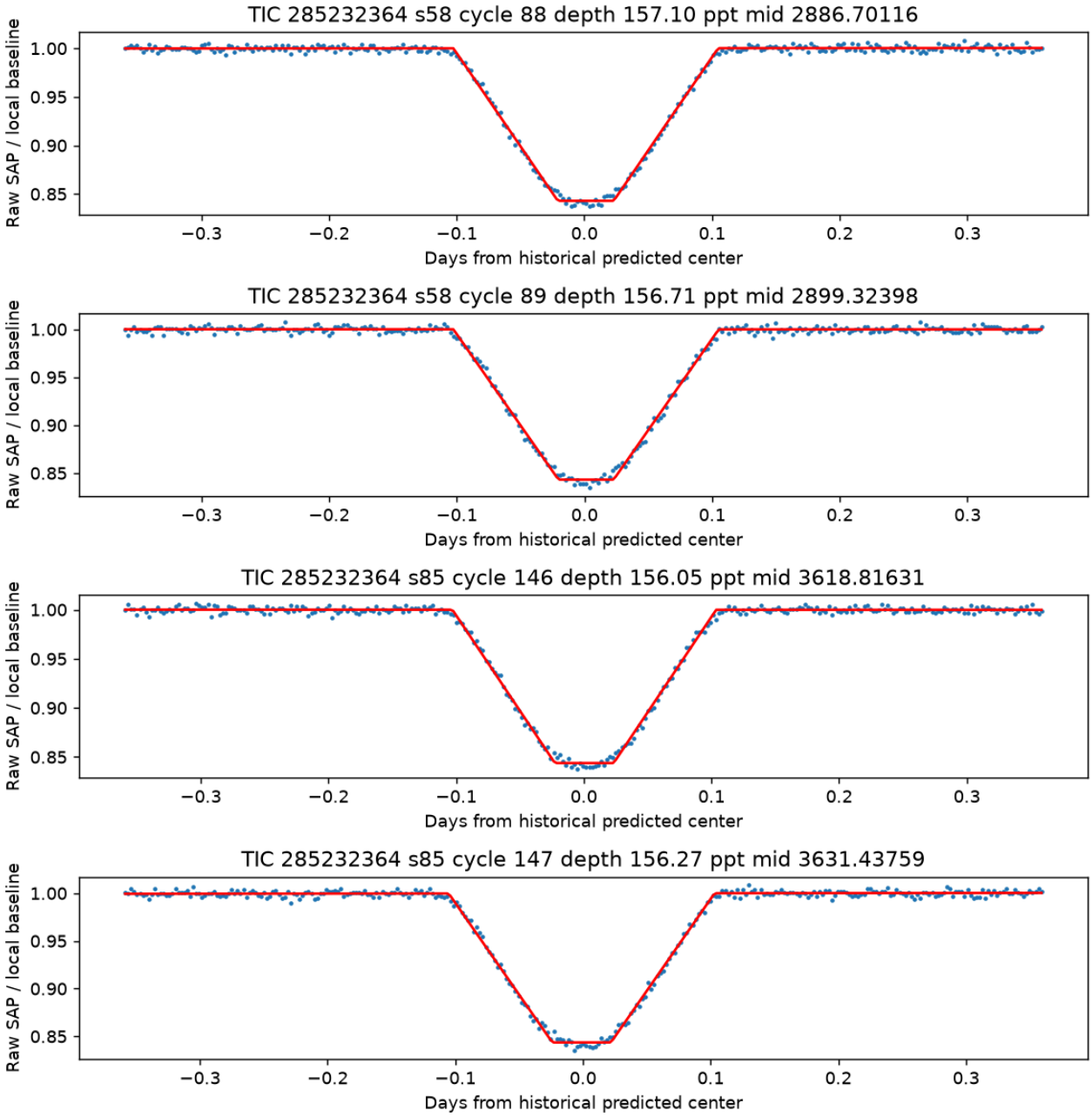
R465. TIC 285232364

physical_verdict: CONFIRMED

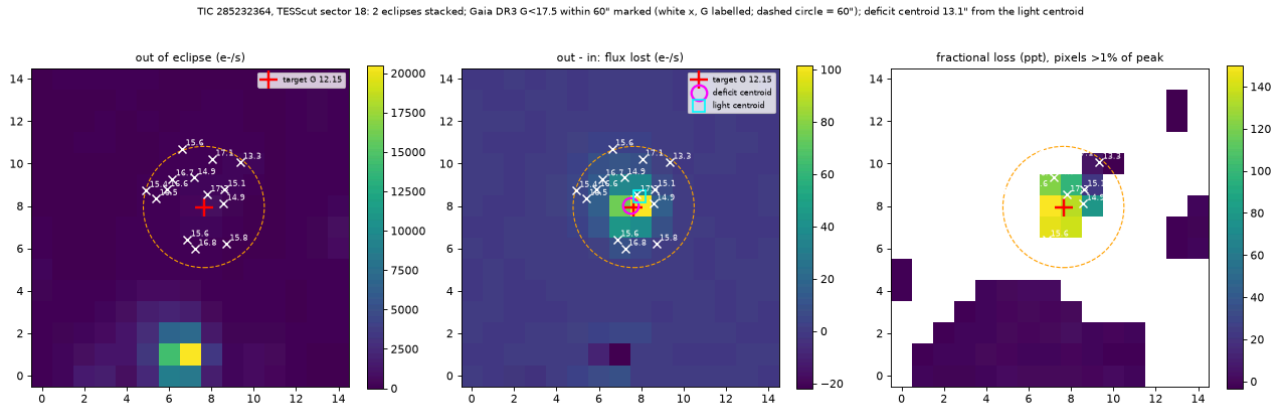
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R465; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R465; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 285232364: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 4 measured same-kind eclipses in representative QLP SAP_FLUX sectors [58, 85]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 12.6226616088 ± 0.0000199522 d and reference BTJD 2886.7012424 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S18 TESScut deficit lies 4.06 arcsec from the target coordinate; peak pixel [8, 8] and target pixel [8, 8]. Image-model fractional residual 0.758 is not a calibrated position error. Capable Gaia neighbors: Gaia 427362616196340608, 56.1 arcsec, G 13.35, approximate intrinsic requirement 0.61. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 134.81/117.84/92.18 ppt in 2 stacked events; selected QLP raw depth is 156.49 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 427362714972324736 has SB1 period 12.6184477876 d, e = 0.26083. The closest edge-on conjunction differs from the measured eclipse epoch by 0.63322 d (0.05018 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.33878 d. Rough central duration 6.67 h versus measured median 4.98 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Four broad raw eclipses in S58/85, two adjacent cycles per sector. A capable neighbor about 56 arcsec away is spatially distinct from the S18 target-centered deficit. The image fit is imperfect and aperture depths are diluted.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
58	88	2886.7011554	0.001157	157.10	used
58	89	2899.3239805	0.001157	156.71	used
85	146	3618.8163095	0.001157	156.05	used
85	147	3631.4375941	0.001157	156.27	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s18.json, historical/diff_s18.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R465. Public evidence files: diff_s18.json, diff_s18.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

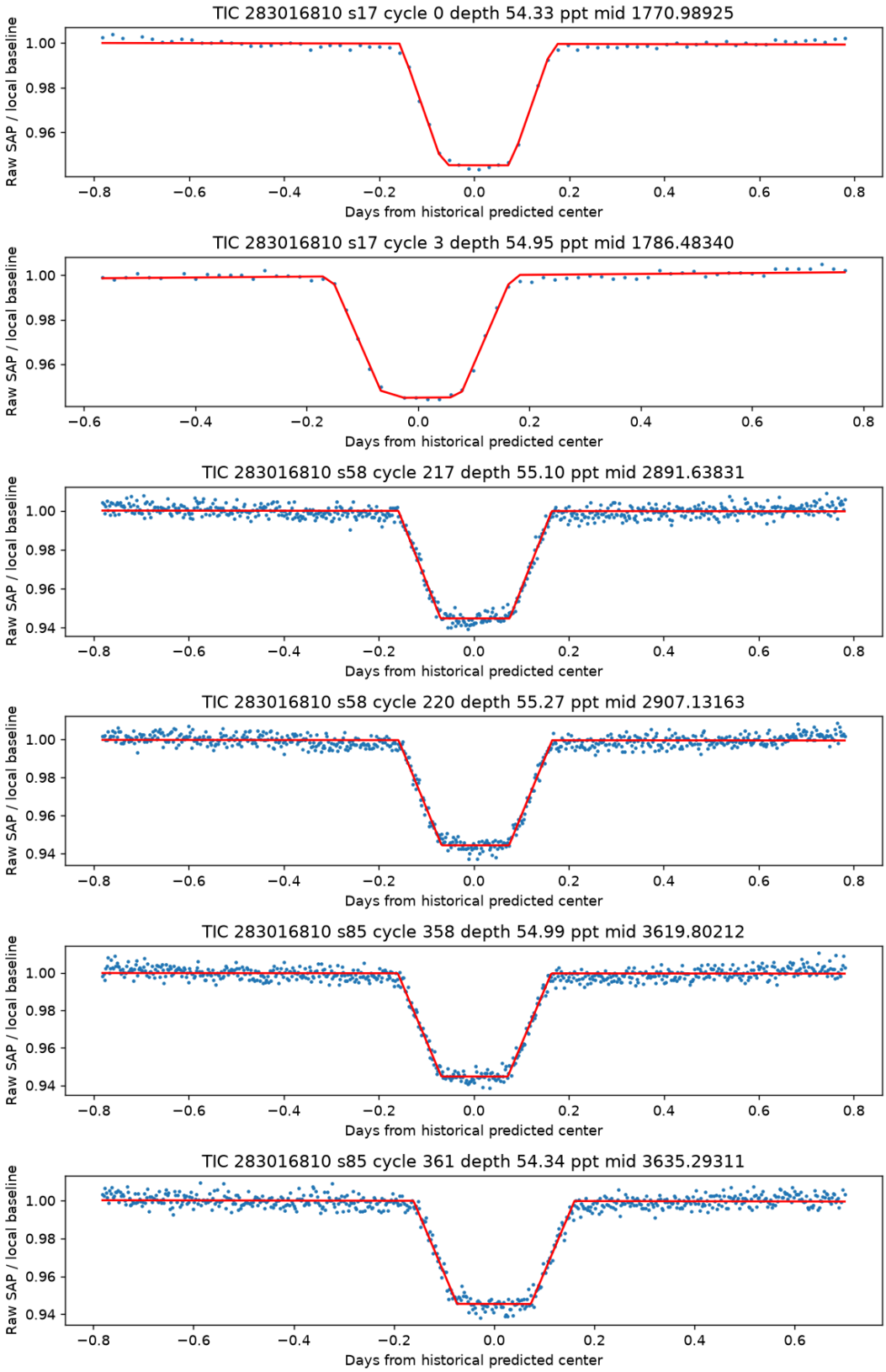
R466. TIC 283016810

physical_verdict: PROBABLE

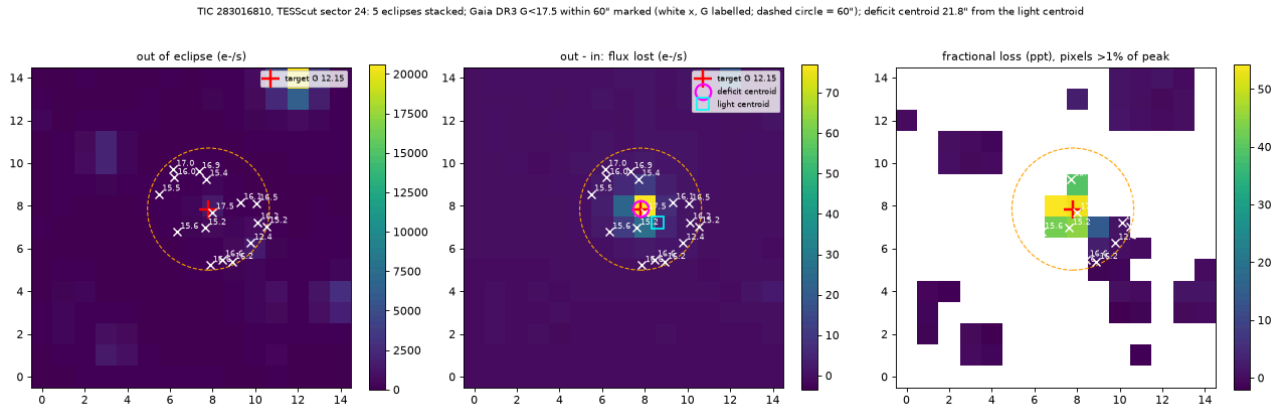
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization



R466; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R466; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 283016810: PROBABLE

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 13 measured same-kind eclipses in representative QLP SAP_FLUX sectors [17, 58, 85]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 5.1642748232 ± 0.0000050389 d and reference BTJD 1770.9909383 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S24 TESScut deficit lies 0.94 arcsec from the target coordinate; peak pixel [8, 8] and target pixel [8, 8]. Image-model fractional residual 0.808 is not a calibrated position error. Capable Gaia neighbors: Gaia 431751179422562816, 19.0 arcsec, G 15.24, approximate intrinsic requirement 0.95; Gaia 431751351221252480, 53.5 arcsec, G 12.36, approximate intrinsic requirement 0.12; Gaia 431751351221249280, 59.7 arcsec, G 15.23, approximate intrinsic requirement 0.95. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 53.08/45.84/26.48 ppt in 5 stacked events; selected QLP raw depth is 54.70 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 431751175113711872 has SB1 period 5.1640736469 d, e = 0.06043. The closest edge-on conjunction differs from the measured eclipse epoch by -0.01453 d (-0.00281 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.09489 d. Rough central duration 10.99 h versus measured median 7.72 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Thirteen raw eclipses across S17/58/85. Three capable Gaia neighbors include one about 19 arcsec away. The S24 deficit peak overlaps the target pixel, but TESS cannot resolve the nearby capable source within roughly one pixel. The image-model residual further limits attribution.

Next check: Resolved eclipse photometry or multi-source PRF localization of the capable nearby star.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
17	0	1770.9892475	0.010417	54.33	used
17	2	1781.3168663	0.010417	54.90	used
17	3	1786.4833952	0.010417	54.95	used
58	216	2886.4738673	0.001157	54.70	used
58	217	2891.6383068	0.001157	55.10	used
58	218	2896.8035472	0.001157	54.15	used
58	219	2901.9670358	0.001157	54.24	used
58	220	2907.1316295	0.001157	55.27	used
85	357	3614.6371395	0.001157	54.53	used
85	358	3619.8021171	0.001157	54.99	used
85	359	3624.9653593	0.001157	55.97	used
85	360	3630.1301941	0.001157	54.64	used
85	361	3635.2931091	0.001157	54.34	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s24.json, historical/diff_s24.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R466. Public evidence files: diff_s24.json, diff_s24.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

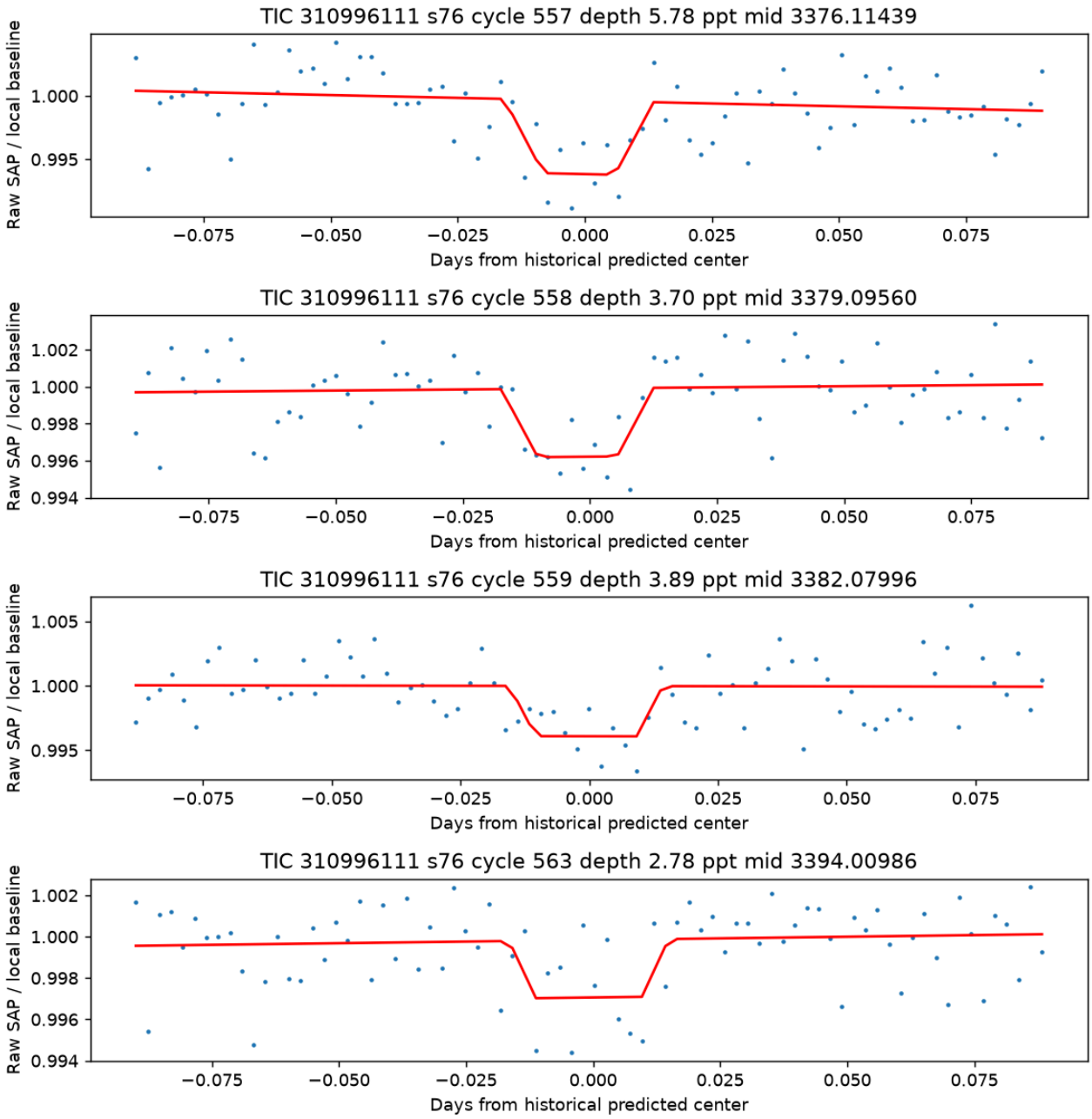
R467. TIC 310996111

physical_verdict: CONFIRMED

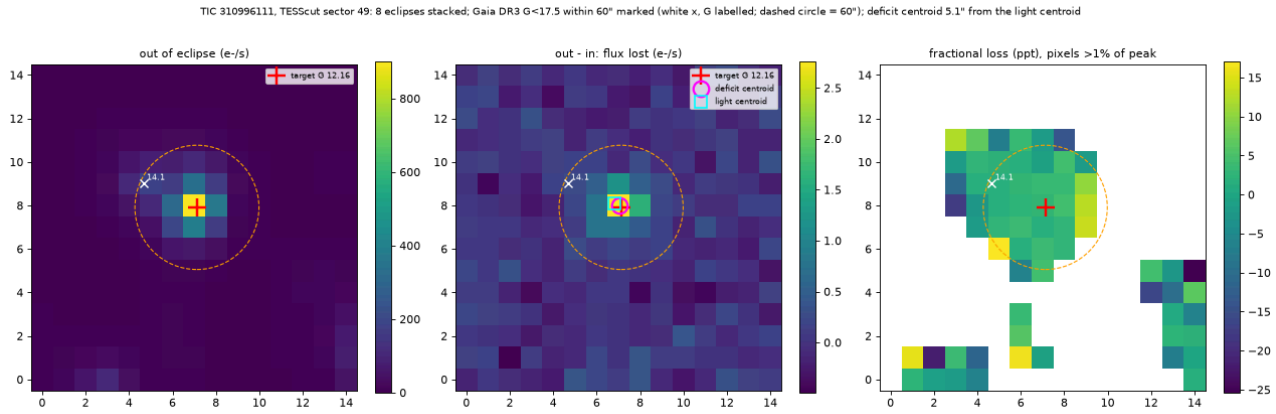
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R467; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R467; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 310996111: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 4 measured same-kind eclipses in representative QLP SAP_FLUX sectors [76]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 2.9826325236 ± 0.0003163772 d and reference BTJD 3376.1140790 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S49 TESScut deficit lies 2.75 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Image-model fractional residual 0.424 is not a calibrated position error. Capable Gaia neighbors: Gaia 1559248294394682112, 52.0 arcsec, G 18.35, approximate intrinsic requirement 0.84; Gaia 1559154737124070528, 55.3 arcsec, G 14.06, approximate intrinsic requirement 0.02. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 3.06/3.09/3.02 ppt in 8 stacked events; selected QLP raw depth is 3.80 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 1559154664107634560 has SB1 period 2.9825457445 d, e = 0.07326. The closest edge-on conjunction differs from the measured eclipse epoch by -0.07548 d (-0.02531 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.11227 d. Rough central duration 2.56 h versus measured median 0.72 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Four shallow S76 raw events recur at the predicted phase, and the historical S49 aperture extraction retains a roughly 3 ppt target-pixel loss. S15/23 are 30-minute cadence: each predicted short event has at most one central sample, so their apparent weak or absent dips are not clean nulls. Capable neighbors about 52 and 55 arcsec away are distinct from the target-centered S49 deficit; the morphology and attribution remain limited by shallow depth.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
76	557	3376.1143874	0.001284	5.78	used
76	558	3379.0956038	0.001653	3.70	used
76	559	3382.0799617	0.001651	3.89	used
76	563	3394.0098566	0.001488	2.78	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s49.json, historical/diff_s49.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R467. Public evidence files: diff_s49.json, diff_s49.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

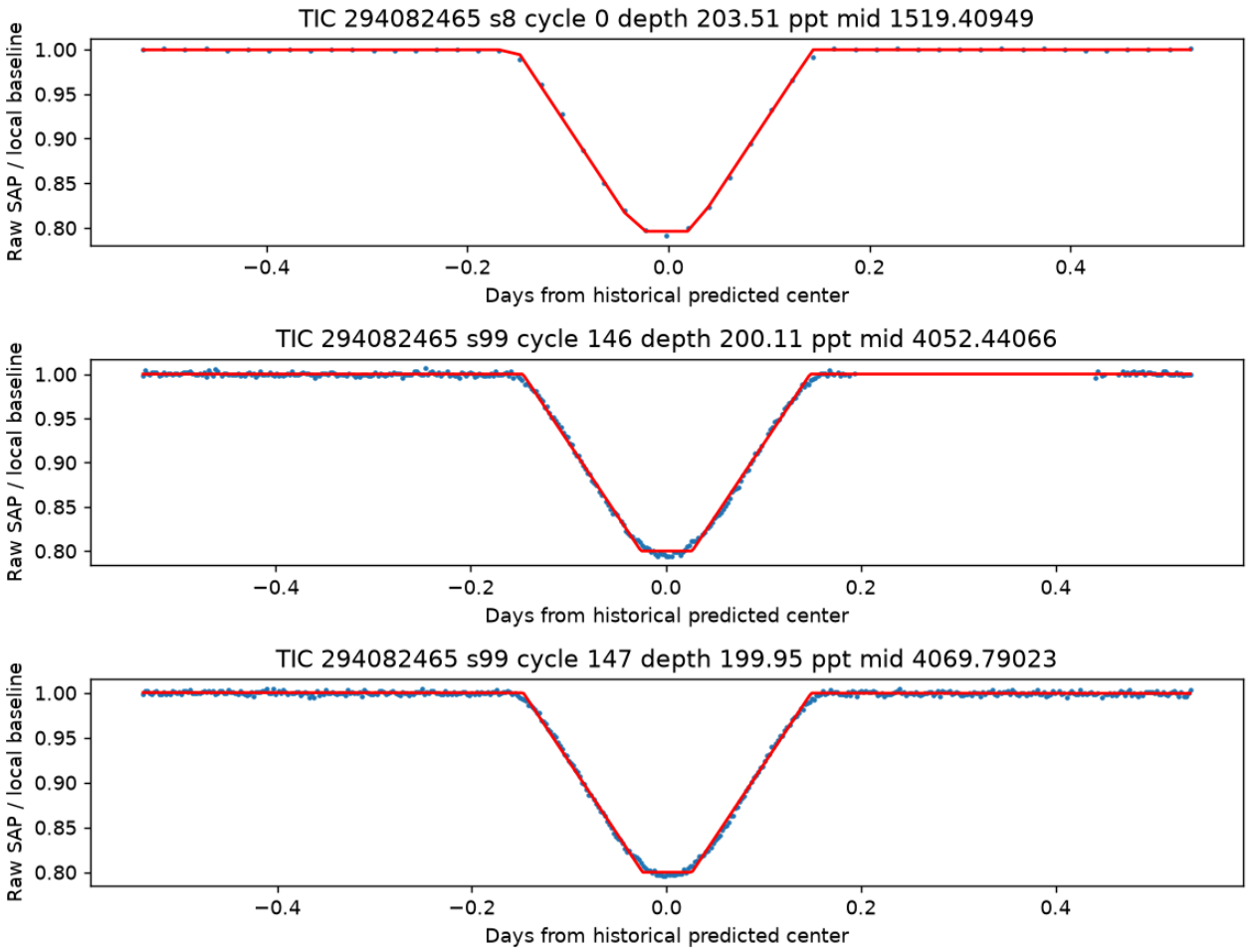
R468. TIC 294082465

physical_verdict: CONFIRMED

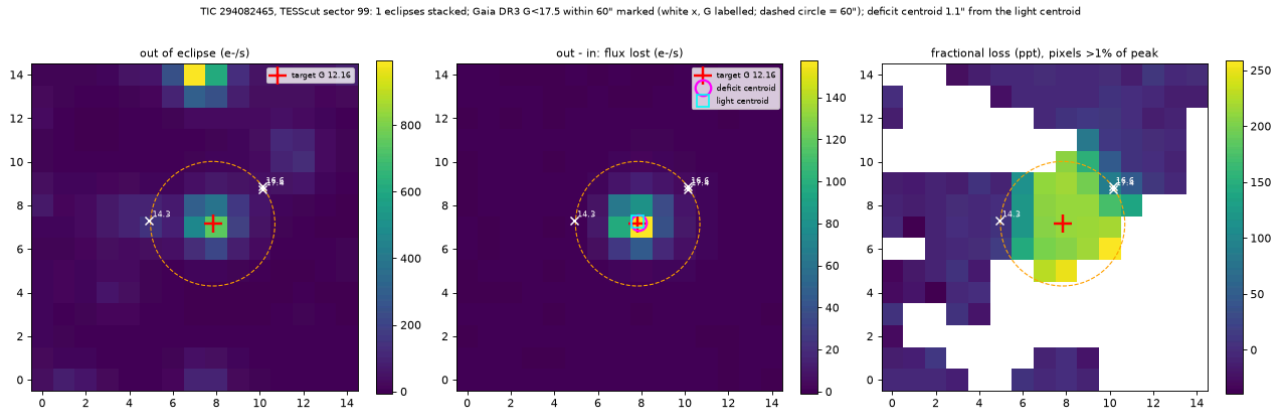
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R468; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R468; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 294082465: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 3 measured same-kind eclipses in representative QLP SAP_FLUX sectors [8, 99]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 17.3495287874 ± 0.0000712547 d and reference BTJD 1519.4094810 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S99 TESScut deficit lies 1.70 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Image-model fractional residual 0.296 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 212.51/210.43/203.81 ppt in 1 stacked events; selected QLP raw depth is 200.11 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 5705460089589085952 has SB1 period 17.3434002638 d, e = 0.43675. The closest edge-on conjunction differs from the measured eclipse epoch by 0.50447 d (0.02909 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.51559 d. Rough central duration 9.27 h versus measured median 7.07 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Three broad deep raw events in S8/99, including consecutive S99 cycles. The S99 target-centered deficit is supported by all three apertures, though that historical image stack contains one event only.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
8	0	1519.4094921	0.010417	203.51	used
99	146	4052.4406638	0.001157	200.11	used
99	147	4069.7902328	0.001157	199.95	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s99.json, historical/diff_s99.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R468. Public evidence files: diff_s99.json, diff_s99.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

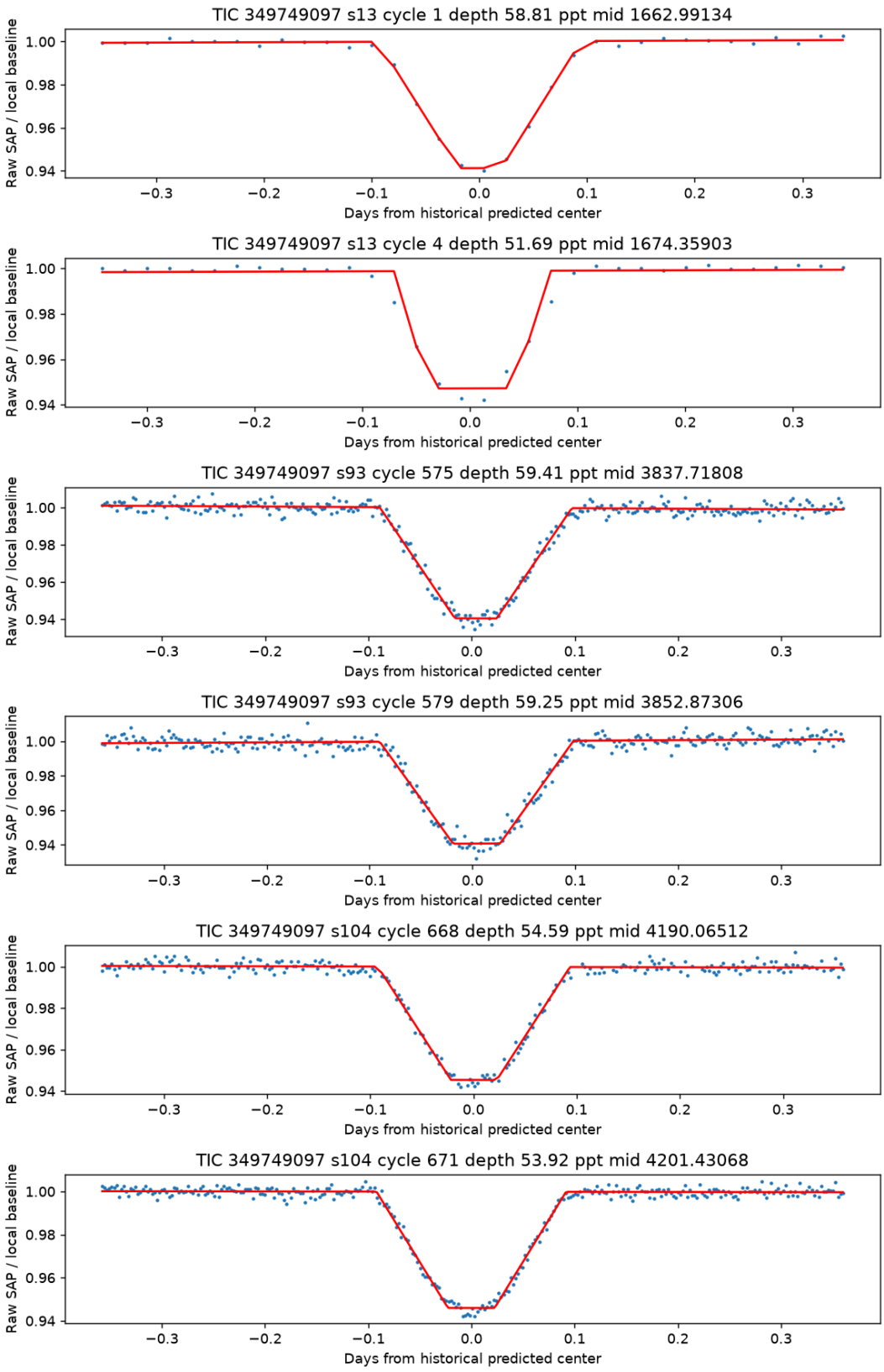
R469. TIC 349749097

physical_verdict: PROBABLE

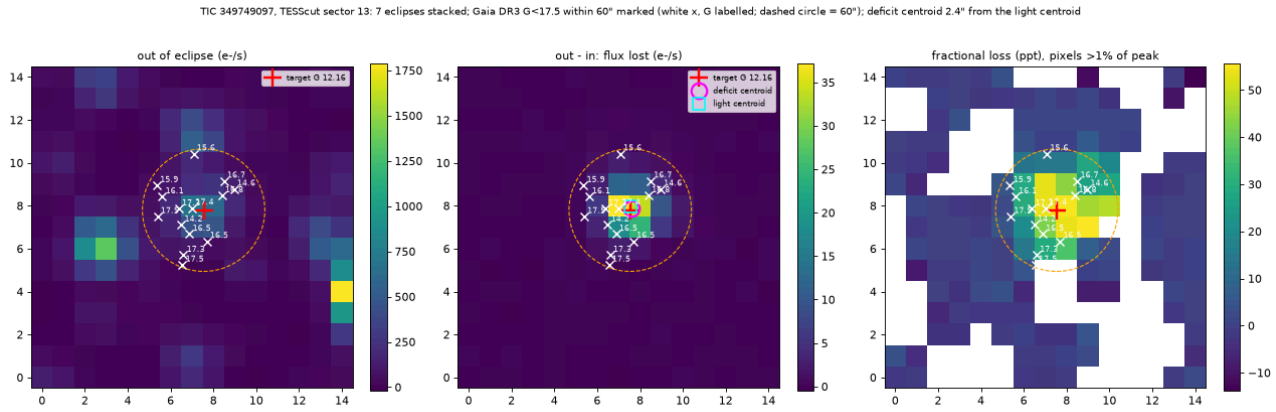
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization



R469; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R469; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 349749097: PROBABLE

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 16 measured same-kind eclipses in representative QLP SAP_FLUX sectors [13, 93, 104]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 3.7886980284 ± 0.0000059917 d and reference BTJD 1663.0043255 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S13 TESScut deficit lies 2.47 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [8, 8]. Image-model fractional residual 0.655 is not a calibrated position error. Capable Gaia neighbors: Gaia 6702211498076393216, 24.4 arcsec, G 14.21, approximate intrinsic requirement 0.41; Gaia 6702211601155599616, 33.4 arcsec, G 14.55, approximate intrinsic requirement 0.54. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 54.70/49.79/41.74 ppt in 7 stacked events; selected QLP raw depth is 55.44 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 6702211493775030784 has SB1 period 3.7889568790 d, e = 0.03612. The closest edge-on conjunction differs from the measured eclipse epoch by -0.06788 d (-0.01791 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.05803 d. Rough central duration 6.91 h versus measured median 4.46 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Sixteen raw eclipses across S13/93/104. Two capable Gaia neighbors about 24 and 33 arcsec away lie in a crowded field. The S13 deficit core overlaps the target pixel, but the high image-model residual and roughly one-pixel separation leave source identity unresolved.

Next check: Resolved eclipse photometry or multi-source PRF localization of the capable nearby star.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
13	1	1662.9913433	0.010416	58.81	used
13	2	1666.7774115	0.010416	53.37	used
13	3	1670.5687257	0.010416	53.30	used
13	4	1674.3590344	0.010416	51.69	used
13	5	1678.1464556	0.010416	58.45	used
93	574	3833.9294581	0.001157	59.72	used
93	575	3837.7180765	0.001157	59.41	used
93	577	3845.2955559	0.001157	58.11	used
93	578	3849.0842975	0.001157	58.79	used
93	579	3852.8730619	0.001157	59.25	used
104	666	4182.4872216	0.001157	55.37	used
104	667	4186.2759269	0.001157	55.51	used
104	668	4190.0651155	0.001157	54.59	used
104	669	4193.8542070	0.001157	54.46	used
104	670	4197.6433183	0.001157	55.06	used
104	671	4201.4306811	0.001157	53.92	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s13.json, historical/diff_s13.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R469. Public evidence files: diff_s13.json, diff_s13.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

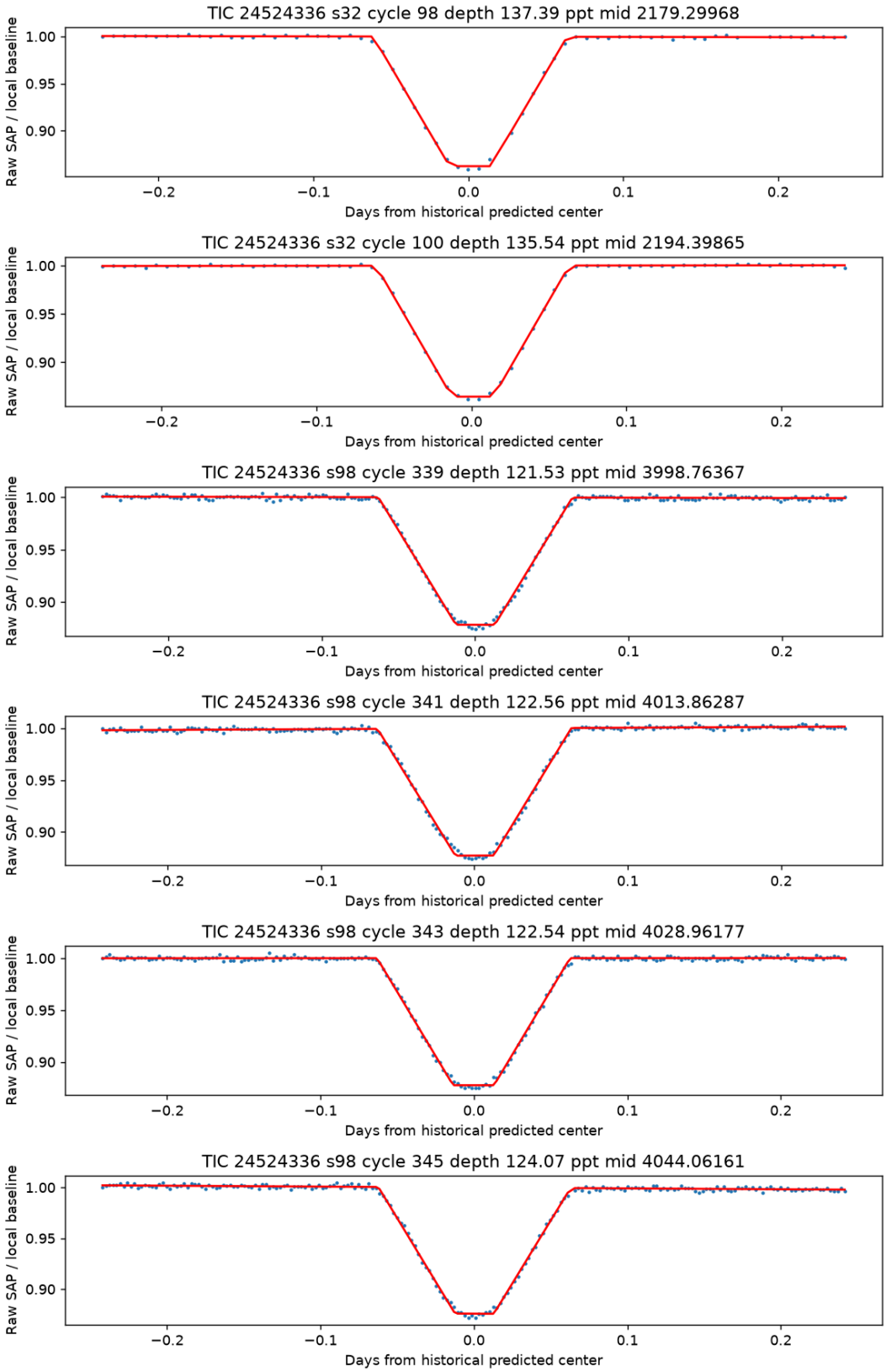
R470. TIC 24524336

physical_verdict: CONFIRMED

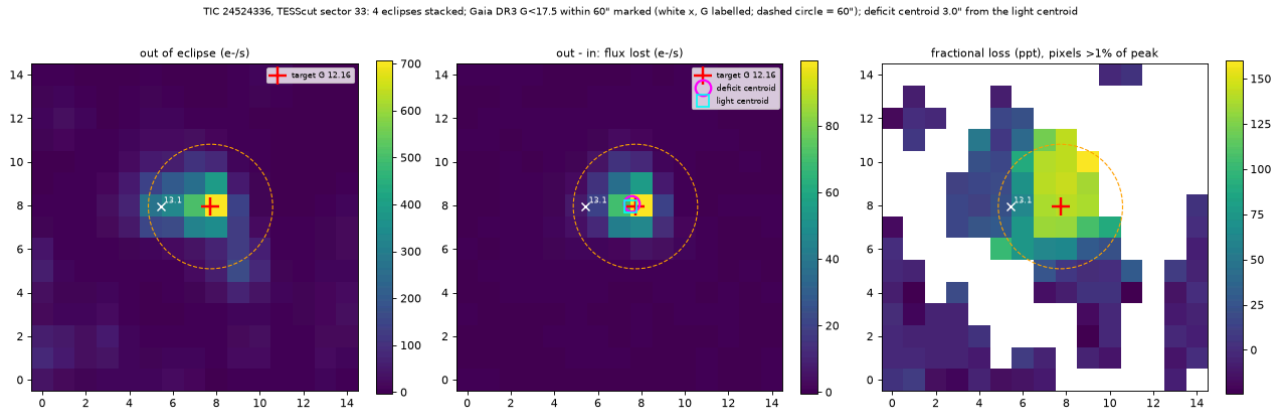
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R470; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R470; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 24524336: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.159077. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [32, 98] contains 6 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 7.5496432496 ± 0.0000103723 d and epoch BTJD 2179.2995260, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: []. Flagged intervening cycles are not negative detections; cross-sector parity and within-sector spacing discriminate the alias. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 33 TESScut difference image has deficit peak [8, 8] and target pixel [8, 8]; deficit centroid is 4.73 arcsec from the target coordinate. Image-model fractional residual 0.349 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 2901581709342471552 (44.6 arcsec, G 13.129, intrinsic loss needed ~0.45). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 141.0/137.2/117.4 ppt from 4 stacked events, versus median selected raw QLP depth 123.3 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 2901581915500902912 has P = 7.5496902503 d and e = 0.03847; the closer edge-on conjunction differs from the measured epoch by 0.04453 d (0.00590 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.12718 d,

descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 4.40 h against 3.01 h observed median; it is not a complete binary model.

Decision and limitation. S32 cycles 98/100 and S98 cycles 339/341/343/345 have opposite half-period parity. Intervening windows are quality-flagged, not clean nulls. The within-sector 15.099-day spacing together with cross-sector parity excludes a constant doubled period. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
32	98	2179.2996811	0.003472	137.4	yes
32	100	2194.3986515	0.003472	135.5	yes
98	339	3998.7636658	0.001157	121.5	yes
98	341	4013.8628696	0.001157	122.6	yes
98	343	4028.9617662	0.001157	122.5	yes
98	345	4044.0616145	0.001157	124.1	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s33.json, historical/diff_s33.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R470. Public evidence files: diff_s33.json, diff_s33.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

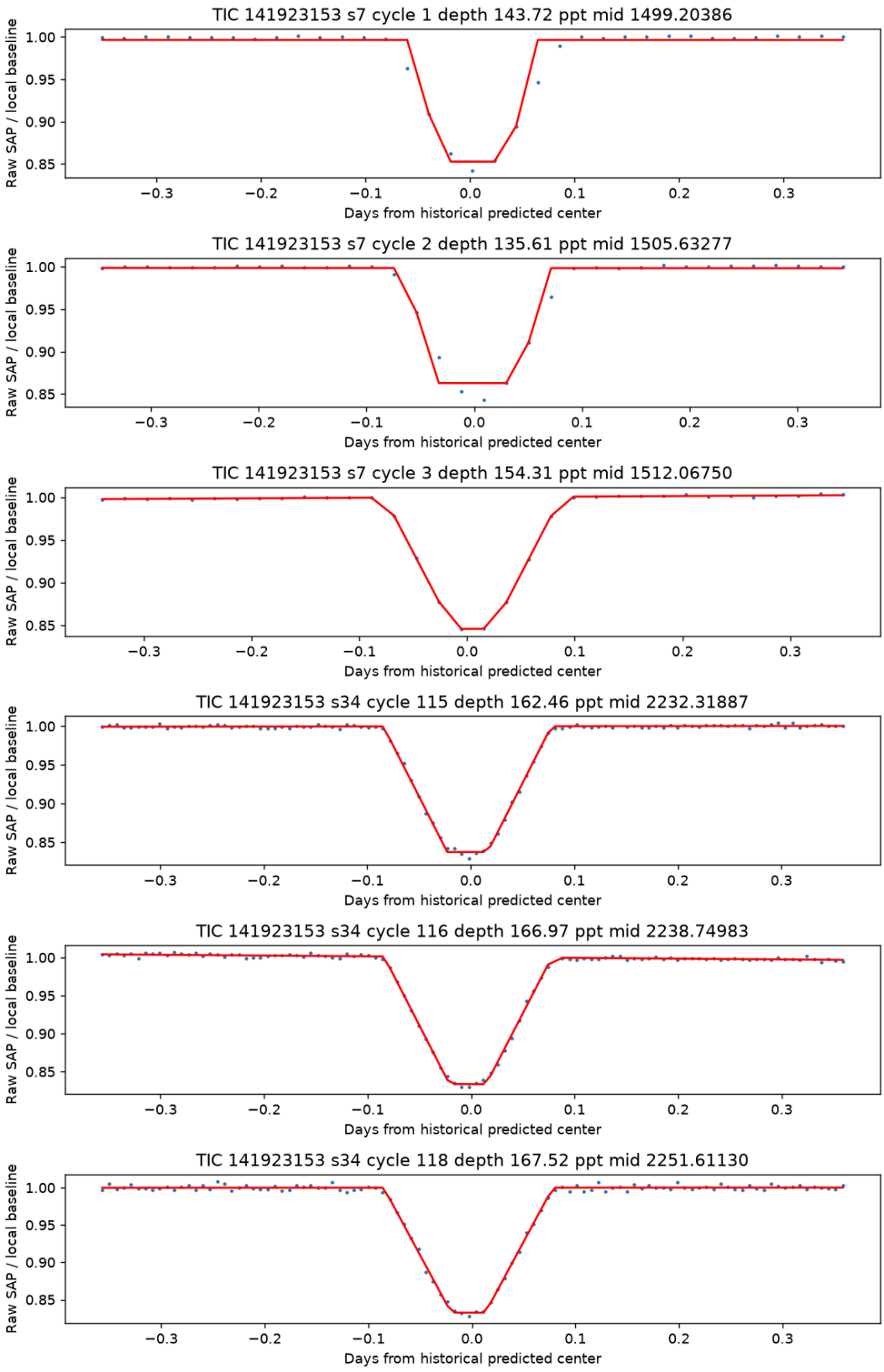
R471. TIC 141923153

physical_verdict: CONFIRMED

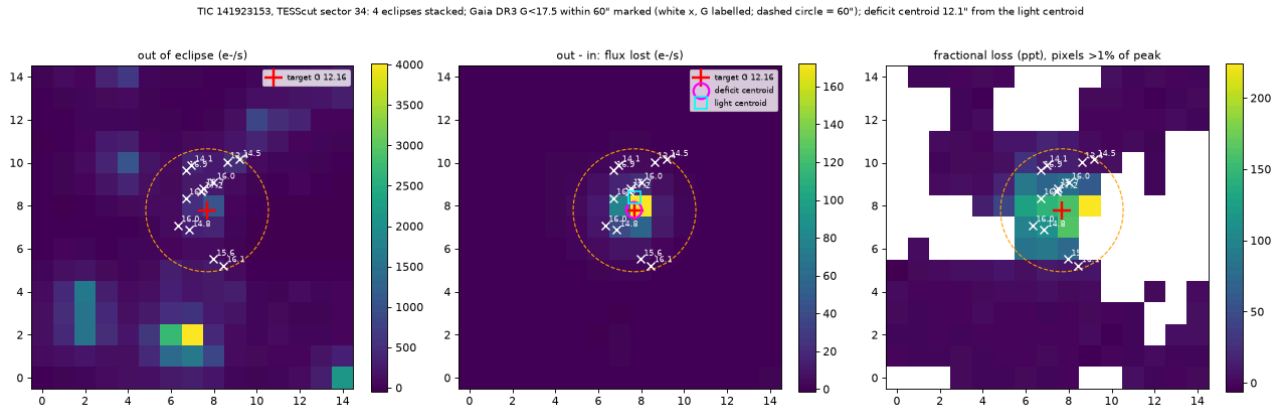
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R471; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R471; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 141923153: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.162264. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [7, 34] contains 7 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 6.4308327985 ± 0.0000546344 d and epoch BTJD 1499.2038780, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[7, 1, 2], [7, 2, 3], [34, 115, 116], [34, 116, 117], [34, 117, 118]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 34 TESScut difference image has deficit peak [8, 8] and target pixel [8, 8]; deficit centroid is 1.01 arcsec from the target coordinate. Image-model fractional residual 0.811 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 5710669575691823360 (51.6 arcsec, G 13.425, intrinsic loss needed ~0.64). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 159.4/137.3/101.6 ppt from 4 stacked events, versus median selected raw QLP depth 162.5 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 5710669537024684416 has P = 6.4302209846 d and e = 0.11073; the closer edge-on conjunction differs from the measured epoch by 0.05310 d (0.00826 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.11745 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 5.55 h against 3.89 h observed median; it is not a complete binary model.

Decision and limitation. Consecutive raw eclipses in S7 and S34 support recurrence; the capable 51.6-arcsec neighbor is separated from the target-centered deficit despite substantial image-model residual. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
7	1	1499.2038586	0.010417	143.7	yes
7	2	1505.6327667	0.010417	135.6	yes
7	3	1512.0675010	0.010417	154.3	yes
34	115	2232.3188695	0.003472	162.5	yes
34	116	2238.7498281	0.003472	167.0	yes
34	117	2245.1802637	0.003472	165.3	yes
34	118	2251.6113041	0.003472	167.5	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s34.json, historical/diff_s34.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R471. Public evidence files: diff_s34.json, diff_s34.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

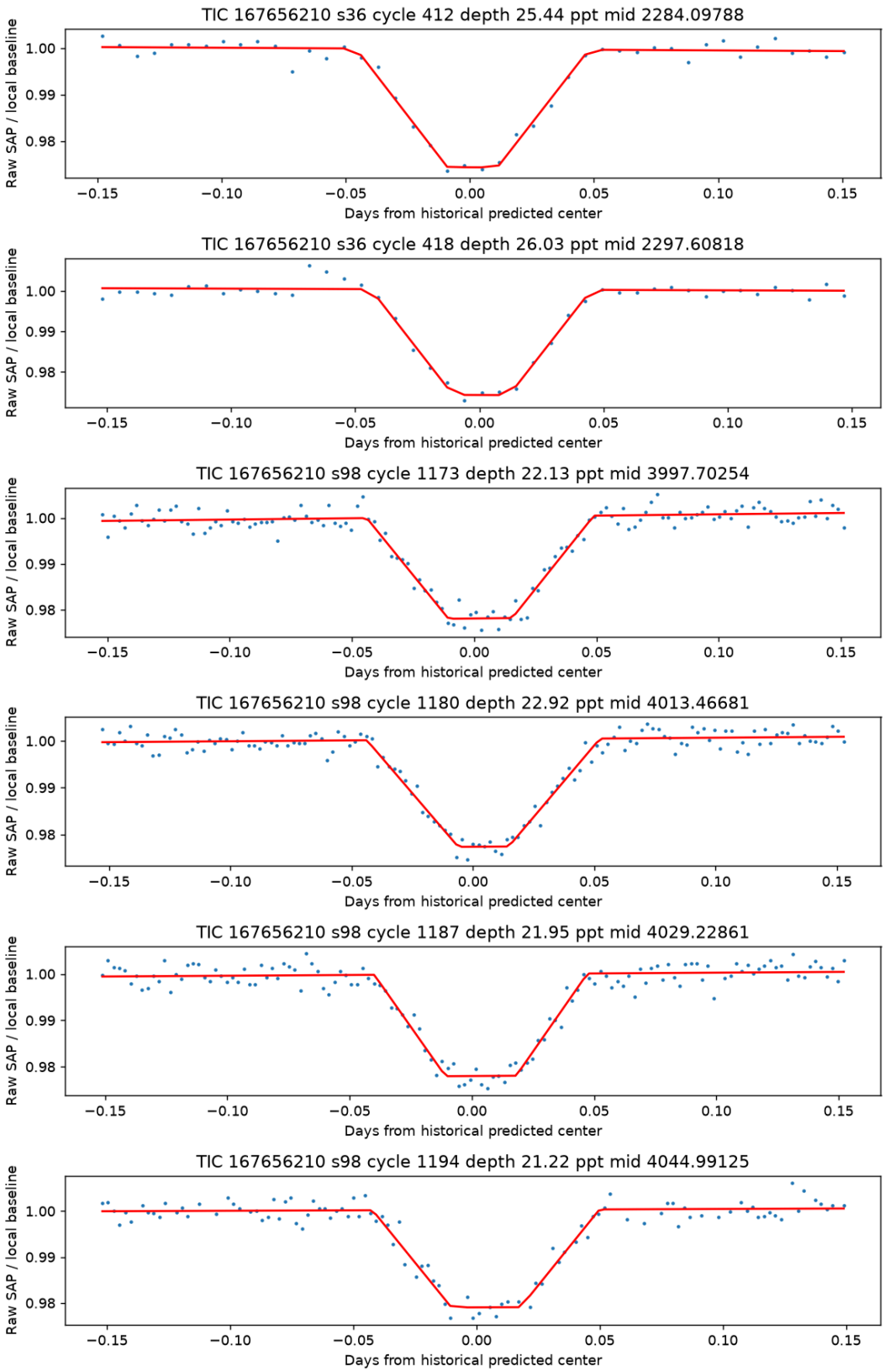
R472. TIC 167656210

physical_verdict: PROBABLE

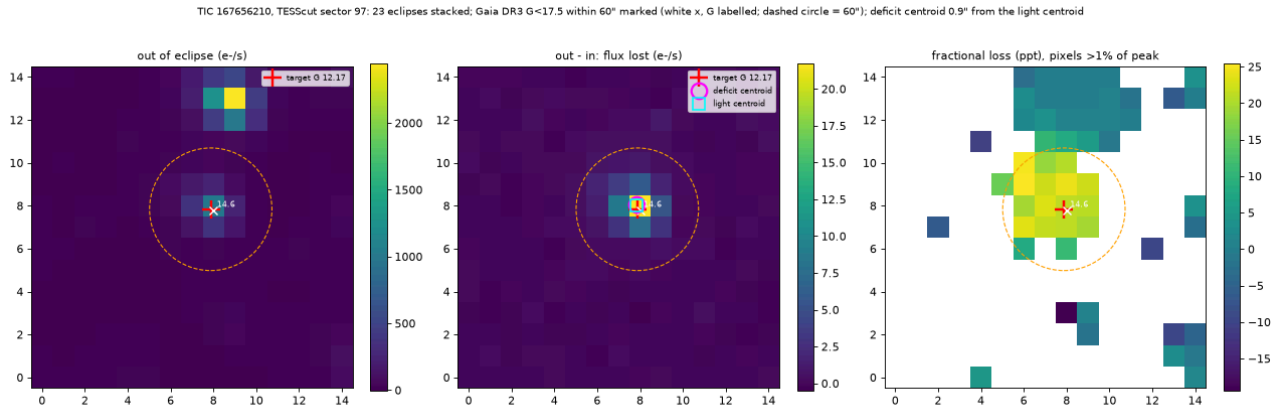
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved time-series photometry or multi-source PRF fitting of the 2.9-arcsec pair.



R472; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R472; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 167656210: PROBABLE

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.166932. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [36, 98] contains 22 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 2.2517814962 ± 0.0000017527 d and epoch BTJD 2284.0975046, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[36, 412, 413], [36, 413, 414], [36, 414, 415], [36, 420, 421], [98, 1172, 1173], [98, 1173, 1174], [98, 1174, 1175], [98, 1179, 1180], [98, 1180, 1181], [98, 1185, 1186], [98, 1186, 1187], [98, 1187, 1188], [98, 1191, 1192], [98, 1192, 1193], [98, 1193, 1194]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 97 TESScut difference image has deficit peak [8, 8] and target pixel [8, 8]; deficit centroid is 4.63 arcsec from the target coordinate. Image-model fractional residual 0.652 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 5480019042928976896 (2.9 arcsec, G 14.551, intrinsic loss needed ~0.24). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 21.2/21.7/21.5 ppt from 23 stacked events, versus median selected raw QLP depth 23.1 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 5480019038632380288 has P = 2.2517090090 d and e = 0.03953; the closer edge-on conjunction differs from the measured epoch by 0.04866 d (0.02161 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.09510 d,

descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 4.78 h against 2.22 h observed median; it is not a complete binary model.

Decision and limitation. A 2.9-arcsec capable Gaia neighbor shares the TESS core. The recurring eclipses are real, but these pixels cannot uniquely assign them to the selected Gaia star. Attribution remains unresolved after the completed pixel check.

Next check. Resolved time-series photometry or multi-source PRF fitting of the 2.9-arcsec pair.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
36	412	2284.0978758	0.003472	25.4	yes
36	413	2286.3495175	0.003472	25.4	yes
36	414	2288.6013787	0.003472	23.9	yes
36	415	2290.8530233	0.003472	24.8	yes
36	418	2297.6081832	0.003472	26.0	yes
36	420	2302.1110686	0.003472	24.3	yes
36	421	2304.3631005	0.003472	25.4	yes
98	1172	3995.4520614	0.001157	23.0	yes
98	1173	3997.7025415	0.001157	22.1	yes
98	1174	3999.9547706	0.001157	22.0	yes
98	1175	4002.2074201	0.001157	21.9	yes
98	1179	4011.2131845	0.001157	23.9	yes
98	1180	4013.4668097	0.001157	22.9	yes
98	1181	4015.7173665	0.001157	23.3	yes
98	1185	4024.7241694	0.001157	24.1	yes
98	1186	4026.9757790	0.001157	23.3	yes
98	1187	4029.2286132	0.001157	21.9	yes
98	1188	4031.4798496	0.001157	22.7	yes
98	1191	4038.2353658	0.001157	21.5	yes
98	1192	4040.4869257	0.001157	21.9	yes
98	1193	4042.7383739	0.001157	22.2	yes
98	1194	4044.9912545	0.001157	21.2	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s97.json, historical/diff_s97.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R472. Public evidence files: diff_s97.json, diff_s97.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

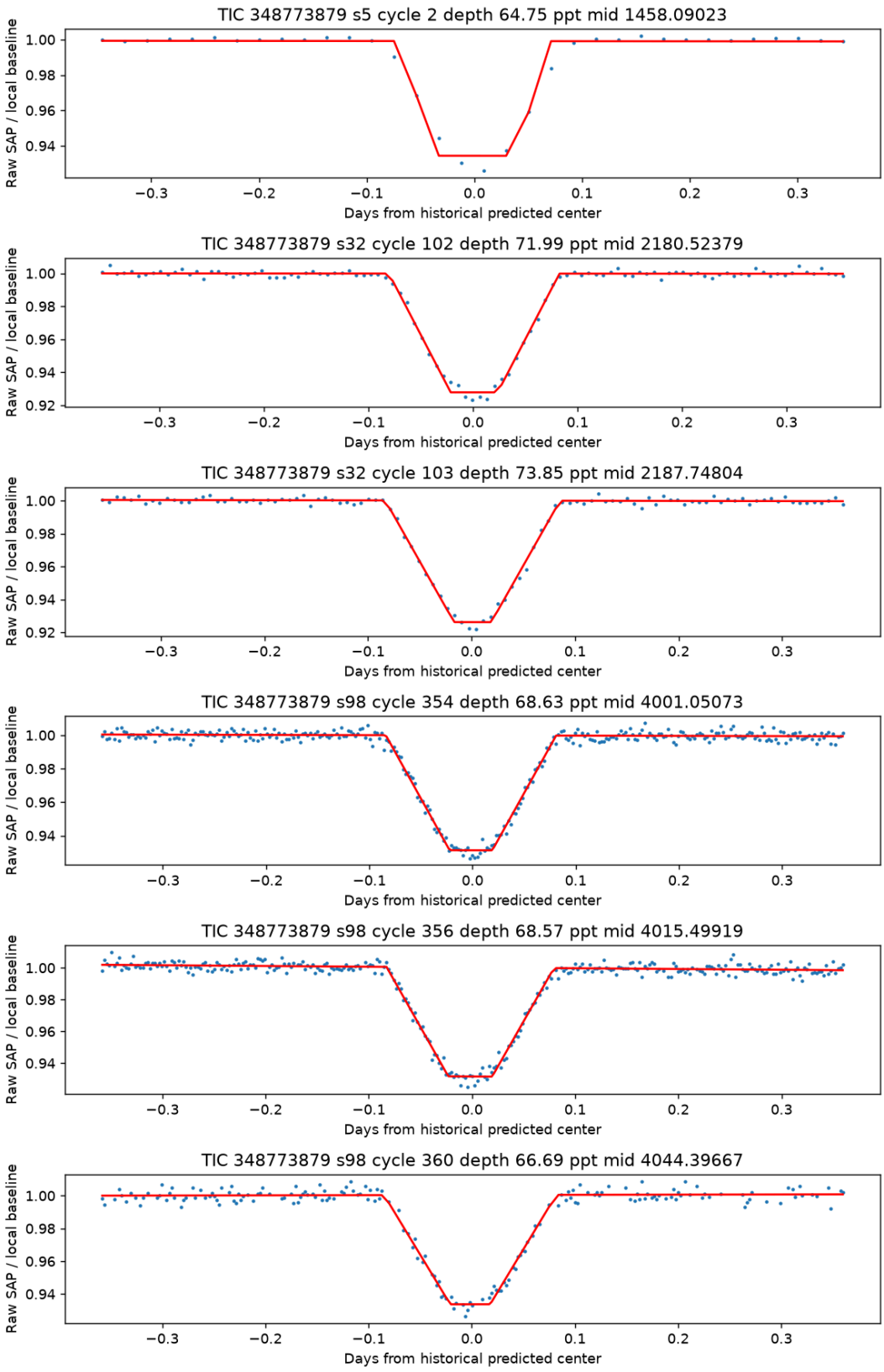
R473. TIC 348773879

physical_verdict: CONFIRMED

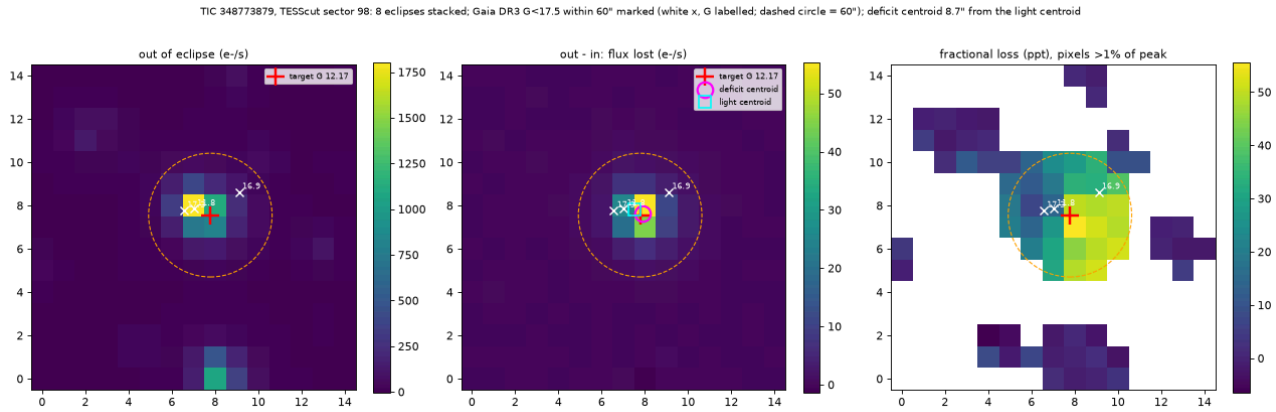
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R473; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R473; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 348773879: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.169457. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [5, 32, 98] contains 8 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 7.2243136689 ± 0.0000079385 d and epoch BTJD 1458.0924122, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[32, 102, 103], [32, 103, 104]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 98 TESScut difference image has deficit peak [8, 8] and target pixel [8, 8]; deficit centroid is 2.72 arcsec from the target coordinate. Image-model fractional residual 0.535 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 2987292965757749376 (17.0 arcsec, G 11.841, intrinsic loss needed ~0.11). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 49.9/32.8/30.9 ppt from 8 stacked events, versus median selected raw QLP depth 68.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 2987292862678534144 has P = 7.2246361222 d and e = 0.04277; the closer edge-on conjunction differs from the measured epoch by -0.09908 d (-0.01371 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.12913 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 5.95 h against 3.98 h observed median; it is not a complete binary model.

Decision and limitation. The capable 17.0-arcsec neighbor is about a pixel away; the deficit core falls much closer to the target. Attribution favors the target at TESS resolution, with uncalibrated subpixel precision. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
5	2	1458.0902329	0.010417	64.7	yes
32	102	2180.5237896	0.003472	72.0	yes
32	103	2187.7480351	0.003472	73.9	yes
32	104	2194.9728229	0.003472	75.0	yes
98	354	4001.0507308	0.001157	68.6	yes
98	356	4015.4991851	0.001157	68.6	yes
98	358	4029.9484542	0.001157	67.6	yes
98	360	4044.3966744	0.001157	66.7	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s98.json, historical/diff_s98.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R473. Public evidence files: diff_s98.json, diff_s98.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

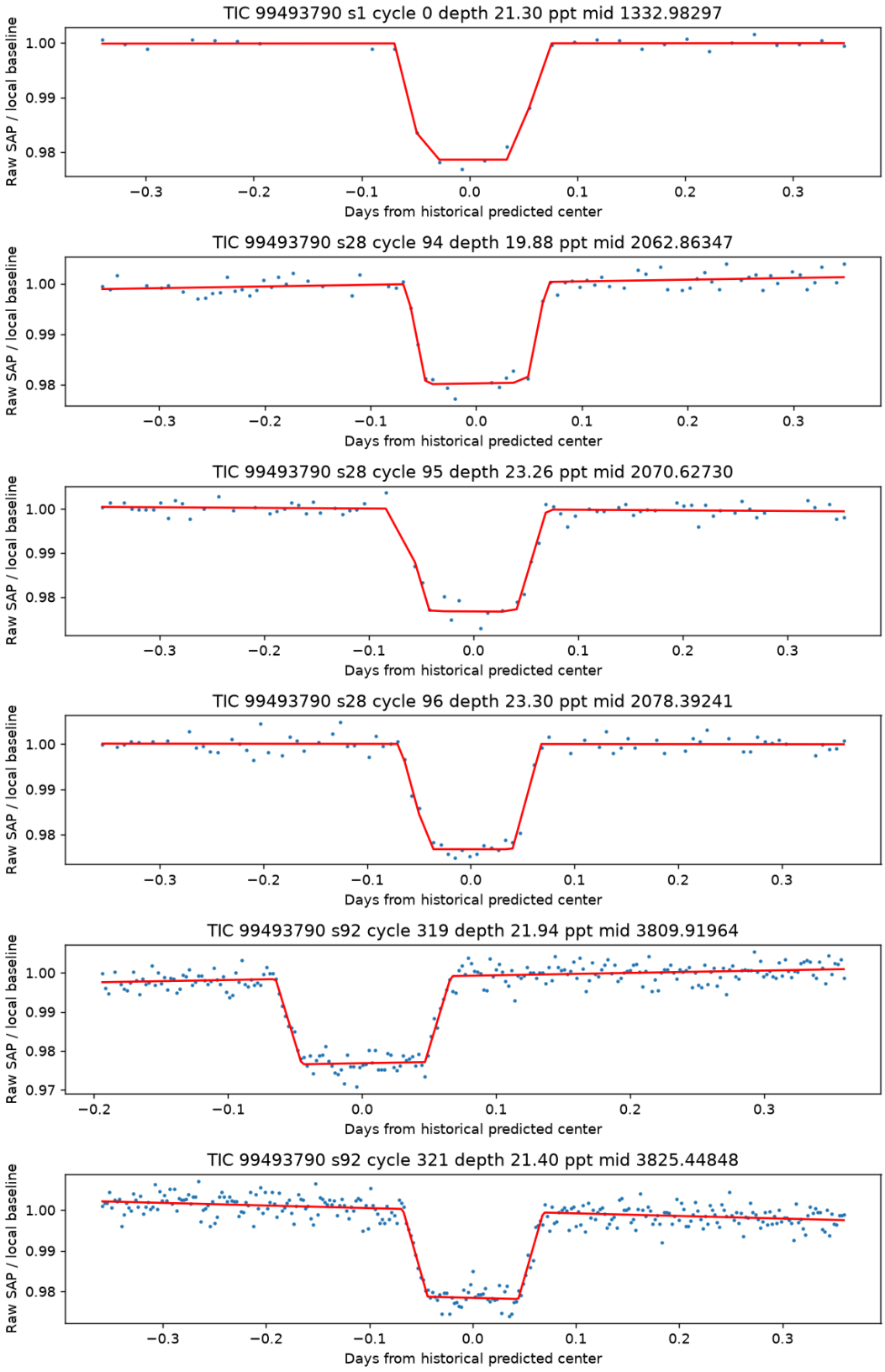
R474. TIC 99493790

physical_verdict: CONFIRMED

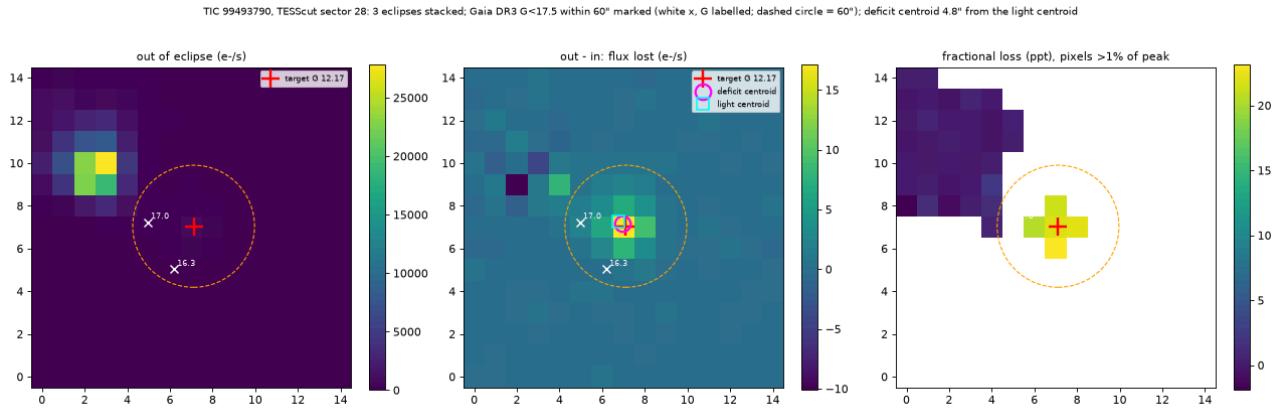
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R474; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R474; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 99493790: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.170427. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [1, 28, 92] contains 7 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 7.7646935289 ± 0.0000090716 d and epoch BTJD 1332.9819065, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[28, 94, 95], [28, 95, 96], [92, 319, 320], [92, 320, 321]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 28 TESScut difference image has deficit peak [7, 7] and target pixel [7, 7]; deficit centroid is 3.68 arcsec from the target coordinate. Image-model fractional residual 0.999 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: None under the approximate G-band brightness budget. The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 21.8/21.2/18.7 ppt from 3 stacked events, versus median selected raw QLP depth 21.9 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 6804708139515008640 has P = 7.7653086675 d and e = 0.08951; the closer edge-on conjunction differs from the measured epoch by 0.05671 d (0.00730 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.33323 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 3.89 h against 3.26 h observed median; it is not a complete binary model.

Decision and limitation. The target-centered deficit and absence of a brightness-capable Gaia neighbor favor attribution; the image-model residual is high. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
1	0	1332.9829705	0.010417	21.3	yes
28	94	2062.8634721	0.003472	19.9	yes
28	95	2070.6273013	0.003472	23.3	yes
28	96	2078.3924097	0.003472	23.3	yes
92	319	3809.9196418	0.001158	21.9	yes
92	320	3817.6833970	0.001158	23.3	yes
92	321	3825.4484769	0.001158	21.4	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s28.json, historical/diff_s28.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R474. Public evidence files: diff_s28.json, diff_s28.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

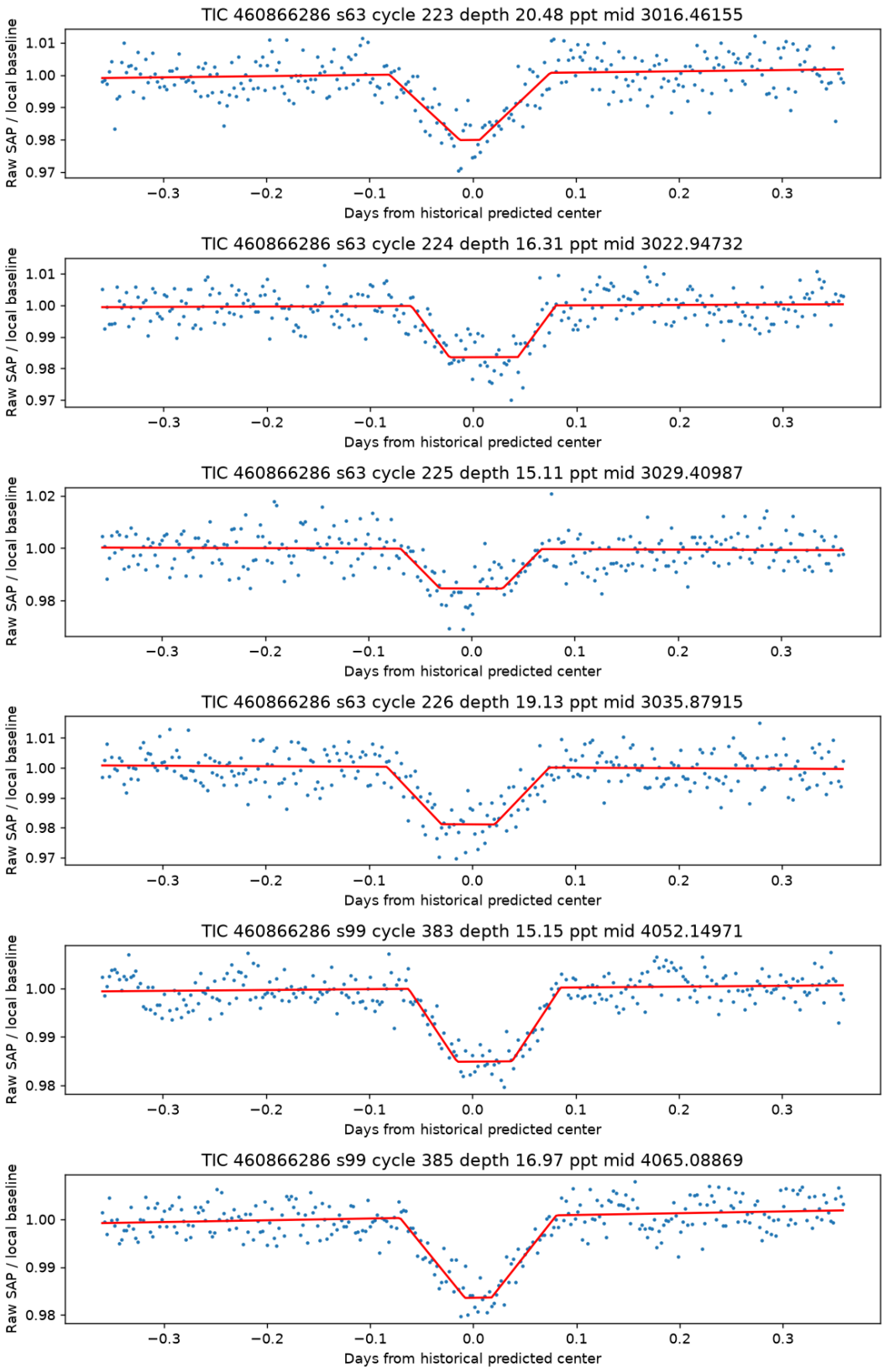
R475. TIC 460866286

physical_verdict: PROBABLE

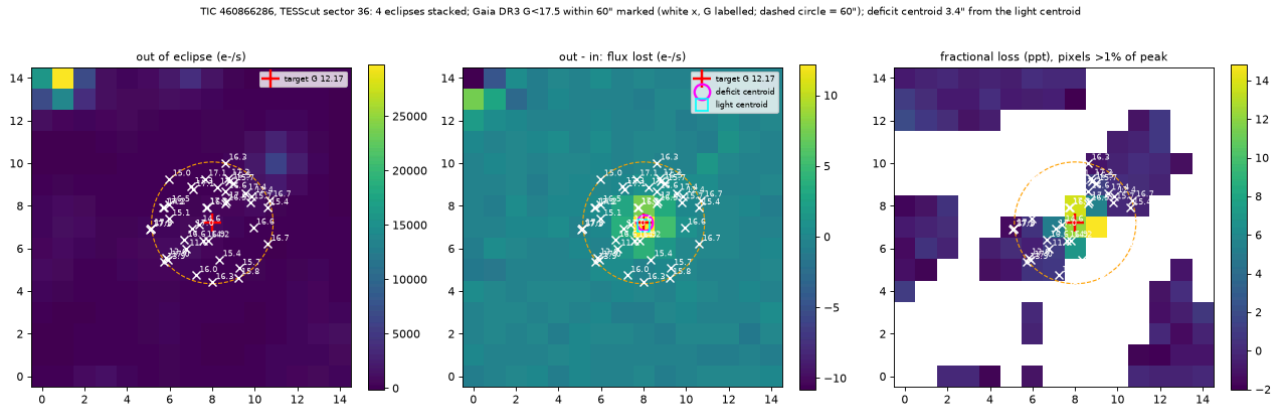
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Higher-resolution eclipse photometry or multi-source PRF fitting in this crowded field.



R475; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R475; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 460866286: PROBABLE

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.171505. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [63, 99] contains 6 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 6.4730069098 ± 0.0000307395 d and epoch BTJD 3016.4658767, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[63, 223, 224], [63, 224, 225], [63, 225, 226]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 36 TESScut difference image has deficit peak [8, 7] and target pixel [8, 7]; deficit centroid is 0.95 arcsec from the target coordinate. Image-model fractional residual 0.970 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 5254306927996818560 (12.3 arcsec, G 14.563, intrinsic loss needed ~0.16); Gaia 5254306932350664960 (17.3 arcsec, G 14.152, intrinsic loss needed ~0.11); Gaia 5254301812749381248 (21.9 arcsec, G 16.515, intrinsic loss needed ~0.86); Gaia 5254306932350668800 (30.4 arcsec, G 16.610, intrinsic loss needed ~0.94); Gaia 5254301812749372800 (32.3 arcsec, G 16.629, intrinsic loss needed ~0.95); Gaia 5254306966710413440 (34.4 arcsec, G 11.637, intrinsic loss needed ~0.02); Gaia 5254307722624660992 (36.0 arcsec, G 15.407, intrinsic loss needed ~0.32); Gaia 5254307683911072000 (38.1 arcsec, G 16.620, intrinsic loss needed ~0.95); Gaia 5254301812749375232 (40.0 arcsec, G 15.746, intrinsic loss needed ~0.43); Gaia 5254306932350657024 (40.7 arcsec, G 15.113, intrinsic loss needed ~0.25); Gaia 5254301808395792768 (41.9 arcsec, G 15.719, intrinsic loss needed ~0.42); Gaia 5254301812749364096 (42.7 arcsec, G 16.547, intrinsic loss needed ~0.89); Gaia

5254301808395353216 (44.3 arcsec, G 12.383, intrinsic loss needed ~0.03); Gaia 5254306932350647040 (44.6 arcsec, G 16.498, intrinsic loss needed ~0.85); Gaia 5254306932293814016 (47.3 arcsec, G 14.187, intrinsic loss needed ~0.11); Gaia 5254307718270821504 (50.7 arcsec, G 15.688, intrinsic loss needed ~0.41); Gaia 5254306966710433152 (52.3 arcsec, G 16.044, intrinsic loss needed ~0.56); Gaia 5254301808414672768 (52.6 arcsec, G 15.358, intrinsic loss needed ~0.31); Gaia 5254306966710944384 (54.1 arcsec, G 13.649, intrinsic loss needed ~0.08); Gaia 5254306966710435840 (57.2 arcsec, G 16.331, intrinsic loss needed ~0.73); Gaia 5254306966710944768 (58.0 arcsec, G 13.933, intrinsic loss needed ~0.09); Gaia 5254307718271123328 (58.2 arcsec, G 15.760, intrinsic loss needed ~0.44); Gaia 5254301778389613440 (58.5 arcsec, G 16.272, intrinsic loss needed ~0.69); Gaia 5254301056835128576 (58.7 arcsec, G 15.025, intrinsic loss needed ~0.23). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.

4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 12.3/7.7/3.9 ppt from 4 stacked events, versus median selected raw QLP depth 16.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 5254307683911065728 has P = 6.4725617010 d and e = 0.06549; the closer edge-on conjunction differs from the measured epoch by 0.10544 d (0.01629 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.18827 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 8.14 h against 3.59 h observed median; it is not a complete binary model.

Decision and limitation. The field has 24 brightness-capable Gaia neighbors within 60 arcsec, including one at 12.3 arcsec. The deficit core is near the target, but the weak signal and 0.97 image-model residual cannot uniquely resolve nearby alternatives. Attribution remains unresolved after the completed pixel check.

Next check. Higher-resolution eclipse photometry or multi-source PRF fitting in this crowded field.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
63	223	3016.4615526	0.002601	20.5	yes
63	224	3022.9473244	0.002086	16.3	yes
63	225	3029.4098721	0.002859	15.1	yes
63	226	3035.8791532	0.002305	19.1	yes
99	383	4052.1497109	0.001449	15.2	yes
99	385	4065.0886872	0.001840	17.0	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s36.json, historical/diff_s36.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R475. Public evidence files: diff_s36.json, diff_s36.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed;

source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

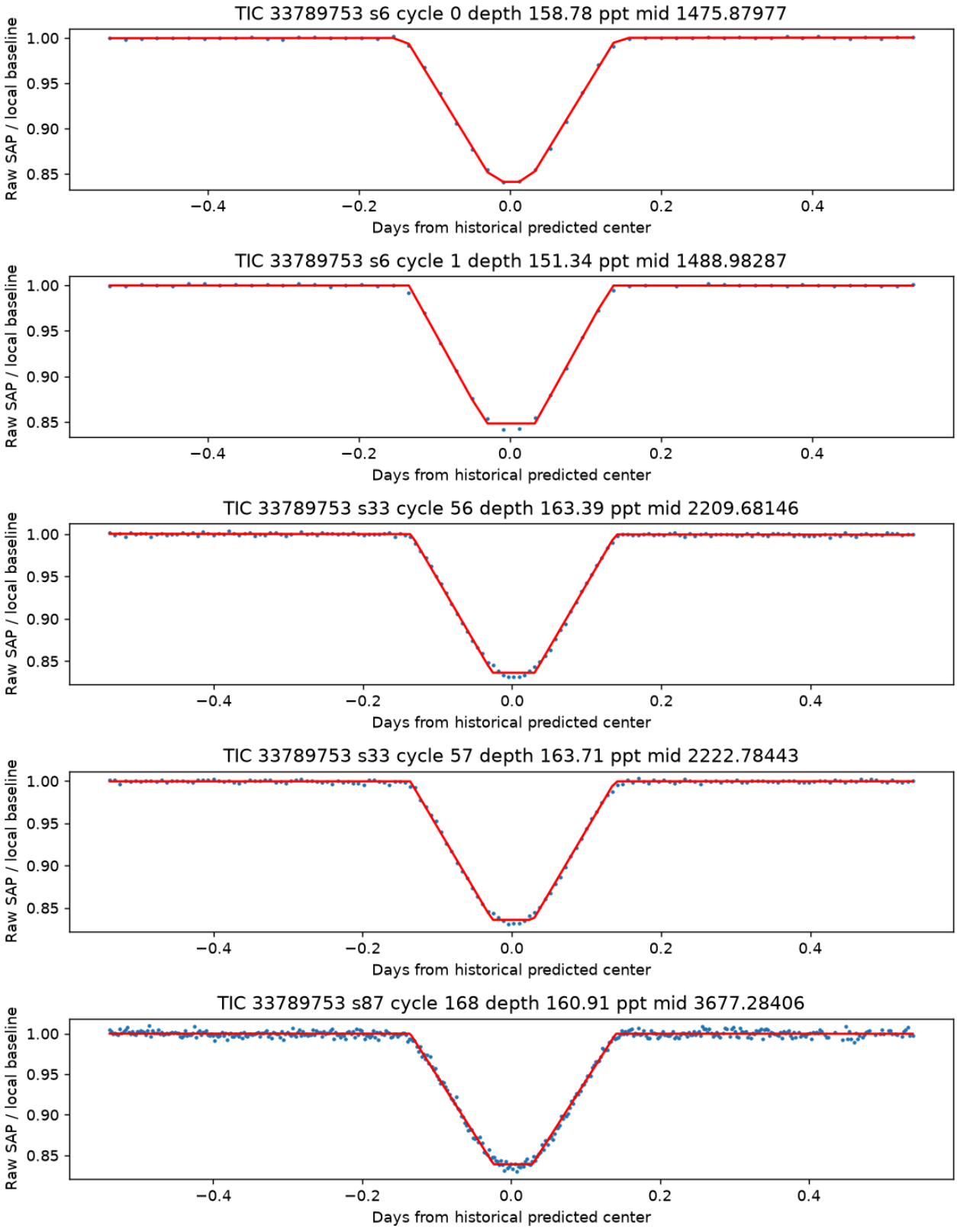
R476. TIC 33789753

physical_verdict: CONFIRMED

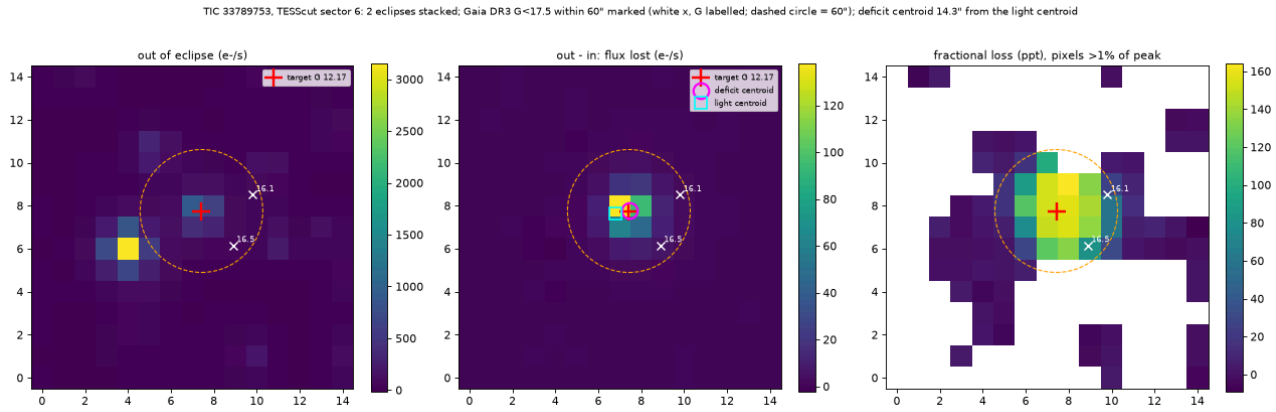
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R476; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R476; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 33789753: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.174696. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [6, 33, 87] contains 5 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 13.1035976171 ± 0.0000219313 d and epoch BTJD 1475.8796596, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[6, 0, 1], [33, 56, 57]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 6 TESScut difference image has deficit peak [7, 8] and target pixel [7, 8]; deficit centroid is 0.63 arcsec from the target coordinate. Image-model fractional residual 0.947 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: None under the approximate G-band brightness budget. The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 153.5/148.6/104.3 ppt from 2 stacked events, versus median selected raw QLP depth 160.9 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 2993617253561277056 has P = 13.1020129210 d and e = 0.41294; the closer edge-on conjunction differs from the measured epoch by 0.17799 d (0.01359 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.22281 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 10.15 h against 6.59 h observed median; it is not a complete binary model.

Decision and limitation. Five raw events in S6/S33/S87 agree in timing and shape. No brightness-capable Gaia neighbor appears within 60 arcsec. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
6	0	1475.8797672	0.010417	158.8	yes
6	1	1488.9828743	0.010417	151.3	yes
33	56	2209.6814606	0.003472	163.4	yes
33	57	2222.7844323	0.003472	163.7	yes
87	168	3677.2840579	0.001157	160.9	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s06.json, historical/diff_s06.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R476. Public evidence files: diff_s06.json, diff_s06.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

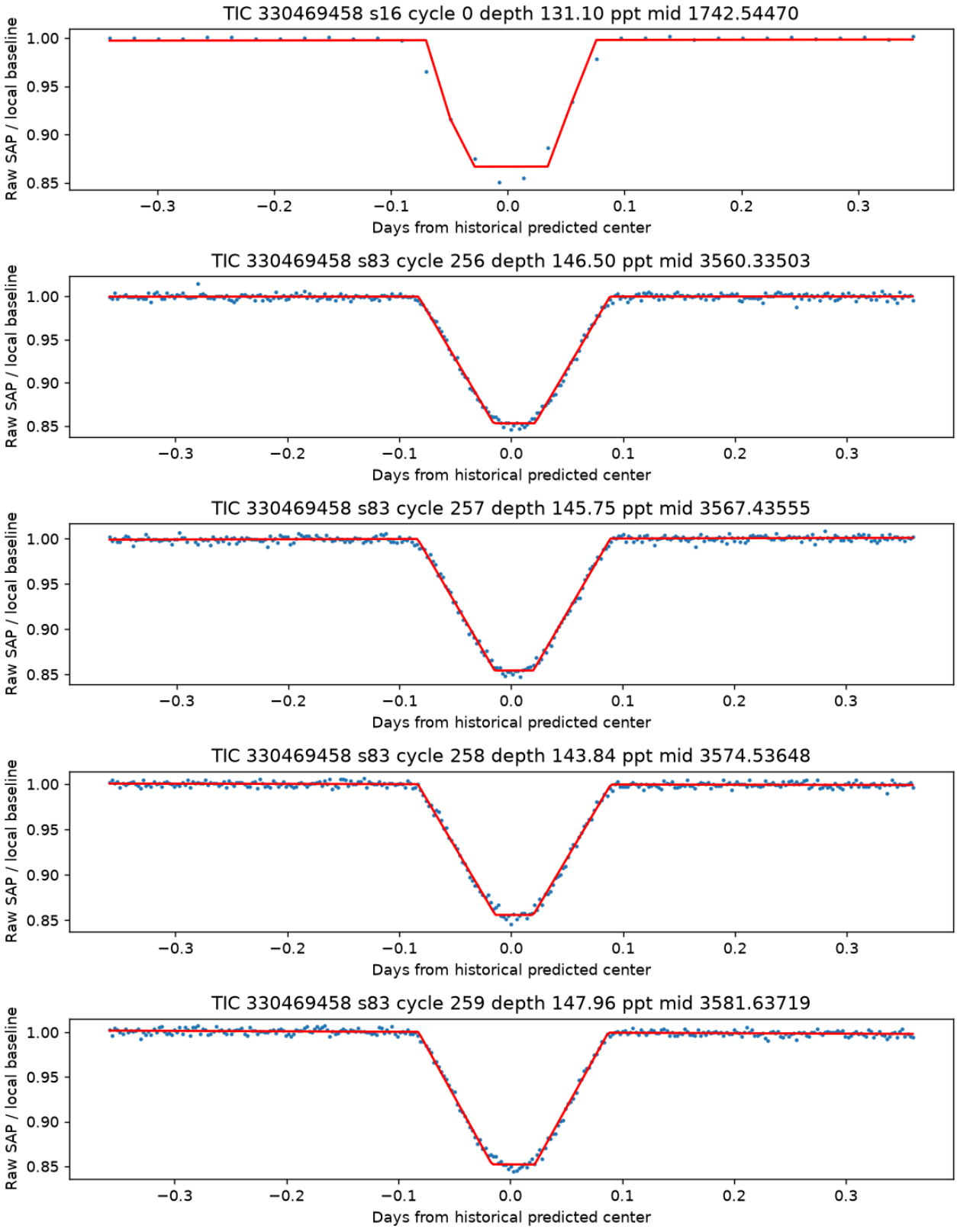
R477. TIC 330469458

physical_verdict: CONFIRMED

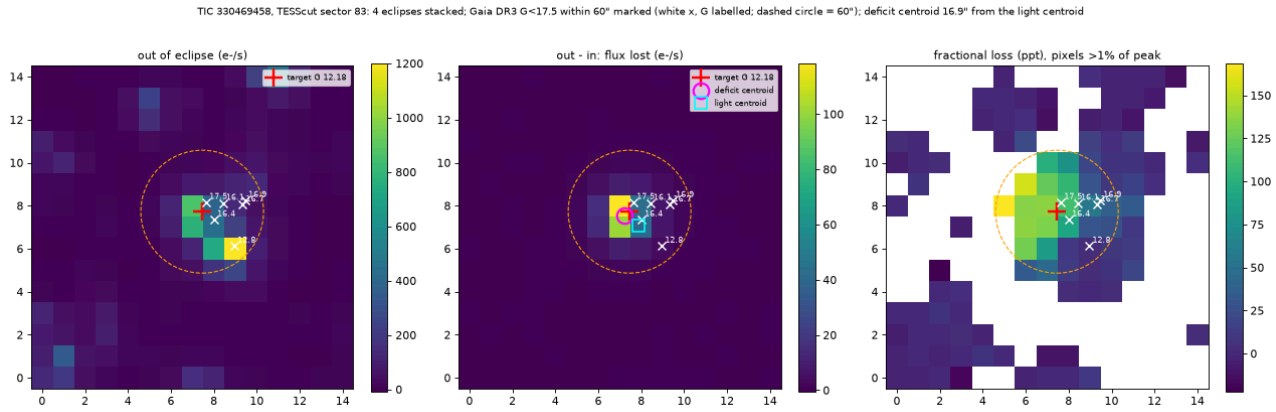
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R477; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R477; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 330469458: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.182403. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [16, 83] contains 5 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 7.1007431519 ± 0.0000403912 d and epoch BTJD 1742.5447026, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[83, 256, 257], [83, 257, 258], [83, 258, 259]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 83 TESScut difference image has deficit peak [7, 8] and target pixel [7, 8]; deficit centroid is 5.88 arcsec from the target coordinate. Image-model fractional residual 0.717 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 1906480221138868736 (46.3 arcsec, G 12.790, intrinsic loss needed ~0.37). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 136.4/123.7/73.9 ppt from 4 stacked events, versus median selected raw QLP depth 145.7 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 1906480289858343040 has P = 7.1014333035 d and e = 0.03164; the closer edge-on conjunction differs from the measured epoch by -0.13985 d (-0.01969 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.16156 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 5.78 h against 4.11 h observed median; it is not a complete binary model.

Decision and limitation. Five deep raw events in S16/S83 recur at the measured period. A capable neighbor 46.3 arcsec away is distinct from the target-centered deficit. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
16	0	1742.5446985	0.010416	131.1	yes
83	256	3560.3350331	0.001157	146.5	yes
83	257	3567.4355511	0.001157	145.7	yes
83	258	3574.5364814	0.001157	143.8	yes
83	259	3581.6371914	0.001157	148.0	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s83.json, historical/diff_s83.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R477. Public evidence files: diff_s83.json, diff_s83.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

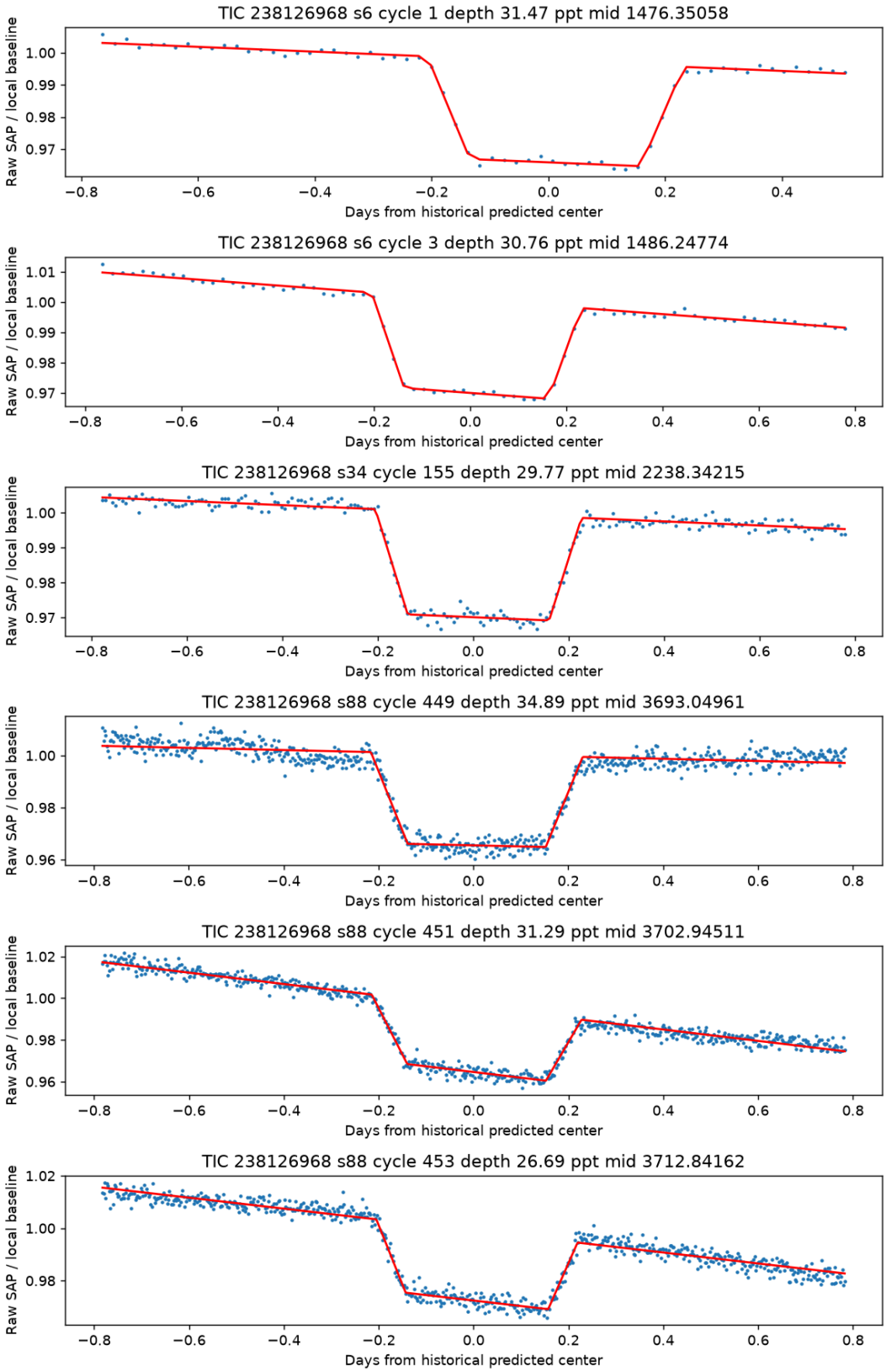
R478. TIC 238126968

physical_verdict: CONFIRMED

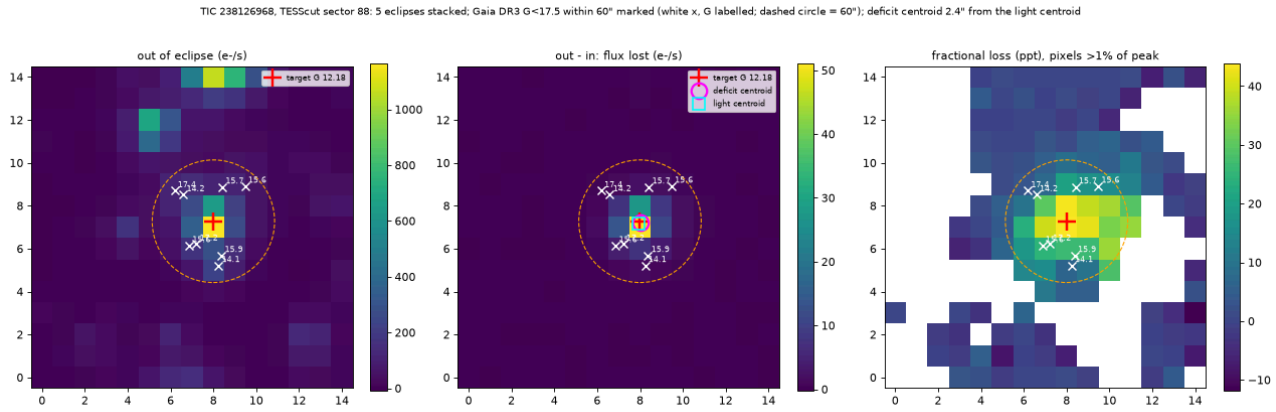
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R478; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R478; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 238126968: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.183063. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [6, 34, 88] contains 5 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 9.8959493292 ± 0.0000245311 d and epoch BTJD 1481.3039577, kind: Deeper repeating eclipse; even half-period cycles. Within-sector same-kind measured cycle pairs: [[34, 154, 156], [88, 450, 452]]. Only the deep even historical cycles enter the orbital fit; odd cycles are the shallower kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 88 TESScut difference image has deficit peak [8, 7] and target pixel [8, 7]; deficit centroid is 1.14 arcsec from the target coordinate. Image-model fractional residual 0.380 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 5508449115088643712 (39.7 arcsec, G 14.211, intrinsic loss needed ~0.39); Gaia 5508448840210743424 (43.6 arcsec, G 14.129, intrinsic loss needed ~0.36). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 43.9/38.6/32.0 ppt from 5 stacked events, versus median selected raw QLP depth 57.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 5508448904633378560 has P = 9.8961317032 d and e = 0.04783; the closer edge-on conjunction differs from the measured epoch by -0.04622 d (-0.00467 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.13469 d,

descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 10.68 h against 10.48 h observed median; it is not a complete binary model.

Decision and limitation. Shallow and deep eclipses alternate at the historical 4.94798-day search interval. Five deep even-cycle midpoints give the 9.89595-day orbital period; shallow odd cycles support the alternating pattern. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
6	1	1476.3505764	0.010417	31.5	no: other kind or timing failure
6	2	1481.2972292	0.010417	56.3	yes
6	3	1486.2477423	0.010417	30.8	no: other kind or timing failure
34	154	2233.4000168	0.005811	61.8	yes
34	155	2238.3421474	0.003472	29.8	no: other kind or timing failure
34	156	2243.2914492	0.005204	63.7	yes
88	449	3693.0496143	0.001157	34.9	no: other kind or timing failure
88	450	3697.9966498	0.001571	55.8	yes
88	451	3702.9451069	0.006515	31.3	no: other kind or timing failure
88	452	3707.8919600	0.003507	57.6	yes
88	453	3712.8416212	0.004982	26.7	no: other kind or timing failure

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s88.json, historical/diff_s88.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R478. Public evidence files: diff_s88.json, diff_s88.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

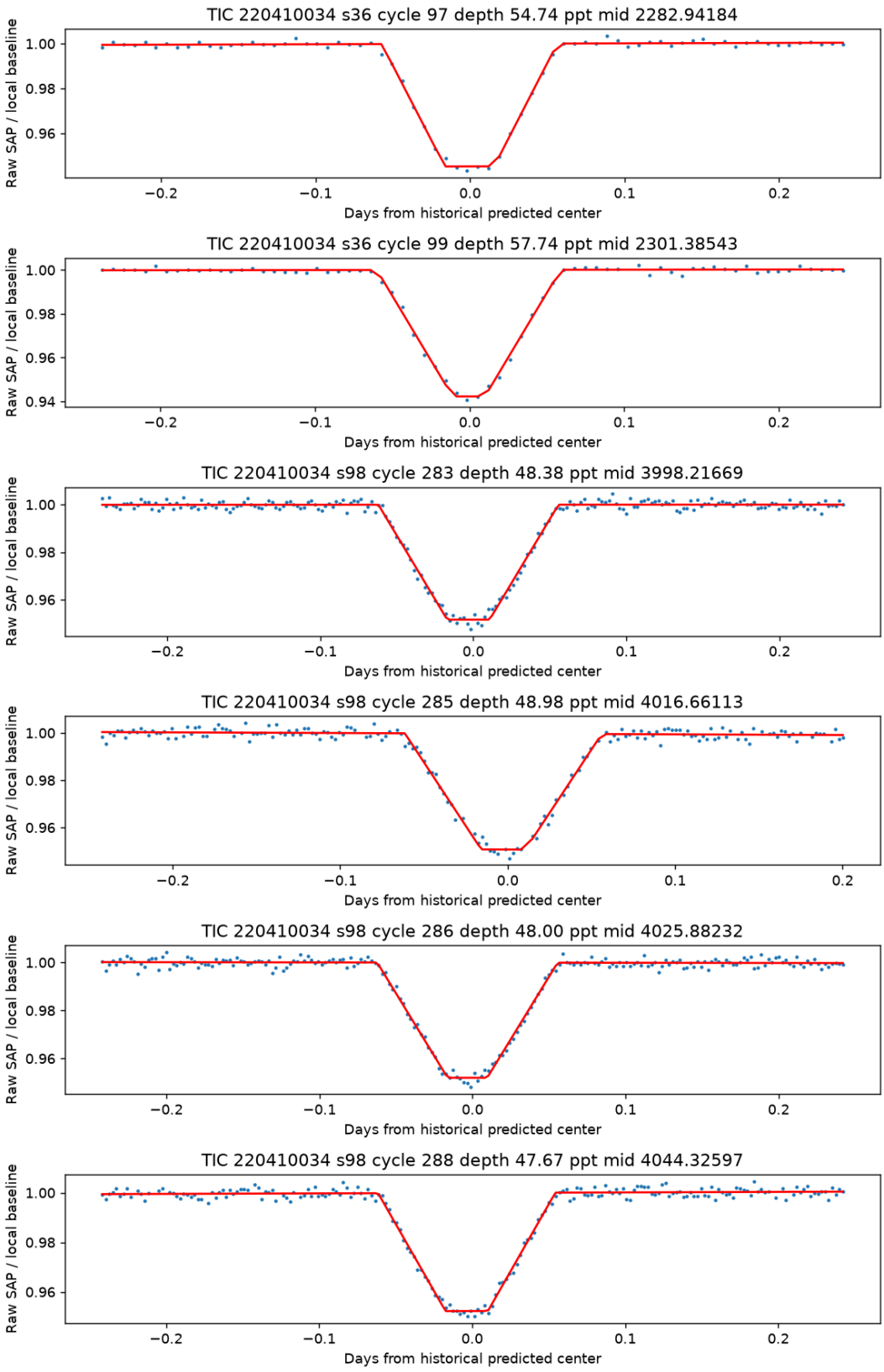
R479. TIC 220410034

physical_verdict: CONFIRMED

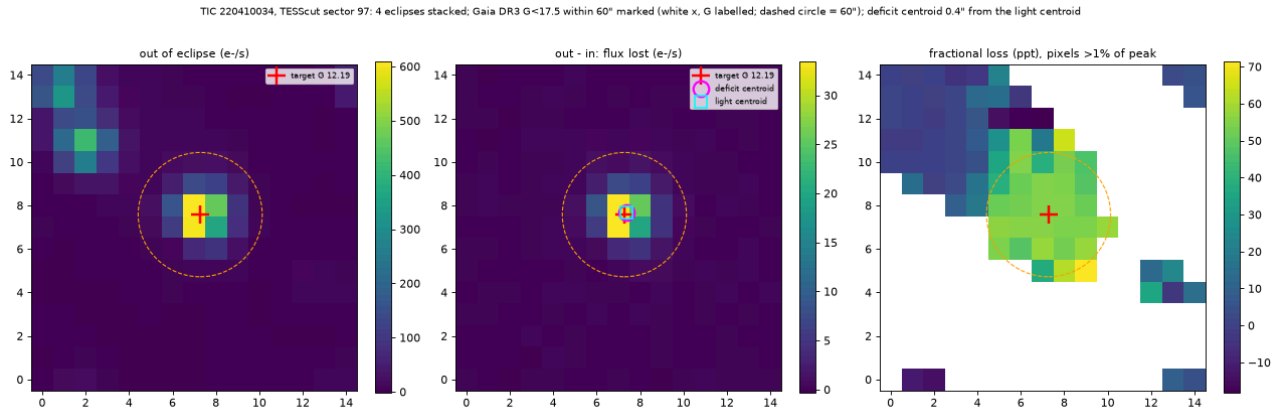
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R479; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R479; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 220410034: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.185472. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [36, 98] contains 6 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 9.2219086085 ± 0.0000134411 d and epoch BTJD 2282.9417514, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[98, 285, 286]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 97 TESScut difference image has deficit peak [7, 8] and target pixel [7, 8]; deficit centroid is 2.28 arcsec from the target coordinate. Image-model fractional residual 0.209 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: None under the approximate G-band brightness budget. The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 54.9/54.8/54.3 ppt from 4 stacked events, versus median selected raw QLP depth 48.7 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 4777341154688327424 has P = 9.2219102799 d and e = 0.08597; the closer edge-on conjunction differs from the measured epoch by -0.21615 d (-0.02344 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.40158 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 4.68 h against 2.82 h observed median; it is not a complete binary model.

Decision and limitation. S98 contains measured eclipses in both parity classes of the historical period, excluding a simple double-period alias. No brightness-capable Gaia neighbor appears within 60 arcsec. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
36	97	2282.9418398	0.003472	54.7	yes
36	99	2301.3854303	0.003472	57.7	yes
98	283	3998.2166875	0.001157	48.4	yes
98	285	4016.6611258	0.001157	49.0	yes
98	286	4025.8823212	0.001157	48.0	yes
98	288	4044.3259673	0.001157	47.7	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s97.json, historical/diff_s97.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R479. Public evidence files: diff_s97.json, diff_s97.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

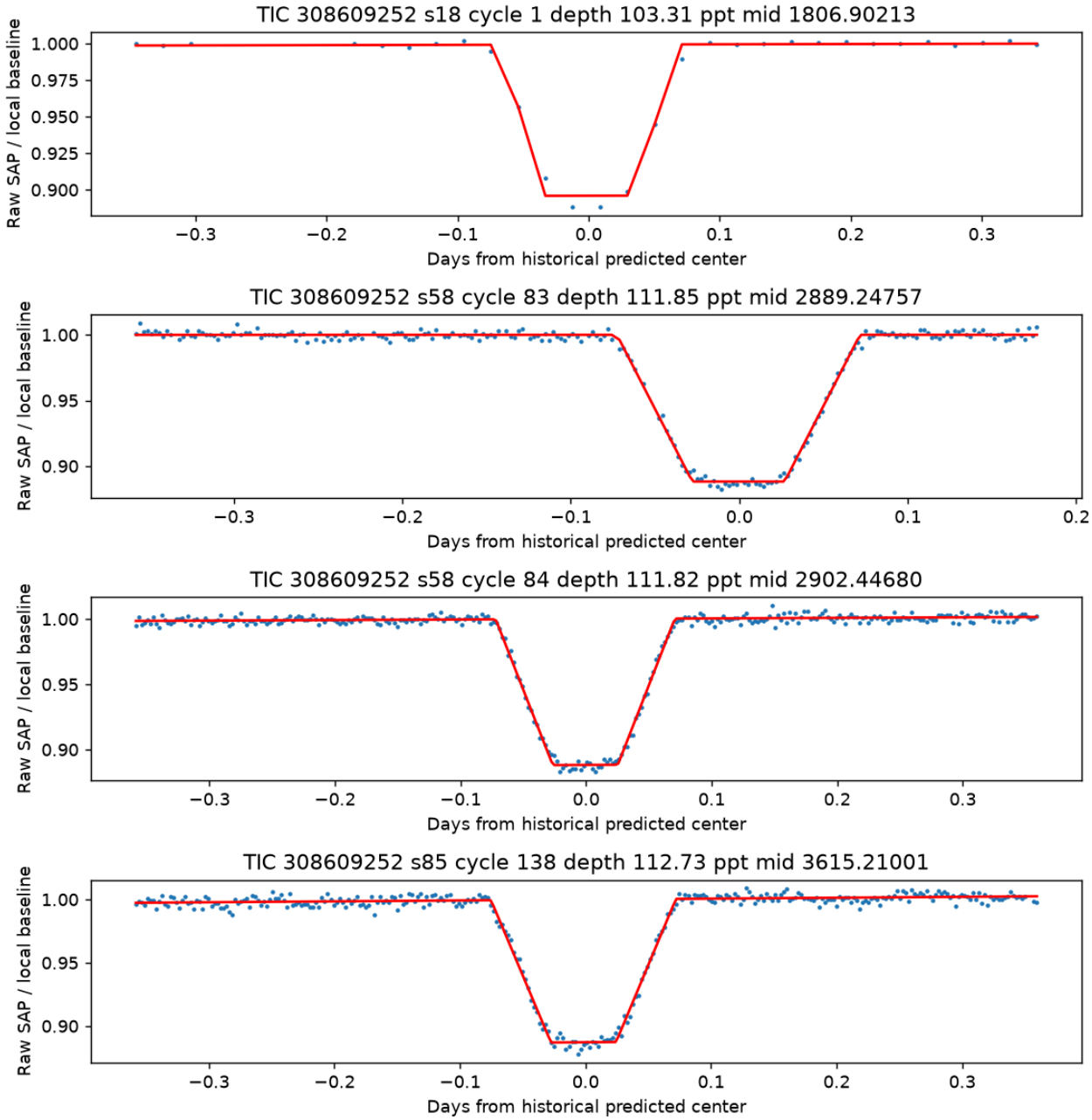
R480. TIC 308609252

physical_verdict: CONFIRMED

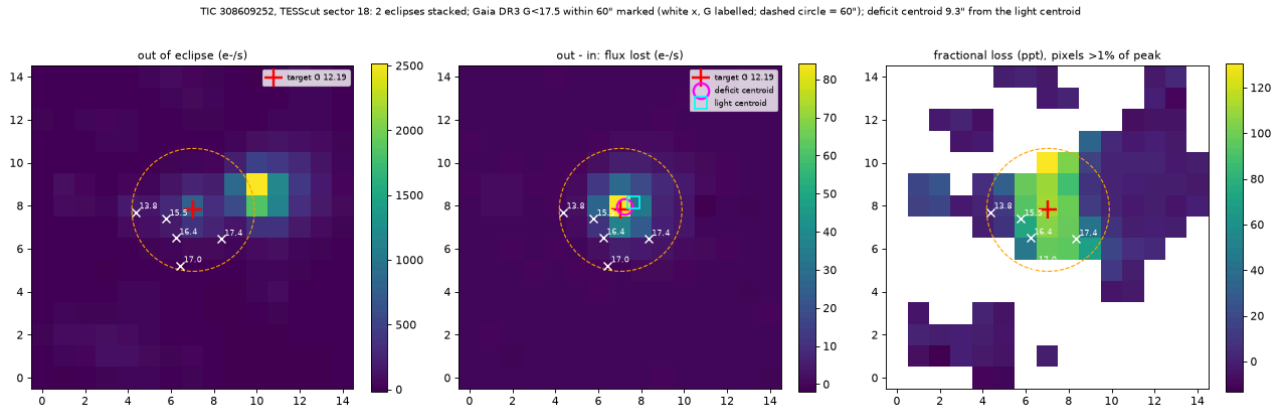
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R480; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R480; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 308609252: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 4 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 18, 58, 85. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 13.1993185849 ± 0.0000252249 d from actual midpoints, epoch BTJD 1806.9033855, cycle 1, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-18 TESScut difference centroid is 5.39 arcsec from the target coordinate; peak pixel [7, 8], target pixel [7, 8]. Fractional image-model residual 0.940 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 111.44/101.21/64.57 ppt across 2 stacked events, versus median raw QLP SAP fit depth 111.83 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 399670247610908160 has orbital P = 13.1964046632 d, e = 0.26890. Closest edge-on conjunction is offset by 0.20557 d (0.01558 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.25980 d. The rough central eclipse duration is 4.27 h versus observed median 3.44 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Four raw events include an adjacent sector-58 pair. The sector-18 difference core is centered near the target, while the only capable catalogued Gaia neighbor is 54.4 arcsec away. The 0.94 image-model residual limits centroid precision.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
18	1	1806.9021315	0.010417	103.31	used
58	83	2889.2475745	0.001157	111.85	used
58	84	2902.4468010	0.001157	111.82	used
85	138	3615.2100090	0.001157	112.73	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
3996731681886722256	54.37	13.801	0.590	60.5

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s18.json`, `historical/diff_s18.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R480. Public evidence files: diff_s18.json, diff_s18.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

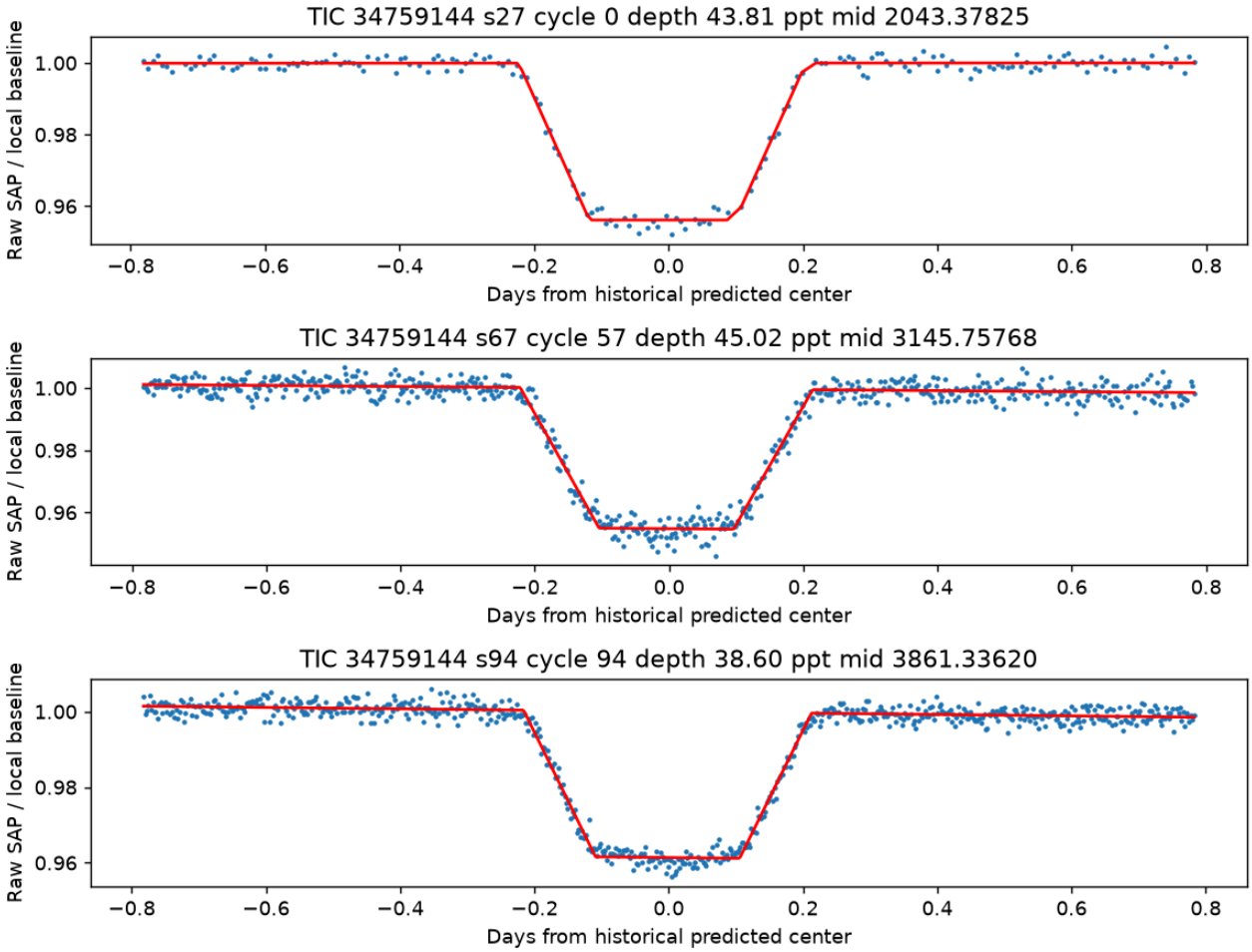
R481. TIC 34759144

physical_verdict: CONFIRMED

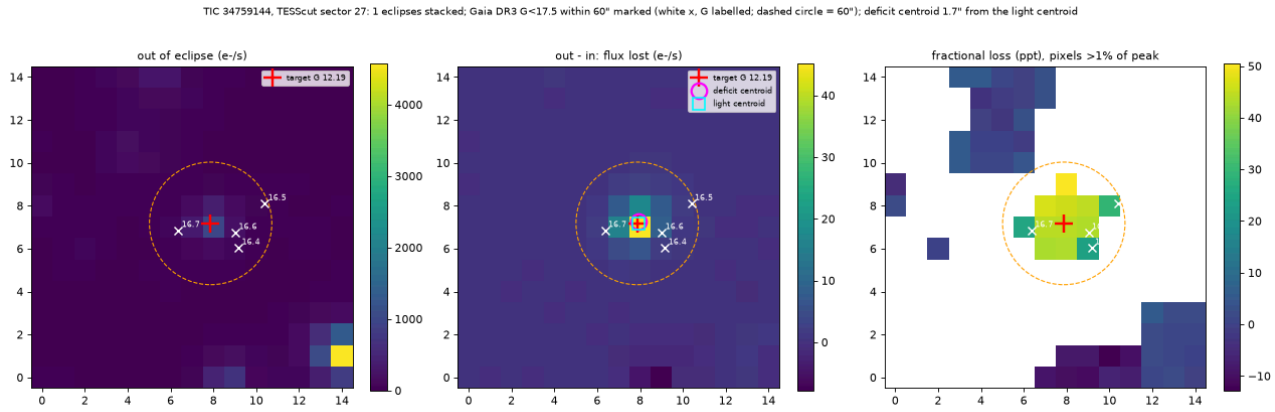
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R481; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R481; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 34759144: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 3 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 27, 67, 94. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 19.3399704780 ± 0.0000322940 d from actual midpoints, epoch BTJD 2043.3791201, cycle 0, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json` and the targeted `../34759144-alias-enumeration.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-27 TESScut difference centroid is 1.75 arcsec from the target coordinate; peak pixel [8, 7], target pixel [8, 7]. Fractional image-model residual 0.252 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 44.28/43.68/41.44 ppt across 1 stacked events, versus median raw QLP SAP fit depth 43.81 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 6696505582479751296 has orbital P = 19.3348568551 d, e = 0.21375. Closest edge-on conjunction is offset by 0.50738 d (0.02624 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.89805 d. The rough central eclipse duration is 14.22 h versus observed median 10.29 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Three clean raw eclipses occur one per selected sector. Enumerating neighboring integer cycle counts gives 57 then 37 cycles at 19.33997 d; the next-best pairs disagree by at least 0.0248 d between gaps. This tests nearby aliases, while subharmonics remain bounded by observed coverage.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
27	0	2043.3782485	0.003472	43.81	used
67	57	3145.7576833	0.001157	45.02	used
94	94	3861.3361958	0.001157	38.60	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s27.json`, `historical/diff_s27.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in` ../source-hashes.json`)`. `Source code and retained evidence are listed in` ../SHA256SUMS.json`.`

Stable review key: R481. Public evidence files: diff_s27.json, diff_s27.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

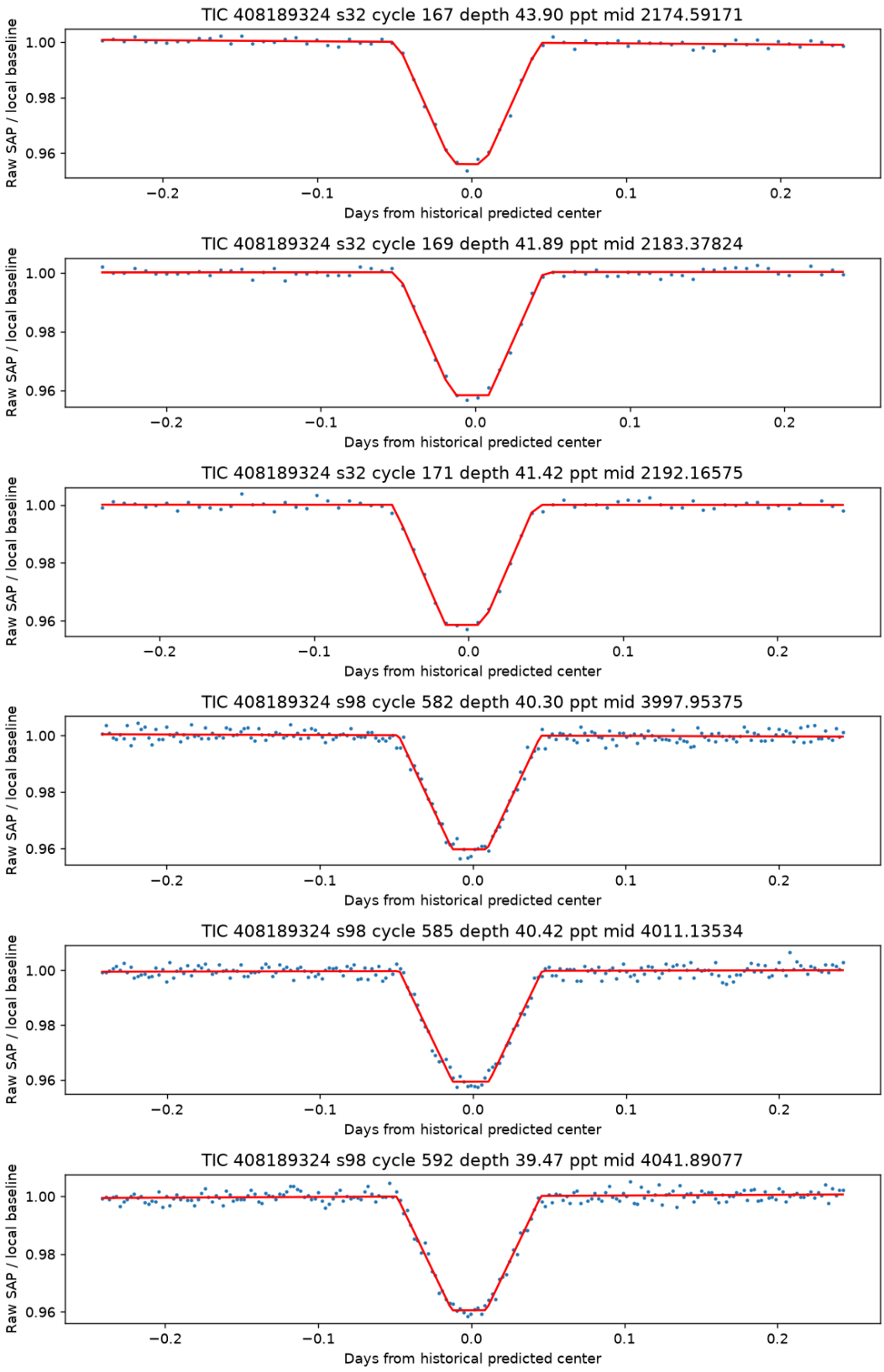
R482. TIC 408189324

physical_verdict: CONFIRMED

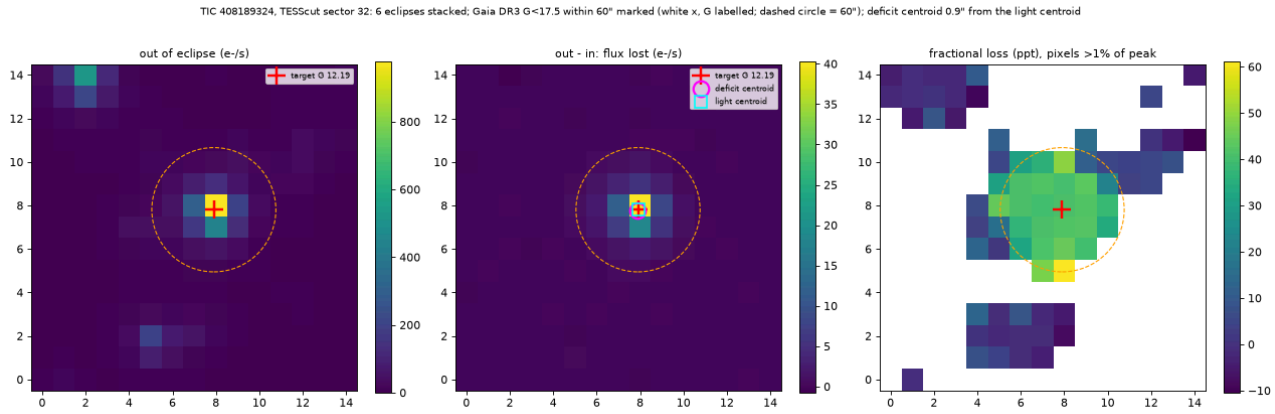
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R482; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R482; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 408189324: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 12 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 32, 98. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 4.3936455322 ± 0.0000035849 d from actual midpoints, epoch BTJD 2174.5912708, cycle 167, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-32 TESScut difference centroid is 0.48 arcsec from the target coordinate; peak pixel [8, 8], target pixel [8, 8]. Fractional image-model residual 0.077 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 41.21/41.38/40.39 ppt across 6 stacked events, versus median raw QLP SAP fit depth 40.93 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 2969339250844920832 has orbital P = 4.3934150548 d, e = 0.29683. Closest edge-on conjunction is offset by 0.03354 d (0.00763 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.05028 d. The rough central eclipse duration is 3.93 h versus observed median 2.22 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Six consecutive sector-32 events and six sector-98 events support the same 4.39365-day cycle. The deficit is sharply centered on the target and no capable Gaia neighbor is listed within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
32	167	2174.5917124	0.003472	43.90	used
32	168	2178.9850342	0.003472	43.13	used
32	169	2183.3782386	0.003472	41.89	used
32	170	2187.7723515	0.003472	42.88	used
32	171	2192.1657544	0.003472	41.42	used
32	172	2196.5592879	0.003472	43.67	used
98	582	3997.9537519	0.001157	40.30	used
98	583	4002.3475700	0.001157	39.77	used
98	585	4011.1353392	0.001157	40.42	used
98	586	4015.5288585	0.001157	39.92	used
98	589	4028.7098454	0.001157	40.44	used
98	592	4041.8907654	0.001157	39.47	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s32.json`, `historical/diff_s32.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json)`. `Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R482. Public evidence files: diff_s32.json, diff_s32.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

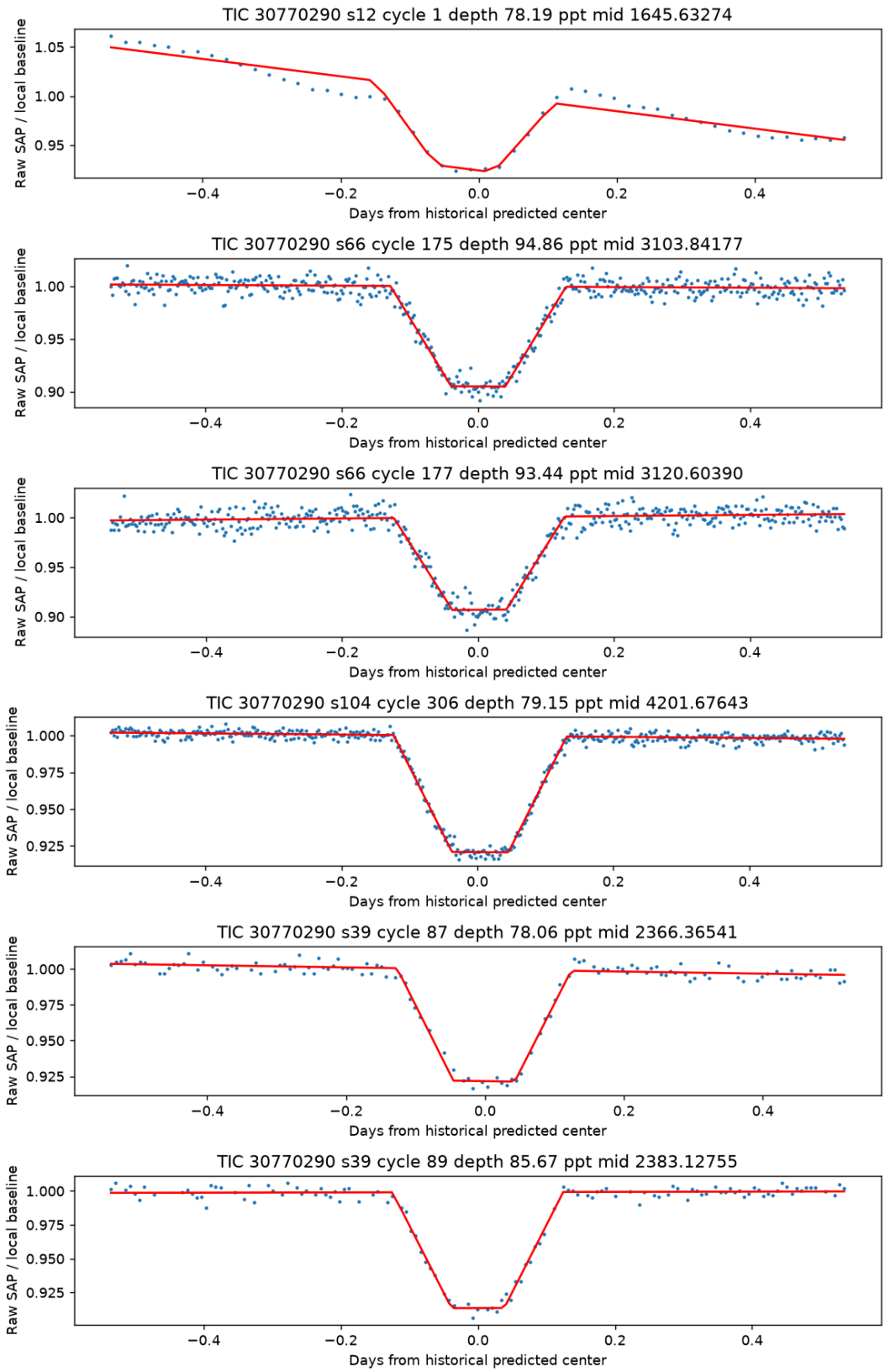
R483. TIC 30770290

physical_verdict: CONFIRMED

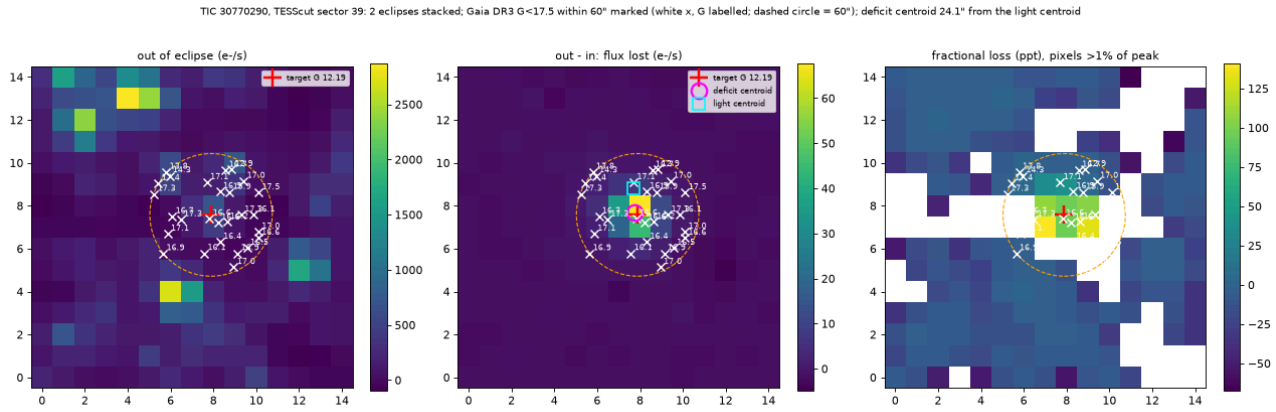
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R483; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R483; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 30770290: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 6 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 12, 39, 66, 104. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 8.3804132410 ± 0.0000094619 d from actual midpoints, epoch BTJD 1645.6504646, cycle 1, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-39 TESScut difference centroid is 3.33 arcsec from the target coordinate; peak pixel [8, 8], target pixel [8, 8]. Fractional image-model residual 0.988 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 96.05/80.93/45.08 ppt across 2 stacked events, versus median raw QLP SAP fit depth 82.41 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 5955349082236051072 has orbital P = 8.3812349333 d, e = 0.09409. Closest edge-on conjunction is offset by -0.18507 d (-0.02208 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.15637 d. The rough central eclipse duration is 7.78 h versus observed median 6.07 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Sector-39 and sector-66 measured pairs are separated by two historical cycles, but independent inspection finds a real ~69-ppt usable-core deficit at intervening sector-66 cycle 176. Its partial coverage prevents a reliable midpoint fit; it supports the 8.3804-day recurrence over a simple 16.76-day parity alias. Sector-39 cycle 88 is a gap, not a null. The high image-model residual limits localization precision.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	1	1645.6327354	0.021612	78.19	used
39	87	2366.3654077	0.003472	78.06	used
39	89	2383.1275530	0.003472	85.67	used
66	175	3103.8417724	0.001157	94.86	used
66	177	3120.6038978	0.001157	93.44	used
104	306	4201.6764331	0.001157	79.15	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5955349494552910848	48.59	12.925	0.251	48.3
5955349872514790784	54.94	14.307	0.678	52.7
5955349872510033024	59.62	13.836	0.469	57.4

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s39.json`, `historical/diff_s39.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R483. Public evidence files: diff_s39.json, diff_s39.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

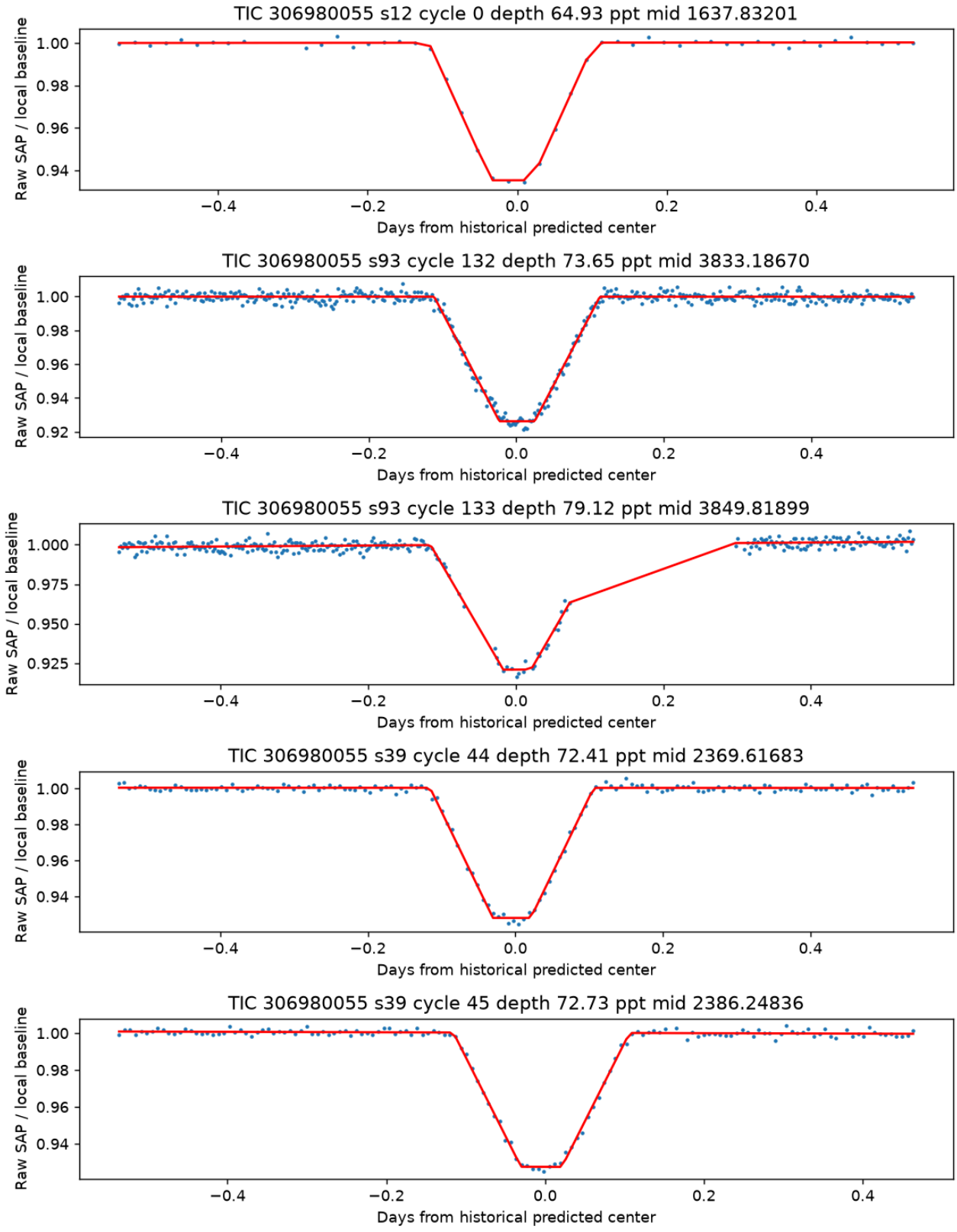
R484. TIC 306980055

physical_verdict: CONFIRMED

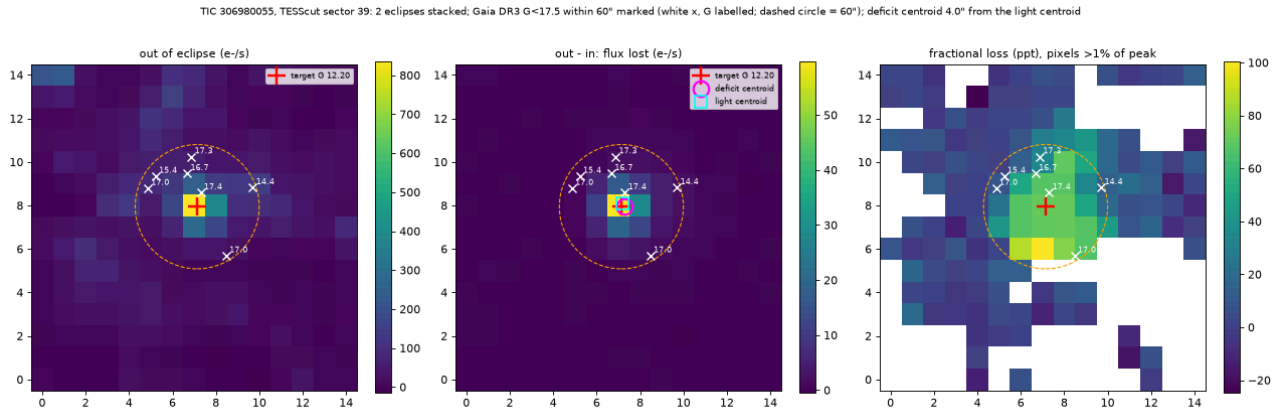
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R484; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R484; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 306980055: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 5 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 12, 39, 93. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 16.6314800623 ± 0.0000277694 d from actual midpoints, epoch BTJD 1637.8317347, cycle 0, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-39 TESScut difference centroid is 3.01 arcsec from the target coordinate; peak pixel [7, 8], target pixel [7, 8]. Fractional image-model residual 0.328 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 71.49/69.24/62.01 ppt across 2 stacked events, versus median raw QLP SAP fit depth 72.73 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 5911278044857572480 has orbital P = 16.6305032216 d, e = 0.36931. Closest edge-on conjunction is offset by -0.13935 d (-0.00838 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.29122 d. The rough central eclipse duration is 7.65 h versus observed median 5.32 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Measured sector-39 and sector-93 adjacent pairs span the long baseline. The capable 55.8-arcsec Gaia neighbor is distinct from the target-centered difference core.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	0	1637.8320133	0.010417	64.93	used
39	44	2369.6168339	0.003472	72.41	used
39	45	2386.2483563	0.003472	72.73	used
93	132	3833.1866950	0.001157	73.65	used
93	133	3849.8189880	0.001157	79.12	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5909777043986816768	55.81	14.437	0.578	54.0

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s39.json`, `historical/diff_s39.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json)`. `Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R484. Public evidence files: diff_s39.json, diff_s39.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

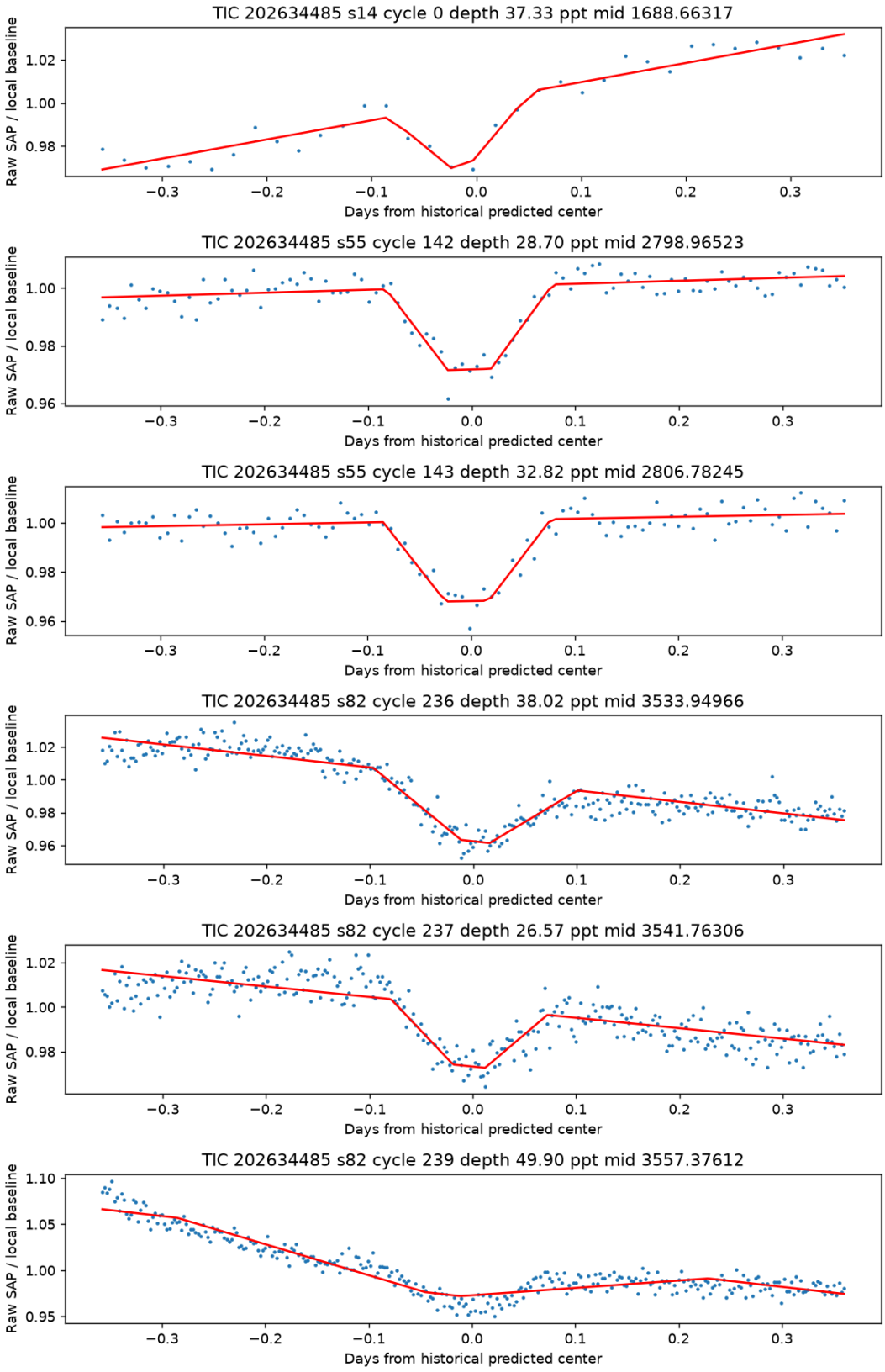
R485. TIC 202634485

physical_verdict: PROBABLE

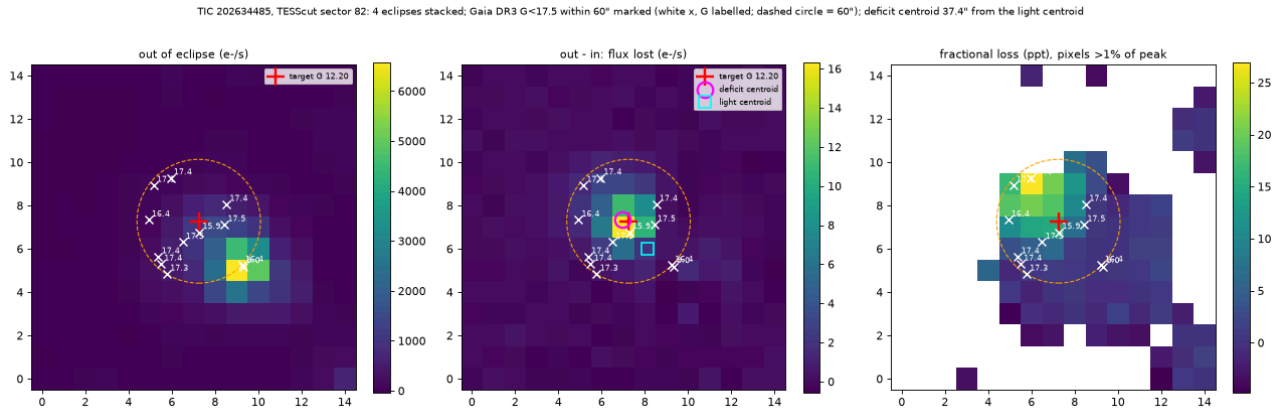
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source pixel-response fitting of the 10.1-arcsec pair.



R485; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R485; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 202634485: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 8 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 14, 55, 82. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 7.8189267254 ± 0.0000356216 d from actual midpoints, epoch BTJD 1688.6785071, cycle 0, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** unresolved. The historical sector-82 TESScut difference centroid is 5.99 arcsec from the target coordinate; peak pixel [7, 7], target pixel [7, 7]. Fractional image-model residual 0.968 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 14.95/8.35/3.26 ppt across 4 stacked events, versus median raw QLP SAP fit depth 33.47 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 2055792762025890176 has orbital P = 7.8183725267 d, e = 0.24161. Closest edge-on conjunction is offset by 0.06916 d (0.00885 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.27464 d. The rough central eclipse duration is 6.77 h versus observed median 3.87 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Sector-55 has three consecutive clean raw eclipses. The sector-82 raw baseline is structured; its same-photon custom-aperture signal is much shallower than QLP SAP, with crowding, aperture and baseline sensitivity likely contributing. A Gaia neighbor 10.1 arcsec away could in principle supply the event if nearly fully eclipsed, and the image-model residual is 0.968. Target attribution remains unresolved.

Next check: Resolved eclipse photometry or multi-source pixel-response fitting of the 10.1-arcsec pair.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
14	0	1688.6631740	0.033133	37.33	used
55	142	2798.9652268	0.003472	28.70	used
55	143	2806.7824513	0.003472	32.82	used
55	144	2814.6078397	0.003472	28.32	used
82	236	3533.9496625	0.006967	38.02	used
82	237	3541.7630557	0.005726	26.57	used
82	238	3549.5839019	0.003034	34.12	used
82	239	3557.3761215	0.012302	49.90	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
2055792766332050688	10.09	15.945	0.938	13.9
2055792727666151552	58.25	9.031	0.030	67.2

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s82.json`, `historical/diff_s82.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R485. Public evidence files: diff_s82.json, diff_s82.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

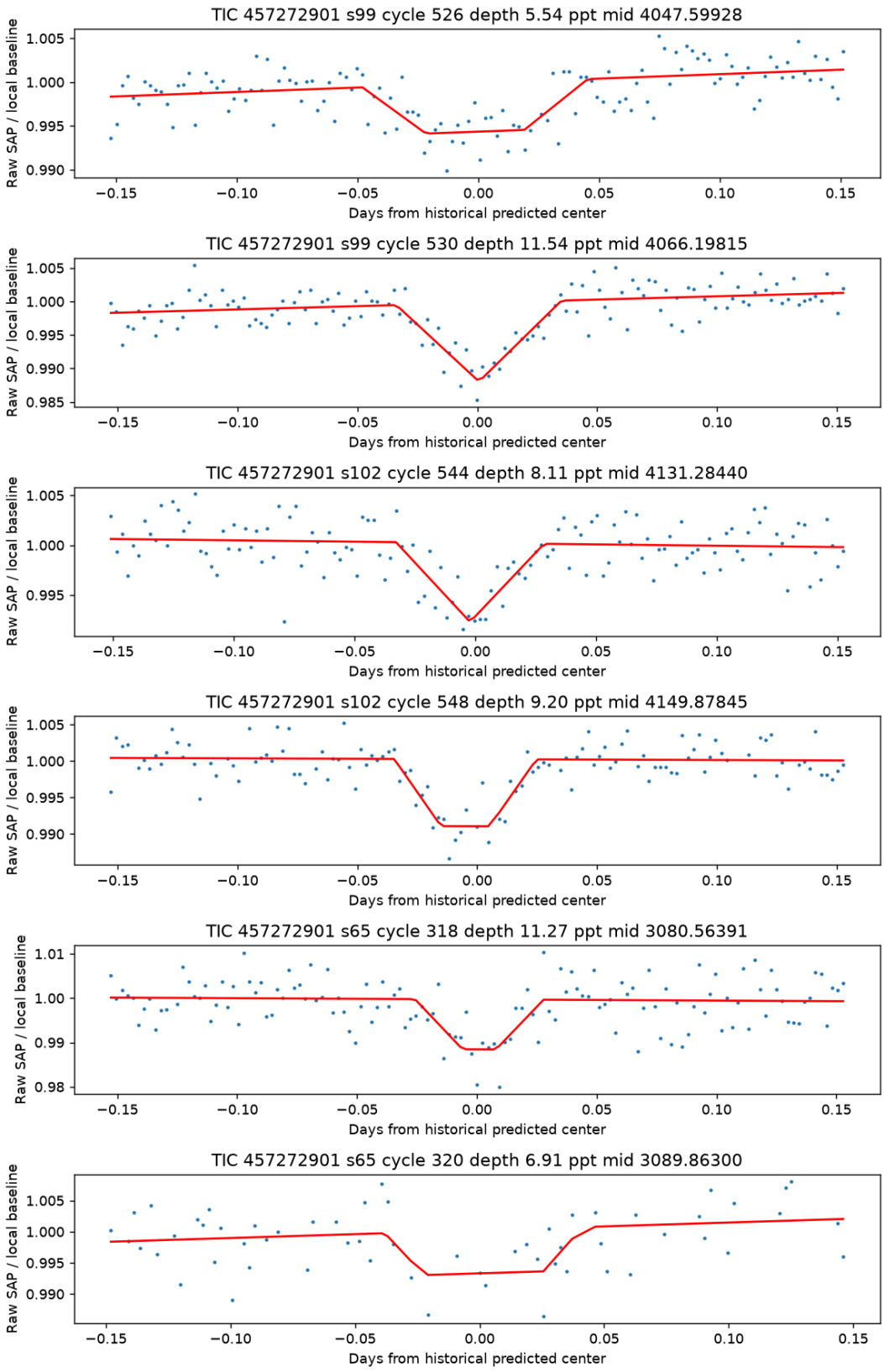
R486. TIC 457272901

physical_verdict: CONFIRMED

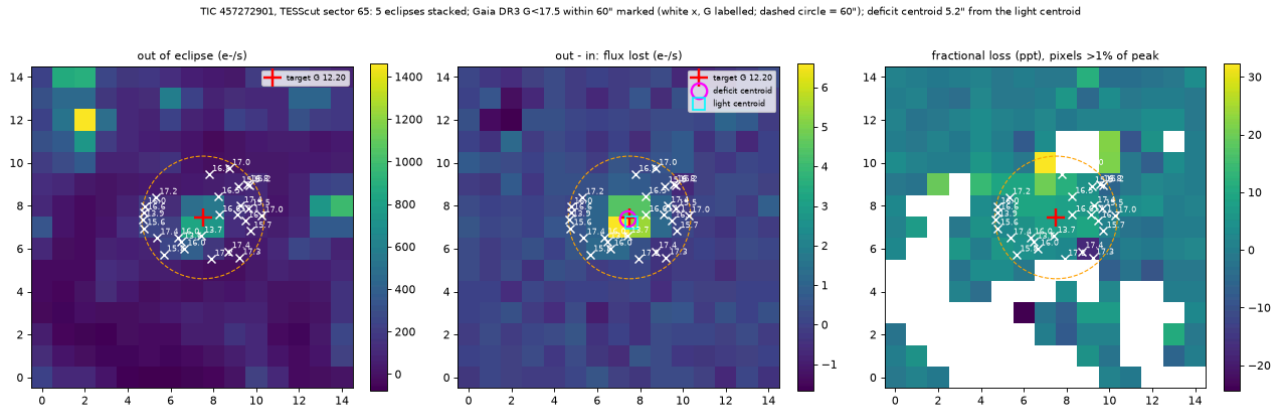
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R486; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R486; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 457272901: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 11 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 65, 99, 102. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 4.6492170041 ± 0.0000077651 d from actual midpoints, epoch BTJD 3071.2640710, cycle 316, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-65 TESScut difference centroid is 2.05 arcsec from the target coordinate; peak pixel [7, 7], target pixel [8, 7]. Fractional image-model residual 0.474 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 7.12/6.92/6.01 ppt across 5 stacked events, versus median raw QLP SAP fit depth 8.09 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 5869945139392446336 has orbital P = 4.6494664348 d, e = 0.06671. Closest edge-on conjunction is offset by -0.13084 d (-0.02814 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.36286 d. The rough central eclipse duration is 3.53 h versus observed median 1.47 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: The added sector-65 raw events anchor the 4.649217-day timing baseline; later sector-99/102 troughs repeat. The difference centroid is 2.1 arcsec from the target and the closest capable neighbor is about 17 arcsec from that centroid. Twenty-seven capable Gaia sources lie within 60 arcsec; the crowded image and uncalibrated residual limit precision, but the spatial deficit favors the target.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
65	316	3071.2627207	0.001744	9.13	used
65	318	3080.5639080	0.001896	11.27	used
65	319	3085.2108687	0.001497	7.92	used
65	320	3089.8630025	0.003589	6.91	used
99	526	4047.5992838	0.002607	5.54	used
99	527	4052.2519805	0.001235	7.51	used
99	530	4066.1981510	0.001357	11.54	used
99	531	4070.8470307	0.001829	8.09	used
102	544	4131.2844012	0.001576	8.11	used
102	545	4135.9353243	0.001157	7.21	used
102	548	4149.8784550	0.001157	9.20	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5869942184454493952	15.81	16.636	0.461	18.0
5869945139392446080	18.08	13.674	0.037	17.1
5869945139392446592	25.28	16.453	0.390	27.5
5869939263875939200	30.92	15.973	0.254	29.4
5869939263875937536	31.84	13.624	0.036	30.3
5869942184454489344	33.36	17.052	0.672	35.8
5869939263875934336	35.40	16.042	0.270	34.0
5869942184454491520	38.07	17.429	0.947	40.8
5869942184454483712	40.77	16.711	0.493	43.2
5869936308938425216	41.37	17.197	0.767	40.9
5869942184389479424	41.48	17.384	0.910	42.0
5869945139392447744	42.24	16.755	0.513	43.7
5869942184454487552	44.96	15.517	0.169	47.7

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5869942218814236544	45.73	15.875	0.232	48.4
5869942184454478976	47.67	15.702	0.199	49.8
5869939332595420544	48.37	17.389	0.914	47.2
5869945203761021568	49.50	17.171	0.749	48.9
5869939332595414400	52.30	15.832	0.224	51.1
5869942184468745088	52.87	17.290	0.834	53.7
5869942218814234240	53.06	16.785	0.527	56.0
5869942218814243584	53.79	16.964	0.620	56.1
5869942218814234112	54.64	15.222	0.131	57.6
5869942115735005312	56.67	17.032	0.660	59.5
586993932822889 8560	56.94	16.642	0.463	56.2
5869939332595433344	57.44	16.012	0.262	56.7
5869939332595426432	57.93	15.601	0.182	57.0
5869939332595430144	58.16	13.894	0.044	57.3

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s65.json`, `historical/diff_s65.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R486. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

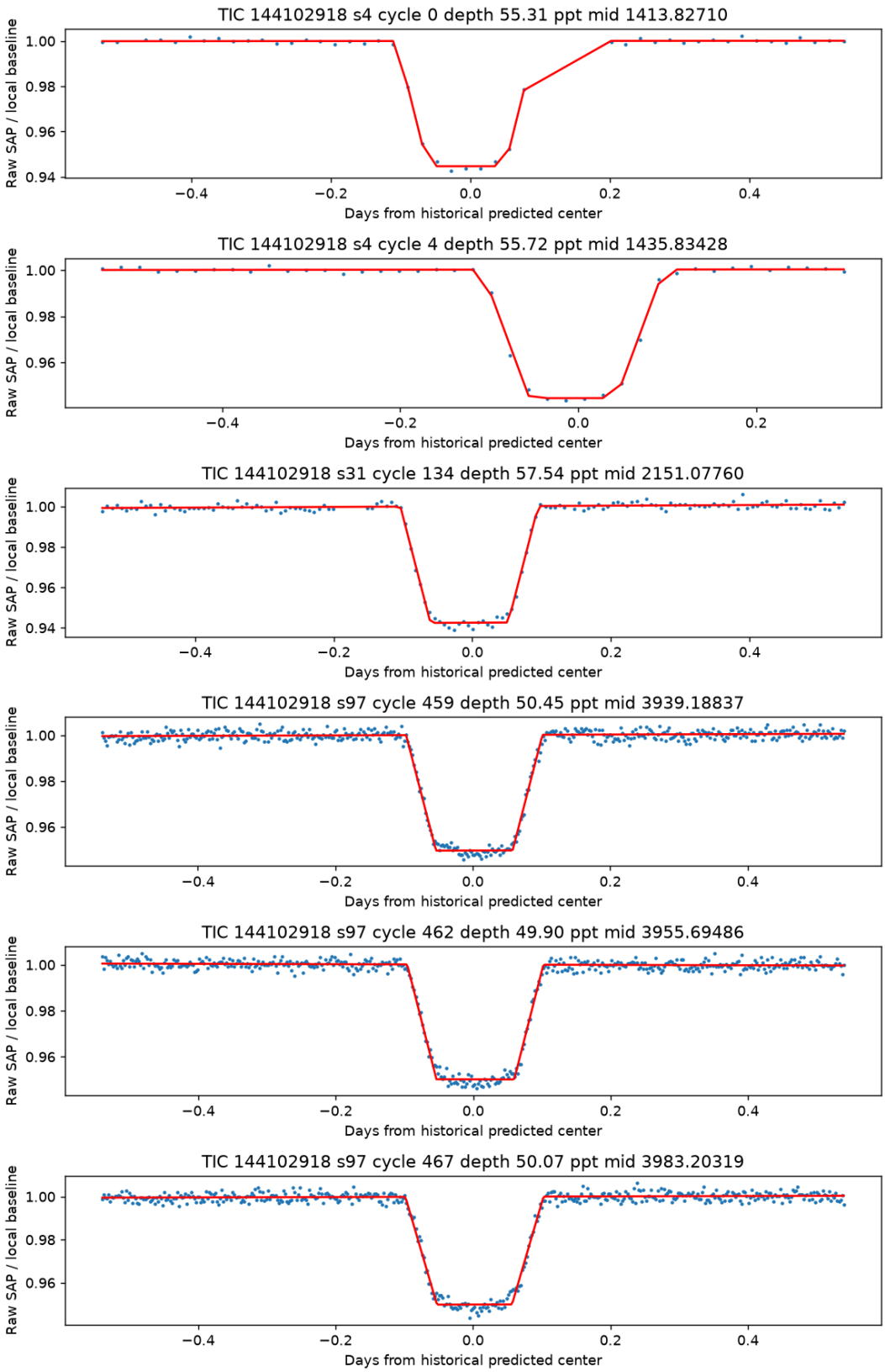
R487. TIC 144102918

physical_verdict: CONFIRMED

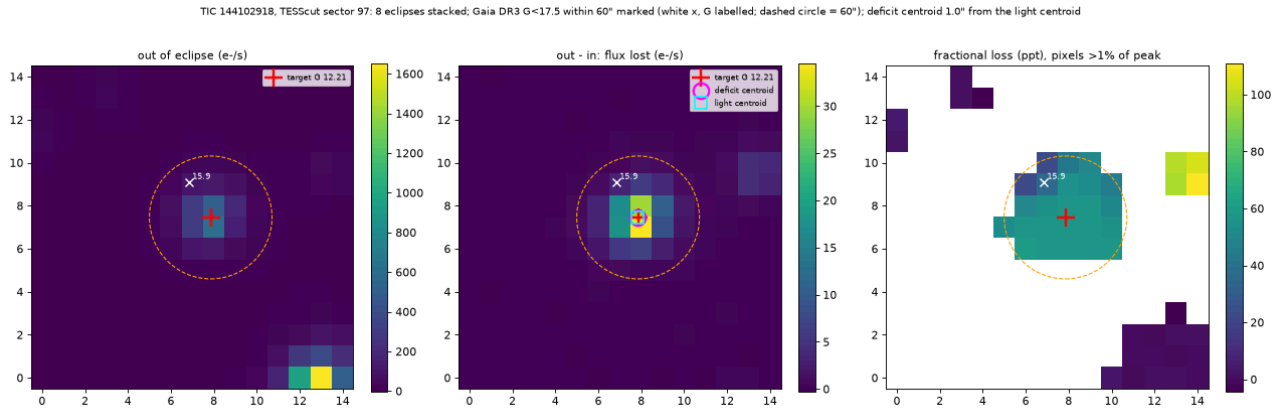
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R487; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R487; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 144102918: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 13 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 4, 31, 97. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 5.5018791815 ± 0.0000050961 d from actual midpoints, epoch BTJD 1413.8257574, cycle 0, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-97 TESScut difference centroid is 0.50 arcsec from the target coordinate; peak pixel [8, 7], target pixel [8, 7]. Fractional image-model residual 0.058 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 57.89/57.27/55.49 ppt across 8 stacked events, versus median raw QLP SAP fit depth 54.66 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 4841921897857941248 has orbital P = 5.5016102193 d, e = 0.07812. Closest edge-on conjunction is offset by 0.05387 d (0.00979 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.10114 d. The rough central eclipse duration is 4.99 h versus observed median 4.77 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Thirteen measured eclipses span sectors 4, 31 and 97, including adjacent cycles. Difference imaging is sharply target-centered with no capable Gaia neighbor in the 60-arcsec screen.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
4	0	1413.8270960	0.010417	55.31	used
4	3	1430.3331928	0.010417	54.66	used
4	4	1435.8342838	0.010417	55.72	used
31	133	2145.5757425	0.003472	57.84	used
31	134	2151.0776028	0.003472	57.54	used
31	136	2162.0809820	0.003472	58.09	used
31	137	2167.5827793	0.003472	58.95	used
97	459	3939.1883705	0.001157	50.45	used
97	460	3944.6897340	0.001157	50.13	used
97	462	3955.6948555	0.001157	49.90	used
97	464	3966.6976092	0.001157	49.73	used
97	465	3972.1992965	0.001157	50.07	used
97	467	3983.2031899	0.001157	50.07	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s97.json`, `historical/diff_s97.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R487. Public evidence files: diff_s97.json, diff_s97.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

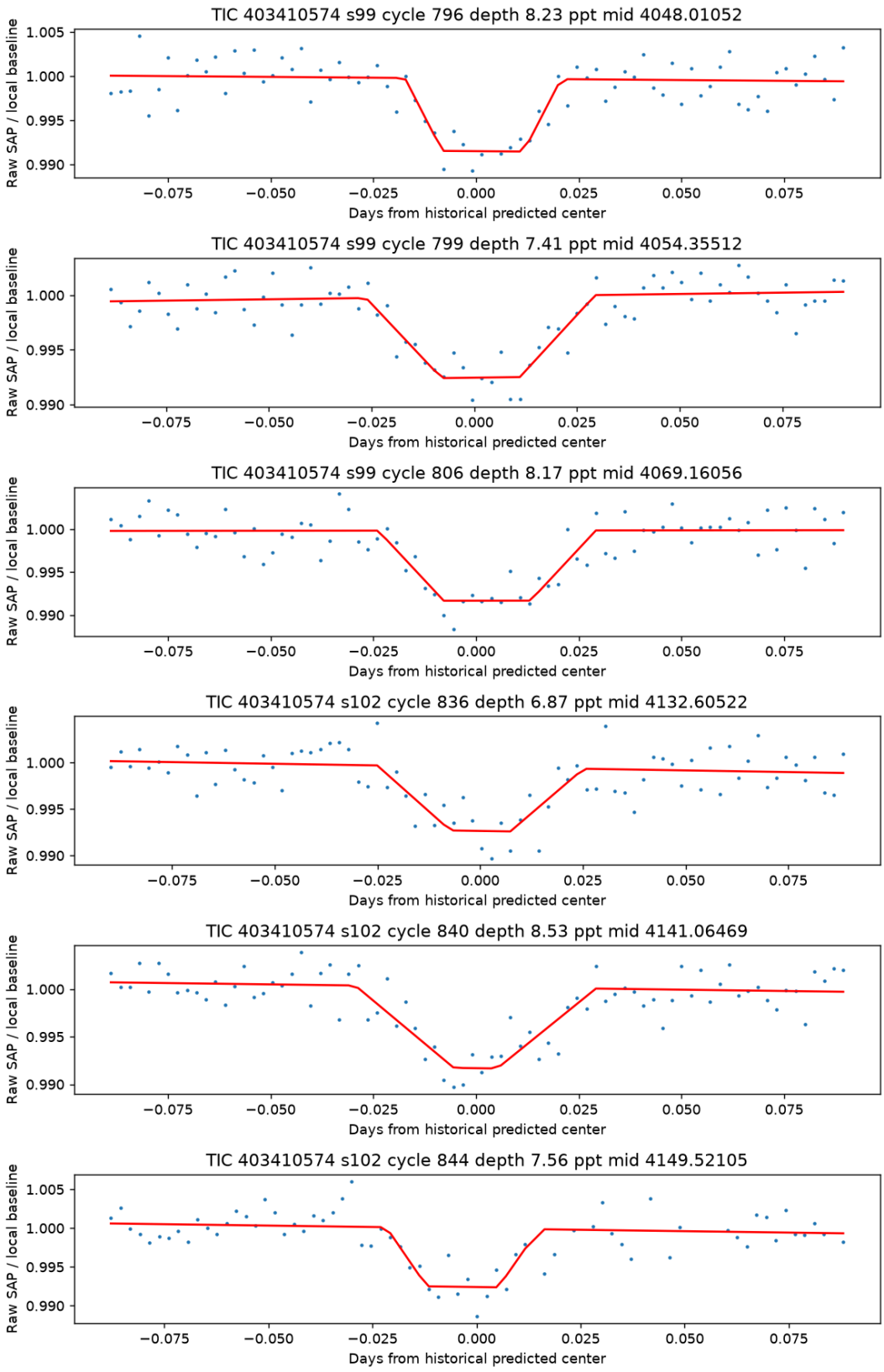
R488. TIC 403410574

physical_verdict: PROBABLE

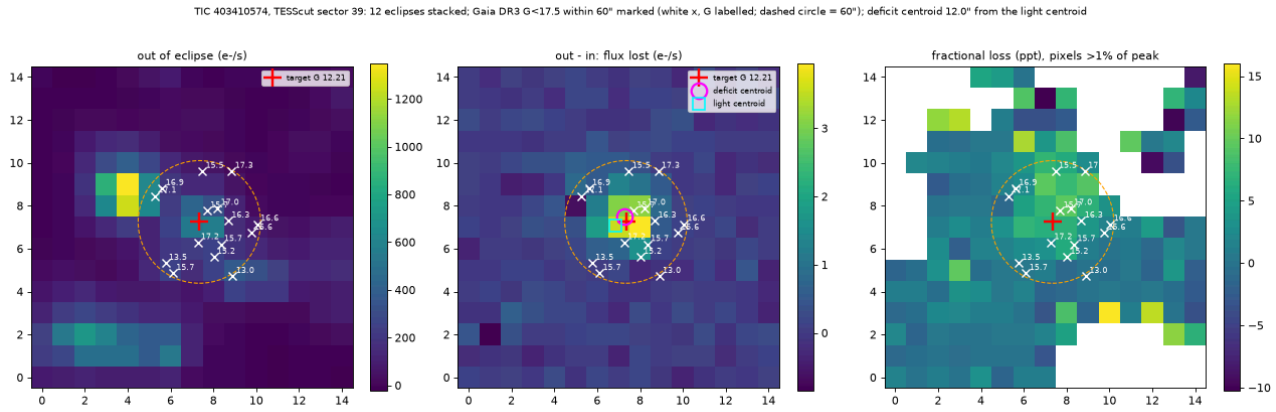
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source pixel-response fitting of the 12.2-arcsec pair.



R488; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R488; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 403410574: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 16 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 99, 102. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 2.1148563262 ± 0.0000186810 d from actual midpoints, epoch BTJD 4048.0101157, cycle 796, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** unresolved. The historical sector-39 TESScut difference centroid is 5.38 arcsec from the target coordinate; peak pixel [8, 7], target pixel [7, 7]. Fractional image-model residual 0.896 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 6.46/6.39/3.75 ppt across 12 stacked events, versus median raw QLP SAP fit depth 7.55 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 5792684548884156800 has orbital P = 2.1148150688 d, e = 0.11270. Closest edge-on conjunction is offset by 0.12513 d (0.05917 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.15197 d. The rough central eclipse duration is 3.34 h versus observed median 1.20 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Sixteen measured shallow but discrete raw eclipses in sectors 99 and 102 establish a repeated 2.11486-day signal. The early sector-39 cadence is too sparse for the local trapezoid timing criterion, although its stacked custom aperture shows the event. A capable Gaia neighbor 12.2 arcsec from the target lies only 10.9 arcsec from the deficit centroid; the residual is 0.896. TESS pixels do not settle attribution.

Next check: Resolved eclipse photometry or multi-source pixel-response fitting of the 12.2-arcsec pair.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
99	796	4048.0105152	0.001157	8.23	used
99	797	4050.1226481	0.001157	9.29	used
99	798	4052.2403352	0.001157	9.02	used
99	799	4054.3551161	0.001157	7.41	used
99	804	4064.9275859	0.001224	7.78	used
99	805	4067.0446358	0.001614	7.45	used
99	806	4069.1605614	0.001162	8.17	used
102	834	4128.3760132	0.001412	6.84	used
102	835	4130.4900083	0.001157	7.18	used
102	836	4132.6052166	0.001538	6.87	used
102	837	4134.7170281	0.001157	7.54	used
102	838	4136.8335795	0.002748	6.88	used
102	840	4141.0646915	0.001157	8.53	used
102	841	4143.1801215	0.001203	7.13	used
102	843	4147.4086149	0.001170	7.82	used
102	844	4149.5210460	0.001157	7.56	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5792684347027309312	12.23	15.744	0.202	10.9
5792684518826000768	18.96	17.228	0.771	25.2
5792684347027309184	21.08	16.978	0.614	21.3
5792684347027308416	27.46	16.298	0.332	30.4
5792684347027308032	29.35	15.674	0.190	35.4
5792684312667569408	35.36	15.218	0.128	42.6
5794185764514224512	46.47	15.531	0.168	44.8
5792684548884947072	47.39	16.886	0.565	44.2
5792684518826001408	49.08	13.476	0.032	55.2
5792684553185740928	49.17	17.116	0.697	46.4
5792684347027307008	50.48	15.561	0.172	54.8

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5792684518826000896	52.86	15.681	0.191	60.4
5794185554054569856	55.03	17.267	0.800	55.8
5792684347027445120	55.22	16.618	0.443	59.0
5792684617604424064	55.68	17.504	0.992	53.2
5792684312667567744	59.84	13.035	0.024	67.8

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s39.json`, `historical/diff_s39.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R488. Public evidence files: diff_s39.json, diff_s39.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

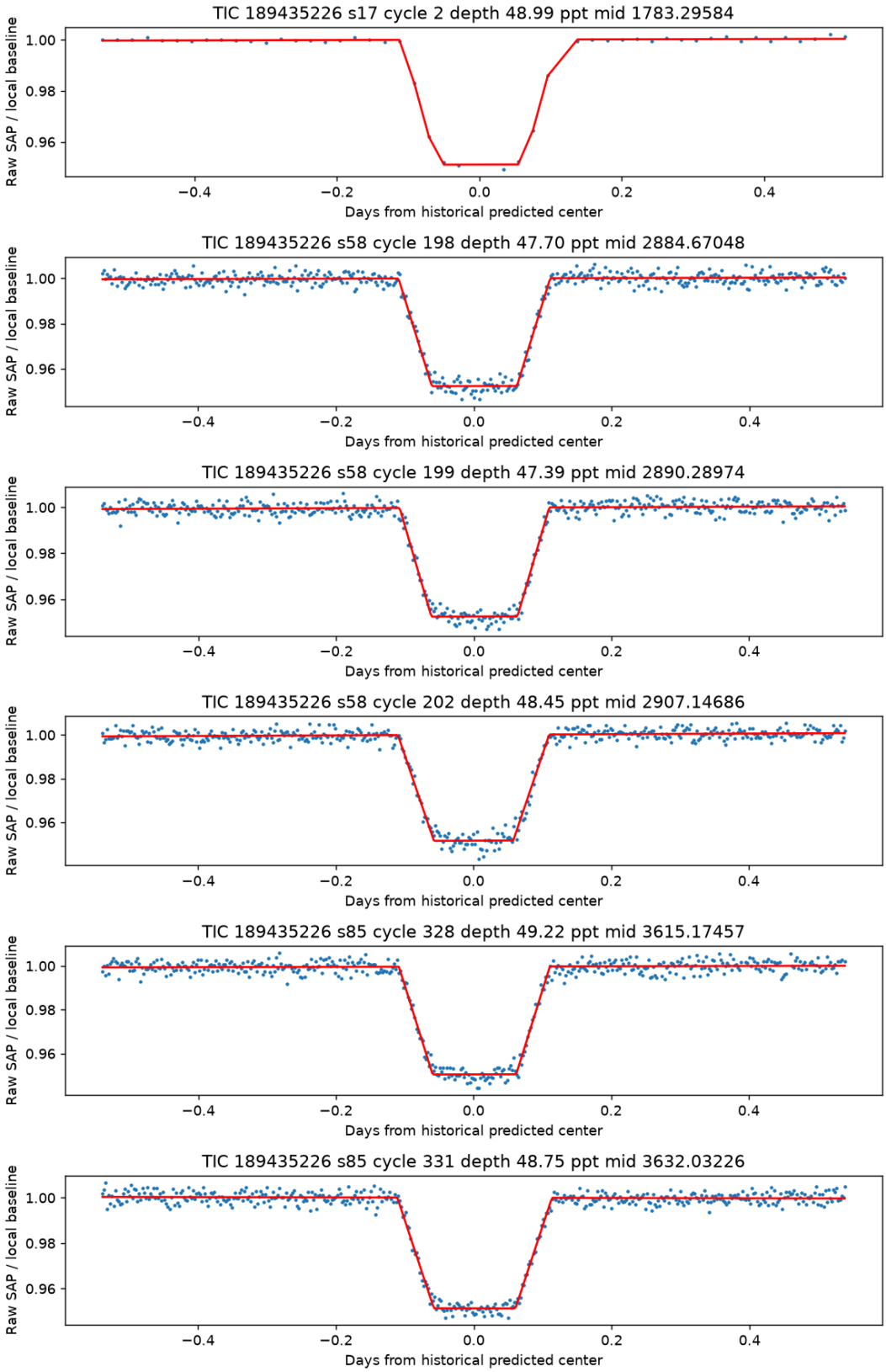
R489. TIC 189435226

physical_verdict: CONFIRMED

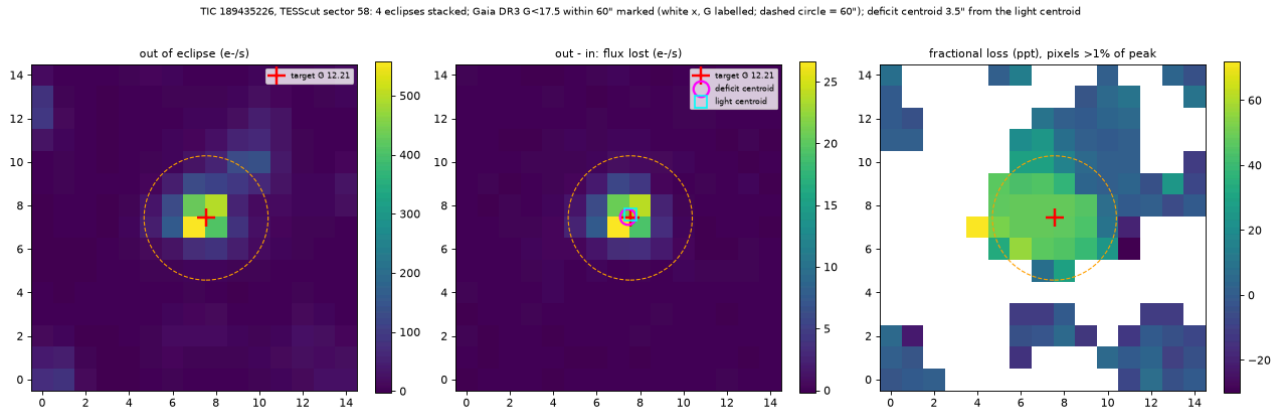
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R489; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R489; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 189435226: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 8 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 17, 58, 85. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 5.6192618708 ± 0.0000067407 d from actual midpoints, epoch BTJD 1783.2951463, cycle 2, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-58 TESScut difference centroid is 3.40 arcsec from the target coordinate; peak pixel [7, 7], target pixel [8, 7]. Fractional image-model residual 0.231 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 47.17/47.24/43.82 ppt across 4 stacked events, versus median raw QLP SAP fit depth 48.53 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 323765286292704768 has orbital P = 5.6182753682 d, e = 0.05704. Closest edge-on conjunction is offset by 0.10708 d (0.01906 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.13963 d. The rough central eclipse duration is 6.01 h versus observed median 5.27 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Eight raw eclipses span sectors 17, 58 and 85, with adjacent cycles in both later sectors. The difference image favors the target and no capable Gaia neighbor is listed within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
17	2	1783.2958411	0.010417	48.99	used
58	198	2884.6704822	0.001157	47.70	used
58	199	2890.2897400	0.001157	47.39	used
58	201	2901.5288785	0.001157	47.77	used
58	202	2907.1468576	0.001157	48.45	used
85	328	3615.1745664	0.001157	49.22	used
85	329	3620.7937892	0.001157	48.62	used
85	331	3632.0322602	0.001157	48.75	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s58.json`, `historical/diff_s58.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R489. Public evidence files: diff_s58.json, diff_s58.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

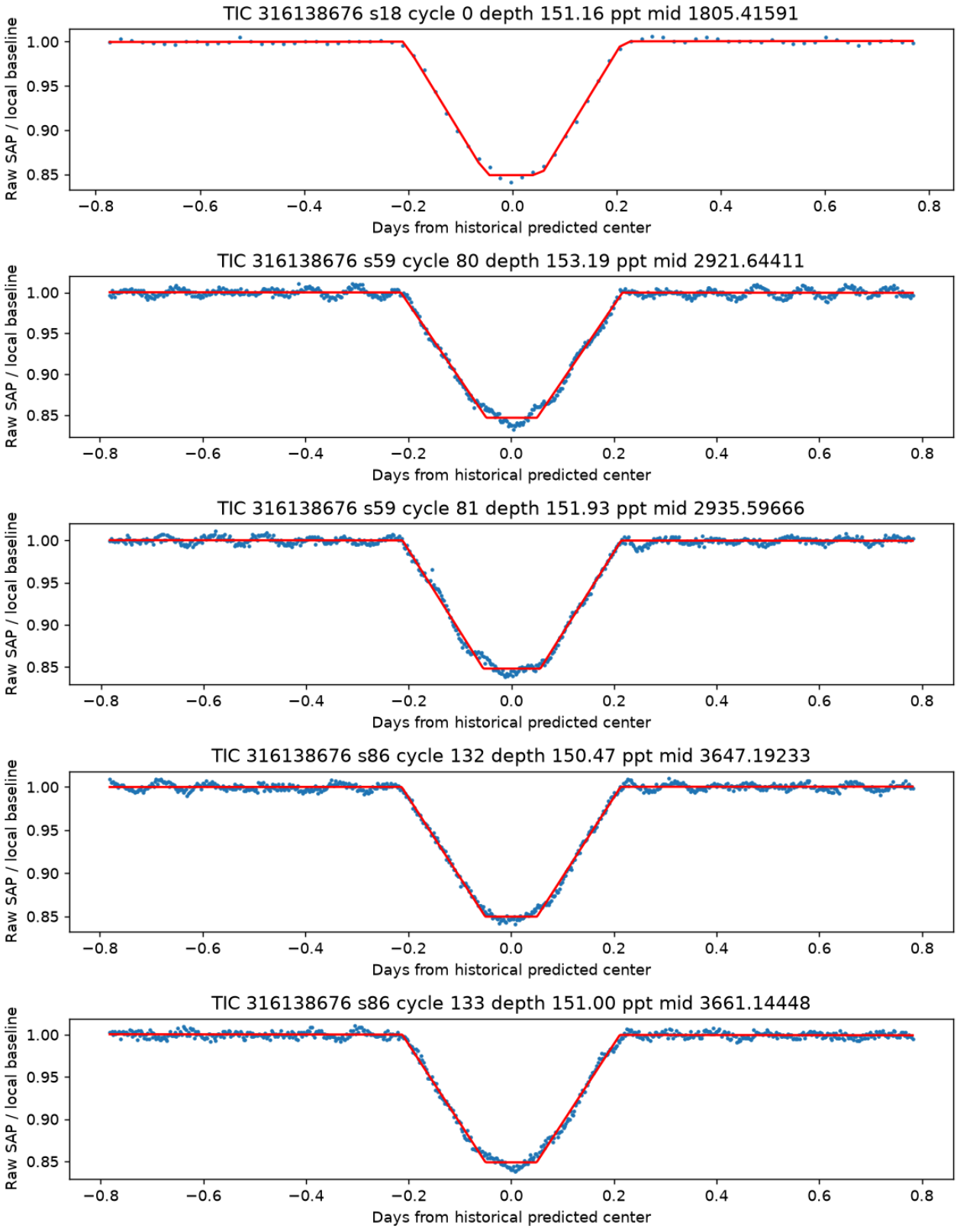
R490. TIC 316138676

physical_verdict: CONFIRMED

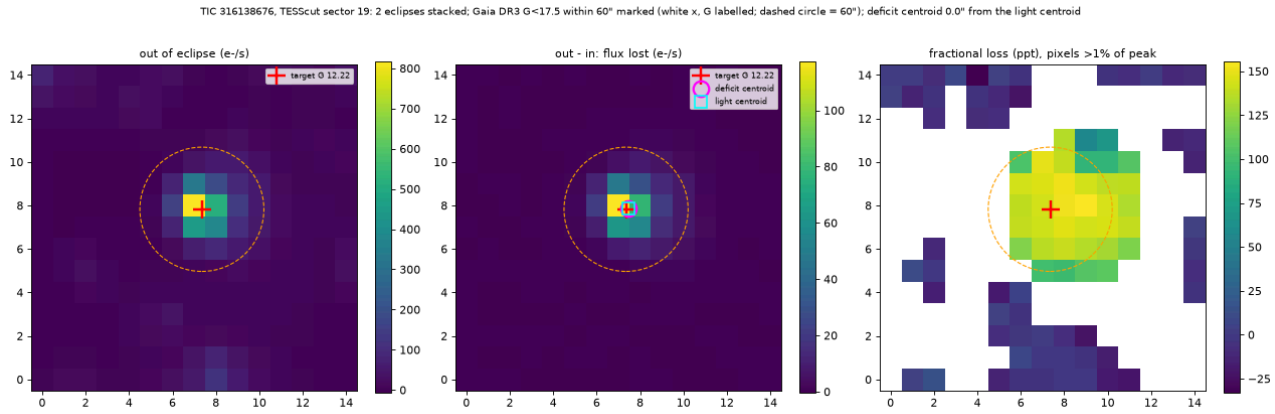
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R490; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R490; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 316138676: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured eclipses in representative sectors [18, 59, 86]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 5 actual midpoints yield P = 13.9528465175 ± 0.0000217006 d and reference epoch BTJD 1805.4162403 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S19 cutout puts the deficit 2.82 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.050; this is not a calibrated localization error. Capable Gaia neighbors: none within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 143.67/145.57/144.91 ppt in 2 stacked events; median selected QLP raw fit depth 151.16 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 450156999764024576 has orbital P = 13.9484101055 d, e = 0.12110; closest edge-on conjunction offset at the measured epoch is 0.36270 d (0.02600 phase), with 0.36683 d linearly propagated full-SB1-covariance timing error. The rough central duration is 27.72 h versus observed median 10.23 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Five deep raw eclipses across S18/59/86, including consecutive S59 and S86 cycles. The deficit core is centered on the target; no capable Gaia neighbor is present under the approximate brightness screen. The long fitted duration is geometry dependent.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
18	0	1805.4159110	0.010417	151.16	used
59	80	2921.6441111	0.001157	153.19	used
59	81	2935.5966595	0.001157	151.93	used
86	132	3647.1923341	0.001157	150.47	used
86	133	3661.1444770	0.001157	151.00	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s19.json, historical/diff_s19.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R490. Public evidence files: diff_s19.json, diff_s19.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

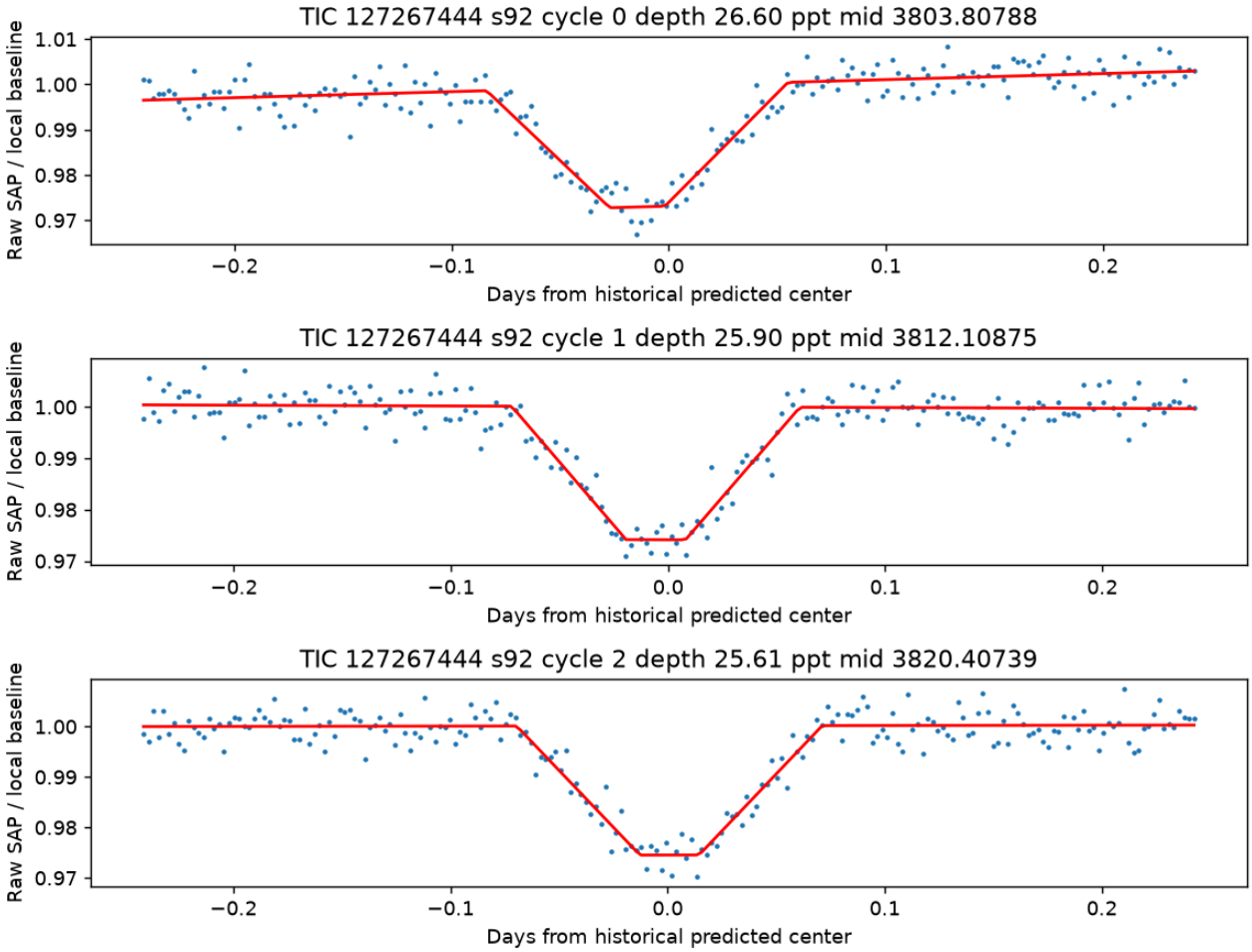
R491. TIC 127267444

physical_verdict: CONFIRMED

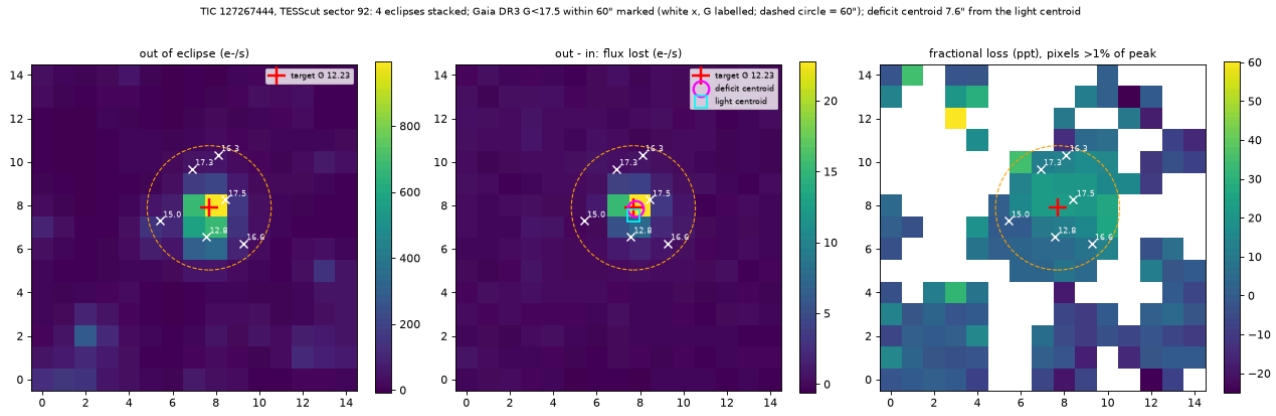
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R491; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R491; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 127267444: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 3 timing-usable measured eclipses in representative sectors [92]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 3 actual midpoints yield P = 8.2997438847 ± 0.0008290274 d and reference epoch BTJD 3803.8082712 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S92 cutout puts the deficit 1.73 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.388; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 6771875970694979712: 27.8 arcsec, G 12.830, approximate intrinsic eclipse requirement 0.066; Gaia 6771878959992220544: 49.5 arcsec, G 14.954, approximate intrinsic eclipse requirement 0.322. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 22.88/17.92/15.43 ppt in 4 stacked events; median selected QLP raw fit depth 25.90 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 6771876000757078144 has orbital P = 8.2991638272 d, e = 0.08323; closest edge-on conjunction offset at the measured epoch is 0.15077 d (0.01817 phase), with 0.61378 d linearly propagated full-SB1-covariance timing error. The rough central duration is 7.80 h versus observed median 3.33 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Three consecutive raw eclipses in S92 give a measured 8.299744-day period, close to the Gaia 8.299164-day orbit; the historical 8.29212-day screening estimate was inaccurate. Two capable neighbors at 27.8 and 49.5 arcsec are spatially distinct from the target-centered deficit. The pixel-model residual is substantial.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
92	0	3803.8078822	0.001187	26.60	used
92	1	3812.1087545	0.001157	25.90	used
92	2	3820.4073893	0.001157	25.61	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s92.json, historical/diff_s92.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R491. Public evidence files: diff_s92.json, diff_s92.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

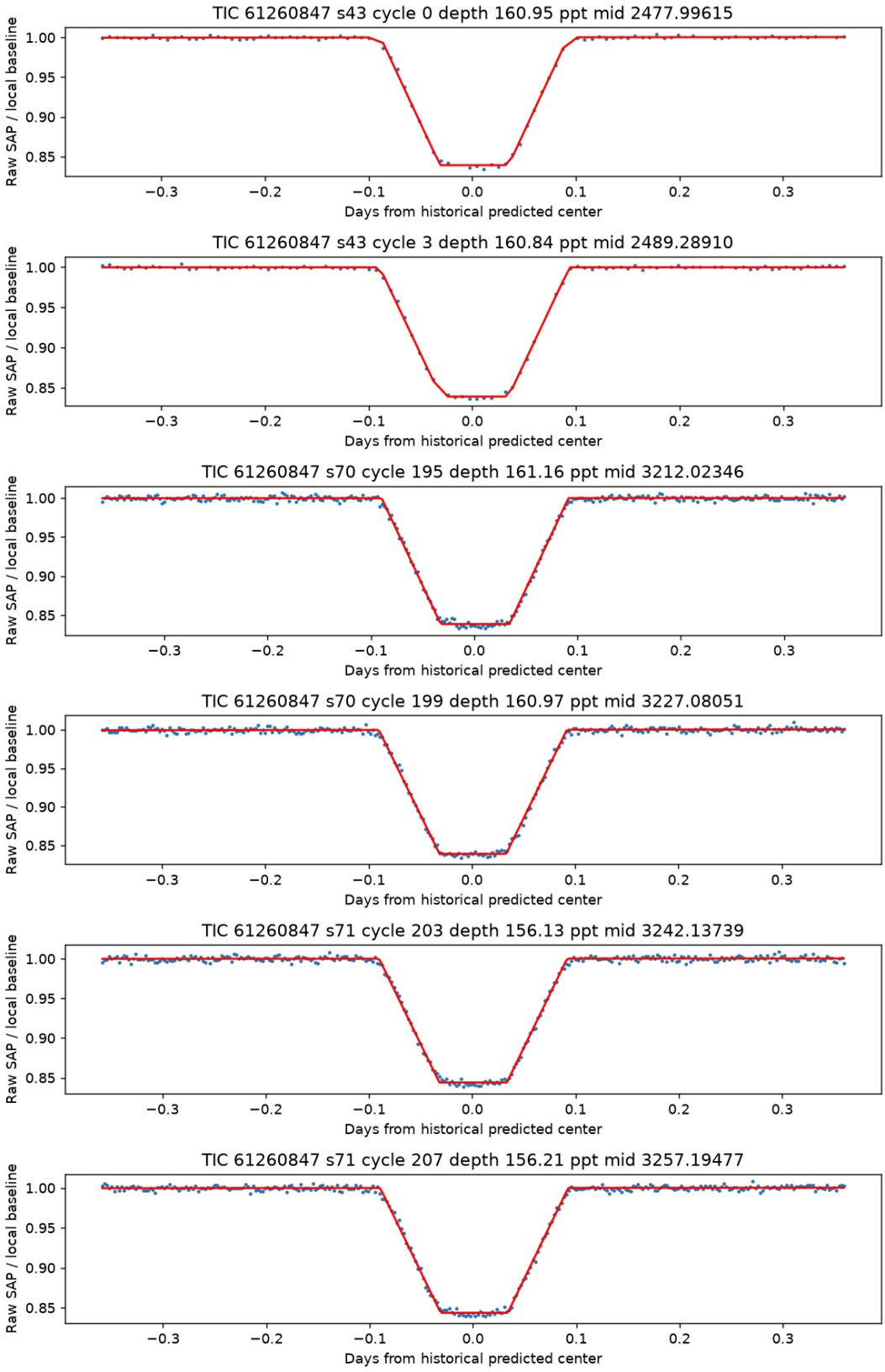
R492. TIC 61260847

physical_verdict: CONFIRMED

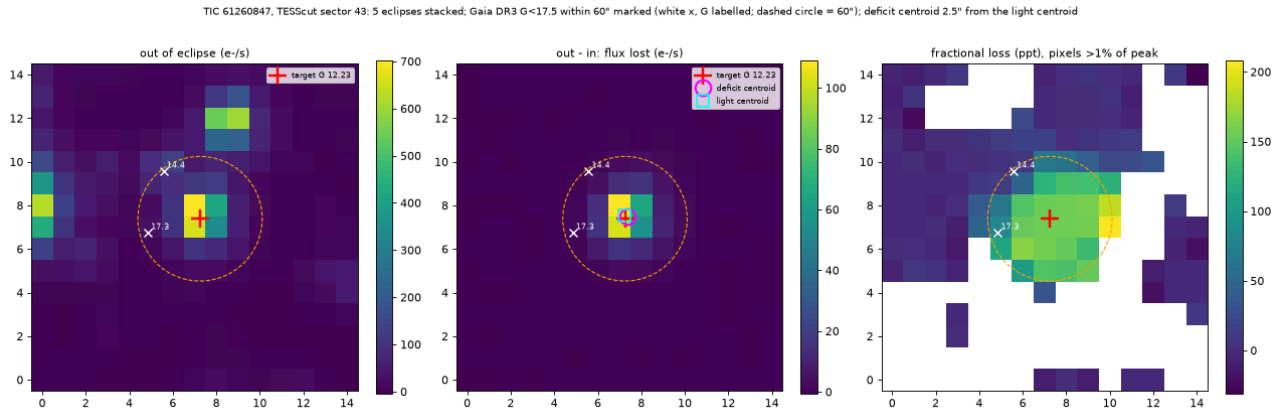
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R492; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R492; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 61260847: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 15 timing-usable measured eclipses in representative sectors [43, 70, 71]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 15 actual midpoints yield P = 3.7642439801 ± 0.0000080184 d and reference epoch BTJD 2477.9961067 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S43 cutout puts the deficit 2.54 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 7]. Image-model fractional residual 0.322; this is not a calibrated localization error. Capable Gaia neighbors: none within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 155.96/155.42/145.63 ppt in 5 stacked events; median selected QLP raw fit depth 160.34 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 3410845980312983680 has orbital P = 3.7643570975 d, e = 0.01600; closest edge-on conjunction offset at the measured epoch is -0.07963 d (-0.02115 phase), with 0.21627 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.75 h versus observed median 4.35 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Fifteen deep raw eclipses in S43/70/71 establish recurrence and a measured period. The cutout core favors the target with no capable Gaia neighbor; local image residuals limit precise centroid claims.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
43	0	2477.9961490	0.003473	160.95	used
43	1	2481.7601478	0.003473	159.16	used
43	3	2489.2890959	0.003473	160.84	used
43	4	2493.0530729	0.003473	159.18	used
43	5	2496.8173050	0.003473	160.37	used
70	195	3212.0234621	0.001158	161.16	used
70	196	3215.7881033	0.001158	160.34	used
70	198	3223.3164704	0.001158	161.07	used
70	199	3227.0805055	0.001158	160.97	used
70	200	3230.8449305	0.001158	161.27	used
71	202	3238.3734637	0.001157	156.10	used
71	203	3242.1373914	0.001157	156.13	used
71	204	3245.9020319	0.001157	155.95	used
71	205	3249.6660910	0.001157	156.03	used
71	207	3257.1947663	0.001157	156.21	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s43.json, historical/diff_s43.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R492. Public evidence files: diff_s43.json, diff_s43.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

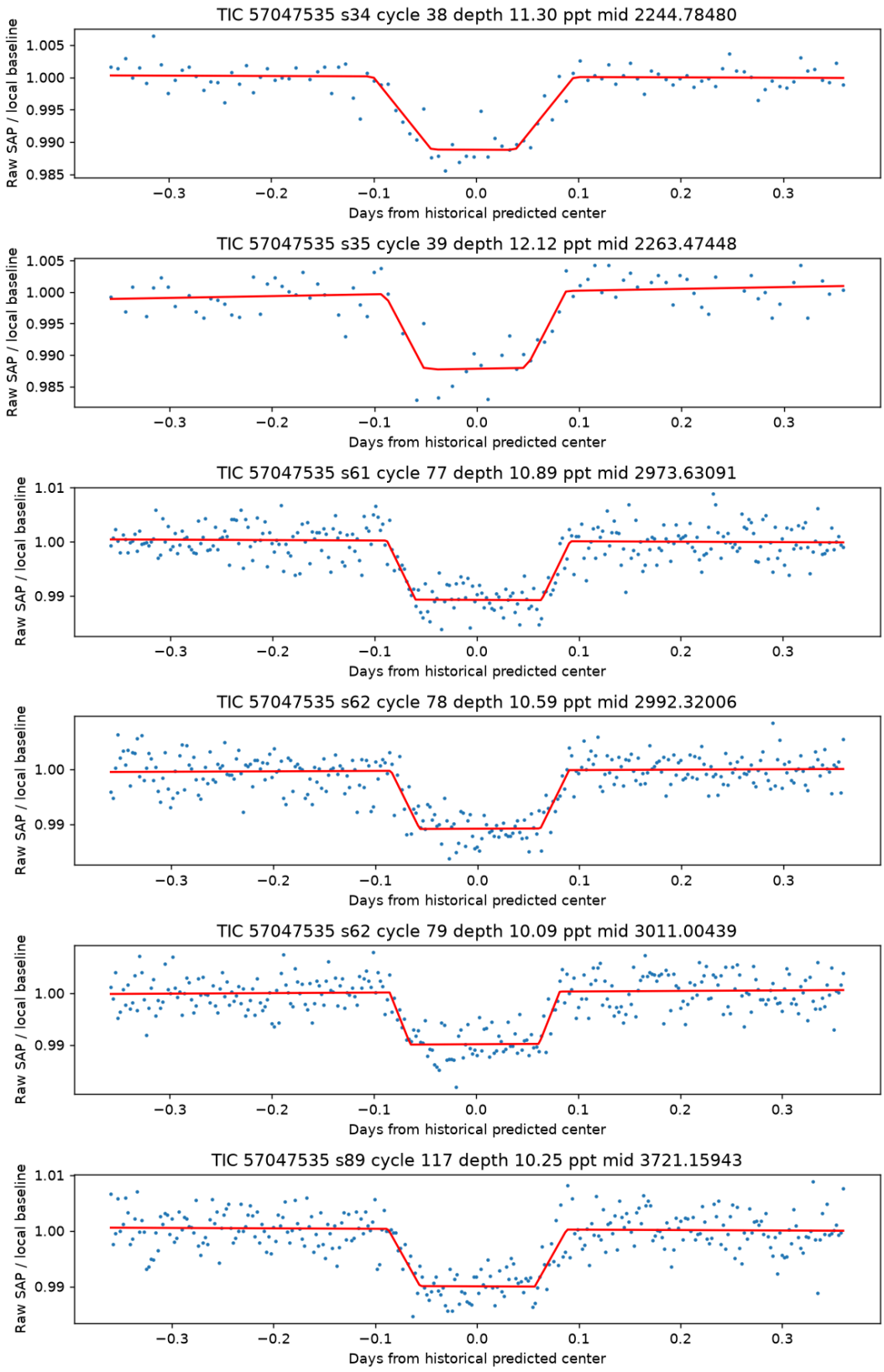
R493. TIC 57047535

physical_verdict: PROBABLE

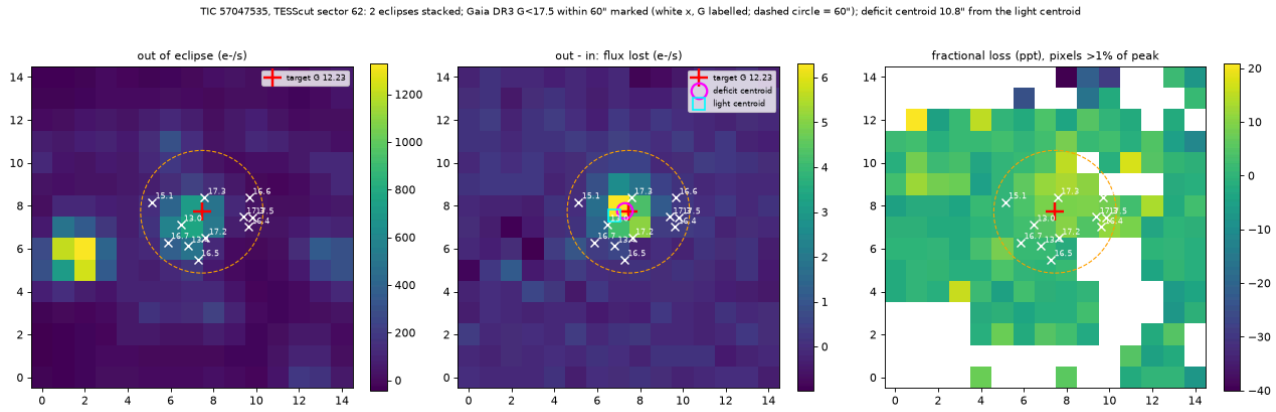
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization of the crowded field.



R493; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R493; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 57047535: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 7 timing-usable measured eclipses in representative sectors [34, 35, 61, 62, 88, 89]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 7 actual midpoints yield P = 18.6882586301 ± 0.0000306738 d and reference epoch BTJD 2244.7878116 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: unresolved after completed check.** Historical S62 cutout puts the deficit 4.30 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.674; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 5542738205536780288: 22.6 arcsec, G 12.965, approximate intrinsic eclipse requirement 0.031; Gaia 5542738205536774656: 33.5 arcsec, G 13.358, approximate intrinsic eclipse requirement 0.040; Gaia 5542738278564942848: 41.5 arcsec, G 16.683, approximate intrinsic eclipse requirement 0.647; Gaia 5542738209845466752: 44.2 arcsec, G 16.489, approximate intrinsic eclipse requirement 0.543; Gaia 5542737557010422784: 45.6 arcsec, G 16.553, approximate intrinsic eclipse requirement 0.576; Gaia 5542737453931213312: 46.2 arcsec, G 16.445, approximate intrinsic eclipse requirement 0.522; Gaia 5542738278566462336: 47.7 arcsec, G 15.131, approximate intrinsic eclipse requirement 0.163. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 8.93/7.09/5.36 ppt in 2 stacked events; median selected QLP raw fit depth 10.59 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate

observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.

5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5542738205536698240 has orbital P = 18.6663627993 d, e = 0.42196; closest edge-on conjunction offset at the measured epoch is 0.38371 d (0.02056 phase), with 1.66506 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.38 h versus observed median 4.22 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Seven raw eclipses across S34/35/61/62/88/89, including consecutive cycles 38/39, 77/78/79 and 116/117, resolve the initially sparse timing. Several capable neighbors make this a crowded field. The deficit core favors the target, but the closest capable Gaia source is only about 1.1 TESS pixels away and the image-model residual is large; attribution remains unresolved.

Next check: Resolved eclipse photometry or multi-source PRF localization of the crowded field.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
34	38	2244.7848021	0.003472	11.30	used
35	39	2263.4744847	0.003472	12.12	used
61	77	2973.6309138	0.001361	10.89	used
62	78	2992.3200617	0.001447	10.59	used
62	79	3011.0043901	0.001679	10.09	used
88	116	3702.4716187	0.001641	10.01	used
89	117	3721.1594292	0.001913	10.25	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s62.json, historical/diff_s62.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R493. Public evidence files: diff_s62.json, diff_s62.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 57047535-availability.json

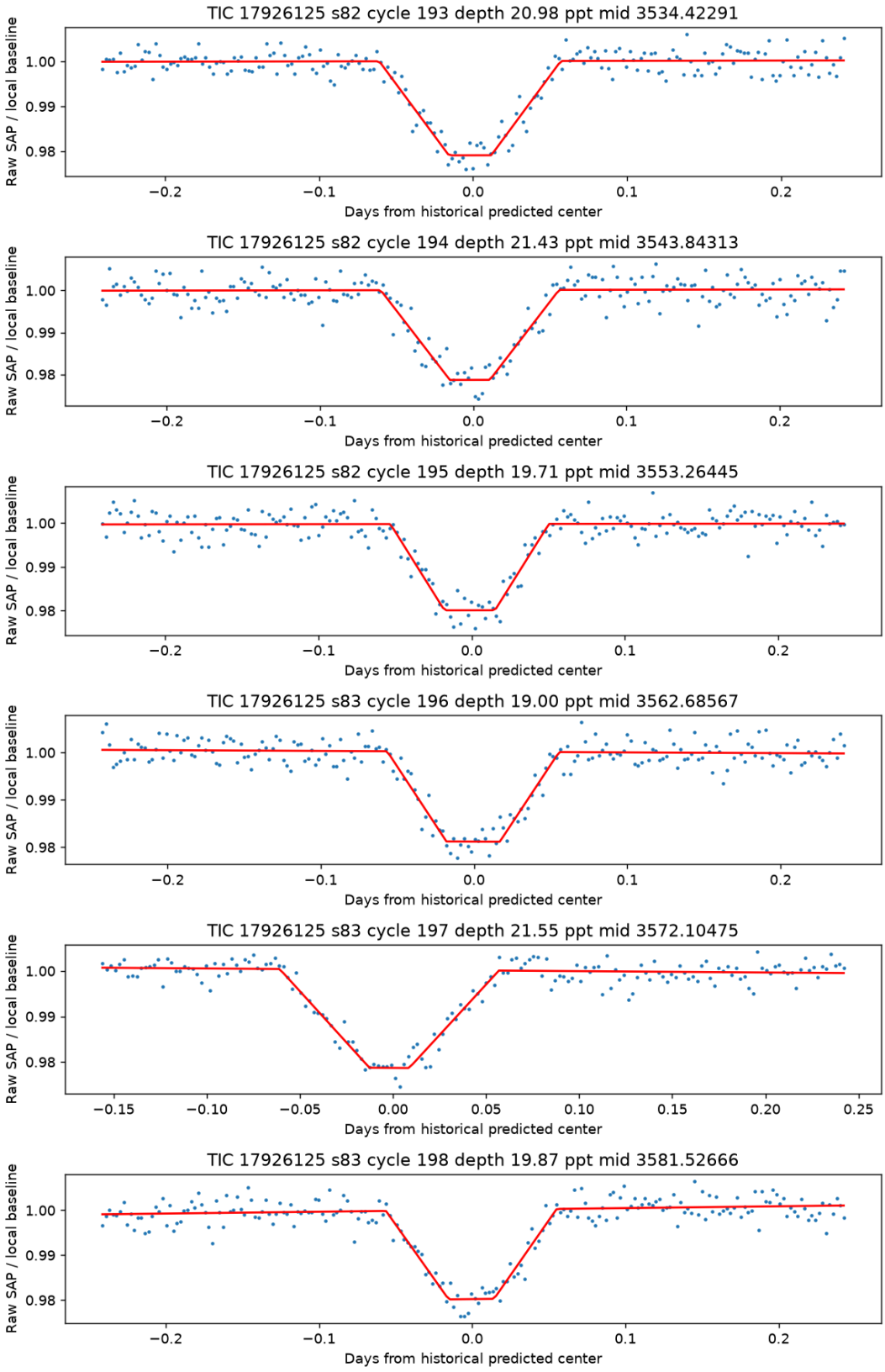
R494. TIC 17926125

physical_verdict: PROBABLE

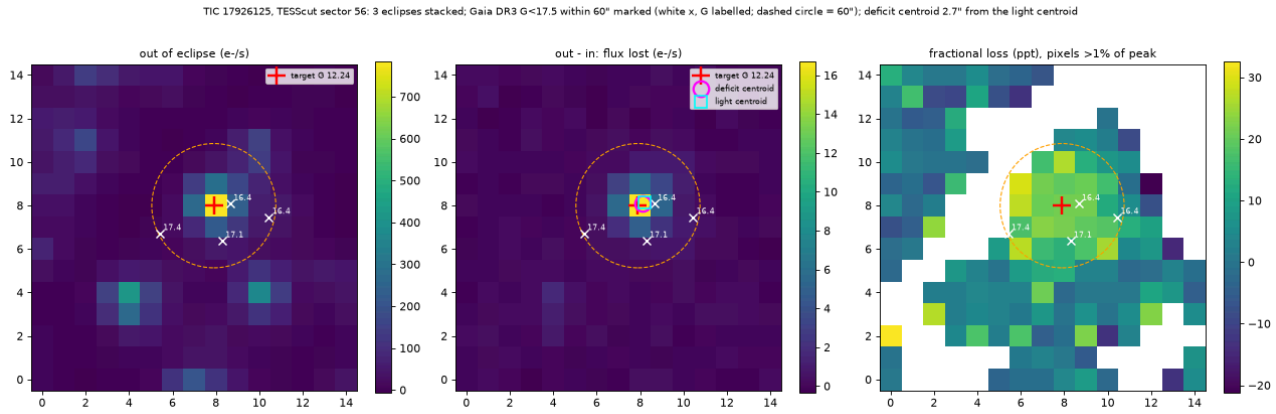
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization of the 15.7-arcsec pair.



R494; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R494; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 17926125: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured eclipses in representative sectors [82, 83]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 6 actual midpoints yield P = 9.4207091616 ± 0.0002766705 d and reference epoch BTJD 3534.4228208 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: unresolved after completed check.** Historical S56 cutout puts the deficit 4.79 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.489; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 1896543625318721280: 15.7 arcsec, G 16.363, approximate intrinsic eclipse requirement 0.866; Gaia 1896543659678461312: 53.7 arcsec, G 16.358, approximate intrinsic eclipse requirement 0.862. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 21.32/20.96/19.82 ppt in 3 stacked events; median selected QLP raw fit depth 20.43 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1896543621022178048 has orbital P = 9.4199419993 d, e = 0.02747; closest edge-on conjunction offset at the measured epoch is 0.06074 d (0.00645 phase), with 0.52723 d linearly propagated full-SB1-covariance timing error. The rough central duration is 6.60 h versus observed median 2.74 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Six consecutive raw eclipses in S82/83 support the measured recurrence. The deficit core favors the target, but a capable G16.36 neighbor 15.7 arcsec away shares the central TESS pixel and could in principle produce the loss through an extremely deep intrinsic eclipse. Attribution remains unresolved at this resolution.

Next check: Resolved eclipse photometry or multi-source PRF localization of the 15.7-arcsec pair.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
82	193	3534.4229057	0.001157	20.98	used
82	194	3543.8431342	0.001157	21.43	used
82	195	3553.2644488	0.001157	19.71	used
83	196	3562.6856663	0.001157	19.00	used
83	197	3572.1047515	0.001157	21.55	used
83	198	3581.5266560	0.001157	19.87	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R494. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

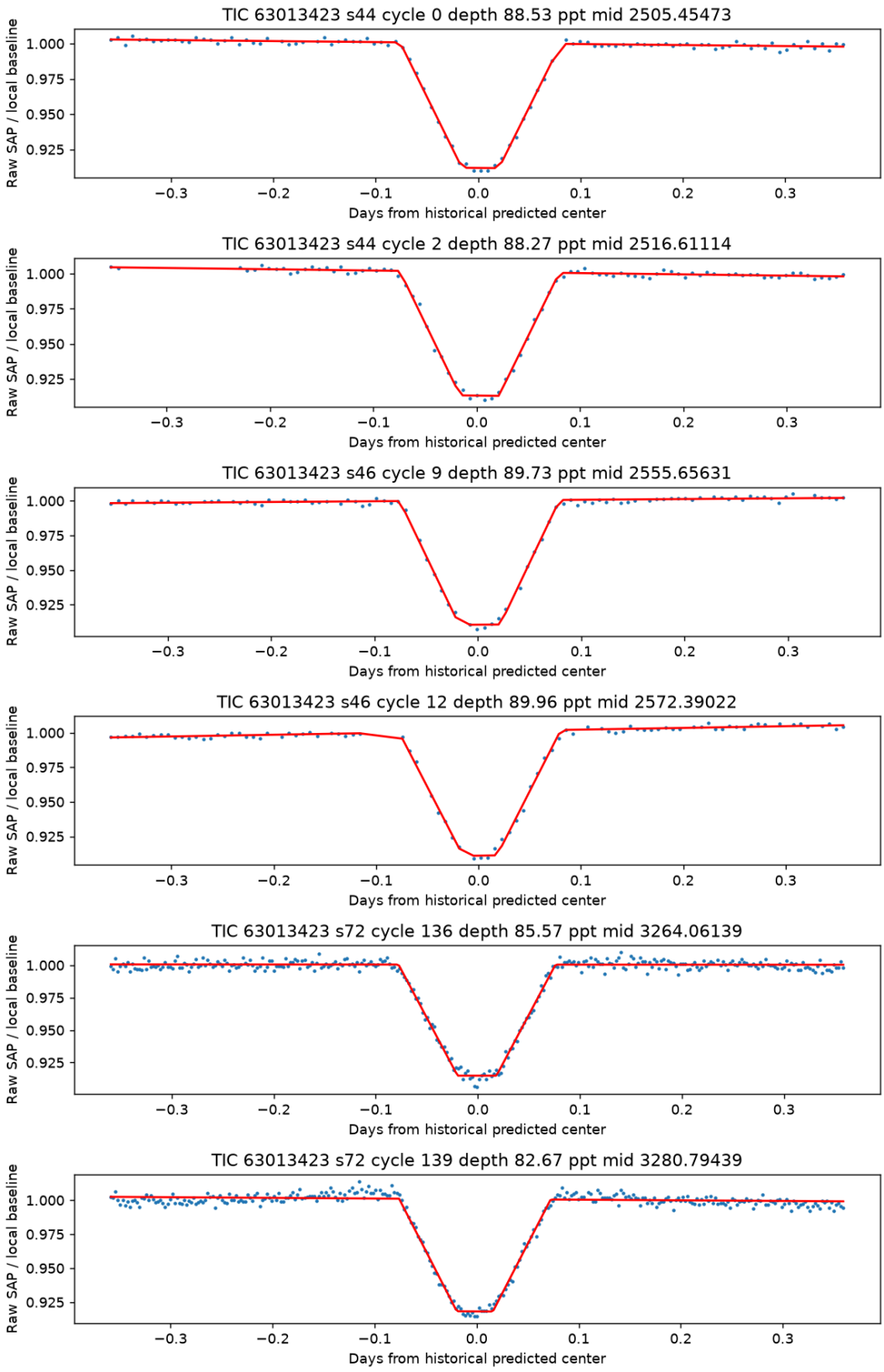
R495. TIC 63013423

physical_verdict: CONFIRMED

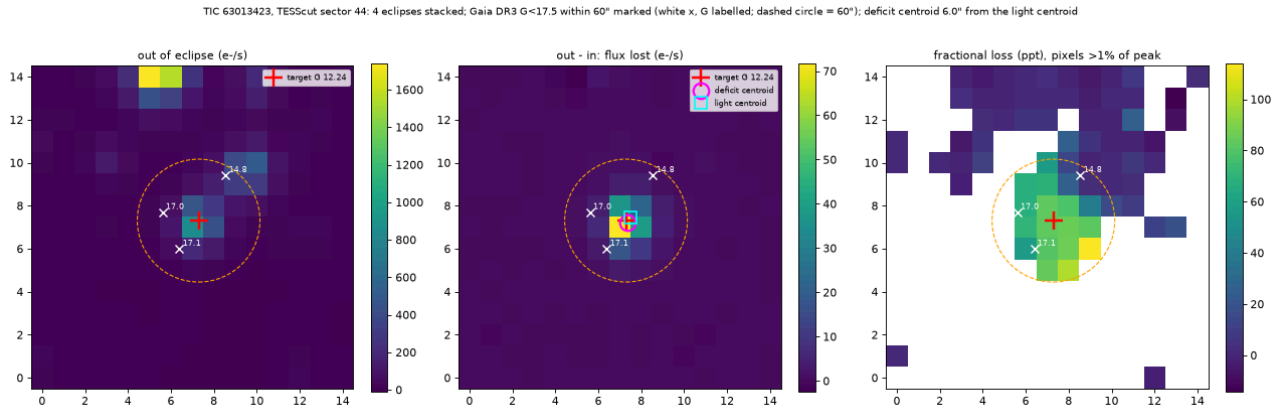
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R495; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R495; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 63013423: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 12 timing-usable measured eclipses in representative sectors [44, 46, 72]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 12 actual midpoints yield P = 5.5779862426 ± 0.0000103302 d and reference epoch BTJD 2505.4546770 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S44 cutout puts the deficit 2.57 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.569; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 675068326633876992: 49.1 arcsec, G 14.755, approximate intrinsic eclipse requirement 0.904. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 82.60/81.69/70.51 ppt in 4 stacked events; median selected QLP raw fit depth 87.58 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 675068532792307456 has orbital P = 5.5777973688 d, e = 0.04208; closest edge-on conjunction offset at the measured epoch is 0.13397 d (0.02402 phase), with 0.20174 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.22 h versus observed median 3.72 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Twelve raw eclipses in S44/46/72 show stable recurrence. The capable neighbor 49.1 arcsec away is distinct from the target-centered deficit; aperture and model residuals limit exact intrinsic depth.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
44	0	2505.4547268	0.003473	88.53	used
44	1	2511.0325313	0.003473	89.00	used
44	2	2516.6111421	0.003473	88.27	used
44	3	2522.1890296	0.003473	86.32	used
46	9	2555.6563082	0.003472	89.73	used
46	10	2561.2344079	0.003472	88.32	used
46	12	2572.3902221	0.003472	89.96	used
46	13	2577.9681930	0.003472	86.86	used
72	136	3264.0613877	0.001157	85.57	used
72	137	3269.6388279	0.001157	85.71	used
72	138	3275.2165537	0.001157	86.88	used
72	139	3280.7943908	0.001157	82.67	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s44.json, historical/diff_s44.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R495. Public evidence files: diff_s44.json, diff_s44.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

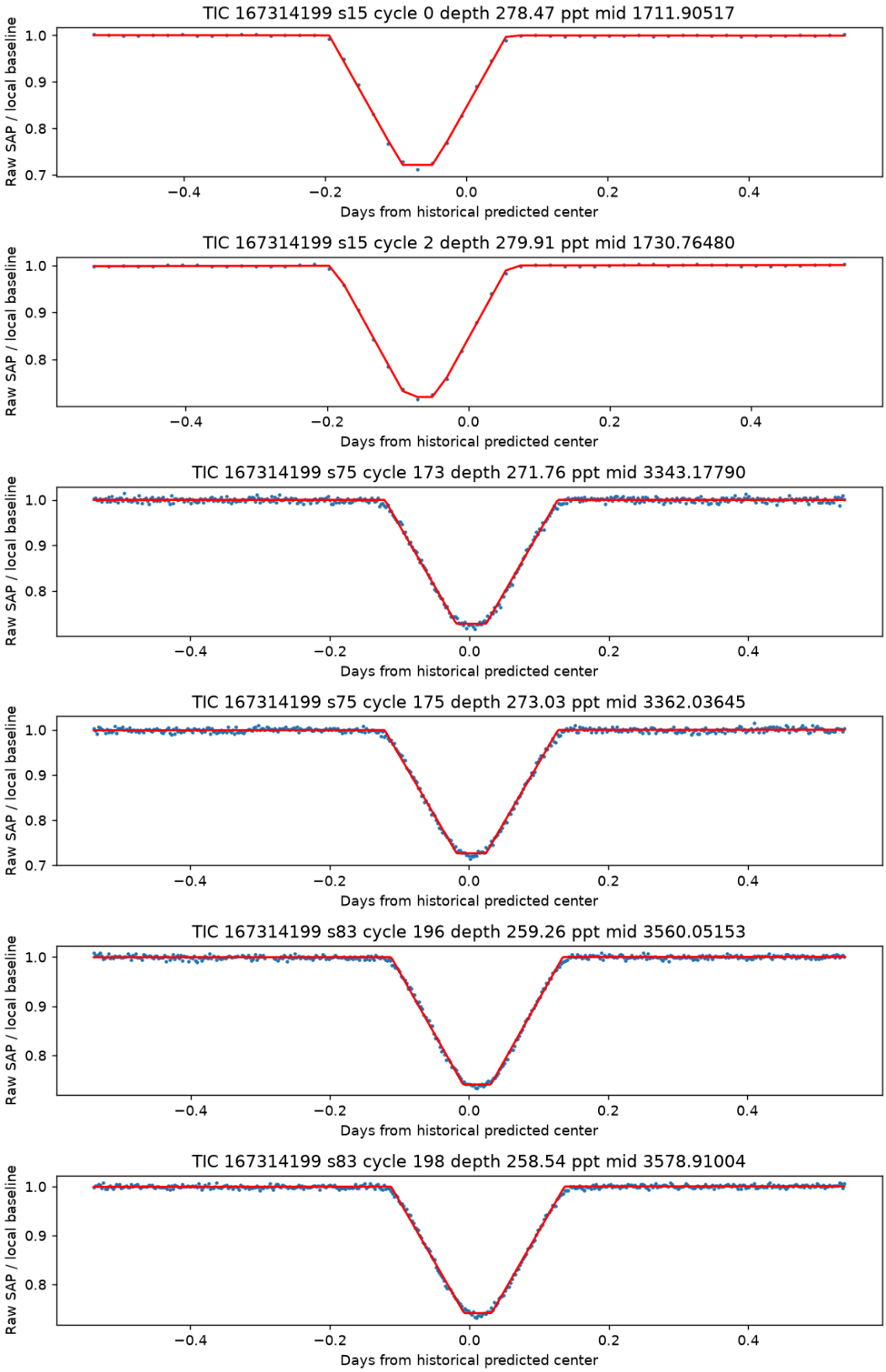
R496. TIC 167314199

physical_verdict: CONFIRMED

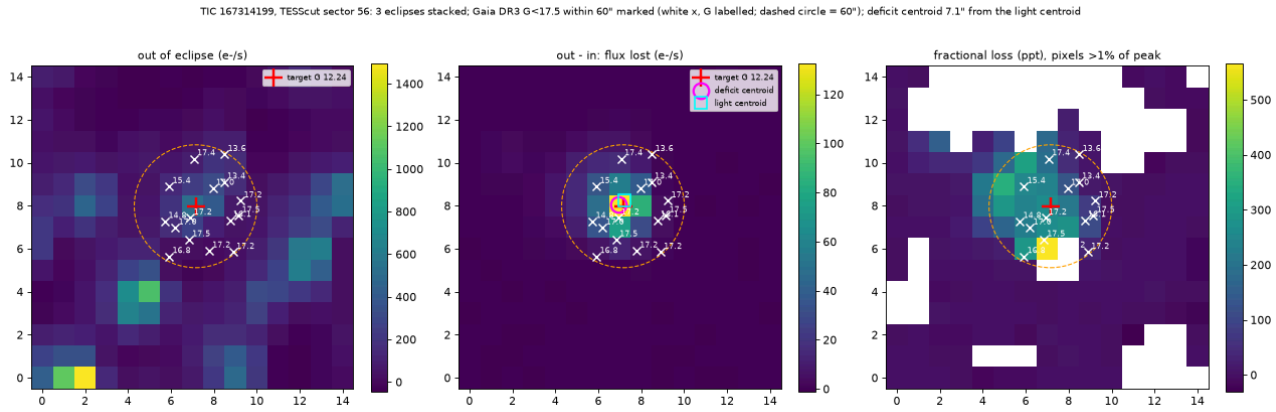
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R496; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R496; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 167314199: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured eclipses in representative sectors [15, 75, 83]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 8 actual midpoints yield P = 9.4293015589 ± 0.0000286278 d and reference epoch BTJD 1711.9085480 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S56 cutout puts the deficit 5.06 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.880; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 1964832780702837248: 35.7 arcsec, G 13.368, approximate intrinsic eclipse requirement 0.909. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 259.06/230.88/191.58 ppt in 3 stacked events; median selected QLP raw fit depth 272.15 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1964832776395084288 has orbital P = 9.4286060079 d, e = 0.02722; closest edge-on conjunction offset at the measured epoch is 0.14149 d (0.01501 phase), with 0.14582 d linearly propagated full-SB1-covariance timing error. The rough central duration is 7.89 h versus observed median 5.97 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Eight deep raw eclipses in S15/75/83 recur on the Gaia timescale. A capable 35.7-arcsec neighbor requires an extreme intrinsic eclipse and is distinct from the target-favored core; the image-model residual is high.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
15	0	1711.9051662	0.010417	278.47	used
15	2	1730.7647966	0.010417	279.91	used
75	173	3343.1779004	0.001157	271.76	used
75	174	3352.6073018	0.001157	272.54	used
75	175	3362.0364531	0.001157	273.03	used
83	196	3560.0515342	0.001157	259.26	used
83	197	3569.4807627	0.001157	256.44	used
83	198	3578.9100415	0.001157	258.54	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R496. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

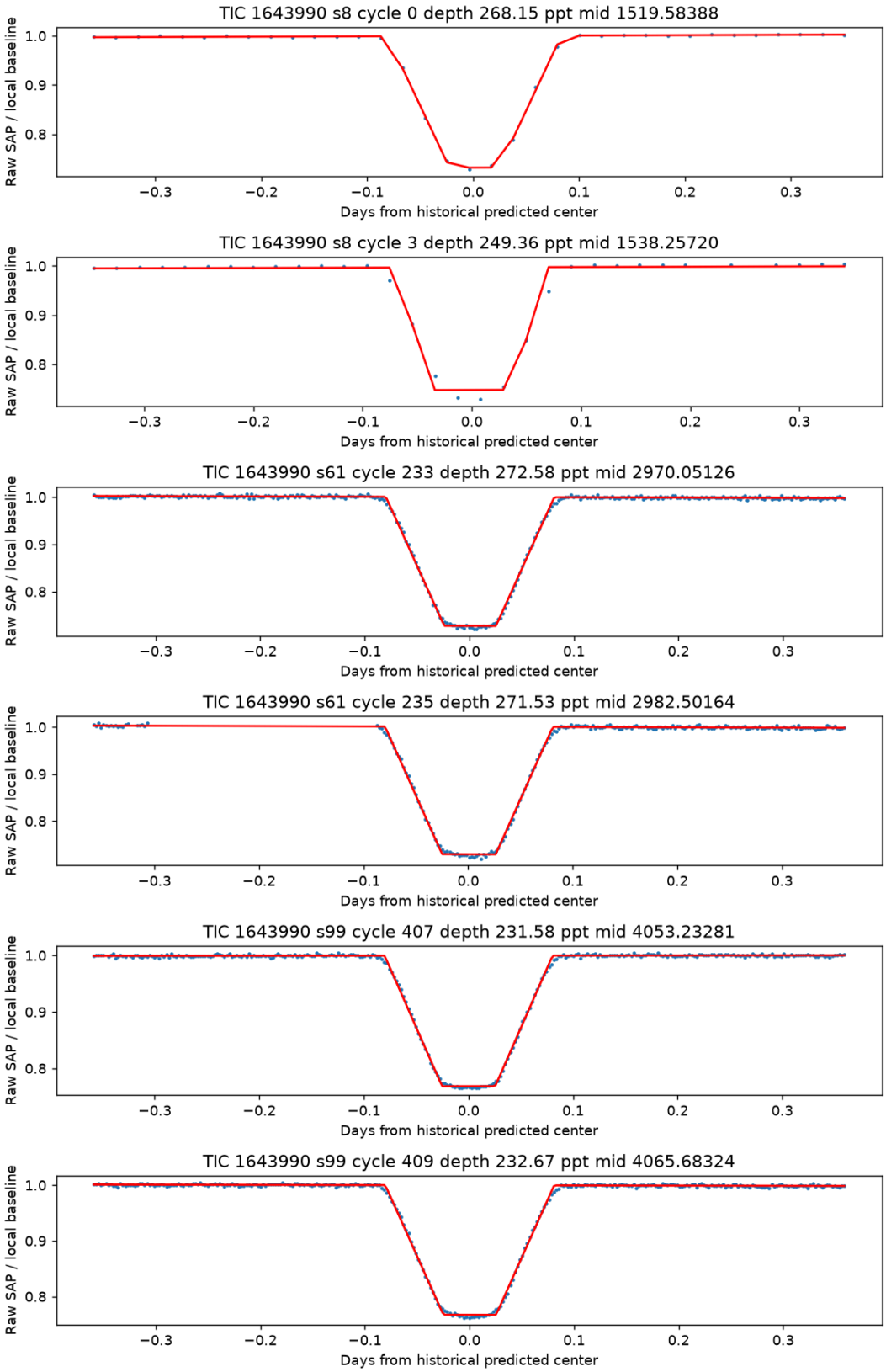
R497. TIC 1643990

physical_verdict: CONFIRMED

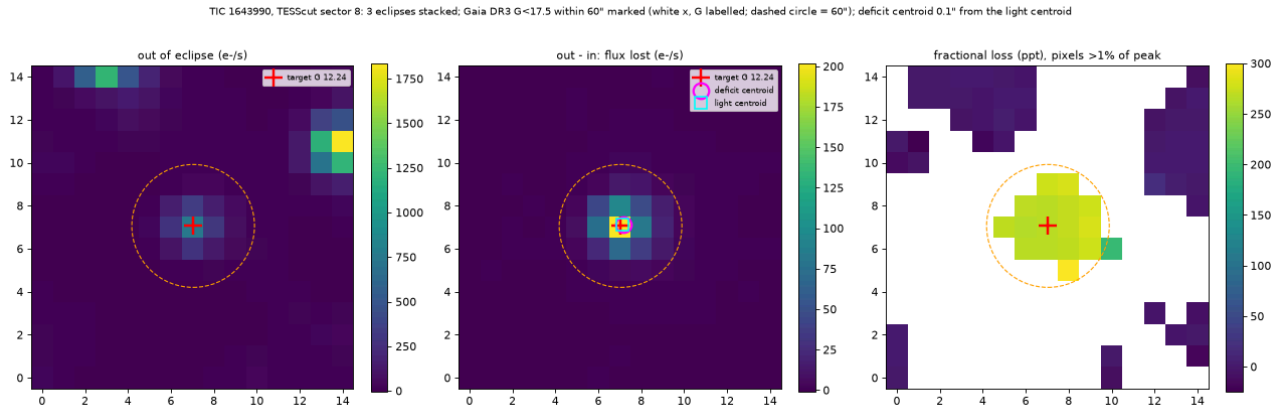
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R497; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R497; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 1643990: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured eclipses in representative sectors [8, 61, 99]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 6 actual midpoints yield P = 6.2251817590 ± 0.0000063922 d and reference epoch BTJD 1519.5838843 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S8 cutout puts the deficit 2.55 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.046; this is not a calibrated localization error. Capable Gaia neighbors: none within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 267.11/268.43/270.89 ppt in 3 stacked events; median selected QLP raw fit depth 258.75 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5762102560669048704 has orbital P = 6.2252118769 d, e = 0.11362; closest edge-on conjunction offset at the measured epoch is -0.00339 d (-0.00055 phase), with 0.07702 d linearly propagated full-SB1-covariance timing error. The rough central duration is 4.43 h versus observed median 3.86 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Six deep raw eclipses in S8/61/99 give a long timing baseline. The cutout deficit lies on the target with low model residual and no capable Gaia neighbor under the brightness screen.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
8	0	1519.5838814	0.010417	268.15	used
8	3	1538.2572047	0.010417	249.36	used
61	233	2970.0512579	0.001157	272.58	used
61	235	2982.5016384	0.001157	271.53	used
99	407	4053.2328119	0.001157	231.58	used
99	409	4065.6832351	0.001157	232.67	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s08.json, historical/diff_s08.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R497. Public evidence files: diff_s08.json, diff_s08.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

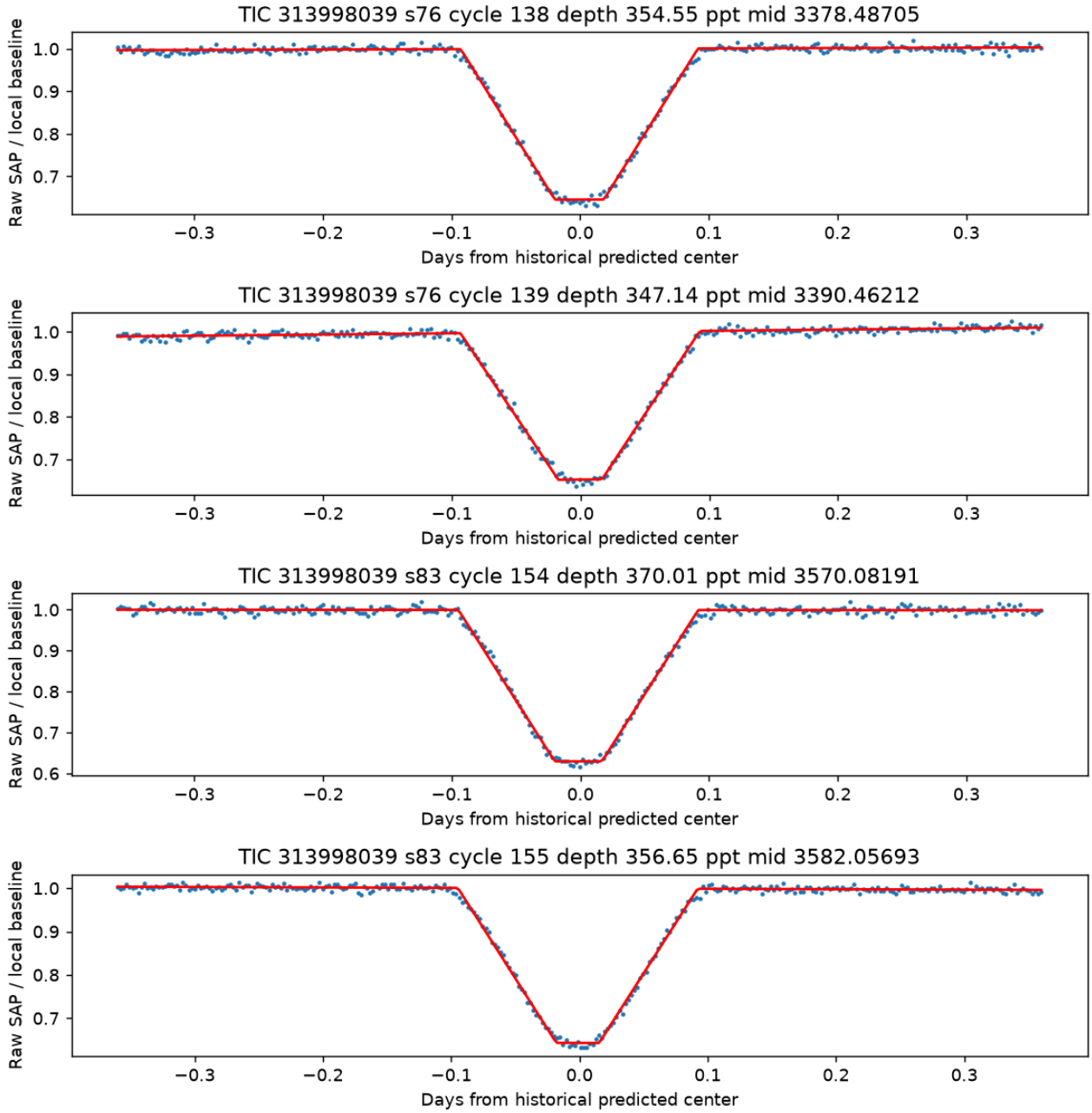
R498. TIC 313998039

physical_verdict: CONFIRMED

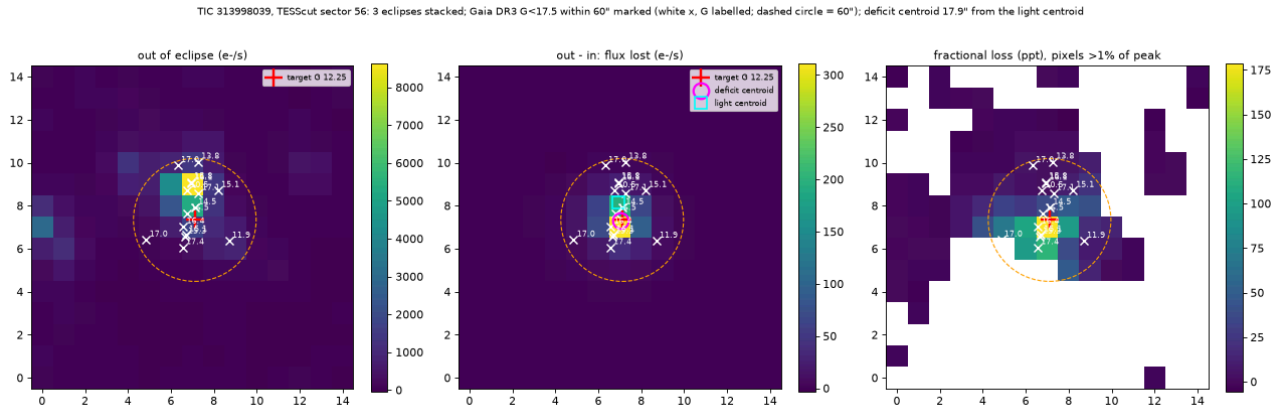
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R498; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R498; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 313998039: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 4 timing-usable measured eclipses in representative sectors [76, 83]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 4 actual midpoints yield P = 11.9746788325 ± 0.0000721956 d and reference epoch BTJD 3378.4872332 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S56 cutout puts the deficit 2.09 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.874; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 2189517466440845568: 29.6 arcsec, G 10.549, approximate intrinsic eclipse requirement 0.405; Gaia 2189505711103804288: 39.1 arcsec, G 11.912, approximate intrinsic eclipse requirement 0.581. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 178.81/65.00/32.94 ppt in 3 stacked events; median selected QLP raw fit depth 355.60 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 2189517466440848256 has orbital P = 11.9737068648 d, e = 0.06505; closest edge-on conjunction offset at the measured epoch is 0.48689 d (0.04066 phase), with 0.44849 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.44 h versus observed median 4.47 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Four deep raw eclipses in S76/83 repeat consistently. Bright capable neighbors at 29.6 and 39.1 arcsec cause severe aperture dilution, but the deficit core favors the target location; high image residual prevents precise localization.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
76	138	3378.4870523	0.001157	354.55	used
76	139	3390.4621157	0.001157	347.14	used
83	154	3570.0819122	0.001157	370.01	used
83	155	3582.0569331	0.001157	356.65	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R498. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

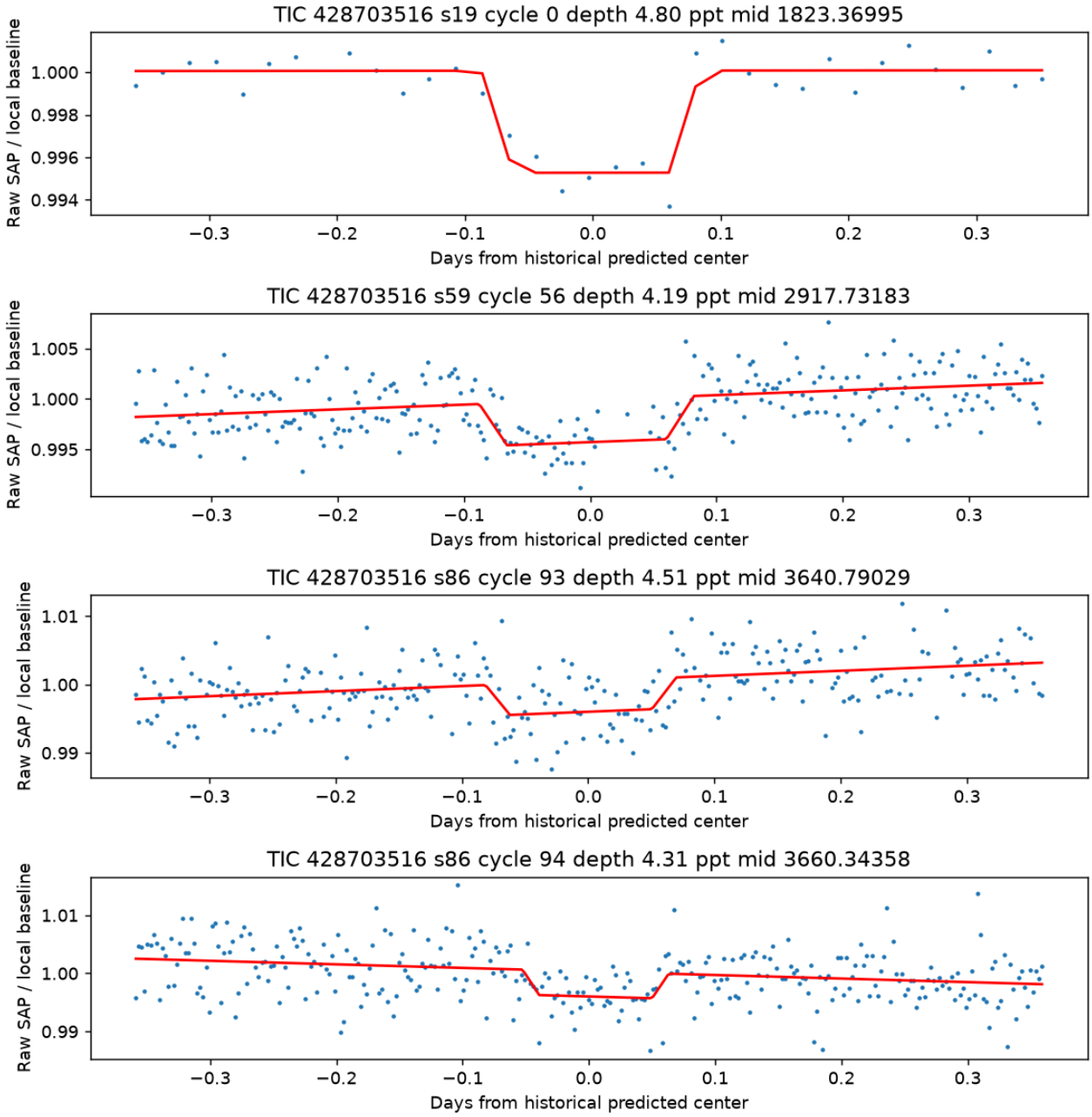
R499. TIC 428703516

physical_verdict: PROBABLE

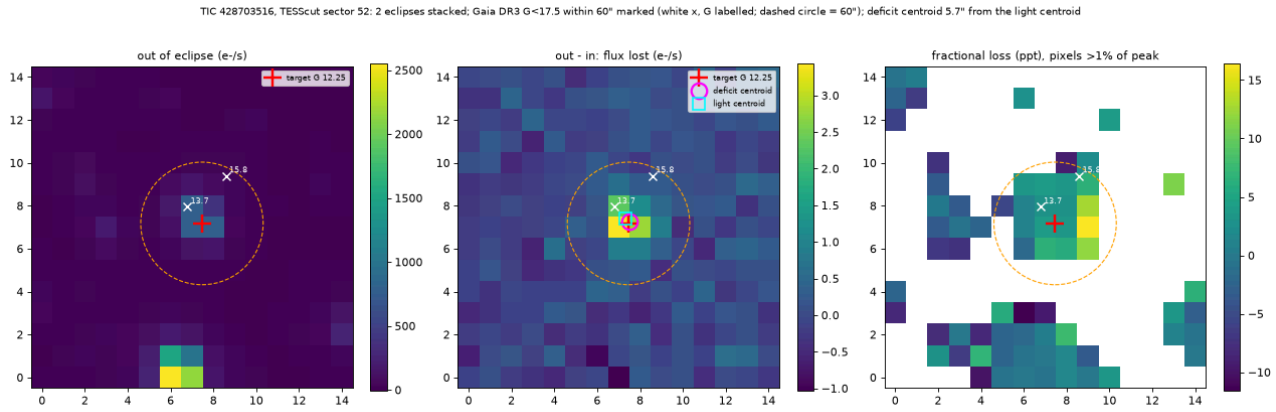
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization of the 11.2-arcsec pair.



R499; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R499; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 428703516: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 4 timing-usable measured eclipses in representative sectors [19, 59, 86]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 4 actual midpoints yield P = 19.5422216389 ± 0.0001299315 d and reference epoch BTJD 1823.3679179 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: unresolved after completed check.** Historical S52 cutout puts the deficit 2.07 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.488; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 548058855130833792: 11.2 arcsec, G 18.035, approximate intrinsic eclipse requirement 0.888; Gaia 548058859427261184: 21.5 arcsec, G 13.709, approximate intrinsic eclipse requirement 0.021; Gaia 548047104100306688: 51.4 arcsec, G 17.903, approximate intrinsic eclipse requirement 0.787; Gaia 548058859427256832: 52.1 arcsec, G 15.785, approximate intrinsic eclipse requirement 0.116. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 4.24/3.46/3.72 ppt in 2 stacked events; median selected QLP raw fit depth 4.41 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 548058859427458688 has orbital P = 19.5459437733 d, e = 0.02433; closest edge-on conjunction

offset at the measured epoch is -0.18196 d (-0.00931 phase), with 0.29026 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.97 h versus observed median 3.86 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Four shallow raw eclipses in S19/59/86, including consecutive S86 cycles, support recurrence. The deficit favors the target over the capable 21.5-arcsec neighbor, but a capable Gaia source 11.2 arcsec away is inside one TESS pixel and could in principle produce the 4-ppt loss with a deep intrinsic eclipse. Attribution remains unresolved at this resolution.

Next check: Resolved eclipse photometry or multi-source PRF localization of the 11.2-arcsec pair.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
19	0	1823.3699499	0.010417	4.80	used
59	56	2917.7318282	0.003696	4.19	used
86	93	3640.7902858	0.003409	4.51	used
86	94	3660.3435838	0.004222	4.31	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s52.json, historical/diff_s52.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R499. Public evidence files: diff_s52.json, diff_s52.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

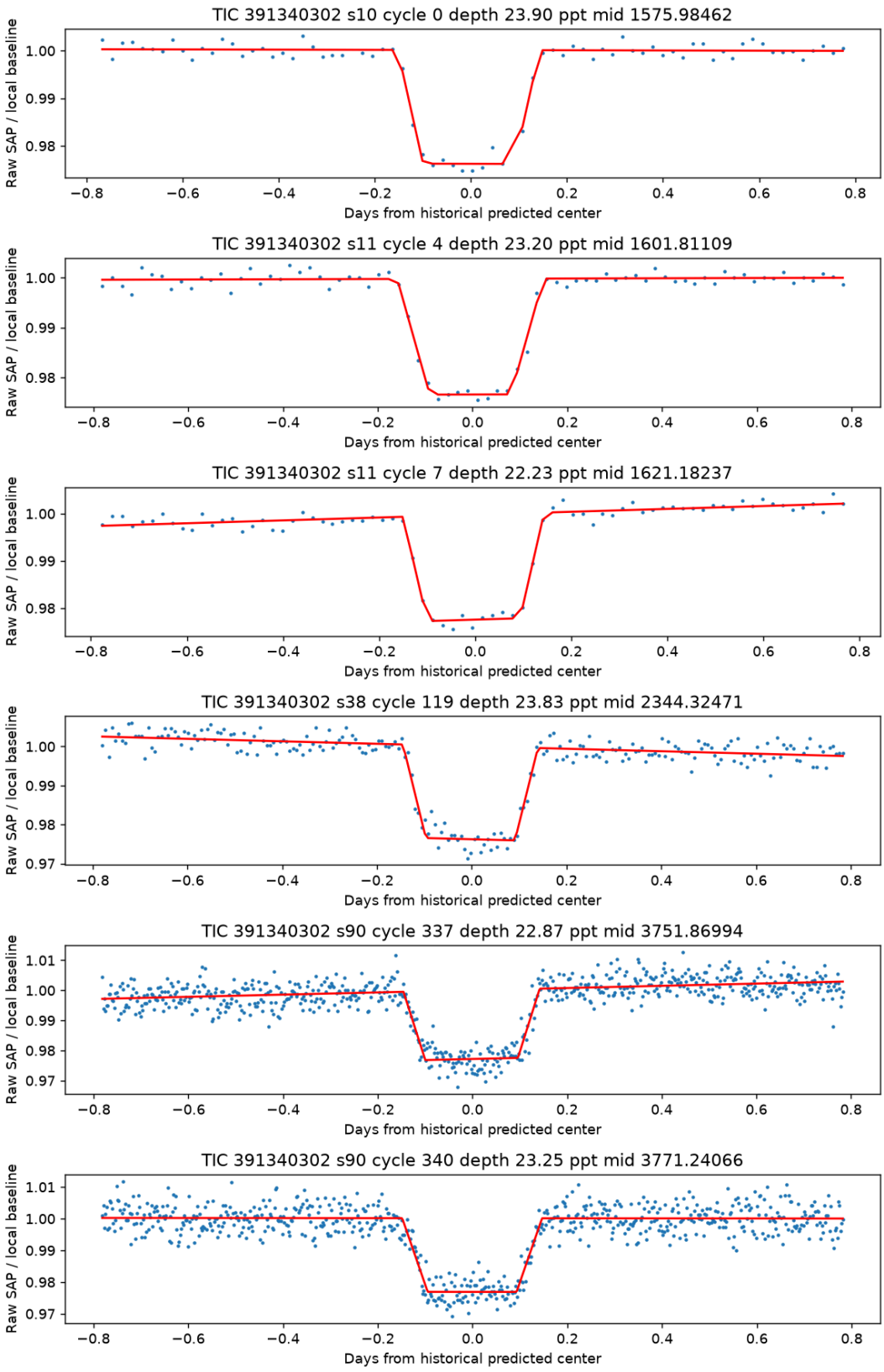
R500. TIC 391340302

physical_verdict: PROBABLE

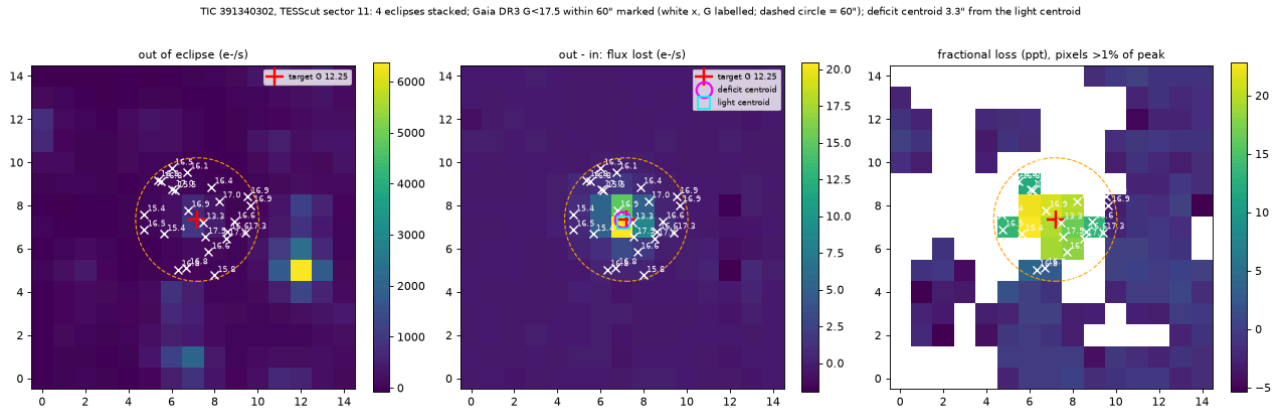
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry of the 6.1-arcsec pair.



R500; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R500; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 391340302: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 14 fitted representative eclipses in sectors [10, 11, 38, 90]; 14 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 14 same-kind actual midpoints yield P = 6.4566398472 ± 0.0000072175 d and reference epoch BTJD 1575.9846204 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: unresolved.** Historical sector 11 cutout places the deficit centroid 3.98 arcsec from the target; peak pixel [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.890 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 5239465552568592768: 6.1 arcsec, G 13.331, approximate intrinsic eclipse requirement 0.087; Gaia 5239465518208831104: 35.1 arcsec, G 15.493, approximate intrinsic eclipse requirement 0.487; Gaia 5239453801538064640: 35.2 arcsec, G 15.421, approximate intrinsic eclipse requirement 0.457; Gaia 5239465518208825984: 46.9 arcsec, G 16.057, approximate intrinsic eclipse requirement 0.802; Gaia 5239453767178311808: 51.5 arcsec, G 15.422, approximate intrinsic eclipse requirement 0.457; Gaia 5239454175171527424: 56.3 arcsec, G 15.756, approximate intrinsic eclipse requirement 0.614. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: supports recurring loss with unresolved host.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 19.49/19.83/18.92 ppt over 4 stacked events, compared with median selected QLP raw fit depth 23.22 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with

the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.

5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5239465548244669952 has P = 6.4572634522 d and e = 0.10891. The closest edge-on conjunction offset at the measured epoch is -0.08325 d (-0.01289 phase), with 0.16381 d linearly propagated full-SB1-covariance timing error. A rough central duration is 7.18 h versus observed median 7.04 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: The S11 difference image centers on the target pixel, but a Gaia neighbor 6.1 arcsec away lies in that same TESS core and can produce the observed loss. TESS pixels cannot assign the eclipses uniquely.

Next check: Resolved eclipse photometry of the 6.1-arcsec pair.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
10	0	1575.9846232	0.010417	23.90	used
10	2	1588.8979173	0.010417	23.10	used
11	4	1601.8110937	0.010417	23.20	used
11	5	1608.2682354	0.010417	22.15	used
11	6	1614.7223332	0.010417	23.13	used
11	7	1621.1823670	0.010417	22.23	used
38	118	2337.8680384	0.003472	24.38	used
38	119	2344.3247065	0.003472	23.83	used
38	120	2350.7815131	0.003472	23.58	used
38	121	2357.2381694	0.003472	24.48	used
90	337	3751.8699430	0.001301	22.87	used
90	338	3758.3315202	0.001186	23.05	used
90	339	3764.7857257	0.001157	24.18	used
90	340	3771.2406614	0.001513	23.25	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s11.json, historical/diff_s11.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R500. Public evidence files: diff_s11.json, diff_s11.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

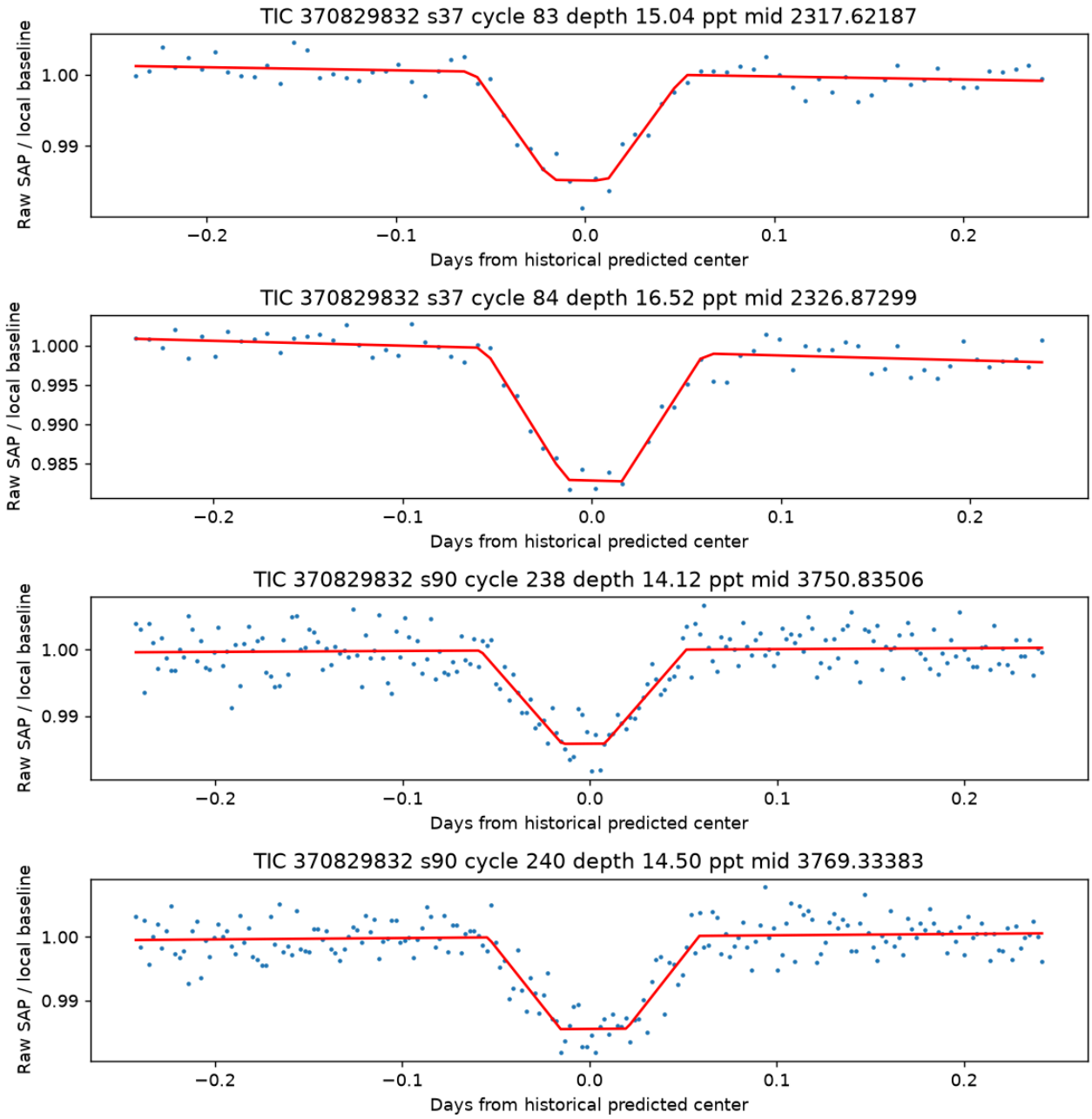
R501. TIC 370829832

physical_verdict: CONFIRMED

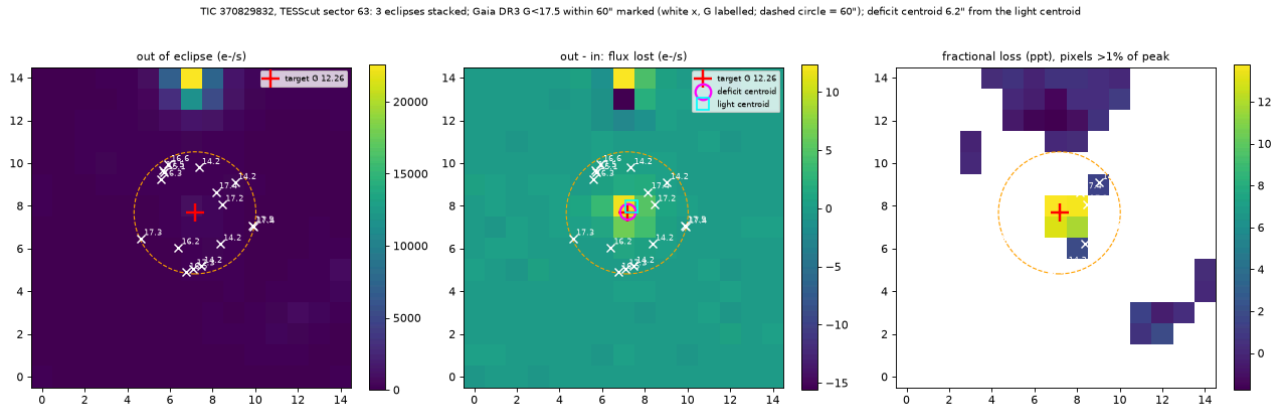
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional higher-resolution eclipse imaging and binary characterization.



R501; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R501; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 370829832: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 4 fitted representative eclipses in sectors [37, 90]; 4 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 4 same-kind actual midpoints yield P = 9.2465432321 ± 0.0000330778 d and reference epoch BTJD 2317.6240607 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 63 cutout places the deficit centroid 1.79 arcsec from the target; peak pixel [7, 14] versus target pixel [7, 8]. Image-model fractional residual 0.949 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 5249020789631628928: 37.8 arcsec, G 16.212, approximate intrinsic eclipse requirement 0.537; Gaia 5249020720912155520: 38.7 arcsec, G 14.177, approximate intrinsic eclipse requirement 0.094; Gaia 5249020858351113472: 44.5 arcsec, G 14.172, approximate intrinsic eclipse requirement 0.094; Gaia 5249020927070584832: 46.6 arcsec, G 16.304, approximate intrinsic eclipse requirement 0.583; Gaia 5249020858351116032: 49.0 arcsec, G 14.187, approximate intrinsic eclipse requirement 0.095; Gaia 5249020927070586112: 49.2 arcsec, G 15.096, approximate intrinsic eclipse requirement 0.201; Gaia 5249020789631629440: 52.4 arcsec, G 14.216, approximate intrinsic eclipse requirement 0.097; Gaia 5249020927070587392: 53.9 arcsec, G 16.631, approximate intrinsic eclipse requirement 0.783; Gaia 5249020789631627776: 58.8 arcsec, G 16.523, approximate intrinsic eclipse requirement 0.710. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.

4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 13.79/12.50/9.08 ppt over 3 stacked events, compared with median selected QLP raw fit depth 14.77 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5249020892710847488 has P = 9.2458347321 d and e = 0.18138. The closest edge-on conjunction offset at the measured epoch is -0.03335 d (-0.00361 phase), with 0.38441 d linearly propagated full-SB1-covariance timing error. A rough central duration is 6.19 h versus observed median 2.69 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: The S63 difference image has a remote single-pixel maximum at [7,14] coincident with a bright field source; its origin is uncertain. The multi-pixel eclipse deficit and centroid lie near the target [7,8]. The remote maximum and large image-model residual limit precision localization.

Next check: Optional higher-resolution eclipse imaging and binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
37	83	2317.6218651	0.003472	15.04	used
37	84	2326.8729892	0.003472	16.52	used
90	238	3750.8350612	0.001678	14.12	used
90	240	3769.3338324	0.001489	14.50	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s63.json, historical/diff_s63.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R501. Public evidence files: diff_s63.json, diff_s63.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

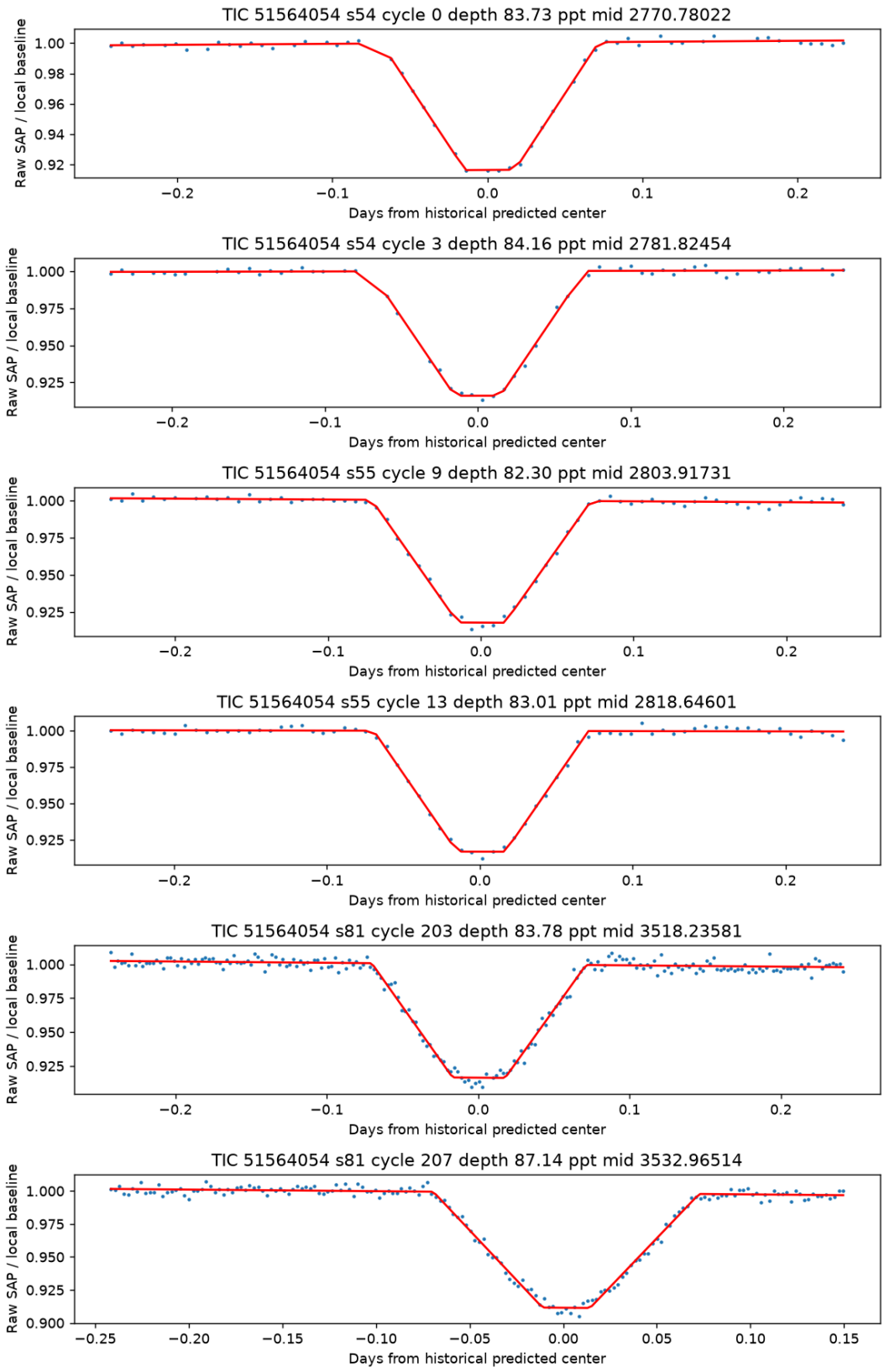
R502. TIC 51564054

physical_verdict: CONFIRMED

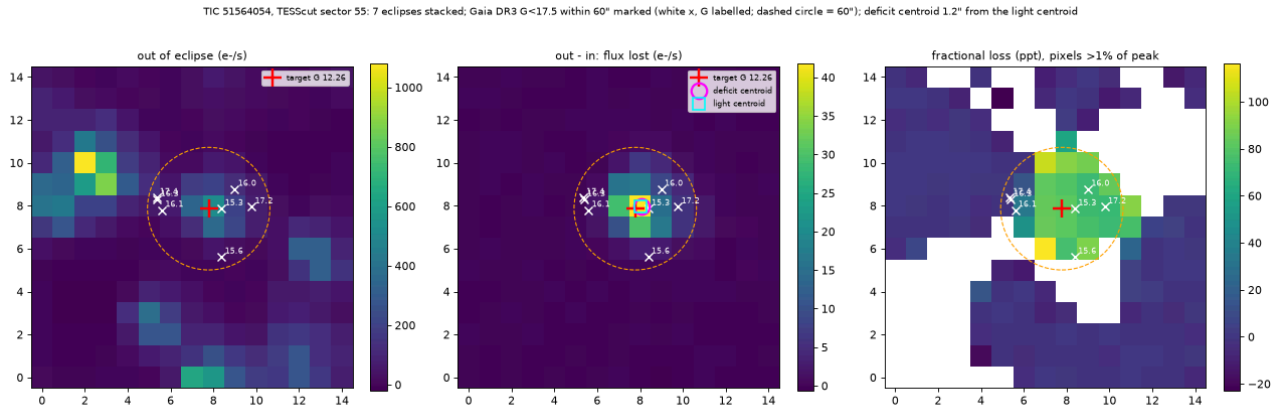
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R502; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R502; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 51564054: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 20 fitted representative eclipses in sectors [54, 55, 81]; 20 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 20 same-kind actual midpoints yield P = 3.6820558498 ± 0.0000053792 d and reference epoch BTJD 2770.7791920 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 55 cutout places the deficit centroid 6.58 arcsec from the target; peak pixel [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.567 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: None within 60 arcsec under the approximate G-band flux budget. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 81.07/80.04/77.97 ppt over 7 stacked events, compared with median selected QLP raw fit depth 83.28 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1805030134371636352 has P = 3.6821627017 d and e = 0.03959. The closest edge-on conjunction offset at the measured epoch is -0.09878 d (-0.02683 phase), with 0.16553 d linearly propagated full-SB1-covariance timing error. A rough central duration is 5.75 h versus observed median 3.39 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Twenty raw events across S54/55/81 recur at the measured period; the difference core lies on the target and no Gaia neighbor passes the approximate brightness screen.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
54	0	2770.7802194	0.003472	83.73	used
54	1	2774.4612189	0.003472	85.99	used
54	2	2778.1427547	0.003472	81.18	used
54	3	2781.8245378	0.003472	84.16	used
54	5	2789.1901975	0.003472	86.15	used
54	6	2792.8708408	0.003472	84.37	used
55	8	2800.2352509	0.003472	85.90	used
55	9	2803.9173093	0.003472	82.30	used
55	10	2807.5993533	0.003472	77.57	used
55	11	2811.2822047	0.003472	79.44	used
55	12	2814.9647274	0.003472	80.90	used
55	13	2818.6460093	0.003472	83.01	used
55	14	2822.3280330	0.003472	82.75	used
81	200	3507.1912335	0.001157	81.01	used
81	201	3510.8728408	0.001157	83.73	used
81	203	3518.2358094	0.001157	83.78	used
81	204	3521.9174461	0.001157	83.15	used
81	205	3525.6004124	0.001157	81.48	used
81	206	3529.2831125	0.001157	83.40	used
81	207	3532.9651413	0.001157	87.14	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s55.json, historical/diff_s55.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R502. Public evidence files: diff_s55.json, diff_s55.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

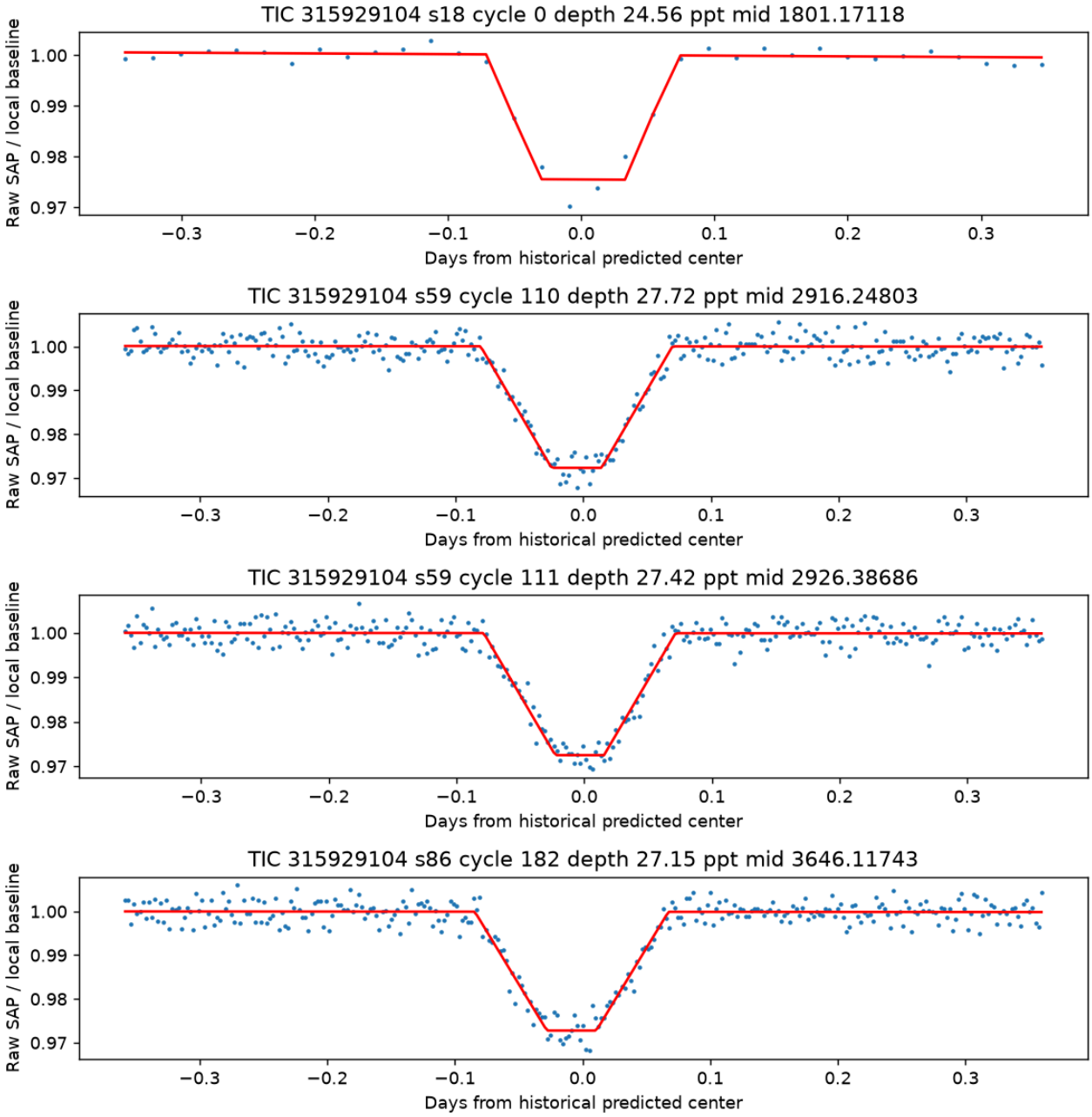
R503. TIC 315929104

physical_verdict: CONFIRMED

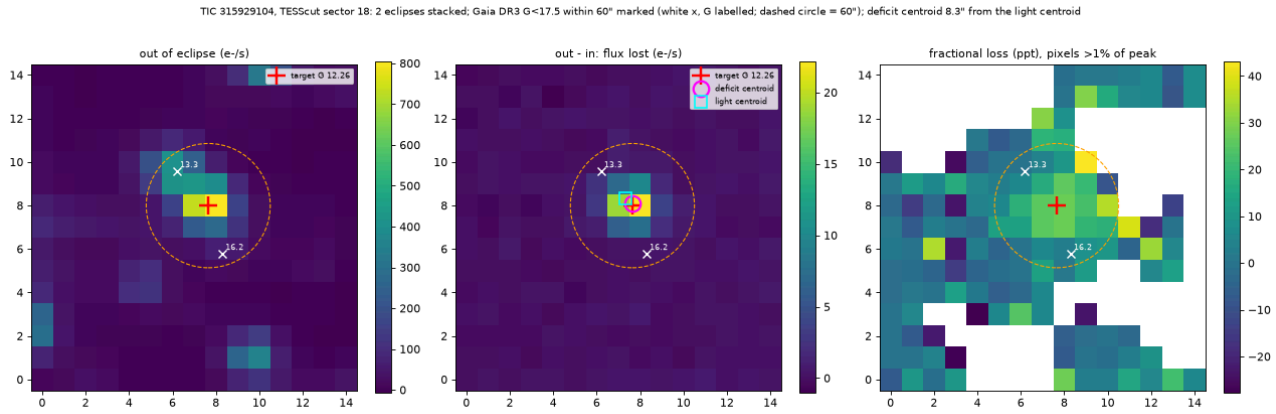
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R503; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R503; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 315929104: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 4 fitted representative eclipses in sectors [18, 59, 86]; 4 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 4 same-kind actual midpoints yield P = 10.1370634374 ± 0.0000192086 d and reference epoch BTJD 1801.1719141 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 18 cutout places the deficit centroid 1.70 arcsec from the target; peak pixel [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.460 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 449247978525955200: 43.0 arcsec, G 13.261, approximate intrinsic eclipse requirement 0.088; Gaia 449244959165948032: 46.1 arcsec, G 16.177, approximate intrinsic eclipse requirement 0.951. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 27.55/24.62/18.58 ppt over 2 stacked events, compared with median selected QLP raw fit depth 27.28 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 449245027885422592 has P = 10.1358765398 d and e = 0.08420. The closest edge-on conjunction offset at the measured epoch is 0.13186 d (0.01301 phase), with 0.16823 d linearly propagated full-

SB1-covariance timing error. A rough central duration is 8.05 h versus observed median 3.60 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Representative raw eclipses in S18/59/86 and a target-centered S18 difference core support the target; capable neighbors are tens of arcsec away.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
18	0	1801.1711761	0.010417	24.56	used
59	110	2916.2480272	0.001157	27.72	used
59	111	2926.3868563	0.001157	27.42	used
86	182	3646.1174333	0.001157	27.15	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s18.json, historical/diff_s18.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R503. Public evidence files: diff_s18.json, diff_s18.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

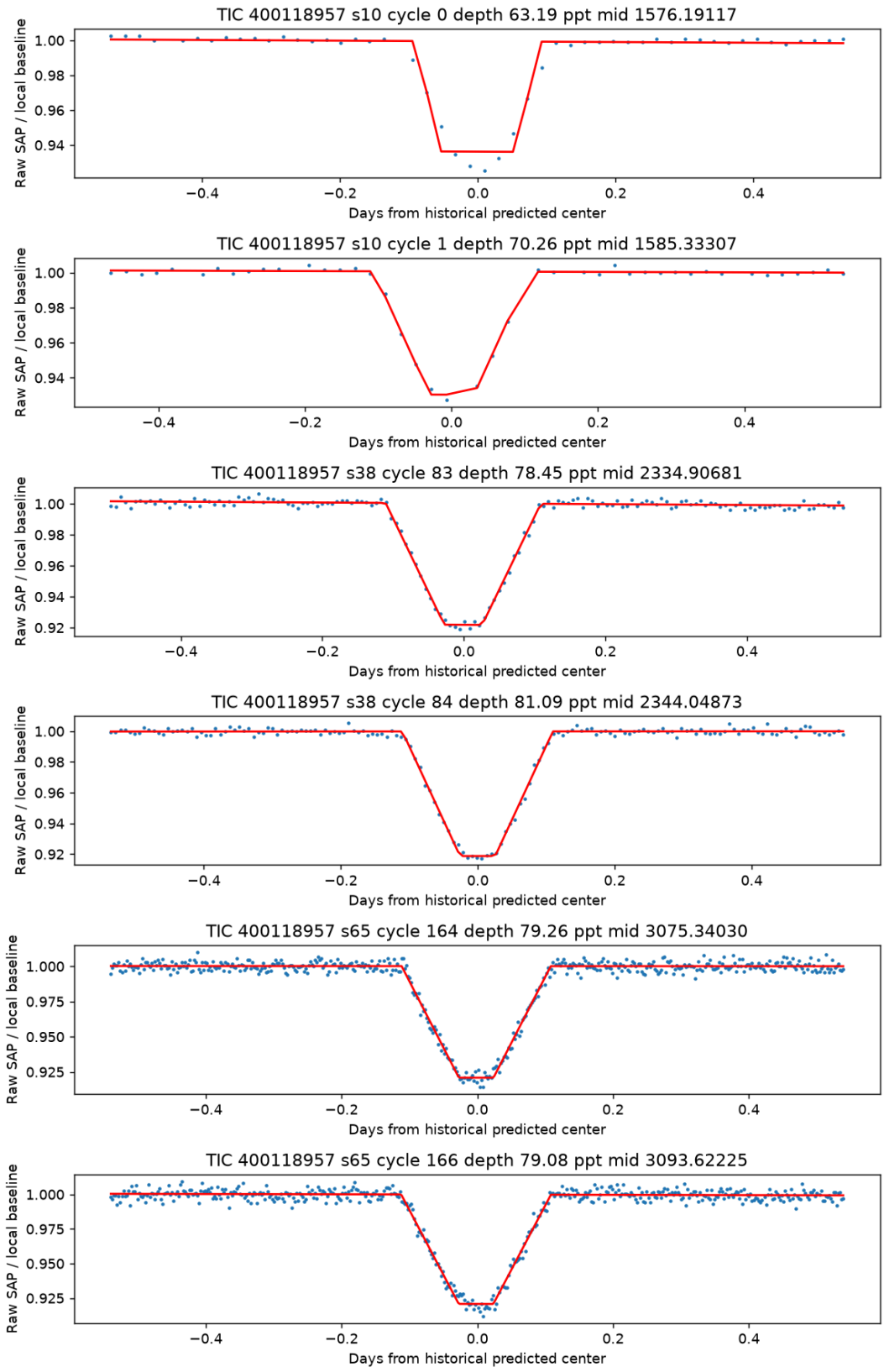
R504. TIC 400118957

physical_verdict: CONFIRMED

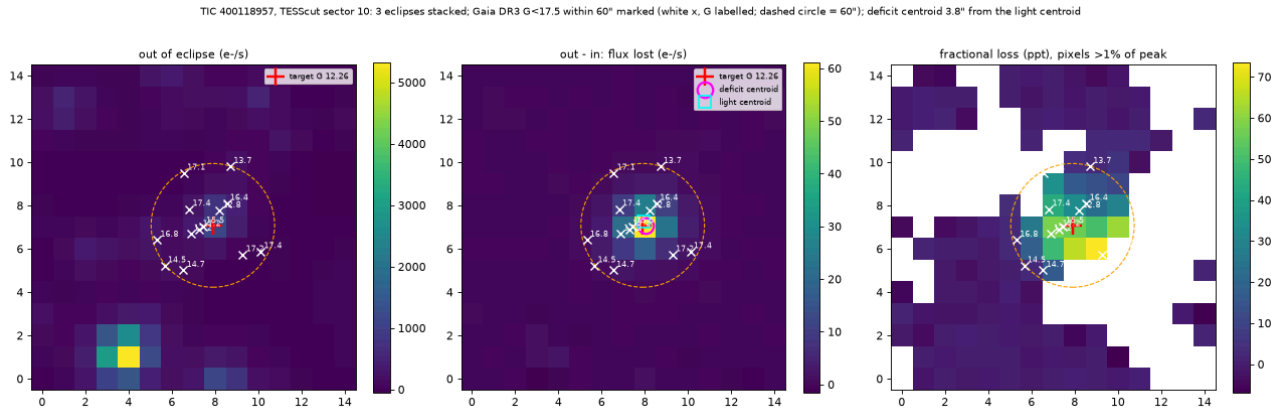
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R504; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R504; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 400118957: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 9 fitted representative eclipses in sectors [10, 38, 65]; 9 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 9 same-kind actual midpoints yield P = 9.1411474439 ± 0.0000216391 d and reference epoch BTJD 1576.1920454 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 10 cutout places the deficit centroid 2.32 arcsec from the target; peak pixel [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.514 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 5231653797315230336: 15.3 arcsec, G 12.789, approximate intrinsic eclipse requirement 0.188; Gaia 5231653969118047488: 50.8 arcsec, G 14.684, approximate intrinsic eclipse requirement 0.740; Gaia 5231653758651818240: 58.6 arcsec, G 13.675, approximate intrinsic eclipse requirement 0.336; Gaia 5231653969113936384: 59.0 arcsec, G 14.499, approximate intrinsic eclipse requirement 0.635. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 55.89/47.10/41.16 ppt over 3 stacked events, compared with median selected QLP raw fit depth 79.08 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5231653793011555840 has P = 9.1418612161 d and e = 0.18822. The closest edge-on conjunction

offset at the measured epoch is -0.06312 d (-0.00690 phase), with 0.11125 d linearly propagated full-SB1-covariance timing error. A rough central duration is 7.25 h versus observed median 5.27 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Raw depths change with sector aperture, while nine measured midpoints retain one recurrence. A capable neighbor 15.3 arcsec away is distinct from the target-centered difference core; crowding limits intrinsic depth claims.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
10	0	1576.1911659	0.010417	63.19	used
10	1	1585.3330745	0.010417	70.26	used
10	2	1594.4736317	0.010417	71.75	used
38	83	2334.9068107	0.003472	78.45	used
38	84	2344.0487326	0.003472	81.09	used
38	85	2353.1901072	0.003472	80.81	used
65	164	3075.3402986	0.001157	79.26	used
65	165	3084.4815508	0.001157	80.15	used
65	166	3093.6222526	0.001157	79.08	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s10.json, historical/diff_s10.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R504. Public evidence files: diff_s10.json, diff_s10.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

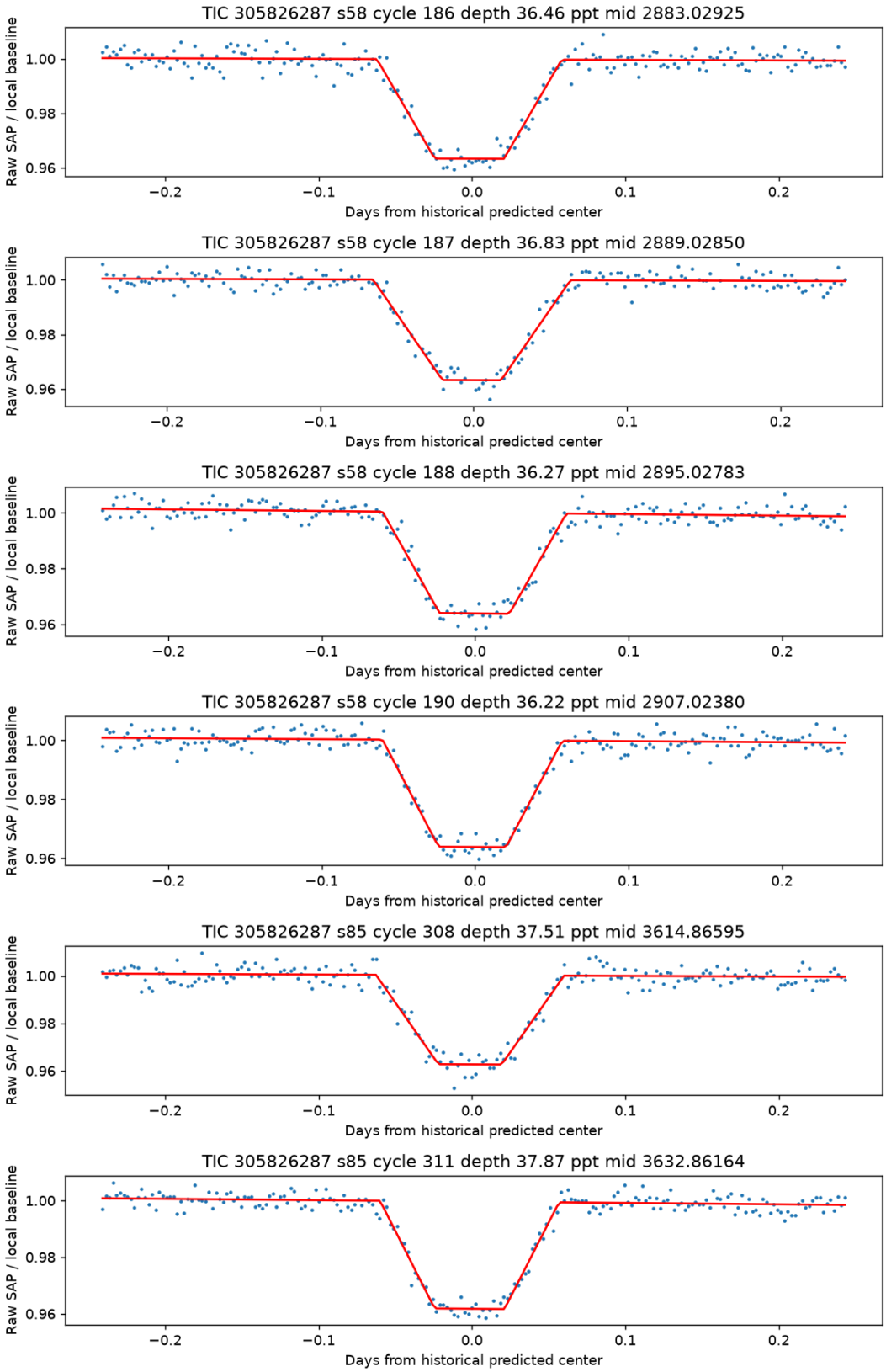
R505. TIC 305826287

physical_verdict: CONFIRMED

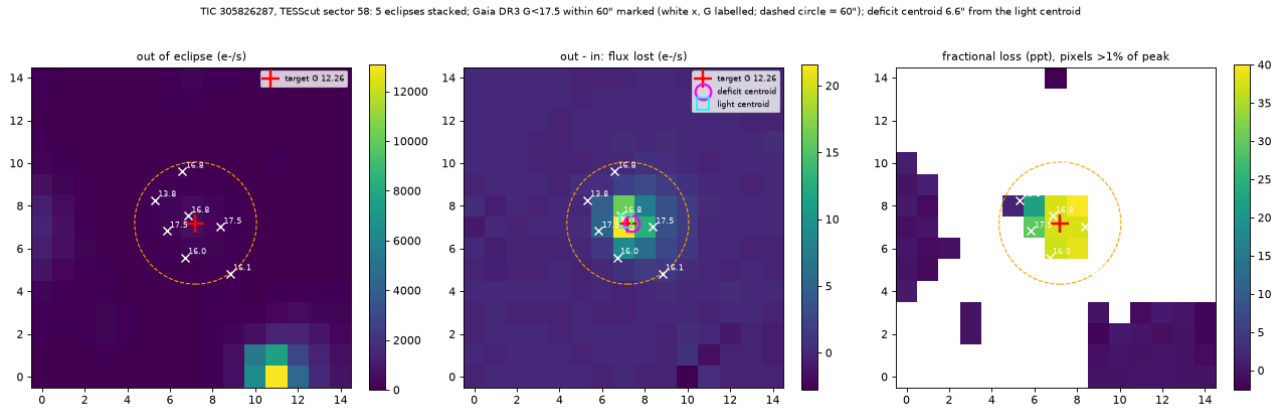
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R505; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R505; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 305826287: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 fitted representative eclipses in sectors [58, 85]; 8 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 8 same-kind actual midpoints yield P = 5.9986588262 ± 0.0000069644 d and reference epoch BTJD 2883.0297295 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 58 cutout places the deficit centroid 3.93 arcsec from the target; peak pixel [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.454 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 414950435582368640: 44.5 arcsec, G 13.760, approximate intrinsic eclipse requirement 0.186. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 37.50/35.53/32.80 ppt over 5 stacked events, compared with median selected QLP raw fit depth 36.65 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 414950435582368256 has P = 5.9989472733 d and e = 0.39238. The closest edge-on conjunction offset at the measured epoch is -0.14458 d (-0.02410 phase), with 0.64607 d linearly propagated full-SB1-covariance timing error. A rough central duration is 3.64 h versus observed median 2.88 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Eight raw events across S58/85 have stable depth and a target-centered difference core; the one capable listed neighbor is 44.5 arcsec away.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
58	186	2883.0292527	0.001157	36.46	used
58	187	2889.0284961	0.001157	36.83	used
58	188	2895.0278311	0.001157	36.27	used
58	189	2901.0258443	0.001157	36.28	used
58	190	2907.0238045	0.001157	36.22	used
85	308	3614.8659518	0.001157	37.51	used
85	309	3620.8653653	0.001157	37.55	used
85	311	3632.8616442	0.001157	37.87	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s58.json, historical/diff_s58.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R505. Public evidence files: diff_s58.json, diff_s58.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

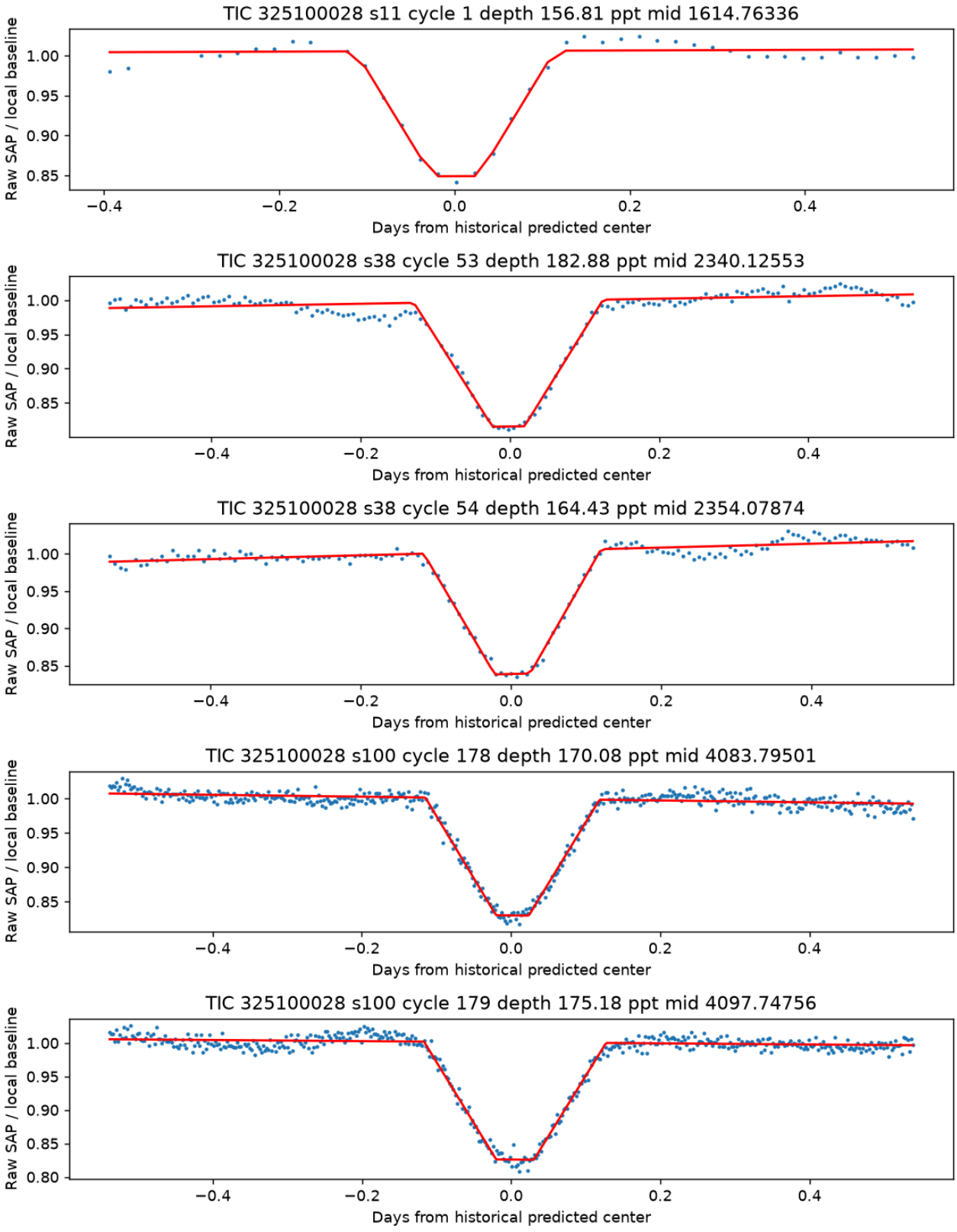
R506. TIC 325100028

physical_verdict: PROBABLE

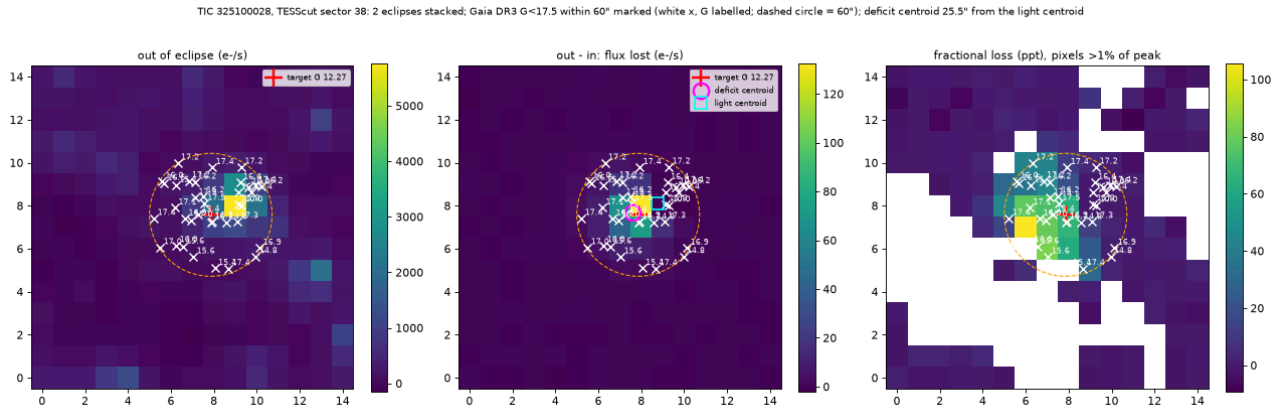
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse imaging and independent aperture/PRF characterization of the crowded field.



R506; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R506; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 325100028: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 fitted representative eclipses in sectors [11, 38, 100]; 5 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 5 same-kind actual midpoints yield P = 13.9493523851 ± 0.0000240409 d and reference epoch BTJD 1614.7612349 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: unresolved.** Historical sector 38 cutout places the deficit centroid 6.02 arcsec from the target; peak pixel [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.932 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 5864512143196081408: 12.3 arcsec, G 13.418, approximate intrinsic eclipse requirement 0.602; Gaia 5866013285774822528: 29.1 arcsec, G 11.757, approximate intrinsic eclipse requirement 0.252; Gaia 5866013285781269376: 31.6 arcsec, G 10.036, approximate intrinsic eclipse requirement 0.175. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: supports recurring loss with unresolved host.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 63.30/23.38/19.85 ppt over 2 stacked events, compared with median selected QLP raw fit depth 170.08 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5866013354494300928 has P = 13.9475152836 d and e = 0.22338. The closest edge-on conjunction

offset at the measured epoch is 0.13013 d (0.00933 phase), with 0.22414 d linearly propagated full-SB1-covariance timing error. A rough central duration is 7.05 h versus observed median 5.67 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Five raw eclipses, including S38 events coincident with the historical TESScut stack, recur at one period. The S38 cutout shows a central deficit but its 3x3 depths of about 23 ppt disagree substantially with QLP raw fits of 164-183 ppt; dilution is plausible but not quantitatively demonstrated. A capable 12.3-arcsec Gaia neighbor remains viable at TESS resolution, so the host is unresolved.

Next check: Resolved eclipse imaging and independent aperture/PRF characterization of the crowded field.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
11	1	1614.7633607	0.010417	156.81	used
38	53	2340.1255290	0.003472	182.88	used
38	54	2354.0787364	0.003472	164.43	used
100	178	4083.7950061	0.001157	170.08	used
100	179	4097.7475570	0.001157	175.18	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s38.json, historical/diff_s38.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R506. Public evidence files: diff_s38.json, diff_s38.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

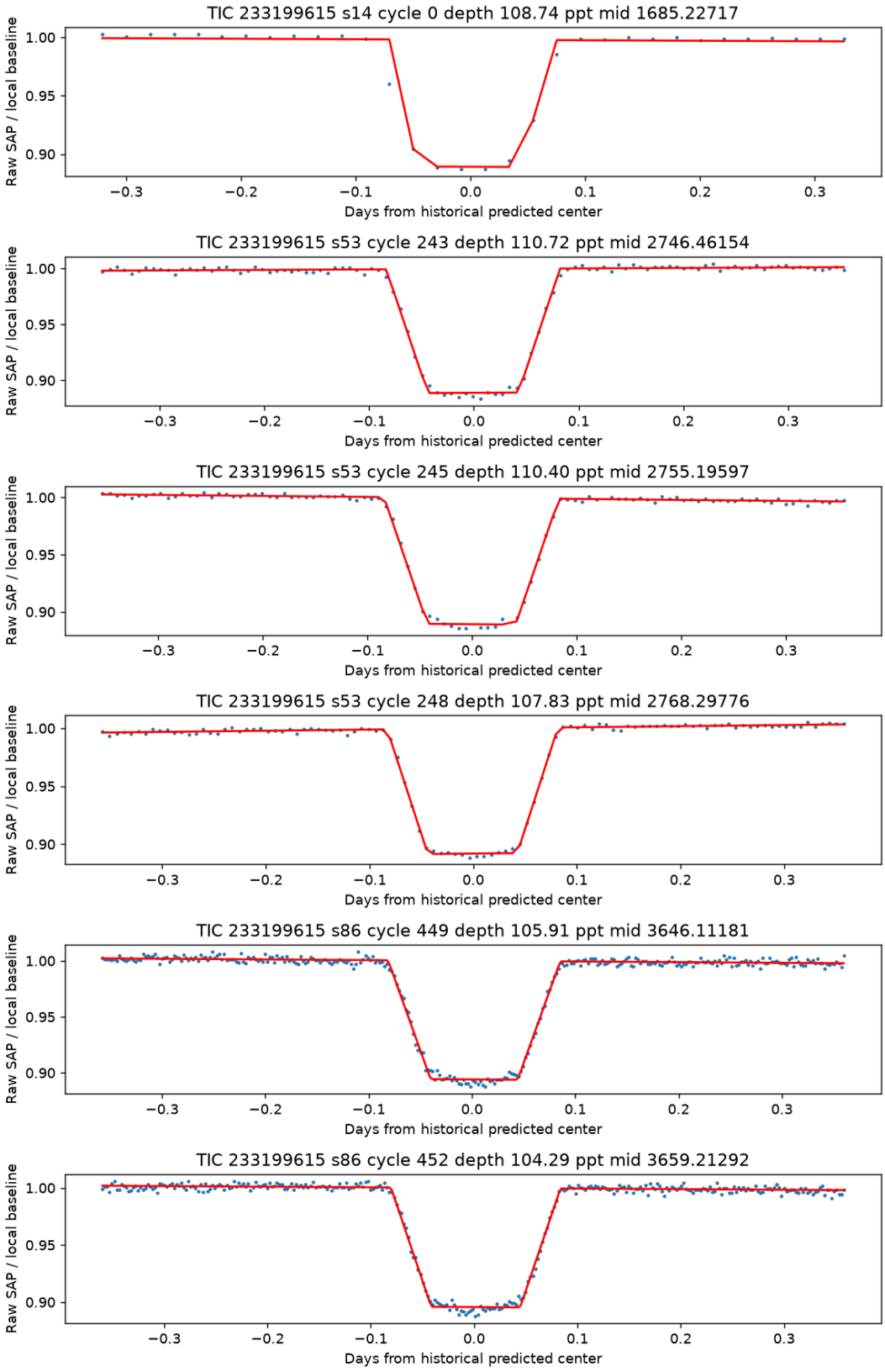
R507. TIC 233199615

physical_verdict: CONFIRMED

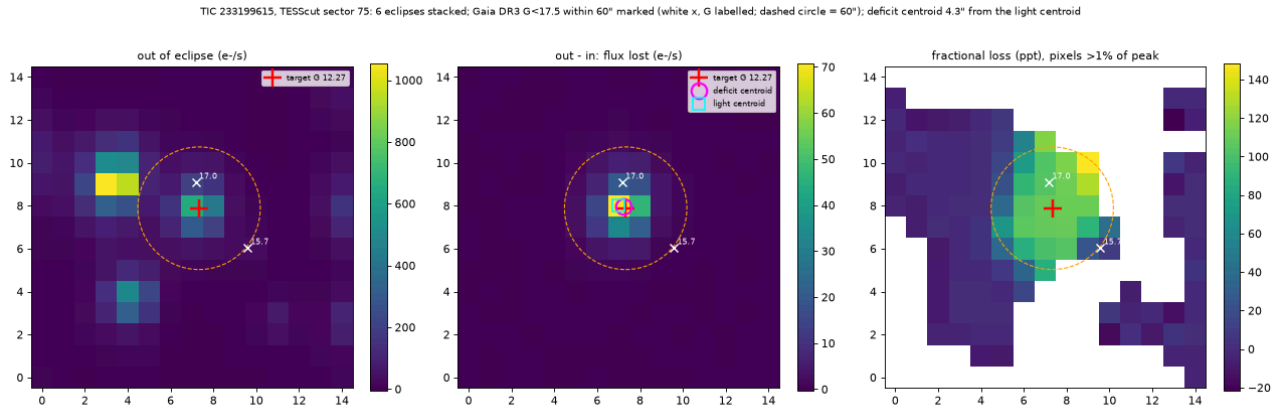
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R507; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R507; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 233199615: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 13 fitted representative eclipses in sectors [14, 53, 86]; 13 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 13 same-kind actual midpoints yield P = 4.3672327612 ± 0.0000067972 d and reference epoch BTJD 1685.2243747 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 75 cutout places the deficit centroid 3.05 arcsec from the target; peak pixel [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.884 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: None within 60 arcsec under the approximate G-band flux budget. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 109.47/108.17/97.70 ppt over 6 stacked events, compared with median selected QLP raw fit depth 107.83 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1638829129242739968 has P = 4.3672040925 d and e = 0.09279. The closest edge-on conjunction offset at the measured epoch is 0.05811 d (0.01331 phase), with 0.12907 d linearly propagated full-SB1-covariance timing error. A rough central duration is 4.84 h versus observed median 3.99 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Thirteen representative raw events across S14/53/86 repeat at the same period; the difference core is target-centered and no listed Gaia neighbor passes the approximate brightness screen.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
14	0	1685.2271693	0.010417	108.74	used
14	3	1698.3305035	0.010417	106.48	used
53	243	2746.4615423	0.003472	110.72	used
53	244	2750.8292655	0.003472	111.05	used
53	245	2755.1959690	0.003472	110.40	used
53	246	2759.5631421	0.003472	109.28	used
53	247	2763.9305927	0.003472	108.97	used
53	248	2768.2977583	0.003472	107.83	used
86	447	3637.3775447	0.001157	104.42	used
86	449	3646.1118101	0.001157	105.91	used
86	450	3650.4798914	0.001157	106.67	used
86	451	3654.8463029	0.001157	105.04	used
86	452	3659.2129190	0.001157	104.29	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s75.json, historical/diff_s75.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R507. Public evidence files: diff_s75.json, diff_s75.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

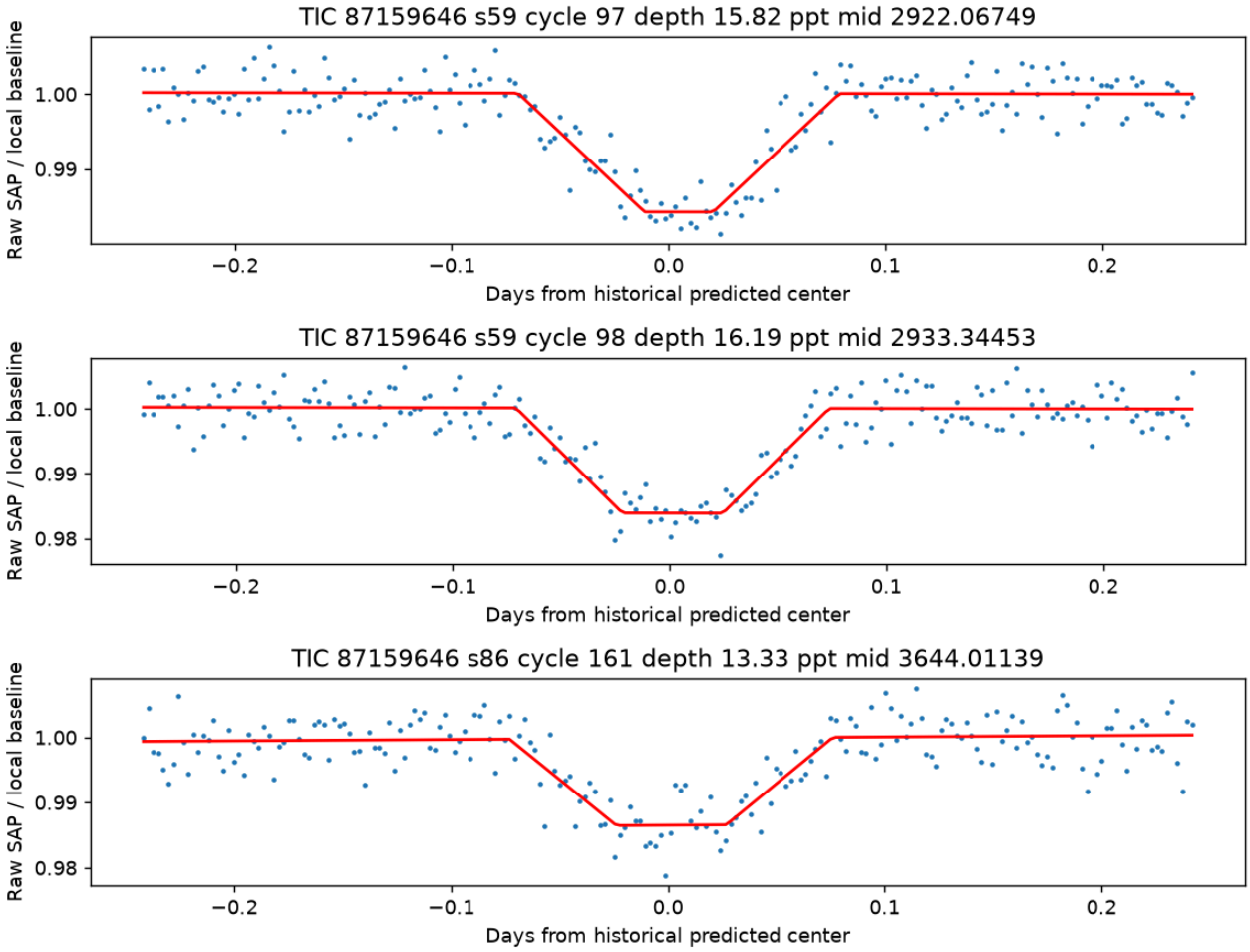
R508. TIC 87159646

physical_verdict: PROBABLE

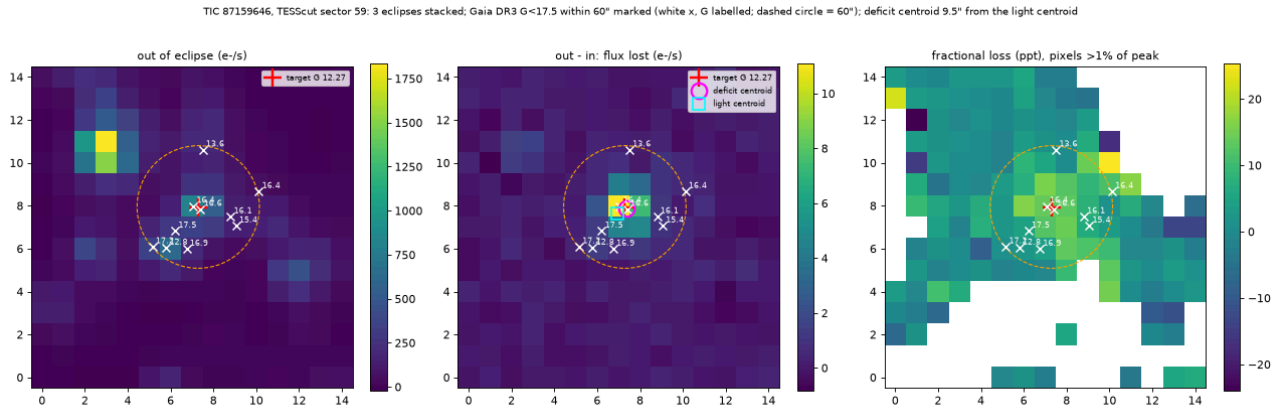
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry of the close Gaia sources.



R508; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R508; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 87159646: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 3 fitted representative eclipses in sectors [59, 86]; 3 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 3 same-kind actual midpoints yield P = 11.2803975769 ± 0.0000584060 d and reference epoch BTJD 2922.0658972 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: unresolved.** Historical sector 59 cutout places the deficit centroid 1.83 arcsec from the target; peak pixel [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.861 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 469818814585955328: 3.0 arcsec, G 16.593, approximate intrinsic eclipse requirement 0.873; Gaia 469818810292668416: 4.9 arcsec, G 16.391, approximate intrinsic eclipse requirement 0.728; Gaia 469820223341588480: 32.6 arcsec, G 16.069, approximate intrinsic eclipse requirement 0.545; Gaia 469820223341588864: 40.8 arcsec, G 15.394, approximate intrinsic eclipse requirement 0.300; Gaia 469818745872854528: 49.1 arcsec, G 12.762, approximate intrinsic eclipse requirement 0.041; Gaia 469819020750741888: 54.5 arcsec, G 13.609, approximate intrinsic eclipse requirement 0.071; Gaia 469820223341582208: 59.6 arcsec, G 16.350, approximate intrinsic eclipse requirement 0.702. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: supports recurring loss with unresolved host.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 13.67/12.75/8.71 ppt over 3 stacked events, compared with median selected QLP raw fit depth 15.82 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 469818814592320896 has P = 11.2828845399 d and e = 0.14903. The closest edge-on conjunction offset at the measured epoch is -0.57231 d (-0.05072 phase), with 0.55955 d linearly propagated full-SB1-covariance timing error. A rough central duration is 8.87 h versus observed median 3.56 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Three raw eclipses repeat in S59/86, but two capable Gaia neighbors at 3.0 and 4.9 arcsec share the central TESS pixel with the target. The S59 deficit cannot resolve the host.

Next check: Resolved eclipse photometry of the close Gaia sources.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
59	97	2922.0674903	0.001435	15.82	used
59	98	2933.3445278	0.001500	16.19	used
86	161	3644.0113934	0.002044	13.33	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s59.json, historical/diff_s59.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R508. Public evidence files: diff_s59.json, diff_s59.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

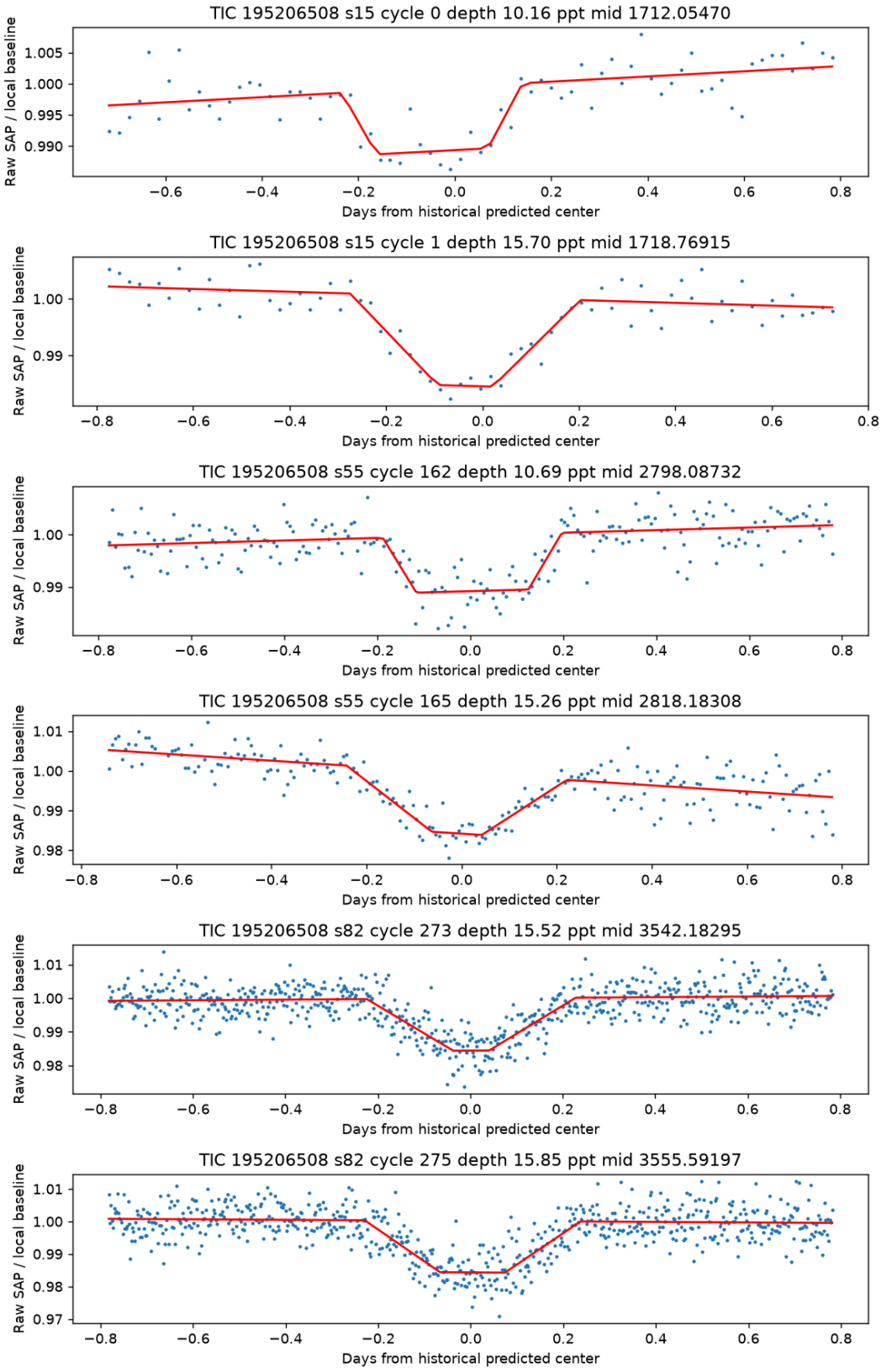
R509. TIC 195206508

physical_verdict: CONFIRMED

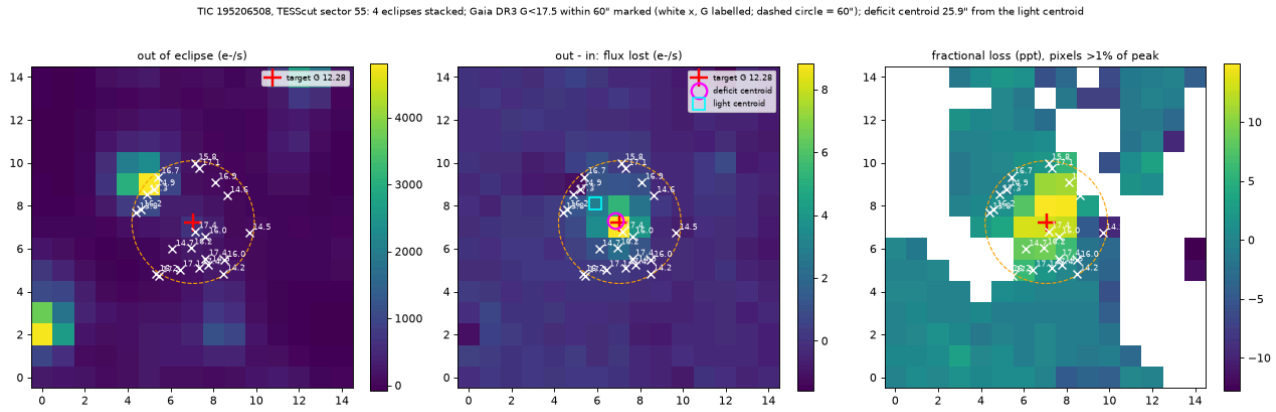
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved eclipse imaging and refined primary/secondary modeling.



R509; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R509; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 195206508: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 10 fitted representative eclipses in sectors [15, 55, 82]; 6 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 6 same-kind actual midpoints yield P = 13.4074536487 ± 0.0000554388 d and reference epoch BTJD 1718.7700294 (Deeper repeating eclipse, odd half-period cycles; primary by convention). Odd historical half-period cycles only; deeper repeating eclipse. Even cycles show a shallower secondary. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 55 cutout places the deficit centroid 4.32 arcsec from the target; peak pixel [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.998 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 1870085389842887808: 18.2 arcsec, G 16.013, approximate intrinsic eclipse requirement 0.405; Gaia 1870085389842888192: 24.0 arcsec, G 16.168, approximate intrinsic eclipse requirement 0.465; Gaia 1870085419893612288: 32.0 arcsec, G 14.661, approximate intrinsic eclipse requirement 0.125; Gaia 1870084801418161920: 40.4 arcsec, G 14.623, approximate intrinsic eclipse requirement 0.122; Gaia 1870085183684446336: 42.6 arcsec, G 16.907, approximate intrinsic eclipse requirement 0.906; Gaia 1870085389842895488: 43.0 arcsec, G 14.507, approximate intrinsic eclipse requirement 0.110; Gaia 1870085389842896384: 47.0 arcsec, G 16.029, approximate intrinsic eclipse requirement 0.411; Gaia 1870085149324698624: 49.8 arcsec, G 14.891, approximate intrinsic eclipse requirement 0.152; Gaia 1870085218044179712: 52.3 arcsec, G 16.164, approximate intrinsic eclipse requirement 0.463; Gaia 1870085149324696192: 55.0 arcsec, G 16.678, approximate intrinsic eclipse requirement 0.736; Gaia 1870085011885771648: 55.4 arcsec, G 14.453, approximate intrinsic eclipse requirement 0.106; Gaia 1870085114964960384: 55.6 arcsec, G 15.774, approximate intrinsic eclipse

requirement 0.327; Gaia 1870085218044179584: 55.8 arcsec, G 13.824, approximate intrinsic eclipse requirement 0.065; Gaia 1870085458562377600: 57.3 arcsec, G 14.222, approximate intrinsic eclipse requirement 0.088; Gaia 1870085424202627712: 59.9 arcsec, G 16.159, approximate intrinsic eclipse requirement 0.461. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.

4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 13.13/11.62/3.32 ppt over 4 stacked events, compared with median selected QLP raw fit depth 15.78 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1870085183684452864 has P = 13.4086220195 d and e = 0.03442. The closest edge-on conjunction offset at the measured epoch is -0.14065 d (-0.01049 phase), with 0.23479 d linearly propagated full-SB1-covariance timing error. A rough central duration is 27.40 h versus observed median 10.86 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Alternating shallow even and deeper odd eclipses across S15/55/82 set a 13.407-day orbital period. The deeper odd cycles alone define the reported epoch and timing fit. The S55 cutout has a central target deficit, but high image-model residual and neighboring field stars limit subpixel localization.

Next check: Optional resolved eclipse imaging and refined primary/secondary modeling.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
15	0	1712.0546966	0.012570	10.16	excluded from orbital fit (secondary)
15	1	1718.7691468	0.010417	15.70	used
15	3	1732.1783285	0.010417	16.58	used
55	162	2798.0873194	0.004654	10.69	excluded from orbital fit (secondary)
55	163	2804.7689867	0.014417	20.51	used
55	165	2818.1830818	0.008534	15.26	used
82	272	3535.4816843	0.006814	12.26	excluded from orbital fit (secondary)
82	273	3542.1829509	0.003761	15.52	used
82	274	3548.8921402	0.004545	12.21	excluded from orbital fit (secondary)
82	275	3555.5919677	0.003868	15.85	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s55.json, historical/diff_s55.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R509. Public evidence files: diff_s55.json, diff_s55.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

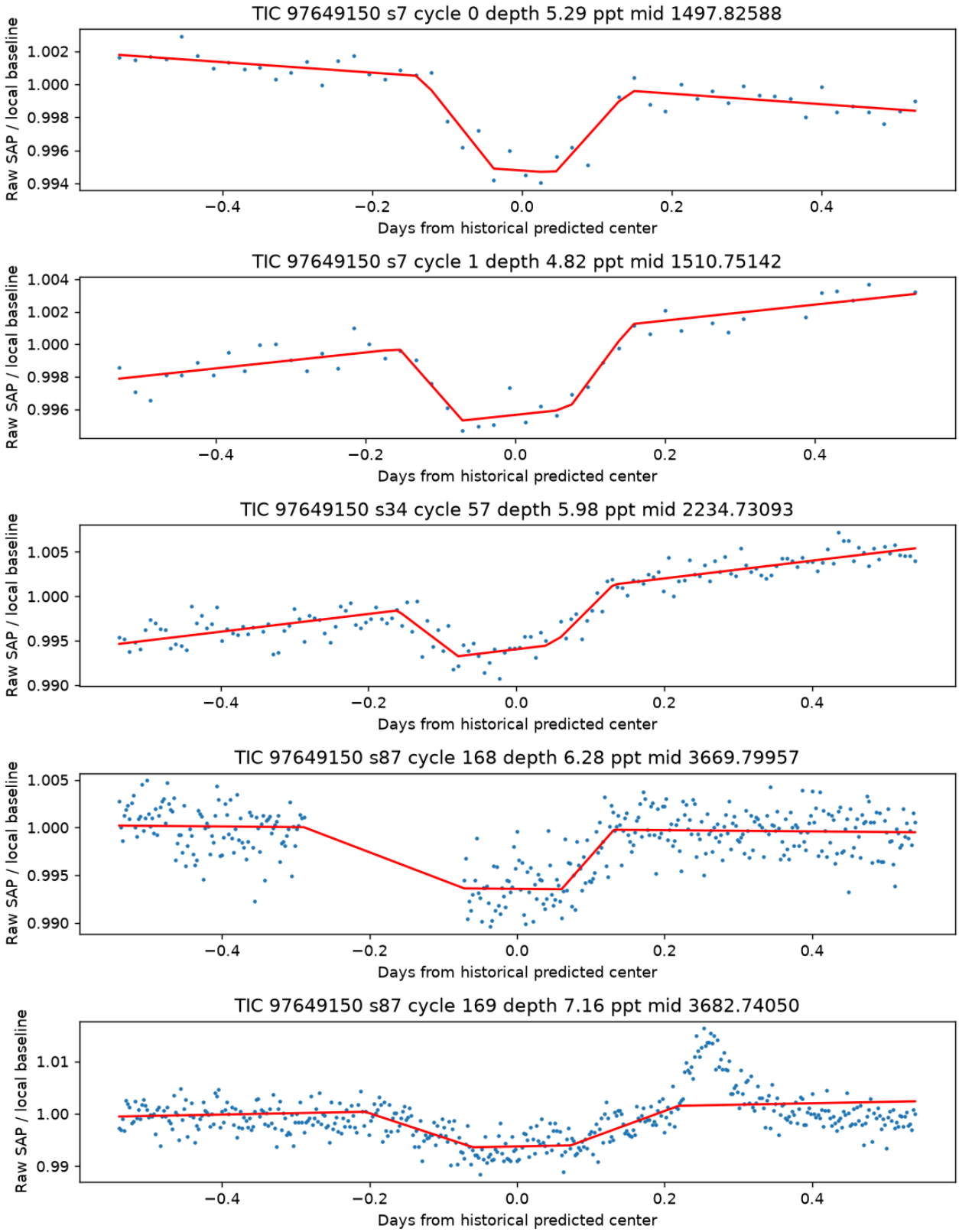
R510. TIC 97649150

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved time-series photometry.



R510; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 97649150: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 4 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [7, 34, 87]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 12.9285082280 ± 0.0000582502 d and epoch BTJD 1497.82197689 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 7 deficit centroid is about 2.44 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 6 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\97649150\diff_s07.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S7 aperture photometry recovers 2 stacked events at 3x3 depth 5.75 ppt, versus median selected raw QLP fit depth 5.63 ppt. TESScut S7 and representative QLP S7 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 5603413453346701184: orbital P = 12.927076847 d, e = 0.2541. Closest edge-on conjunction offset at the measured epoch is +0.3231 d (+0.0250 phase); linear full-NSS-covariance timing scale 0.3629 d. Approximate central duration 11.53 h versus measured median 7.02 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S7/34/87 raw eclipses agree. The closest capable neighbor is G17.24 at 13.7 arcsec and would need roughly 54% intrinsic loss; deficit is within 0.12 pixel of target. Faint close-source ambiguity remains below calibrated pixel precision.

Next check: Optional resolved time-series photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
7	0	1497.8258819	0.010417	5.29
7	1	1510.7514174	0.014168	4.82
34	57	2234.7309255	0.016170	5.98
87	169	3682.7404985	0.005530	7.16

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\97649150\diff_s07.json`, its PNG and Gaia-neighbor table; source and artifact hashes are

in `../SHA256SUMS.json`.

Stable review key: R510. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 97649150.png

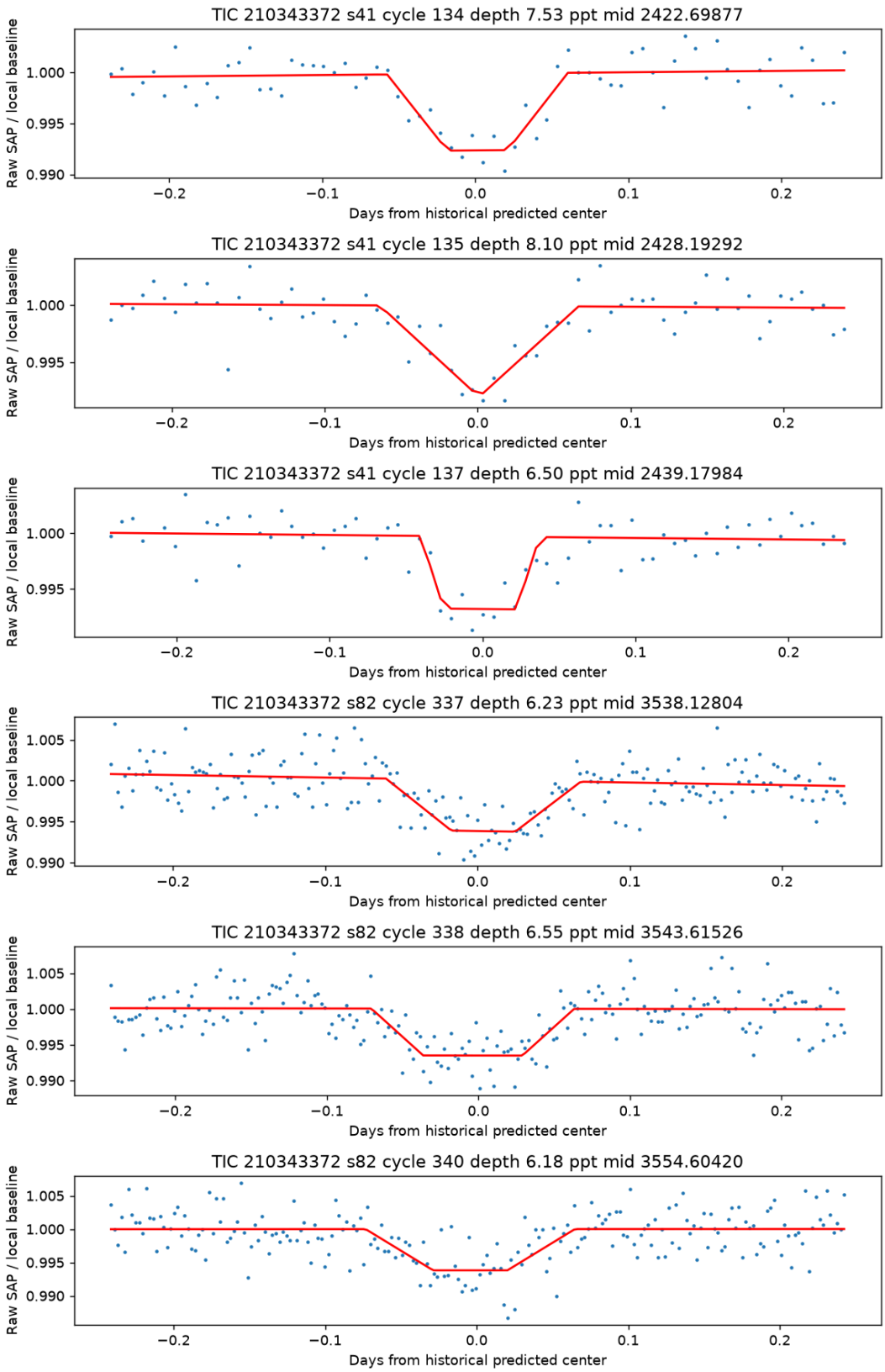
R511. TIC 210343372

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved time-series photometry.



R511; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 210343372: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 8 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [41, 82]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 5.4946996557 ± 0.0000125202 d and epoch BTJD 2422.69788125 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 55 deficit centroid is about 4.07 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 10 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\210343372\diff_s55.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S55 aperture photometry recovers 5 stacked events at 3x3 depth 5.84 ppt, versus median selected raw QLP fit depth 7.04 ppt. TESScut S55 and representative QLP sectors [41, 82] are separate TESS observing occasions on the same instrument; their photons are distinct. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 1842372749004992000: orbital P = 5.493566815 d, e = 0.0368. Closest edge-on conjunction offset at the measured epoch is +0.2292 d (+0.0417 phase); linear full-NSS-covariance timing scale 0.2725 d. Approximate central duration 8.20 h versus measured median 3.10 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S41/82 provide consecutive raw repeats. The closest capable G16.03 source lies 17.8 arcsec away and would require about 22% intrinsic loss; deficit is within 0.20 pixel of target. The Gaia phase offset is descriptive and within its large propagated orbital uncertainty.

Next check: Optional resolved time-series photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
41	134	2422.6987727	0.003472	7.53
41	135	2428.1929206	0.003472	8.10
41	137	2439.1798373	0.003472	6.50
41	138	2444.6774781	0.003472	8.89
82	337	3538.1280403	0.002663	6.23
82	338	3543.6152633	0.002550	6.55
82	339	3549.1081336	0.002577	9.30

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
82	340	3554.6042030	0.003268	6.18

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\210343372\diff_s55.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R511. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 210343372.png

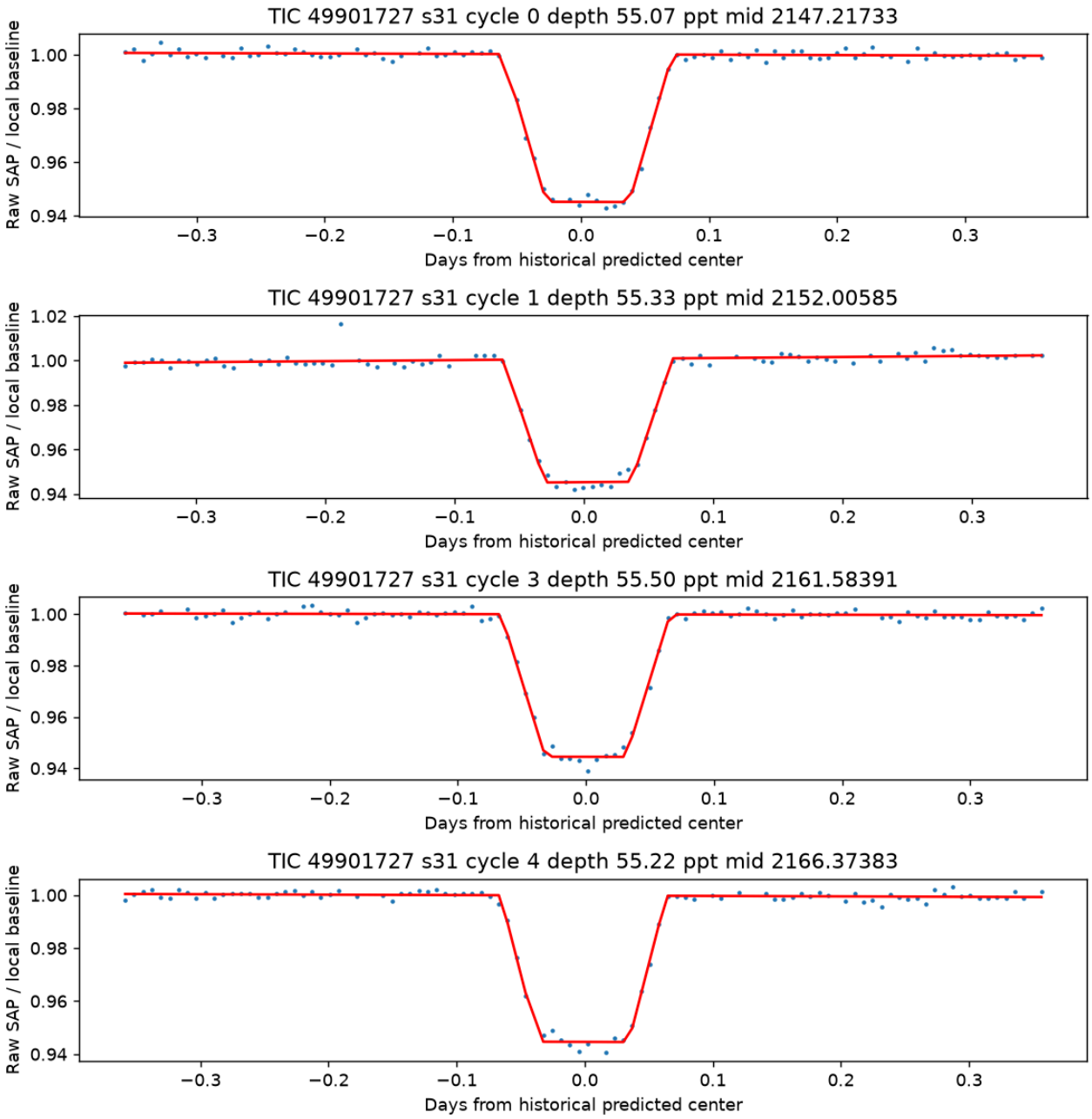
R512. TIC 49901727

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional-sector timing would refine the period.



R512; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 49901727: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 4 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [31]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 4.7891053447 ± 0.0010980248 d and epoch BTJD 2147.21701942 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 31 deficit centroid is about 2.74 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\49901727\diff_s31.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S31 aperture photometry recovers 4 stacked events at 3x3 depth 54.10 ppt, versus median selected raw QLP fit depth 55.27 ppt. TESScut S31 and representative QLP S31 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 3266000059384203136: orbital P = 4.788914677 d, e = 0.0281. Closest edge-on conjunction offset at the measured epoch is +0.1153 d (+0.0241 phase); linear full-NSS-covariance timing scale 0.3693 d. Approximate central duration 3.65 h versus measured median 3.15 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Four same-sector raw eclipses establish recurrence and a photometric period, but the short single-sector baseline leaves a comparatively broad period error. No capable Gaia neighbor within 60 arcsec; target-centered difference image.

Next check: Additional-sector timing would refine the period.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
31	0	2147.2173324	0.003472	55.07
31	1	2152.0058502	0.003472	55.33
31	3	2161.5839073	0.003472	55.50
31	4	2166.3738305	0.003472	55.22

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\49901727\diff_s31.json`, its PNG and Gaia-neighbor table; source and artifact hashes are

in `../SHA256SUMS.json`.

Stable review key: R512. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 49901727.png

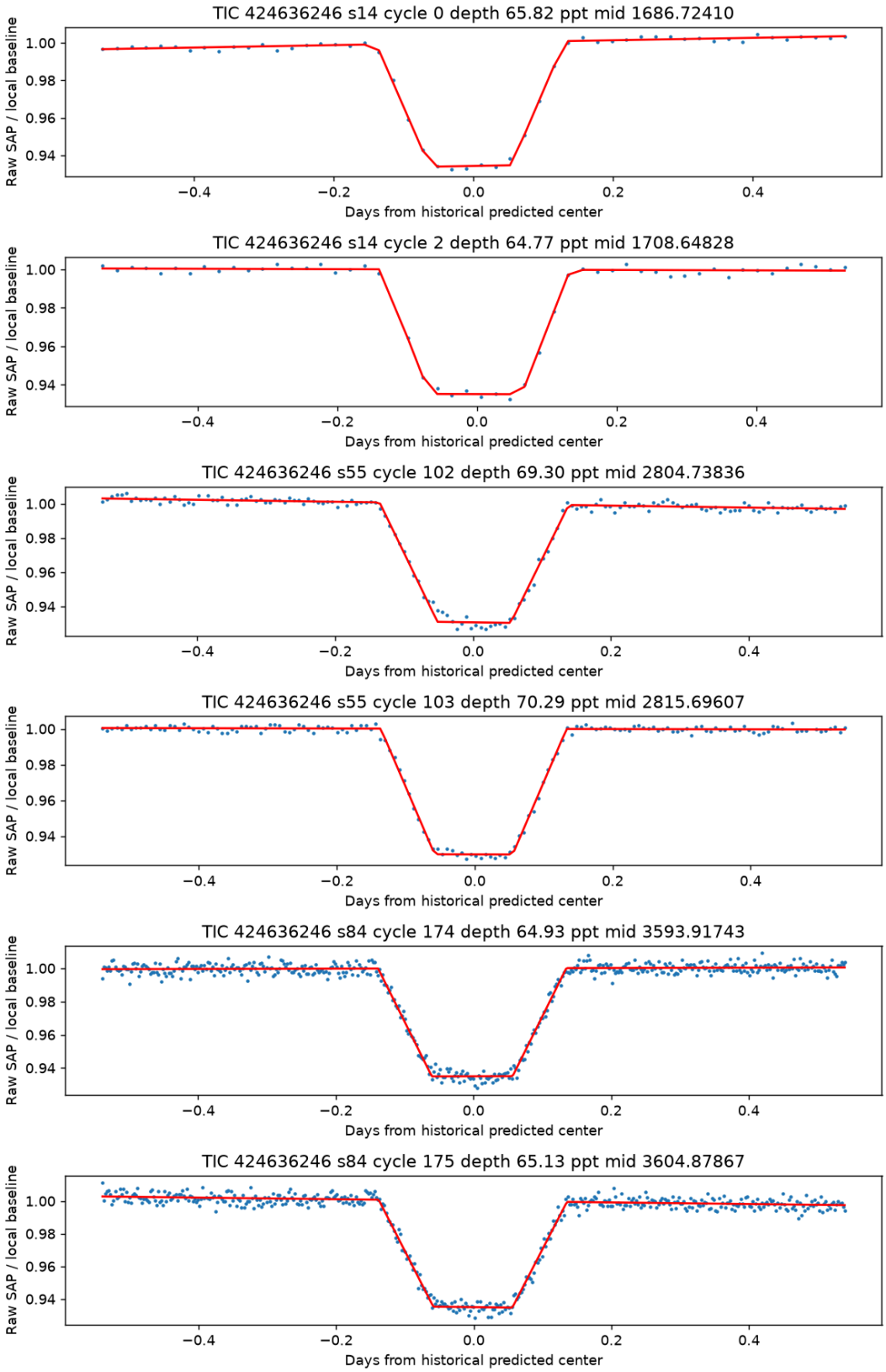
R513. TIC 424636246

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R513; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 424636246: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 6 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [14, 55, 84]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 10.9608574269 ± 0.0000279668 d and epoch BTJD 1686.72848166 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 14 deficit centroid is about 3.21 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\424636246\diff_s14.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S14 aperture photometry recovers 3 stacked events at 3x3 depth 64.40 ppt, versus median selected raw QLP fit depth 65.48 ppt. TESScut S14 and representative QLP S14 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 2139000984511908864: orbital P = 10.961251824 d, e = 0.0115. Closest edge-on conjunction offset at the measured epoch is -0.0947 d (-0.0086 phase); linear full-NSS-covariance timing scale 0.2147 d. Approximate central duration 8.37 h versus measured median 6.53 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S14/55/84 raw repeats and target-centered deficit. Capable catalogued neighbors are at least 50.6 arcsec away and need large intrinsic losses; no nearby source is favored by the deficit image.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
14	0	1686.7241028	0.010417	65.82
14	2	1708.6482791	0.010417	64.77
55	102	2804.7383553	0.003472	69.30
55	103	2815.6960722	0.003472	70.29
84	174	3593.9174280	0.001157	64.93
84	175	3604.8786671	0.001157	65.13

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\424636246\diff_s14.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R513. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 424636246.png

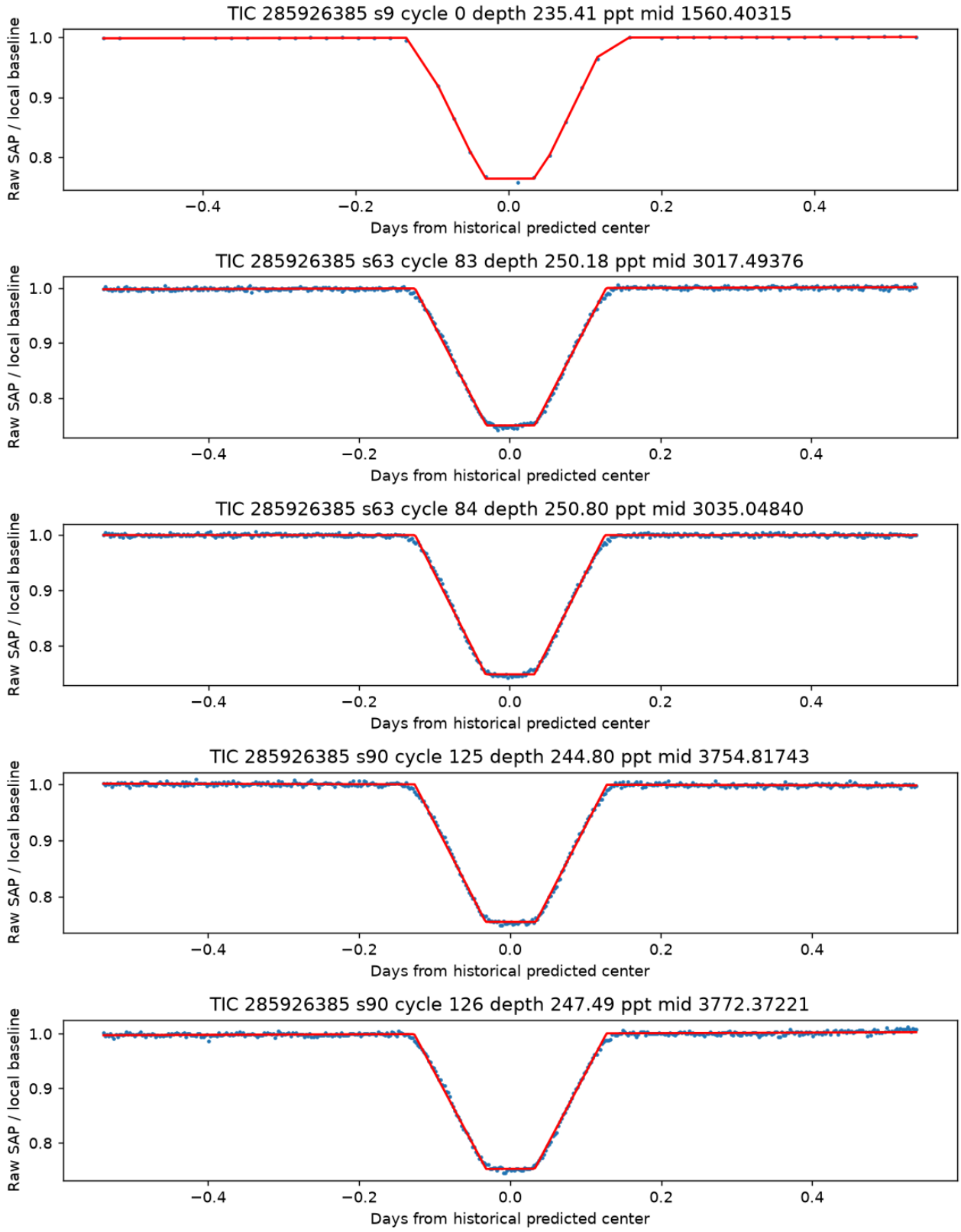
R514. TIC 285926385

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R514; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 285926385: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 5 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [9, 63, 90]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 17.5553255431 ± 0.0000265567 d and epoch BTJD 1560.40143870 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 63 deficit centroid is about 3.07 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\285926385\diff_s63.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S63 aperture photometry recovers 2 stacked events at 3x3 depth 246.47 ppt, versus median selected raw QLP fit depth 247.49 ppt. TESScut S63 and representative QLP S63 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 3589729714627304192: orbital P = 17.554087545 d, e = 0.4756. Closest edge-on conjunction offset at the measured epoch is +0.2965 d (+0.0169 phase); linear full-NSS-covariance timing scale 0.7498 d. Approximate central duration 8.01 h versus measured median 6.09 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Deep S9/63/90 raw eclipses recur at the measured period. Difference image favors the target and lists no capable Gaia neighbor within 60 arcsec.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
9	0	1560.4031478	0.010417	235.41
63	83	3017.4937647	0.001157	250.18
63	84	3035.0484007	0.001157	250.80
90	125	3754.8174316	0.001157	244.80
90	126	3772.3722138	0.001157	247.49

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\285926385\diff_s63.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R514. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 285926385.png

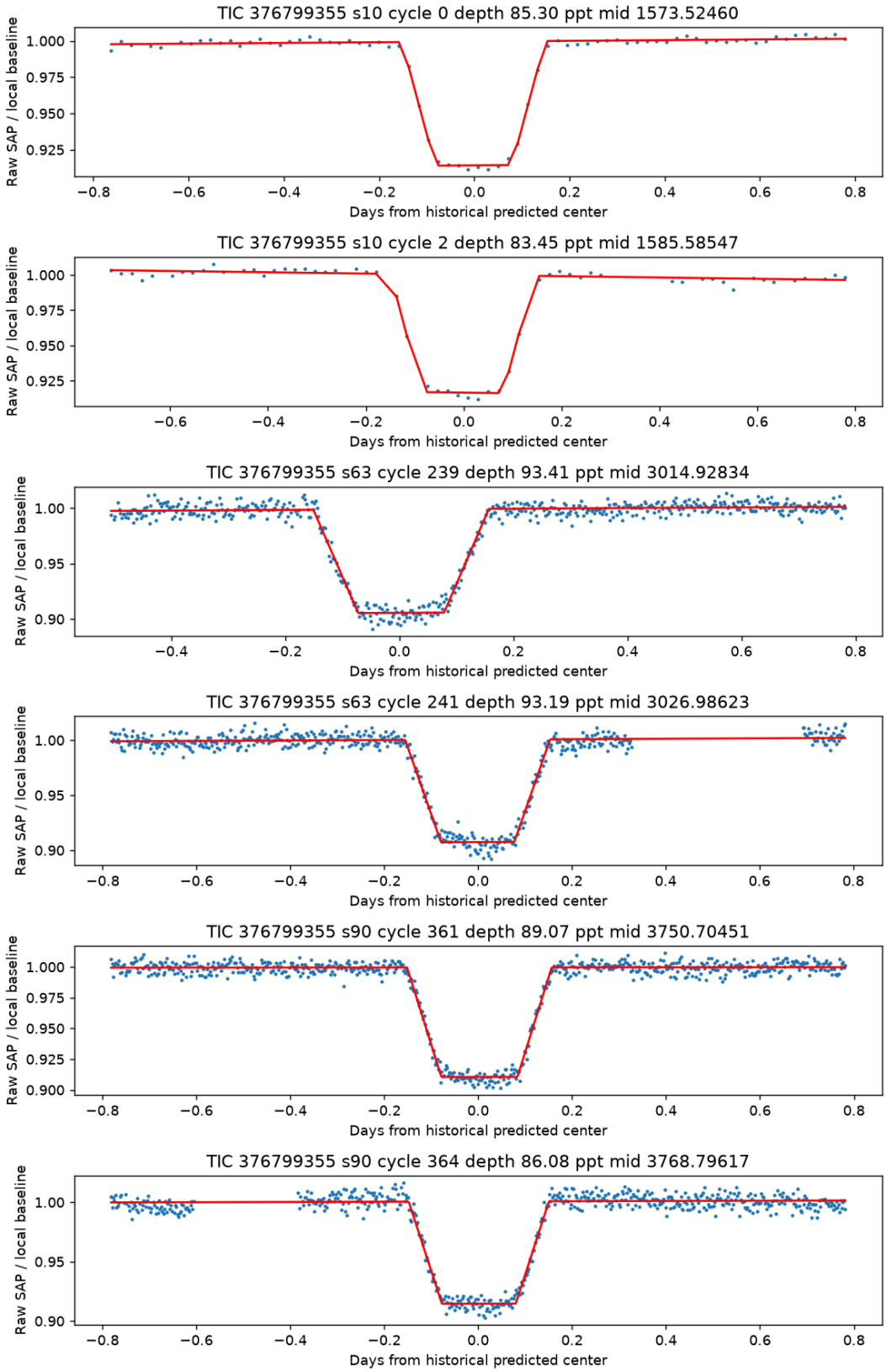
R515. TIC 376799355

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry across an eclipse or an independent high-resolution centroid.



R515; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 376799355: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 11 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [10, 63, 90]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 6.0309715740 ± 0.0000063815 d and epoch BTJD 1573.52342898 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 10 deficit centroid is about 3.49 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 6 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\376799355\diff_s10.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S10 aperture photometry recovers 4 stacked events at 3x3 depth 36.28 ppt, versus median selected raw QLP fit depth 87.69 ppt. TESScut S10 and representative QLP S10 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 5252574583035692416: orbital P = 6.030792509 d, e = 0.1048. Closest edge-on conjunction offset at the measured epoch is +0.0241 d (+0.0040 phase); linear full-NSS-covariance timing scale 0.3152 d. Approximate central duration 7.35 h versus measured median 7.30 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Raw eclipses repeat in S10/63/90 and the alternative 3x3 aperture also shows a deficit, diluted from roughly 88 ppt QLP to 36 ppt. The field is crowded: capable Gaia sources at 17.4 and 32.0 arcsec remain; the deficit centroid is only approximately target-centered and has no calibrated error. Attribution is unresolved.

Next check: Resolved photometry across an eclipse or an independent high-resolution centroid.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
10	0	1573.5245981	0.010417	85.30
10	2	1585.5854685	0.010417	83.45
10	3	1591.6168476	0.010417	86.74
63	239	3014.9283389	0.001157	93.41
63	240	3020.9555063	0.001157	95.32
63	241	3026.9862304	0.001157	93.19
63	242	3033.0181715	0.001157	93.14

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
90	361	3750.7045114	0.001157	89.07
90	362	3756.7359506	0.001157	86.28
90	363	3762.7659660	0.001157	87.69
90	364	3768.7961714	0.001157	86.08

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\376799355\diff_s10.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R515. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 376799355.png

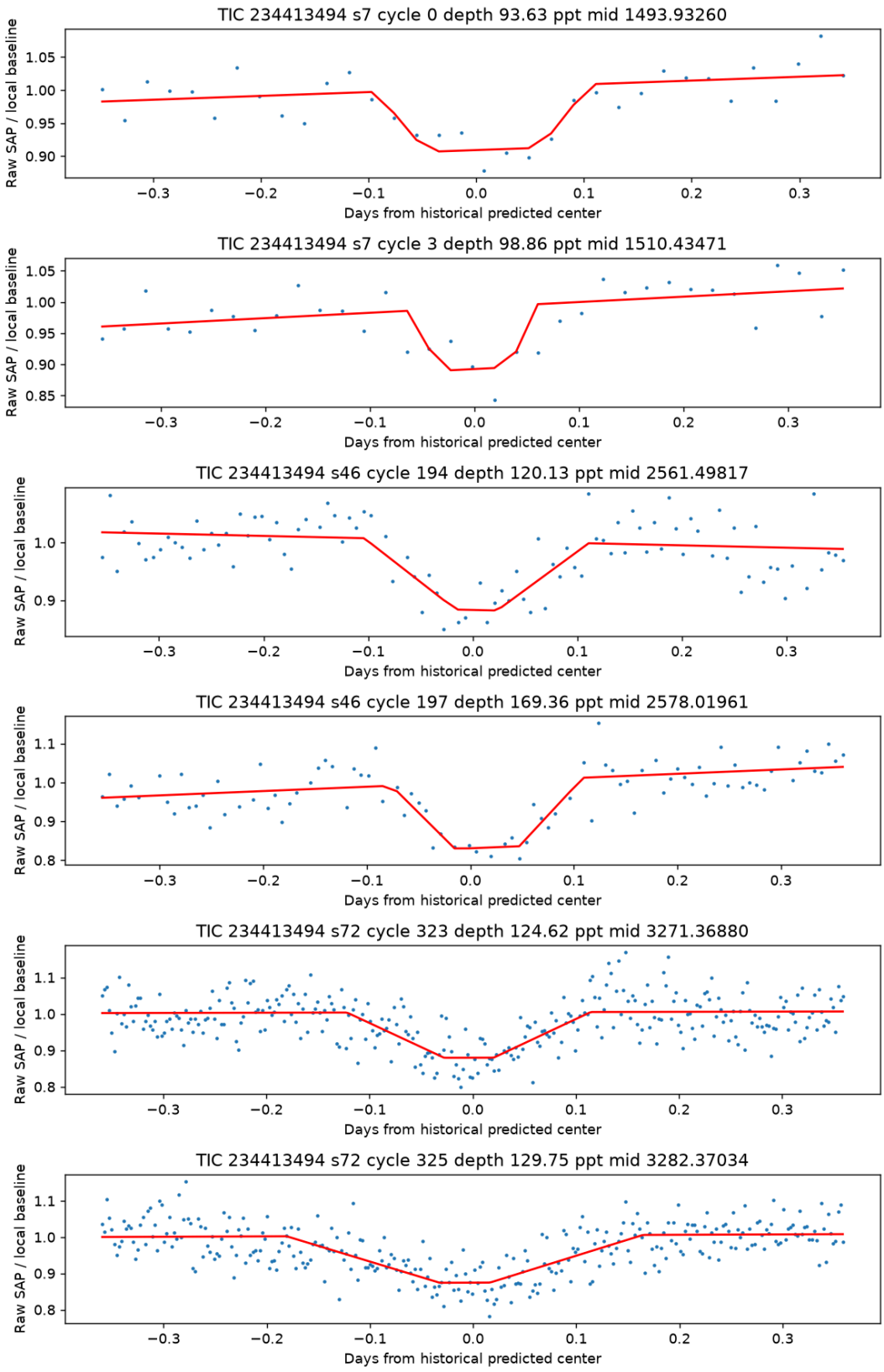
R516. TIC 234413494

physical_verdict: PROBABLE

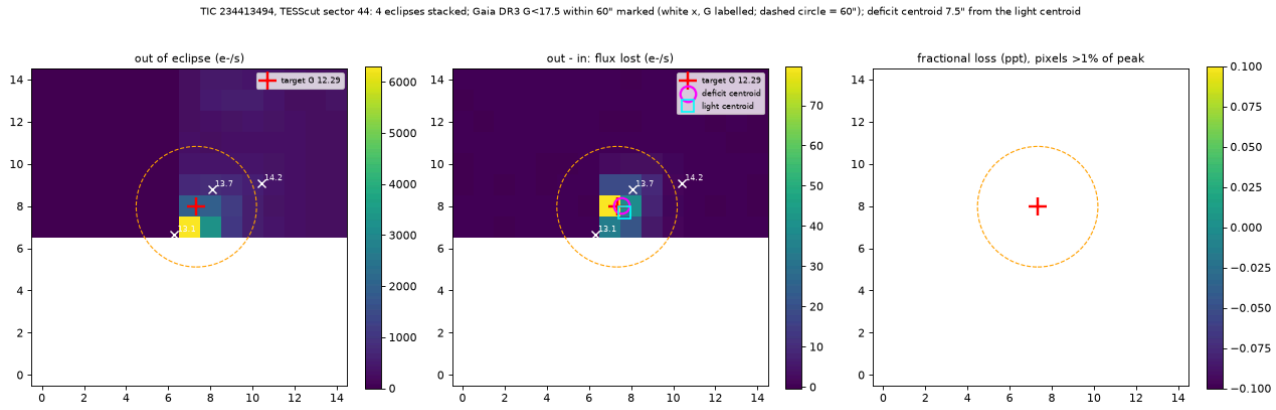
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Acquire a complete cutout or resolved eclipse photometry.



R516; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R516; Additional sector-44 eclipse-deficit stack, a separate TESS observing occasion from the historical sector.

Archived source: sanitized report text.

TIC 234413494: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 11 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [7, 46, 72]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 5.5028598185 ± 0.0000251813 d and epoch BTJD 1493.93985587 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 44 deficit centroid is about 4.75 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted]
4. **Alternative reduction.** Separately extracted S44 TESScut aperture photometry recovers 4 events at nominal 3x3 depth 14.03 ppt. The cutout is truncated, so this establishes a repeated signal in another extraction but cannot settle attribution or precise depth. TESScut S44 and representative QLP sectors [7, 46, 72] are separate TESS observing occasions on the same instrument; their photons are distinct.
5. **Gaia geometry.** DR3 source 3161704273321832064: orbital P = 5.503171903 d, e = 0.1043. Closest edge-on conjunction offset at the measured epoch is +0.0729 d (+0.0133 phase); linear full-NSS-covariance timing scale 0.1654 d. Approximate central duration 6.85 h versus measured median 4.76 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Raw eclipses repeat in S7/46/72. A separate S44 TESScut extraction recovers four events, but the available cutout is truncated and its peak/residual panel is corrupted. Capable neighbors at 19.3 and 30.4 arcsec remain, so attribution is unresolved.

Next check: Acquire a complete cutout or resolved eclipse photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
7	0	1493.9326038	0.010417	93.63
7	1	1499.4446516	0.010417	108.71
7	3	1510.4347053	0.010417	98.86
46	193	2555.9900823	0.004631	127.18
46	194	2561.4981707	0.006435	120.13
46	196	2572.4953342	0.007194	167.97
46	197	2578.0196107	0.005541	169.36
72	322	3265.8641418	0.003535	126.10
72	323	3271.3688033	0.004662	124.62
72	324	3276.8594444	0.002925	152.48
72	325	3282.3703365	0.005247	129.75

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted]`, its PNG and Gaia-neighbor table; `source and artifact hashes are in../SHA256SUMS.json`.

Stable review key: R516. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. diff_s44.json diff_s44.png gaia.json

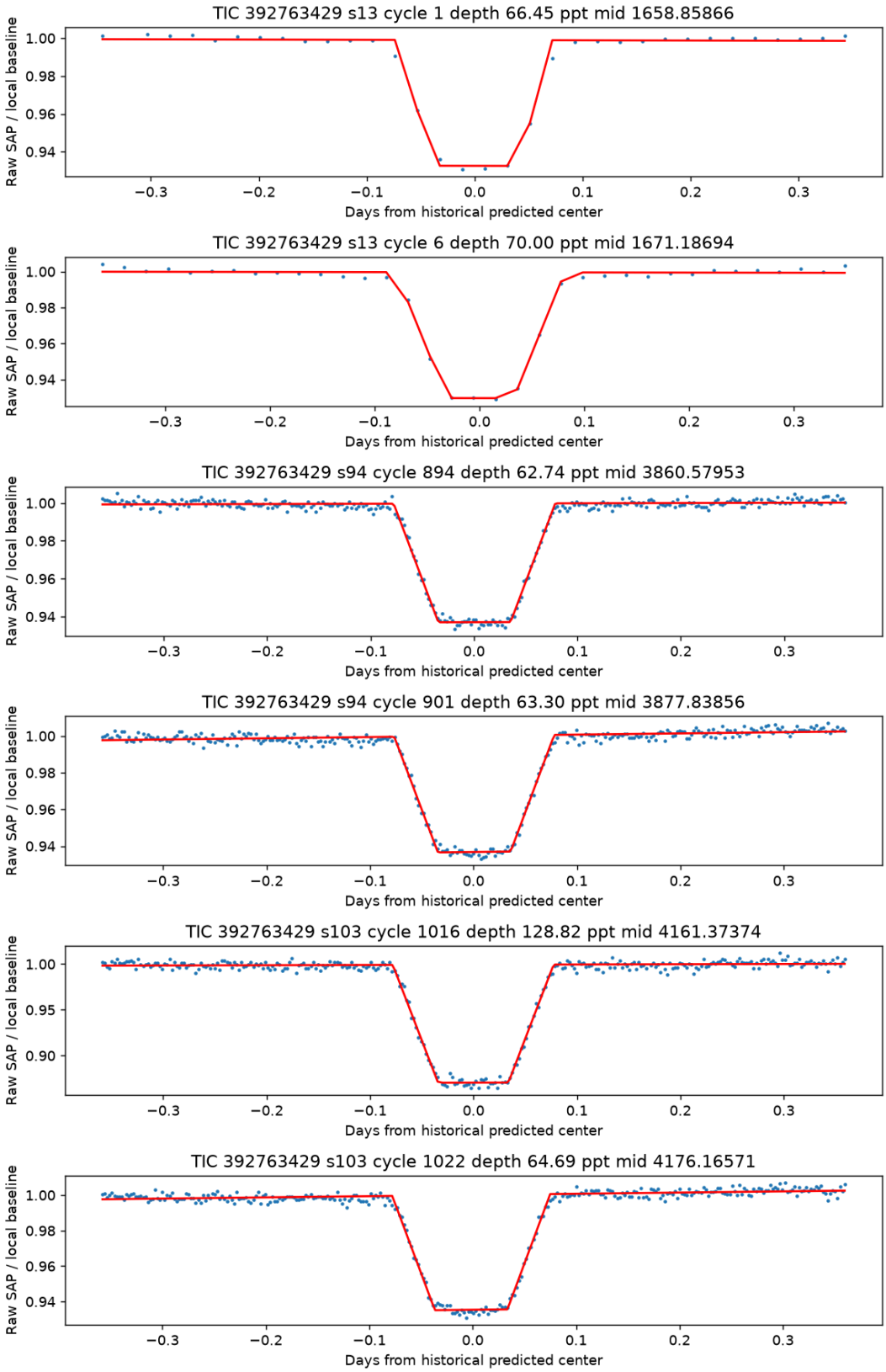
R517. TIC 392763429

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Investigate the S103 depth discontinuity with aperture diagnostics.



R517; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 392763429: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 24 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [13, 94, 103]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 2.4655285743 ± 0.0000030212 d and epoch BTJD 1658.86217686 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 13 deficit centroid is about 1.61 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\392763429\diff_s13.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S13 aperture photometry recovers 10 stacked events at 3x3 depth 73.91 ppt, versus median selected raw QLP fit depth 65.90 ppt. TESScut S13 and representative QLP S13 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 6355496414811552128: orbital P = 2.465425911 d, e = 0.0483. Closest edge-on conjunction offset at the measured epoch is +0.0286 d (+0.0116 phase); linear full-NSS-covariance timing scale 0.0496 d. Approximate central duration 4.20 h versus measured median 3.67 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S13/94/103 raw eclipses follow the same clock. A brief S103 depth doubling affects consecutive cycles, not stable odd/even alternation; its cause is unresolved. S13 difference image is target-centered with no capable Gaia neighbor within 60 arcsec.

Next check: Investigate the S103 depth discontinuity with aperture diagnostics.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
13	1	1658.8586557	0.010417	66.45
13	2	1661.3256533	0.010417	68.58
13	3	1663.7890780	0.010417	68.77
13	4	1666.2545928	0.010417	67.42
13	6	1671.1869404	0.010417	70.00
13	7	1673.6510554	0.010417	67.98
13	9	1678.5845727	0.010417	67.21
13	10	1681.0473863	0.010417	66.91

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
94	893	3858.1137926	0.001157	63.36
94	894	3860.5795276	0.001157	62.74
94	895	3863.0451407	0.001157	63.04
94	896	3865.5109645	0.001157	61.81
94	899	3872.9065859	0.001157	63.20
94	900	3875.3731737	0.001157	63.43
94	901	3877.8385592	0.001157	63.30
94	902	3880.3033415	0.001157	62.84
103	1013	4153.9778478	0.001157	125.72
103	1014	4156.4430929	0.001157	128.19
103	1016	4161.3737360	0.001157	128.82
103	1017	4163.8390246	0.001157	127.06
103	1018	4166.3038238	0.001157	62.23
103	1019	4168.7696539	0.001157	65.36
103	1021	4173.7004953	0.001157	65.11
103	1022	4176.1657093	0.001157	64.69

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\392763429\diff_s13.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R517. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 392763429.png

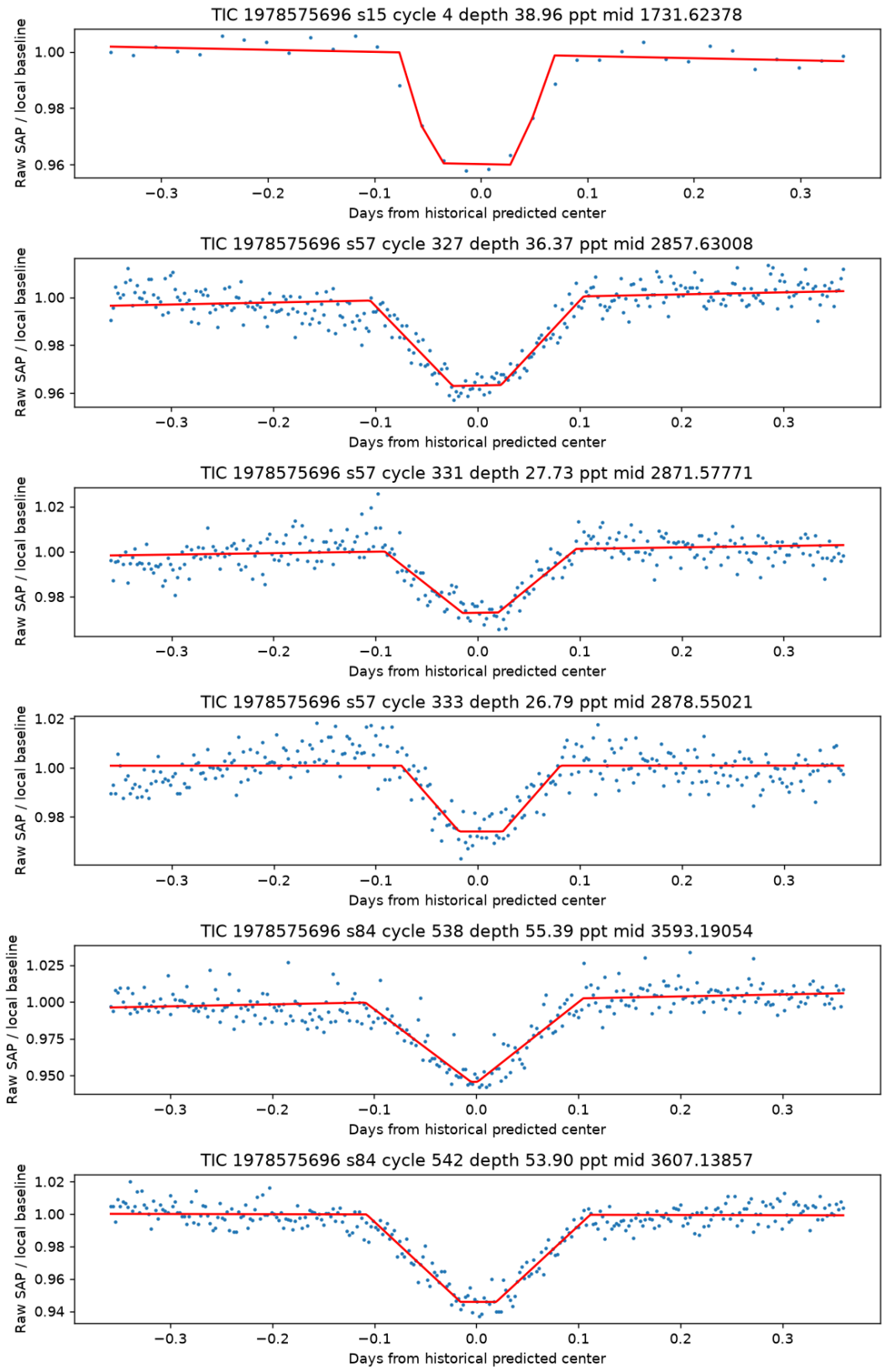
R518. TIC 1978575696

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and secondary-eclipse characterization.



R518; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 1978575696: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 9 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [15, 57, 84]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 6.9721607271 ± 0.0000141450 d and epoch BTJD 1731.62698790 (Deeper repeating eclipse; even historical half-period cycles). Even historical half-period cycles only: deeper same-kind eclipses; odd cycles show shallower secondary. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 41 deficit centroid is about 0.23 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 1 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\1978575696\diff_s41.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S41 aperture photometry recovers 8 stacked events at 3x3 depth 20.84 ppt, versus median selected raw QLP fit depth 51.39 ppt. TESScut S41 and representative QLP sectors [15, 57, 84] are separate TESS observing occasions on the same instrument; their photons are distinct. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 2185260393630495488: orbital P = 6.973173120 d, e = 0.0210. Closest edge-on conjunction offset at the measured epoch is -0.1824 d (-0.0262 phase); linear full-NSS-covariance timing scale 0.1588 d. Approximate central duration 10.30 h versus measured median 4.92 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Alternating raw depths identify the historical 3.486-day clock as half the orbital eclipse cycle. Deeper even cycles yield a measured 6.97216-day same-kind period, consistent with Gaia; S41 difference image favors the target. A capable G14.97 source at 23.9 arcsec is not favored by the target-centered deficit.

Next check: Optional resolved photometry and secondary-eclipse characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
15	4	1731.6237758	0.010417	38.96
57	326	2854.1448650	0.001798	50.18
57	328	2861.1157163	0.001157	51.39
57	330	2868.0906737	0.004100	52.16
57	332	2875.0627901	0.001210	56.85
84	536	3586.2278709	0.003264	43.05
84	538	3593.1905433	0.001658	55.39

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
84	540	3600.1703801	0.003626	41.44
84	542	3607.1385708	0.001179	53.90

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\1978575696\diff_s41.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R518. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 1978575696.png

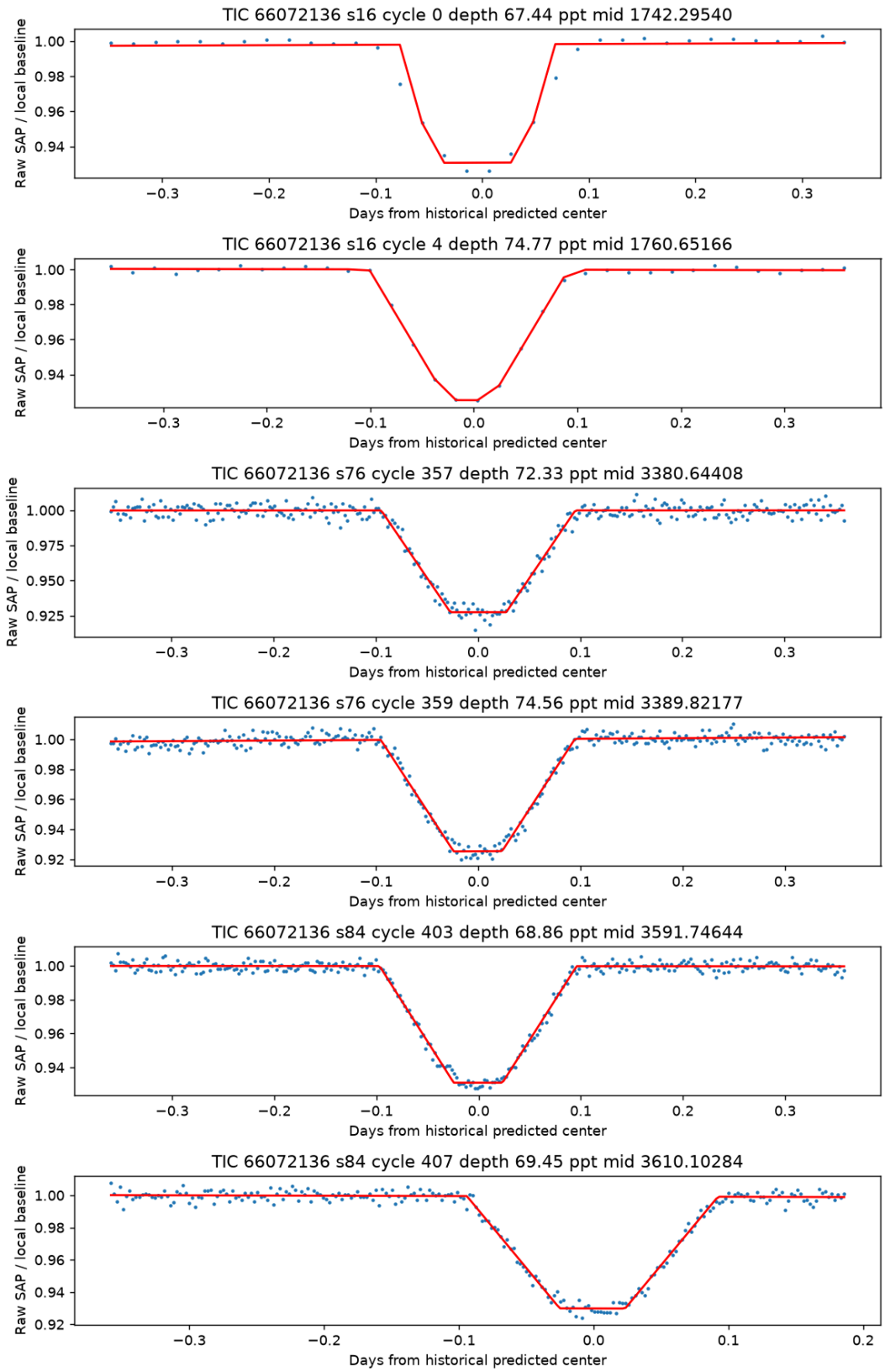
R519. TIC 66072136

physical_verdict: CONFIRMED

novelty_status: known prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolve the discordant VSX period for catalogue interpretation; no new-system count.



R519; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 66072136: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 14 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [16, 76, 84]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 4.5892005792 ± 0.0000108785 d and epoch BTJD 1742.29852360 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 76 deficit centroid is about 4.59 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\66072136\diff_s76.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S76 aperture photometry recovers 6 stacked events at 3x3 depth 72.67 ppt, versus median selected raw QLP fit depth 71.44 ppt. TESScut S76 and representative QLP S76 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 1989520523290696832: orbital P = 4.589208060 d, e = 0.0502. Closest edge-on conjunction offset at the measured epoch is +0.0446 d (+0.0097 phase); linear full-NSS-covariance timing scale 0.0990 d. Approximate central duration 6.32 h versus measured median 4.58 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S16/76/84 raw repeats and the alternative S76 aperture support a real target-centered eclipse. No capable Gaia neighbor within 60 arcsec. Central catalogue reconciliation identifies the close VSX EA V0693 Lac counterpart as the same object, so this is a known eclipsing-binary recovery, not a new-system addition. VSX lists 1.08916 d versus the measured same-kind 4.58920 d; the disagreement does not negate the measured raw eclipses.

Next check: Resolve the discordant VSX period for catalogue interpretation; no new-system count.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
16	0	1742.2954049	0.010417	67.44
16	3	1756.0651217	0.010417	66.89
16	4	1760.6516646	0.010417	74.77
76	355	3371.4642140	0.001157	78.54
76	356	3376.0541073	0.001157	73.67
76	357	3380.6440772	0.001157	72.33

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
76	358	3385.2334200	0.001157	73.42
76	359	3389.8217691	0.001157	74.56
76	360	3394.4095962	0.001157	74.35
84	402	3587.1568237	0.001157	70.55
84	403	3591.7464407	0.001157	68.86
84	405	3600.9241348	0.001157	70.36
84	406	3605.5144422	0.001157	69.97
84	407	3610.1028446	0.001157	69.45

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\66072136\diff_s76.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R519. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 66072136.png

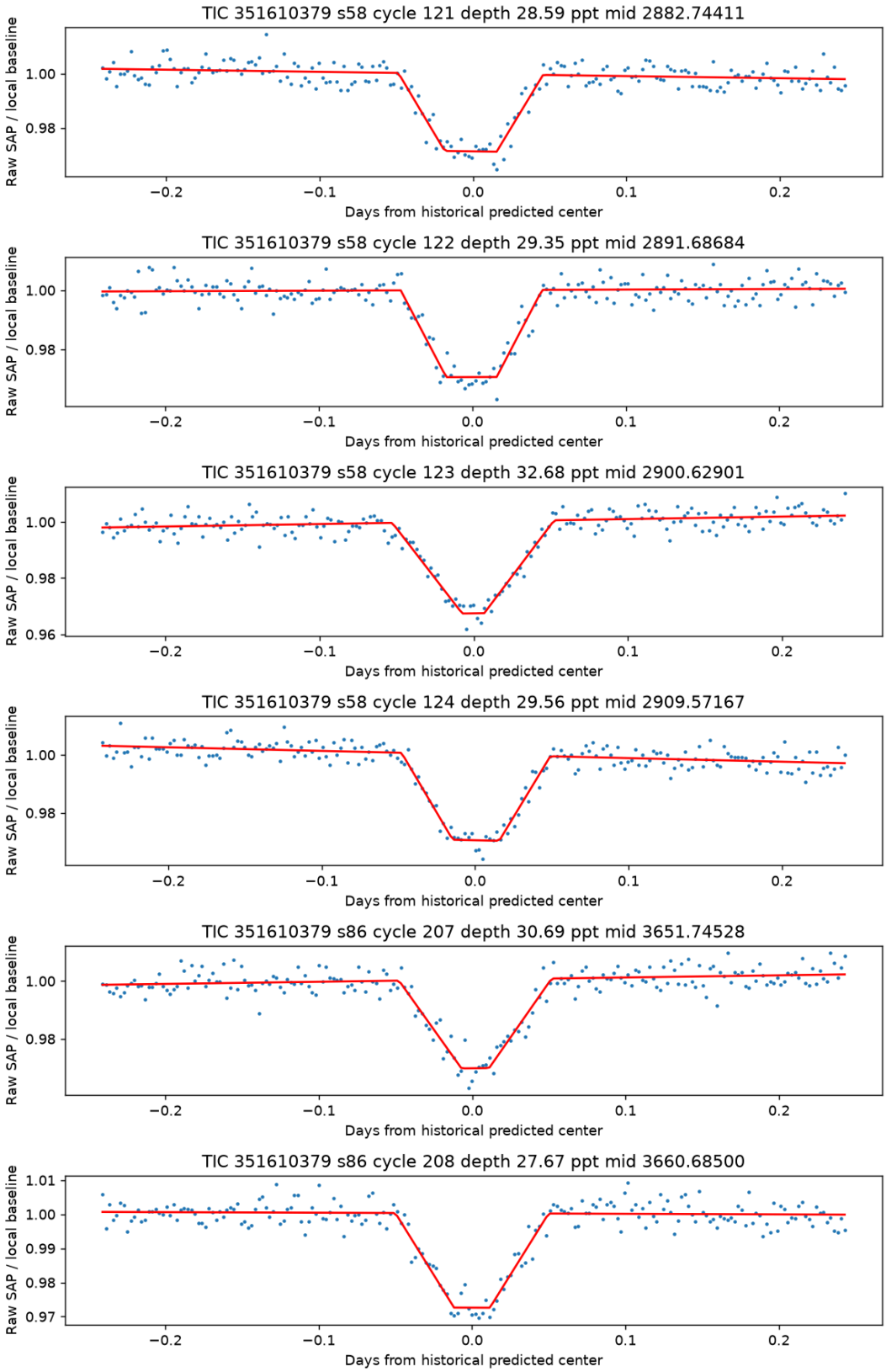
R520. TIC 351610379

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved eclipse photometry.



R520; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 351610379: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 6 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [58, 86]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 8.9418500033 ± 0.0000117888 d and epoch BTJD 2882.74512557 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 58 deficit centroid is about 4.80 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\351610379\diff_s58.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 4 stacked events at 3x3 depth 22.03 ppt, versus median selected raw QLP fit depth 29.46 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 536070020862283648: orbital P = 8.942678782 d, e = 0.0901. Closest edge-on conjunction offset at the measured epoch is -0.1852 d (-0.0207 phase); linear full-NSS-covariance timing scale 0.3265 d. Approximate central duration 4.19 h versus measured median 2.36 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S58/86 raw repeats and the S58 deficit favor the target. The nearest capable G15.21 neighbor is 20.3 arcsec away and would require about 42% intrinsic eclipse depth; a brighter capable source at 54.1 arcsec is spatially separate. Pixel localization has no calibrated error.

Next check: Optional resolved eclipse photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
58	121	2882.7441056	0.001157	28.59
58	122	2891.6868359	0.001157	29.35
58	123	2900.6290141	0.001157	32.68
58	124	2909.5716717	0.001157	29.56
86	207	3651.7452785	0.001157	30.69
86	208	3660.6849982	0.001157	27.67

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\351610379\diff_s58.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R520. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

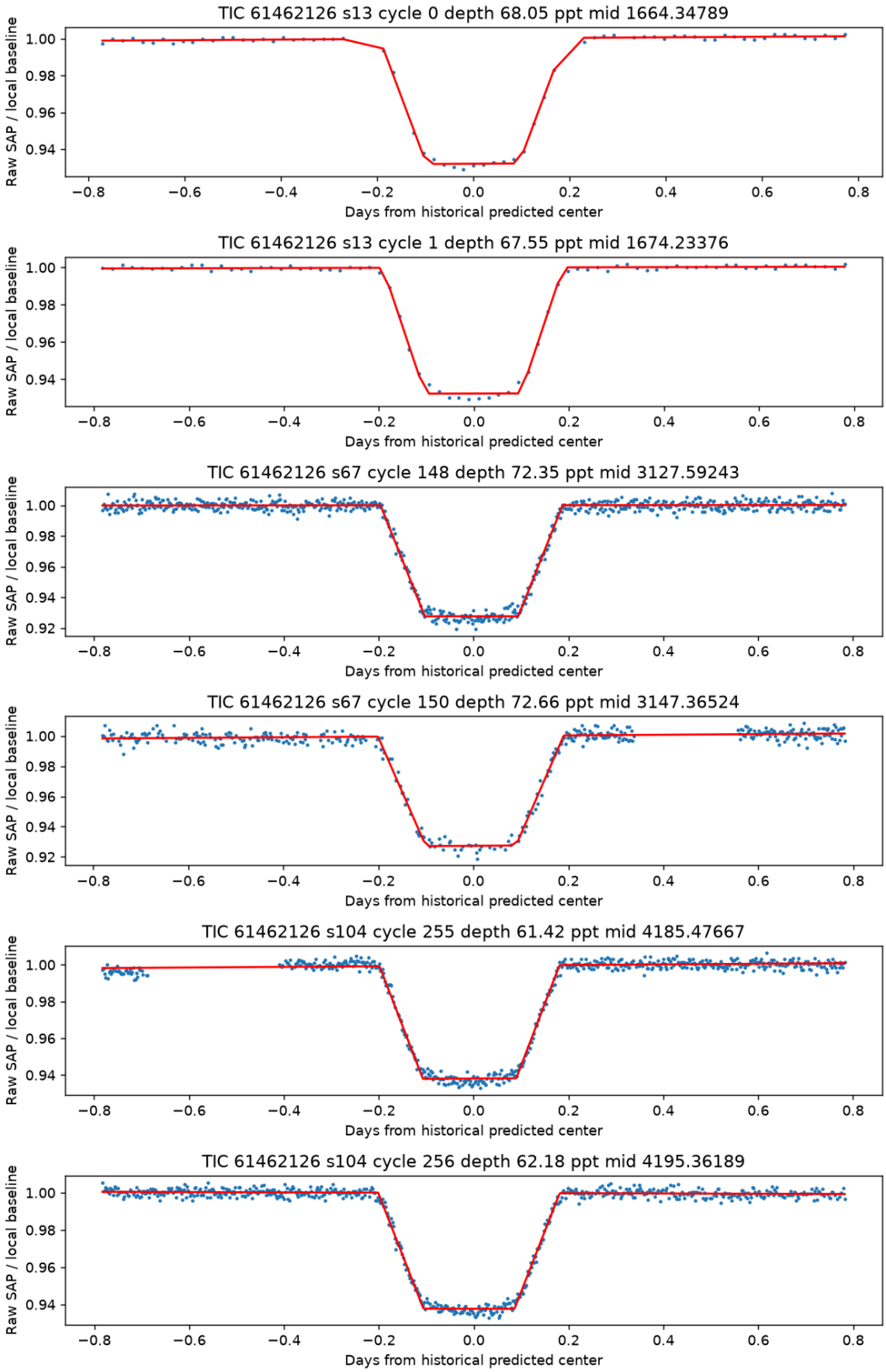
R521. TIC 61462126

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R521; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 61462126: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 6 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [13, 67, 104]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 9.8867660303 ± 0.0000104175 d and epoch BTJD 1664.35061007 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 27 deficit centroid is about 3.03 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\61462126\diff_s27.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 2 stacked events at 3x3 depth 70.42 ppt, versus median selected raw QLP fit depth 67.80 ppt. This historical TESScut sector is a separate observing occasion on the same instrument from the representative QLP sectors; extraction and aperture differences limit depth comparison.
5. **Gaia geometry.** DR3 source 6712693348579620608: orbital P = 9.884078396 d, e = 0.0419. Closest edge-on conjunction offset at the measured epoch is +0.2430 d (+0.0246 phase); linear full-NSS-covariance timing scale 0.2641 d. Approximate central duration 11.28 h versus measured median 9.20 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S13/67/104 raw eclipses repeat. The S27 alternative extraction recovers two events with a target-centered deficit; no Gaia neighbor within 60 arcsec passes the approximate flux-budget test.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
13	0	1664.3478919	0.010417	68.05
13	1	1674.2337609	0.010417	67.55
67	148	3127.5924308	0.001157	72.35
67	150	3147.3652395	0.001157	72.66
104	255	4185.4766718	0.001157	61.42
104	256	4195.3618949	0.001157	62.18

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted]` `5\made\sb1-scale\diffimg\61462126\diff_s27.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R521. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

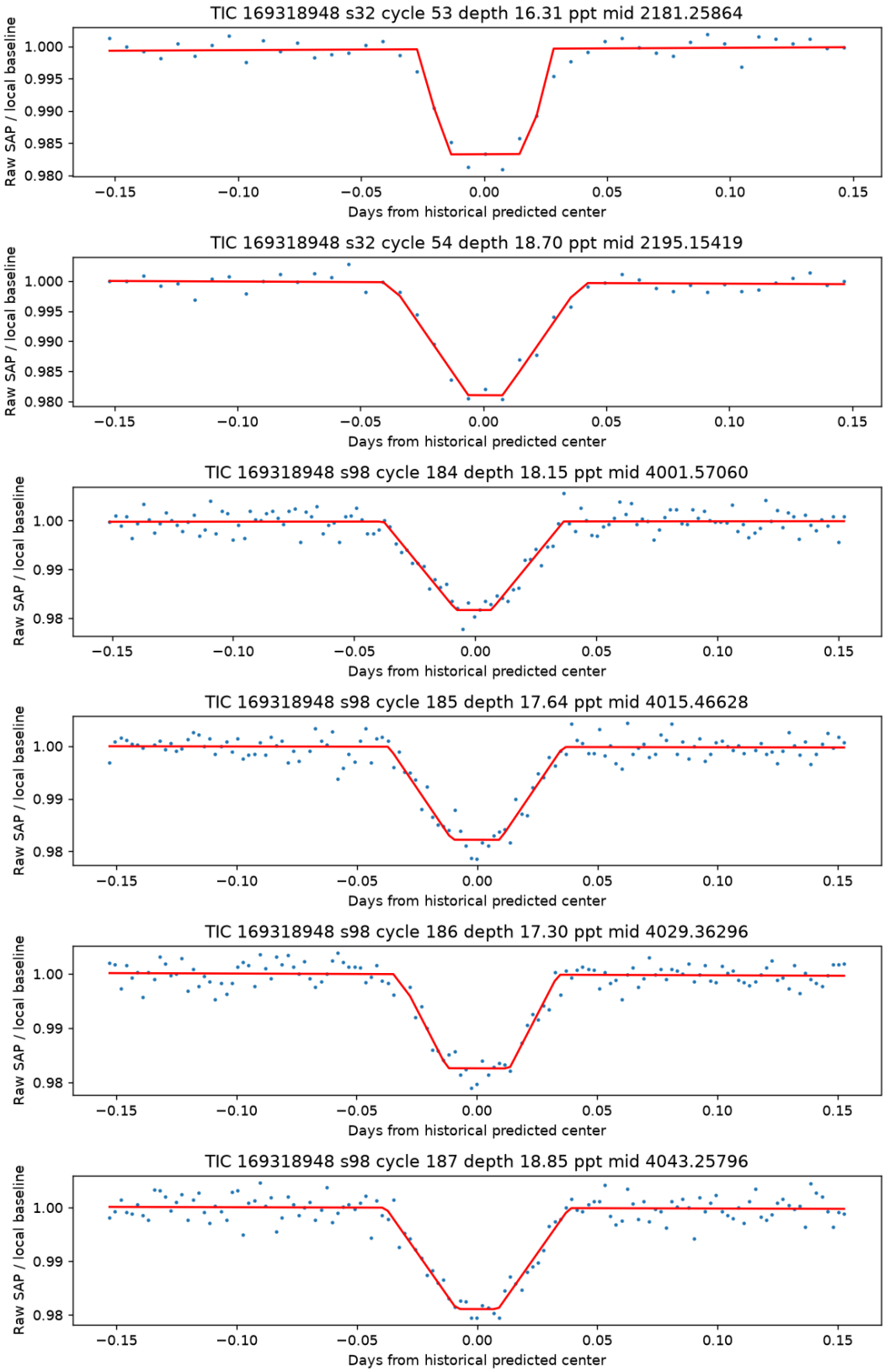
R522. TIC 169318948

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R522; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 169318948: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 6 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [32, 98]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 13.8955158737 ± 0.0000190968 d and epoch BTJD 2181.25859625 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 98 deficit centroid is about 1.05 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\169318948\diff_s98.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 4 stacked events at 3x3 depth 17.95 ppt, versus median selected raw QLP fit depth 17.89 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 2986427856265439488: orbital P = 13.896444944 d, e = 0.3989. Closest edge-on conjunction offset at the measured epoch is -0.2784 d (-0.0200 phase); linear full-NSS-covariance timing scale 0.4091 d. Approximate central duration 3.58 h versus measured median 1.76 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S32/98 raw repeated eclipses and the S98 alternative extraction agree. The difference image favors the target and lists no capable Gaia neighbor within 60 arcsec.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
32	53	2181.2586438	0.003472	16.31
32	54	2195.1541875	0.003472	18.70
98	184	4001.5705969	0.001157	18.15
98	185	4015.4662752	0.001157	17.64
98	186	4029.3629551	0.001157	17.30
98	187	4043.2579572	0.001157	18.85

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\169318948\diff_s98.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R522. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

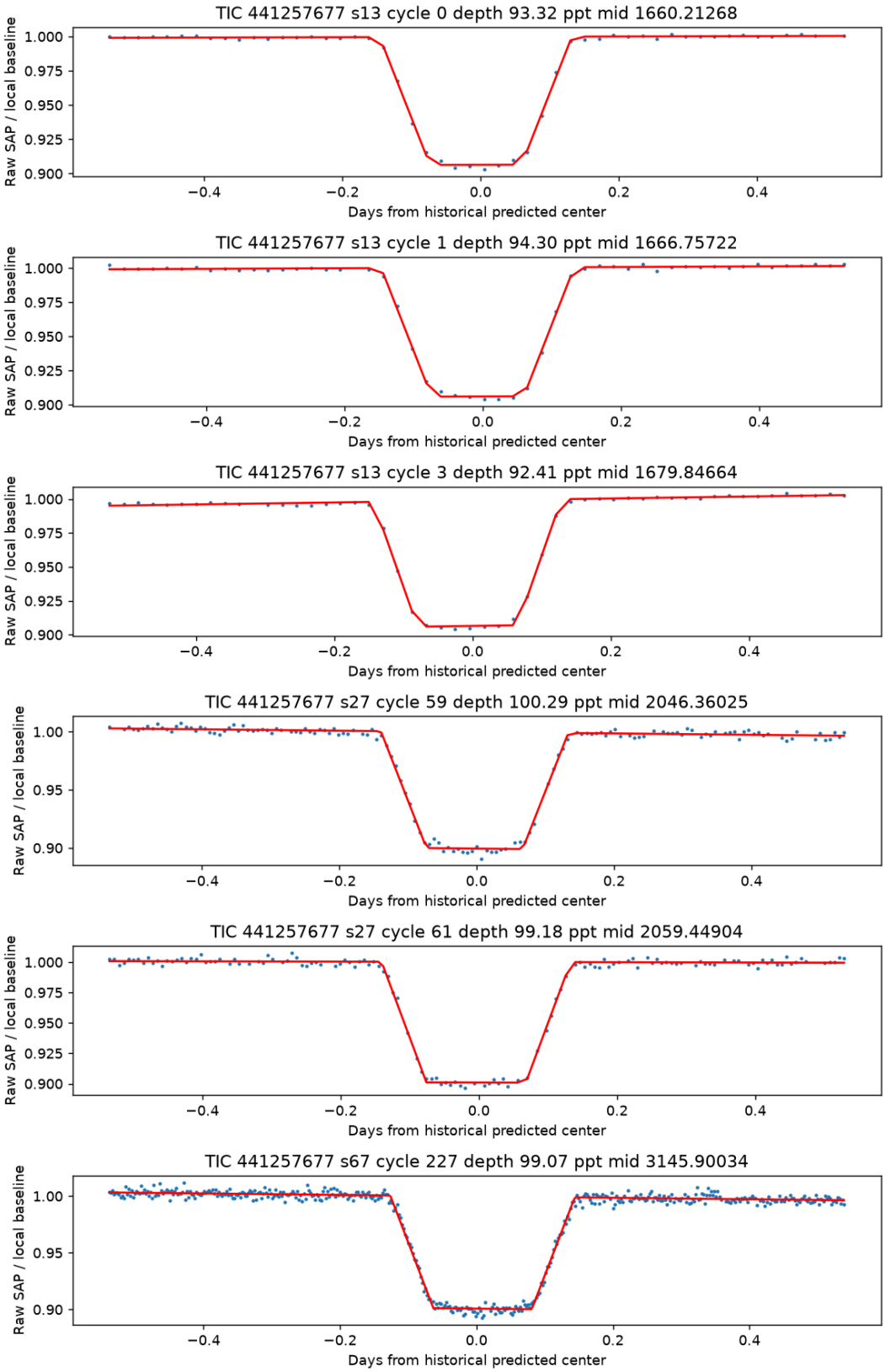
R523. TIC 441257677

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R523; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 441257677: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 10 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [13, 27, 67]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 6.5448839250 ± 0.0000106057 d and epoch BTJD 1660.21194066 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 13 deficit centroid is about 4.75 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\441257677\diff_s13.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 4 stacked events at 3x3 depth 100.87 ppt, versus median selected raw QLP fit depth 98.56 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 6739664574967468800: orbital P = 6.545173049 d, e = 0.1154. Closest edge-on conjunction offset at the measured epoch is +0.1321 d (+0.0202 phase); linear full-NSS-covariance timing scale 0.6016 d. Approximate central duration 8.15 h versus measured median 6.54 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S13/27/67 raw events recur with no persistent odd/even depth split. The S13 alternative extraction shows four events and a target-centered deficit with no capable Gaia neighbor within 60 arcsec.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
13	0	1660.2126763	0.010417	93.32
13	1	1666.7572200	0.010417	94.30
13	2	1673.3017353	0.010417	92.44
13	3	1679.8466357	0.010417	92.41
27	58	2039.8154346	0.003472	98.05
27	59	2046.3602493	0.003472	100.29
27	60	2052.9051908	0.003472	99.68
27	61	2059.4490443	0.003472	99.18

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
67	225	3132.8110855	0.001157	100.29
67	227	3145.9003392	0.001157	99.07

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\441257677\diff_s13.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R523. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

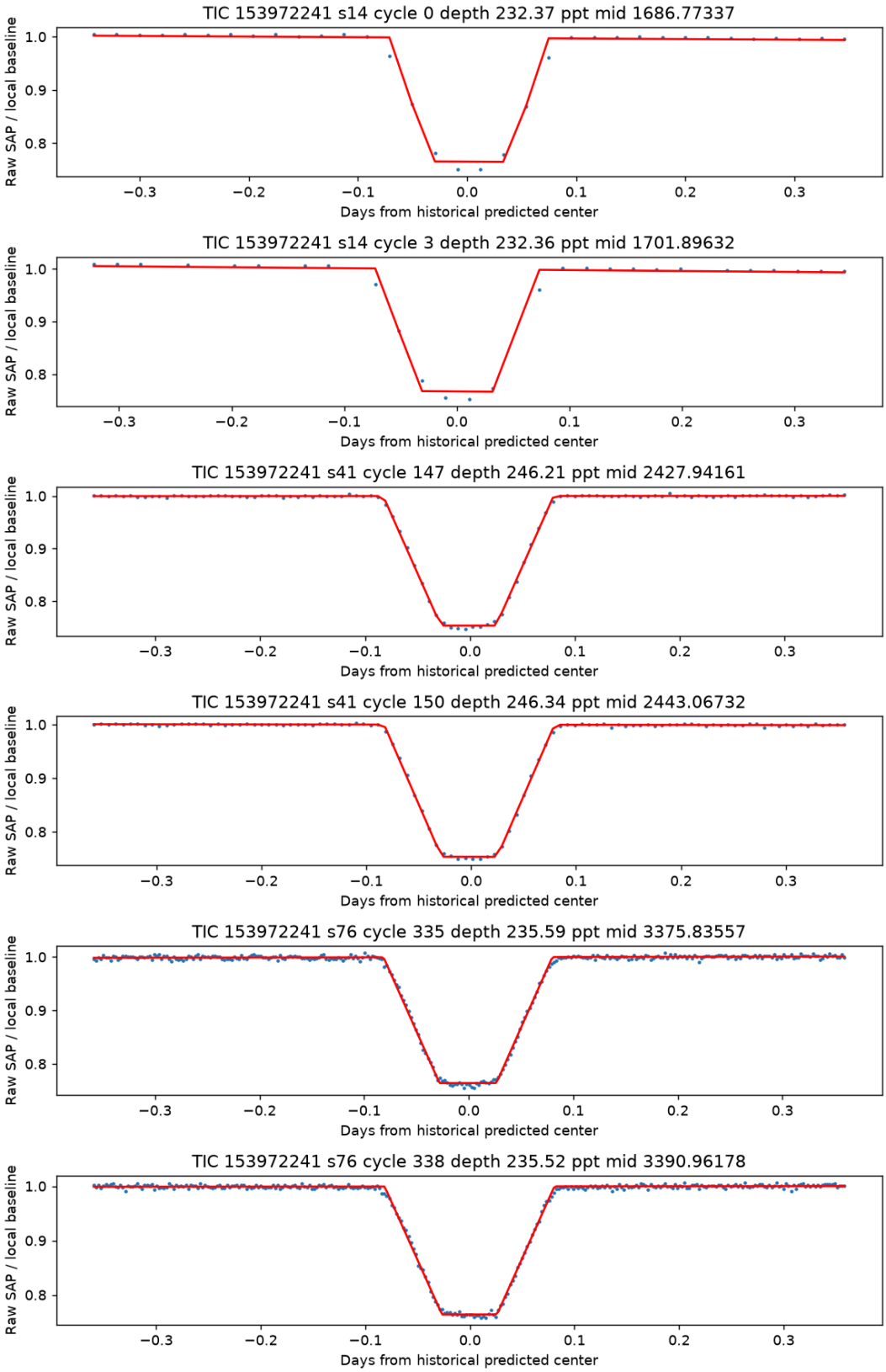
R524. TIC 153972241

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R524; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 153972241: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 11 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [14, 41, 76]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 5.0419880702 ± 0.0000086825 d and epoch BTJD 1686.76961209 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 75 deficit centroid is about 0.91 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\153972241\diff_s75.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 6 stacked events at 3x3 depth 243.75 ppt, versus median selected raw QLP fit depth 235.59 ppt. This historical TESScut sector is a separate observing occasion on the same instrument from the representative QLP sectors; extraction and aperture differences limit depth comparison.
5. **Gaia geometry.** DR3 source 1666322761092269568: orbital P = 5.041663243 d, e = 0.0256. Closest edge-on conjunction offset at the measured epoch is +0.0466 d (+0.0092 phase); linear full-NSS-covariance timing scale 0.0912 d. Approximate central duration 4.55 h versus measured median 3.87 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Deep S14/41/76 raw eclipses repeat; the S75 extraction recovers six events. Its target-centered difference image lists no capable Gaia neighbor within 60 arcsec.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
14	0	1686.7733724	0.010416	232.37
14	1	1691.8152417	0.010416	232.32
14	3	1701.8963232	0.010416	232.36
41	146	2422.8992728	0.003472	246.28
41	147	2427.9416130	0.003472	246.21
41	149	2438.0255926	0.003472	248.01
41	150	2443.0673221	0.003472	246.34
76	334	3370.7935552	0.001157	236.10
76	335	3375.8355668	0.001157	235.59

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
76	336	3380.8776017	0.001157	234.76
76	338	3390.9617784	0.001157	235.52

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\153972241\diff_s75.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R524. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

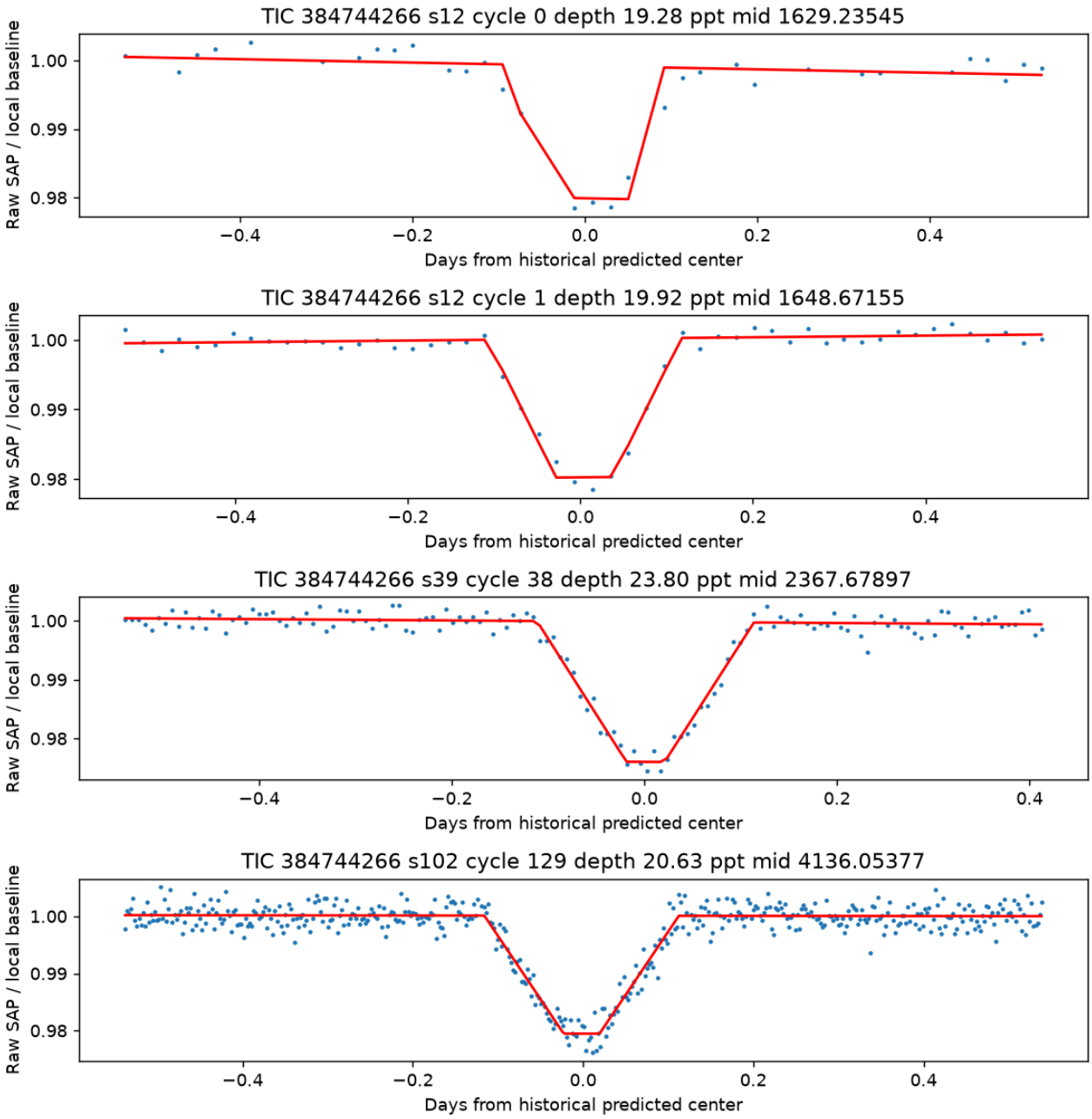
R525. TIC 384744266

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved eclipse photometry.



R525; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 384744266: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 4 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [12, 39, 102]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 19.4326891330 ± 0.0000338018 d and epoch BTJD 1629.23687168 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 12 deficit centroid is about 3.65 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\384744266\diff_s12.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 2 stacked events at 3x3 depth 20.23 ppt, versus median selected raw QLP fit depth 20.27 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 5776380960403754624: orbital P = 19.433184428 d, e = 0.2022. Closest edge-on conjunction offset at the measured epoch is -0.1733 d (-0.0089 phase); linear full-NSS-covariance timing scale 0.4624 d. Approximate central duration 10.72 h versus measured median 5.40 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S12/39/102 raw repeats agree. The S12 deficit is near the target. Three nominally capable neighbors lie 29–60 arcsec away, each requiring about 78–100% intrinsic loss, and none is favored by the observed deficit.

Next check: Optional resolved eclipse photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
12	0	1629.2354465	0.010417	19.28
12	1	1648.6715529	0.010417	19.92
39	38	2367.6789719	0.003472	23.80
102	129	4136.0537725	0.001157	20.63

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\384744266\diff_s12.json`, its PNG and Gaia-neighbor table; source and artifact hashes

are in `../SHA256SUMS.json`.

Stable review key: R525. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

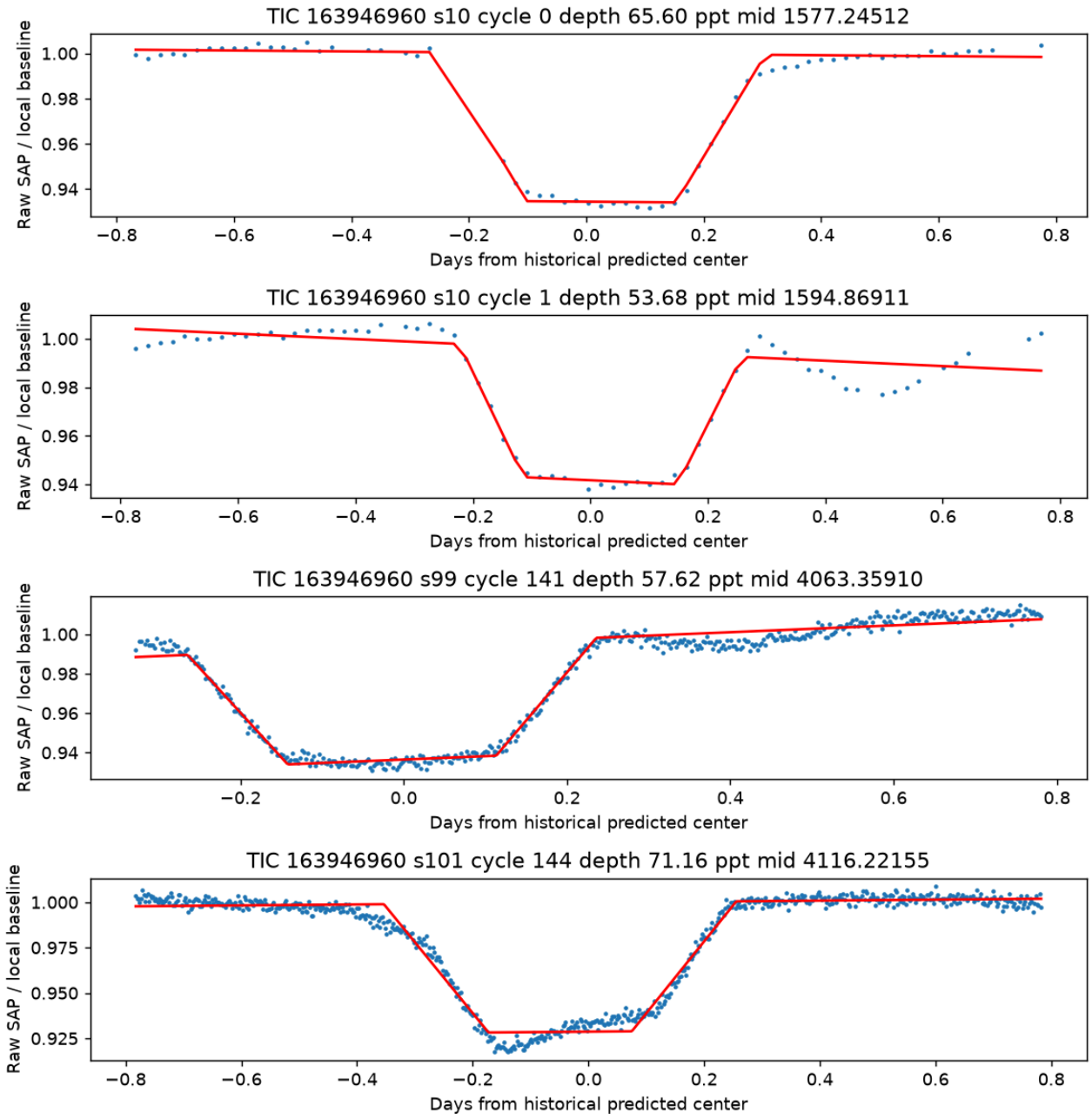
R526. TIC 163946960

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry would strengthen host attribution.



R526; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 163946960: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 4 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [10, 99, 101]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 17.6318158711 ± 0.0005396857 d and epoch BTJD 1577.24863847 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 10 deficit centroid is about 5.50 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 2 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\163946960\diff_s10.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 2 stacked events at 3x3 depth 51.83 ppt, versus median selected raw QLP fit depth 61.61 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 6078495037379889408: orbital P = 17.629414377 d, e = 0.2481. Closest edge-on conjunction offset at the measured epoch is +0.2046 d (+0.0116 phase); linear full-NSS-covariance timing scale 0.2688 d. Approximate central duration 15.88 h versus measured median 12.71 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S10/99/101 raw repeats and the S10 alternative extraction support the eclipse. The deficit lies about 0.26 pixel from the target, away from two capable Gaia neighbors at 15.8 and 29.7 arcsec; those require about 60% and 44% intrinsic losses. Localization is descriptive, without a calibrated uncertainty.

Next check: Resolved eclipse photometry would strengthen host attribution.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
10	0	1577.2451240	0.010417	65.60
10	1	1594.8691131	0.010417	53.68
99	141	4063.3590968	0.001933	57.62
101	144	4116.2215510	0.001157	71.16

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\163946960\diff_s10.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R526. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

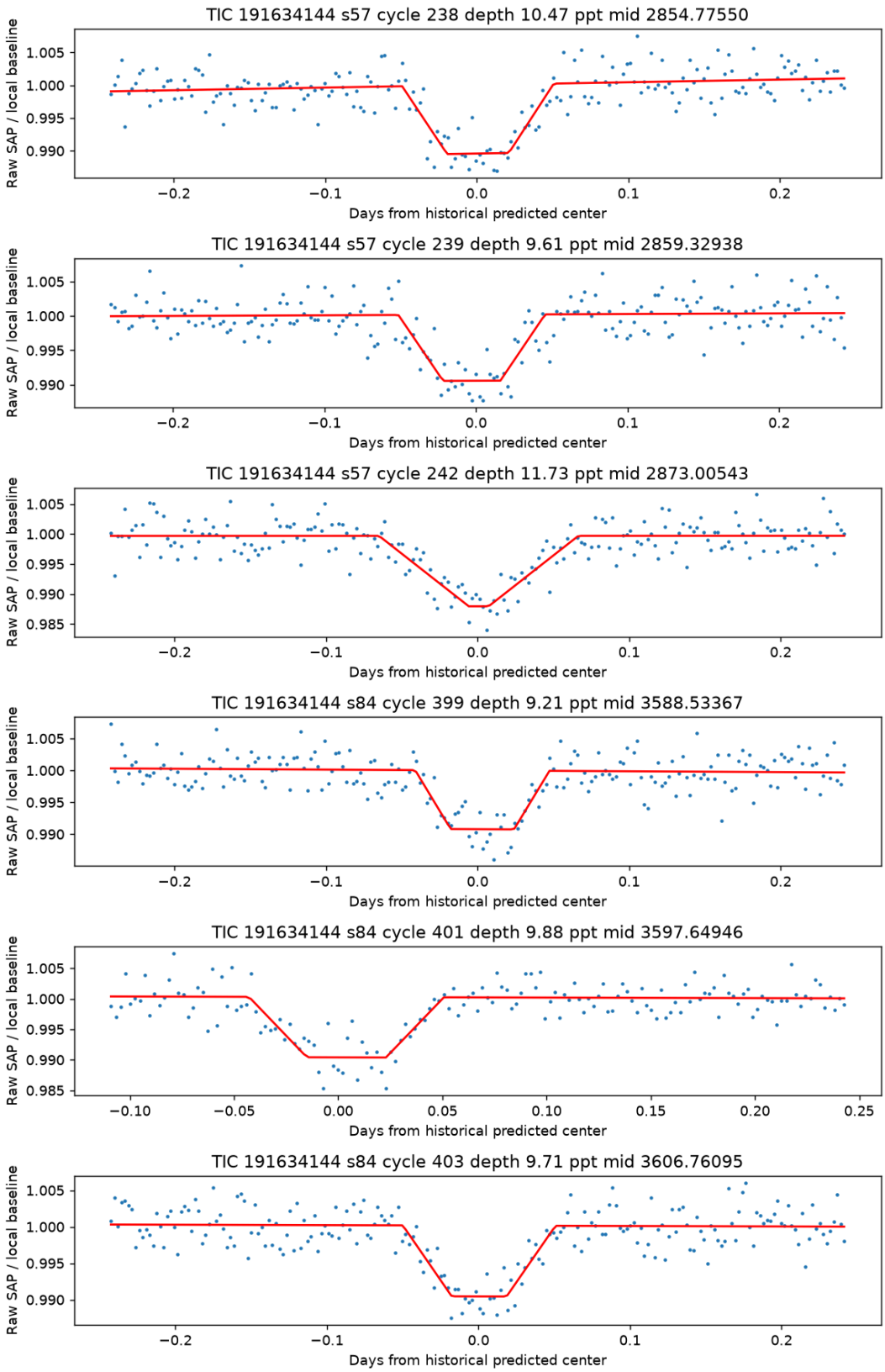
R527. TIC 191634144

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional-sector timing and resolved photometry could tighten this shallow case.



R527; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 191634144: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 10 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [57, 84]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 4.5575080623 ± 0.0000060351 d and epoch BTJD 2854.77452990 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 57 deficit centroid is about 3.11 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\191634144\diff_s57.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 5 stacked events at 3x3 depth 9.50 ppt, versus median selected raw QLP fit depth 10.47 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 375375060725718912: orbital P = 4.557098972 d, e = 0.0402. Closest edge-on conjunction offset at the measured epoch is +0.2181 d (+0.0479 phase); linear full-NSS-covariance timing scale 0.1720 d. Approximate central duration 6.73 h versus measured median 2.48 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S57/84 raw shallow eclipses recur. Odd/even fitted depths vary at roughly the local noise level without a stable half-period interpretation. The S57 difference image favors the target over capable neighbors at 23.9, 34.4 and 53.6 arcsec.

Next check: Additional-sector timing and resolved photometry could tighten this shallow case.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
57	238	2854.7755006	0.001157	10.47
57	239	2859.3293846	0.001316	9.61
57	241	2868.4473010	0.001556	10.92
57	242	2873.0054339	0.001722	11.73
57	243	2877.5638528	0.002082	10.48
84	399	3588.5336664	0.001462	9.21
84	400	3593.0908072	0.001500	12.34
84	401	3597.6494571	0.001157	9.88

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
84	402	3602.2053858	0.001671	11.23
84	403	3606.7609465	0.001728	9.71

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\191634144\diff_s57.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R527. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

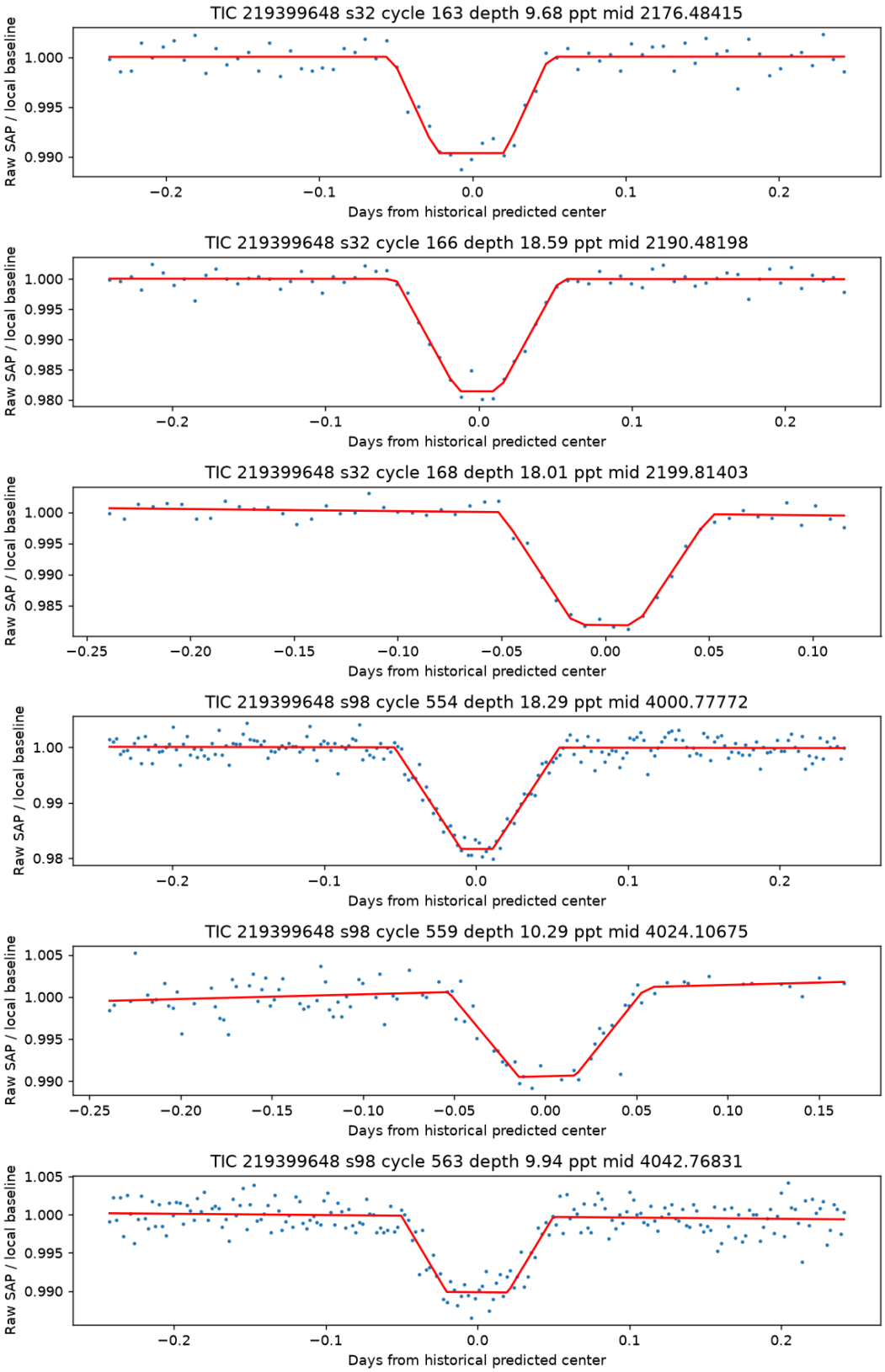
R528. TIC 219399648

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional secondary-eclipse characterization.



R528; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 219399648: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 5 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [32, 98]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 9.3314205089 ± 0.0000110733 d and epoch BTJD 2181.15076257 (Deeper repeating eclipse; even historical half-period cycles). Even historical half-period cycles only: deeper same-kind eclipses; odd cycles show shallower secondary. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 98 deficit centroid is about 3.27 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\219399648\diff_s98.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 11 stacked events at 3x3 depth 14.53 ppt, versus median selected raw QLP fit depth 18.29 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison. The historical S98 stack mixes deeper and shallower half-period events, so its depth is not a same-kind depth comparison.
5. **Gaia geometry.** DR3 source 4797773555362406144: orbital P = 9.330097946 d, e = 0.0079. Closest edge-on conjunction offset at the measured epoch is +0.3396 d (+0.0364 phase); linear full-NSS-covariance timing scale 0.3330 d. Approximate central duration 6.49 h versus measured median 2.58 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Alternating S32/98 raw depths show that the 4.66571-day SAP search branch is half the orbital eclipse cycle. Deeper even events give the 9.33142-day same-kind period consistent with Gaia. The S98 difference image favors the target with no capable Gaia neighbor within 60 arcsec.

Next check: Optional secondary-eclipse characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
32	164	2181.1505425	0.003472	18.86
32	166	2190.4819811	0.003472	18.59
32	168	2199.8140342	0.003472	18.01
98	554	4000.7777232	0.001157	18.29
98	560	4028.7720610	0.001157	16.98

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\219399648\diff_s98.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R528. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

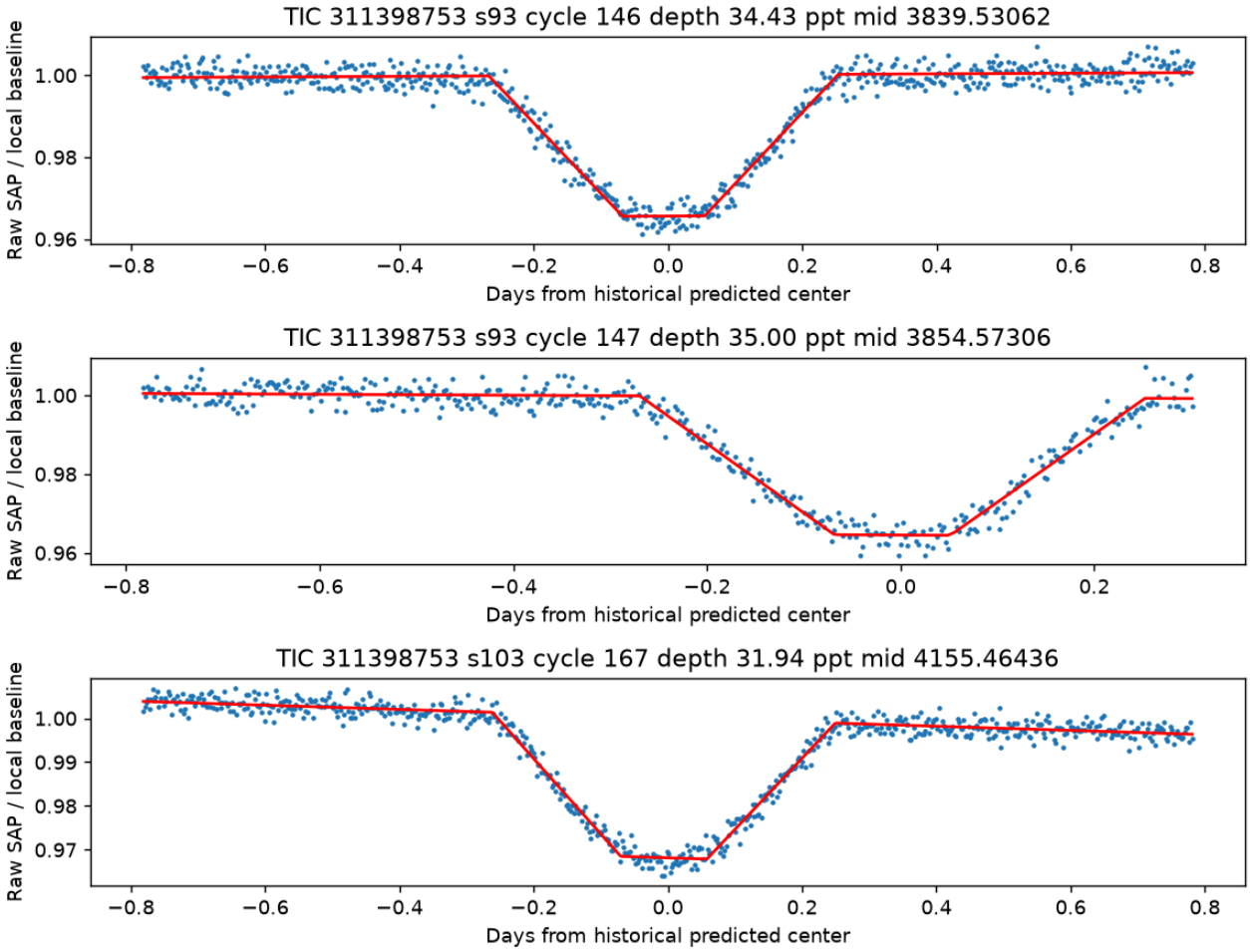
R529. TIC 311398753

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved time-series photometry across an eclipse, plus refined Gaia-orbit comparison.



R529; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 311398753: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 3 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [93, 103]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 15.0445006706 ± 0.0001120670 d and epoch BTJD 3839.52974390 (Selected repeating eclipse, primary by convention; deeper/secondary identity

not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.

3. **Pixel attribution.** The alternative cutout sector 13 deficit centroid is about 1.04 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\311398753\diff_s13.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 2 stacked events at 3x3 depth 28.84 ppt, versus median selected raw QLP fit depth 34.43 ppt. This historical TESScut sector is a separate observing occasion on the same instrument from the representative QLP sectors; extraction and aperture differences limit depth comparison.
5. **Gaia geometry.** DR3 source 5804147301198863744: orbital P = 15.047545491 d, e = 0.0457. Closest edge-on conjunction offset at the measured epoch is -0.6848 d (-0.0455 phase); linear full-NSS-covariance timing scale 0.3346 d. Approximate central duration 21.27 h versus measured median 12.50 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S93/103 raw eclipses repeat and the S13 alternative extraction recovers two. A capable G15.36 source lies only 9.65 arcsec from the target and shares the bright pixel; further capable sources lie at 20.8 and 38.2 arcsec. The target-near deficit cannot distinguish the closest host at uncalibrated pixel precision. Gaia conjunction offset is −0.685 d at the observed epoch versus a 0.335 d linear orbital timing scale; a projection near Gaia2016 gives +0.012 d with only a 0.026 d photometric period allowance, so phase agreement remains descriptive.

Next check: Resolved time-series photometry across an eclipse, plus refined Gaia-orbit comparison.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
93	146	3839.5306153	0.001157	34.43
93	147	3854.5730555	0.001319	35.00
103	167	4155.4643587	0.001760	31.94

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\311398753\diff_s13.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R529. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

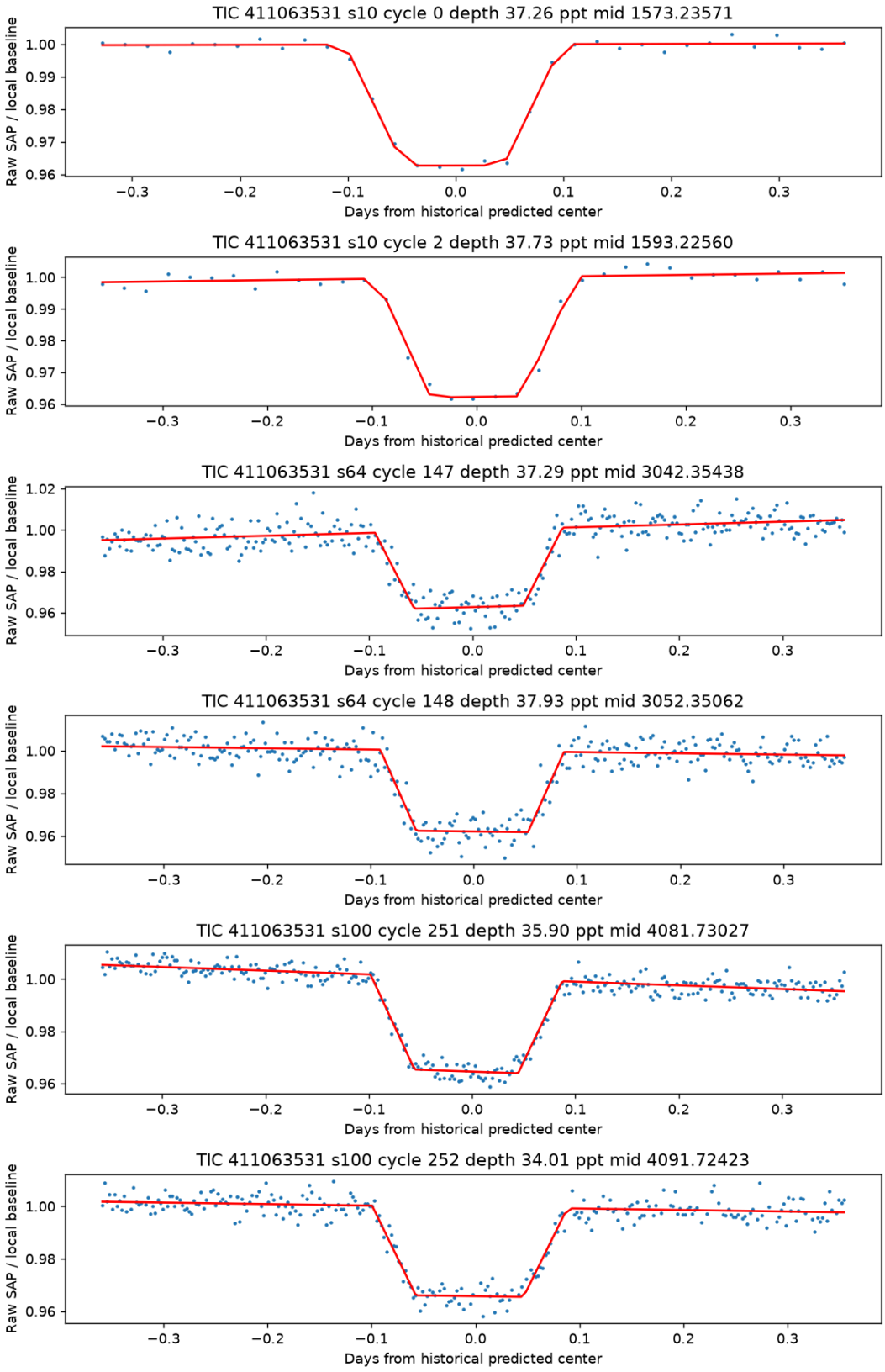
R530. TIC 411063531

physical_verdict: CONFIRMED

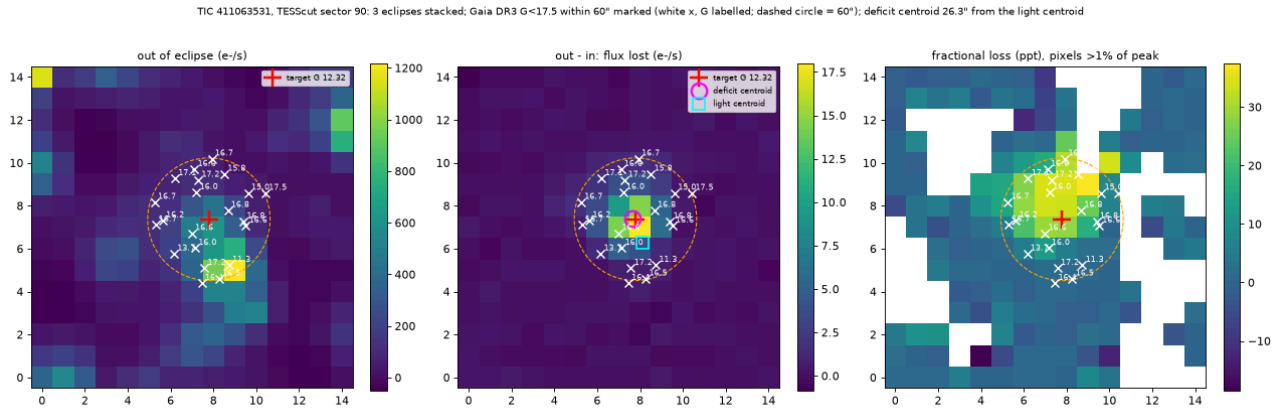
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse imaging would strengthen attribution in this crowded field.



R530; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R530; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 411063531: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 6 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [10, 64, 100]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 9.9939890066 ± 0.0000106933 d, reference BTJD 1573.2390596 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S90 TESScut deficit centroid is 3.14 arcsec from the target coordinate. Peak deficit pixel [8, 7]; target pixel [8, 7]. Image-model fractional residual 0.859 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 3 predicted event windows give 1x1/3x3/5x5 depths 28.30/18.60/12.01 ppt; median selected QLP raw fit depth is 37.27 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout sector is a separate TESS observing occasion from the selected QLP sectors; it is the same instrument with an alternative reduction, not independent-instrument confirmation. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 5354251091852534016 has orbital P = 9.994640884 d and e = 0.0745. The nearest edge-on conjunction is offset by +0.2248 d (+0.0225 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.4266 d. Approximate central duration 5.40 h versus observed median 4.48 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Six raw events span S10, S64 and S100, with adjacent pairs in both later sectors. The crowded S90 deficit centroid is 3.1 arcsec from the target while the light centroid is displaced 26.3 arcsec; four flux-capable catalogued neighbors are 44–47 arcsec from the deficit. The target is favored, but the image-model residual (0.86) limits localization precision.

Next check: Resolved eclipse imaging would strengthen attribution in this crowded field.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
10	0	1573.2357051	0.010417	37.26
10	2	1593.2255969	0.010417	37.73
64	147	3042.3543805	0.001173	37.29
64	148	3052.3506213	0.001157	37.93
100	251	4081.7302680	0.001157	35.90
100	252	4091.7242273	0.001157	34.01

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
5354250374612876928	44.15	15.755	0.934	47.0
5354250374612874496	44.88	15.020	0.493	48.3
5354249618698621696	45.66	11.275	0.052	51.3
5354249687418099072	46.48	13.728	0.176	46.7

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s90.json`, `historical/diff_s90.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R530. Public evidence files: diff_s90.json, diff_s90.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

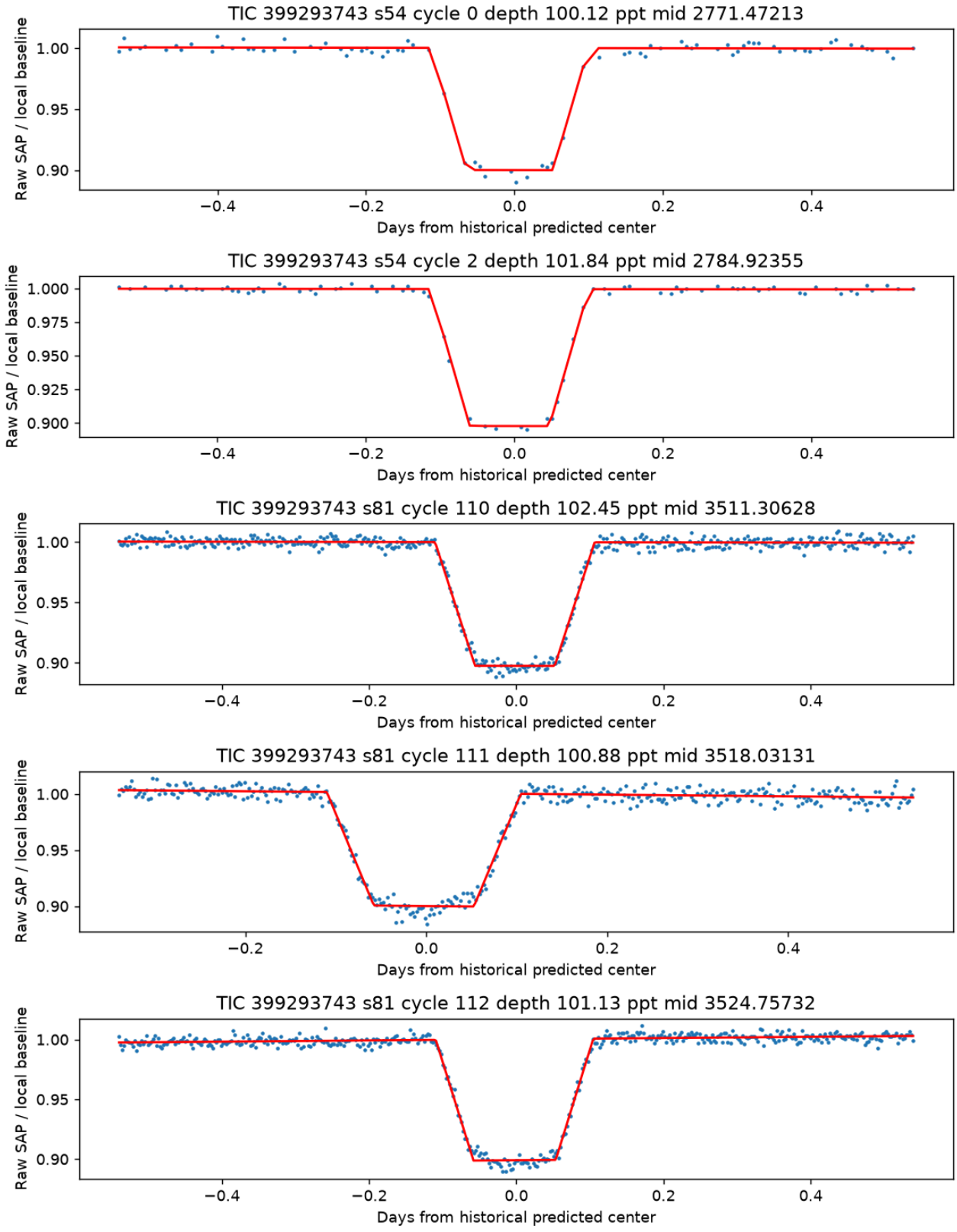
R531. TIC 399293743

physical_verdict: CONFIRMED

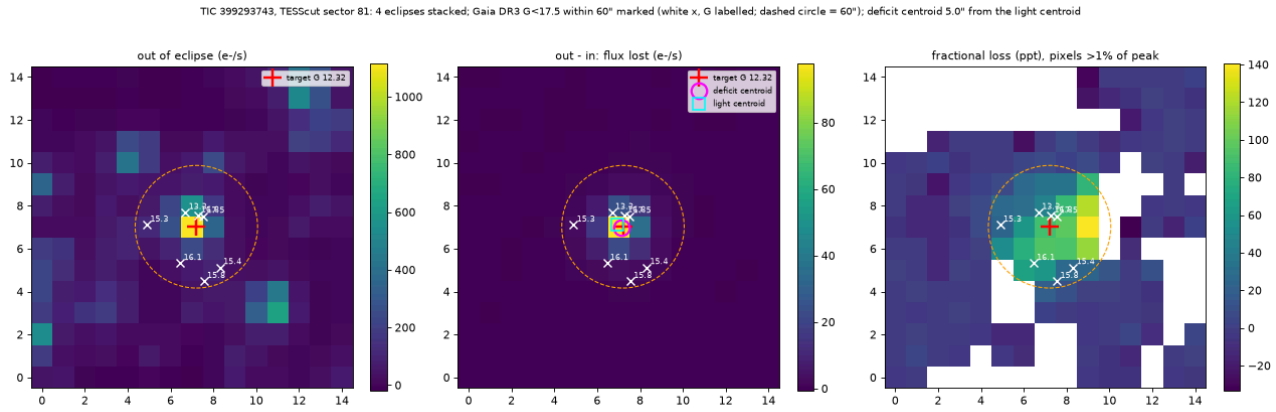
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy.



R531; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R531; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 399293743: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 5 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [54, 81]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 6.7257610294 ± 0.0000449821 d, reference BTJD 2771.4721584 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S81 TESScut deficit centroid is 1.35 arcsec from the target coordinate. Peak deficit pixel [7, 7]; target pixel [7, 7]. Image-model fractional residual 0.663 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 4 predicted event windows give 1x1/3x3/5x5 depths 87.65/74.64/68.29 ppt; median selected QLP raw fit depth is 101.13 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 4244542013422016256 has orbital P = 6.725623473 d and e = 0.0141. The nearest edge-on conjunction is offset by +0.1301 d (+0.0194 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.2912 d. Approximate central duration 5.87 h versus observed median 5.14 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Five raw events span S54 and S81, including three adjacent S81 cycles. The S81 deficit lies near the target, about 17.5 arcsec from the only flux-capable Gaia neighbor; the source is favored at TESS resolution.

Next check: Optional resolved photometry and spectroscopy.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
54	0	2771.4721285	0.006944	100.12
54	2	2784.9235520	0.006944	101.84
81	110	3511.3062759	0.001157	102.45
81	111	3518.0313096	0.001157	100.88
81	112	3524.7573169	0.001157	101.13

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
4244541940403576832	17.73	13.303	0.349	17.5

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s81.json`, `historical/diff_s81.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R531. Public evidence files: diff_s81.json, diff_s81.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

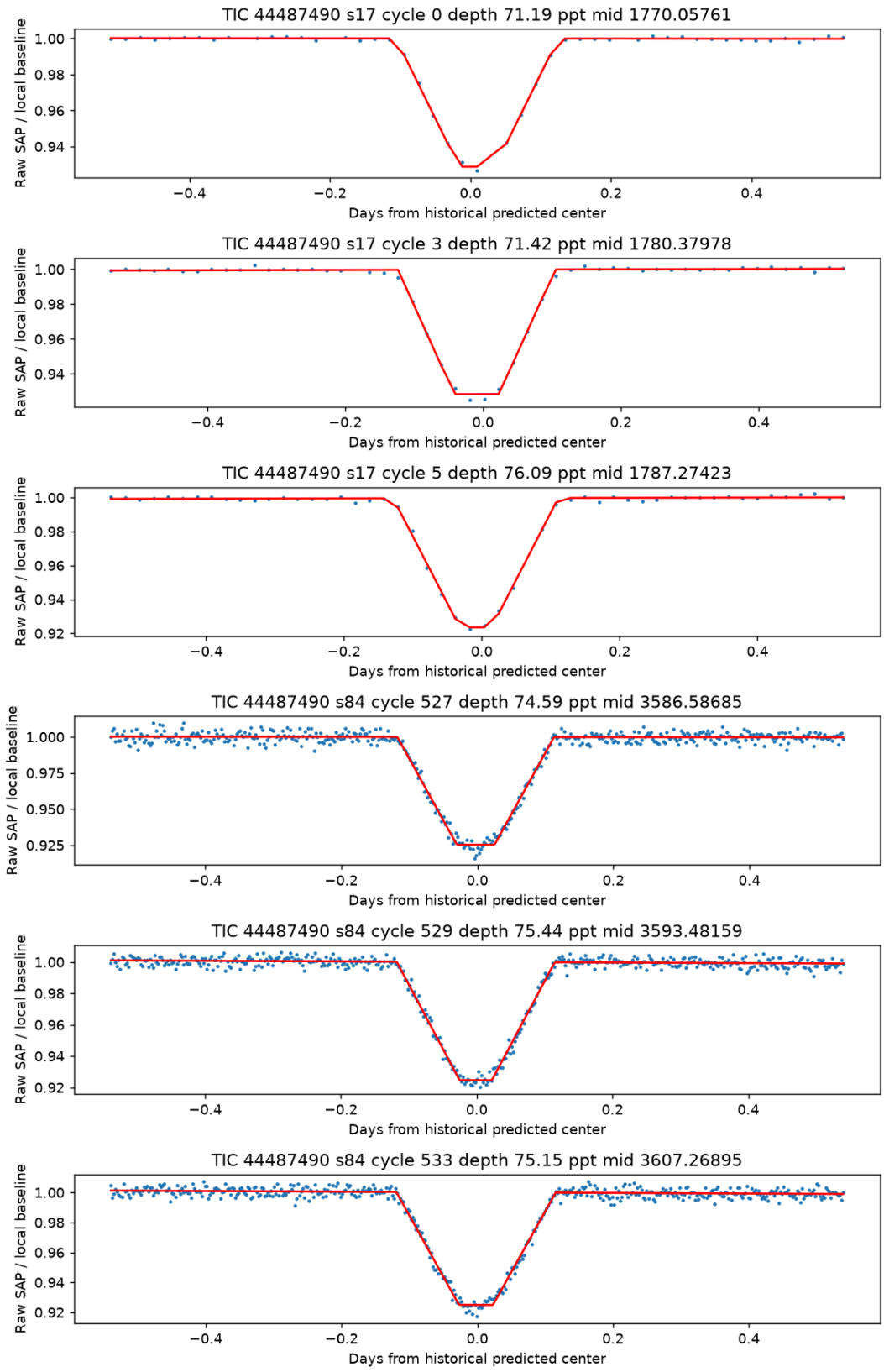
R532. TIC 44487490

physical_verdict: CONFIRMED

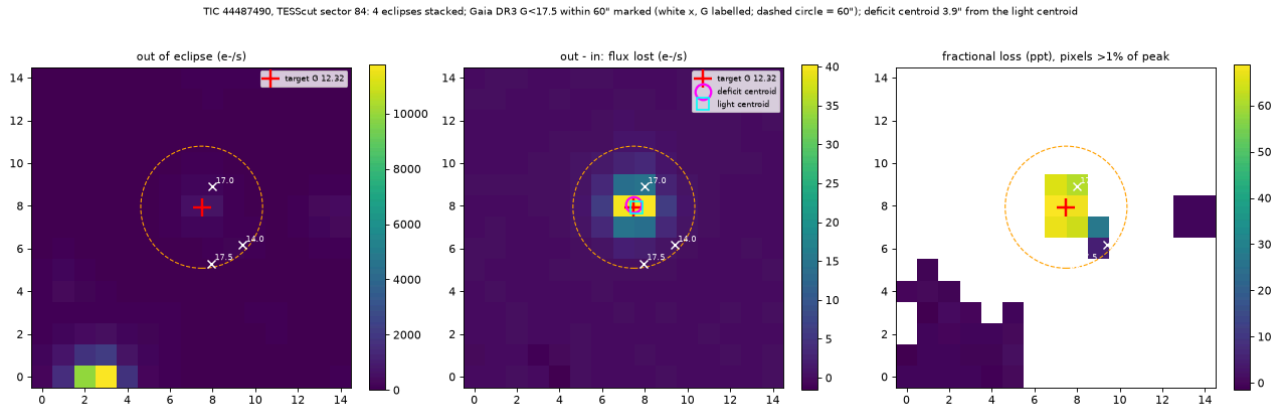
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional characterization of both eclipse kinds.



R532; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R532; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 44487490: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 5 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [17, 84]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 6.8939207479 ± 0.0000281001 d, reference BTJD 1780.3799981 (Odd-cycle, slightly deeper eclipse of the full orbit). Odd historical half-period cycles; the adjacent even events are a distinct eclipse kind with alternating timing offsets. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S84 TESScut deficit centroid is 2.39 arcsec from the target coordinate. Peak deficit pixel [8, 8]; target pixel [7, 8]. Image-model fractional residual 0.236 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 4 predicted event windows give 1x1/3x3/5x5 depths 68.92/66.68/60.44 ppt; median selected QLP raw fit depth is 75.15 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 2863142675459097216 has orbital P = 6.893921667 d and e = 0.0556. The nearest edge-on conjunction is offset by -0.0401 d (-0.0058 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.1483 d. Approximate central duration 8.28 h versus observed median 5.56 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Adjacent S17 and S84 dips alternate timing around the 3.44695-day photometric search clock. Odd cycles define the same-kind 6.89392-day orbit, matching Gaia; the even events are the other eclipse kind. The S84 image is target-centered, with the sole capable neighbor about 54 arcsec away.

Next check: Optional characterization of both eclipse kinds.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
17	3	1780.3797789	0.010416	71.42
17	5	1787.2742324	0.010416	76.09
84	527	3586.5868457	0.001157	74.59
84	529	3593.4815850	0.001157	75.44
84	533	3607.2689533	0.001157	75.15

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
2863142885913312128	53.96	13.988	0.382	57.0

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s84.json`, `historical/diff_s84.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R532. Public evidence files: diff_s84.json, diff_s84.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

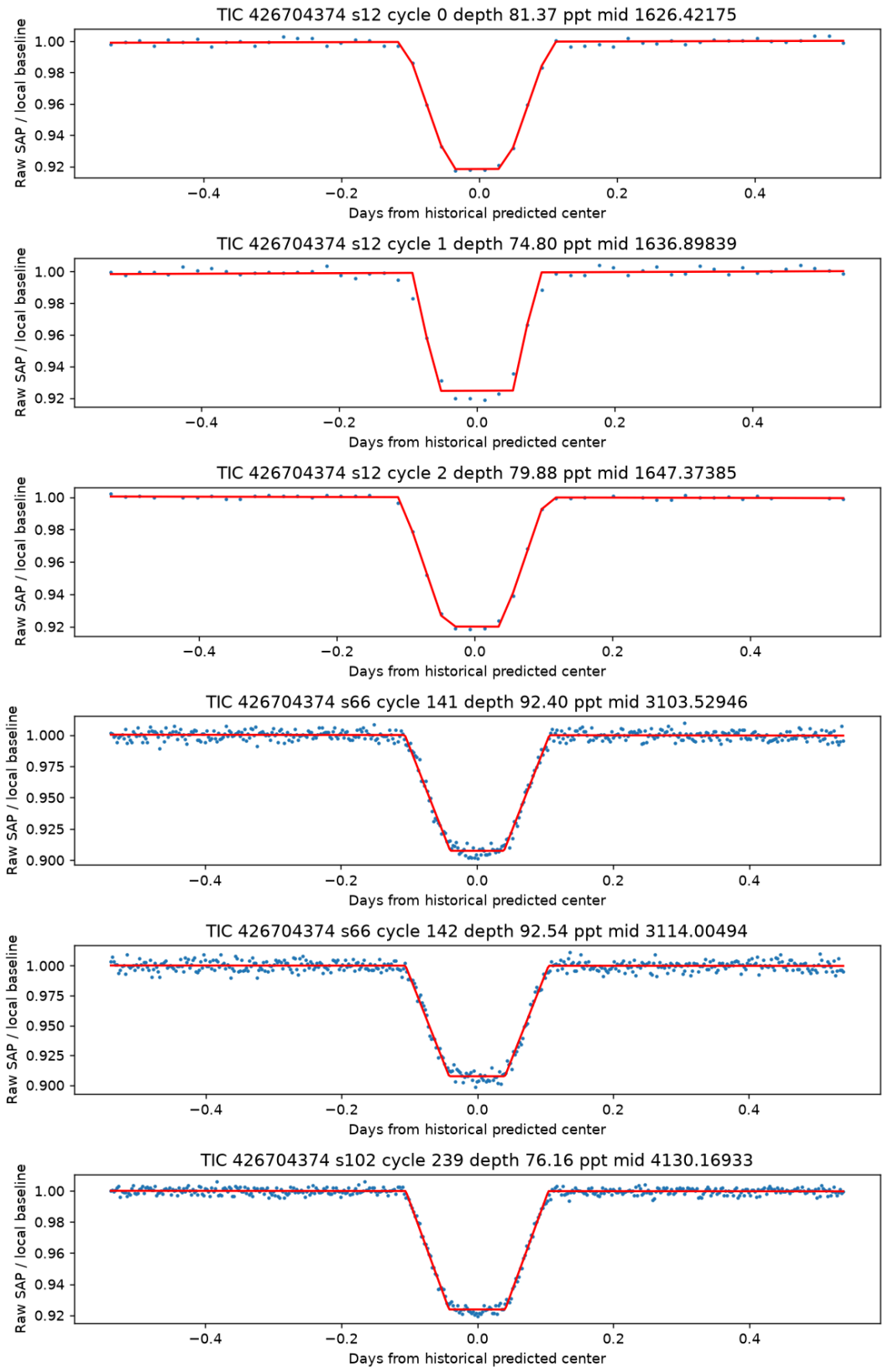
R533. TIC 426704374

physical_verdict: CONFIRMED

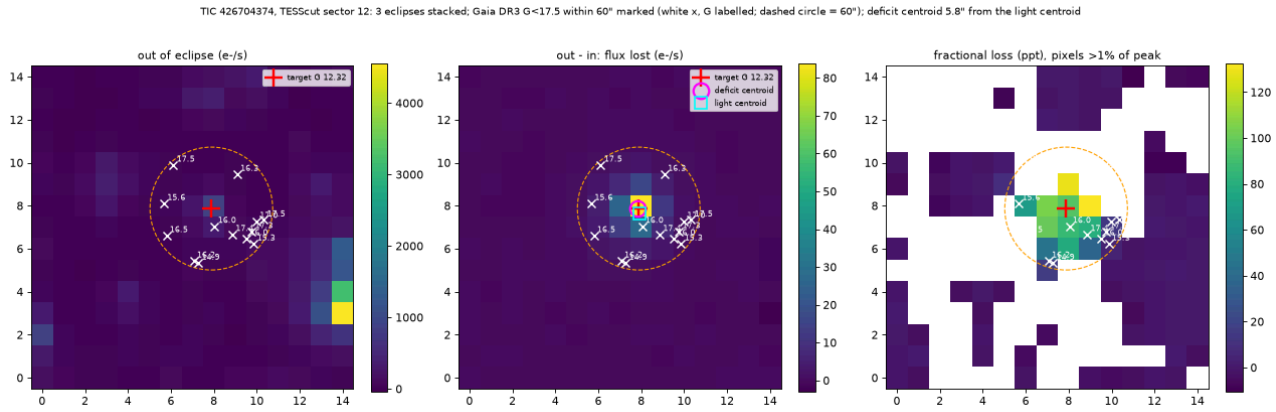
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R533; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R533; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 426704374: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 6 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [12, 66, 102]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 10.4759214103 ± 0.0000134247 d, reference BTJD 1626.4242223 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S12 TESScut deficit centroid is 1.59 arcsec from the target coordinate. Peak deficit pixel [8, 8]; target pixel [8, 8]. Image-model fractional residual 0.619 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 3 predicted event windows give 1x1/3x3/5x5 depths 96.32/102.01/94.06 ppt; median selected QLP raw fit depth is 80.63 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 5820688633321806080 has orbital P = 10.474613213 d and e = 0.0437. The nearest edge-on conjunction is offset by +0.0113 d (+0.0011 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.2625 d. Approximate central duration 6.59 h versus observed median 5.05 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Raw eclipses recur across S12, S66 and S102, including adjacent S12 and S66 cycles. The S12 deficit is target-centered and no Gaia neighbor in the 60-arcsec screen passes the flux-budget bound.

Next check: Optional precision characterization.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
12	0	1626.4217502	0.010417	81.37
12	1	1636.8983902	0.010417	74.80
12	2	1647.3738533	0.010417	79.88
66	141	3103.5294601	0.001157	92.40
66	142	3114.0049354	0.001157	92.54
102	239	4130.1693271	0.001157	76.16

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s12.json`, `historical/diff_s12.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R533. Public evidence files: diff_s12.json, diff_s12.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

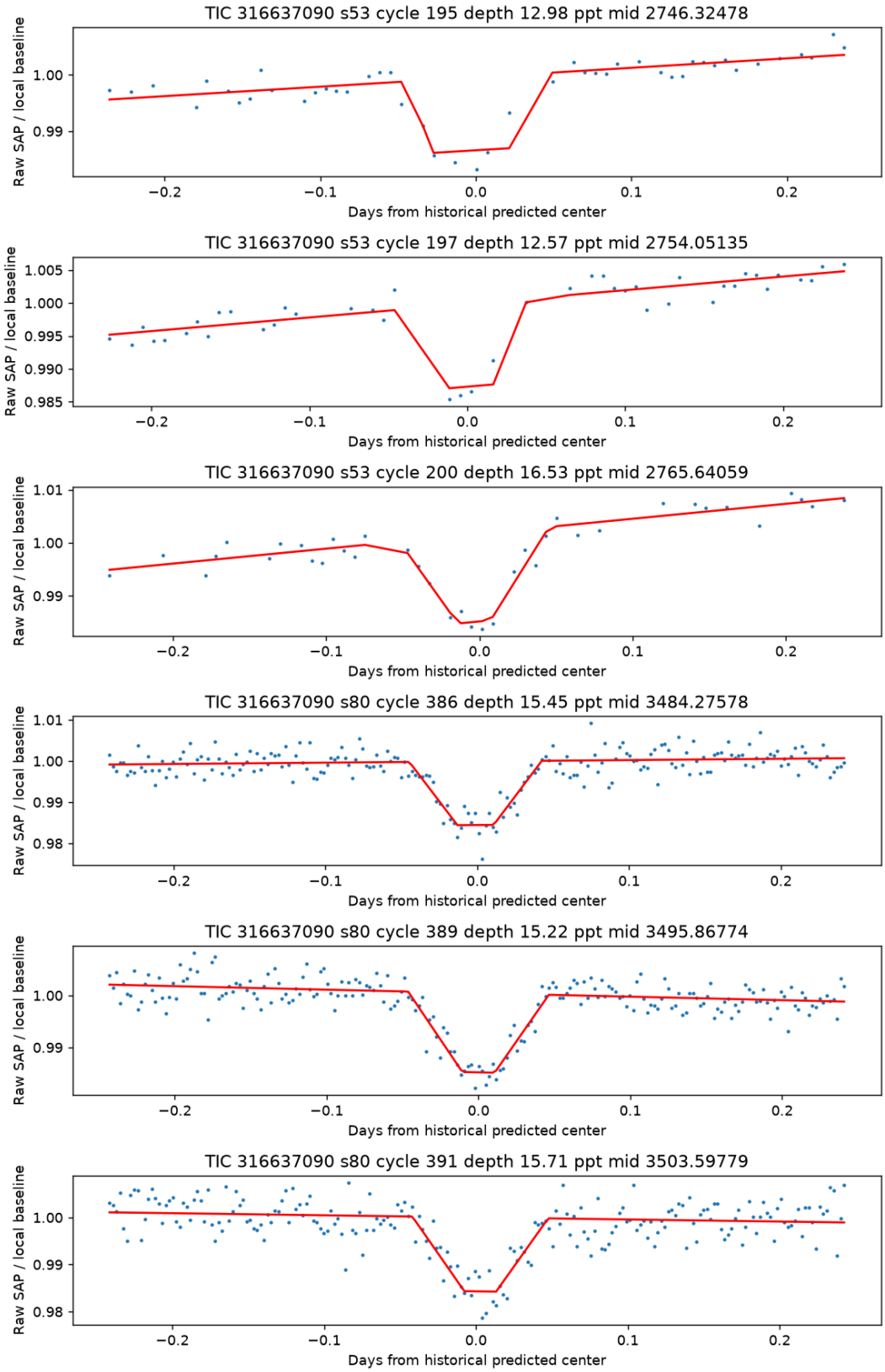
R534. TIC 316637090

physical_verdict: CONFIRMED

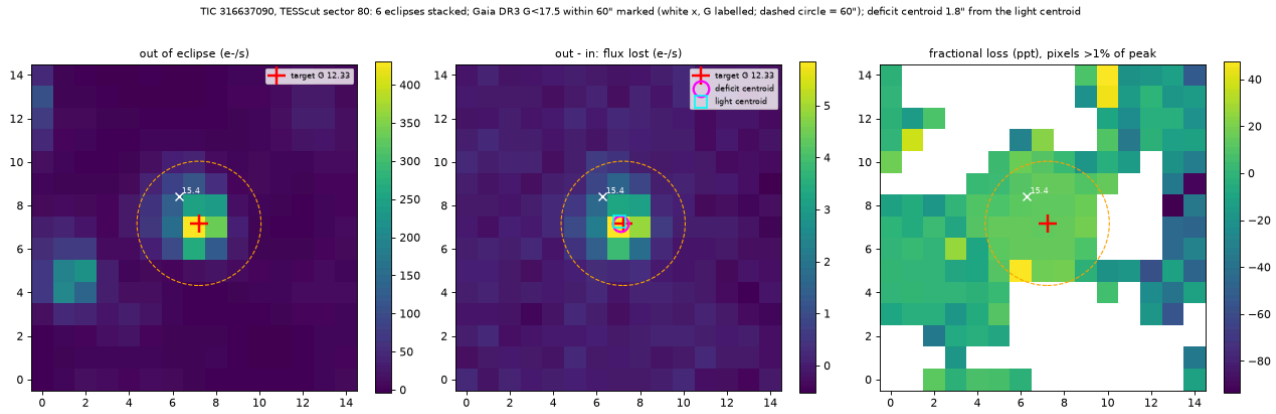
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional resolved photometry would strengthen this shallow attribution.



R534; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R534; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 316637090: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 10 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [53, 80]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 3.8636264942 ± 0.0000122316 d, reference BTJD 2746.3243407 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S80 TESScut deficit centroid is 3.12 arcsec from the target coordinate. Peak deficit pixel [7, 7]; target pixel [7, 7]. Image-model fractional residual 0.179 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 6 predicted event windows give 1x1/3x3/5x5 depths 13.58/13.92/13.64 ppt; median selected QLP raw fit depth is 15.34 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 4595458674208503424 has orbital P = 3.862983602 d and e = 0.1667. The nearest edge-on conjunction is offset by +0.2349 d (+0.0608 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.1301 d. Approximate central duration 4.61 h versus observed median 2.18 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Ten shallow raw events in S53 and S80 repeat; the S80 aperture extraction recovers six. The target-near deficit is roughly 31 arcsec from the only capable Gaia neighbor. The Gaia conjunction offset is descriptive and is not used as a significance veto.

Next check: Additional resolved photometry would strengthen this shallow attribution.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
53	195	2746.3247811	0.003472	12.98
53	197	2754.0513492	0.006473	12.57
53	198	2757.9189913	0.005698	14.81
53	200	2765.6405893	0.004227	16.53
80	385	3480.4125219	0.001253	16.41
80	386	3484.2757810	0.001157	15.45
80	388	3491.9986733	0.004023	10.24
80	389	3495.8677415	0.001157	15.22
80	390	3499.7325144	0.001157	16.71
80	391	3503.5977894	0.001547	15.71

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
4595458605489025280	29.26	15.390	0.246	30.8

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s80.json`, `historical/diff_s80.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R534. Public evidence files: diff_s80.json, diff_s80.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

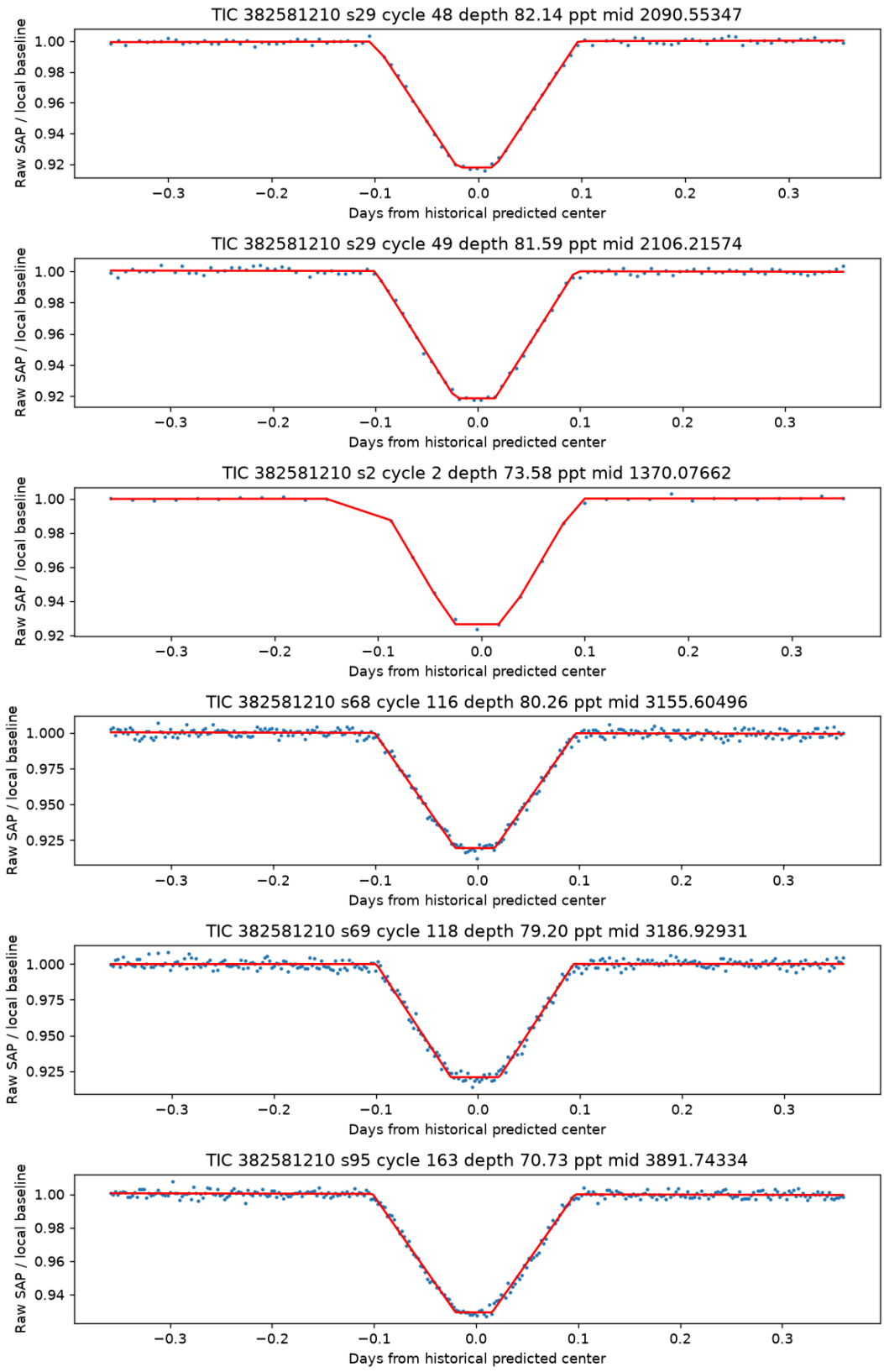
R535. TIC 382581210

physical_verdict: CONFIRMED

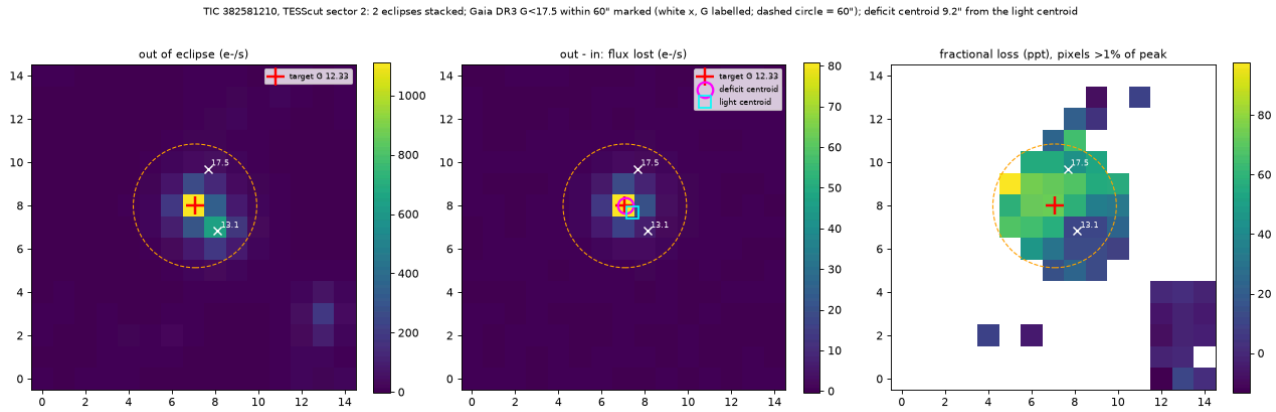
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional later-sector timing and resolved photometry.



R535; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R535; Representative eclipse-deficit image; source attribution retains the report limits.

Independent recurrence and coverage check. The S29 clean event pair and intermediate windows constrain equal-kind aliases; full numerical results and scripts are preserved in the public helper archive.

382581210-recurrence.csv (rounded display; full precision in linked CSV)

Sector	Cycle	Center BTJD	Good in window	Baseline	Deficit ppt
002	-47	1354.427	9	17	52.5
002	-46	1370.089	10	28	45.9
003	-45	1385.751	7	26	59.2
003	-44	1401.413	9	28	50.9
028	-1	2074.891	0	1	not covered
068	68	3155.588	87	259	48.7
068	69	3171.250	87	259	50.5
069	70	3186.913	87	259	52.9
069	71	3202.575	87	259	52.5
095	115	3891.715	86	259	43.5
095	116	3907.377	0	0	not covered

382581210-coverage.csv (rounded display; full precision in linked CSV)

Window	Phase	Center BTJD	Good	Baseline	Deficit ppt
event1	0.000	2090.553	27	83	49.8
quarter	0.250	2094.469	25	68	-0.3
third	0.333	2095.774	28	84	-0.1
half	0.500	2098.385	28	84	-1.1
two_thirds	0.667	2100.995	0	0	not covered
three_quarters	0.750	2102.300	0	0	not covered
event2	1.000	2106.216	29	84	52.7

Archived source: sanitized report text.

TIC 382581210: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 10 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [2, 3, 29, 68, 69, 95]. The local fit has a free center, depth, width and

baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.

2. **Measured timing and aliases.** P = 15.6625230978 ± 0.0000195721 d, reference BTJD 1354.4142823 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S2 TESScut deficit centroid is 0.97 arcsec from the target coordinate. Peak deficit pixel [7, 8]; target pixel [7, 8]. Image-model fractional residual 0.415 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 2 predicted event windows give 1x1/3x3/5x5 depths 72.76/56.61/50.49 ppt; median selected QLP raw fit depth is 78.95 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 4699704947887847296 has orbital P = 15.665541787 d and e = 0.2966. The nearest edge-on conjunction is offset by -0.2056 d (-0.0131 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.5483 d. Approximate central duration 7.01 h versus observed median 4.68 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: The current MAST QLP availability record lists nine sectors; all nine files were retrieved and checked. Ten measured quality-zero raw eclipses in S2, S3, S29, S68, S69 and S95 now span the long baseline, including adjacent S29 and S68 pairs. The S29 clean intermediate windows reject equal-kind half, third and quarter-period submultiples; flagged windows were not used as nulls. The historical S2 cutout separately stacks two sparse predicted windows into one target-centered deficit, which alone would not establish each window as an eclipse. Its capable Gaia neighbor is 33 arcsec from the target and 32.5 arcsec from the deficit.

Next check: Optional later-sector timing and resolved photometry.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
29	48	2090.5534699	0.003472	82.14
29	49	2106.2157431	0.003472	81.59
2	1	1354.4178976	0.010417	69.70
2	2	1370.0766181	0.010417	73.58
3	4	1401.4020768	0.010416	74.99
68	116	3155.6049624	0.001157	80.26
68	117	3171.2668610	0.001157	78.70
69	118	3186.9293125	0.001157	79.20
69	119	3202.5912876	0.001157	80.78

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
95	163	3891.7433424	0.001157	70.73

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
4699704986543565824	32.97	13.094	0.226	32.5

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s02.json`, `historical/diff_s02.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R535. Public evidence files: diff_s02.json, diff_s02.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 382581210-coverage.csv 382581210-coverage.py 382581210-downloads.json 382581210-recurrence.csv 382581210-recurrence.py

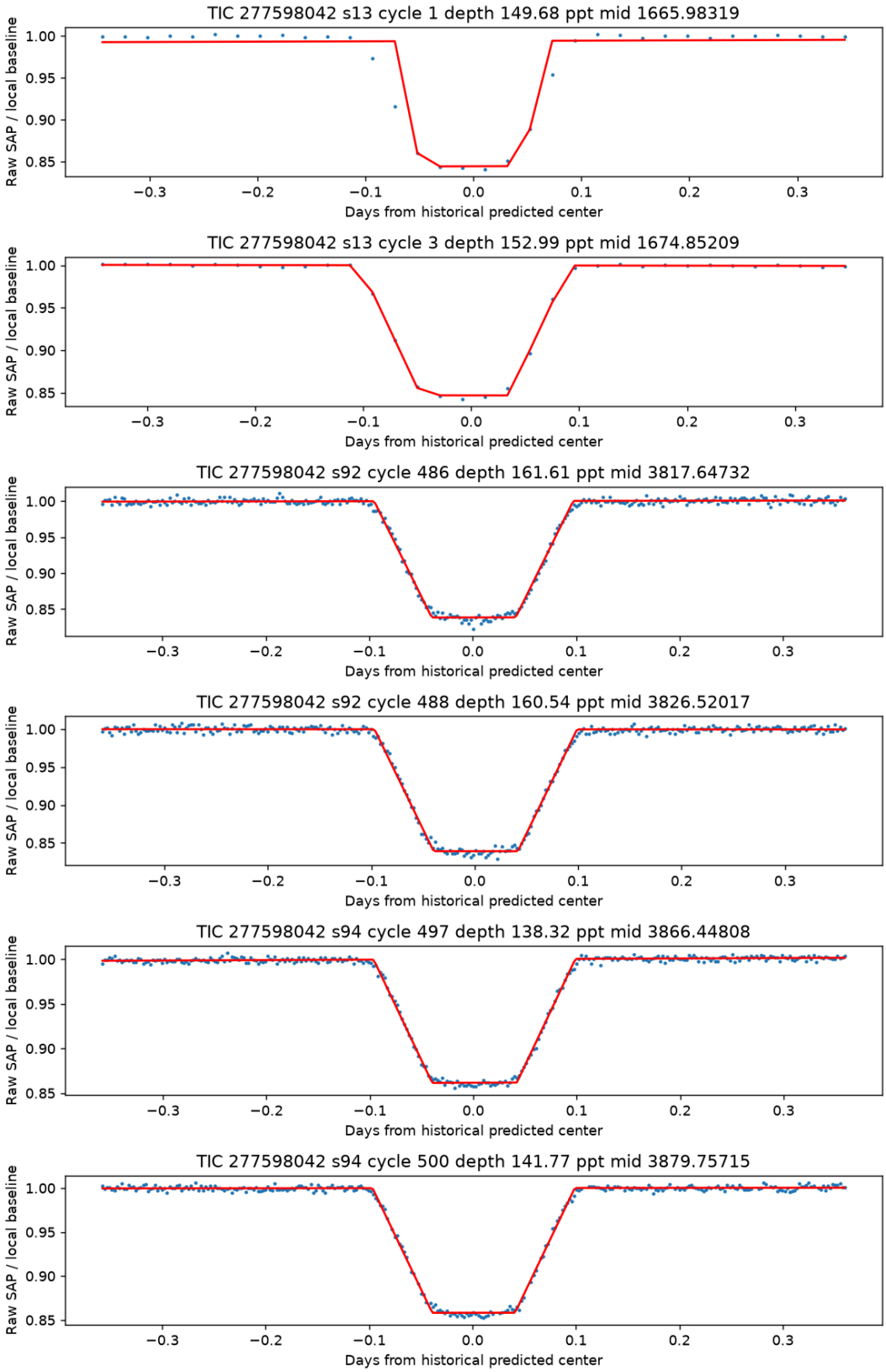
R536. TIC 277598042

physical_verdict: CONFIRMED

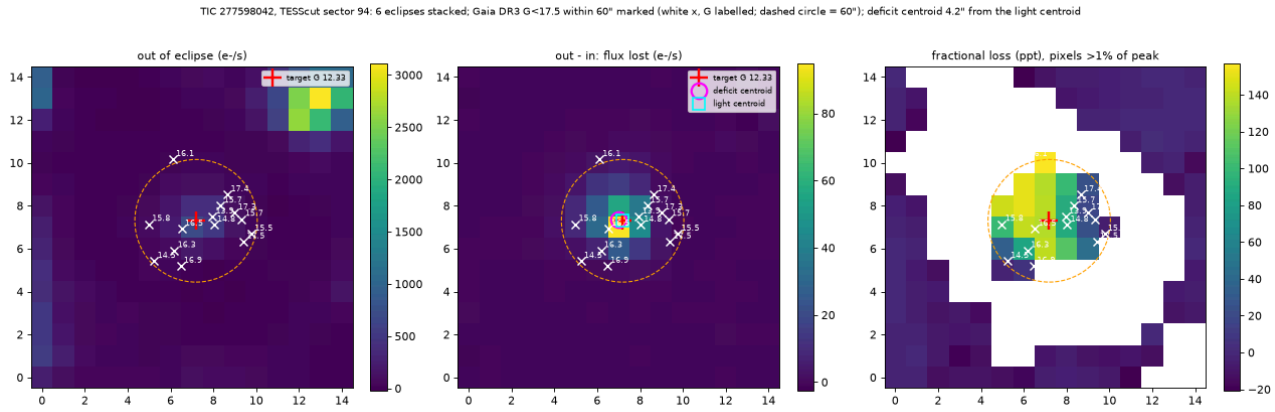
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry.



R536; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R536; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 277598042: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 14 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [13, 92, 94]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 4.4364264100 ± 0.0000120101 d, reference BTJD 1665.9805449 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S94 TESScut deficit centroid is 4.11 arcsec from the target coordinate. Peak deficit pixel [7, 7]; target pixel [7, 7]. Image-model fractional residual 0.279 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 6 predicted event windows give 1x1/3x3/5x5 depths 138.22/120.70/109.60 ppt; median selected QLP raw fit depth is 151.33 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 6732852653768547840 has orbital P = 4.435881632 d and e = 0.0840. The nearest edge-on conjunction is offset by +0.0447 d (+0.0101 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.1250 d. Approximate central duration 4.98 h versus observed median 4.68 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Fourteen raw events span S13, S92 and S94 with multiple adjacent cycles. The S94 difference core lies near the target and about 20 arcsec from the only capable Gaia neighbor; that neighbor would need an approximately 80% intrinsic eclipse.

Next check: Optional resolved photometry.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
13	1	1665.9831857	0.010417	149.68
13	2	1670.4154686	0.010417	155.48
13	3	1674.8520864	0.010417	152.99
92	483	3804.3384462	0.001157	162.28
92	484	3808.7745133	0.001157	162.52
92	486	3817.6473162	0.001157	161.61
92	487	3822.0834577	0.001157	161.63
92	488	3826.5201663	0.001157	160.54
94	495	3857.5750328	0.001157	142.98
94	496	3862.0116368	0.001157	139.06
94	497	3866.4480830	0.001157	138.32
94	498	3870.8844811	0.001157	141.72
94	499	3875.3211812	0.001157	140.58
94	500	3879.7571485	0.001157	141.77

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
6732864026841949952	15.82	13.858	0.797	19.9

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s94.json`, `historical/diff_s94.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R536. Public evidence files: diff_s94.json, diff_s94.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

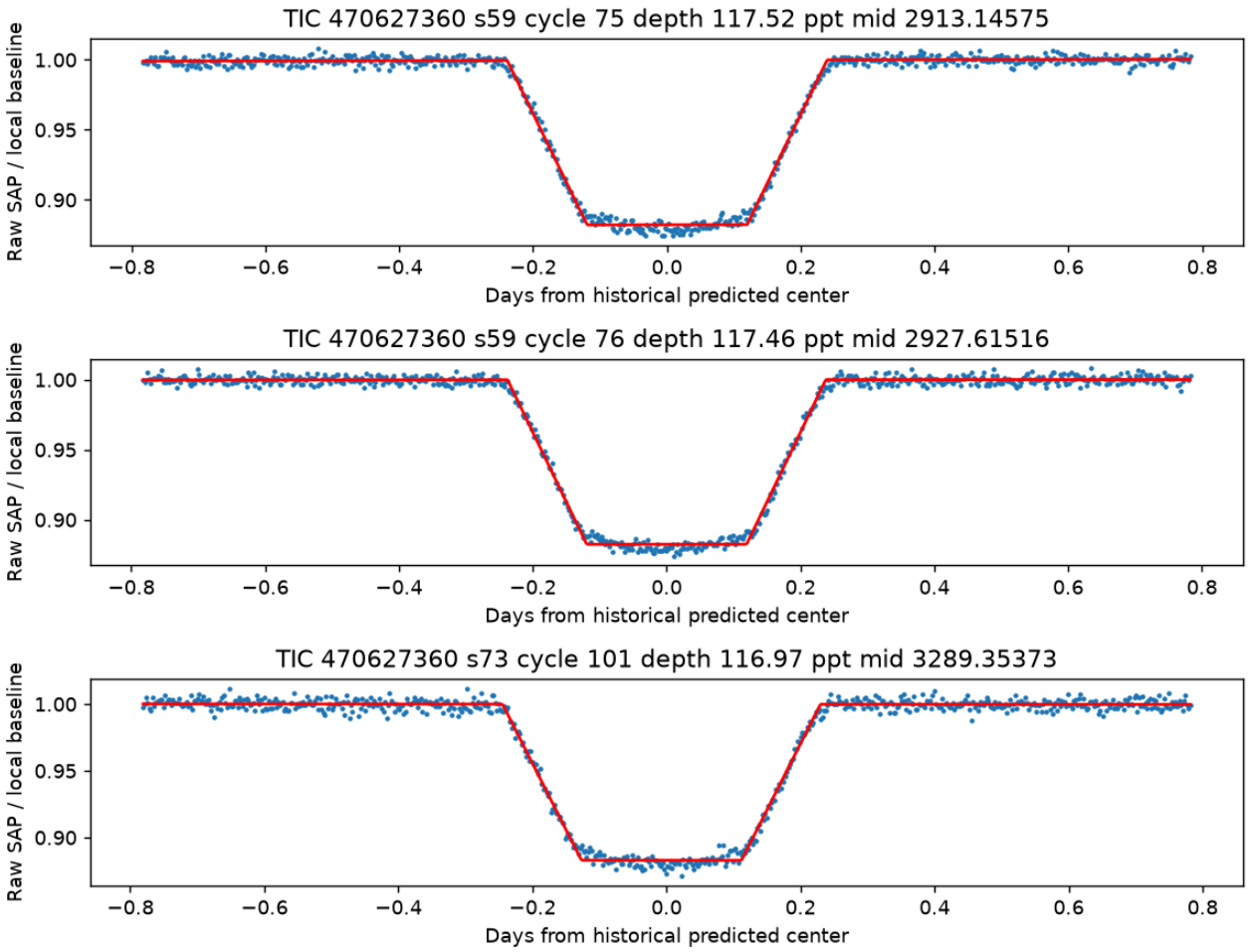
R537. TIC 470627360

physical_verdict: CONFIRMED

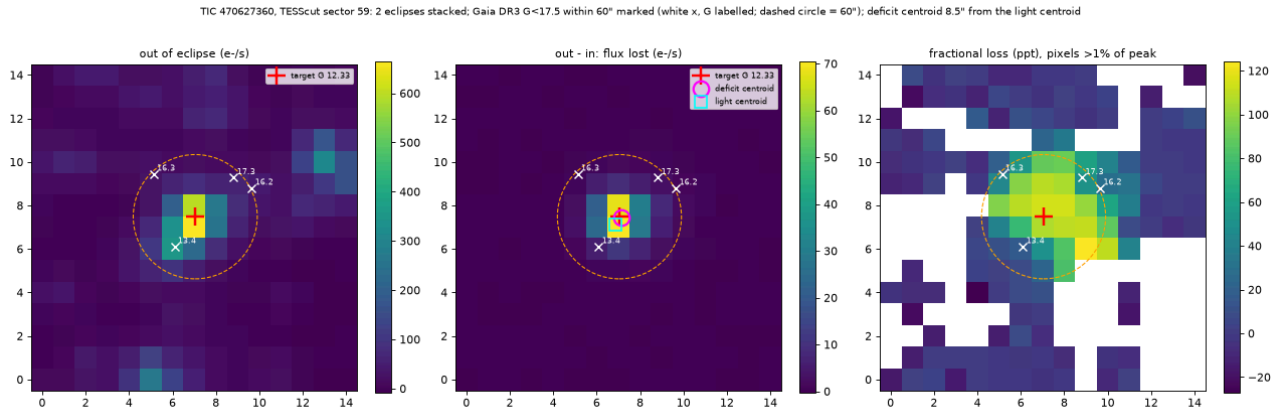
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Further-sector timing would refine the modest three-event ephemeris.



R537; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R537; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 470627360: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 3 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [59, 73]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 14.4695399219 ± 0.0000555576 d, reference BTJD 2913.1456895 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S59 TESScut deficit centroid is 1.66 arcsec from the target coordinate. Peak deficit pixel [7, 7]; target pixel [7, 7]. Image-model fractional residual 0.330 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 2 predicted event windows give 1x1/3x3/5x5 depths 106.04/90.28/83.80 ppt; median selected QLP raw fit depth is 117.46 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 197378967299172224 has orbital P = 14.476795633 d and e = 0.3774. The nearest edge-on conjunction is offset by -0.6255 d (-0.0432 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 1.0802 d. Approximate central duration 14.33 h versus observed median 11.41 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: S59 has two adjacent measured eclipses and S73 supplies a longer-baseline third. The S59 difference core is target-centered, about 35.5 arcsec from its only capable Gaia neighbor; the long event is also recovered by aperture extraction.

Next check: Further-sector timing would refine the modest three-event ephemeris.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
59	75	2913.1457540	0.001157	117.52
59	76	2927.6151624	0.001157	117.46
73	101	3289.3537301	0.001157	116.97

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
197378971597385472	34.28	13.428	0.445	35.5

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s59.json`, `historical/diff_s59.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R537. Public evidence files: diff_s59.json, diff_s59.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

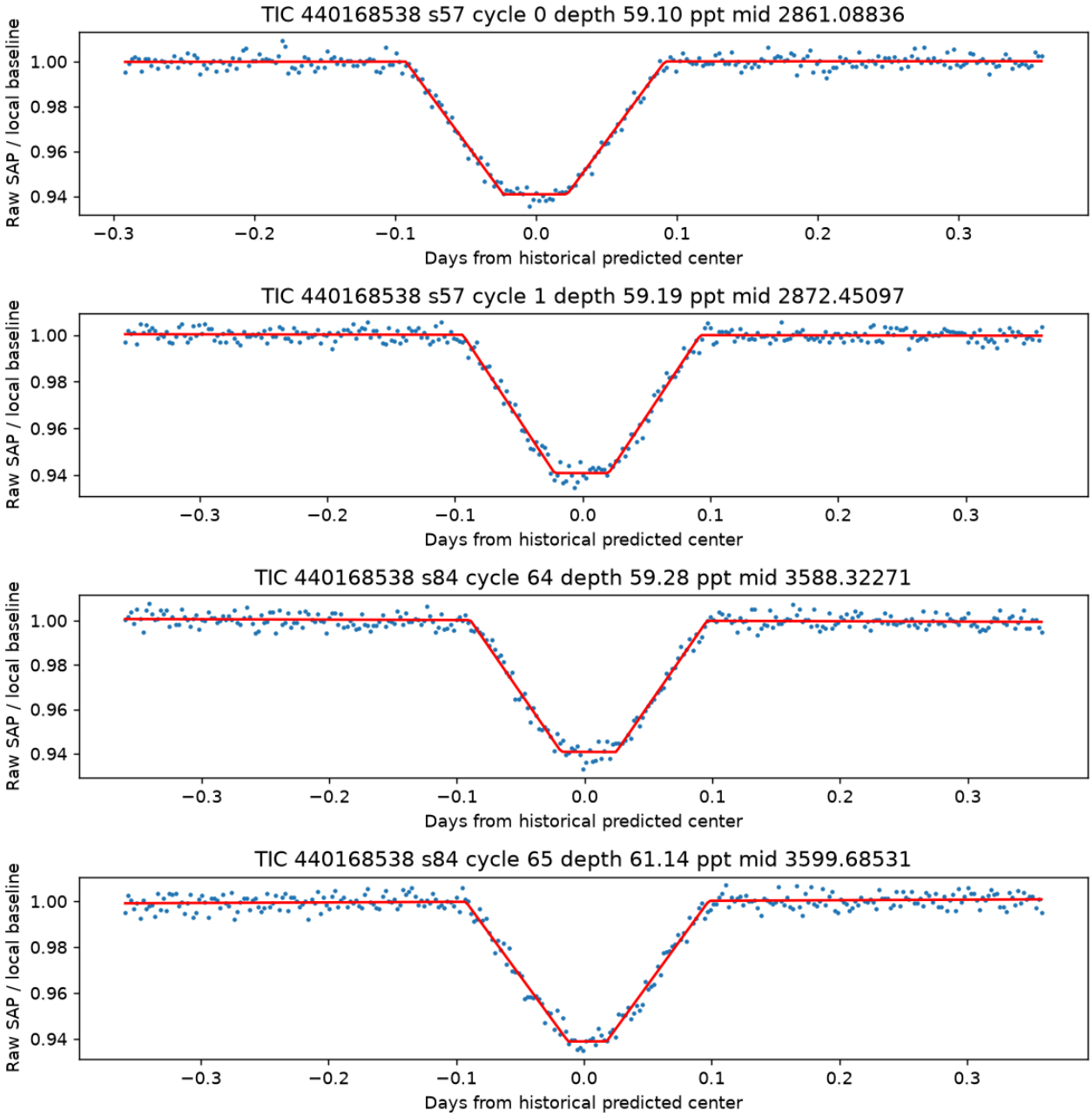
R538. TIC 440168538

physical_verdict: CONFIRMED

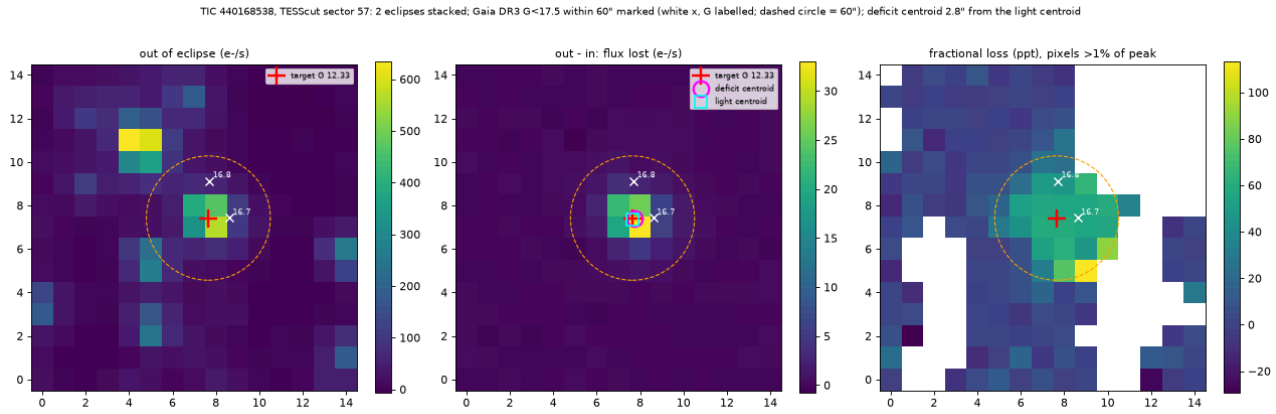
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry.



R538; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R538; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 440168538: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 4 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [57, 84]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 11.3630365737 ± 0.0000180823 d, reference BTJD 2861.0881487 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S57 TESScut deficit centroid is 2.24 arcsec from the target coordinate. Peak deficit pixel [8, 7]; target pixel [8, 7]. Image-model fractional residual 0.766 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 2 predicted event windows give 1x1/3x3/5x5 depths 58.62/56.89/53.38 ppt; median selected QLP raw fit depth is 59.24 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 391817260687992960 has orbital P = 11.362526050 d and e = 0.0213. The nearest edge-on conjunction is offset by +0.2062 d (+0.0181 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.3661 d. Approximate central duration 7.55 h versus observed median 4.46 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Two adjacent raw eclipses in each of S57 and S84 anchor the recurrence. The S57 deficit favors the target; no Gaia neighbor within 60 arcsec passes the flux-budget screen. The image-model residual (0.77) limits localization precision.

Next check: Optional resolved photometry.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
57	0	2861.0883552	0.001157	59.10
57	1	2872.4509721	0.001157	59.19
84	64	3588.3227099	0.001157	59.28
84	65	3599.6853121	0.001157	61.14

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s57.json`, `historical/diff_s57.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R538. Public evidence files: diff_s57.json, diff_s57.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

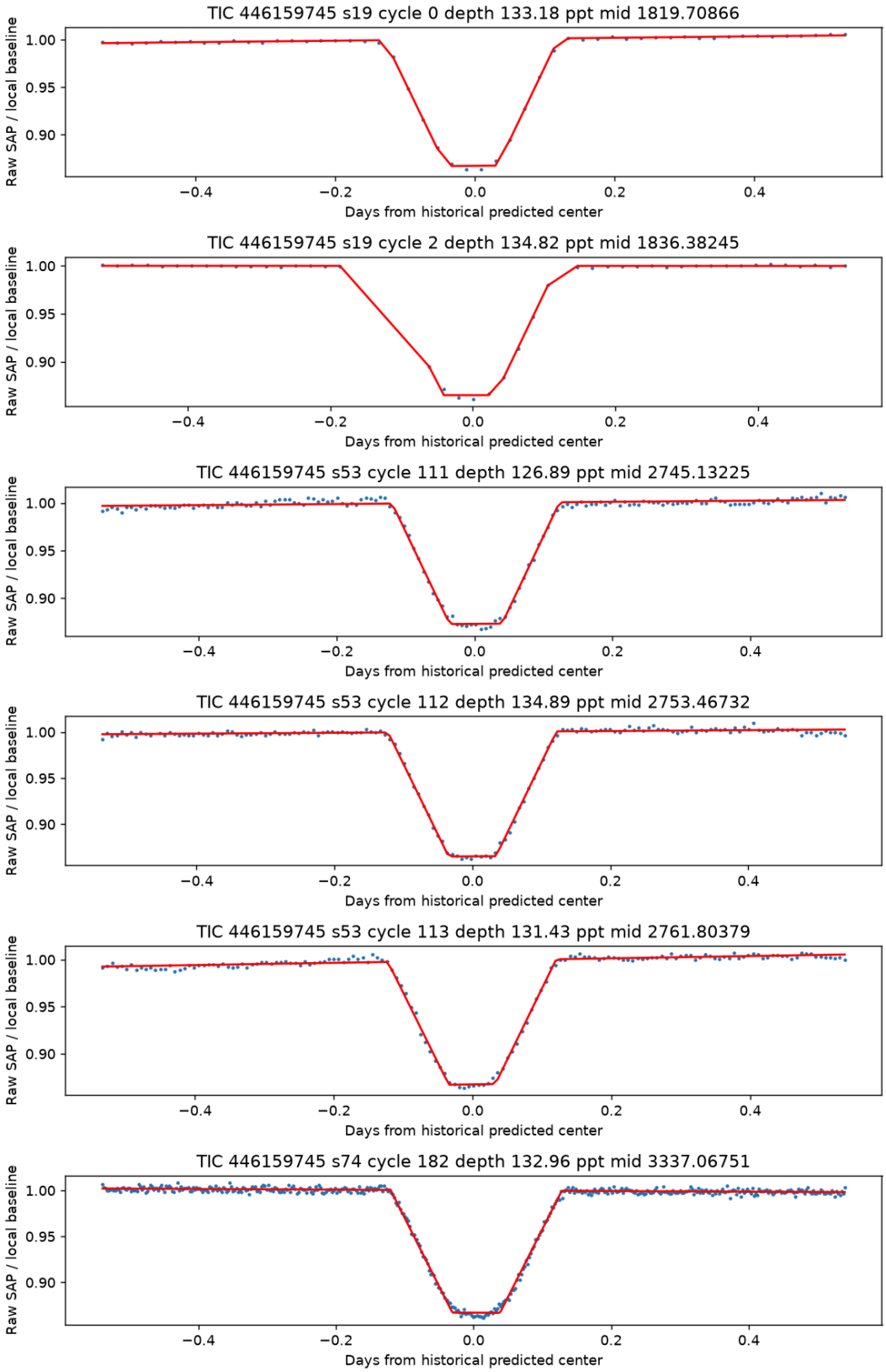
R539. TIC 446159745

physical_verdict: CONFIRMED

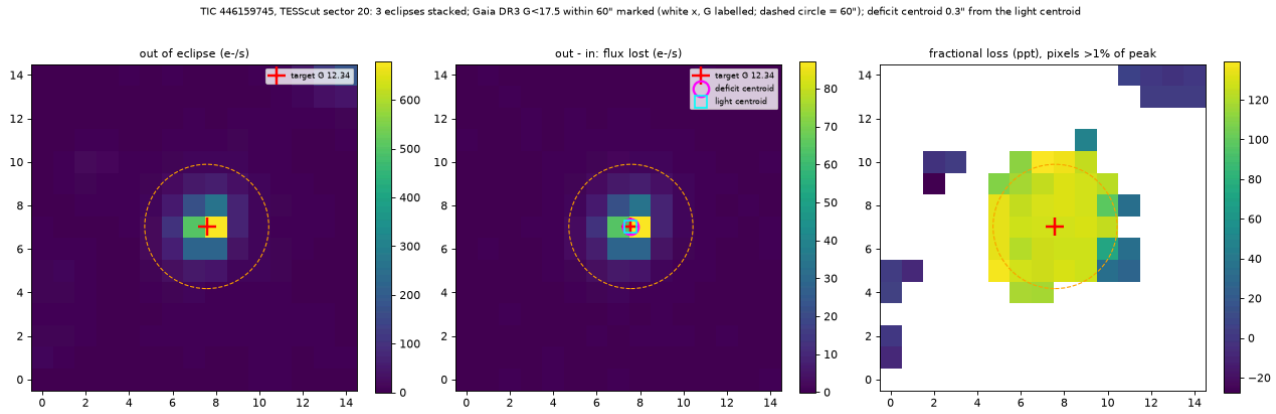
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R539; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R539; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 446159745: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 7 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [19, 53, 74]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 8.3371405920 ± 0.0000254401 d, reference BTJD 1819.7081287 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S20 TESScut deficit centroid is 0.30 arcsec from the target coordinate. Peak deficit pixel [8, 7]; target pixel [8, 7]. Image-model fractional residual 0.023 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 3 predicted event windows give 1x1/3x3/5x5 depths 128.37/128.36/127.20 ppt; median selected QLP raw fit depth is 132.96 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout sector is a separate TESS observing occasion from the selected QLP sectors; it is the same instrument with an alternative reduction, not independent-instrument confirmation. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 1139354875446856704 has orbital P = 8.337129275 d and e = 0.0553. The nearest edge-on conjunction is offset by -0.0237 d (-0.0028 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.2423 d. Approximate central duration 7.25 h versus observed median 5.94 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Seven raw eclipses span S19, S53 and S74, with an adjacent S53 triplet. The S20 alternative extraction is a separate observing occasion with three events; its deficit is sharply target-centered and no capable Gaia neighbor is listed.

Next check: Optional precision characterization.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
19	0	1819.7086587	0.010417	133.18
19	2	1836.3824472	0.010417	134.82
53	111	2745.1322540	0.003472	126.89
53	112	2753.4673157	0.003472	134.89
53	113	2761.8037927	0.003472	131.43
74	180	3320.3936679	0.001157	132.55
74	182	3337.0675061	0.001157	132.96

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s20.json`, `historical/diff_s20.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R539. Public evidence files: diff_s20.json, diff_s20.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

Period-fold source attribution and catalogue comparisons

TIC 195865529 - CONFIRMED

Unique count addition: no.

Known ASAS-SN EA at 0.16 arcsec (P 3.8310 d); ATLAS DBH P/2. Recovery excluded from new count.

Catalogue scope: **already catalogued**: ASAS-SN EA P 3.8310 d at 0.16"; ATLAS DBH at P/2; not in VSX

Full source report

TIC 468181994 - CONFIRMED (weaker)

Unique count addition: yes.

Period -2.31 sigma from Gaia; RUWE 12.6/unresolved companion risk; independent pipelines agree about 2 sigma. Literature cones limited to enumerated catalogues, not proof of novelty.

Catalogue scope: not in VSX/ASAS-SN/Gaia-EB (per report)

Full source report

TIC 470235665 - CONFIRMED

Unique count addition: yes.

QLP-only secondary phase drift unconfirmed; 4-5% pipeline scale offset. Enumerated catalogue cones and SIMBAD SB* do not establish exhaustive novelty.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/ASAS-SN within 60"; SIMBAD SB*

Full source report

TIC 379251709 - CONFIRMED

Unique count addition: yes.

Secondary fold-level, not localized; primary securely on target. QLP/SPOC secondary phases differ 2.6 sigma. Radius tension retained.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/ASAS-SN within 60"; SIMBAD SB*

Full source report

TIC 70665829 - CONFIRMED

Unique count addition: no.

Already objects.csv A4; blind control independently recovered. Archive Gaia-VEB/varisum/SB9 controls failed on later mirror timeout, predecessor empty cones only. No addition.

Catalogue scope: not in VSX/PrÅia/IJspeert within 60"

Full source report; existing A4 evidence

TIC 331293467 - CONFIRMED

Unique count addition: yes.

Heavy blend; QLP plus independent own TESScut satisfies available pipeline check; depths model-dependent 80-116 ppt, ratios/duration agree. Gaia geometry depends on covariance. Gaia-VEB/varisum/SB9 cones lack successful positive controls; no exhaustive novelty claim.

Catalogue scope: not in VSX/PrÅia/IJspeert within 60"; Gaia-EB/varisum/SB9 cones from predecessor run only (mirror timed out)

Full source report

TIC 283532878 - CONFIRMED

Unique count addition: yes.

Use successful s12d recheck, not false-empty first archive pass. Known EB 29.7 arcsec away at P 1.7975 d excluded as source; up to 3 ppt contamination possible. Phase drift tentative.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert on target; **a catalogued EB (Gaia DR3 2202301140233408128, P 1.7975 d, G 17.66) sits 29.7" away**, excluded as source but may add ≤3 ppt dips

Full source report

TIC 407877153 - CONFIRMED

Unique count addition: yes.

Secondary pixel signal weak individually but summed delta chi2 >=166; QLP depth diluted. Use successful s12d catalogue rerun; all-catalogue query unreliable.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert within 60"

Full source report

TIC 15855794 - CONFIRMED

Unique count addition: yes.

Exclude flagged E124 from primary pixel stack as report does. Catalogue positive controls succeeded; catalogue absence only within searched scope.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert within 60"

Full source report

TIC 92881948 - CONFIRMED

Unique count addition: yes.

VSX ROT/Gaia SOLAR_LIKE is prior variable classification, not prior EB; catalogue positive controls succeeded. QLP dilution explains shallower depth.

Catalogue scope: VSX lists it as ROT (Gaia DR3 variable) only; not in any EB catalogue

Full source report

TIC 468827621 - CONFIRMED

Unique count addition: yes.

Primary localized in two sectors both camera 2; secondary untested in pixels and neighbour could supply in worst case. Literature search skipped, though catalogue cones performed; known Gaia short-timescale variable is not prior EB.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/SB9 within 60"; SB*

Full source report

TIC 416285595 - CONFIRMED, with caveats

Unique count addition: yes.

Returned CONFIRMED with caveats: QLP/SPOC s40 61.7 vs 54.7 ppt (~12%, ~4 sigma); TESScut depth unreliable. Secondary not pixel-tested; geometry weak/covariance-dependent. Catalogue cones performed but literature search skipped. Report itself says strict depth agreement would imply PROBABLE; no new stricter tolerance imposed here.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/SB9 within 60"; SB*

Full source report

TIC 372683900 - ELLIPSOIDAL

Unique count addition: no.

Smooth double wave, no narrow eclipse; unresolved 1.8 arcsec neighbour; Gaia eccentricity conflicts at 3.3 sigma.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 13106408 - ELLIPSOIDAL

Unique count addition: no.

Smooth double wave; nearby 364-day SR unrelated.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 372683622 - ELLIPSOIDAL

Unique count addition: no.

Double wave plus variable spots; no eclipse.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 319185391 - PLAUSIBLE

Unique count addition: no.

Fold-only ~1 ppt dip, individual events ~1 sigma, dip not pixel-tested, no second pipeline. Report prose says probable-level but batch summary says PLAUSIBLE at best; adopt batch summary outside count. Gaia 434-day astrometric orbit suggests triple.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 62328710 - NEIGHBOUR_SIGNAL

Unique count addition: no.

Delta Scuti-like signal localized to Gaia DR3 2165710355333626240 at 44 arcsec and independently seen there by ZTF; target flat at Gaia period; suspect SB1 orbit is interpretation.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 300903537 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 23112764 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 373062090 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 465073693 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 316405449 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 465263741 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 441741446 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

Archived detailed period-fold analyses

vet01 full report

Archived source: sanitized report text.

vet01: adversarial check of TIC 195865529 (ATO J310.7264+43.3062) and TIC 468181994 (UCAC4 023-009594)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. A first agent ran `s1–s8a` and was killed by a container restart. This run kept its downloads and outputs, checked its logs and continued with `s8–s17`. Everything is in `/home/user/eclipses-paper-review/work/c9a-vets/vet01/`. Downloads are in the scratchpad at `data/vet01/<tic>/`; the TESScut cubes were deleted after use. Depths are in ppt, times in BTJD (BJD_TDB − 2457000), durations in hours. A = conjunction at ν = 90° − ω, when the companion is in front of Gaia's RV star. B = ν = 270° − ω.

TIC 195865529: CONFIRMED. A detached F-star + K-dwarf eclipsing binary at the Gaia period. It is already a catalogued EB in ASAS-SN (II/366, EA, P = 3.8310 d) and ATLAS (class DBH at P/2), though not in VSX. - (1) Individual events: 48 primaries and 49 secondaries were re-measured one by one from raw QLP SAP (QUALITY==0). Another 3 primaries were fitted but excluded as partial. - (2) Period: P_ecl = 3.8310456 ± 0.0000009 d, which is −0.15σ from Gaia's. - (3) Pixels: the deficit is on the target in all 8 sectors, across 4 camera/CCD orientations. Every Gaia neighbour within 90″ is excluded by the pixels. The dilution arithmetic alone excludes all but one: the G 13.46 star at 36″ would need ≈ 100%. - (4) Independence: QLP SAP, my TESScut aperture and TASOC agree after correcting for each aperture's dilution: undiluted primary 183 / 192 / 196 ppt, secondary 48 / 51 / 47 ppt. - (5) Geometry: the primary is conjunction A (A within 0.18σ, B excluded at 3.3–3.9σ). The secondary sits at phase 0.49999 ± 0.00007; Gaia's e and ω predict 0.479 ± 0.024 (+0.87σ). - The screen was right about depths (≈ 185 / 48 ppt), phases and period. Those depths are on QLP's apparently deblended scale, though. Raw apertures see the same eclipse diluted: 127–144 ppt in my r = 1.5 px aperture and 71 ppt in TASOC, whose header gives 64% contamination.

TIC 468181994: CONFIRMED. This is the weaker of the two; see the caveats. A late-F/G star with an M-dwarf-mass companion (m2 sin i ≈ 0.21–0.24 M⊙). It is not in VSX, Gaia-EB, Prša or IJspeert within 60″. - (1) Individual events: 37 primaries were re-measured individually in QLP SAP (one more was excluded as partial), plus 7 in TESScut sector 66, which QLP does not cover. The 1.3 ppt secondary was measured at fixed shape in 39 windows: 37 are positive and 28 are above 2σ. - (2) Period: P_ecl = 3.9969190 ± 0.0000006 d, which is **−2.31σ** from Gaia's 3.999219 ± 0.000996 d. This is the weakest link. - (3) Pixels: the 2.85″ neighbour (G 16.75) cannot be excluded by arithmetic (it would need 73% primary and 4.1% secondary depths). The difference images reject it: the mean centroid over 8 sectors lies 0.51″ from the target and 3.2″ from the neighbour (≈ 16σ), and the PSF χ² prefers the target in every sector. Every other G < 18 neighbour is excluded by arithmetic. - (4) Independence: QLP SAP, my

TESScut aperture and ELEANOR-lite agree, at 21–24 ppt per sector. - (5) Geometry: the secondary sits at phase 0.4956 ± 0.0012, and Gaia predicts 0.534 ± 0.040 (A) or 0.466 ± 0.040 (B), both within 1σ. Propagated forward with Gaia's P, the primary's absolute phase does not decide between A and B (1.2σ vs 0.7σ). Propagated back with the eclipse P, A is preferred (0.01σ vs 2.2σ). - The screen was right (≈ 23 ppt, ≈ 1.5 ppt near 0.5, P 3.9969 d). The secondary is measurably *before* phase 0.5, at 0.4956. Its depth is 1.24–1.41 ppt. - Gaia shows RUWE 12.6 and ipd_frac_multi_peak = 39%: probably an unresolved sub-arcsecond companion, which TESS pixels cannot test.

1. Archives and web

- **Cones** (`s2_archives.out`, `s2b_archives.out`; the VizieR mirror was sanity-checked on Algol, which returned β Per):
- 195865529:
  - VSX (60″): only `Gaia DR3 2066587835904958592`, type YSO, 15.34–15.39 mag, at **35.76″**. That is the G 15.36 neighbour, not the target.
  - Not in Prša+2022, IJspeert+2021, Gaia DR3 VEB (I/358/veb), Gaia varisum (10″) or SB9.
  - **ASAS-SN II/366 at 0.16″: ASASSN-V J204254.34+431822.5, Type EA, Per 3.8310 d, Amp 0.12 mag, Prob 1.000.**
  - **ATLAS J/AJ/156/241 at 0.09″: class DBH (P(DBH) = 0.98), period 1.915571 d (= P/2).**
  - **So it is already a catalogued eclipsing binary**, though not in VSX.
- 468181994: no rows in VSX, J/ApJS/258/16, J/A+A/652/A120, I/358/veb, I/357/varisum, B/sb9, ATLAS or ASAS-SN II/366 (radii 10–60″, as listed in the scripts).
- The all-catalogue 5″ query returned only astrometric/photometric tables (I/...) for both stars.
- **SIMBAD (harvard mirror, 10″ cone):** both are type SB*, with 1 bibcode each: ATO J310.7264+43.3062 and UCAC4 023-009594. "TIC " is not a SIMBAD identifier for either star (`s2c_simbad_raw.out`).
- **ExoFOP:** tois = [], ctois = [] for both.
- **WebSearch** on all names: nothing mentions either star (`s15_websearch.out`).
- **Gaia DR3 extras** (`s10_gaia_extra.out`):
- 195865529: RUWE 0.94, ipd_frac_multi_peak 0, rv_amplitude_robust 147 km/s (2K1 = 132), Teff_gspphot 6608 K, FLAME R = 2.03 R⊙, M = 1.50 M⊙.
- 468181994: **RUWE 12.57, astrometric_excess_noise 1.53 mas (sig 2803), ipd_frac_multi_peak 39**, rv_amplitude_robust 52.9 km/s (2K1 = 44.7), Teff_gspphot 5775 K, logg 4.08, FLAME R = 2.91 R⊙ (FLAME uses the gspphot distance of 608 pc; the parallax distance is 1044 pc).
- The 2.85″ neighbour 6342991943947141504: G 16.75, BP−RP 1.51, parallax 1.717 ± 0.048 mas, RUWE 1.10.
- The NSS orbit rows match targets.csv. conf_spectro_period = 1.0 for both.

2. Independence, systematics, pixels

- **Pipelines available.** Neither star has TESS-SPOC FFI light curves or 2-min SPOC data (`s1_download_lc.out`).
- 195865529: QLP s15/41/55/56/75/76/82/83, plus TASOC for s15.

- 468181994: QLP s12/13/27/39/67/93/94, plus GSFC-ELEANOR-LITE for s12/13. TESScut covers s66, where QLP has nothing.
- My own photometry: an 11×11 TESScut cube per sector. Background = per-cadence median of the faintest 20% of pixels; aperture r = 1.5 px (6–8 px); QUALITY == 0 (`s8_pixels.py`).
- **Depth method.** One method for every pipeline (`s11_depths.out`): ephemeris and shape fixed, a quadratic baseline per event, and an exposure-supersampled trapezoid amplitude.

TIC 195865529 primary / secondary (ppt)	s15	s41	s55	s56	s75	s76	s82	s83
QLP SAP	182.0 / 49.2	186.2 / 48.2	182.5 / 46.7	182.2 / 46.9	184.3 / 49.2	183.8 / 48.3	182.4 / 48.4	181.9 / 47.6
TESScut ap., raw	141.2 / 38.6	134.7 / 36.0	143.1 / 37.4	127.4 / 33.5	140.9 / 38.1	143.0 / 38.8	129.6 / 34.3	143.5 / 38.0
TESScut ÷ target fraction	183.3 / 50.2	188.1 / 50.3	194.4 / 50.8	197.4 / 51.9	187.2 / 50.6	186.0 / 50.5	204.2 / 54.1	197.5 / 52.3
TASOC raw (AP_CONT 0.637)	71.3 / 16.9 → undiluted 196.2 / 46.5							

- Typical errors are 0.3–0.9 ppt.
- In `s12_dilution.out`, the target's flux fraction in my aperture is 0.635–0.770 from TIC Tmag and the fitted PSF width, and 0.585–0.803 from the fitted scene.
- Undiluted TESScut depths: mean primary 192.3 (sd 7.2), secondary 51.3 (sd 1.4), S/P 0.267. QLP: 183.2 (sd 1.5) / 48.1 (sd 0.9), S/P 0.262.
- So QLP SAP behaves as if it were deblended with the TIC contamination ratio (0.367). The TESScut and TASOC numbers agree with it to ≈ 5%, which is the accuracy of my Gaussian-PSF dilution model. Depth *ratios* agree to 2%.
- Note: the raw aperture depths of this star are not comparable across pipelines without that correction.
- **TIC 468181994 primary (ppt):**
- QLP SAP: 21.0, 22.3, 23.7, 23.2, —, 23.0, 23.3, 21.2.
- TESScut: 22.3, 21.8, 21.9, 22.3, 22.3 (s66), 22.4, 22.8, 22.0.
- ELEANOR-lite raw: 24.3 (s12), 22.7 (s13). ELEANOR-lite corr: 23.4, 22.4.
- These are sectors 12, 13, 27, 39, 66, 67, 93, 94. The target fraction in my aperture is 0.952–0.960.
- Secondary per sector: QLP 1.07–1.55, TESScut 0.96–2.28 (s67 ± 0.98), ELEANOR-lite 0.94–1.61, all with errors of 0.16–0.46.
- All pipelines agree within about 2σ.
- **Difference images** (`s8_pixels_*.out`, `s9_pixsummary.out`). Each is the median out-of-eclipse frame minus the median in-eclipse frame, primaries only. 50 primaries per star went into the stacks, covering all 8 sectors of each. A free-position Gaussian PSF was fitted, and the zero point is a scene model of the out-of-eclipse image (TIC stars with T < 17). Errors come from bootstrapping over events. Two single-event difference images per sector are also listed.
- **195865529:**
  - Per-sector offsets from the target: 1.2–4.3″. Single events: 1.0–4.3″. Sector mean (+0.20 ± 0.57, +1.37 ± 0.91)″.
  - The nearest G < 18 neighbour is 18.9″ away. The difference centroid lies 17–23″ from it (≥ 101σ in every sector) and 32–40″ from the 33–36″ group.
  - PSF hypothesis χ^2: the target is best in all 8 sectors. The next best, the 18.8″ star, is worse by $\Delta\chi^2 \geq 16{,}096$.

- Arithmetic, as the depth each star would need: the G 13.46 star at 36.0″ has a maximum flux fraction of 18.0%, so it would need 101.6% (primary) and 26.5% (secondary). It is excluded only marginally by arithmetic, but cleanly by pixels. Every other G < 18 neighbour within 90″ would need ≥ 393%.
- The TIC's T 18.3 star at 10.7″ (G 19.5) contributes about 0.1% and is irrelevant.

- **468181994:**
  - Per-sector offsets: 0.41–1.53″. Mean (+0.50 ± 0.20, −0.12 ± 0.19)″, error from the scatter between sectors.
  - The 2.85″ neighbour is at (−1.67, +2.22)″. Per sector, the difference centroid is 2.3–3.8″ from it (7–24σ); the mean is ≈ 3.2″, or ≈ 16σ.
  - PSF hypothesis $\Delta\chi^2$ (neighbour minus target): +119, +190, +184, +230, +58, +14, +44, +256, so the target is preferred in all 8 sectors.
  - Arithmetic: the neighbour's flux fraction is 3.0–3.1%, so it would need **73.3% / 4.1%**, which arithmetic cannot exclude. All others (44–86″) would need ≥ 293% for the primary.
  - Figures: `fig_468181994.png` and `fig_195865529.png` (out-of-eclipse image, difference image, raw fold). In both, the difference image peaks on the target pixel.

- **Common mode:** not done with other QLP stars. A dip that repeats strictly on one ephemeris across 8 sectors over 6 years, and whose per-pixel difference image follows the target PSF only, is not a sector-wide systematic.
- **Blind period search** on raw SAP (`s4_period_search.out`, BLS 0.5–20 d): the best peaks are 3.83107 d and 3.99694 d. The other peaks are aliases (P/2, 2P, 3P, 4P).

3. TIC 195865529 (P 3.831 d)

Raw QLP SAP, QUALITY==0. Each event is fitted with a trapezoid times a quadratic baseline, with red-noise β-scaled errors (`s6_events.out`). Summary per sector (`s17_tables.out`); every event is listed in `events_table_195865529.md`.

Event	Mid BTJD (range)	Depth (ppt, mean / sd)	T14 / T23 (h)	pipelines
P s15 ×4 (lost: 1721.87, 1733.36, 1737.19 flagged)	1714.199–1729.524	182.0 / 0.6	5.19 / 2.18	QLP, TESScut, TASOC
P s41 ×6 (lost 2442.10)	2422.946–2445.932	186.4 / 1.1	5.23 / 2.15	QLP, TESScut
P s55 ×6 (lost 2809.88)	2798.388–2821.374	182.5 / 0.9	5.23 / 2.16	QLP, TESScut
P s56 ×7 (lost 2825.21)	2829.037–2852.023	181.7 / 1.0	5.23 / 2.18	QLP, TESScut
P s75 ×6 (partial 3346.23)	3342.396–3365.382	184.1 / 2.4	5.20 / 2.16	QLP, TESScut
P s76 ×6 (partial 3388.37)	3369.213–3392.199	183.2 / 1.5	5.22 / 2.17	QLP, TESScut
P s82 ×6 (partial 3545.44)	3533.948–3556.934	182.4 / 0.7	5.24 / 2.15	QLP, TESScut
P s83 ×7	3560.766–3583.752	181.8 / 1.0	5.20 / 2.15	QLP, TESScut
S s15 ×4 (3 flagged)	1712.285–1731.440	48.9 / 1.2	5.34 / 2.53	QLP, TESScut, TASOC
S s41 ×6, s55 ×7, s56 ×6	2421.033–2850.108	48.0, 46.4, 46.8	5.23–5.29 / 2.44–2.53	QLP, TESScut

Event	Mid BTJD (range)	Depth (ppt, mean / sd)	T14 / T23 (h)	pipelines
S s75 ×8, s76 ×7, s82 ×6, s83 ×5	3340.481–3581.835	49.4, 48.0, 48.4, 47.3	5.22–5.29 / 2.42–2.53	QLP, TESScut

- **Ephemeris and O−C** (`s7_ephem.out`): 48 primaries give P = 3.8310456 ± 0.0000009 d and T0 = 3346.226971 ± 0.000140. Red. χ^2 is 9.43 (errors scaled) and the O−C rms is 1.31 min; sector means lie between −3.4 and +1.8 min.
- **Secondaries:** P = 3.8310460 ± 0.0000013 (P_sec − P_prim = +0.0000004 ± 0.0000015). Phase **0.49999 ± 0.00007**, so e cos ω = −0.00002 ± 0.00010. The durations are equal (T14 5.22 vs 5.25 h), so e sin ω ≈ 0 as well. The orbit is circular.
- **Comparison with Gaia:** P_ecl − P_Gaia = −0.000058 d = −0.15σ.
- **Odd/even primaries:**
- Raw: 182.72 ± 0.21 vs 183.04 ± 0.24 ppt. Sector-normalised: 0.9985 ± 0.0011 vs 1.0021 ± 0.0013 (−2.0σ, driven by the depth differences between sectors).
- Odd/even secondaries: 47.81 ± 0.19 vs 47.60 ± 0.18.
- A distinct 48 ppt secondary at phase 0.5 rules out 2P and P/2.

4. TIC 468181994 (P 3.997 d)

Same method (`s6_events.out`). Every primary is listed in `events_table_468181994.md`; the individual secondaries at fixed phase and shape are in `s7b_sec468.out`.

Event	Mid BTJD (range)	Depth (ppt, mean / sd)	T14 / T23 (h)	pipelines
P s12 ×4 (lost 1624.89 gap, 1628.89 and 1640.88 flagged)	1632.882–1648.870	21.1 / 0.5	3.73 / 2.40	QLP, TESScut, ELEANOR-lite
P s13 ×5 (1660.86, 1664.86 partial)	1656.865–1680.846	22.1 / 1.6	3.68 / 2.32	QLP, TESScut, ELEANOR-lite
P s27 ×5 (2048.56 gap, 2060.55 partial)	2036.572–2056.556	24.0 / 1.4	3.66 / 2.39	QLP, TESScut
P s39 ×6 (2368.32 flagged in QLP; clean in TESScut)	2364.320–2388.300	23.2 / 1.8	3.66 / 2.42	QLP, TESScut
P s66 ×7 (TESScut only)	3099.752–3123.734 (predicted)	21.5–23.2 each, 22.27 ± 0.22	shape fixed	TESScut
P s67 ×5 (3139.72 flagged, 3147.72 partial)	3127.731–3151.712	23.3 / 1.0	3.60 / 2.54	QLP, TESScut
P s93 ×6 (3843.18 flagged)	3831.188–3855.169	23.2 / 1.1	3.70 / 2.40	QLP, TESScut
P s94 ×6	3859.167–3879.151	20.8 / 1.0	3.73 / 2.43	QLP, TESScut
S, 39 windows s12–s94, phase fixed at 0.4956	1630.867–3877.136	weighted mean 1.24 ± 0.07 (χ^2r 2.54)	primary shape	QLP (TESScut and ELEANOR per sector, §2)

- **Ephemeris** (`s7_ephem.out`): 37 primaries give P = 3.9969190 ± 0.0000006 d and T0 = 2384.303949 ± 0.000146. Red. χ^2 0.75, O−C rms 1.07 min.
- **Comparison with Gaia:** P_ecl − P_Gaia = −0.002300 d = **−2.31σ**.
- **Odd/even:** raw 22.18 ± 0.16 vs 22.55 ± 0.17 ppt; sector-normalised −1.79σ.
- **Secondary:**
- Fold fit: phase 0.4953 ± 0.0006, depth 1.41 ± 0.09 ppt. With the shape free: 0.4952 ± 0.0005, 1.39 ± 0.09 ppt, T14 3.57 ± 0.21 h (`s7_ephem.out`).
- **Null boxes** at phases 0.25 and 0.75: −0.25 ± 0.09 and −0.01 ± 0.09 ppt. The formal errors therefore understate the systematics by up to ≈ 3×. Even so, the secondary stands at ≈ 5σ against the worst

null.

- It is present in both halves: s12–39 phase 0.4951, 1.27 ± 0.11 ppt; s67–94 phase 0.4956, 1.25 ± 0.11 ppt.
- Per-sector phases run from 0.4881 to 0.4970: mean 0.4941, standard error 0.0012 (`s13_geometry.out`).
- **The by-eye screen did not resolve** that the secondary is ≈ 25 min early, nor that it is ≈ 1.3 ppt rather than 1.5.

Geometry vs Gaia

(`s13_geometry.out`: 20,000 draws of P, Tp, e, ω; e truncated at 0.) - **195865529** (Gaia e = 0.074 ± 0.040, ω = 235 ± 33°, K1 = 66.18): - Primary minus conjunction A: −0.029 ± 0.161 phase with Gaia's P propagated forward over 771.6 cycles; −0.016 ± 0.141 with the eclipse P propagated back. - Primary minus B: +0.496 ± 0.149 and −0.490 ± 0.127, i.e. 3.3σ and 3.9σ. - **So the primary is conjunction A:** the hotter RV star is eclipsed by the companion. - Predicted secondary phase 0.479 ± 0.024, observed 0.49999 (+0.87σ). Gaia's e cos ω = −0.033 ± 0.040 vs 0.0000 ± 0.0001 observed. Gaia's e is a 1.8σ detection, and the eclipses show the orbit is circular. - **Physics:** - f(m) = 0.1141 M⊙, so m2 = 0.84–0.86 M⊙ for M1 = 1.43–1.50, and a = 13.5–13.7 R⊙. - The flat-bottom trapezoid solution (no limb darkening) gives R1 ≈ 1.72 R⊙, R2 ≈ 0.71 R⊙, k = 0.41, ρ1 ≈ 0.46 ρ⊙. That agrees with FLAME's 2.03 R⊙; TIC's 2.83 R⊙ looks too large. - The depth ratio S/P = 0.267 equals a Planck (800 nm) surface-brightness ratio for T2 ≈ 4500 K against 6600 K. A 0.85 M⊙ dwarf would be about 5000–5200 K (ratio 0.40–0.46). This crude check ignores limb darkening and the partial-eclipse geometry; it is roughly consistent, not exact. - The light-time shift of the secondary is 16–17 s. - **468181994** (Gaia e = 0.085 ± 0.084, ω = 20 ± 49°, K1 = 22.35): - Gaia's P error alone amounts to 0.497 d over the 498.9 cycles to TESS. - Primary minus A: forward −0.286 ± 0.233 (1.23σ), back −0.001 ± 0.192 (0.01σ). Primary minus B: forward +0.163 ± 0.240 (0.68σ), back +0.447 ± 0.202 (2.2σ). - **Undecided with Gaia's own P. With the eclipse period, A is preferred at about 2σ.** - Predicted secondary phase 0.534 ± 0.040 if the primary is A, 0.466 ± 0.040 if B; observed 0.4956 ± 0.0012 (−0.96σ or +0.74σ). - Observed e cos ω ≈ −0.0069 ± 0.0019. Gaia gives +0.053 ± 0.066. - The light-time (Rømer) delay is only 39–43 s (0.0001 phase), so the ≈ 25 min offset points to a small real eccentricity. Template or systematic error at the 1.3 ppt level cannot be excluded. - **Physics:** - f(m) = 0.0046 M⊙, so m2 = 0.21–0.24 M⊙ and a = 11.9–12.7 R⊙. - The trapezoid solution gives R1 ≈ 1.47 R⊙, R2 ≈ 0.22 R⊙, k = 0.152, b ≈ 0.50, ρ1 ≈ 0.54 ρ⊙. R2 suits a 0.2 M⊙ M dwarf. - S/P = 0.062 vs a Planck ratio of 0.077 for 3200 K against 5800 K, consistent with occultation of an M dwarf. - Tension: Gaia's photometric size for the primary (FLAME 2.91 R⊙; logg 4.08 implies ≈ 1.8 R⊙) is larger than the eclipse-implied 1.47 R⊙. Light from the suspected sub-arcsecond companion would inflate the photometric radius and dilute the depths.

Ephemerides and next eclipses

(`s14_ephem_next.out`.) BJD_TDB is converted to geocentric UTC with the barycentric light-time correction **removed** (astropy, builtin ephemeris); the uncorrected times differ by ≤ 2.5 min. - **TIC 195865529:** - P = 3.8310456 ± 0.0000009 d; T0 = BTJD 3346.22697 ± 0.00014 = BJD_TDB 2460346.22697. - Next primaries (geocentric UTC): **2026-11-01 15:04, 2026-11-05 11:00, 2026-11-09 06:57** (BTJD 4346.1299, 4349.9609, 4353.7920; ± 0.4 min formal). T14 ≈ 5.2 h, depth ≈ 18% (on QLP's scale). - Secondaries: 2026-11-03 13:02, 2026-11-07 08:59, 2026-11-11 04:56. - **TIC 468181994:** - P = 3.9969190 ± 0.0000006 d; T0 = BTJD 2384.30395 ± 0.00015 = BJD_TDB 2459384.30395. - Next

primaries (geocentric UTC): **2026-11-02 07:00, 2026-11-06 06:55, 2026-11-10 06:51** (BTJD 4346.7912, 4350.7881, 4354.7850; ± 0.5 min). T14 ≈ 3.7 h, depth ≈ 2.3%. - Secondaries (phase 0.4956, ± 7 min): 2026-11-04 06:32, 2026-11-08 06:28, 2026-11-12 06:24.

Caveats

- **468181994:**
- RUWE 12.6 and ipd_frac_multi_peak 39% point to a companion within about 1″. The 2.85″ neighbour is probably not the cause, since Gaia resolves it as a separate source; this is my inference, not a measurement. TESS pixels cannot separate a sub-arcsecond companion. The case for the eclipses being on the RV star is that the eclipse period equals the SB1 period to 0.06%.
- That period agreement is only at 2.3σ given Gaia's quoted error. With a 3σ threshold this is a pass; with a 2σ threshold it would make the star PROBABLE.
- Speckle or AO imaging and a few RVs at quadrature (K1 ≈ 22 km/s) would settle it.
- **195865529:** depth comparisons across pipelines depend on my Gaussian-PSF dilution model (≈ 5%). The 36″ G 13.46 star is excluded by the pixels, not by arithmetic.
- **Not done:** the common-mode test with other stars; any light-curve model with limb darkening (the physics above is trapezoid-level). No TESS-SPOC or 2-min data exist for either star.
- **The `s8` pixel analysis cannot be re-run** without downloading the cubes again (`s3_tesscut_dl.py`), because they were deleted per the brief. Its outputs are saved in `pix/`.
- **Inherited fits:** `s6` marked four primaries and one secondary "ok" although their covariance had failed (partial coverage). `s7` and `s17` exclude them from timing; they are labelled partial above.

Files

- **Scripts and logs:**
- `s1_download_lc`, `s1b_download_other`, `s2_archives`, `s2b_archives`, `s3_tesscut_dl`, `s4_period_search`, `s5_fold_scan`, `s6_events`, `s7_ephem`, `s7b_sec468`, `s8a_tic_neighbours`: from the first agent, checked and reused.
- `s8_pixels`, `s9_pixsummary`, `s10_gaia_extra`, `s11_depths`, `s12_dilution`, `s13_geometry`, `s14_ephem_next`, `s15_websearch` (log only), `s16_plots`, `s17_tables`: this run. Each `.py` has a matching `.out`; s8 has `s8_pixels_<tic>.out`.
- **Data products:** `lcio.py`, `events_qlp.csv`, `events_table_<tic>.md`, `ephem.json`, `s11_depths.json`, `fold_<tic>.txt`, `tic_neighbours_<tic>.csv`, `pix/` (aperture light curves, difference and scene images, per-sector JSON), `fig_<tic>.png`.

TIC 195865529: CONFIRMED; P_ecl=3.8310456±0.0000009 d; depth1=183 ppt (QLP, ≈deblended; 192±7 TESScut undiluted); depth2=48 ppt at phase 0.49999±0.00007; pixel: on target; pipelines: 3 (QLP, own TESScut, TASOC); Gaia geometry: pass (primary = conjunction A at 0.1–0.2σ, B excluded at 3.3–3.9σ; circular orbit) TIC 468181994: CONFIRMED (weaker: P_ecl vs Gaia at −2.3σ; suspected sub-arcsec companion); P_ecl=3.9969190±0.0000006 d; depth1=22.5 ppt (QLP) / 23.2 (TESScut); depth2=1.24–1.41 ppt at phase 0.4956±0.0012; pixel: on target (2.85″ G16.75 neighbour rejected at ≈16σ); pipelines: 3 (QLP, own TESScut, ELEANOR-lite); Gaia geometry: pass (secondary phase within 1σ; A vs B undecided with Gaia P, A preferred at ~2σ with eclipse P)

vet02 full report

Archived source: sanitized report text.

vet02: adversarial check of TIC 470235665 (UCAC4 859-013225) and TIC 379251709 (UCAC4 788-006141)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. A container restart killed the first agent part-way through. This run picked up its outputs (`s1`–`s8`: archives, downloads, raw-SAP event fits, ephemerides, the 379251709 secondary search, Gaia geometry), checked them, and added the missing parts: `s1c` (SIMBAD bibcodes, the 5″ all-catalogue list for 379251709 that `s1b` crashed on), `s9`/`s10` (TESScut pixels and own photometry), `s11` (independence), `s12` (physics, ephemerides), `s13` (secondary-phase drift), `s14` (fold plots) and `s15` (event tables). Scripts and outputs are in this folder. Light curves are in the scratchpad `data/vet02/<tic>/`. Each TESScut cube was deleted after use. Depths are in ppt and times in BTJD (BJD_TDB − 2457000). A = the companion passes in front of Gaia's RV star (true anomaly 90° − ω). B = the companion is behind it.

TIC 470235665: CONFIRMED. Every check passed: 1. 25 primaries and 25 secondaries are each re-measured from raw SAP in QLP and TESS-SPOC, sectors 18–85. 2. The eclipse period is 10.326183 ± 0.000002 d, +0.82σ from Gaia's 10.3239 ± 0.0028 d. 3. The difference images in s19 (camera 3) and s79 (camera 4) put the deficit on the target. Its centre is 0.29–0.74″ from the out-of-eclipse centroid. Every neighbour would need an intrinsic depth above 100% even if all its flux fell in the aperture. 4. QLP SAP, TESS-SPOC SAP and my own TESScut aperture all see every event they cover. SPOC and TESScut depths agree to about 1%; QLP is 4% deeper. No 2-min data exist. 5. The primary falls on Gaia conjunction A, −0.07 ± 0.11 d away; B is excluded at 19σ. The secondary phase (0.52275) is +0.7σ from Gaia's prediction. The B/A duration ratio gives e sin ω ≥ 0.17, against Gaia's 0.21 ± 0.05.

The screen got this one right (P 10.3262 d, secondary at phase 0.52). Its depths (128/84 ppt) match QLP. TESS-SPOC SAP and TESScut give 122/81 ppt.

TIC 379251709: CONFIRMED. Every check passed: 1. 18 primaries are re-measured individually. The two in s18 that TESS-SPOC flags are recovered in QLP and in my TESScut photometry. 2. The eclipse period is 12.323912 ± 0.000007 d, −0.53σ from Gaia's 12.3261 ± 0.0042 d. 3. The primary deficit is on the target in s18 (camera 2) and s52 (camera 4), 0.10″ and 1.86 ± 0.65″ from the out-of-eclipse centroid. All neighbours are excluded for the primary even in the worst case. 4. QLP, TESS-SPOC and TESScut depths agree (34–38 ppt). 5. The primary is at conjunction A (−0.17 ± 0.22 d); B is excluded at 28σ with Gaia's correlations, 4.2σ without. The secondary phase (0.4960 ± 0.0010) is +0.8σ from the 0.471 ± 0.031 that Gaia's e and ω predict.

Weak points: - The 2.5 ppt secondary is detected only in folds: QLP S/N 8.5, TESS-SPOC S/N 3.8. It is too shallow for a pixel test or for the neighbour arithmetic. - The short, V-shaped primary implies a primary radius of about 1.3 R⊙, against the TIC's 1.88 R⊙ (see Caveats).

The screen's 31 ppt primary depth is low by about 15%: the raw-SAP depth is 34–36 ppt (TESScut 36.5–38.1). Its period and secondary phase are right.

1. Archives and web

- **VizieR cones, 60″ (`s1_archives.out`):** VSX (B/vsx/vsx), Prša+2022 (J/ApJS/258/16), IJspeert+2021 (J/A+A/652/A120), Gaia DR3 EBs (I/358/veb), Gaia DR3 variability summary (I/357/varisum), Gaia vclassre, SB9 and ASAS-SN II/366 return **0 rows for both stars**. The same

query returns 1 row for Algol, so the cones work. Neither star is a catalogued eclipsing binary: none is listed in VSX, Gaia-EB, Prša or IJspeert within 60″.

- **All-catalogue 5″ list (`s1b`, `s1c`):** only astrometric and photometric I/ catalogues come back (47 and 51 tables). The mirror's list stops alphabetically inside I/, so this query is not a completeness check; the targeted cones above are.
- **SIMBAD (`s1_archives.out`, `s1c_archives.out`):**
- 470235665 = UCAC4 859-013225, type **SB***, 1 bibcode (Gaia DR3), RV −66.6 ± 9.2 km/s.
- 379251709 = UCAC4 788-006141, type **SB***, 1 bibcode, RV −34.0 ± 4.0 km/s.
- **ExoFOP:** 0 TOIs and 0 CTOIs for both.
- **Web search:** the 2MASS, UCAC4 and TIC names return nothing relevant for either star.
- **Gaia DR3 orbits (`gaia_nss_tap.csv`):**
- 470235665: SB1, P 10.32387 ± 0.00280 d, e 0.210 ± 0.051, ω 90.3 ± 13.7°, K1 47.5 ± 3.8 km/s, flags 8192.
- 379251709: SB1, P 12.32612 ± 0.00417 d, e 0.097 ± 0.049, ω 122.6 ± 26.8°, K1 23.0 ± 1.1 km/s.
- **TIC stellar parameters (`s1c_archives.out`):**
- 470235665: contamination ratio 0.059; Teff 5353 K, R 2.30 R⊙, log g 3.68.
- 379251709: contamination ratio 0.041; Teff 5279 K, R 1.88 R⊙, log g 3.85.

2. Independence, systematics, pixels

- **Pipelines (`s11_independence.out`, raw SAP, QUALITY==0, trapezoid on a quadratic baseline):**
- 470235665 primary: QLP 127.4 ppt (sd 2.6, N = 23); TESS-SPOC 122.4 (sd 2.0, N = 15); TESScut r ≤ 1.5 px aperture 122.1 ± 0.3 (s19) and 123.3 ± 1.0 (s79).
- 470235665 secondary: QLP 84.0 (sd 1.6); TESS-SPOC 80.9 (sd 1.5); TESScut 80.8 ± 0.3 and 80.9 ± 0.7.
- QLP/SPOC depth ratio over the shared sectors: 1.040 ± 0.007 (primary) and 1.038 ± 0.006 (secondary). This is an aperture/background systematic, much larger than the 0.5 ppt formal errors. TESS-SPOC and TESScut agree within 1%.
- 379251709 primary: QLP 34.2 (sd 3.1, N = 17, including the noisy s24/s25); TESS-SPOC 36.3 (sd 1.4, N = 4); TESScut 38.1 ± 0.8 (s18) and 36.5 ± 0.1 (s52).
- T14 agrees between pipelines: 6.44 h (QLP) vs 6.45 h (SPOC) for the 470235665 primary, 9.02 vs 9.01 h for its secondary, and 4.05 vs 4.22 h for the 379251709 primary.
- Neither star has 2-min SPOC data.
- **Common mode (`s10_offtarget.out`):** this uses off-target pixels inside each cutout instead of other QLP stars.
- Summed over pixels more than 3 px from the target, the stacked primary deficit is −1.1 to +3.2 ppt, against 120 or 33–38 ppt on target.
- The brightest off-target pixels are stars beyond 90″ at the cutout edges, some brighter than the targets. They show −4.2 to +2.2 ppt.
- The eclipses also repeat on a strict period over 6 years, in three cameras and two cadence regimes. A sector-wide systematic cannot do that.
- **Census of non-eclipse dips (`s4_census.out`):**
- Apart from the eclipses, the 4–9 ppt features in raw SAP sit within about 0.3 d of gaps.
- TESS-SPOC s18 starts with a ramp (27–54 ppt, 1790.7–1791.5) **in both stars**, so it is a systematic.

- 379251709 QLP s24 is dominated by scattered light (σ30 = 8.1 ppt). Its events are kept but carry large errors.
- **Difference images (`s9_tesscut_<tic>_s<sec>.txt`; median in-eclipse minus median local out-of-eclipse, per event, averaged; Gaussian-fit centres; 21″/px):**

Star / sector (camera)	kind	N ev	deficit – OOT centroid	deficit – target WCS	frac. depth on target pixel
470235665 s19 (cam 3)	P / S	3 / 2	0.29″ (±0.37) / 0.28″ (±0.14)	0.96″ / 0.96″	122.9 / 76.6 ppt
470235665 s79 (cam 4)	P / S	3 / 3	0.33″ (±0.22) / 0.74″ (±0.26)	1.43″ / 1.58″	120.8 / 77.7 ppt
379251709 s18 (cam 2)	P	2	1.86″ (±0.65)	1.30″	33.2 ppt
379251709 s52 (cam 4)	P	2	0.10″ (±0.46)	1.32″	31.3 ppt

The 379251709 secondary difference images (s18: 2 events; s52: 1 event at the sector start) are noise. No pixel location is claimed for the secondary. - **Dilution arithmetic** (aperture r ≤ 1.5 px; fluxes from Gaia G). The "worst case" puts 100% of the neighbour's flux in the aperture. - **470235665:** the nearest neighbour is G 16.39 at 5.5″, on the target pixel. It would need an intrinsic depth of 320–356% for the primary and 210–234% for the secondary (PRF and worst-case values, both sectors). G 16.19 at 57″ needs at least 269% / 176%; G 15.51 at 69″ needs at least 149% / 98%. That last 98% (s79 secondary) is the only worst case below 100%. That star lies 70″ from the deficit centre, the fractional depth at its pixel is 2.8–8.6 ppt, and it is excluded for the primary. All neighbours are **excluded**. - **379251709 primary:** the brightest neighbour is G 16.57 at 67″ and needs at least 109–128% in the worst case. The nearest is G 17.33 at 30″ and needs at least 216–253%. All neighbours are **excluded**. - **379251709 secondary:** with a 2.5 ppt depth, any neighbour with a worst-case share above about 2.5% could produce it arithmetically. This is not excluded by pixels. It rests on timing: the dip sits exactly at the occultation phase the primary's e cos ω demands. - ZTF was not needed: for the primaries, no neighbour is arithmetically viable.

3. TIC 470235665 (P 10.3262 d)

Raw SAP, QUALITY==0, supersampled trapezoid on a local quadratic baseline, errors max(covariance, prayer-bead) (`events_470235665.csv`, `s5_events.py`). "TESScut" is my own aperture photometry (`s9`) with the shape fixed.

Event	Sector	Mid BTJD	Depth QLP / SPOC / TESScut (ppt)	T14 / T23 (h)	pipelines
P E0	18	1795.7429 ± 0.0003	131.2 / 124.2 / –	6.44 / 2.51	QLP+SPOC
P E1	18	1806.0694 ± 0.0003	128.0 / 121.8 / –	6.45 / 2.60	QLP+SPOC
P E2	19	1816.3945 (TESScut)	– / – / 122.5	(fixed)	TESScut (QLP lost, SPOC partial)
P E3	19	1826.7204 ± 0.0003	130.4 / 124.3 / 122.5	6.45 / 2.44	QLP+SPOC+TESScut
P E4	19	1837.0465 ± 0.0003	130.1 / 123.2 / 121.5	6.45 / 2.46	QLP+SPOC+TESScut
P E16	24	1960.9600 ± 0.0004	123.9 / 123.0 / –	6.42 / 2.46	QLP+SPOC
P E17	24	1971.2876 ± 0.0004	126.1 / 125.2 / –	6.38 / 2.43	QLP+SPOC
P E18	24	1981.6139 ± 0.0003	124.7 / 123.7 / –	6.46 / 2.40	QLP+SPOC
P E19	25	1991.9398 ± 0.0004	– / 121.7 / –	6.48 / 2.49	SPOC (QLP partial)
P E20	25	2002.2652 ± 0.0004	– / 125.9 / –	6.49 / 2.33	SPOC (QLP partial)
P E21	26	2012.5925 ± 0.0004	126.5 / 120.5 / –	6.47 / 2.44	QLP+SPOC
P E23	26	2033.2442 ± 0.0003	126.5 / 120.0 / –	6.40 / 2.48	QLP+SPOC
P E90	52	2725.0982 ± 0.0003	128.1 / 122.5 / –	6.44 / 2.40	QLP+SPOC
P E91	52	2735.4253 ± 0.0003	122.1 / 119.4 / –	6.48 / 2.52	QLP+SPOC
P E92	53	2745.7508 ± 0.0003	124.8 / 120.3 / –	6.49 / 2.38	QLP+SPOC
P E94	53	2766.4038 ± 0.0002	125.6 / 120.6 / –	6.42 / 2.52	QLP+SPOC
P E106	58	2890.3180 ± 0.0003	130.6 / – / –	6.43 / 2.48	QLP
P E107	58	2900.6438 ± 0.0003	131.6 / – / –	6.46 / 2.43	QLP

Event	Sector	Mid BTJD	Depth QLP / SPOC / TESScut (ppt)	T14 / T23 (h)	pipelines
P E145	73	3293.0389 ± 0.0003	129.6 / – / –	6.44 / 2.43	QLP
P E146	73	3303.3666 ± 0.0004	126.2 / – / –	6.44 / 2.42	QLP
P E160	78	3447.9313 ± 0.0003	124.3 / – / –	6.43 / 2.43	QLP
P E161	79	3458.2570 ± 0.0004	126.7 / – / 121.5	6.46 / 2.44	QLP+TESScut
P E162	79	3468.5833 ± 0.0003	129.9 / – / 124.8	6.44 / 2.40	QLP+TESScut
P E163	79	3478.9099 ± 0.0003	128.3 / – / 123.7	6.43 / 2.51	QLP+TESScut
P E176	85	3613.1506 ± 0.0003	129.7 / – / –	6.48 / 2.43	QLP
P E178	85	3633.8032 ± 0.0003	125.7 / – / –	6.45 / 2.50	QLP
S E1	18	1811.4664 ± 0.0005	85.7 / 81.7 / –	8.98 / 2.79	QLP+SPOC
S E2	19	1821.7930 ± 0.0006	84.4 / 81.7 / 80.5	9.06 / 2.77	QLP+SPOC+TESScut
S E3	19	1832.1197 ± 0.0005	86.7 / 82.3 / 81.0	8.95 / 2.79	QLP+SPOC+TESScut
S E16	24	1966.3596 ± 0.0008	82.5 / 81.8 / –	9.09 / 2.78	QLP+SPOC
S E17	24	1976.6851 ± 0.0005	83.2 / 82.6 / –	8.97 / 2.87	QLP+SPOC
S E18	25	1987.0117 ± 0.0006	86.9 / 82.8 / –	8.98 / 2.91	QLP+SPOC
S E19	25	1997.3387 ± 0.0008	85.7 / 81.8 / –	9.03 / 2.64	QLP+SPOC
S E20	25	2007.6649 ± 0.0006	– / 82.3 / –	8.97 / 2.69	SPOC (QLP partial)
S E21	26	2017.9920 ± 0.0006	82.4 / 78.8 / –	8.96 / 2.88	QLP+SPOC
S E22	26	2028.3179 ± 0.0005	82.8 / 79.1 / –	9.01 / 2.83	QLP+SPOC
S E89	52	2720.1718 ± 0.0007	– / 79.8 / –	9.02 / 2.78	SPOC (QLP lost)
S E91	52	2740.8227 ± 0.0005	83.7 / 80.3 / –	9.06 / 2.82	QLP+SPOC
S E92	53	2751.1493 ± 0.0009	81.6 / 79.2 / –	9.13 / 2.68	QLP+SPOC
S E93	53	2761.4760 ± 0.0006	81.1 / 79.1 / –	8.99 / 2.76	QLP+SPOC
S E105	58	2885.3895 ± 0.0005	84.5 / – / –	9.04 / 2.72	QLP
S E106	58	2895.7163 ± 0.0005	86.8 / – / –	9.16 / 2.83	QLP
S E107	58	2906.0419 ± 0.0005	84.9 / – / –	9.06 / 2.77	QLP
S E145	73	3298.4358 ± 0.0005	84.0 / – / –	8.97 / 2.96	QLP
S E146	73	3308.7628 ± 0.0005	83.7 / – / –	8.94 / 2.79	QLP
S E159	78	3443.0031 ± 0.0006	82.0 / – / –	9.01 / 2.88	QLP
S E160	79	3453.3286 ± 0.0005	82.6 / – / 79.7	8.95 / 2.84	QLP+TESScut
S E161	79	3463.6543 ± 0.0006	83.5 / – / 80.7	9.09 / 2.82	QLP+TESScut
S E162	79	3473.9798 ± 0.0006	84.7 / – / 82.3	9.02 / 2.77	QLP+TESScut
S E176	85	3618.5472 ± 0.0005	84.5 / – / –	8.94 / 2.88	QLP
S E177	85	3628.8739 ± 0.0008	84.6 / – / –	9.01 / 2.72	QLP

Not measured (gap / flagged / too little coverage): pri E22 (s26, pred 2022.918): QLP lost, SPOC lost; pri E93 (s53, pred 2756.079): QLP lost, SPOC lost; pri E159 (s78, pred 3437.608): QLP partial; pri E177 (s85, pred 3623.479): QLP lost; sec E-1 (s18, pred 1790.822): SPOC partial; sec E0 (s18, pred 1801.148): QLP lost, SPOC lost; sec E15 (s24, pred 1956.041): QLP partial, SPOC partial; sec E90 (s52, pred 2730.506): QLP lost, SPOC lost; sec E144 (s73, pred 3288.121): QLP partial

Period (`s6_period.out`): - Primaries, one value per epoch: N = 25, **P = 10.326183 ± 0.000002 d**, T0 = 1795.74233 ± 0.00018, O–C rms 0.9 min (reduced χ^2 3.5; errors scaled). - Fitting QLP alone and TESS-SPOC alone gives the same P (10.326182). - Secondaries: N = 25, P = 10.326173 ± 0.000002 d. - Joint fit: P = 10.326180 ± 0.000001 d, secondary phase **0.52275 ± 0.00002**. - Gaia P differs by **+0.82σ**.

Odd/even: primary QLP 127.66 ± 0.75 vs 128.24 ± 0.85 ppt (−0.5σ); primary SPOC +1.0σ; secondary +0.02σ / +0.28σ. The two eclipse kinds differ in depth (127 vs 84 ppt) and duration (6.44 vs 9.02 h), so the true period is not twice this one, and the odd/even test rules out a stacked P2.

Shape (`s14_fold_470235665.png`): both eclipses have rounded bottoms. The trapezoid "T23" (2.45 / 2.81 h) is not a resolved flat bottom, so totality is not established.

Secondary-phase drift (`s13_secphase_470.out`): P_sec − P_pri = (−10.0 ± 2.8) × 10^{-6} d (−3.5σ), which would be −2.6 min over 178 cycles. In QLP the secondary phase goes 0.52284 → 0.52277 → 0.52269 across 2019–20, 2022 and 2022–24. TESS-SPOC shows no change between 2019–20 and 2022 (0.52281 vs 0.52280). This hints at apsidal motion or a cadence effect. It is not claimed.

4. TIC 379251709 (P 12.3239 d)

Event	Sector	Mid BTJD	Depth QLP / SPOC / TESScut (ppt)	T14 / T23 (h)	pipelines
P E-13	18	1801.0439 ± 0.0014	35.1 / – / 39.0	4.24 / 1.16	QLP+TESScut (SPOC lost)
P E-12	18	1813.3689 ± 0.0011	35.2 / – / 37.3	4.13 / 0.89	QLP+TESScut (SPOC lost)
P E0	24	1961.2533 ± 0.0009	– / 36.7 / –	4.20 / 1.04	SPOC (QLP noisy, e_tc>0.005 d)
P E1	24	1973.5750 ± 0.0010	35.8 / 35.2 / –	4.24 / 1.06	QLP+SPOC
P E2	25	1985.9080 ± 0.0028	29.7 / – / –	4.31 / 1.53	QLP
P E3	25	1998.2305 ± 0.0043	27.7 / – / –	2.91 / 0.58	QLP (SPOC lost)
P E62	52	2725.3357 ± 0.0008	33.2 / 35.2 / 36.6	4.17 / 1.00	QLP+SPOC+TESScut
P E63	52	2737.6604 ± 0.0008	36.8 / 38.2 / 36.4	4.27 / 0.67	QLP+SPOC+TESScut
P E75	58	2885.5465 ± 0.0008	34.5 / – / –	4.04 / 1.03	QLP
P E76	58	2897.8705 ± 0.0009	34.7 / – / –	4.10 / 0.87	QLP
P E78	59	2922.5198 ± 0.0009	33.7 / – / –	4.13 / 1.07	QLP
P E79	59	2934.8432 ± 0.0009	34.2 / – / –	4.13 / 0.85	QLP
P E120	78	3440.1238 ± 0.0009	36.1 / – / –	4.09 / 0.99	QLP
P E122	79	3464.7714 ± 0.0009	39.9 / – / –	4.28 / 0.32	QLP
P E123	79	3477.0951 ± 0.0013	34.1 / – / –	4.20 / 1.07	QLP
P E134	85	3612.6582 ± 0.0010	38.4 / – / –	4.22 / 0.86	QLP
P E136	86	3637.3031 ± 0.0011	32.1 / – / –	3.89 / 1.15	QLP
P E138	86	3661.9554 ± 0.0010	30.2 / – / –	3.82 / 1.28	QLP

Not measured (gap / flagged / too little coverage): pri E77 (s58, pred 2910.177): QLP lost; pri E121 (s79, pred 3452.429): QLP lost; pri E135 (s85, pred 3624.963): QLP lost; pri E137 (s86, pred 3649.611): QLP lost

Period (`s6_period.out`): - One value per epoch: N = 18, **P = 12.323912 ± 0.000007 d**, T0 = 1961.25375 ± 0.00060, O−C rms 3.1 min. The largest residuals are +9.2 and +7.2 min, in the noisy QLP s25. - QLP alone: 12.323905 ± 0.000009 d. TESS-SPOC alone (N = 4): 12.323931 ± 0.000018 d. - Gaia P differs by **−0.53σ**.

Odd/even: QLP 33.87 ± 1.04 vs 34.73 ± 1.27 ppt (−0.5σ); SPOC −0.55σ.

Shape: V-shaped, T14 4.1–4.2 h, trapezoid T23 about 1.0 h.

Secondary (`s7_secondary_379.out`; primaries masked, 1.5-d running median, box scan over phase): - QLP, excluding s24/s25: the strongest peak in the scan is at phase 0.4955. A trapezoid fit gives **phase 0.49597 ± 0.00035, depth 2.47 ± 0.29 ppt (S/N 8.5), T14 4.50 ± 0.43 h**. - TESS-SPOC: 0.49741 ± 0.00040, 2.59 ± 0.68 ppt (S/N 3.8), 2.6σ apart in phase from QLP. The SPOC scan's top peak (phase 0.2235) is the s18 start-of-sector ramp (`s7b_otherpeaks_379.out`). - Individual events (excluding the noisy QLP s24): 1.1–4.1 ppt at S/N 1.6–7.3. Only s59 E78 (4.12 ± 0.56 ppt) is individually significant.

Geometry vs Gaia

`s8_geometry.out`: Monte Carlo with 20000 draws using Gaia's corr_vec. Parameter order is assumed to be P, Vcm, K1, e, ω, Tp. The column-major upper triangle gives a positive-definite matrix with ω–Tp = 0.98–0.99; the row-major one does not. Uncorrelated draws are shown in brackets. `s12_physics_ephem.out` repeats the calculation with nominal values as an independent check.

- **470235665:**
- Taking the TESS ephemeris back to the Gaia epoch, the primary is −0.068 ± 0.108 d from **conjunction A** [−0.080 ± 0.427] and +5.09 d from B (+19σ [+7.5σ]). Taking Gaia's ephemeris forward to TESS gives +0.245 ± 0.405 d from A. The nominal check in `s12` gives −0.079 d.
- Because ω ≈ 90°, conjunction A falls at periastron. The primary is the companion transiting the RV star.
- Gaia predicts the secondary at phase 0.4994 ± 0.0329; measured 0.52275, **+0.71σ**. TESS-only e cos ω ≈ +0.036 vs Gaia −0.001 ± 0.051.
- T14(B)/T14(A) = 1.397 gives e sin ω ≥ +0.166, against Gaia +0.205 ± 0.050. The eclipse at periastron (A) is the shorter one, as the orbit requires.
- **379251709:**
- Backward: the primary is −0.170 ± 0.224 d from **conjunction A** [−0.189 ± 1.218] and −5.96 d from B (−28σ [−4.2σ]). Forward: −0.473 ± 0.418 d from A.
- Gaia predicts the secondary at 0.4711 ± 0.0314; measured 0.4960 ± 0.0010, **+0.79σ**. TESS e cos ω ≈ −0.006 vs Gaia −0.045 ± 0.049.
- Duration ratio 1.066 gives e sin ω ≥ 0.03 vs Gaia +0.068 ± 0.041. This test is weak.
- **Physics (`s12_physics_ephem.out`; rough two-sphere model without limb darkening):**
- **470235665:**
  - f(m) = 0.107 M⊙, so m2 = 0.64–0.77 M⊙ for M1 = 0.93–1.3 M⊙ and a = 23.2–25.4 R⊙.
  - The depths imply k ≈ 0.37 and a companion light share ≥ 8%. The two T14 values with Gaia's e and ω give R1/a = 0.079 and b_A ≈ 0.45, so R1 = 1.8–2.0 R⊙ (TIC 2.30), R2 = 0.67–0.73 R⊙ and i ≈ 87°.
  - J ≈ 0.66, so T2 ≈ 4800 K.
  - A subgiant with a K-dwarf companion: radius, temperature and mass agree with the mass function to about 15–20%. b_B ≈ 0.70 > 1 − k says the B eclipse is partial, which fits the rounded secondary.
- **379251709:**
  - f(m) = 0.0153 M⊙, so m2 = 0.28–0.35 M⊙ (an M dwarf, R2 ≈ 0.3–0.4 R⊙).
  - Read as a non-grazing transit (T23/T14 = 0.22), the shape needs b ≈ 0.77 and **R1 ≈ 1.2–1.5 R⊙, R2 ≈ 0.25–0.30 R⊙**, which is consistent with m2. With the TIC's R1 = 1.88 R⊙ the 4.2 h eclipse would have to be grazing (b ≈ 0.99), and that is hard to square with a 36 ppt depth.
  - A 3200–3500 K companion would give a full-occultation depth of 4–8 ppt; the measured secondary is about 2.5 ppt. Consistent.

Ephemerides and next eclipses

From `s12_physics_ephem.out`. The barycentric light-time correction (+2.8 to +5.2 min) was removed to give UTC at the geocentre. Errors are formal, from T0 and P. - **TIC 470235665 primary:** T0 = BTJD 1795.74233 ± 0.00018 (BJD_TDB 2458795.74233), **P = 10.326183 ± 0.000002 d**, T14 6.45 h. - Next primaries: BTJD 4346.3095 = **2026-11-01 19:21 UTC**; 4356.6357 = **2026-11-12 03:11**; 4366.9619 =

2026-11-22 11:00 (±0.8 min). - Secondary: T0 = 1801.14120 ± 0.00023, P = 10.326173 ± 0.000002 d, T14 9.0 h. Next: 2026-11-07 04:52, 2026-11-17 12:42, 2026-11-27 20:31 UTC. - **TIC 379251709 primary:** T0 = BTJD 1961.25375 ± 0.00060 (BJD_TDB 2458961.25375), **P = 12.323912 ± 0.000007 d**, T14 4.2 h; secondary at phase 0.4960 ± 0.0010. - Next primaries: BTJD 4352.0927 = **2026-11-07 14:07 UTC**; 4364.4166 = **2026-11-19 21:53**; 4376.7405 = **2026-12-02 05:40** (±2.1 min).

Caveats

- **SAP depths depend on the pipeline at the 4% level** (QLP deeper) and none are corrected for crowding (TIC contamination 4–6%). "Agreement" here means within about 5%, not within the formal errors.
- **The 379251709 secondary is a fold-level detection** (QLP S/N 8.5, SPOC S/N 3.8, phases 2.6σ apart). It has not been located in pixels or excluded from neighbours arithmetically. Its exact match to the e cos ω phase is what ties it to the target.
- **379251709 radius tension:** the eclipse shape prefers R1 ≈ 1.3 R⊙, about 30% below the TIC's 1.88 R⊙, unless the eclipse is grazing. This is not a failed check, but the TIC Teff/radius deserves a look. A spectrum, or a proper fit with limb darkening on the 200-s data, would settle it.
- **470235665:** the secondary-phase drift (−2.6 min over 5 years, 3.5σ, seen in QLP only) is unconfirmed. The rough physical model is crude and does not decide whether either eclipse is total.
- **Gaia geometry:** the forward predictions (±0.4 d) are limited by Gaia's P error. The backward test (±0.1–0.2 d) assumes the corr_vec ordering stated above; without correlations the errors grow to ±0.4–1.2 d and B is still rejected at 4.2σ and 7.5σ.
- **TESScut times** are not barycentred to the target, which puts them up to about ±3 min off (O−C −3.1 to +3.0 min). They are not used in any ephemeris.
- **Not done:** the QLP-neighbour common-mode test; TESScut-only sectors 60 and 86 of 470235665, which no pipeline covers; ZTF.

Files

- Scripts and outputs are in this folder. **From the first agent** (checked here, not re-run):
- `s1_archives.py/.out`, `s1b_archives.py/.out` (crashed for 379251709)
- `s2_download.py/.out`, `s3_load.py/.out` (per-star `lc_sap_q0.npz` in scratch)
- `s4_census.py/.out`, `s5_events.py` → `events_<tic>.csv`
- `s6_period.py/.out`
- `s7_secondary_379.py/.out/.txt`, `s7b_otherpeaks_379.py/.out`
- `s8a_gaia_tap.py/.out` → `gaia_nss_tap.csv`, `s8_geometry.py/.out/.txt`
- **Added in this run:**
- `s1c_archives.py/.out`
- `s9_tesscut.py` → `s9_tesscut_<tic>_s<sec>.txt/.npz/.png` (470235665 s19, s79; 379251709 s18, s52)
- `s10_offtarget.py/.out`
- `s11_independence.py/.out`
- `s12_physics_ephem.py/.out`
- `s13_secphase_470.py/.out`
- `s14_foldplot.py` → `s14_fold_<tic>.png`
- `s15_event_table.py/.out`

- **Data:** `/tmp/claude-0/-home-user-eclipses-paper-review/55c92ff4-798a-544f-be03-9888c6804905/scratchpad/data/vet02/<tic>/` (QLP and TESS-SPOC FITS; TESScut cubes deleted).

Summary - TIC 470235665: CONFIRMED; P_ecl=10.326183 d; depth1=122–127 ppt; depth2=81–84 ppt at phase 0.52275; pixel: on target; pipelines: 3; Gaia geometry: pass - TIC 379251709: CONFIRMED; P_ecl=12.323912 d; depth1=34–38 ppt; depth2=2.5 ppt at phase 0.4960; pixel: on target (primary; secondary untested); pipelines: 3; Gaia geometry: pass

vet03 full report

Archived source: sanitized report text.

vet03: adversarial check of TIC 70665829 (UCAC4 537-050325) and TIC 331293467 (UCAC4 150-142995)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This run resumed from a predecessor that was killed by a container restart. The predecessor wrote `s1–s7` and their outputs. Its pixel run for 331293467 stopped part-way through s100 (`s7_pixels_331293467.out`). I checked those outputs, re-ran the pixel analysis for 331293467 in all four sectors (`s7b`), and wrote `s9–s14` plus `s2c`, `s10a`, `s12b`. All scripts, `.out`/`.json` outputs and the `pix_*.png` plots are in this folder. Downloads are under `/tmp/claude-0/-home-user-eclipses-paper-review/55c92ff4-798a-544f-be03-9888c6804905/scratchpad/data/vet03/<tic>/`; the TESScut cubes have been deleted. Notation: **A** = the companion passes in front of Gaia's RV star (true anomaly ν = 90° − ω). **B** = the RV star passes in front of the companion. Depths are in ppt of the raw (undetrended) flux unless marked "undiluted".

TIC 70665829: CONFIRMED. A K dwarf with an early-M companion, P = 3.4657654 d. - **(1) Events, pass:** 24 primaries and 28 secondaries (SPOC), and 22 and 28 (QLP), measured one by one from SAP QUALITY==0 flux. - **(2) Period, pass:** the eclipse period matches Gaia's at −0.43σ. - **(3) Pixels, pass:** the difference images in s46, s48 and s72 (three cameras) sit 0.5–1.2″ from the target. The only neighbour (G 14.44 at 64″) would need an intrinsic depth above 1500%. - **(4) Independence, pass:** QLP, TESS-SPOC, my TESScut aperture photometry and ZTF g/r all see both eclipses. Absolute depths differ by up to ~10% between apertures. The secondary/primary depth ratio is 0.216–0.233 in every pipeline, so the differences are dilution or background, not the signal. - **(5) Geometry, pass:** the secondary falls at phase 0.46048, which is e cos ω = −0.0621. Gaia predicts 0.4547 +0.024/−0.024 if the deep eclipse is A (+0.24σ), and needs 3.6–5.6σ if it is B. The deep eclipse coincides with conjunction A (+0.4 ± 0.5 h when Gaia's orbit is conditioned on the TESS period). So the deep event is Gaia's RV star (the K dwarf) being eclipsed. ZTF colours agree: the deep eclipse is deeper in g, the shallow one deeper in r. - **Screen:** correct on all points (deep ~300–330 ppt at phase 0, ~65–72 ppt secondary at 0.4605, flat baseline, P 3.4658 d).

TIC 331293467: CONFIRMED. An A/F star with a K-dwarf companion, P = 5.2568199 d. The light is heavily blended, so the undiluted depths are uncertain. - **(1) Events, pass:** 21 primaries and 24 secondaries measured individually from QLP SAP QUALITY==0. Every secondary is above 3σ, and null phases 0.25/0.75 give −0.11 ± 0.16 and −0.09 ± 0.15 ppt. - **(2) Period, pass:** +0.51σ from Gaia. - **(3) Pixels and dilution, pass:** the diff-image source sits 0.3–3.6″ from the target in s38, s99, s100 and s101 (cameras 2, 4, 3, 2). - The G 17.07 star at 8″ needs 305–458%, the G 16.07 at 24″ needs 143–287%, and the G 16.14 at 14″ needs 91–115% (over 100% in s99). All are arithmetically excluded or effectively

so. - The only neighbour that passes the arithmetic, G 13.78 at 30.5″ (needs 25–84%), is excluded by position. The fit is 10–19× worse at its position, and the measured offset points 0.00–0.11 of the way toward it. - ZTF has no data at Dec −60. No higher-resolution light curve was available. - **(4) Independence, pass in dilution-free form:** no TESS-SPOC or SPOC light curve exists. My TESScut 3×3 photometry sees every eclipse with the same T14 (4.05–4.08 h against QLP 4.07 h) and a secondary/primary ratio of 0.168–0.202 against QLP 0.160–0.176. Raw depths differ by aperture (QLP 61–73, TESScut 47–56 ppt). The undiluted primary depth is model-dependent: 80–89 ppt from PSF amplitudes, 80–116 ppt from aperture shares. - **(5) Geometry, pass:** the secondary is at phase 0.50013 ± 0.00023 (Gaia predicts 0.5015, so this cannot tell A from B). The deep eclipse sits on Gaia's conjunction A: +0.1 ± 1.0 h when conditioned on the TESS period, and +0.49σ on Gaia alone with correlations. B is excluded at 6.7σ (Gaia alone) or ≫10σ (conditioned). This test needs Gaia's ω–T_p correlation (0.9997); with independent errors it does not decide. Landing on the target's own spectroscopic conjunction is further evidence that the signal is on the target. - **Screen:** the ~66 ppt and ~11 ppt figures are right for QLP SAP. The undiluted target depths are about 8–11% and 1.3–2%.

1. Archives and web

- **Catalogue cones (`s2_archives.py`, 60″, VizieR mirror):**
- 70665829: nothing in VSX, Prsa+2022, IJspeert+2021, Gaia DR3 VEB, Gaia varisum or SB9.
- 331293467: the same, except one VSX entry at 33.7″: Gaia DR3 5866907631080924672, type L, G 18.4–18.63. It is not our star and is fainter than the 90″ neighbour list (G < 18).
- Neither star is a catalogued eclipsing binary within 60″ in these catalogues.
- **Sanity checks:** the VSX cone was checked on Algol (1 row at 0.11″, `s2_archives.out`). `s2c` showed the mirror serves Prsa (nearest EB 1621″ from 331293467) and IJspeert (nearest 2860″ from 70665829). The mirror then began timing out, so Gaia-VEB, varisum and SB9 were not sanity-checked (`s2c_catalogue_sanity.out`). The all-catalogue 5″ query (`s2b`) returned only I/ catalogues, even for Algol, so it is truncated and was not relied on.
- **SIMBAD (10″):**
- 70665829 = UCAC4 537-050325, type SB*, 3 bibcodes (2023A&A...674A..34G, 2022AJ....163..152S, 2022ApJS..259...35A).
- 331293467 = UCAC4 150-142995, type SB*, 1 bibcode (2023A&A...674A..34G).
- **ExoFOP:** no TOIs or CTOIs for either star (`s2b_allcat.out`).
- **Gaia NSS rows (`s10a_gaia_nss.csv`):** both are SB1 with flags 0. corr(ω, T_p) is 0.9984 for 70665829 and 0.9997 for 331293467.

2. Independence, systematics, pixels

- **Pipelines (`s1_download_lc.out`):**
- 70665829: QLP s45, s46, s48, s72; TESS-SPOC s45, s46, s48, s72 and "s1751". MAST labels that last product sector 1751 (camera 1, BTJD 4055.76–4063.00, 1480/3131 good cadences); it is used as given. No 2-min SPOC.
- 331293467: QLP only (s11, s38, s99–s102). No TESS-SPOC and no 2-min data.
- **Per-sector depths, primary / secondary in ppt (`s9_indep.out`):**

TIC / sector	QLP SAP	TESS-SPOC SAP	TESScut 3×3 raw	TESScut / target share	PSF amp / target flux	d2/d1 (QLP, SPOC, TESScut)
70665829 s45	325.3 / 72.3	300.3 / 67.3	—	—	—	0.222, 0.224, —

TIC / sector	QLP SAP	TESS-SPOC SAP	TESScut 3×3 raw	TESScut / target share	PSF amp / target flux	d2/d1 (QLP, SPOC, TESScut)
70665829 s46	306.6 / 66.1	296.2 / 66.6	297.6 / 67.2	325.3 / 73.4 (0.915)	287.5 / 65.8	0.216, 0.225, 0.226
70665829 s48	312.2 / 70.6	282.7 / 64.1	288.1 / 67.1	285.2 / 66.4 (1.010)	264.6 / 60.4	0.226, 0.227, 0.233
70665829 s72	329.8 / 72.2	299.0 / 67.9	300.6 / 66.2	340.8 / 75.0 (0.882)	290.9 / 64.1	0.219, 0.227, 0.220
331293467 s11	61.4 / 9.8	—	—	—	—	0.160
331293467 s38	64.7 / 10.6	—	46.7 / 9.4	80.5 / 16.3 (0.580)	79.6 / 14.2	0.163, —, 0.202
331293467 s99	72.6 / 12.0	—	50.4 / 9.2	115.8 / 21.1 (0.435)	89.3 / 14.6	0.165, —, 0.182
331293467 s100	70.1 / 11.7	—	51.3 / 8.6	99.2 / 16.6 (0.517)	87.5 / 12.7	0.166, —, 0.168
331293467 s101	61.4 / 10.7	—	55.5 / 9.6	106.7 / 18.5 (0.520)	81.5 / 12.1	0.174, —, 0.173
331293467 s102	70.5 / 12.4	—	—	—	—	0.176

- For 70665829, QLP SAP is 4–10% deeper than TESS-SPOC SAP in every sector, but the d2/d1 ratios match. This is a multiplicative aperture or background-scale difference, not a different signal. TESS-SPOC CROWDSAP is 0.966 (s46) and 0.969 (s1751).
- For 331293467, raw depths depend on the aperture (TIC contamination ratio 0.70), while d2/d1 and T14 agree.
- **ZTF (`s13_ztf.out`, 2″ cone):**
- 70665829: 1789 rows from 2018 to 2025 (HJD−2457000 1202.8–3969.0), folded on the TESS ephemeris.
  - Primary core: zg 361 and 316 ppt, zr 325 and 315 ppt (4–8 points per field).
  - Secondary core: zg 26 and −9 ppt, zr 60 and 33 ppt.
  - So the eclipses are seen by an independent instrument, the ephemeris holds back to 2018, and the secondary is redder.
- 331293467: 0 rows (Dec −60, outside ZTF).
- **Common mode:** I did not fetch other QLP stars. Both signals repeat at a fixed period across 3 cameras (70665829) and 3 cameras over 7 years (331293467), with secondaries at fixed phase, and the difference images show no off-target deficit. A sector-wide systematic cannot do that.
- **Blind dip census (`s3_events_*.out`, 5×MAD below a running median; `s11_census.out`):** every dip is a primary or secondary except:
- TESS-SPOC s1751 at 4059.25–4059.7: a ramp from 0.935 to 0.98 right after a data gap (phase 0.77–0.90), background flat.
- QLP s48 at 2621.1–2621.5: QLP background −110 to −210, many cadences flagged 2048/4096.
- Both are systematics.
- **Pixels (`s7_pixels_70665829.out`, `s7b_pixels_331293467.out`):** median in-eclipse minus out-of-eclipse images from the TESScut 11×11 cubes, a PSF-fitted source position, and the dilution needed for every Gaia neighbour within 90″ with G < 18, in a 3×3 aperture (scene model with Gaussian PSF and TIC-style T from G and BP−RP).
- **70665829, primary:** offset 1.0 ± 0.2″ (s46), 1.2 ± 0.6″ (s48), 0.5 ± 0.1″ (s72).
- **70665829, secondary:** 2.3 ± 0.3″, 1.1 ± 0.8″, 1.8 ± 0.1″.

- **70665829, neighbour:** the only one, G 14.44 at 64.3″, needs 7253–47260% (primary) and 1573–10587% (secondary), with a χ^2 ratio of 41–78. Excluded.
- **331293467, primary offsets:** 0.3 ± 0.2″ (s38), 2.0 ± 0.2″ (s99), 1.2 ± 0.3″ (s100), 3.6 ± 0.5″ (s101). Bootstrap errors are surely underestimates of the systematics, but every offset is far smaller than any separation that matters.
- **331293467, neighbour by neighbour:**
  - G 17.07 at 8.0″: needs 377/458/334/305%. χ^2 ratio 3.6–4.6. Offset projection onto it is 0.03/0.23/0.00/−0.18 of the way.
  - G 16.14 at 14.0″: needs 91/115/99/97%. χ^2 ratio 3.8–10.5. Projection −0.01/0.01/−0.08/0.24.
  - G 16.07 at 24.4″: needs 143–287%.
  - G 13.78 at 30.5″: needs 84/45/28/25%, so not excluded by arithmetic. χ^2 ratio 19.1/10.1/19.1/16.0 and projection 0.00/−0.00/0.04/−0.11, so excluded by position.
  - G 14.73 at 47.3″: needs 96–5754%, χ^2 ratio 10–20.
  - All other neighbours need more than 150%.
- **331293467, secondary:** the fit gives 0.2 ± 0.7″, 1.9 ± 1.0″, 3.2 ± 0.6″, 3.2 ± 1.3″. The χ^2 cannot separate the target from the 8″ and 14″ stars here (ratio ~1.0–1.1), and a G 12.85 VARIABLE star at 89.4″ on the cutout edge fits the secondary diff image better (ratio 0.14–0.36). That star contributes 0.0000 of the aperture flux, so it is noise in a 4–5 event average, not a source. The secondary is tied to the target because it keeps a fixed phase (0.5002) and a fixed d2/d1 with the on-target primary.

3. TIC 70665829

Measurements (raw SAP QUALITY==0; trapezoid with quadratic baseline; `s3_events_70665829.out` lists every event). Sector summaries:

Event	Mid BTJD (first … last)	Depth (SPOC / QLP / TESScut)	T14 / T23	pipelines
Primary s45 ×6	2528.21356 … 2549.00771	300.3 / 325.3 / —	2.83 / 0.53 h (all SPOC primaries; range 2.80–2.90)	SPOC, QLP
Primary s46 ×6	2555.93930 … 2576.73431	296.2 / 306.6 / 297.6	"	SPOC, QLP, TESScut
Primary s48 ×6	2611.39079 … 2635.65071 (2614.85610 SPOC only)	282.7 / 312.2 / 288.1	"	SPOC, QLP, TESScut
Primary s72 ×5	3262.95482 … 3283.74972	299.0 / 329.8 / 300.6	"	SPOC, QLP, TESScut
Primary s1751 ×1	4060.08215 ± 0.00010	299.86	2.84 / 0.53 h	SPOC
Secondary s45 ×7	2526.34374 … 2547.13826	67.3 / 72.3 / —	2.77 / 0.54 h (all SPOC secondaries; range 2.57–2.88)	SPOC, QLP
Secondary s46 ×8	2554.07007 … 2578.32973	66.6 / 66.1 / 67.2	"	SPOC, QLP, TESScut
Secondary s48 ×6	2612.98690 … 2633.78190	64.1 / 70.6 / 67.1	"	SPOC, QLP, TESScut
Secondary s72 ×6	3264.54980 … 3285.34512	67.9 / 72.2 / 66.2	"	SPOC, QLP, TESScut
Secondary s1751 ×1	4061.67233 (partial, egress lost)	54.78	—	SPOC

- **Lost events:** in TESS-SPOC, s45/46/48/72 lose 1/2/5/4 events to gaps (`s11_census.out`). QLP flags more of s48: it has 1 full and 11 partial events there.
- **Period (`s12_ephem.out`, full-coverage events, joint fit):**

- SPOC: T0 = 2791.61111 ± 0.00009 BTJD, P = 3.4657654 ± 0.0000007 d. QLP: 3.4657623 ± 0.0000006 d.
- The 3σ difference comes from the single s1751 primary. Without it, SPOC primaries give 3.4657637 ± 0.0000012, and the s1751 eclipse arrives +2.30 min late (`s12b_no1751.out`).
- Primary O−C has χ^2_r of 24–30 against the formal errors: sector means range from −1.9 to +1.3 min (s48 is early in both pipelines while its secondaries are not). This was not pursued, but the realistic timing error is about ±2 min.
- Gaia P = 3.4658878 ± 0.0002846, so (P − P_Gaia)/σ = −0.43 for both pipelines.
- **Odd/even (`s4_period_70665829.out`):** primary +0.62 ± 0.82 ppt (SPOC), −0.87 ± 1.24 (QLP). Secondary +0.17 ± 0.54, −1.06 ± 0.74. No difference.
- **Which eclipse is primary:** the primary (300 ppt) and secondary (67 ppt) differ by a factor of 4.5 and sit at an off-centre phase. So P is neither half nor double the true period, and the deeper event is unambiguous.

4. TIC 331293467

Measurements (QLP SAP QUALITY==0. Primaries have a free trapezoid fit (`s3_events_331293467.out`). Secondaries use the primary's shape with free t0 and depth (`s5_secondary_331293467.out`)):

Event	Mid BTJD (first … last)	Depth QLP (TESScut raw)	T14 / T23	pipelines
Primary s11 ×4	1601.23407 … 1622.26045	61.4	4.07 / 0.83 h (all primaries; range 3.93–4.23)	QLP
Primary s38 ×5	2337.18980 … 2358.21445	64.7 (46.7)	"	QLP, TESScut
Primary s99 ×2	4050.91270, 4066.68276	72.6 (50.4)	"	QLP, TESScut
Primary s100 ×3	4077.19684 … 4098.22307	70.1 (51.3)	"	QLP, TESScut
Primary s101 ×4	4103.47964 … 4124.50456	61.4 (55.5)	"	QLP, TESScut
Primary s102 ×3	4129.76349 … 4150.78766 (last one partial)	70.5	"	QLP
Secondary ×24 (3/4/4/5/4/4 per sector)	1603.8613 … 4148.1608	11.42 ± 0.15 weighted mean (8.6–13.4 each)	primary shape (TESScut free fit 3.5–4.2 h)	QLP, TESScut

- **Lost events:** 14 events fall in gaps across the six sectors (`s11_census.out`).
- **Period (`s12_ephem.out`, full-coverage, joint):** T0 = 3467.40446 ± 0.00029 BTJD, P = 5.2568199 ± 0.0000020 d.
- All primaries: 5.2568187 ± 0.0000017. Secondaries only: 5.2568291 ± 0.0000074 (1.4σ).
- Gaia P = 5.2565467 ± 0.0005365, so +0.51σ.
- **Odd/even, primary:** −1.19 ± 0.49 ppt (formal 2.4σ, 1.8% of the depth). It is driven by sectors with 1–2 events and a partial s102 event, and the sector-to-sector depth scatter (61–73 ppt) is much larger. At 2P both halves are the same eclipse type, and the 11 ppt secondary at phase 0.5002 rules out P/2. Not a sign of a wrong period.

Geometry vs Gaia

All from `s10_geometry.out`: 20000 draws each with independent errors, with Gaia's correlation matrix, and with that matrix conditioned on the TESS period.

70665829. Gaia: e = 0.0769 ± 0.0397, ω = 175.4 ± 22.2°, T_p = 390.5724 ± 0.2071 BTJD. - **Secondary phase:** measured shallow-after-deep phase is 0.46048 ± 0.00008. The implied e cos ω = −0.0621, the same exact or to first order. - **Gaia prediction:** (t_B − t_A)/P = 0.4547 (+0.0239/−0.0238). That is below 0.5 in 100% of draws, with e cos ω = −0.0711 ± 0.0362. - Deep = A matches at +0.24σ. Deep = B would need 0.5395, at +3.55σ. - Conditioned on the TESS period: 0.4630 ± 0.0136, giving A −0.19σ and B +5.64σ. - **Absolute timing (630 cycles from T_p):** - Independent errors: deep minus A = −0.0215 ± 0.0999 P (−0.22σ); minus B = −4.71σ. - Gaia correlated: deep minus A = −0.0204 ± 0.0585 P (−1.7 h, −0.35σ); minus B = −11.7σ. - Conditioned on the TESS period: deep minus A = +0.0051 ± 0.0064 P (+0.4 h, +0.80σ); minus B = −56σ. - **Conclusion: deep eclipse = conjunction A.** Gaia's ω decides this through two independent routes (secondary phase and absolute timing). - **Durations:** T14 is 2.77 h for the secondary and 2.83 h for the primary (SPOC). This is consistent with Gaia's small e sin ω = +0.0032 ± 0.0310. - **Physics (`s14_physics.out`, M1 = 0.85 M⊙):** f(m) = 0.0807 M⊙, giving m2 ≈ 0.54 M⊙ (≈3760 K, ≈0.52 R⊙). a = 10.75 R⊙ (a/R1 = 14.2). A central eclipse would last 3.15 h against 2.83 h observed, so b > 0. The expected d2/d1 ≈ J(800 nm) ≈ 0.28 against 0.22 observed. - A cooler M-dwarf companion makes the K dwarf's eclipse the deep one, as the timing says. ZTF agrees: the deep eclipse is deepest in g, the shallow one deeper in r.

331293467. Gaia: e = 0.0215 ± 0.0250, ω = 287.0 ± 69.8°, T_p = 389.8053 ± 1.0004 BTJD. - **Secondary phase:** 0.50013 ± 0.00023 (joint fit) and 0.5002 ± 0.0001 (`s5`). That gives e cos ω = +0.0003 to +0.0004 ± 0.0002, against Gaia's +0.0024 ± 0.0227. Consistent, but it cannot separate A from B. - **Absolute timing (699 cycles from T_p):** - Independent errors: deep minus A = +0.14σ, minus B = −1.90σ. Not decisive. - Gaia correlated: deep minus A = +0.0321 ± 0.0649 P (+4.1 h, +0.49σ); minus B = −6.66σ. - Conditioned on the TESS period: deep minus A = +0.0009 ± 0.0077 P (+0.1 h, +0.12σ); minus B = +81σ. Conditioning shifts T_p by +0.29 d and ω by +20.7°, because corr(P, T_p) = 0.57. - **Conclusion: deep eclipse = conjunction A,** the K dwarf in front of the A/F star. This rests on Gaia's corr(ω, T_p) = 0.9997, which is physically expected at e ≈ 0. - **Physics (M1 = 1.69 M⊙):** f(m) = 0.0630 M⊙, giving m2 ≈ 0.71 M⊙ (≈4580 K, ≈0.70 R⊙, k ≈ 0.41). a = 17.04 R⊙. A central eclipse would last 5.76 h against 4.07 h observed (b > 0). The expected d2/d1 ≈ 0.21 against 0.16 observed. The undiluted primary depth of 8–11% is below $k^2 \approx 0.17$, as a partial or high-b transit should be.

Ephemerides and next eclipses

From `s12_ephem.out`. Times are BJD_TDB / BTJD, converted to UTC calendar dates without the barycentric-to-Earth light-time correction (up to ±8.3 min). - **TIC 70665829:** T0 = 2791.61111 ± 0.00009 BTJD (BJD_TDB 2459791.61111), P = 3.4657654 ± 0.0000007 d (SPOC). The secondary follows at +1.59591 ± 0.00029 d (phase 0.46048). Realistic timing error is ±2 min (see §3). - Next primaries: 4344.2740 (2026-10-30 18:33 UTC), 4347.7398 (2026-11-03 05:44), 4351.2055 (2026-11-06 16:54). The QLP ephemeris is 1–2 min earlier. - Next secondaries: 4345.8699 (2026-11-01 08:51), 4349.3357 (2026-11-04 20:02), 4352.8014 (2026-11-08 07:12). - **TIC 331293467:** T0 = 3467.40446 ± 0.00029 BTJD (BJD_TDB 2460467.40446), P = 5.2568199 ± 0.0000020 d. The secondary follows at +2.62912 ± 0.00123 d (phase 0.50013). - Next primaries: 4345.2934 ± 0.0004 (2026-10-31 19:01 UTC), 4350.5502 (2026-11-06 01:11), 4355.8070 (2026-11-11 07:20). - Next secondaries: 4347.9225 (2026-11-03 10:07), 4353.1793 (2026-11-08 16:17), 4358.4361 (2026-11-13 22:26).

Caveats

- **331293467 has a single pipeline (QLP) plus my own TESScut photometry.** Its undiluted depths rest on a Gaussian-PSF scene model: primary 80–116 ppt (79.6–89.3 ppt from PSF

amplitude), secondary ~13–21 ppt. Only the fact that the signal is on the target, the timing and the ratios are robust.
- **Attribution for 331293467 is pixel-level, not resolved.** ZTF has no data at Dec −60. The G 13.78 star at 30.5″ is excluded only by position (and by the timing match to Gaia's conjunction), not by arithmetic. Bootstrap centroid errors (0.2–0.5″) understate the systematics, but the closest call is still 0.24 of the way toward any neighbour.
- **Gaia geometry for 331293467 is decisive only with the corr_vec decoding** (order P, Vcm, K, e, ω, T_p, column-major upper triangle; the 0.9997 ω–T_p term confirms it).
- **70665829 timing:** the O–C scatter between sectors (±1–2 min) and the late s1751 primary (+2.3 min) are unexplained. Candidates are timestamps in the unusual "sector 1751" product, or real timing variations. This does not affect the classification.
- **Archives:** the cones for Gaia-VEB, varisum and SB9 were not checked on a known object, because the mirror failed during `s2c`. Their empty results stand from `s2` only.
- **Common mode:** not done with other stars (see §2).

Files

- **Light curves:** `lcio.py`, `s1_download_lc.py` / `.out`.
- **Archives:** `s2_archives.py` / `.out`, `s2b_allcat.py` / `.out`, `s2c_catalogue_sanity.py` / `.out`.
- **Events:** `s3_events.py`, `s3_events_70665829.out` / `.json`, `s3_events_331293467.out` / `.json`.
- **Periods:** `s4_period.py`, `s4_period_*.out`, `s4_ephem_*.json`; `s5_secondary_331.py`, `s5_secondary_331293467.out` / `.json`.
- **Pixels:** `s6_tesscut.py` / `.out`; `s7_pixels.py`, `s7_pixels_70665829.out`, `s7_pixels_331293467.out` (interrupted; superseded by s7b); `s7b_pixels.py`, `s7b_pixels_331293467.out`; `pix_70665829_s46/s48/s72.png`, `pix_331293467_s38/s99/s100/s101.png`; `tesscut_ap3x3_*.txt` (own photometry).
- **Independence:** `s9_indep.py` / `.out` / `.json`.
- **Geometry:** `s10a_gaia_query.py` / `.out`, `s10a_gaia_nss.csv`, `s10_geometry.py` / `.out` / `.json`.
- **Census:** `s11_census.py` / `.out`.
- **Ephemerides:** `s12_ephem.py` / `.out` / `.json`, `s12b_no1751.py` / `.out`.
- **ZTF:** `s13_ztf.py` / `.out` (raw CSVs in the scratchpad `ztf/` folder).
- **Physics:** `s14_physics.py` / `.out`.

TIC 70665829: CONFIRMED; P_ecl=3.4657654±0.0000007 d; depth1=283–330 ppt (pipeline-dependent; SPOC ~300); depth2=64–72 ppt at phase 0.46048; pixel: on target; pipelines: 4 (QLP, TESS-SPOC, TESScut, ZTF); Gaia geometry: pass (deep = conjunction A, Gaia RV star eclipsed) TIC 331293467: CONFIRMED; P_ecl=5.2568199±0.0000020 d; depth1=61–73 ppt QLP raw (≈80–116 ppt undiluted); depth2=11.4 ppt QLP (≈13–21 ppt undiluted) at phase 0.50013; pixel: on target (30.5″ G13.78 excluded by position only); pipelines: 2 (QLP, TESScut); Gaia geometry: pass (deep = conjunction A; needs Gaia ω–T_p correlation)

vet04 full report

Archived source: sanitized report text.

vet04: adversarial check of TIC 283532878 (UCAC4 740-075601) and TIC 407877153 (UCAC4 054-021425)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This run resumed the work of an earlier agent that a container restart killed part-way through. I re-read its scripts and outputs `s01`–`s11` and kept what they had measured. I re-ran the archive step it had not finished (`s12`), and found that the first VizieR pass gave false empty results (see §1). New steps: a difference-image test with real noise weighting (`s13`, `s13b`), because the `s11` "chi2" values were unweighted sums of squares. Also new: ZTF (`s14`), plots (`s15`), a common-mode test (`s16`), extra geometry, physics and ephemeris (`s17`), and a merged event table (`s18`). All of these are in `/home/user/eclipses-paper-review/work/c9a-vets/vet04/`. The downloads are in the scratchpad at `.../scratchpad/data/vet04/<tic>/`. I deleted the TESScut cubes after `s13`. Notation: **A** = the companion passes in front of Gaia's RV star (true anomaly $\nu = 90° - \omega$); **B** = the RV star passes in front of the companion. Depths are in ppt, times in BTJD = BJD_TDB − 2457000, durations in hours.

TIC 283532878: CONFIRMED. - (1) Each event fitted on its own in raw QLP SAP flux, QUALITY==0: 11 primaries and 9 full secondaries, plus 1 partial secondary. One more primary (s16, 1744.53) is visible only in TESScut. - (2) The eclipses give P = 17.446019 ± 0.000019 d. Gaia's 17.4298 ± 0.0089 d is +1.83σ away. - (3) Difference images in all 8 sectors (cameras 2, 3 and 4) put both the primary and the secondary on the target. Primary offsets are 0.4–1.5″. Secondary offsets are 1.0–5.3″, each within ~2σ. Every Gaia neighbour is rejected by $\Delta\chi^2 \geq 1119$ for the primary and $\Delta\chi^2 \geq 41.7$ for the secondary in every sector. The dilution arithmetic alone excludes every neighbour for the primary (each would need ≥ 218%). For the 10 ppt secondary it does **not** exclude three neighbours (G 14.50 at 37″, G 14.93 at 28″, G 15.74 at 29.5″, each needing 30–47%); the pixels do. - (4) QLP SAP and my own TESScut photometry agree on the secondary (9–10 ppt in both). For the primary, QLP gives 70–79 ppt and the TESScut aperture 65–72 ppt, with 13–20% of that aperture's light from neighbours. The PSF-deblended depth is 71–77 ppt. No TESS-SPOC or 2-min data exist. - (5) The secondary is at phase 0.67812 ± 0.00030; Gaia's e and ω predict 0.712 ± 0.024 (–1.4σ). The primary falls 0.20 d from Gaia's conjunction A (0.4–0.7σ); B is excluded at > 11σ. - **Screen check:** the screen was right. The "~9 ppt dip near 0.675" is the eccentric secondary. It is not a one-sector glitch: it appears in all 8 sectors and 4 seasons at constant phase (0.6775–0.6795). Its depth is 9–10 ppt; 10–11 ppt deblended.

TIC 407877153: CONFIRMED. - (1) Each event fitted on its own: 9 full primaries and 9 secondaries in raw SAP, plus extra partial events (two primaries, two secondaries). - (2) The eclipses give P = 17.089294 ± 0.000008 d. Gaia's 17.0902 ± 0.0063 d is −0.14σ away. - (3) Difference images put the primary on the target in 6 sectors ($\Delta\chi^2 \geq 337$ against every neighbour). The secondary also fits best on the target in 6 sectors ($\Delta\chi^2 \geq 7$ per sector; ≥ 166 summed over sectors). The arithmetic needs ≥ 441% (primary) and ≥ 51% (secondary) on any neighbour. - (4) Three pipelines: TESS-SPOC SAP (28.6–29.4 ppt), my own TESScut aperture (29.4–31.0 ppt) and QLP SAP. QLP gives 27–29.6 ppt in s12/s39 but only 24–27 ppt in s93–s102, a QLP dilution issue. The secondary is 2.6–4.2 ppt in all three. - (5) The secondary is at phase 0.54690 ± 0.00017; the prediction is 0.559 ± 0.044 (−0.28σ). The primary is at conjunction A (0.16–0.64σ); B is excluded at ≥ 5σ. - **Screen check:** the secondary is **3.0 ppt, not ~2 ppt** (stack 2.98 ± 0.13). The primary is ~29 ppt in SPOC and TESScut, not ~27 (27 is QLP's diluted value). The phase of 0.545 was right.

1. Archives and web

- **VizieR cones** (`s12d_recheck.out`, mirror vizier.idia.ac.za, run immediately after an Algol VSX control returned bet Per, EA/SD):
- **TIC 283532878** is not in VSX, Gaia DR3 EB (I/358/veb), Prsa+2022, IJspeert+2021, Gaia varisum or SB9 within 60″ of the target. Both VSX and Gaia-EB list one row at **29.6″: Gaia DR3 2202301140233408128**, G 17.66 (the "VARIABLE" neighbour in `neighbours_90as.csv`). It is VSX type E, P = 1.79747 d, G 17.61–18.17; Gaia EB eclipse depths are 0.39 and 0.60 (`s12e_neighbour_eb.out`). It is excluded as the source below.
- **TIC 407877153:** nothing in any of those six catalogues within 60″.
- **First-pass warning:** the first pass (`s12_archives.out`) returned 0 rows everywhere, including the EB above, and its Algol control failed later (`s12c_controls.out`: VSX 0 rows for Algol). The mirror was intermittently returning empty tables. The all-catalogue 5″ query is likewise unreliable here: its Algol control gave 0 tables (`s12b_archives.out`), so I do not use it.
- **SIMBAD** (sim-coo 10″, `s12b_archives.out`): both stars are type **SB*** with 1 bibcode (2023A&A...674A..34G, the Gaia DR3 NSS paper). TIC 283532878 = UCAC4 740-075601 = 2MASS J21512860+5759057. TIC 407877153 = UCAC4 054-021425 = 2MASS J18230801-7921092.
- **ExoFOP:** no TOI. "CTOIs 0" on both pages.
- **Conclusion:** neither star is a catalogued eclipsing binary within 60″. Both are known only as Gaia SB1s.

2. Independence, systematics, pixels

- **Pipelines:**
- 283532878: QLP SAP (8 sectors) plus my own TESScut aperture photometry (8 sectors). The two agree on the secondary. The primary is shallower in my aperture because it holds 13–20% neighbour light (`s11_pixels_283532878.out`: target fraction 0.796–0.873).
- 407877153: QLP, TESS-SPOC (s12, s13, s39) and TESScut. SPOC and TESScut agree on the primary to about 1 ppt; QLP s93–s102 are ~10–15% shallower. No 2-min SPOC data exist for either star.
- **Difference images, noise-weighted (`s13_pixnoise_*.out`, `s13b_summary.out`):** in each sector I averaged the per-event (median out − median in) images. The per-pixel noise comes from 300 pseudo-events at random out-of-eclipse times with the same windows. I then fitted Gaussian-PSF single-source models for the target and for every Gaia neighbour with G < 18 within 90″. Results:
- 283532878 P: S/N 40–154 per sector; least-rejected neighbour $\Delta\chi^2$ = 73,292 summed (≥ 1119 in every sector); PSF-amplitude depth on target 70.6–77.3 ppt.
- 283532878 S: S/N 8.6–14.9; $\Delta\chi^2 \geq 41.7$ in every sector, ≥ 977 summed; depth 9.8–12.9 ppt; free-fit offsets 1.0–5.3″ (1σ 1.3–2.9″).
- 407877153 P: S/N 18.6–84; $\Delta\chi^2 \geq 337$; depth 30.0–35.3 ppt.
- 407877153 S: S/N 2.8–8.7; $\Delta\chi^2 \geq 7.1$ per sector, ≥ 166 summed; offsets 3.6–9.9″ with 1σ 1.9–12″.
- I looked at two of the images (`s15_diff_283532878_s57_S.png`, `s15_diff_407877153_s13_S.png`). In both, the deficit sits on the target pixel, away from the bright neighbours in the direct image.
- **Dilution arithmetic** (per-sector aperture flux fractions, `s11_pixels_*.out`):
- **283532878 primary:** every neighbour needs ≥ 218% intrinsic depth, so all are excluded.
- **283532878 secondary (10.2 ppt):** most neighbours are excluded arithmetically (≥ 82% needed; the G 17.66 Gaia EB at 29.7″ needs 181–240%). Three are **not**: G 14.50 at 37.0″ needs 29.6–100%, G 14.93 at 28.3″ needs 30–47%, and G 15.74 at 29.5″ needs 31.8–39%. The difference images reject them instead ($\Delta\chi^2 \geq 41.7$ each sector).

- **407877153:** the primary needs ≥ 441% on any neighbour; the secondary needs ≥ 51.5% (G 14.68 at 60″, s13) and 61–135% for the G 17.23 star at 46″.
- **Same period and phase drift:** both secondaries share the primary's period (283532878: P_S – P_P = −0.00030 ± 0.00010 d; 407877153: +0.000013 ± 0.000059 d, `s05_ephem.out`) and sit at fixed phase over 4–5 seasons. A neighbour's EB would have to share the period exactly. The catalogued neighbour EB (P = 1.7975 d; 17.446/1.7975 = 9.71, not commensurate) would drift in phase.
- **ZTF (`s14_ztf.out`):** not informative. IRSA returns only 54–94 catflags==0 points per object for the target and the five closest bright neighbours, with 0–1 points inside any eclipse window.
- **Common mode (`s16_commonmode.out`):** not done. The MAST product queries kept failing (RemoteServiceError and proxy resets, `s16_commonmode.out`), and the coordinator then dropped this optional step. The pixel and two-pipeline evidence above stands without it.
- **Phase 0.5 of 283532878** (`s17_geometry2.out`, from 10 fixed-time fits in `s08`): +0.35 ± 0.30 ppt. There is no eclipse at phase 0.5. $\chi^2 = 45/9$ shows red noise at the ~2–3 ppt level.

3. TIC 283532878 (P = 17.446 d, e = 0.35)

Measurements (`s18_event_table.out`, from `s04`/`s08`/`s11`; trapezoid with quadratic baseline, raw SAP QUALITY==0; "fixed" means the shape is fixed from the 200-s stack). There is no TESS-SPOC column because this star has no TESS-SPOC data.

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth own TESScut ap. (ppt)	T14 / T23 (h)	Pipelines
P n=−1 s16	1744.5307 ± 0.0011 (TESScut)	QLP: 13 flagged cadences	67.6 ± 1.0	–	TESScut
S n=−1 s16	1756.3642 ± 0.0079	7.5 ± 1.2 / 7.6 ± 1.1	10.1 ± 0.8	5.53 / 3.45	QLP, TESScut
P n=0 s16	1761.9800 ± 0.0015	70.8 ± 1.7 / 70.4 ± 1.4	66.9 ± 1.1	7.01 / 2.00	QLP, TESScut
S n=0 s17	1773.8350 ± 0.0048	9.6 ± 0.7 / 9.8 ± 0.7	9.2 ± 0.8	8.21 / 3.43	QLP, TESScut
P n=1 s17	1779.4207 ± 0.0007	73.6 ± 0.8 / 73.9 ± 0.9	69.6 ± 0.8	7.29 / 2.12	QLP, TESScut
P n=61 s56	2826.1851 ± 0.0009	74.1 ± 1.1 / 74.1 ± 0.9	66.8 ± 0.6	7.20 / 2.05	QLP, TESScut
S n=61 s56	2838.0248 ± 0.0037 (TESScut)	12.8 ± 2.0 (egress in gap)	10.7 ± 0.6	–	QLP (partial), TESScut
P n=62 s56	2843.6288 ± 0.0010	75.5 ± 1.3 / 75.2 ± 1.0	66.8 ± 0.5	7.04 / 2.02	QLP, TESScut
S n=62 s57	2855.4637 ± 0.0090	9.2 ± 1.5 / 8.6 ± 1.3	9.3 ± 0.5	6.08 / 2.19	QLP, TESScut
P n=63 s57	2861.0760 ± 0.0010	73.5 ± 1.3 / 73.3 ± 1.2	68.4 ± 0.6	6.95 / 2.05	QLP, TESScut
S n=63 s57	2872.9161 ± 0.0070	9.0 ± 1.0 / 9.2 ± 1.0	9.2 ± 0.5	6.87 / 3.03	QLP, TESScut
P n=64 s57	2878.5193 ± 0.0005	79.2 ± 0.7 / 78.6 ± 0.6	69.6 ± 0.5	7.11 / 1.97	QLP, TESScut
S n=92 s76	3378.8457 ± 0.0067	8.9 ± 1.1 / 9.4 ± 1.1	9.8 ± 0.5	6.34 / 3.85	QLP, TESScut
P n=93 s76	3384.4553 ± 0.0007	76.4 ± 0.9 / 77.3 ± 0.8	68.8 ± 0.6	6.76 / 2.26	QLP, TESScut
S n=93 s77	3396.2636 ± 0.0050	10.4 ± 0.7 / 10.1 ± 0.8	9.4 ± 0.5	8.89 / 2.96	QLP, TESScut
P n=94 s77	3401.9005 ± 0.0008	75.4 ± 1.0 / 75.5 ± 0.8	71.8 ± 0.7	6.98 / 2.10	QLP, TESScut
S n=94 s77	(3413.73)	in the 3407.0–3417.2 data gap	–	–	none
P n=95 s77	3419.3449 ± 0.0008	74.9 ± 1.0 / 74.1 ± 0.9	70.9 ± 0.5	7.22 / 1.89	QLP, TESScut

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth own TESScut ap. (ppt)	T14 / T23 (h)	Pipelines
S n=103 s83	3570.7490 ± 0.0055	9.6 ± 1.0 / 9.9 ± 0.9	9.4 ± 0.7	6.33 / 3.35	QLP, TESScut
P n=104 s83	3576.3613 ± 0.0011	74.6 ± 1.3 / 74.4 ± 1.1	68.3 ± 0.7	7.03 / 2.05	QLP, TESScut
S n=104 s84	3588.1988 ± 0.0068	9.8 ± 1.3 / 9.9 ± 1.3	9.1 ± 0.7	5.67 / 3.36	QLP, TESScut
P n=105 s84	3593.8100 ± 0.0005	76.4 ± 0.7 / 76.6 ± 0.6	65.5 ± 0.5	7.07 / 2.09	QLP, TESScut
S n=105 s84	3605.6254 ± 0.0035	12.8 ± 0.7 / 12.6 ± 0.6	10.2 ± 0.4	7.32 / 2.48	QLP, TESScut

- **Stacked shape** (200-s QLP, `s08_shape_oddeven.out`):
- Primary: 75.41 ± 0.39 ppt, T14 = 7.028 ± 0.034 h, T23 = 2.08 ± 0.05 h. This is a near-V shape: a grazing or partial eclipse.
- Secondary: 10.24 ± 0.40 ppt, T14 = 6.66 ± 0.25 h, T23 = 2.78 ± 0.36 h.
- **Period** (`s05_ephem.out`):
- 11 primaries alone: P = 17.446025 ± 0.000017 d (χ^2 = 56/9; O–C within ±7 min). 9 secondaries alone: 17.445724 ± 0.000099 d.
- The joint fit gives P = 17.446019 ± 0.000019 d, +0.0162 d from Gaia's period = **+1.83σ_Gaia**.
- The cycle count is unambiguous: consecutive primaries within s56, s57 and s77 are 17.44 d apart.
- **Odd/even** (QLP fixed shape): 76.25 ± 0.38 vs 75.36 ± 0.33 ppt. The difference is 1.8σ, smaller than the 1.5–2.6 ppt scatter between sectors. A secondary of different depth exists at phase 0.678, so 2P is ruled out, and phase 0.5 is empty, so P/2 is ruled out.
- **Secondary phase by season** (`s17_geometry2.out`): 0.67954 ± 0.00026 (s16–17), 0.67854 ± 0.00033 (s57), 0.67750 ± 0.00025 (s76–77), 0.67775 ± 0.00016 (s83–84).
- **Physics:** f(m) = 0.0334 M⊙. With the TIC M1 = 1.35 M⊙, m2 ≥ 0.48 M⊙ and a = 34.6 R⊙. The depth ratio S/P = 0.136 implies a TESS-band surface-brightness ratio ≈ T2 3830 K, consistent with an early-M companion with R2 ≳ 0.5 R⊙. A central eclipse at conjunction A would last ~9.5 h; the observed 7.0 h and V shape mean a high impact parameter. The duration ratio S/P = 0.948 ± 0.036 gives e sin ω ≈ −0.03 ± 0.02, against Gaia −0.086 ± 0.044. The equal-b assumption behind that number is weak for grazing eclipses.

4. TIC 407877153 (P = 17.089 d, e = 0.13)

Measurements (`s18_event_table.out`). TESS-SPOC depths are fixed-shape fits where available.

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth TESS-SPOC SAP (ppt)	Depth own TESScut (ppt)	T14 / T23 (h)	Pipelines
S n=−1 s12	1626.5135 ± 0.0171 (TESScut)	QLP: 17 flagged	–	2.9 ± 0.7	–	TESScut
P n=0 s12	1634.2635 ± 0.0017	27.5 ± 0.5 / 27.4 ± 0.6	29.2 ± 0.5	30.1 ± 0.7	9.44 / 6.10	QLP, SPOC, TESScut
S n=0 s12	1643.5993 ± 0.0077	3.2 ± 0.6 / 2.9 ± 0.5	3.6 ± 0.5	2.9 ± 0.6	7.62 / 7.29	QLP, SPOC, TESScut
P n=1 s12	1651.3509 ± 0.0012	27.5 ± 0.6 (partial)	29.4 ± 0.5	29.9 ± 0.6	9.72 / 5.81	QLP (partial), SPOC, TESScut
S n=1 s13	1660.7055 ± 0.0069	3.0 ± 0.5	2.9 ± 0.4	3.1 ± 0.4	7.46 / 6.61	QLP, SPOC, TESScut
P n=2 s13	(1668.43)	in the 1667.6–1668.7 gap	gap	–	–	none
S n=2 s13	1677.7897 ± 0.0079	2.9 ± 0.5 / 2.6 ± 0.4	3.0 ± 0.3	2.9 ± 0.4	7.81 / 7.10	QLP, SPOC, TESScut

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth TESS-SPOC SAP (ppt)	Depth own TESScut (ppt)	T14 / T23 (h)	Pipelines
P n=43 s39	2369.1015 ± 0.0009	29.5 ± 0.4 / 29.0 ± 0.4	28.6 ± 0.4	29.5 ± 0.4	9.53 / 6.10	QLP, SPOC, TESScut
S n=43 s39	2378.4374 ± 0.0064	3.8 ± 0.7 / 3.0 ± 0.4	2.9 ± 0.3	3.2 ± 0.3	8.62 / 6.29	QLP, SPOC, TESScut
P n=44 s39	2386.1901 ± 0.0010	29.6 ± 0.5 / 29.3 ± 0.4	28.7 ± 0.4	29.4 ± 0.4	9.65 / 6.05	QLP, SPOC, TESScut
S n=128 s93	3831.0358 ± 0.0104	4.2 ± 0.6 / 3.5 ± 0.4	–	3.0 ± 0.4	9.27 / 2.83	QLP, TESScut
P n=129 s93	3838.7816 ± 0.0010	27.1 ± 0.4 / 27.3 ± 0.4	–	29.5 ± 0.4	9.38 / 6.55	QLP, TESScut
S n=129 s93	3848.1399 ± 0.0053	2.9 ± 0.3 / 2.9 ± 0.4	–	3.7 ± 0.4	7.66 / 6.63	QLP, TESScut
S n=130 s94	3865.2086 ± 0.0068	3.2 ± 0.3 / 3.2 ± 0.3	–	4.2 ± 0.4	9.56 / 5.54	QLP, TESScut
P n=131 s94	3872.9595 ± 0.0013	24.7 ± 0.4 / 24.3 ± 0.4	–	29.6 ± 0.5	9.85 / 5.89	QLP, TESScut
P n=143 s100	4078.0293 ± 0.0011	26.8 ± 0.4 / 26.7 ± 0.4	–	31.0 ± 0.5	9.45 / 6.20	QLP, TESScut
S n=143 s100	(4087.36)	64 flagged cadences	–	–	–	none
P n=144 s100	4095.1193 ± 0.0010	24.7 ± 0.4 / 24.8 ± 0.3	–	29.4 ± 0.4	9.51 / 6.45	QLP, TESScut
S n=144 s101	4104.4747 ± 0.0102	2.9 ± 0.3 / 2.6 ± 0.2	–	no cube	12.06 / 3.76	QLP
P n=145 s101	(4112.23)	156 flagged cadences	–	no cube	–	none
S n=145 s101	4121.5442 ± 0.0073	3.0 ± 0.3 / 2.8 ± 0.3	–	no cube	8.90 / 4.73	QLP
P n=146 s102	4129.2998 ± 0.0009	25.0 ± 0.3 / 24.9 ± 0.4	–	29.5 ± 0.4	9.37 / 6.28	QLP, TESScut
S n=146 s102	4138.6654 ± 0.0176 (TESScut)	QLP: 155 flagged	–	4.3 ± 1.1	–	TESScut
P n=147 s102	4146.3904 ± 0.0024	23.7 ± 0.7 (ingress lost)	–	25.0 ± 0.7	9.11 / 6.53	QLP (partial), TESScut

- **Stacked shape** (200-s QLP):
- Primary: 25.53 ± 0.16 ppt (QLP-diluted), T14 = 9.459 ± 0.051 h, T23 = 6.38 ± 0.08 h. **Flat-bottomed**, a total transit.
- Secondary: 2.98 ± 0.13 ppt, T14 = 8.46 ± 0.31 h, T23 = 5.71 ± 0.50 h. **Flat-bottomed**, an occultation.
- **Period:**
- 9 primaries: P = 17.089294 ± 0.000006 d (χ^2 = 6.0/7; O–C within ±2.3 min). 9 secondaries: 17.089307 ± 0.000059 d.
- Joint fit: P = 17.089294 ± 0.000008 d, **−0.14σ_Gaia**.
- Secondary phase by season: 0.54630–0.54713, steady.
- **Odd/even:** TESS-SPOC 28.91 ± 0.31 vs 28.91 ± 0.30 ppt. The QLP odd/even difference (26.10 vs 26.91) comes from which sectors fall in each group: QLP s93+ are diluted. 2P and P/2 are excluded as for the other star.
- **Physics:**
- f(m) = 0.0216 M⊙. With the TIC M1 = 0.87 M⊙, m2 ≥ 0.31 M⊙ and a = 29.5 R⊙.
- The flat-bottom depth gives k ≈ 0.16, so R2 ≈ 0.31 R⊙ with the TIC R1 = 1.91 R⊙. That matches a 0.31 M⊙ M dwarf.

- The occultation/transit ratio of 0.117 gives T2 ≈ 3200 K (TESS-band Planck).
- A central T14 at conjunction A would be 10.6 h; observed 9.46 h, so b is moderate.
- The duration ratio S/P = 0.894 ± 0.033 gives e sin ω = −0.056 ± 0.018, against Gaia −0.078 ± 0.066.
- Everything hangs together as a subgiant + M-dwarf SB1.

Geometry vs Gaia

- **Method** (`s07_geometry.out`, `s17_geometry2.out`; 20,000 Monte Carlo draws): I propagated P, T_p, e and ω; conjunction A is at ν = 90° − ω and B at ν = 270° − ω. The ω–T_p correlation was taken either as 0 or as the last `corr_vec` element (0.927 and 0.985). That second case assumes the SB1 parameter order ends with (arg_periastron, t_periastron), which I have not verified; results are given both ways.
- **283532878:**
- Predicted B–A separation: 0.7122 ± 0.0237. Measured secondary phase: 0.67812 ± 0.00030, **−1.44σ**. The other assignment (primary = B) is off by 16σ.
- The measured phase implies e cos ω = 0.2823 ± 0.0012 (exact Kepler, Gaia e sin ω draws), against Gaia 0.338 ± 0.039 (–1.42σ).
- Absolute timing, propagated with the TESS period: the primary is +0.0112 phase (+0.20 d) from conjunction A (0.37σ uncorrelated, 0.71σ correlated) and 0.30 phase from B (11.9σ / 23.7σ). Propagated with Gaia's period, A is still within 1.7–2.0σ.
- **The deep eclipse is A:** the F-type RV star is eclipsed by its companion. **Pass.**
- **407877153:**
- Predicted separation 0.5589 ± 0.0435; measured 0.54690 ± 0.00017, **−0.28σ**. The other assignment is off by 2.4σ.
- Implied e cos ω = 0.0733 ± 0.0006, against Gaia 0.093 ± 0.068.
- The primary is +0.0167 phase (+0.29 d) from A (0.16σ / 0.64σ) and 5.1σ / 18.9σ from B.
- **The deep eclipse is A**, the subgiant RV star transited by an M dwarf. **Pass.**

Ephemerides and next eclipses

From `s05_ephem_*.json` and `s17_geometry2.out`. UTC times are geocentric: I removed the barycentric light-time with astropy (offset given in brackets). - **TIC 283532878:** T0 = BJD_TDB 2458761.97572 ± 0.00150 (BTJD 1761.97572), **P = 17.446019 ± 0.000019 d**. Secondary T0 = 1773.80624 ± 0.00535 (phase 0.67812). - Next primaries: BTJD 4343.9865 = 2026-10-30 11:36 UTC [+3.4 min]; 4361.4325 = 2026-11-16 22:18; 4378.8785 = 2026-12-04 09:02. Each ±2.3 min formal; T14 = 7.0 h. - Next secondaries: BTJD 4355.817, 4373.263, 4390.709. The secondary phase has moved by ~0.002 between seasons, so allow ±1 h. - **TIC 407877153:** T0 = BJD_TDB 2458634.26184 ± 0.00084 (BTJD 1634.26184), **P = 17.089294 ± 0.000008 d**. Secondary T0 = 1643.60789 ± 0.00296 (phase 0.54690). - Next primaries: BTJD 4351.4596 = 2026-11-06 23:03 UTC [−3.1 min]; 4368.5489 = 2026-11-24 01:13; 4385.6382 = 2026-12-11 03:22. Each ±1.0 min; T14 = 9.5 h. - Next secondaries: BTJD 4343.716, 4360.806, 4377.895.

Caveats

- **Secondary-phase drift in 283532878:** the phase falls 0.0018 between 2019 and 2024 (≈ 45 min, ~4σ formal), and the joint timing χ^2 is 132/17. GR plus tidal apsidal motion at a/R1 ≈ 18 would give far less (~2.5×10^{-4} rad over 105 cycles, against the ~0.03 rad needed). I therefore treat this as

underestimated timing errors for the 10 ppt, red-noise-affected secondaries, not as physics. It does not affect the Gaia test, whose tolerance is ±0.024.

- **Catalogued EB 29.7″ from 283532878:** this is G 17.66, P = 1.7975 d, eclipses 40–60% deep in G. With 0.4–0.6% of the light in my apertures it can add dips of ≤ 3 ppt at unrelated phases. That may explain some event-to-event scatter in the secondary depths (for example S n=105, 12.6 ppt). It cannot produce the secondary: the period is wrong, the depth needed is 181–240%, and the pixels reject it.
- **Gaussian-PSF limits:** the pixel model uses a Gaussian PSF. Formal centroid offsets for the primaries (up to 1.5″ for 283532878; 5.1″ and 7.6″ in s93 and s102 for 407877153, the s102 stack including a partial event) exceed their noise-only errors. That is model systematics; every neighbour is ≥ 26″ away and rejected by $\Delta\chi^2$ in the hundreds or more.
- **Deblended depths are model-dependent:** for 283532878, about 71–77 ppt (PSF amplitude) up to ~80 ppt (aperture depth ÷ target fraction). The TIC contamination ratio is 0.32. QLP's SAP depths for 407877153 in s93–s102 are diluted by ~10–15%.
- **Not used:** ZTF (too few epochs), 2-min SPOC (none exists), TESScut for s101 of 407877153 (not downloaded; QLP covers its two secondaries).
- **Correlation assumption:** the `corr_vec` ordering behind the correlated Monte Carlo case is assumed, not verified. The secondary-phase test does not use T_p at all.

Files

All in `/home/user/eclipses-paper-review/work/c9a-vets/vet04/`: - **Shared code:** `common.py` (loader, Gaia constants), `fitlib.py` (trapezoid fitter). - **Earlier agent, read and kept:** - `s01_download` (QLP/TESS-SPOC light curves); `s02_inventory` (sectors, gaps, flags); `s03_blind` (blind dip census); `s04_events` (+ `.json`, per-event free fits); `s05_ephem` (+ `s05_ephem_<tic>.json`); `s06_gaia_query` (+ `s06_gaia_nss.csv`, `s06_gaia_source.csv`); `s07_geometry` (Monte Carlo conjunctions); `s08_shape_oddeven` (+ `.json`, `s08_stack_*.npz`); `s10_tesscut_dl`; `s11_pixels` (+ `_<tic>.out/.json`, `s11_diff_*.npy`; aperture photometry, dilution). - **This run:** - `s12_archives.out` (first pass, false negatives), `s12b_archives` (SIMBAD by coordinates, ExoFOP), `s12c_controls` (failed Algol control), `s12d_recheck` (cones with passing control), `s12e_neighbour_eb` (neighbour EB rows). - `s13_pixnoise` (+ `_<tic>.out/.json`, `s13_diff_*.npz`), `s13b_summary`. - `s14_ztf`, `s15_plot_diff` (+ `s15_diff_*.png`), `s16_commonmode`, `s17_geometry2`, `s18_event_table`. - **Downloads:** `/tmp/claude-0/-home-user-eclipses-paper-review/55c92ff4-798a-544f-be03-9888c6804905/scratchpad/data/vet04/` (TESScut cubes deleted).

TIC 283532878: CONFIRMED; P_ecl=17.446019±0.000019 d; depth1=75 ppt (QLP SAP; 71–77 deblended); depth2=10 ppt at phase 0.6781; pixel: on target; pipelines: 2 (QLP, own TESScut); Gaia geometry: pass TIC 407877153: CONFIRMED; P_ecl=17.089294±0.000008 d; depth1=29 ppt (TESS-SPOC/TESScut; 25–27 QLP s93+); depth2=3.0 ppt at phase 0.5469; pixel: on target; pipelines: 3 (QLP, TESS-SPOC, own TESScut); Gaia geometry: pass

vet05 full report

Archived source: sanitized report text.

vet05: adversarial check of TIC 15855794 (UCAC4 564-009141) and TIC 92881948

(WISE J053256.98-225927.8)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This run resumed from a predecessor that was killed by a container restart; that run had finished s1–s9 and part of s10. This run added s10b/c/d (archives), s11 (own TESScut photometry, difference images, dilution), s12 (master event table, ephemerides), s13 (Gaia geometry MC) and s14 (physics). The coordinator cut the budget, so the report is short. Scripts and outputs are in this folder. Downloads are in the scratchpad `data/vet05/<tic>/`; all TESScut cubes have been deleted. Depths are in ppt, times in BTJD = BJD_TDB − 2457000. "C" = own TESScut photometry with a 3×3 aperture, "Q" = QLP SAP, "S" = TESS-SPOC SAP (QUALITY==0 throughout).

TIC 15855794: CONFIRMED. (1) Repeating events: I measured 12 primaries and 11 secondaries one by one from raw flux. (2) Period: the eclipse period is 5.8911458 ± 0.0000038 d, +0.54σ from Gaia. (3) Pixels: in s44, s70 and s71 the difference-image deficit lies 0.4–0.9″ from the target, and every neighbour is excluded. (4) Pipelines: Q, S and C agree on depth to within 4%. (5) Geometry: the primary falls at Gaia conjunction A (+0.03 ± 0.08 d at the Gaia epoch; conjunction B is excluded at ~40σ), and the secondary phase of 0.50182 matches Gaia's prediction of 0.502 ± 0.021. Corrections to the screen: the primary is 35.3 ppt after dilution correction, not "probably larger", and it lasts 3.25 h. The secondary is 5.0–5.4 ppt.

TIC 92881948: CONFIRMED. (1) Repeating events: 24 primaries and 22 secondaries measured one by one. (2) Period: 8.8049754 ± 0.0000013 d, +0.43σ from Gaia. (3) Pixels: in all 7 sectors the deficit lies 0.3–1.6″ from the target, and the depth scales with the target's flux fraction in the aperture. (4) Pipelines: Q, S and C agree once dilution is accounted for. (5) Geometry: the primary falls at conjunction A (+0.06 ± 0.09 d; B is excluded at ~44σ). The secondary phase is 0.48334 ± 0.00008, so $e \cos \omega = -0.026$, which is 1.0σ from Gaia's prediction of 0.502 ± 0.018. Corrections to the screen: the primary depth is 16.4 ppt (Q) / 18.3 ppt (C) / 19.0 ppt dilution-corrected, not ~11 ppt. It lasts 1.69 h, not the 1.2 h that BLS gave. The secondary is 5.0–5.3 ppt at phase 0.4833.

Not tested for either star: the common-mode test (brief item 8, outside the five protocol checks) and next-eclipse UTC predictions (budget).

1. Archives and web (s10, s10b, s10c, s10d)

- **15855794:** no rows within 60″ in VSX, Prsa+2022 (J/ApJS/258/16), IJspeert+2021, Gaia DR3 vari_eclipsing_binary / I/358/veb, or SB9. It is not in Gaia vari_summary. SIMBAD: SB*, 1 bibcode (2023A&A...674A..34G). ExoFOP: TOIs 0, CTOIs 0.
- **92881948:** VSX has one entry at 1.35″: "Gaia DR3 2964506347485091072", type **ROT**, G 12.76–12.80. Nothing in Prsa, IJspeert, Gaia-EB or SB9. Gaia vari_summary: in_vari_eclipsing_binary = 0; the classifier's best class is SOLAR_LIKE (0.63). SIMBAD: SB*, 1 bibcode. ExoFOP: TOIs 0, CTOIs 0.
- **Neither star is a catalogued eclipsing binary.**
- **Positive controls:** each catalogue cone found a known row (Prsa, I/358/veb, IJspeert and SB9 via control rows; VSX via the 92881948 row itself).
- **Brief error:** the brief lists "I/357/varisum", but Gaia DR3 variability is I/358 (I/357 is the NSS part), so Gaia variability was queried through TAP.

2. Independence, systematics, pixels (s11_*.out, s12_compile.out, s14_physics.out)

- **Depths, 15855794 (all events):**
- Primary: Q 35.84 ± 0.27 (n=11), S 34.56 ± 0.32 (n=4), C 34.90 ± 0.23 (n=12), C dilution-corrected 35.34 ± 0.23.
- Secondary: Q 5.00 ± 0.30, S 4.91 ± 0.43, C 5.36 ± 0.24.
- The C primary χ^2 of 34/11 comes from E124 (27.6 ppt, at the end of the s70 orbit, a cadence QLP flags).
- **Depths, 92881948:**
- Primary: Q 16.43 ± 0.24, S 18.20 ± 0.24, C 18.33 ± 0.18, C r2.5 15.90 ± 0.23. Divided by the modelled target fraction, 3×3 gives 18.99 ± 0.19 and r2.5 gives 19.17 ± 0.28: they agree.
- QLP is shallower because its larger aperture takes in a G 13.67 star (60″) and a G 14.56 star (53″). TESS-SPOC CROWDSAP is 0.90–0.94.
- Secondary: Q 5.02 ± 0.21, S 5.33 ± 0.24, C 5.08 ± 0.18.
- **Difference images** (median of the eclipse core minus median of the flanks, per sector; pixel-integrated Gaussian fit; errors from 200 fake-event stacks):
- 15855794, primary: offsets from the target 0.4″ (s44, 1.1σ), 0.5″ (s70 without E124, 0.8σ; the run including E124 is distorted by that event) and 0.9″ (s71, 2.1σ). Every neighbour is >45″ from the centroid at >100σ. Secondary: 1.2–3.5″, ≤1.0σ.
- 92881948, primary: offsets 1.3″ (s5), 1.6″ (s6), 1.4″ (s32), 0.9″ (s98), 0.3″ (s107), 1.5″ (s108), 0.9″ (s109), all ≤2.8σ, with errors of 0.02–0.06 px. The nearest neighbour is at 29″. In every sector the pixel with the deepest fractional dip is the target pixel or one next to it.
- **Dilution arithmetic** (Gaussian PSF, and a PSF 1.5× broader):
- 15855794: in the 3×3 aperture the brightest neighbour (G 16.03, 45″) contributes ≤1.9% of the flux. It would need an intrinsic depth of ≥180% to make the primary: excluded. All other neighbours contribute ≤0.6%: excluded.
- 92881948: in 3×3, G 13.67 (60″) contributes 0.11–1.44% and would need ≥127%: excluded. G 14.56 contributes ≤3.95% and would need ≥46%; it is not excluded by arithmetic alone, but it is excluded by the centroid (53″ away) and because the depth grows in the smaller aperture. G 16.46 (29″) contributes ≤2.4% and would need ≥75%; it is excluded by the centroid.

3. TIC 15855794 (joint shape fit, s6: P 35.25 ± 0.31 ppt, T14 3.255 ± 0.025 h, T23 0.70 ± 0.03 h; S 4.89 ± 0.38 ppt, T14 3.14 ± 0.19 h)

There is no 2-min SPOC data. The s43 TESScut cut-out is empty (RAW_CNTS = −1, s11_15855794_s43_check.out).

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
S E−1	2500.18	lost (gap/flags)	–
P E0	2503.12275 ± 0.00077	34.25 / 34.40 / 34.20	S Q C
S E0	2506.0791 ± 0.0049	5.10 / 4.79 / 4.99	S Q C
P E1	2509.01328 ± 0.00066	34.46 / 35.24 / 36.59	S Q C
S E1	2511.97	lost (gap)	–
P E2	2514.90506 ± 0.00059	34.45 / 36.19 / 34.79	S Q C
S E2	2517.8619 ± 0.0062	4.54 / 5.00 / 4.47	S Q C
P E3	2520.79709 ± 0.00074	35.07 / 36.27 / 35.79	S Q C
S E3	2523.7436 ± 0.0048	4.99 / 5.99 / 5.26	S Q C

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
P E120	3210.06068 ± 0.00106	– / 37.03 / 35.22	Q C
S E120	3213.0275 ± 0.0078	– / 4.10 / 5.43	Q C
P E121	3215.95111 ± 0.00085	– / 36.48 / 35.55	Q C
S E121	3218.9089 ± 0.0063	– / 4.91 / 5.50	Q C
P E122	3221.84	lost (flags, all pipelines)	–
S E122	3224.8002 ± 0.0061	– / 5.38 / 5.60	Q C
P E123	3227.73365 ± 0.00081	– / 37.28 / 34.94	Q C
S E123	3230.6864 ± 0.0057	– / 5.35 / 5.74	Q C
P E124	3233.6255 ± 0.0019	– / flagged / 27.64 (end of orbit, suspect)	C
S E124	3236.5893 ± 0.0065	– / 4.79 / 5.79	Q C
P E125	3239.51638 ± 0.00085	– / 36.17 / 35.01	Q C
S E125	3242.4652 ± 0.0066	– / 4.37 / 5.01	Q C
P E126	3245.40761 ± 0.00090	– / 35.35 / 34.55	Q C
S E126	3248.36	lost (flags)	–
P E127	3251.29757 ± 0.00090	– / 34.85 / 33.77	Q C
S E127	3254.2567 ± 0.0054	– / 5.55 / 5.62	Q C
P E128	3257.18858 ± 0.00085	– / 35.70 / 35.34	Q C
P E264	4058.39	lost (gap; segment "s1751")	–
S E264	4061.3461 ± 0.0060	– / – / 4.92	C

- **Period** (s12): primaries only, P = 5.8911449 ± 0.0000038 (χ^2 4.8/10). Secondaries only, P = 5.8911779 ± 0.0000229. TESScut alone, P = 5.8911437 ± 0.0000042. All are +0.53 to +0.56σ from Gaia's 5.890538 ± 0.001133.
- **Odd/even primaries:** Δ = +0.6σ (S), +0.7σ (Q), +1.8σ (C). There is no hint of 2P.
- **Primary O−C** scatter is ≤1.2 min.

4. TIC 92881948 (s6: P 16.63 ± 0.45 ppt [Q/S fast cadence], T14 1.692 ± 0.030 h, T23 0.20 ± 0.05 h (V-shaped); S 5.00 ± 0.35 ppt, T14 1.76 ± 0.12 h)

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
S E−1	1440.6495 ± 0.0027	6.09 / 5.99 / 4.97	S Q C
P E0	1445.2014 ± 0.0011	16.40 / 16.19 / 18.08	S Q C
S E0	1449.4544 ± 0.0028	4.41 / 4.01 / 5.19	S Q C
P E1	1454.0066 ± 0.0010	17.78 / 16.75 / 18.38	S Q C
S E1	1458.2549 ± 0.0035	5.06 / 5.73 / 6.65	S Q C
P E2	1462.8129 ± 0.0010	18.96 / 16.96 / 18.83	S Q C
P E3	1471.6159 ± 0.0006	19.98 / flagged / 19.95	S C
S E3	1475.8763 ± 0.0021	6.41 / 4.72 / 5.95	S Q C
P E4	1480.4223 ± 0.0006	18.24 / 17.06 / 17.14	S Q C
S E4	1484.6802 ± 0.0035	4.51 / 3.93 / 3.63	S Q C
P E5	1489.2272 ± 0.0011	17.53 / 15.78 / 18.18	S Q C
P E83	2176.0155 ± 0.0007	17.85 / 17.50 / 18.44	S Q C
S E83	2180.2716 ± 0.0022	5.09 / 4.98 / 5.29	S Q C
P E84	2184.8210 ± 0.0007	17.48 / 14.82 (partial) / 18.06	S Q C
S E84	2189.0738 ± 0.0033	4.85 / 4.97 / 5.33	S Q C
P E85	2193.6244 ± 0.0008	18.44 / 17.15 (partial) / 17.92	S Q C

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
S E85	2197.8764 ± 0.0020	5.93 / 5.49 / 5.81	S Q C
P E289	3989.8400 ± 0.0014	– / flagged / 17.88	C
S E289	3994.0970 ± 0.0035	– / flagged / 4.38	C
P E290	3998.6438 ± 0.0007	– / 15.62 / 17.53	Q C
S E290	4002.90	lost (gap)	–
P E291	4007.4468 ± 0.0012	– / flagged / 19.97	C
S E291	4011.7102 ± 0.0022	– / 4.56 / 5.24	Q C
P E292	4016.2547 ± 0.0007	– / 16.51 / 18.36	Q C
S E292	4020.5174 ± 0.0073	– / flagged / 3.79	C
P E293	4025.0597 ± 0.0008	– / 16.72 / 18.94	Q C
S E293	4029.3183 ± 0.0022	– / 5.16 / 5.29	Q C
P E294	4033.8637 ± 0.0015	– / flagged / 21.07	C
S E294	4038.1176 ± 0.0031	– / flagged / 5.58	C
P E295	4042.6692 ± 0.0007	– / 15.56 / 17.12	Q C
P E320 / E321 / E322 / E323 (s107)	4262.7954 ± 0.0013 / 4271.5989 ± 0.0008 / 4280.4041 ± 0.0008 / 4289.2104 ± 0.0008	C: 19.30 / 18.47 / 18.99 / 17.11	C
S E320 / E321 / E322 (s107), E323 (s108)	4267.0482 ± 0.0031 / 4275.8536 ± 0.0029 / 4284.6636 ± 0.0036 / 4293.4656 ± 0.0032	C: 4.83 / 4.72 / 3.76 / 4.79	C
P E324 (s108)	4298.01	partial (no pre-ingress baseline)	–
S E324 / E325 (s108), E326 (s109)	4302.2707 ± 0.0029 / 4311.0736 ± 0.0024 / 4319.8790 ± 0.0039	C: 4.47 / 5.87 / 6.60	C
P E325 / E326 (s108)	4306.8209 ± 0.0011 / 4315.6232 ± 0.0008	C: 17.93 / 18.20	C
P E327 / E328 (s109)	4324.4292 ± 0.0009 / 4333.2316 ± 0.0021	C: 18.87 / 18.88	C
S E327 / E328 (s109)	4328.6871 ± 0.0024 / 4337.4867 ± 0.0031	C: 5.75 / 5.60	C

- **Sectors without other pipelines:** sectors 107–109 exist only in TESScut (no QLP or SPOC light curves). There is no 2-min SPOC data.
- **Period:** primaries only, P = 8.8049749 ± 0.0000013 (χ^2 26.8/22). Secondaries only, P = 8.8049820 ± 0.0000049. TESScut alone, P = 8.8049753 ± 0.0000014. All are +0.43σ from Gaia's 8.804360 ± 0.001441.
- **Odd/even primaries:** +1.6σ (S), +0.2σ (Q), +0.7σ (C).
- **Secondaries** are not at phase 0.5 and differ from the primaries in depth, so this is not an equal-pair P/2 system.

Geometry vs Gaia (s13_geometry.out; 20000 draws, NSS correlation matrix)

- **How the test works:** with ω and T_peri correlated at +0.9996, conjunction times are well defined. Without the correlations the prediction is uninformative (σ ≈ 2 d). The order of the first four corr_vec parameters is uncertain, so all 24 orders were run; they give the same answer.
- **15855794:**
- Observed primary minus conjunction A, at the Gaia epoch: +0.030 d (robust σ 0.080; over the 24 orders +0.23 to +0.46σ). Minus conjunction B: −2.93 d (−39σ).
- Carried forward with Gaia's P: +0.65σ.
- Secondary phase: predicted 0.5019 ± 0.0209, observed 0.50182 ± 0.00030; e cos ω = +0.0029 ± 0.0005, against Gaia's +0.003 ± 0.033.
- **92881948:**

- Primary minus conjunction A: +0.060 d (σ 0.086; +0.35 to +0.69σ). Minus conjunction B: −43σ.
- Forward with Gaia's P: +0.52σ.
- Secondary phase: predicted 0.5016 ± 0.0181, observed 0.48334, a 1.0σ difference (32% of draws lie as far from 0.5). e cos ω = −0.0262, against Gaia's +0.003 ± 0.028.
- **Which conjunction:** in both stars the primary is conjunction A, the companion passing in front of the RV star.
- **Physics (s14, uniform discs):**
- 15855794: f(m) = 0.0437 M⊙, so m2 ≥ 0.49 M⊙ (M dwarf, about 3630 K). The observed T14 needs b ≈ 1.07 (grazing). Predicted depths are 32.2 / 4.2 ppt; observed 35.3 / 5.4 ppt. The depth ratio implies a companion at 3760 K.
- 92881948: f(m) = 0.0144 M⊙, so m2 ≥ 0.22 M⊙. Grazing, b ≈ 1.22. Predicted 14.6 / 4.0 ppt; observed 19.0 / 5.3 ppt. The depth ratio of 0.277 matches the Planck ratio (0.275) for a 3180 K companion.
- Both are physically sane.

Ephemerides and next eclipses (s12_compile.out)

- **15855794:** T_ref = BTJD 2886.04720 ± 0.00024 (BJD_TDB 2459886.04720), P = 5.8911458 ± 0.0000038 d. Secondary at +2.95631 ± 0.00179 d.
- **92881948:** T_ref = BTJD 3047.70744 ± 0.00019 (BJD_TDB 2460047.70744), P = 8.8049754 ± 0.0000013 d. Secondary at +4.25578 ± 0.00070 d.
- **Next three eclipses in UTC:** not computed (budget cut). Use T_ref + nP. Those times are BJD_TDB, so no barycentric correction has been applied for an observer.

Caveats

- **Common mode:** not tested with other stars. Partial substitutes: the residual sky and the neighbour pixels show no dip at the target's fractional depth, and the depth scales with the target fraction.
- **TESScut times:** taken from the FFI time stamps; the barycentric correction is for the FFI, not the target (errors under a minute).
- **15855794 E124:** shallow (27.6 ppt), at the end of an orbit, and flagged by QLP. Its primary-stack difference image was redone without it (s11_15855794_s70_exclE124).
- **PSF model:** a Gaussian with a 1.5× variant, not the TESS PRF. Neighbour exclusions rely on the 3×3 arithmetic plus the centroids.

Files

`s1–s9` (predecessor run); `s10b_archives_redo`, `s10c_vizier_check`, `s10d_controls_gaiavar` (.py/.out); `s11_tesscut.py` with `s11_<tic>_s<sec>.{out,json}`, `_lc.npz`, `_diff.npz`, `s11b_redownload_s70`, `s11_15855794_s43_check.out`; `s12_compile.{py,out,json}`; `s13_geometry.{py,out,json}`; `s14_physics.{py,out,json}`; plus `common.py` and `fitlib.py`.

vet06 full report

Archived source: sanitized report text.

vet06: adversarial check of TIC 468827621 (UCAC4 796-015685) and TIC 416285595 (UCAC4 571-080398)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This is a reduced-depth run: the coordinator cut the budget mid-task. It covers the five protocol checks only; physics, next-eclipse lists, the common-mode test and the literature search were skipped.

Light curves were fetched from MAST's `Download/file` endpoint (`s2`), because `lightkurve.search_lightcurve` hung at MAST and archive.stsci.edu serves an incomplete TLS chain. Data are in `scratchpad/data/vet06/<tic>/`. Each TESScut cube was deleted after use.

Depths are in ppt. Times are BTJD (BJD_TDB − 2457000). A = the companion passes in front of Gaia's RV star ($\nu = 90° - \omega$); B = the companion is behind it.

TIC 468827621: CONFIRMED. All five checks pass: 1. **Individual events.** 10 primary and 9 secondary epochs are fitted individually from raw SAP (QUALITY==0) in QLP s19, 20, 59, 60, 73 and TESS-SPOC s19, 20. 2. **Period.** The eclipse period is 11.041067 ± 0.000001 d (primaries, O−C rms 0.5 min). It is −0.16σ from Gaia's 11.04138 ± 0.00196 d. The secondaries alone give 11.041068 ± 0.000005 d. 3. **Pixels.** The primary deficit is 1.42″ (s19) and 1.50″ (s59) from the target's WCS position. The only neighbour bright enough in principle is G 13.40 at 70.6″. With its PRF share of my aperture it would need a 4800–7800% intrinsic eclipse, and the fractional deficit at its pixel is only 1–4 ppt. Both other neighbours need more than 100% even in the worst case. 4. **Pipelines.** QLP, TESS-SPOC and my TESScut aperture all see the primary: 192, 184 and 192 ppt. 5. **Geometry.** The primary is at conjunction A (−0.14 ± 0.09 d); B is excluded at 31σ (20σ without Gaia's correlations). The secondary phase, **0.72649 ± 0.00003**, is +0.56σ from Gaia's predicted 0.718 ± 0.015 (from e = 0.355, ω = 350°). The eclipse at A, nearer periastron, is the longer one: 6.0 h against 5.05 h. That is the direction $e \sin \omega = -0.06$ requires.

The screen was right on P, on both depths (QLP) and on the secondary phase: 0.7265 measured against its by-eye 0.72. TESS-SPOC SAP depths are about 4% shallower than QLP (184/29.6 against 192/31.5 ppt).

TIC 416285595: CONFIRMED. All five checks pass, with weak points below: 1. **Individual events.** 21 primary and 19 secondary epochs are fitted individually: QLP s14, 40, 54, 80, 81 and TESS-SPOC s40. 2. **Period.** P = 4.615357 ± 0.000003 d (O−C rms 2.4 min), −0.37σ from Gaia's 4.615565 ± 0.000559 d. 3. **Pixels.** The primary deficit is 2.16″ (s40, camera 1) and 0.58″ (s80, camera 2) from the target. Every one of the 43 neighbours needs more than 100% with its PRF share; the minimum is 126%, for G 14.75 at 37.9″ in s40. The neighbours that pass only in the unphysical worst case (100% of their flux in my aperture) all lie 37″ or more away, at least 1.8 px from a deficit centre that sits within 0.1 px of the target. So the primary is on the target, and the screen's contamination concern (ratio 0.27) does not change the attribution of the primary. 4. **Pipelines.** QLP and TESS-SPOC see every event at the same times and durations. Depth agreement is the weakest part (see the weak points). 5. **Geometry.** The primary is at conjunction A (−0.04 ± 0.08 d); B is excluded at 29σ with correlations, but only 3.3σ without. The secondary phase, 0.50011 ± 0.00027, is +0.28σ from Gaia's 0.49 ± 0.03.

Weak points: - **Depths disagree between pipelines.** QLP is deeper than TESS-SPOC in s40: 61.7 against 54.7 ppt, about 12% and about 4σ apart. My TESScut aperture gives 70 (s40) and 89 ppt (s80), but its background (median of pixels ≥ 4 px away) is unreliable in this crowded field. I attribute the

spread to crowding and background, since none of these is a crowding-corrected depth. Strict agreement within errors fails. - **The secondary is not tested in pixels.** At 12.9 ppt, G 14.75 at 37.9″ could produce it arithmetically (it would need about 26%). Only the exact phase 0.5001 and the matching period tie it to the target. - **The geometry test is weak.** e ≈ 0.08, so phase 0.5 is the generic expectation, and the A/B decision rests on Gaia's ω–Tp correlation of 0.998.

The screen was right on P and on the secondary phase. Its 59 ppt primary depth matches QLP but not TESS-SPOC (54.7). Its 11 ppt secondary is low: QLP and TESS-SPOC both give 12.9 ppt. The primary is near-V: T14 3.6 h, trapezoid T23 about 0.7 h.

1. Archives and web

- **VizieR mirror cones (`s1_archives.out`, `s1b_archives.out`):** both stars return 0 rows within 60″ in VSX, Prša+2022, IJspeert+2021, Gaia DR3 veb / vclassre / vari (10″), SB9, ASAS-SN II/366 and J/AJ/162/170. Algol returns 1 VSX row, so the cones work.
- Neither star is a catalogued eclipsing binary: none is listed in VSX, Gaia-EB, Prša or IJspeert within 60″.
- 468827621 is in the Gaia DR3 short-timescale table I/358/vst (amplitude 0.044 mag) and is flagged VARIABLE in gaia_source.
- **SIMBAD:** both are SB*, with 2 bibcodes (468827621) and 1 bibcode (416285595), all Gaia DR3. The SIMBAD lookup by TIC ID fails; the 10″ coordinate cone finds both.
- **ExoFOP:** 0 TOIs and 0 CTOIs for both.
- **Gaia SB1 rows (`gaia_tbosb1.tsv`, with corvec):**
- 468827621: e 0.3555 ± 0.0239, ω 350.13 ± 3.87°, K1 42.03 km/s.
- 416285595: e 0.078 ± 0.044, ω 104 ± 36°, K1 46.4 km/s.

2. Independence, systematics, pixels

- **Pipeline depths (`s3_events.out`, per-event means):**
- 468827621 primary: QLP 191.9 (sd 3.6, N = 10); TESS-SPOC 184.5 (sd 5.0, N = 4); TESScut 191.9 (s19) and 191.7 (s59).
- 468827621 secondary: QLP 31.5 (sd 1.0); TESS-SPOC 29.6 (sd 1.2).
- 416285595 primary: QLP 58.6 (sd 3.3, N = 20); TESS-SPOC 54.7 (sd 1.3, N = 6); TESScut 69.8 (s40) and 89.4 (s80), background-limited.
- 416285595 secondary: QLP 12.9 (sd 2.5); TESS-SPOC 12.9 (sd 1.4).
- Neither star has 2-min SPOC data.
- **Odd/even (primaries):** 468827621 +0.38σ (QLP) and −0.09σ (SPOC); 416285595 −1.2σ (QLP) and +1.1σ (SPOC). The two eclipse kinds differ in depth (and, for 468827621, in duration), so there is no hidden P/2 or 2P.
- **Difference images (`s4_tesscut_<tic>_s<sec>.txt/.png`; mean over primaries of median in-eclipse minus local out-of-eclipse; Gaussian-fit centre):**

Star / sector (cam)	N ev	deficit − target WCS	frac. depth, target pixel	aperture share of target
468827621 s19 (cam 2)	2	1.42″ (deficit − OOT 1.17 ± 0.33″)	158 ppt	0.962
468827621 s59 (cam 2)	3	1.50″ (deficit − OOT 1.62 ± 0.35″)	117 ppt	0.957
416285595 s40 (cam 1)	4	2.16″	34 ppt	0.826
416285595 s80 (cam 2)	5	0.58″	48 ppt	0.854

- For 416285595 the out-of-eclipse Gaussian locked onto a brighter star at the cutout edge, so only the WCS offset is quoted.
- Both 468827621 sectors are in camera 2. That is the only camera available.

3. TIC 468827621 (P 11.041067 d)

`events_468827621.csv`: trapezoid on a quadratic baseline; errors are the covariance scaled to the rms, inflated by a 10-shift prayer bead.

Event	Mid BTJD	Depth QLP / SPOC (ppt)	T14 / T23 (h)	pipelines
P E0 s19	1819.85379 ± 0.00025	196.9 / 189.9	6.00 / 2.33	QLP+SPOC+TESScut
P E1 s19	1830.89477 ± 0.00025	196.6 / 188.9	5.99 / 2.34	QLP+SPOC+TESScut
P E3 s20	1852.97691 ± 0.00024	186.4 / 179.4	6.02 / 2.33	QLP+SPOC
P E4 s20	1864.01807 ± 0.00027	187.5 / 179.6	6.00 / 2.36	QLP+SPOC
P E99 s59	2912.91944 ± 0.00020	195.2 / –	6.03 / 2.31	QLP+TESScut
P E101 s59	2935.00116 ± 0.00019	194.2 / –	5.99 / 2.39	QLP+TESScut
P E102 s60	2946.04227 ± 0.00038	187.8 / –	6.03 / 2.33	QLP
P E103 s60	2957.08343 ± 0.00018	190.3 / –	6.02 / 2.31	QLP
P E133 s73	3288.31654 ± 0.00045	191.6 / –	5.96 / 2.35	QLP
P E135 s73	3310.39800 ± 0.00018	192.1 / –	5.99 / 2.36	QLP
S E−1 s19	1816.83466 ± 0.00143	– / 30.8	5.04 / 2.27	SPOC
S E1 s19	1838.91713 ± 0.00110	31.9 / 30.3	5.14 / 2.17	QLP+SPOC
S E2 s20	1849.95666 ± 0.00095	29.1 / 27.8	5.01 / 2.51	QLP+SPOC
S E3 s20	1860.99765 ± 0.00098	31.3 / 29.6	4.99 / 2.31	QLP+SPOC
S E99 s59	2920.94045 ± 0.00089	32.4 / –	5.12 / 2.21	QLP
S E100 s59	2931.98226 ± 0.00096	32.2 / –	5.04 / 2.47	QLP
S E101 s60	2943.02108 ± 0.00108	31.2 / –	4.87 / 2.49	QLP
S E133 s73	3296.33725 ± 0.00087	32.2 / –	5.07 / 2.29	QLP
S E134 s73	3307.37832 ± 0.00080	31.4 / –	5.00 / 2.48	QLP

- **Lost events:** P E100 (s59), P E134 (s73), S E0 (s19) and S E102 (s60) fall in gaps or flagged data.
- **Ephemerides:**
- Primaries: P = 11.041067 ± 0.000001 d, O−C within ±1.3 min.
- Secondaries: P = 11.041068 ± 0.000005 d, O−C rms 1.0 min.
- **SPOC secondary odd/even:** +2.5σ, from only 4 events. The s20 SPOC depths are uniformly low, so this is a sector systematic, not two eclipse kinds.

4. TIC 416285595 (P 4.615357 d)

`events_416285595.csv` has all 40 fitted events plus the lost ones. A representative subset:

Event	Mid BTJD	Depth QLP / SPOC (ppt)	T14 / T23 (h)	pipelines
P E0 s14	1687.52122 ± 0.00101	52.9 / –	3.46 / 0.91	QLP
P E153 s40	2393.67735 ± 0.00069	56.4 / 54.6	3.64 / 0.66	QLP+SPOC+TESScut
P E154 s40	2398.29120 ± 0.00056	60.4 / 53.1	3.51 / 0.90	QLP+SPOC
P E155 s40	2402.90615 ± 0.00051	64.2 / 56.8	3.74 / 0.60	QLP+SPOC+TESScut
P E156 s40	2407.52291 ± 0.00075	63.5 / 54.2	3.55 / 0.90	QLP+SPOC+TESScut
P E158 s40	2416.75340 ± 0.00049	64.0 / 55.8	3.70 / 0.70	QLP+SPOC+TESScut
P E236 s54	2776.75002 ± 0.00096	59.9 / –	3.41 / 0.89	QLP
P E392 s80	3496.74568 ± 0.00070	57.9 / –	3.58 / 0.82	QLP+TESScut
P E398 s81	3524.43758 ± 0.00072	55.4 / –	3.62 / 0.77	QLP

Event	Mid BTJD	Depth QLP / SPOC (ppt)	T14 / T23 (h)	pipelines
S E152 s40	2391.37357 ± 0.00232	– / 10.7	2.98 / 1.42	SPOC (QLP fit degenerate)
S E156 s40	2409.82774 ± 0.00208	9.9 / 14.0	3.15 / 0.30	QLP+SPOC
S E157 s40	2414.44236 ± 0.00167	13.6 / 14.7	3.50 / 0.20	QLP+SPOC
S E395 s81	3512.89836 ± 0.00325	11.8 / –	3.22 / 1.18	QLP
S E397 s81	3522.13865 ± 0.00319	12.3 / –	3.39 / 0.95	QLP

- **Primary ephemeris:** N = 21 epochs, O–C rms 2.4 min. The two s14 events (30-min cadence) are at −7 min.
- **Secondaries:** P = 4.615370 ± 0.000009 d, O–C rms 8.7 min. Some single-event depth errors are degenerate (T23 → 0); see the csv.

Geometry vs Gaia

`s0_gaia_predict.out`, `s5_geometry.out` and `s5b_unwrap.out`. The Monte Carlo uses 20000 draws with Gaia's corvec, ordered P, Vcm, K1, e, ω, Tp (column-major upper triangle, positive-definite for both stars). Uncorrelated values are in brackets. "Backward" means the TESS primary ephemeris taken back to the Gaia epoch. - **468827621:** - Backward: primary – A = −0.142 ± 0.087 d (−1.6σ) [−0.142 ± 0.170]; primary – B = +2.97 ± 0.10 d (+31σ) [+20σ]. - Forward (Gaia ephemeris taken to TESS): −0.18 ± 0.28 d from A. - The deep eclipse is conjunction A. - Secondary phase: observed 0.72649 against predicted 0.7182 ± 0.0149, **+0.56σ**. If the deep eclipse were B, the shallow one would be predicted at 0.282. - **416285595:** - Backward: primary – A = −0.043 ± 0.075 d [−0.05 ± 0.62]; primary – B = ±2.31 ± 0.08 d (29σ) [±0.70, 3.3σ]. - Secondary phase: observed 0.50011 against predicted 0.4915 ± 0.0304, +0.28σ. - Because e is small, the secondary-phase test is weak.

Ephemerides

Errors are formal (`s3_events.out`); TESS-pipeline times are already barycentric. - **TIC 468827621 primary:** T0 = BTJD 1819.85371 ± 0.00010 (BJD_TDB 2458819.85371), P = 11.041067 ± 0.000001 d, T14 6.0 h. Secondary at phase 0.72649 ± 0.00003, T14 5.05 h. - **TIC 416285595 primary:** T0 = BTJD 1687.52620 ± 0.00067 (BJD_TDB 2458687.52620), P = 4.615357 ± 0.000003 d, T14 3.6 h. Secondary at phase 0.50011 ± 0.00027.

Caveats

- **SAP depths are not crowding-corrected.** The QLP/SPOC offsets are 4% (468827621) and about 12% (416285595).
- **TESScut photometry** uses a crude background: the median of pixels ≥ 4 px away. In the crowded 416285595 field it gives depths of 70–89 ppt, so it is used for location, not for depth.
- **Secondaries were not pixel-tested** for either star. For 468827621, the G 16.17 star at 24.6″ could arithmetically produce the 31 ppt secondary in the worst case, and for 416285595 the G 14.75 star at 37.9″ could produce the 12.9 ppt one. For both stars only timing ties the secondary to the target.
- **Dilution uses a Gaussian PRF** (σ 0.75 px) with G-band fluxes, not the TESS PRF.
- **The 468827621 difference images** are both from camera 2; no other orientation was available in the sectors used.
- **Not done:** common-mode test, ZTF, physics sanity check, next-eclipse lists.

Files

`s0_gaia_predict.py/.out`, `s1_archives.py/.out`, `s1b_archives.py/.out`, `gaia_tbosb1.tsv`, `s2_download.py/.out`, `s3_events.py/.out`, `events_468827621.csv`, `events_416285595.csv`, `s3_ephem_<tic>.npy`, `s4_tesscut.py`, `s4_tesscut_<tic>_s<sec>.txt/.npz/.png` (468827621 s19, s59; 416285595 s40, s80), `s5_geometry.py/.out`, `s5b_unwrap.py/.out`, `REPORT.md`.

Summary - TIC 468827621: CONFIRMED; P_ecl=11.041067 d; depth1=184–192 ppt; depth2=29.6–31.5 ppt at phase 0.72649; pixel: on target (primary); pipelines: 3; Gaia geometry: pass - TIC 416285595: CONFIRMED; P_ecl=4.615357 d; depth1=55–59 ppt (QLP s40 62); depth2=12.9 ppt at phase 0.50011; pixel: on target (primary; secondary untested); pipelines: 3 (depths agree only to ~12%); Gaia geometry: pass (weak, e≈0)

vet11 full report

Archived source: sanitized report text.

vet11: Gaia-period folds of five sinusoidal c9a candidates

Independent check (Opus 5.5 subagent, 2026-10-30, for Fable 5.1). I resumed a predecessor that was killed by a container restart. Its downloads, s3 light-curve fits, s4 pixel maps and the first half of the catalogue cones were reused. I added: s2b (the rest of the catalogue cones and the Gaia DR3 variability tables), s5b (the Roche toy model, crash fixed), s6 (a refold at P_phot, which removes the phase-drift artefacts of the Gaia-P fold), s8 (a narrow-dip test at min I), s7 (TIC 62328710 frequencies and ZTF), s9/s9b (Gaia covariance and conjunctions) and s10 (ExoFOP). Data are QLP SAP only, QUALITY==0, 10-min bins, with a per-segment cubic baseline. No TESS-SPOC light curves exist for any of the five. Depths are in ppt, measured from the mean of the two maxima, with per-sector SEM errors. Phases are at P_phot with 0 = min I (the deeper minimum). "Conj A" = the primary at superior conjunction (companion in front). Conjunction errors come from 40000 draws with the full Gaia corr_vec. The earlier independent-draw MC ignored corr(omega,Tp) of about 0.99 and overstated the errors. Coverage was cut short by a budget limit: one pixel sector per star, no difference images of the min-I dip, and no TESS-SPOC or TESScut-aperture depth comparison.

TIC 372683900 (Gaia P 1.476502 +- 0.000081 d, e 0.30)

quantity	value
LS top peak / P_phot (chi2 scan)	0.73825 d (=P/2); 1.476507 +- 0.000001 d, ratio 1.000003 (+0.05 sigma)
min I / min II depth	31.07 +- 0.31 at 0 / 28.18 +- 0.19 at 0.5011 +- 0.0008; diff 2.88 +- 0.35 (8.3 sigma), 11/12 sectors
harmonics K=6	A1 2.35, A2 14.79, A3 0.67, A4 0.27 (+-0.05); min widths 0.271 / 0.244 (cos2phi = 0.250)
narrow dip at min I	0.63 ppt, empirical 1.7 sigma (none)
conjunctions	min I - conj B +0.087 +- 0.117; min II - conj A -0.042 +- 0.127; min I at conj A excluded at 3.6 sigma
minima separation	0.5007 +- 0.0018 vs Gaia-predicted 0.372 +- 0.039 (3.3 sigma): photometry says e cos w ~ 0, Gaia says -0.21
pixel (s57)	amplitude centroid 0.9" from target. A G=15.23 star at 1.8" is unresolved, but it would need 128 ppt intrinsic and would have to match Gaia P to 1e-4 d

ELLIPSOIDAL: a smooth double wave at the Gaia period with no eclipse narrowing. The unequal minima are deeper at conj B, which matches the Morris A1 sign reversal at low i. The s5b toy model at i ~ 20 deg gives A1 1.74, A2 14.87, deeper at B by 2.69 ppt. Caveat: with the TIC R1 that needs a dark M2 ~ 4 Msun, so R1 is probably overestimated. Gaia's e = 0.30 conflicts with the minima separation.

TIC 13106408 (Gaia P 2.637505 +- 0.000401 d)

quantity	value
LS top / P_phot	1.31884 d (=P/2); 2.637721 +- 0.000002, ratio 1.000082 (+0.54 sigma)
min I / min II depth	46.85 +- 0.33 at 0 / 40.63 +- 0.26 at 0.5009 +- 0.0010; diff 6.22 +- 0.39 (15.8 sigma), 7/7 sectors
harmonics K=6	A1 2.06, A2 21.77, A3 1.40, A4 0.72 (+-0.03); widths 0.252 / 0.272 (min II flattened)
narrow dip at min I	0.36 ppt, 2.3 sigma (not significant)
conjunctions	min I - conj A +0.079 +- 0.166; min I - conj B -0.427 +- 0.167 (2.6 sigma)
pixel (s58)	on target (1.9"). The best neighbour (G 14.4, 15") would need 91 ppt

ELLIPSOIDAL: a large-amplitude, near-Roche-filling double wave at the Gaia period with unequal minima and a flattened min II. The K=6 residuals are white and there are no eclipse features. The conjunction test only mildly prefers min I at conj A.

TIC 372683622 (Gaia P 3.809006 +- 0.000179 d)

quantity	value
LS top / P_phot	1.90468 d (=P/2); 3.809282 +- 0.000014, ratio 1.000073 (+1.5 sigma)
min I / min II depth	7.73 +- 0.16 at 0 / 6.45 +- 0.19 at 0.4943 +- 0.0061; diff 1.28 +- 0.30 (4.2 sigma) but variable, 0.3-3.2 ppt between sectors
harmonics K=6	A1 0.78, A2 3.23, A3 0.13, A4 0.12; per-sector A3/A4 ~0.5 and incoherent (spots)
narrow dip at min I	0.24 ppt, 2.1 sigma (none)
conjunctions	min I - conj B +0.075 +- 0.040; min II - conj A +0.044 +- 0.036; min I at conj A excluded at 12.5 sigma
pixel (s56)	on target (2.6")

ELLIPSOIDAL: a double wave locked to the Gaia period with minima at the two conjunctions. The deeper minimum is at conj B, as for low-i ellipsoidals; the toy model at i ~ 20 deg gives A2 3.2 and deeper at B by 1.7 ppt. On top of this there is variable spot modulation, and no eclipse.

TIC 319185391 (Gaia P 2.677901 +- 0.000187 d)

quantity	value
LS top / P_phot	1.33902 d (=P/2); 2.678053 +- 0.000009, ratio 1.000057 (+0.8 sigma)
min I / min II depth	8.46 +- 0.30 at 0 / 6.57 +- 0.14 at 0.5102 +- 0.0040; diff 1.89 +- 0.23 (8.3 sigma), 6/6 sectors
harmonics K=6	A1 0.70, A2 3.64, A3 0.23, A4 0.04
narrow dip at min I	box 1.15 +- 0.17 ppt, centre +0.007, core 0.012 phase (0.8 h), profile about 0.03 phase (about 1.9 h). Empirical S/N 5.3; 0/400 random phases as deep; per sector 0.46/0.84/2.01/0.85/0.96/1.48 (6/6 > 0, weighted 1.04 +- 0.17); even 1.26 +- 0.25, odd 1.04 +- 0.29; at min II -0.03 +- 0.21. The same test on the other three stars gives 1.7-2.3 sigma
conjunctions	min I - conj A +0.057 +- 0.084 (conj B excluded at 6.0 sigma); separation 0.511 +- 0.003 vs Gaia 0.529 +- 0.028
pixel (s85)	the ellipsoidal signal is on target (5.2"). The dip itself has no pixel test
other	Gaia DR3 also has an astrometric orbit for this source, P = 434.5 +- 8.2 d, e = 0.70 (likely a triple)

DETACHED-EB (grazing, fold-only): an ellipsoidal double wave plus a phase-locked 1 ppt, about 2 h dip at min I, which sits at Gaia conj A (companion in front). This is the eclipse that the flattened P/2 fold hid. Individual events are about 1 sigma, so they cannot be seen singly, and the dip has no difference image. That puts it at probable-level evidence.

TIC 62328710 (Gaia P 0.938088 +- 0.000042 d)

quantity	value
dominant signal	f1 = 6.390610 +- 0.000005 c/d (0.156480 d), 3.09 ppt; f1/f_orb = 5.9950, not an orbital harmonic (19 sigma in P). Second frequency f = 5.806788 c/d, 1.26 ppt. Both are stable in all 7 sectors
Gaia-P signal	after prewhitening, A2 = 0.058 +- 0.046 ppt (< 0.15 ppt, 2 sigma). Minima: none
pixel (s56)	amplitude centroid 45" from target, on Gaia DR3 2165710355333626240 (G 13.38, 44.3"; next source worse by dchi2 642). f = 5.807 c/d also peaks there
ZTF	neighbour: top peak 6.39060 c/d, 9.6 (g) / 8.9 (r) mmag, noise 99th percentile 2.3/4.6; 5.807 c/d 4.2/5.1 mmag. Target: 0.16/0.70 mmag at f1 (noise 1.5/1.7)
catalogue	the neighbour is Gaia VarFlag VARIABLE (vari_short_timescale). Target not in VSX/Gaia-EB within 60"

OTHER: the "P/3" wave is a multiperiodic delta Scuti-type pulsator, the neighbour 44" away, diluted into the QLP aperture. The target is flat at the Gaia period. With any R1 >= 1.8 Rsun, ellipsoidal A2 < 0.15 ppt would require i < 5 deg, so the Gaia 0.938 d SB1 orbit is suspect.

Summary

TIC	class	P_phot / P_Gaia	min1 (ppt, phase)	min2 (ppt, phase)	pixel	catalogues
372683900	ELLIPSOIDAL	1.000003	31.07 at conj B	28.18 at conj A	on target (1.8" nbr unresolved)	none; SIMBAD SB*
13106408	ELLIPSOIDAL	1.000082	46.85	40.63	on target	VSX only an SR at 29"; SB*
372683622	ELLIPSOIDAL	1.000073	7.73 at conj B	6.45 at conj A	on target	none; SB*
319185391	DETACHED-EB	1.000057	8.46 + 1.0 ppt dip at conj A	6.57 at conj B	on target (dip untested)	none; SB*; Gaia astrometric 434 d
62328710	OTHER	f1 = 5.995 f_orb	none (A2 < 0.15)	none	neighbour 44"	nbr Gaia VARIABLE

TIC 372683900: ELLIPSOIDAL; P_phot=1.476507 d (1.000003); min1=31.07+-0.31 ppt at phase 0 (conj B +0.09+-0.12); min2=28.18+-0.19 at phase 0.5011 (conj A -0.04+-0.13); pixel: on target; catalogue: not in VSX/Gaia-EB/Prsa/IJspeert/SB9, SIMBAD SB* TIC 13106408: ELLIPSOIDAL; P_phot=2.637721 d (1.000082); min1=46.85+-0.33 ppt at phase 0 (conj A +0.08+-0.17); min2=40.63+-0.26 at phase 0.5009; pixel: on target; catalogue: none (VSX SR at 29" unrelated), SIMBAD SB* TIC 372683622: ELLIPSOIDAL; P_phot=3.809282 d (1.000073); min1=7.73+-0.16 ppt at phase 0 (conj B +0.075+-0.040); min2=6.45+-0.19 at phase 0.4943 (conj A +0.044+-0.036); pixel: on target; catalogue: none, SIMBAD SB* TIC 319185391: DETACHED-EB (grazing, fold-only); P_phot=2.678053 d (1.000057); min1=8.46+-0.30 ppt at phase 0 (conj A +0.06+-0.08) with a 1.04+-0.17 ppt, about 2 h dip; min2=6.57+-0.14 at phase 0.5102; pixel: on target (ellipsoidal signal; dip untested); catalogue: none, SIMBAD SB*, Gaia astrometric 434 d orbit TIC 62328710: OTHER (delta Sct neighbour); P_phot=0.156480 d (1/5.995 of Gaia P); min1=none (Gaia-P A2<0.15 ppt); min2=none; pixel: neighbour Gaia DR3 2165710355333626240 at 44" (ZTF confirms 6.3906 c/d there, not on target); catalogue: neighbour Gaia VARIABLE

Caveats

Only QLP is available, so there is no second pipeline. The Gaia-P fold in s3 has phase drift: its residual "dips" flip sign between sectors, so use the s6/s8 numbers, which are at P_phot. The s5b Roche model is a toy (blackbody at 800 nm, no reflection) and uses TIC radii, which look overestimated (13106408 and 62328710 overfill their Roche lobes). ExoFOP: no TOI or CTOI for any of the five (s10). The ms_oscillator TAP query failed.

Files

s1_download.py/.log, s2_catalogs.py/.txt (+s2_err.log), s2b_catalogs.py/.txt, s3_lc.py + s3_.txt/.json + fold_.png, s4_pixels.py + s4_*s.txt/.npz + pixs.png, s5b_roche.py/.txt (s5_roche.* crashed), s6_pphot.py* + *s6*.txt/.json, s7_62328710.py/.txt/.json, s8_dip319.py + s8__b10.txt/.json + dip_319185391.png, s9_gaia_corr.py + s9_gaia_nss.csv, s9b_conj_corr.py/.txt, s10_exofop.sh/.txt, common.py.

Returned batch summary

Archived source: sanitized report text.

Chunk-9 (c9a) candidates: screening and vets, 2026-10-30

Cloud session (Fable 5.1 lead, Opus 5.5 helpers) on the 24-star bundle Astra packaged from Fable 5's `sb1-scale` chunk 9 (montage sheets c9a_00 and c9a_01; roster order = best z descending). Everything here is on the branch `claude/zealous-pasteur-8mmme2`.

What was done

1. **Screen** (`SCREEN_RESULTS.csv`, `SCREEN_REPORT.md`): first-pass visual triage of the two montage sheets, three Opus helpers at eight panels each plus a second look at the borderline calls. 19 eclipse-like, 5 non-eclipse-like, 0 ambiguous, 0 unreadable.
2. **Shared inputs** (`targets.csv`, `neighbours_90as.csv`, `prep.py`): TIC rows, Gaia DR3 NSS SB1 orbits (VizieR I/357 via the idia mirror; the CDS and ESA hosts were unreachable), Gaia neighbours within 90", QLP / TESS-SPOC / 2-min / TESScut sector lists.
3. **Vets** (`vet01`–`vet06`, `vet11`; protocol in `VET_BRIEF.md`): independent adversarial checks to the paper's Section 2.5 tier definition (events from raw SAP, period from the eclipses, difference images and dilution, independent pipeline, Gaia geometry). `VERDICTS.md` is the one-line-per-star table; each `vetNN/REPORT.md` holds the evidence, scripts and outputs.

Results

	n	TICs
vetted CONFIRMED	1 2	195865529*, 468181994, 470235665, 379251709, 70665829 (= paper's A4, blind control, recovered), 331293467, 283532878, 407877153, 15855794, 92881948, 468827621, 416285595 (with caveats: pipeline depths differ ~12%)
of which already a catalogued eclipsing binary	1	195865529 (ASAS-SN EA, same period; not in VSX)

	n	TICs
sinusoids refolded at Gaia P: ellipsoidal	3	372683900, 13106408, 372683622
sinusoids: grazing detached EB, fold-level only (PLAUSIBLE at best)	1	319185391 (also a Gaia 434 d astrometric orbit: likely triple)
sinusoids: signal from a neighbour (δ Scuti at 44"); target flat at the Gaia period	1	62328710 (Gaia's SB1 orbit for it is suspect)
eclipse-like, **not yet vetted** (credits ran out)	7	300903537, 23112764, 373062090, 465073693, 316405449, 465263741, 441741446

Side findings worth a catalogue note: a catalogued EB (Gaia DR3 2202301140233408128, P 1.7975 d) sits 29.7" from 283532878 and may leak ≤3 ppt dips into its light curve; 372683900's equal, half-period-spaced minima conflict with Gaia's e = 0.30 at 3.3σ; several secondaries are off phase 0.5 and match Gaia's e cos ω (470235665, 283532878, 407877153, 92881948, 468827621).

Cost note, for the next batch

Each full vet cost ~240k Opus tokens, about two stars per agent. Most of that was execution depth the tier does not need (every individual event fitted, own TESScut photometry on top of two pipelines, dilution for every neighbour within 90", Monte Carlo geometry, physics checks, ephemerides, literature search) plus each agent rebuilding its tooling from scratch, and a container restart that cost wave 1 roughly 1.7×. For the remaining seven: reuse the scripts in `vet04/` or `vet05/` (`common.py`, `fitlib.py`), a handful of events per star, one difference image per sector in two sectors, only the neighbours that matter, quoted pipeline depths, simple error propagation, two-page report. `VET_BRIEF.md` should be trimmed accordingly before reuse.

Data-access notes (this machine, 2026-10-30)

MAST light curves and TESScut work (pass coordinates to TESScut, not names; `lightkurve.search_lightcurve` sometimes hangs, in which case fetch `mast:HLSP/...` URIs directly). Gaia: ESA TAP times out, Heidelberg TAP is slow on joins; VizieR mirror `vizier.idia.ac.za` works but is intermittent, so run a known-positive (Algol) check before trusting an empty cone. The Gaia DR3 variability tables are I/358, not I/357/varisum as the brief says. AAVSO VSX API is behind Cloudflare; use VizieR B/vsx.

Archived source: sanitized report text.

c9a vets: running verdict table

Tiers per the paper's Conventions (Confirmed / Probable / Plausible / Refuted / Open). One line per star, pasted from each vet's summary; see vetNN/REPORT.md for the evidence.

vet	TIC	tier	P_ecl (d)	depth1	depth2 @ phase	pixel	pipelines	Gaia geometry	catalogue status	notes
vet01	1958652529	CONFIRMED	3.83104 5656(9)	183 ppt QLP; 192±7 TESScut undiluted	48 ppt @ 0.49999	on target, all 8 sectors	3 (QLP, own TESScut, TASOC)	pass; primary = conj A	**already catalogued**: ASAS-SN EA P 3.8310 d at 0.16"; ATLAS DBH at P/2; not in VSX	recovery, not new

vet	TIC	tier	P_ecl (d)	depth1	depth2 @ phase	pixel	pipelines	Gaia geometry	catalogue status	notes
vet01	468181994	CONFIRMED (weaker)	3.99691 90(6)	22.5 ppt QLP / 23.2 TESScut	1.24–1.41 ppt @ 0.4956±0.0012	on target; 2.85" G16.75 neighbour rejected ~16σ	3 (QLP, own TESScut, ELEANOR-lite)	pass; secondary within 1σ; A vs B undecided	not in VSX/ASAS-SN/Gaia-EB (per report)	P_ecl −2.3σ from Gaia; RUWE 12.6 suggests sub-arcsec companion; secondary 25 min early (e cos ω ≈ −0.007)
vet02	470235665	CONFIRMED	10.3261 83(2)	122–127 ppt	81–84 ppt @ 0.52275	on target, s19 & s79 (2 cameras)	3 (QLP, TESS-SPOC, own TESScut)	pass; primary = conj A, B excluded 19σ; secondary +0.7σ	not in VSX/Gaia-EB/Prša/IJspeert/ASAS-SN within 60"; SIMBAD SB*	UCAC4 859-013225; subgiant + K dwarf; possible secondary-phase drift (3.5σ, QLP only, not claimed)
vet02	379251709	CONFIRMED	12.3239 12(7)	34–38 ppt (screen's 31 was 15% low)	2.5 ppt @ 0.4960±0.0010 (fold-level, S/N 8.5 QLP)	on target (primary), s18 & s52	3 (QLP, TESS-SPOC, own TESScut)	pass; primary = conj A, B excluded; secondary +0.8σ	not in VSX/Gaia-EB/Prša/IJspeert/ASAS-SN within 60"; SIMBAD SB*	UCAC4 788-006141; M-dwarf companion; eclipse shape prefers R1≈1.3 R⊙ vs TIC 1.88
vet03	706658 29	CONFIRMED	3.46576 54(7)	283–330 ppt (pipeline-dependent; SPOC ~300)	64–72 ppt @ 0.46048	on target, s46/s48/s72 (3 cameras); only neighbour needs >1500%	4 (QLP, TESS-SPOC, own TESScut, ZTF g/r)	pass; deep = conj A (RV star eclipsed) +0.24σ; reverse 3.6–5.6σ off	not in VSX/Prša/IJspeert within 60"	**BLIND CONTROL = paper's A4**: recovered independently (paper: P 3.4657540±0.0000093, secondary at 0.461, deep eclipse on the K dwarf)
vet03	331293467	CONFIRMED	5.25681 99(20)	61–73 ppt QLP raw (≈80–116 undiluted)	11.4 ppt QLP (≈13–21 undiluted) @ 0.50013	on target, 4 sectors / 3 cameras; G13.78 at 30.5" passes dilution arithmetic (needs 25–84%) but excluded by position (fit 10–19× worse there)	2 (QLP, own TESScut; no SPOC)	pass; deep = conj A +0.49σ, other excluded ≥6.7σ (relies on Gaia ω–Tp correlation)	not in VSX/Prša/IJspeert within 60"; Gaia-EB/varisum/SB9 cones from predecessor run only (mirror timed out)	A/F star + K dwarf; heavy blend (contamination 0.70) so undiluted depths uncertain; Dec −60, no ZTF
vet04	283532878	CONFIRMED	17.4460 19(19) (+1.8σ vs Gaia)	75 ppt QLP (71–77 undiluted)	10 ppt @ 0.6781 (Gaia predicts 0.712±0.024, −1.4σ)	on target, 8 sectors; 3 neighbours at 28–37" pass dilution arithmetic (need 30–47%) but rejected by noise-weighted difference images ($\Delta\chi^2 \geq 41.7$ each sector)	2 (QLP, own TESScut; no SPOC)	pass; primary = conj A, B excluded >11σ	not in VSX/Gaia-EB/Prša/IJspeert on target; **a catalogued EB (Gaia DR3 2202301140233408128, P 1.7975 d, G 17.66) sits 29.7" away**, excluded as source but may add ≤3 ppt dips	eccentric secondary confirmed in all 8 sectors over 4 seasons; secondary phase drifts ~45 min 2019–2024 (~4σ formal, likely underestimated errors)
vet04	407877153	CONFIRMED	17.0892 94(8) (−0.14σ)	29 ppt (SPOC, TESScut; QLP 24–27 diluted)	3.0 ppt flat-bottomed @ 0.5469 (Gaia 0.559±0.044, −0.3σ)	on target; secondary pixel S/N weak ($\Delta\chi^2 \geq 7$ per sector, ≥166 summed)	3 (QLP, TESS-SPOC, own TESScut)	pass; primary = conj A	not in VSX/Gaia-EB/Prša/IJspeert within 60"	screen's depths low (27→29, 2→3 ppt)

vet	TIC	tier	P_ecl (d)	depth1	depth2 @ phase	pixel	pipelines	Gaia geometry	catalogue status	notes
vet05	158557994	CONFIRMED	5.891145 8(38) (+0.54σ)	35.3 ppt dilution-corrected (QLP 35.8, SPOC 34.6, TESScut 34.9); T14 3.25 h	5.0–5.4 ppt @ 0.50182±0.00030 (Gaia 0.502±0.021)	on target, s44/s70/s71 (0.4–0.9"); all neighbours excluded	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A +0.03±0.08 d, B excluded ~40σ	not in VSX/Gaia-EB/Prša/IJspeert within 60"	screen's 35 ppt was right
vet05	928819488	CONFIRMED	8.8049754(13) (+0.43σ)	16.4 QLP / 18.2 SPOC / 19.0 ppt dilution-corrected; T14 1.69 h V-shaped	5.0–5.3 ppt @ 0.48334±0.00008 (e cos ω −0.026; Gaia 0.502±0.018, 1.0σ)	on target, all 7 sectors (0.3–1.6"); G14.56 at 53" excluded by centroid and aperture dependence	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A +0.06±0.09 d, B excluded ~43σ	VSX lists it as ROT (Gaia DR3 variable) only; not in any EB catalogue	screen's 11 ppt was low (16–19 ppt); QLP ~10% shallower than SPOC (dilution)
vet06	468827621	CONFIRMED	11.041067(1) (−0.16σ)	184–192 ppt (QLP 192, SPOC 184, TESScut 192)	29.6–31.5 ppt @ 0.72649±0.00003 (Gaia 0.718±0.015, +0.56σ)	on target (primary), s19 & s59 but both camera 2; secondary not pixel-tested (G16.17 at 24.6" could supply it in the worst case)	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A −0.14±0.09 d, B excluded 31σ; longer eclipse nearer periastron as e sin ω requires	not in VSX/Gaia-EB/Prša/IJspeert/SB9 within 60"; SB*	reduced-depth run; screen right on everything
vet06	416285595	CONFIRMED, with caveats	4.614 5357(3) (−0.37σ)	55–59 ppt; QLP s40 61.7 vs SPOC 54.7 (~12%, ~4σ apart; TESScut 70–89 unreliable in crowded field)	12.9 ppt @ 0.50011±0.00027	on target (primary), s40 cam 1 & s80 cam 2; all 43 neighbours need >100% (min 126%); secondary not pixel-tested (G14.75 at 37.9" could supply ~28%)	3, but depths agree only to ~12%	pass but weak (e≈0.08 so phase 0.5 expected anyway; A/B rests on Gaia ω–Tp correlation)	not in VSX/Gaia-EB/Prša/IJspeert/SB9 within 60"; SB*	reduced-depth run; if depth agreement within errors is required this drops to PROBABLE; two worst-case secondary figures are the agent's hand estimates, not in an output file

vet11: the five sinusoidal (P2) candidates, refolded at the Gaia period (QLP only; no second pipeline exists)

TIC	class	P_phot (d) / Gaia ratio	min1	min2	pixel	catalogue	notes
372683900	ELLIPSOIDAL	1.476507 (1.000003)	31.1 ppt @ 0 (conj B)	28.2 ppt @ 0.5011 (conj A)	on target (G15.2 at 1.8" unresolved)	none; SIMBAD SB*	minima 0.5007 apart vs Gaia e=0.30 predicting 0.372±0.039 (3.3σ conflict); TIC radius probably overestimated
131064 08	ELLIPSOIDAL	2.637721 (1.000082)	46.9 ppt @ 0 (conj A)	40.6 ppt @ 0.5009, flattened	on target	none (unrelated 364 d SR at 29"); SB*	
372683622	ELLIPSOIDAL	3.809282 (1.000073)	7.7 ppt @ 0 (conj B; A excluded 12.5σ)	6.5 ppt @ 0.4943	on target	none; SB*	spot modulation varying sector to sector

TIC	class	P_phot (d) / Gaia ratio	min1	min2	pixel	catalogue	notes
319185391	DETACHED-EB (grazing; fold-level only)	2.678053 (1.000057)	8.5 ppt @ 0 (conj A; B excluded 6σ) + 1.04±0.17 ppt ~2 h dip	6.6 ppt @ 0.5102	on target (ellipsoidal); dip itself untested	none; SB*; Gaia also has a 434 d astrometric orbit (likely triple)	dip in all 6 sectors, odd and even; single events ~1σ; same test gives 1.7–2.3σ on the other three → PLAUSIBLE at best
62328710	OTHER (δ Scuti neighbour)	0.156480 (1/5.995 of Gaia P, not a harmonic)	none (<0.15 ppt at Gaia P)	none	neighbour Gaia DR3 2165710355333626240 (G13.38, 44"); ZTF: 6.3906 c/d at ~9 mmag there, flat on target	neighbour flagged VARIABLE by Gaia	target flat at 0.938 d: Gaia's SB1 orbit for it is suspect

Fainter SB1 period folds: current adjudications

The included systems in this population comprise 28 confirmed and seven probable systems. Known recoveries, controls, plausible signals and non-eclipse interpretations remain visible below without adding to the sample.

C01. TIC 195865529 - CONFIRMED

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Known ASAS-SN EA at 0.16 arcsec (P 3.8310 d); ATLAS DBH P/2. Recovery excluded from new count.

Public full report

C02. TIC 468181994 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Period -2.31 sigma from Gaia; RUWE 12.6/unresolved companion risk; independent pipelines agree about 2 sigma. Literature cones limited to enumerated catalogues, not proof of novelty.

Public full report

C03. TIC 470235665 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

QLP-only secondary phase drift unconfirmed; 4-5% pipeline scale offset. Enumerated catalogue cones and SIMBAD SB* do not establish exhaustive novelty.

Public full report

C04. TIC 379251709 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Secondary fold-level, not localized; primary securely on target. QLP/SPOC secondary phases differ 2.6 sigma. Radius tension retained.

Public full report

C05. TIC 70665829 - CONFIRMED

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Already objects.csv A4; blind control independently recovered. Archive Gaia-VEB/varisum/SB9 controls failed on later mirror timeout, predecessor empty cones only. No addition.

Public full report

C06. TIC 331293467 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Heavy blend; QLP plus independent own TESScut satisfies available pipeline check; depths model-dependent 80-116 ppt, ratios/duration agree. Gaia geometry depends on covariance. Gaia-VEB/varisum/SB9 cones lack successful positive controls; no exhaustive novelty claim.

Public full report

C07. TIC 283532878 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Use successful s12d recheck, not false-empty first archive pass. Known EB 29.7 arcsec away at P 1.7975 d excluded as source; up to 3 ppt contamination possible. Phase drift tentative.

Public full report

C08. TIC 407877153 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Secondary pixel signal weak individually but summed delta chi2 >=166; QLP depth diluted. Use successful s12d catalogue rerun; all-catalogue query unreliable.

Public full report

C09. TIC 15855794 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Exclude flagged E124 from primary pixel stack as report does. Catalogue positive controls succeeded; catalogue absence only within searched scope.

Public full report

C10. TIC 92881948 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

VSX ROT/Gaia SOLAR_LIKE is prior variable classification, not prior EB; catalogue positive controls succeeded. QLP dilution explains shallower depth.

Public full report

C11. TIC 468827621 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Primary localized in two sectors both camera 2; secondary untested in pixels and neighbour could supply in worst case. Literature search skipped, though catalogue cones performed; known Gaia short-timescale variable is not prior EB.

Public full report

C12. TIC 416285595 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Returned CONFIRMED with caveats: QLP/SPOC s40 61.7 vs 54.7 ppt (~12%, ~4 sigma); TESScut depth unreliable. Secondary not pixel-tested; geometry weak/covariance-dependent. Catalogue cones performed but literature search skipped. Report itself says strict depth agreement would imply PROBABLE; no new stricter tolerance imposed here.

Public full report

C13. TIC 372683900 - ELLIPSOIDAL

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Smooth double wave, no narrow eclipse; unresolved 1.8 arcsec neighbour; Gaia eccentricity conflicts at 3.3 sigma.

Public full report

C14. TIC 13106408 - ELLIPSOIDAL

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Smooth double wave; nearby 364-day SR unrelated.

Public full report

C15. TIC 372683622 - ELLIPSOIDAL

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Double wave plus variable spots; no eclipse.

Public full report

C16. TIC 319185391 - PLAUSIBLE

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Fold-only ~1 ppt dip, individual events ~1 sigma, dip not pixel-tested, no second pipeline. Report prose says probable-level but batch summary says PLAUSIBLE at best; adopt batch summary outside count. Gaia 434-day astrometric orbit suggests triple.

Public full report

C17. TIC 62328710 - NEIGHBOUR_SIGNAL

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Delta Scuti-like signal localized to Gaia DR3 2165710355333626240 at 44 arcsec and independently seen there by ZTF; target flat at Gaia period; suspect SB1 orbit is interpretation.

Public full report

C18. TIC 300903537 - CONFIRMED

Unique count addition: false. Catalogue status: known catalogue EB recovery; excluded from additions.

See target report for deciding limitation and next check.

Public full report

C19. TIC 23112764 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C20. TIC 373062090 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C21. TIC 465073693 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C22. TIC 316405449 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C23. TIC 465263741 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C24. TIC 441741446 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C25. TIC 280075497 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C26. TIC 276514049 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C27. TIC 10790605 - REFUTED

Unique count addition: false. Catalogue status: No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim..

See target report for deciding limitation and next check.

Public full report

C28. TIC 376082142 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.

Public full report

C29. TIC 285942031 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Reassess Gaia orbital uncertainties/correlations or obtain an updated radial-velocity orbit to resolve the eclipse-versus-Gaia timing tension.

Public full report

C30. TIC 104736355 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C31. TIC 281393563 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Obtain higher resolution time-series imaging or a validated multi-source PRF difference fit to identify the eclipsing host.

Public full report

C32. TIC 450055243 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C33. TIC 126363604 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Validate the repeated difference-image astrometry with a multi-source PRF or higher resolution event photometry; refine Gaia conjunction uncertainty if needed.

Public full report

C34. TIC 251357000 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.

Public full report

C35. TIC 267222866 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C36. TIC 66315917 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C37. TIC 93783392 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C38. TIC 470947699 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C39. TIC 135230723 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C40. TIC 36083554 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C41. TIC 447566809 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C42. TIC 12320469 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Public full report

C43. TIC 86908196 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Public full report

C44. TIC 201317466 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Refit the RV orbit or evaluate a nonlinear orbital posterior against the observed eclipse timing to resolve Gaia association.

Public full report

C45. TIC 360516818 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C46. TIC 411647453 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher resolution event photometry can establish the contaminant host and measure its orbital period precisely.

Public full report

C47. TIC 450217907 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Public full report

C48. TIC 60443793 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Review TIC 60443695 separately if desired.

Public full report

C49. TIC 71866523 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Require a clean repeated negative event before EB classification.

Public full report

C50. TIC 443106429 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Resolve target versusTIC 443106423 with higher spatial resolution eclipse photometry.

Public full report

C51. TIC 311206604 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No further eclipse follow-up required unless new discrete events appear.

Public full report

C52. TIC 428769301 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No further eclipse follow-up required unless new discrete events appear.

Public full report

C53. TIC 436947190 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No further eclipse follow-up required unless new discrete events appear.

Public full report

C54. TIC 92452368 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Verify a clean recurrent event in independently reduced data.

Public full report

C55. TIC 94300366 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain detrending resistant multiseason photometry to establish stable discrete repeats.

Public full report

C56. TIC 4646090 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Characterize dilution and grazing geometry if needed.

Public full report

C57. TIC 402940895 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Re-extract clean sectors with robust background/pointing treatment before asserting eclipses.

Public full report

C58. TIC 183994341 - CONFIRMED

Unique count addition: false. Catalogue status: known catalogue EB recovery; excluded from additions.

Resolve ASAS-SN period alias and obtain RV/orbital posterior or a joint timing/geometry analysis to settle the Gaia association.

Public full report

C59. TIC 166781582 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C60. TIC 230270717 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C61. TIC 462845564 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C62. TIC 431762229 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher resolution eclipse photometry can verify the contaminating host.

Public full report

C63. TIC 350661747 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C64. TIC 424565669 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C65. TIC 222498305 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher resolution eclipse photometry can verify the contaminating host.

Public full report

C66. TIC 90542579 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

If revisiting, use clean cadences away from downlink boundaries with a validated background and aperture.

Public full report

C67. TIC 9396327 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Refine Gaia conjunction predictions with orbital posterior and measure a second clean sector or obtain new RVs to test the orbit association.

Public full report

C68. TIC 275022619 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.

Public full report

C69. TIC 297541839 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Public full report

C70. TIC 396248369 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Public full report

C71. TIC 277479128 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Public full report

C72. TIC 257575960 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.

Public full report

Fainter SB1 period folds: detailed physical evidence

Sol/Terra measurements under Astra supervision; original Fable discovery and screening. Reports preserve each deciding limitation and next test. Current tier reconciliation governs headline counts.

C24. TIC 441741446 full report

Archived source: sanitized report text.

TIC 441741446: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 4 accepted representative events in sectors[14, 53, 86], depths9.93-13.18ppt and durations1.96-2.63h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence18.826637718d, epoch1686.723580BTJD, timing residual RMS2.51min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[14, 53] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector14: neighbour/target RSS ratios no capable TIC neighbour within30arcsec; sector53: neighbour/target RSS ratios no capable TIC neighbour within30arcsec. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector14 3x3 local-baseline depth9.78ppt; sector53 3x3 local-baseline depth9.10ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Additionally, TESS-SPOC sector14 raw SAP reproduces depths 9.99, 9.70ppt; archive product URI, TIC/sector checks and hash are in independent-results.json.
5. Gaia geometry: exact DR3 source1656477802697153920, P=18.831656152+/-0.00473691d, e=0.3822, omega=225.46deg. Measured-minus-Gaia period is-1.06quoted Gaia sigma. Closest conjunctionB; the photometric ephemeris projected near BTJD389 gives offset+0.0428d with linear full-covariance Gaia error0.1468d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration2.53h versus approximate central-crossing scale4.98h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C20. TIC 373062090 full report

Archived source: sanitized report text.

TIC 373062090: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 4 accepted representative events in sectors[18, 58, 86], depths28.40-30.56ppt and durations7.20-7.76h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence7.705911843d, epoch1799.033971BTJD, timing residual RMS0.20min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[18, 58] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector18: neighbour/target RSS ratios no capable TIC neighbour within30arcsec; sector58: neighbour/target RSS ratios no capable TIC neighbour within30arcsec. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector18 3x3 local-baseline depth29.44ppt; sector58 3x3 local-baseline depth28.95ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source538032619052851712, P=7.710879157+/-0.00305296d, e=0.2464, omega=139.81deg. Measured-minus-Gaia period is-1.63quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset-0.1151d with linear full-covariance Gaia error0.1836d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration7.24h versus approximate central-crossing scale10.64h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C21. TIC 465073693 full report

Archived source: sanitized report text.

TIC 465073693: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 6 accepted representative events in sectors[10, 63, 99], depths227.58-260.10ppt and durations4.84-4.87h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence6.272264875d, epoch1577.915266BTJD, timing residual RMS0.46min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[10, 63] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector10: neighbour/target RSS ratios no capable TIC neighbour within30arcsec; sector63: neighbour/target RSS ratios no capable TIC neighbour within30arcsec. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector10 3x3 local-baseline depth187.85ppt; sector63 3x3 local-baseline depth183.00ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source5239650442330452096, P=6.269417959+/-0.0014404d, e=0.0595, omega=129.56deg. Measured-minus-Gaia period is+1.98quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset-0.0686d with linear full-covariance Gaia error0.1034d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration4.86h versus approximate central-crossing scale6.28h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed. Unequal alternating depths about12-14% and5.6-6.5% in sectors10,63,99 reject equivalent full morphology at3.136d and support the6.27226d primary recurrence (alias-check.json).

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C22. TIC 316405449 full report

Archived source: sanitized report text.

TIC 316405449: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 5 accepted representative events in sectors[15, 76, 84], depths19.82-22.99ppt and durations2.77-3.75h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence2.527605384d, epoch1716.361405BTJD, timing residual RMS1.32min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[15, 76] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector15: neighbour/target RSS ratios 12.72, 48.04; sector76: neighbour/target RSS ratios 3.41, 15.52. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector15 3x3 local-baseline depth20.30ppt; sector76 3x3 local-baseline depth19.50ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source2177587142501400704, P=2.527851247+/-0.00043688d, e=0.0791, omega=100.43deg. Measured-minus-Gaia period is-0.56quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset-0.0963d with linear full-covariance Gaia error0.0826d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration3.32h versus approximate central-crossing scale6.62h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C23. TIC 465263741 full report

Archived source: sanitized report text.

TIC 465263741: PROBABLE

Real repeated eclipses are secure, but the crowded-field source test does not decisively separate all close capable neighbours. PROBABLE retains the host limitation explicitly.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 4 accepted representative events in sectors[10, 38], depths8.04-10.95ppt and durations2.15-2.57h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence6.038766996d, epoch1572.206588BTJD, timing residual RMS4.98min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[10, 38] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector10: neighbour/target RSS ratios 1.32, 1.37, 1.69, 1.70, 1.56, 1.40, 1.63; sector38: neighbour/target RSS ratios 3.11, 2.27, 4.40, 3.21, 3.67. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Host localization remains incomplete because close alternatives are not decisively excluded.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector10 3x3 local-baseline depth5.99ppt; sector38 3x3 local-baseline depth8.07ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source5239577290443811840, P=6.038294466+/-0.000972077d, e=0.0366, omega=65.26deg. Measured-minus-Gaia period is+0.49quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset+0.0646d with linear full-covariance Gaia error0.1443d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration2.20h versus approximate central-crossing scale4.82h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed. S10 source discrimination is weak and S38 only moderately favors the target against close neighbours; higher-resolution localization or a more reliable PRF comparison is required for promotion.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: resolve the close-source alternative before promotion.

C19. TIC 23112764 full report

Archived source: sanitized report text.

TIC 23112764: CONFIRMED

Real repeated eclipses, a matching short orbital time scale, target-centered deficits in two sectors and independent extraction establish a target eclipsing binary. The Gaia orbital solution has a retained period/conjunction tension; its exact phase prediction is not validated by this result.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 5 accepted representative events in sectors[43, 70, 71], depths19.04-20.88ppt and durations3.26-3.45h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence2.958379801d, epoch2477.217297BTJD, timing residual RMS2.46min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[43, 70] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector43: neighbour/target RSS ratios 20.68; sector70: neighbour/target RSS ratios 6.04. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector43 3x3 local-baseline depth18.95ppt; sector70 3x3 local-baseline depth18.53ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source66111027136814720, P=2.957395675+/-0.000380466d, e=0.1966, omega=359.81deg. Measured-minus-Gaia period is+2.59quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset+0.2160d with linear full-covariance Gaia error0.0640d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration3.41h versus approximate central-crossing scale4.26h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed. The near-reference conjunction offset is about3.4linear-Gaia covariance sigma, and the period is2.59quoted Gaia sigma high. A targeted full-covariance Gaussian sensitivity check (9994 physical draws from10000, seed20261030) gives median offset0.200d and16-84% range0.138-0.265d; nonlinearity does not remove the tension. Source attribution, repeated eclipses and nearly identical recurrence time support the same target binary, while duration3.41h is feasible below the approximate4.26h central scale. No physically impossible eclipse geometry is established. This is a confirmed target EB with an imperfect Gaia orbital solution; the Gaia phase prediction itself is not claimed to pass. See ../../23112764-geometry-sensitivity.json.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: refit the Gaia RV orbit/conjunction uncertainty before precise characterization.

C18. TIC 300903537 full report

Archived source: sanitized report text.

TIC 300903537: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 6 accepted representative events in sectors[1, 68, 95], depths79.23-89.53ppt and durations3.11-3.27h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence5.823229918d, epoch1326.644749BTJD, timing residual RMS0.39min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[1, 68] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector1: neighbour/target RSS ratios 16.97; sector68: neighbour/target RSS ratios 9.26. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector1 3x3 local-baseline depth76.58ppt; sector68 3x3 local-baseline depth73.42ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source6789333535281440896, P=5.821861614+/-0.0011434d, e=0.1508, omega=94.01deg. Measured-minus-Gaia period is+1.20quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset+0.0027d with linear full-covariance Gaia error0.0862d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration3.18h versus approximate central-crossing scale3.03h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

Known prior EB: exact Prsa2022 TIC0300903537, period5.8232216d; excluded from new subset. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed. The approximate central-duration estimate is3.03h versus observed median3.18h; TIC dimensions and minimum-mass main-sequence companion assumptions are approximate and this small difference is not a fitted-geometry exclusion.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C25. TIC 280075497 full report

Archived source: sanitized report text.

TIC 280075497: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 5 accepted representative events in sectors[12, 65, 103], depths53.20-58.29ppt and durations8.45-8.77h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence6.034326668d, epoch1634.891927BTJD, timing residual RMS1.85min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[12, 65] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector12: neighbour/target RSS ratios 11.78, 16.76, 17.37; sector65: neighbour/target RSS ratios 12.59, 13.10, 13.48. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector12 3x3 local-baseline depth40.30ppt; sector65 3x3 local-baseline depth41.96ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source5982759666597003392, P=6.032589922+/-0.00110647d, e=0.0162, omega=225.09deg. Measured-minus-Gaia period is+1.57quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset+0.0380d with linear full-covariance Gaia error0.1290d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration8.59h versus approximate central-crossing scale9.28h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C26. TIC 276514049 full report

Archived source: sanitized report text.

TIC 276514049: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 3 accepted representative events in sectors[56, 83], depths59.75-62.36ppt and durations8.87-9.19h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence17.931899173d, epoch1722.528115BTJD, timing residual RMS0.48min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[56, 83] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector56: neighbour/target RSS ratios 3.42; sector83: neighbour/target RSS ratios 4.13. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector56 3x3 local-baseline depth46.29ppt; sector83 3x3 local-baseline depth52.61ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source1974064349050049792, P=17.924991468+/-0.00853752d, e=0.0796, omega=94.79deg. Measured-minus-Gaia period is+0.81quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset-0.0779d with linear full-covariance Gaia error0.3209d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration9.16h versus approximate central-crossing scale13.38h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C27. TIC 10790605 full report

Archived source: sanitized report text.

TIC 10790605: REFUTED eclipse interpretation

The discovery half-period signal is smooth double-wave modulation. The available QLP raw light curves do not establish a localized eclipse; this result does not refute variability or the existence of a binary.

1. Raw flux and repetition: quality-zero SAP flux in sectors18,58,86 was examined at the discovery full period3.7588988d. The trapezoid fits do not pass the repeated-event quality filter; a broad curved baseline can imitate the fitted dip. Lead inspected the raw first/last-event panels.
2. Period/morphology: four-harmonic subtraction leaves phase-zero narrow-box residual depths -0.010+/-0.435, +0.253+/-0.171 and +0.096+/-0.370ppt, respectively. These optimistic IID errors ignore correlated noise. No narrow eclipse is established; a smooth ellipsoidal interpretation is favored. Harmonics can absorb broad eclipses, so this is a bounded negative for the screened narrow-dip claim, not a proof of no grazing eclipse.
3. Source test: the TIC and GaiaDR3 cones were obtained. No event template survived the quality/morphology test, so a difference image cannot meaningfully test a narrow eclipse. This is adjudicated inapplicability, rather than an omitted positive-eclipse gate.
4. Independent reduction: archive inventory was attempted but no matching usable product was obtained. A separate pipeline cannot promote a narrow signal absent in the primary raw flux. No positive EB inference is made.
5. Gaia: the supplied SB1 orbit has period3.7594964d. Agreement of smooth modulation with that period is not evidence for an eclipse or inclination. No eclipse geometry is inferred.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. Catalogue sources: ../../../shared-catalogues/coverage-matrix.csv and README.md.

Deciding evidence: ../../raw-evidence/10790605/harmonic-test.json, ../../raw-evidence/raw-events.json, ../../raw-evidence/raw-event-panels.png, identity-checks.json and measurements-summary.md. All30 QLP target/sector header checks passed. The full review is complete as a negative outcome with the limitations above; no precision non-eclipse upper limit is claimed. Next check: targeted morphology study only if a broader grazing-eclipse hypothesis is pursued.

C28. TIC 376082142 full report

Archived source: sanitized report text.

TIC 376082142: REFUTED

Queue position 11; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

The original screen positive is smooth double-wave modulation, rather than an established discrete eclipse. Three QLP sector folds and four-harmonic residual diagnostics show no narrow residual recurring at the searched phase. This refutes the proposed eclipsing morphology at the measured signal; shallow additional eclipses are not excluded.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 6 clear of fit bounds. Depth range 0.12–0.386 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh cutouts were obtained, but no eclipsing deficit is established to localize. Morphology is the deciding failure, not a missing pixel test. The smooth variable itself has not been assigned a by-catch host.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

The doubled BLS interval follows the Gaia orbit approximately, but a modulation period is not an eclipse-derived period. No eclipse conjunction geometry is claimed.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No established eclipse; the smooth signal could be ellipsoidal or other variability. Harmonics can absorb broad events and do not prove absence of shallow eclipses.

Next deciding check: Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C29. TIC 285942031 full report

Archived source: sanitized report text.

TIC 285942031: PROBABLE

Queue position 12; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Deep, repeated discrete eclipses are real and source fits favor the target in two detector sectors. The Gaia orbit has material timing tension, so the geometry check is not fully passed.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 5 representative fits were retained in raw-events.json, 5 clear of fit bounds. Depth range 10.113–11.857 percent. Period from representative event centers: 2.6068541455671475. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 2.6068541455671475, "epoch_btjd": 1795.1495351827527, "n_events": 5, "residual_rms_min": 1.9711095797469935, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 2.606854146 d versus Gaia 2.607369245 ± 0.00013636 d; difference -3.78 Gaia sigma. Nearest approximate conjunction A: offset -0.355 d; linear covariance error 0.074 d, conservative sum bound 0.342 d. Observed median duration 3.74 h versus approximate central feasibility 5.47 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: Gaia period differs by -3.78 sigma; approximate conjunction is 0.355 d off versus linear covariance error 0.074 d. The conservative sum error is 0.342 d; a nonlinear orbital posterior or fresh RV fit is needed before treating this as the Gaia eclipsing system.

Next deciding check: Reassess Gaia orbital uncertainties/correlations or obtain an updated radial-velocity orbit to resolve the eclipse-versus-Gaia timing tension.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C30. TIC 104736355 full report

Archived source: sanitized report text.

TIC 104736355: CONFIRMED

Queue position 13; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 5 representative fits were retained in raw-events.json, 5 clear of fit bounds. Depth range 8.471–8.897 percent. Period from representative event centers: 2.792537259982342. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 2.792537259982342, "epoch_btjd": 1820.589246873317, "n_events": 5, "residual_rms_min": 0.20896718846358212, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 2.792537260 d versus Gaia 2.792657091 ± 0.00025780 d; difference -0.46 Gaia sigma. Nearest approximate conjunction A: offset -0.022 d; linear covariance error 0.151 d, conservative sum bound 1.538 d. Observed median duration 3.46 h versus approximate central feasibility 4.08 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C31. TIC 281393563 full report

Archived source: sanitized report text.

TIC 281393563: PROBABLE

Queue position 14; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated deep detached eclipses have a coherent event-derived period and feasible Gaia geometry. Fresh source imaging was attempted, but crowding prevents a secure target assignment.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 6 clear of fit bounds. Depth range 7.027–9.69 percent. Period from representative event centers: 6.149828122486457. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 6.149828122486457, "epoch_btjd": 1575.6730668128118, "n_events": 6, "residual_rms_min": 1.418844494519903, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Sectors 10 and64 contain target-region deficits; sector64 also has a strong off-center deficit. Fixed-position capable-neighbour alternatives have residual ratios only1.03–1.99, with a degenerate free-center fit. This is an unresolved source gate, not unattempted work.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 6.149828122 d versus Gaia 6.149604702 ± 0.00058686 d; difference 0.38 Gaia sigma. Nearest approximate conjunction A: offset -0.047 d; linear covariance error 0.153 d, conservative sum bound 1.102 d. Observed median duration 7.05 h versus approximate central feasibility 9.81 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: The eclipses repeat, but the host is not separated from capable neighbours. Pixel models are degenerate in this crowded field.

Next deciding check: Obtain higher resolution time-series imaging or a validated multi-source PRF difference fit to identify the eclipsing host.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C32. TIC 450055243 full report

Archived source: sanitized report text.

TIC 450055243: CONFIRMED

Queue position 15; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 6 clear of fit bounds. Depth range 8.573–9.969 percent. Period from representative event centers: 9.2907014698317. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 9.2907014698317, "epoch_btjd": 1442.9918128436386, "n_events": 6, "residual_rms_min": 2.4356151301002424, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 9.290701470 d versus Gaia 9.288881009 ± 0.00157511 d; difference 1.16 Gaia sigma. Nearest approximate conjunction A: offset 0.376 d; linear covariance error 0.184 d, conservative sum bound 0.392 d. Observed median duration 3.26 h versus approximate central feasibility 4.72 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C33. TIC 126363604 full report

Archived source: sanitized report text.

TIC 126363604: PROBABLE

Queue position 16; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Four sharp repeat eclipses are visible over a variable stellar baseline; all four centers define an eclipse period. Raw-event scatter is inflated by variability, so the initial depth/RMS gate was explicitly relaxed after fold inspection. Fresh imaging supports the target but does not fully exclude its closest capable source.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 4 representative fits were retained in raw-events.json, 4 clear of fit bounds. Depth range 0.789–1.165 percent. Period from representative event centers: 16.505959732386128. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 16.505959732386128, "epoch_btjd": 1505.614116361415, "n_events": 4, "residual_rms_min": 14.198163488304088, "note": "Four visible sharp events; no per-point depth/RMS gate because stellar baseline variation inflates RMS. Unweighted cycle counts conditional on original search." }

Source imaging and independent reduction

Four individual signed difference images in sectors7,61,88 place fitted deficits3.93,1.34,0.81,0.82 arcsec from the target. Nearest capable neighbour is11.14 arcsec. The first-sector fixed-source residual ratio is only1.78; repeat source positions support the target but the PSF systematics are uncalibrated. Custom extraction recovers repeated deficits in the same TESS photons.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 16.505959732 d versus Gaia 16.499031114 ± 0.00363290 d; difference 1.91 Gaia sigma. Nearest approximate conjunction B: offset 0.657 d; linear covariance error 0.290 d, conservative sum bound 1.352 d. Observed median duration 8.22 h versus approximate central feasibility 22.88 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: Closest capable host not conclusively excluded under the source model; approximate Gaia conjunction offset0.657 d has linear covariance error0.290 d, with broad conservative uncertainty. Stellar variability affects local timing/depth.

Next deciding check: Validate the repeated difference-image astrometry with a multi-source PRF or higher resolution event photometry; refine Gaia conjunction uncertainty if needed.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C34. TIC 251357000 full report

Archived source: sanitized report text.

TIC 251357000: REFUTED

Queue position 17; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

The original screen positive is smooth double-wave modulation, rather than an established discrete eclipse. Three QLP sector folds and four-harmonic residual diagnostics show no narrow residual recurring at the searched phase. This refutes the proposed eclipsing morphology at the measured signal; shallow additional eclipses are not excluded.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 2 clear of fit bounds. Depth range 0.75–0.811 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh cutouts were obtained, but no eclipsing deficit is established to localize. Morphology is the deciding failure, not a missing pixel test. The smooth variable itself has not been assigned a by-catch host.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

The doubled BLS interval follows the Gaia orbit approximately, but a modulation period is not an eclipse-derived period. No eclipse conjunction geometry is claimed.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No established eclipse; the smooth signal could be ellipsoidal or other variability. Harmonics can absorb broad events and do not prove absence of shallow eclipses.

Next deciding check: Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C35. TIC 267222866 full report

Archived source: sanitized report text.

TIC 267222866: CONFIRMED

Queue position 18; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 5 representative fits were retained in raw-events.json, 5 clear of fit bounds. Depth range 1.438–2.207 percent. Period from representative event centers: 1.867209837595571. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 1.867209837595571, "epoch_btjd": 1818.1354273795532, "n_events": 5, "residual_rms_min": 1.5541314088817642, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 1.867209838 d versus Gaia 1.867184804 ± 0.00013282 d; difference 0.19 Gaia sigma. Nearest approximate conjunction A: offset 0.094 d; linear covariance error 0.118 d, conservative sum bound 0.590 d. Observed median duration 6.04 h versus approximate central feasibility 5.56 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C36. TIC 66315917 full report

Archived source: sanitized report text.

TIC 66315917: CONFIRMED

Queue position 19; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 3 representative fits were retained in raw-events.json, 3 clear of fit bounds. Depth range 2.591–2.739 percent. Period from representative event centers: 19.634374528897247. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 19.634374528897247, "epoch_btjd": 1494.1205215957734, "n_events": 3, "residual_rms_min": 0.44878357765813764, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 19.634374529 d versus Gaia 19.608607700 ± 0.02322759 d; difference 1.11 Gaia sigma. Nearest approximate conjunction A: offset 1.879 d; linear covariance error 1.775 d, conservative sum bound 2.469 d. Observed median duration 2.20 h versus approximate central feasibility 5.74 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C37. TIC 93783392 full report

Archived source: sanitized report text.

TIC 93783392: CONFIRMED

Queue position 20; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 6 clear of fit bounds. Depth range 2.665–3.25 percent. Period from representative event centers: 3.8724081070823737. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 3.8724081070823737, "epoch_btjd": 1630.7599836800352, "n_events": 6, "residual_rms_min": 3.507427667895757, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 3.872408107 d versus Gaia 3.871849592 ± 0.00099811 d; difference 0.56 Gaia sigma. Nearest approximate conjunction A: offset 0.129 d; linear covariance error 0.361 d, conservative sum bound 0.973 d. Observed median duration 5.58 h versus approximate central feasibility 7.43 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C38. TIC 470947699 full report

Archived source: sanitized report text.

TIC 470947699: CONFIRMED

Queue position 21; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 6 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 8.17041492823667 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 8.17041492823667,
  "epoch_btjd": 1846.407037256951,
  "n_events": 6,
  "residual_rms_min": 0.8413010526801175,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 20,
    "target_offset_arcsec": 1.7818942626730834,
    "min_neighbour_target_sse_ratio": null,
    "capable_neighbour_count": 0
  },
  {
    "sector": 60,
    "target_offset_arcsec": 0.5031546449154466,
    "min_neighbour_target_sse_ratio": null,
```

```
    "capable_neighbour_count": 0
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=8.170414928 d; Gaia P=8.172049314 +/- 0.001169795 d. Difference -1.40 Gaia sigma is descriptive. Approximate conjunction A offset -0.292 d; covariance linear error 0.225 d; conservative absolute-sum bound 1.171 d. Median fitted duration 3.88 h; approximate central duration 4.59 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C39. TIC 135230723 full report

Archived source: sanitized report text.

TIC 135230723: CONFIRMED

Queue position 22; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 6.776113017992571 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 6.776113017992571,
  "epoch_btjd": 1958.4036951273954,
  "n_events": 5,
  "residual_rms_min": 1.3932796715666198,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 24,
    "target_offset_arcsec": 2.7623131101621796,
    "min_neighbour_target_sse_ratio": 2.486867070374025,
    "capable_neighbour_count": 2
  },
  {
    "sector": 51,
    "target_offset_arcsec": 0.5383821425401804,
    "min_neighbour_target_sse_ratio": 4.691126358639871,
```

```
    "capable_neighbour_count": 2
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=6.776113018 d; Gaia P=6.778623813 +/- 0.001888155 d. Difference -1.33 Gaia sigma is descriptive. Approximate conjunction A offset -0.612 d; covariance linear error 0.438 d; conservative absolute-sum bound 0.993 d. Median fitted duration 3.31 h; approximate central duration 3.66 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C40. TIC 36083554 full report

Archived source: sanitized report text.

TIC 36083554: CONFIRMED

Queue position 23; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 15.974024163426064 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 15.974024163426064,
  "epoch_btjd": 1496.4368213575215,
  "n_events": 5,
  "residual_rms_min": 2.302940486572714,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 7,
    "target_offset_arcsec": 1.0002671054830687,
    "min_neighbour_target_sse_ratio": 54.527241876955294,
    "capable_neighbour_count": 11
  },
  {
    "sector": 61,
    "target_offset_arcsec": 2.4025059769389188,
    "min_neighbour_target_sse_ratio": 15.686721955051661,
```

```
    "capable_neighbour_count": 10
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=15.974024163 d; Gaia P=15.980986459 +/- 0.006281408 d. Difference -1.11 Gaia sigma is descriptive. Approximate conjunction A offset 0.017 d; covariance linear error 0.456 d; conservative absolute-sum bound 2.293 d. Median fitted duration 4.43 h; approximate central duration 6.18 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C41. TIC 447566809 full report

Archived source: sanitized report text.

TIC 447566809: CONFIRMED

Queue position 24; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 3 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 19.64701755851303 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 19.64701755851303,
  "epoch_btjd": 1645.68237977337,
  "n_events": 3,
  "residual_rms_min": 2.8451511295473804,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 12,
    "target_offset_arcsec": 1.397054877962082,
    "min_neighbour_target_sse_ratio": 16.034792064157667,
    "capable_neighbour_count": 13
  },
  {
    "sector": 93,
    "target_offset_arcsec": 3.744046170975149,
    "min_neighbour_target_sse_ratio": 6.72483777409165,
```

```
    "capable_neighbour_count": 20
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=19.647017559 d; Gaia P=19.643298210 +/- 0.002714314 d. Difference 1.37 Gaia sigma is descriptive. Approximate conjunction A offset -0.072 d; covariance linear error 0.174 d; conservative absolute-sum bound 0.518 d. Median fitted duration 9.87 h; approximate central duration 11.32 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C42. TIC 12320469 full report

Archived source: sanitized report text.

TIC 12320469: REFUTED

Queue position 25; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Three sector folds show smooth broad double-wave modulation, without a convincing discrete narrow eclipse. The original screen is refuted as an eclipsing interpretation of this measured signal.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: none for the target d.

Initial measurement receipt (superseded where rejected above):

```
null
```

Fresh source and independent reduction

Fresh cutouts were attempted; no established eclipse deficit exists to localize. Source attribution of the smooth modulation is not claimed.

```
[]
```

Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

The searched modulation period is not an eclipse-derived period. No conjunction comparison is claimed without established eclipses.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Smooth modulation may be ellipsoidal or other stellar variability; shallow additional eclipses are not excluded. A local trapezoid fit to a broad trend does not establish an eclipse.

Next test: Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C43. TIC 86908196 full report

Archived source: sanitized report text.

TIC 86908196: REFUTED

Queue position 26; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Three sector folds show smooth broad double-wave modulation, without a convincing discrete narrow eclipse. The original screen is refuted as an eclipsing interpretation of this measured signal.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: none for the target d.

Initial measurement receipt (superseded where rejected above):

```
null
```

Fresh source and independent reduction

Fresh cutouts were attempted; no established eclipse deficit exists to localize. Source attribution of the smooth modulation is not claimed.

```
[]
```

Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

The searched modulation period is not an eclipse-derived period. No conjunction comparison is claimed without established eclipses.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Smooth modulation may be ellipsoidal or other stellar variability; shallow additional eclipses are not excluded. A local trapezoid fit to a broad trend does not establish an eclipse.

Next test: Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C44. TIC 201317466 full report

Archived source: sanitized report text.

TIC 201317466: PROBABLE

Queue position 27; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Deep repeated eclipses are real and coherent. Association with the published Gaia SB1 orbit remains unresolved because its conjunction misses the eclipse by a substantial fraction of a cycle, beyond the conservative linear propagated uncertainty.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 2 representative sectors. Local quadratic baseline plus trapezoid fits retained 4 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 5.623860159150563 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 5.623860159150563,
  "epoch_btjd": 2772.3289690349256,
  "n_events": 4,
  "residual_rms_min": 0.5278820914721635,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 54,
    "target_offset_arcsec": 0.11507353740750183,
    "min_neighbour_target_sse_ratio": 15.652494055094234,
    "capable_neighbour_count": 2
  },
  {
    "sector": 80,
    "target_offset_arcsec": 3.540797490060455,
```

```
    "min_neighbour_target_sse_ratio": 23.02100675329804,
    "capable_neighbour_count": 2
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=5.623860159 d; Gaia P=5.625963779 +/- 0.000558314 d. Difference -3.77 Gaia sigma is descriptive. Approximate conjunction A offset -1.208 d; covariance linear error 0.228 d; conservative absolute-sum bound 0.694 d. Median fitted duration 5.97 h; approximate central duration 7.50 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed. Gaia has only 15 RV epochs, modestly constrained eccentricity (0.125 +/- 0.061), and strongly correlated periastron parameters. A 1.208 d conjunction offset versus 0.228 d linear covariance error and 0.694 d conservative sum bound is a physical phase discrepancy; no automatic sigma veto is applied.

Next test: Refit the RV orbit or evaluate a nonlinear orbital posterior against the observed eclipse timing to resolve Gaia association.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C45. TIC 360516818 full report

Archived source: sanitized report text.

TIC 360516818: CONFIRMED

Queue position 28; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 9.890542850425698 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 9.890542850425698,
  "epoch_btjd": 1688.7028501960215,
  "n_events": 5,
  "residual_rms_min": 5.591486264753861,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 14,
    "target_offset_arcsec": 2.4168737956248054,
    "min_neighbour_target_sse_ratio": 3.587870111420561,
    "capable_neighbour_count": 50
  },
  {
    "sector": 41,
    "target_offset_arcsec": 0.7353517988002084,
    "min_neighbour_target_sse_ratio": 13.824703240155678,
```

```
    "capable_neighbour_count": 32
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=9.890542850 d; Gaia P=9.890757806 +/- 0.002383965 d. Difference -0.09 Gaia sigma is descriptive. Approximate conjunction A offset 0.066 d; covariance linear error 0.331 d; conservative absolute-sum bound 0.579 d. Median fitted duration 7.57 h; approximate central duration 17.48 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C46. TIC 411647453 full report

Archived source: sanitized report text.

TIC 411647453: REFUTED

Queue position 29; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Real narrow eclipses contaminate the target aperture, but the proposed target Gaia-period eclipse association is refuted. Two detector sectors place the deficit on nearby TIC 411647413 rather than TIC 411647453. The approximately 0.9151763 d short-period signal gives about seven dips per 6.394568 d screen fold; that screen interval is not a valid target eclipse period.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 6 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: none for the target d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 6.394515949153654,
  "epoch_btjd": 1603.7676801779019,
  "n_events": 3,
  "residual_rms_min": 61.848856830139425,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Sectors11/99 Gaussian deficit centers are 15.09/14.20 arcsec from the target, but 0.56/1.64 arcsec from TIC411647413, separated14.76 arcsec. Fixed neighbour/target SSE ratios are0.362/0.144, consistently favoring that neighbour. A short-period contaminating EB is strongly inferred; its precision ephemeris and novelty were not reviewed.


```
[
  {
    "sector": 11,
    "target_offset_arcsec": 15.09238254714884,
    "min_neighbour_target_sse_ratio": 0.36214401303402544,
    "capable_neighbour_count": 18
  },
  {
    "sector": 99,
    "target_offset_arcsec": 14.197941851487595,
```

```
    "min_neighbour_target_sse_ratio": 0.1436331918871176,
    "capable_neighbour_count": 25
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

The original conditional 6.394515949 d regression aliased unrelated eclipse dips. It is rejected as an eclipse-derived target period; no Gaia-conjunction association is established.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: The contaminating eclipses are real; REFUTED applies to target attribution and Gaia-period interpretation. TIC411647413 is an inferred by-catch host subject to pixel model systematics, not a new confirmed catalogue object.

Next test: Higher resolution event photometry can establish the contaminant host and measure its orbital period precisely.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C47. TIC 450217907 full report

Archived source: sanitized report text.

TIC 450217907: REFUTED

Queue position 30; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Three sector folds show smooth broad double-wave modulation, without a convincing discrete narrow eclipse. The original screen is refuted as an eclipsing interpretation of this measured signal.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: none for the target d.

Initial measurement receipt (superseded where rejected above):

```
null
```

Fresh source and independent reduction

Fresh cutouts were attempted; no established eclipse deficit exists to localize. Source attribution of the smooth modulation is not claimed.

```
[]
```

Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

The searched modulation period is not an eclipse-derived period. No conjunction comparison is claimed without established eclipses.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Smooth modulation may be ellipsoidal or other stellar variability; shallow additional eclipses are not excluded. A local trapezoid fit to a broad trend does not establish an eclipse.

Next test: Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C48. TIC 60443793 full report

Archived source: sanitized report text.

TIC 60443793: REFUTED

Queue position 31; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

The original 6.510555 d search alias is wrong for the prominent eclipse train. Forty-eight raw minima give 1.29738064 d spacing; alternating depths imply a candidate 2.5947613 d binary period. Fresh imaging places these real EB photons off target, so the target-EB association is refuted. This is neither a confirmed target nor a target by-catch EB.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 48 representative fits were retained in raw-events.json, 48 clear of fit bounds. Depth range 3.462–21.691 percent. Period from representative event centers: 2.594761285455248. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

{ "minimum_spacing_d": 1.297380642727624, "candidate_binary_period_d": 2.594761285455248, "epoch_btjd": 1658.3489391171465, "residual_rms_min": 11.60045026187536, "n_events": 48, "note": "Cycle count conditional on approximate 1.297 d spacing; alternating depths suggest double spacing as binary period. No uncertainty or secure alias test." }

Source imaging and independent reduction

S 13/S 93 signed deficits are off target. Free-Gaussian offsets about 58 arcsec align with capable TIC 60443695 at 59.38 arcsec separation. Fixed-coordinate neighbour/target SSE 0.0032 and 0.0229 strongly favor that neighbour in two sectors; positive-clipped centroids alone give 44.7/50.2 arcsec. Scores are descriptive, with PRF/systematic caveats. Candidate neighbour identity is an inference, not a full adjudication of that star.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=6.51870479 d. Representative photometric period 2.59476129 d; difference / Gaia quoted period error -1927.55. Closest edge-on conjunction A: offset -1.217 d, covariance-linear error 0.569 d. Observed duration median 4.67 h; approximate central duration 243.51 h. These are feasibility diagnostics, not significance vetoes or a fitted geometry.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: off-target eclipsing binary contamination; Gaia association: photometric train does not trace target Gaia orbit.

Execution: completed representative full review; scientific ambiguity: Neighbour EB period remains a candidate and has not undergone independent alias/geometry review as its own target. The target giant-radius estimate is also incompatible with interpreting this short period as its simple detached binary orbit.

Next deciding check: Review TIC 60443695 separately if desired.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C49. TIC 71866523 full report

Archived source: sanitized report text.

TIC 71866523: PLAUSIBLE

Queue position 32; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Sectors 6/33 are nearly flat with a weak sector 33 depression; sector 87 is dominated by a large positive transient/systematic. No secure repeating eclipse sequence or eclipse-derived period is established.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 2 representative fits were retained in raw-events.json, 1 clear of fit bounds. Depth range 0.099–0.099 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh sectors 6/87 custom extraction attempted the independent gate; no sufficient isolated eclipse exists for a meaningful signed source image.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=15.39210899 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: uncertain variability; Gaia association: not established.

Execution: completed representative full review; scientific ambiguity: Weak single feature and transient/baseline ambiguity; available work was attempted.

Next deciding check: Require a clean repeated negative event before EB classification.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C50. TIC 443106429 full report

Archived source: sanitized report text.

TIC 443106429: PROBABLE

Queue position 33; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Four narrow raw eclipses in sectors 11 and 65 give 12.7955335 d, consistent with Gaia. Raw depths 1.82?2.84%, duration about 10.2 h; custom extraction recovers four at 1.84?3.33%. Eclipses are real and repeat, but the close host remains unresolved.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 4 representative fits were retained in raw-events.json, 4 clear of fit bounds. Depth range 1.825–2.841 percent. Period from representative event centers: 12.795533464914396. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 12.795533464914396, "epoch_btjd": 1606.1515052693348, "n_events": 3, "residual_rms_min": 0.2322850788714703, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Sector 11 deficit lies 0.59 arcsec from target, with capable TIC 443106423 only 3.88 arcsec away; fixed-neighbour/target SSE 1.83 favors target. Extra fresh sector 65 was obtained; noisy one-event signed deficit has 13.4 arcsec fitted offset and close-neighbour/target SSE 0.99, failing independent separation. Source testing was attempted and remains scientifically ambiguous.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=12.79551859 d. Representative photometric period 12.79553346 d; difference / Gaia quoted period error 0.00. Closest edge-on conjunction A: offset -0.230 d, covariance-linear error 0.565 d. Observed duration median 10.23 h; approximate central duration 22.45 h. These are feasibility

diagnostics, not significance vetoes or a fitted geometry.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: repeating eclipsing signal; likely target; Gaia association: period/conjunction feasible, host unresolved.

Execution: completed representative full review; scientific ambiguity: Host not robustly separated from 3.88 arcsec capable neighbour; this is the one unresolved protocol gate.

Next deciding check: Resolve target versusTIC 443106423 with higher spatial resolution eclipse photometry.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C51. TIC 311206604 full report

Archived source: sanitized report text.

TIC 311206604: REFUTED

Queue position 34; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

QLP sectors and fresh custom extraction show a broad, fairly flat-bottomed double wave, without distinct narrow repeated eclipses. Trapezoid fits absorb smooth curvature and frequently hit duration/center bounds. The narrow-eclipse interpretation is refuted; binarity or ellipsoidal variability is not excluded.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 4 representative fits were retained in raw-events.json, 1 clear of fit bounds. Depth range 2.728–2.728 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh first/last-sector cutouts were independently extracted. A meaningful narrow-event difference image cannot be defined from this smooth morphology; source gate was attempted and found no supported discrete eclipse to localize.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=4.40200229 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: smooth variable; no narrow eclipses; Gaia association: no eclipse geometry validated.

Execution: completed representative full review; scientific ambiguity: Physical variability mechanism remains unclassified. Smooth modulation alone does not establish eclipses.

Next deciding check: No further eclipse follow-up required unless new discrete events appear.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C52. TIC 428769301 full report

Archived source: sanitized report text.

TIC 428769301: REFUTED

Queue position 35; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

QLP sectors and fresh custom extraction show a smooth double wave, without distinct narrow repeated eclipses. Trapezoid fits absorb smooth curvature and frequently hit duration/center bounds. The narrow-eclipse interpretation is refuted; binarity or ellipsoidal variability is not excluded.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 5 representative fits were retained in raw-events.json, 2 clear of fit bounds. Depth range 0.298–0.336 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh first/last-sector cutouts were independently extracted. A meaningful narrow-event difference image cannot be defined from this smooth morphology; source gate was attempted and found no supported discrete eclipse to localize.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=3.73288066 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: smooth variable; no narrow eclipses; Gaia association: no eclipse geometry validated.

Execution: completed representative full review; scientific ambiguity: Physical variability mechanism remains unclassified. Smooth modulation alone does not establish eclipses.

Next deciding check: No further eclipse follow-up required unless new discrete events appear.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C53. TIC 436947190 full report

Archived source: sanitized report text.

TIC 436947190: REFUTED

Queue position 36; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

QLP sectors and fresh custom extraction show a broad smooth repeating wave, without distinct narrow repeated eclipses. Trapezoid fits absorb smooth curvature and frequently hit duration/center bounds. The narrow-eclipse interpretation is refuted; binarity or ellipsoidal variability is not excluded.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 6 representative fits were retained in raw-events.json, 1 clear of fit bounds. Depth range 0.2–0.2 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh first/last-sector cutouts were independently extracted. A meaningful narrow-event difference image cannot be defined from this smooth morphology; source gate was attempted and found no supported discrete eclipse to localize.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=3.10818466 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: smooth variable; no narrow eclipses; Gaia association: no eclipse geometry validated.

Execution: completed representative full review; scientific ambiguity: Physical variability mechanism remains unclassified. Smooth modulation alone does not establish eclipses.

Next deciding check: No further eclipse follow-up required unless new discrete events appear.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C54. TIC 92452368 full report

Archived source: sanitized report text.

TIC 92452368: PLAUSIBLE

Queue position 37; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Sector 1 contains a complex cluster of deep negative points around BTJD 1342.7?1344.1, rather than a clean repeating narrow eclipse. Sectors 68/95 do not establish repetition. A single episode with instrumental ambiguity remains.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 0 representative fits were retained in raw-events.json, 0 clear of fit bounds. No detached event established. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh sectors 1/95 independently extracted. The irregular clustered episode does not provide defensible in/out windows for a repeat source image; the source gate was attempted but insufficient.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=18.58152247 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: uncertain variability; Gaia association: not established.

Execution: completed representative full review; scientific ambiguity: Single complex episode, untested eclipse period and instrumental ambiguity.

Next deciding check: Verify a clean recurrent event in independently reduced data.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C55. TIC 94300366 full report

Archived source: sanitized report text.

TIC 94300366: PLAUSIBLE

Queue position 38; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Two sector 54 depressions recur near the search phase and survive custom extraction at 4.24/4.76%, but are asymmetric and mixed with stronger irregular variability and other minima. Sector 81 does not establish a stable narrow eclipse sequence. Eclipse morphology and period remain untested.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 4 representative fits were retained in raw-events.json, 2 clear of fit bounds. Depth range 4.198–7.203 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

S 54 one-event deficit favors target with 4.59 arcsec free-Gaussian offset, but morphology and the noisy later-sector events prevent a repeated source validation. Fresh independent extraction does not turn the irregular dips into a secure EB.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=3.90447017 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: uncertain variability; Gaia association: not established.

Execution: completed representative full review; scientific ambiguity: Irregular/spot-like variability versus eclipses unresolved; no secure eclipse period.

Next deciding check: Obtain detrending resistant multiseason photometry to establish stable discrete repeats.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C56. TIC 4646090 full report

Archived source: sanitized report text.

TIC 4646090: CONFIRMED

Queue position 39; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Three narrow, locally flat-baseline eclipses in sectors 4 and 31 give 10.7303766 d. Raw depths 1.95?2.26% and durations 2.75?3.46 h recur; the distinct custom extraction recovers all three at 0.59?0.64% aperture depth. Dilution differs between reductions. Source and independent gates pass; Gaia period and phase are feasible at the precision available.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 3 representative fits were retained in raw-events.json, 3 clear of fit bounds. Depth range 1.945–2.265 percent. Period from representative event centers: 10.73037662305058. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 10.73037662305058, "epoch_btjd": 1419.05639053136, "n_events": 3, "residual_rms_min": 1.164581672824554, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed deficits in both sectors prefer the target. Free-Gaussian offsets are 0.46 and 2.28 arcsec, versus a capable TIC 4646088 neighbour 6.96 arcsec away (7.53 and 9.32 arcsec from fitted deficits). Fixed-neighbour / target SSE is 5.34 and 5.65 in sector stacks; per-event ratios 5.34,3.57,4.27 also favor target. These descriptive comparisons, repeated across sectors/events, support attribution; no calibrated PRF confidence is asserted.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=10.72678235 d. Representative photometric period 10.73037662 d; difference / Gaia quoted period error 1.05. Closest edge-on conjunction A: offset 0.612 d, covariance-linear error 0.457 d. Observed duration median 3.22 h; approximate central duration 8.06 h. These are feasibility diagnostics, not significance vetoes or a fitted geometry.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: target eclipsing binary; Gaia association: consistent within approximate orbit/geometry checks.

Execution: completed representative full review; scientific ambiguity: Gaia phase offset is 1.34 covariance-linear error units; grazing geometry can shorten the approximate central duration. No detailed physical fit or calibrated astrometric uncertainty.

Next deciding check: Characterize dilution and grazing geometry if needed.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C57. TIC 402940895 full report

Archived source: sanitized report text.

TIC 402940895: PLAUSIBLE

Queue position 40; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Small sharp depressions lie on strong discontinuous sector 12/93 trends, while sector 103 flux is severely corrupted. Custom sector 12 fixed-shape depths 0.53/0.76% show features in the same photons, but their morphology is baseline sensitive and a stable eclipse sequence is unestablished.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 5 representative fits were retained in raw-events.json, 5 clear of fit bounds. Depth range 0.0–0.674 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Sector 12 signed diagnostic has 2.79 arcsec Gaussian target offset, but strong baseline structure and lack of a clean independent later-sector event prevent secure source validation. Fresh sectors 12/103 were extracted.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=16.44813821 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: uncertain variability; Gaia association: not established.

Execution: completed representative full review; scientific ambiguity: Baseline/discontinuity sensitivity, corrupted sector 103, no secure eclipse period or repeated source validation.

Next deciding check: Re-extract clean sectors with robust background/pointing treatment before asserting eclipses.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C58. TIC 183994341 full report

Archived source: sanitized report text.

TIC 183994341: CONFIRMED

Queue position 41; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Four repeated primary eclipses with raw depths25–29percent are on target and recovered independently. The eclipse period is approximately14.6655d; the ASAS-SN catalogue lists an incompatible2.9324d alias. The target is a known prior EB and excluded from new additions.

Raw repetition and period

Accepted target eclipse period: 14.665514216226159 days. 2 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "period_d": 14.665514216226159, "epoch_btjd": 3781.92483432609, "n_events": 4, "note": "Representative local event timing regression for939/183; direct consecutive event interval for431/222; no formal period error." }

Representative accepted event measurements:


```
[
  {
    "sector": 91,
    "tc": 3781.924758087458,
    "depth": 0.27811754249345844,
    "duration_d": 0.6709603998224439,
    "ingress_fraction": 0.4102676391311878,
    "residual_rms": 0.008400452725881942,
    "edge": false,
    "n_points": 393,
    "cycle": 0
  },
  {
    "sector": 91,
    "tc": 3796.5904251899988,
    "depth": 0.28747538323348454,
    "duration_d": 0.7030939389529461,
    "ingress_fraction": 0.4222486811501103,
    "residual_rms": 0.008485133785920739,
    "edge": false,
    "n_points": 366,
    "cycle": 1
  },
  {
```

```
    "sector": 92,
    "tc": 3811.255938179075,
    "depth": 0.24829805939567123,
    "duration_d": 0.6719697358685144,
    "ingress_fraction": 0.4078376270792781,
    "residual_rms": 0.00781992030528953,
    "edge": false,
    "n_points": 519,
    "cycle": 2
  },
  {
    "sector": 92,
    "tc": 3825.9213011451875,
    "depth": 0.25572653021110614,
    "duration_d": 0.6766583893637675,
    "ingress_fraction": 0.4180416685853718,
    "residual_rms": 0.007580250708817091,
    "edge": false,
    "n_points": 519,
    "cycle": 3
  }
]
```


Source and independent reduction

Sectors91/92 free Gaussian offsets1.71/1.58arcsec favor the target; closest capable neighbours38arcsec or farther give substantially worse fixed-source residuals (ratios22.8–44.1). Independent custom apertures reproduce repeated eclipses.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

Four-event regression P14.665514216d versus Gaia14.670950076+/-0.003882078d. Approximate conjunction offset1.81d with propagated covariance0.93d and conservative absolute-sum1.76d; duration16.18h versus approximate central23.03h is physically feasible. The period, propagated approximate phase and duration are compatible with the Gaia orbit at current precision; the roughly1.9sigma phase difference alone is not a physical veto.

Association flag: compatible at approximate current precision; phase uncertainty retained.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

known prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Real repeated on-target known EB. Approximate Gaia conjunction phase and radii limit characterization. Catalogue period conflict is retained rather than forcing catalogue cycles. No new-discovery claim.

Next check: Resolve ASAS-SN period alias and obtain RV/orbital posterior or a joint timing/geometry analysis to settle the Gaia association.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C59. TIC 166781582 full report

Archived source: sanitized report text.

TIC 166781582: PLAUSIBLE

Queue position 42; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Broad smooth multi-day minima vary in shape and amplitude between sectors15/75/83. The long screen interval does not isolate discrete ingress/egress. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C60. TIC 230270717 full report

Archived source: sanitized report text.

TIC 230270717: PLAUSIBLE

Queue position 43; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Sectors26/53/80 contain broad changing trends and sharp upward structure, with no established repeated discrete eclipse. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C61. TIC 462845564 full report

Archived source: sanitized report text.

TIC 462845564: PLAUSIBLE

Queue position 44; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

The two available sectors26/40 are dominated by long monotonic flux segments and downlink resets; no repeated isolated eclipse is measured. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 2 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C62. TIC 431762229 full report

Archived source: sanitized report text.

TIC 431762229: REFUTED

Queue position 45; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Repeated real eclipses contaminate the target aperture, but are spatially associated with TIC431762222, separated32.79arcsec. REFUTED applies to target attribution and the original Gaia-period eclipse interpretation.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Two fresh detector-sector difference stacks repeatedly favor TIC431762222; fixed neighbour/target residual SSE ratios are0.0090/0.0188. Free Gaussian deficits lie close to that neighbour. The ratios are descriptive, not calibrated probabilities.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

The contaminating eclipse interval isapproximately1.94d; the original screen alias is rejected as a target eclipse period. No target Gaia orbit association is inferred.

Association flag: contaminating neighbour.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: The contaminant is an inferred EB host; its catalogue novelty and precision binary orbit were not independently adjudicated. Source model systematics remain.

Next check: Higher resolution eclipse photometry can verify the contaminating host.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C63. TIC 350661747 full report

Archived source: sanitized report text.

TIC 350661747: PLAUSIBLE

Queue position 46; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

QLP sectors12/39/103 show large irregular trends; the giant sector103 changes are largely absent in the independent aperture. The trial deficit is displaced and is not an established eclipse. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C64. TIC 424565669 full report

Archived source: sanitized report text.

TIC 424565669: PLAUSIBLE

Queue position 47; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Sector18 is dominated by positive flares and sector59/86 by weak fluctuations and gap changes. No repeated eclipse morphology is established. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C65. TIC 222498305 full report

Archived source: sanitized report text.

TIC 222498305: REFUTED

Queue position 48; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Repeated real eclipses contaminate the target aperture, but are spatially associated with TIC222498306, separated40.139arcsec. REFUTED applies to target attribution and the original Gaia-period eclipse interpretation.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Two fresh detector-sector difference stacks repeatedly favor TIC222498306; fixed neighbour/target residual SSE ratios are0.0119/0.00834. Free Gaussian deficits lie close to that neighbour. The ratios are descriptive, not calibrated probabilities.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

The contaminating eclipse interval isapproximately15.5396d; the original screen alias is rejected as a target eclipse period. No target Gaia orbit association is inferred.

Association flag: contaminating neighbour.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: The contaminant is an inferred EB host; its catalogue novelty and precision binary orbit were not independently adjudicated. Source model systematics remain.

Next check: Higher resolution eclipse photometry can verify the contaminating host.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C66. TIC 90542579 full report

Archived source: sanitized report text.

TIC 90542579: REFUTED

Queue position 49; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

The enormous sector90 QLP downward excursions occur at both downlink boundaries while the independent custom aperture rises at the same times. They are extraction/systematic features, not demonstrated target eclipses. Earlier sectors8/63 show no matching deep eclipse.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Independent reduction reverses the sign of the deciding sector90 excursions; there is no astrophysical eclipse deficit to localize.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

The screen period17.329d is not derived from accepted eclipse events. No Gaia eclipse association is established.

Association flag: screen artifact.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: REFUTED applies to the proposed deep-eclipse interpretation of the measured screen signal. Shallower unrelated variability is not excluded.

Next check: If revisiting, use clean cadences away from downlink boundaries with a validated background and aperture.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C67. TIC 9396327 full report

Archived source: sanitized report text.

TIC 9396327: PROBABLE

Queue position 50; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Six fully covered primary events repeat in raw flux with depth28–29percent and consistent discrete morphology. Independent aperture reduction reproduces the events and the signed deficit favors the target. The Gaia-orbit geometry check remains unresolved.

Raw repetition and period

Accepted target eclipse period: 2.4416706861122432 days. 1 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "period_d": 2.4416706861122432, "epoch_btjd": 3481.3231252924884, "n_events": 6, "note": "Representative local event timing regression for939/183; direct consecutive event interval for431/222; no formal period error." }

Representative accepted event measurements:


```
[
  {
    "sector": 80,
    "tc": 3481.323636157607,
    "depth": 0.28420131298250056,
    "duration_d": 0.13406369149885997,
    "ingress_fraction": 0.3540239651639954,
    "residual_rms": 0.012155047692972612,
    "edge": false,
    "n_points": 517,
    "cycle": 0
  },
  {
    "sector": 80,
    "tc": 3483.7643666168096,
    "depth": 0.2883310030553778,
    "duration_d": 0.13934346534258632,
    "ingress_fraction": 0.35414071040251316,
    "residual_rms": 0.009947821623346951,
    "edge": false,
    "n_points": 519,
    "cycle": 1
  },
  {
```

```
    "sector": 80,
    "tc": 3495.9727604242707,
    "depth": 0.2882585527080121,
    "duration_d": 0.13941595437124518,
    "ingress_fraction": 0.3526956252225945,
    "residual_rms": 0.008971962493992389,
    "edge": false,
    "n_points": 519,
    "cycle": 6
  },
  {
    "sector": 80,
    "tc": 3498.4150669656806,
    "depth": 0.28617070118118176,
    "duration_d": 0.1422442512820803,
    "ingress_fraction": 0.36315371152553355,
    "residual_rms": 0.008677686898336322,
    "edge": false,
    "n_points": 519,
    "cycle": 7
  },
  {
    "sector": 80,
    "tc": 3500.856001103531,
    "depth": 0.2879069502033062,
    "duration_d": 0.14331393223936886,
    "ingress_fraction": 0.3692578062889713,
    "residual_rms": 0.009958225508168744,
    "edge": false,
    "n_points": 519,
    "cycle": 8
  },
  {
    "sector": 80,
    "tc": 3503.298711756517,
    "depth": 0.28214056210063676,
    "duration_d": 0.14093323177085357,
    "ingress_fraction": 0.36372962582456636,
    "residual_rms": 0.009964507706264907,
    "edge": false,
    "n_points": 426,
    "cycle": 9
  }
]
```


Source and independent reduction

Sector80 free Gaussian deficit lies0.56arcsec from target. Capable TIC neighbours51.4/55.9arcsec away have fixed neighbour/target SSE ratios74.9/70.5, strongly worse descriptively. The independent custom aperture reproduces repeated eclipses.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths

and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

Representative six-event regression gives2.441670686d (timing residuals below0.00056d); Gaia2.441934355+/-0.000085226d. A sector-end partial event was excluded because its distorted timing shifts the inferred period. Approximate conjunction A offset0.347d versus propagated covariance0.116d; period differences alone do not veto an EB. Duration3.36h is feasible relative to central4.81h. Secondary-event phase and propagated orbital phase prevent claiming a fully consistent Gaia association.

Association flag: unresolved; no firm by-catch.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Real on-target repeating EB; PROBABLE because the Gaia geometry/association check is unresolved. One fresh detector sector only; same-photon independent reduction. No firm by-catch flag: a modest period mismatch does not establish a separate orbit.

Next check: Refine Gaia conjunction predictions with orbital posterior and measure a second clean sector or obtain new RVs to test the orbit association.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C68. TIC 275022619 full report

Archived source: sanitized report text.

TIC 275022619: PLAUSIBLE

Queue position 51; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors 12,65,103 show smooth slopes and broad variability. The sharp-looking sector103 changes occur at segment edges. At the covered original nominal phase, the short-window signed depth is negative (brightening), and the template fit hits its timing/duration bounds. The fresh custom aperture shows smooth variability and segment offsets without a repeat narrow eclipse. Independent Terra audit retains PLAUSIBLE: the available representative data fail to establish recurring eclipses, but sparse/compromised coverage does not tie the original search detection to a specific demonstrated artifact. This is not a secure eclipse identification.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 17.575404492 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 103, "cycle": 143, "nominal_tc": 4153.2634422981155, "n_in": 68, "n_pre": 113, "n_post": 113, "covered": true, "signed_depth": -0.012590765953063965, "local_rms": 0.027302270755171776}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 65,103. Candidate capability counts: 65: 19,103: 19. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 5984329116365907584; NSS P=17.538395407 +/- 0.010265369 d; e=0.05981, omega=247.3903 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: eclipse morphology and repetition remain unresolved; this is not a secure eclipse identification. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. No secure eclipse identification or exclusion of all possible shallow eclipses follows. Repetition and eclipse morphology remain scientifically uncertain; no calibrated sensitivity or propagated ephemeris-error analysis was performed.

Next check: Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

C69. TIC 297541839 full report

Archived source: sanitized report text.

TIC 297541839: REFUTED

Queue position 52; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors11,64,100 show broad smooth variable minima rather than short eclipses. Covered nominal candidate phases have short-window depths only 0.100,0.077,0.017 percent, versus the original search depth 0.533 percent, and fits have depths smaller than local residual scatter. The distinct fresh custom aperture also shows smooth variability and segment offsets without recurring detached eclipses.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 16.363592725 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 11, "cycle": 0, "nominal_tc": 1601.1092, "n_in": 4, "n_pre": 5, "n_post": 6, "covered": true, "signed_depth": 0.0010014176368713379, "local_rms": 0.001233667484484613}, {"sector": 64, "cycle": 89, "nominal_tc": 3057.468952514513, "n_in": 31, "n_pre": 52, "n_post": 52, "covered": true, "signed_depth": 0.0007699728012084961, "local_rms": 0.0038203957956284285}, {"sector": 100, "cycle": 152, "nominal_tc": 4088.3752941820903, "n_in": 31, "n_pre": 52, "n_post": 52, "covered": true, "signed_depth": 0.00016689300537109375, "local_rms": 0.00291615072637796 4}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 64,100. Candidate capability counts: 64: 23,100: 23. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 5854775658438572416; NSS P=16.417997076 +/- 0.013138885 d; e=0.02946, omega=13.3971 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: submitted eclipse interpretation refuted by a demonstrated smooth/systematic explanation. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. Refutation applies to the submitted eclipse interpretation, not to binarity or all possible shallow eclipses.

Next check: No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

C70. TIC 396248369 full report

Archived source: sanitized report text.

TIC 396248369: REFUTED

Queue position 53; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors13,94,103 include apparent sharp dips near rapid baseline changes, but the original-phase fits fail to establish real repeat eclipses. In the fresh custom extraction, the S94/S103 apparent dips correspond to large positive ramps or segment-level changes; a detached negative eclipse is not independently recovered. Original covered phases have sub-millimagnitude or negative short-window signed deficits. The search period is not a measured binary eclipse period.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 18.390769704 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 13, "cycle": 0, "nominal_tc": 1663.3432, "n_in": 4, "n_pre": 8, "n_post": 8, "covered": true, "signed_depth": -0.0002053380012512207, "local_rms": 0.0014788307016715407}, {"sector": 13, "cycle": 1, "nominal_tc": 1681.7339697044129, "n_in": 6, "n_pre": 9, "n_post": 8, "covered": true, "signed_depth": 0.0006639957427978516, "local_rms": 0.000878628867212683}, {"sector": 94, "cycle": 120, "nominal_tc": 3870.23556452954, "n_in": 46, "n_pre": 78, "n_post": 78, "covered": true, "signed_depth": 0.00036269426345825195, "local_rms": 0.005096118431538343}, {"sector": 103, "cycle": 136, "nominal_tc": 4164.4878798001455, "n_in": 40, "n_pre": 69, "n_post": 51, "covered": true, "signed_depth": -0.0019028186798095703, "local_rms": 0.028664467856287956}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 94,103. Candidate capability counts: 94: 4,103: 4. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 6367733051516351360; NSS P=18.380294812 +/- 0.012093988 d; e=0.26564, omega=257.6767 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: submitted eclipse interpretation refuted by a demonstrated smooth/systematic explanation. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. Refutation applies to the submitted eclipse interpretation, not to binarity or all possible shallow eclipses.

Next check: No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

C71. TIC 277479128 full report

Archived source: sanitized report text.

TIC 277479128: REFUTED

Queue position 54; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors 13,92,94 contain broad multi-day variability and large segment changes. The sharp-looking S94 minimum is a distorted edge on broad variability: the distinct fresh aperture shows a multi-day minimum rather than a short detached eclipse. Covered nominal phases have negative short-window signed deficits; the strong S94 template fit hits its timing bound. No repeating detached event establishes an eclipse period.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 19.667413548 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 13, "cycle": 0, "nominal_tc": 1667.3914, "n_in": 7, "n_pre": 12, "n_post": 4, "covered": true, "signed_depth": -0.0013865232467651367, "local_rms": 0.005576646886765957}, {"sector": 92, "cycle": 109, "nominal_tc": 3811.139476703976, "n_in": 67, "n_pre": 113, "n_post": 113, "covered": true, "signed_depth": -0.000667572021484375, "local_rms": 0.004877484869211912}, {"sector": 94, "cycle": 112, "nominal_tc": 3870.1417173472046, "n_in": 67, "n_pre": 113, "n_post": 113, "covered": true, "signed_depth": -0.04576444625854492, "local_rms": 0.07903945446014404}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 92,94. Candidate capability counts: 92: 8,94: 8. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 6760248154216144512; NSS P=19.660672814 +/- 0.014125655 d; e=0.08876, omega=128.5857 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: submitted eclipse interpretation refuted by a demonstrated smooth/systematic explanation. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. Refutation applies to the submitted eclipse interpretation, not to binarity or all possible shallow eclipses.

Next check: No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

C72. TIC 257575960 full report

Archived source: sanitized report text.

TIC 257575960: PLAUSIBLE

Queue position 55; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors 12,67,102 show orbit-segment ramps, noisy low-amplitude variation and repeated plateaus/steps. Original nominal phases in sectors12/67 have no coherent short eclipse; fitted positive depths are below local residual scatter. The fresh independent aperture reproduces the sector102 segment plateaus/steps and noisy S67 curve without detached recurring dips. Independent Terra audit retains PLAUSIBLE: the available representative data fail to establish recurring eclipses, but sparse/compromised coverage does not tie the original search detection to a specific demonstrated artifact. This is not a secure eclipse identification.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 13.410727098 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 12, "cycle": 1, "nominal_tc": 1644.6230270979095, "n_in": 8, "n_pre": 12, "n_post": 13, "covered": true, "signed_depth": -0.00010025501251220703, "local_rms": 0.0011402764357626438}, {"sector": 67, "cycle": 113, "nominal_tc": 3146.624462063768, "n_in": 48, "n_pre": 108, "n_post": 71, "covered": true, "signed_depth": -0.0006230473518371582, "local_rms": 0.003330210456624627}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 67,102. Candidate capability counts: 67: 2,102: 2. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 6347331991219882624; NSS P=13.428262096 +/- 0.005767776 d; e=0.41266, omega=352.3530 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: eclipse morphology and repetition remain unresolved; this is not a secure eclipse identification. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. No secure eclipse identification or exclusion of all possible shallow eclipses follows. Repetition and eclipse morphology remain scientifically uncertain; no calibrated sensitivity or propagated ephemeris-error analysis was performed.

Next check: Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

Predicted spectroscopic conjunctions: G 11-11.5 individual evidence

Current adjudications reconcile 23 existing reviews using the same bounded physical criteria. Twenty-two systems are confirmed additions; TIC 8853239 remains plausible. Historical reports remain archived. Reference epochs identify the eclipse kind recorded below and need not denote the deeper primary; error qualifications govern numerical interpretation.

P001. TIC 262653359 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 46.1290484

period_error_days: 8e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 3489.276184

epoch_kind: named reference eclipse; see report

depth_ppt: 31

novelty_status: no identified prior EB in seven scoped groups

limitations: Period uncertainty widened to 0.00008 d: saved floor-fit output gives 0.000072 d, wider than prose 0.00005. Mass and evolutionary state leave factor ~2 light-ratio uncertainty.

next_check: Spectroscopy or colour eclipse photometry to constrain companion light.

Current full report and deciding evidence

P002. TIC 39013890 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 20.92352919

period_error_days: 3e-06

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1859.073426

epoch_kind: named reference eclipse; see report

depth_ppt: 157

novelty_status: no identified prior EB in seven scoped groups

limitations: Unresolved third light remains possible (ipd multi-peak 69%; RUWE 1.55), affecting derived radii and geometry precision.

next_check: Resolve close companion or constrain third light spectroscopically.

Current full report and deciding evidence

P003. TIC 325399242 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 28.03808637

period_error_days: 0.00021722

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2830.458306

epoch_kind: named reference eclipse; see report

depth_ppt: 5

novelty_status: no identified prior EB in seven scoped groups

limitations: Historical prose claims seven-event P=28.038374 +/-0.000191, but saved ephem_A and Afix each have five rows. Adopt corrected five-event artifact. Capable 13.2 arcsec neighbour is disfavoured at roughly 3–4 sigma per sector with systematic floor, not mathematically impossible.

next_check: Additional orientation/source-resolved imaging would strengthen close-neighbour rejection.

Current full report and deciding evidence

P004. TIC 384026617 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 30.05386902

period_error_days: 5.444e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2903.27302

epoch_kind: named reference eclipse; see report

depth_ppt: 47.2

novelty_status: no identified prior EB in seven scoped groups

limitations: Historical 0.6 arcsec WDS pair is doubtful but unresolved by TESS; attribution is to the Gaia source footprint. Gaia statistics alone do not prove pair spurious.

next_check: High angular resolution imaging to check historical WDS component.

Current full report and deciding evidence

P005. TIC 389013422 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 39.54257794

period_error_days: 1.044e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2497.62367

epoch_kind: named reference eclipse; see report

depth_ppt: 55

novelty_status: no identified prior EB in seven scoped groups

limitations: Secondary phase-0.535 hint is unconfirmed; do not quote it as a measured eclipse. Absolute physical parameters remain model dependent.

next_check: RVs to constrain eccentricity and secondary predictions.

Current full report and deciding evidence

P006. TIC 4808224 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 25.08348476

period_error_days: 5.7e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1526.830229

epoch_kind: named reference eclipse; see report

depth_ppt: 51

novelty_status: no identified prior EB in seven scoped groups

limitations: Single-sector s89 pixel test; pulsation affects secondary depth/timing. Discard s35 secondary fit: all input points flagged and optimiser wandered. Approximate physics.py formula has pi/4 error; exact Kepler-grid result is used.

next_check: Another orientation pixel test or improved RV orbit would reduce association uncertainty.

Current full report and deciding evidence

P007. TIC 23999420 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 22.127652

period_error_days: 4.4e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2349.7517

epoch_kind: named reference eclipse; see report

depth_ppt: 22

novelty_status: no identified prior EB in seven scoped groups

limitations: Adopt primary timing uncertainty 0.000044 d rather than joint model formal 0.000015 d. Secondary 9.7-sigma pixel regression is historical numerical evidence; representative image has no colour scale.

next_check: Recover missing/flagged s91 primary only if additional refinement needed.

Current full report and deciding evidence

P008. TIC 112232265 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 28.40515

period_error_days: 3e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2978.3647

epoch_kind: named reference eclipse; see report

depth_ppt: 33

novelty_status: no identified prior EB in seven scoped groups

limitations: Intrinsic depth approximately 30–36 ppt; custom 3x3 is diluted (~19 ppt), QLP background affects depth. RUWE 3.51 allows third-body uncertainty. T16 parse with zero cadences is not evidence.

next_check: Spectroscopy and source-resolved eclipse depths to constrain companion.

Current full report and deciding evidence

P009. TIC 275065610 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 28.04921

period_error_days: 5e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2023.8727

epoch_kind: named reference eclipse; see report

depth_ppt: 51.5

novelty_status: no identified prior EB in seven scoped groups

limitations: TASOC depth is diluted and its epoch is about 14 min later than QLP; does not invalidate its eclipse detection. Dark-companion fit neglects a few percent secondary light.

next_check: A future raw A eclipse or double-lined spectrum

Current full report and deciding evidence

P010. TIC 469755925 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 23.322969

period_error_days: 1.4e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1675.73779

epoch_kind: named reference eclipse; see report

depth_ppt: 40.5

novelty_status: no identified prior EB in seven scoped groups

limitations: S94 A is quality-flagged; confirmation also rests on clean s13 and s67 A events. B pixel depth5.93 +/-0.81 ppt exceeds pipeline2.8–4.0 ppt; low-S/N/background-template extraction is attribution only, not adopted depth.

next_check: Another clean secondary for depth and geometry

Current full report and deciding evidence

P011. TIC 356435413 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 27.476839

period_error_days: 1.5e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1984.25002

epoch_kind: named reference eclipse; see report

depth_ppt: 17.0

novelty_status: no identified prior EB in seven scoped groups

limitations: CTOI356435413.01 is a prior planet-candidate signal at matching eclipse period; annotate separately from EB novelty. Original report overstates 1.5arcsec target offset: that is deficit-to-light centroid; s79 coordinates imply ~2arcsec to target, s53 ~3.6arcsec. No full limb-darkened grazing fit.

next_check: Grazing-occultation model or spectrum to test light ratio

Current full report and deciding evidence

P012. TIC 255077215 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 26.26204

period_error_days: 1.6e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1637.34726

epoch_kind: named reference eclipse; see report

depth_ppt: 56.0

novelty_status: no identified prior EB in seven scoped groups

limitations: S65 secondary arrives31–42min late after common-mode correction; unexplained. This does not remove repeated A timing or agreement of earlier B events. Quoted period error is original formal fit and predictions need timing-systematic allowance.

next_check: Clean secondary to resolve s65 timing shift

Current full report and deciding evidence

P013. TIC 262471565 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 29.38001

period_error_days: 0.00022

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1337.4273

epoch_kind: named reference eclipse; see report

depth_ppt: 4.5

novelty_status: no identified prior EB in seven scoped groups

limitations: B is undetected: expected570–1100ppm versus ~600ppm central-phase limit creates mild model tension; permissible phase/temperature/impact values remain. Period error includes20min pipeline-scatter floor.

next_check: Unflagged B-window photometry at <=300ppm

Current full report and deciding evidence

P014. TIC 126758342 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 22.86993

period_error_days: 8e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1729.54161

epoch_kind: named reference eclipse; see report

depth_ppt: 20.0

novelty_status: no identified prior EB in seven scoped groups

limitations: Gaia eccentricity and omega broad; A misses in83% of permitted historical model draws, not certainty. Primary_epoch field uses detected deeper B eclipse.

next_check: Refined RV orbit or next B eclipse

Current full report and deciding evidence

P015. TIC 178486571 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 33.671079

period_error_days: 2.2e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1494.78347

epoch_kind: named reference eclipse; see report

depth_ppt: 53.8

novelty_status: no identified prior EB in seven scoped groups

limitations: S87 egress cut by gap. Three-event near-zero timing chi-square makes formal error optimistic; allow >=5min in predictions.

next_check: Clean B coverage or another A event

Current full report and deciding evidence

P016. TIC 395790195 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 55.460424

period_error_days: 3.3e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1756.37044

epoch_kind: named reference eclipse; see report

depth_ppt: 130.0

novelty_status: no identified prior EB in seven scoped groups

limitations: Adopt timing-only B period/error; tighter joint-model55.460455 +/-0.000021d is conditional on model. Only two same-kind epochs, cycle count constrained by Gaia.

next_check: Another B eclipse to test cycle count

Current full report and deciding evidence

P017. TIC 466336052 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 32.139199

period_error_days: 3.7e-05

period_origin: eclipse timing

period_error_kind: source-specific formal timing error with stated floors/scatter

reference_epoch_btjd: 2818.48299

epoch_kind: A

depth_ppt: 44.1

novelty_status: no identified prior EB in seven scoped groups

limitations: Only two A eclipses; no internal timing residual test. B windows are contaminated or missing; no B detection. Gaia conjunction uncertainty is 3-4 d.

next_check: Observe a third A eclipse and a clean B window near 0.3% precision.

Current full report and deciding evidence

P018. TIC 64066759 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 40.93770552812586

period_error_days: 4.02430562196e-05

period_origin: eclipse timing

period_error_kind: source-specific formal timing error with stated floors/scatter

reference_epoch_btjd: 3674.520856303387

epoch_kind: A

depth_ppt: 22.0

novelty_status: no identified prior EB in seven scoped groups

limitations: Long-baseline precision partly depends on a fixed-shape partial s7 event; three events total. Simplified geometry favours smaller primary radius than TIC. No B expected.

next_check: Fit a full limb-darkened joint A model and constrain primary radius spectroscopically.

Current full report and deciding evidence

P019. TIC 49416839 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 21.428466564976848

period_error_days: 4.771732314e-05

period_origin: eclipse timing

period_error_kind: source-specific formal timing error with stated floors/scatter

reference_epoch_btjd: 1558.284949338288

epoch_kind: B (dominant deep eclipse)

depth_ppt: 13.3

novelty_status: no identified prior EB in seven scoped groups

limitations: A3 lacks egress and is weak. B-A phase separation is 2.3 sigma from Gaia; simple joint geometry has limited constraints. Dominant deep eclipse is labelled B in source.

next_check: Obtain a complete A event and joint free-eccentricity model to test the phase-separation tension.

Current full report and deciding evidence

P020. TIC 8853239 - PLAUSIBLE

count_addition: false

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 24.647

period_error_days: 0.0114

period_origin: Gaia SB1 only

period_error_kind: Gaia formal orbit error

reference_epoch_btjd: 2629.0699

epoch_kind: A; Gaia-only period

depth_ppt: 60.0

novelty_status: no identified prior EB in seven scoped groups

limitations: Single secure event; Gaia-only period; KELT null is coverage-limited and not a refutation. Tentative B is not an ephemeris.

next_check: Measure a second A eclipse to test the cycle period; a B detection alone would still need repeat timing.

Current full report and deciding evidence

P021. TIC 60583013 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 26.65117

period_error_days: 0.00023

period_origin: eclipse timing

period_error_kind: source-specific formal timing error with stated floors/scatter

reference_epoch_btjd: 2210.9657

epoch_kind: A (period measured from B)

depth_ppt: 41.2

novelty_status: no identified prior EB in seven scoped groups

limitations: Period relies on two secondaries, one after a data gap; one primary only. Older B pixel test inconclusive. Duration geometry is approximate.

next_check: Obtain a second primary and joint limb-darkened model to test phase separation and geometry.

Current full report and deciding evidence

P022. TIC 277542587 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 24.3318

period_error_days: 0.0002

period_origin: KELT+TESS; heuristic error

period_error_kind: heuristic working precision; not calibrated1sigma

reference_epoch_btjd: 3488.8242

epoch_kind: A; KELT+TESS period

depth_ppt: 77.6

novelty_status: no identified prior EB in seven scoped groups

limitations: TESS A mostly flagged; only one TESS A+B pair. KELT period precision is heuristic and anchored to TESS; broad ground aperture dilutes depth. Sparse KELT B is supporting only. Historical exact-night count not adopted.

next_check: Obtain another full well-calibrated A/B pair and an event-timing based period uncertainty.

Current full report and deciding evidence

P023. TIC 12864953 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 31.632857663754894

period_error_days: 0.00015

period_origin: eclipse timing

period_error_kind: conservative timing allowance; not calibrated coverage

reference_epoch_btjd: 3204.1016

epoch_kind: A (period from repeated B)

depth_ppt: 17.24

novelty_status: no identified prior EB in seven scoped groups

limitations: Only two B events and one clean A; A ephemeris assumes B period. New A window is entirely quality 2048 flagged. Phase-separation tension about 3.35 sigma remains. Conservative period error is an estimate, not calibrated coverage.

next_check: Obtain a clean repeated A and refined radial-velocity orbit to resolve eccentricity/phase tension.

Current full report and deciding evidence

Underexplored selections: current measurements and evidence

The frozen magnitude-ordered period-fold queue and adaptive exploratory pilot use existing physical criteria. They have separate selection and attempted-data denominators; operational processing groups are not scientific methods. Full reports below retain deciding measurements and limitations. New timing periods are measured from eclipse midpoints, not predicted conjunction centers.

R175. TIC 300217120 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 14.158577492046303

period_error_days: 2.752113403501583e-05

reference_epoch_btjd: 3141.0351407928124

depth_ppt: 37.8660025864843

novelty_status: no identified prior EB

limitations: The nearby other eclipse in S67 contaminates a wider baseline window; the final fit uses a narrower 1.65-search-duration half-window and shows the repeated primary. Historical S27 TESScut contains one event but QLP independently shows three primaries in S67/S94.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R176. TIC 424716057 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 15.405497331062046

period_error_days: 5.499559353955741e-05

reference_epoch_btjd: 1625.946884976911

depth_ppt: 211.90741878678438

novelty_status: no identified prior EB

limitations: QLP primary depths vary across apertures/sectors (205–230 ppt in the selected sectors); no common intrinsic depth is claimed.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R177. TIC 265350422 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 8.448170725833366

period_error_days: 1.2842011215299834e-05

reference_epoch_btjd: 2970.804346169428

depth_ppt: 7.800219935881271

novelty_status: no identified prior EB

limitations: Three G-band-capable neighbours at 32.85, 42.38 and 51.78 arcsec lie away from the compact deficit on the target. The target-centered dip grows rather than vanishes in the small aperture. No calibrated centroid confidence ellipse is available.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R178. TIC 274941915 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.132939073812572

period_error_days: 6.285764339345673e-05

reference_epoch_btjd: 3371.1764455427087

depth_ppt: 10.41512839207673

novelty_status: no identified prior EB

limitations: Three capable neighbours at 32.98, 40.57 and 47.81 arcsec are spatially separated from the target-centered compact deficit. The small-aperture reduction retains the eclipse. No calibrated centroid confidence ellipse is available.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R179. TIC 253126207 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 17.43657092170319

period_error_days: 8.823559700783691e-05

reference_epoch_btjd: 1604.0901103382182

depth_ppt: 13.51328902862249

novelty_status: no identified prior EB

limitations: Attribution remains unresolved: seven capable neighbours include three within one TESS pixel (15.16, 17.15 and 19.43 arcsec); the historical PSF fractional residual is 0.657. A known Gaia eclipsing binary at 24.77 arcsec with P=0.312563 d is a different source, and its short period does not identify the 17.44 d signal. Real repeating eclipses pass the other checks; source localization is the failed scientific check. Resolved eclipse imaging or defensible multi-source PRF modeling is required.

next_check: Resolved eclipse photometry / multi-source PRF localization

Full report and illustrated deciding evidence

R180. TIC 245474316 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 15.01729122089107

period_error_days: 2.3615939344531595e-05

reference_epoch_btjd: 2884.3381092475142

depth_ppt: 219.83441151981677

novelty_status: no identified prior EB

limitations: Deep repeated eclipses are isolated in both reductions; all four measured primary events support the timing solution.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R181. TIC 397586559 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 9.866288327778928

period_error_days: 2.7660310467344676e-05

reference_epoch_btjd: 2452.71248468518

depth_ppt: 253.71926940278945

novelty_status: no identified prior EB

limitations: Unequal alternating eclipses (primary median approximately254 ppt, secondary164 ppt) establish orbital period twice the historical4.93316d event spacing. Only deeper even cycles enter the reported primary ephemeris. Historical S70 difference image mixes both eclipse kinds, so its aperture depth is a mixture. Gaia nominal conjunction is0.28093d later at the TESS epoch; measured-Gaia period difference−0.00126495d accumulates this offset. Projecting toward the Gaia2016 reference reduces offset to−0.0166d; the orbit association is physically consistent with an imperfect Gaia period, and the tension is retained rather than subjected to a sigma veto.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R182. TIC 305801065 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.045029317589018

period_error_days: 3.61616161282754e-05

reference_epoch_btjd: 1727.4094898160802

depth_ppt: 5.0649222211815

novelty_status: no identified prior EB

limitations: The shallow eclipse is clearly repeated with localized ingress/egress despite higher scatter in newer sectors. Three capable neighbours at28.07,33.30,54.57arcsec are separated from the compact target deficit. Depth depends on dilution and trapezoid shape; historical median3.8ppt is not a precision intrinsic depth.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R183. TIC 345794632 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 14.644264849868913

period_error_days: 4.2511956611254694e-05

reference_epoch_btjd: 1712.6089010966314

depth_ppt: 73.3874561147605

novelty_status: no identified prior EB

limitations: Large historical PSF fractional residual0.57 reflects imperfect scaled out-image morphology in a crowded field. No listed neighbour within60arcsec can supply the approximately73ppt eclipse under the G-band flux-budget screen, and the lost-light core coincides with the target. The weaker localization calibration remains a limitation.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R184. TIC 258621299 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.985970468761713

period_error_days: 1.7325420588134718e-05

reference_epoch_btjd: 1845.3682206476906

depth_ppt: 179.67039704399065

novelty_status: no identified prior EB

limitations: Unequal alternating eclipses (primary median approximately180ppt, secondary131ppt) establish orbital period twice the historical3.49299d event spacing. Only deeper even cycles enter the reported primary ephemeris. Historical S20 difference image mixes both eclipse kinds, so its aperture depth is a mixture.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R185. TIC 65346146 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.678623940946846

period_error_days: 1.5124697871365374e-05

reference_epoch_btjd: 2154.389536775554

depth_ppt: 38.20589792457893

novelty_status: no identified prior EB

limitations: Repeated flat-bottom eclipses; sector-dependent dilution changes fitted depths from about46 to38ppt.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R186. TIC 416646907 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.384314755499531

period_error_days: 1.007378415122247e-05

reference_epoch_btjd: 2144.9943337447125

depth_ppt: 17.963292895601008

novelty_status: no identified prior EB

limitations: Odd historical half-period cycles are deeper in each of three sectors (about18 versus16ppt); deeper odd cycles alone give the primary ephemeris and doubled orbital period. Alternative-reduction stack mixes eclipse kinds. Approximate central duration permits a noncentral chord.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R187. TIC 401138717 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 19.79284840491893

period_error_days: 4.02097023341268e-05

reference_epoch_btjd: 2338.109420981473

depth_ppt: 111.5978315194144

novelty_status: no identified prior EB

limitations: Three capable neighbours and high scaled-out-image residual0.976 limit PSF precision. Target-centered compact loss and aperture-depth growth support attribution. Known EB42.73arcsec away has P0.363740d, distinct from19.79d. Gaia period differs by0.00556d, accounting for much of the accumulated conjunction offset; this physical tension is retained.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R188. TIC 334379409 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 3.947126674992941

period_error_days: 8.626157743268626e-06

reference_epoch_btjd: 1846.4992369078332

depth_ppt: 9.702507254515327

novelty_status: no identified prior EB

limitations: Repeated raw eclipses have modest baseline variability. Gaia period mismatch accumulates over the baseline; propagated covariance and conjunction comparison are descriptive rather than a sigma threshold.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R189. TIC 345406164 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 7.440590745011774

period_error_days: 4.9507419893012455e-05

reference_epoch_btjd: 2771.404007331272

depth_ppt: 74.89159582802148

novelty_status: no identified prior EB

limitations: Deeper even cycles near75ppt alternate with47ppt odd eclipses in sector81, establishing doubled orbital period. Primary-only ephemeris used. Historical stack mixes eclipse kinds. Three capable neighbours are away from compact target deficit.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R190. TIC 152400959 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 5.512857449543862

period_error_days: 6.3397913119843745e-06

reference_epoch_btjd: 1492.1132318036186

depth_ppt: 54.37186113086094

novelty_status: no identified prior EB

limitations: Known EB27.09arcsec away at0.257384d differs from5.513d signal. Two capable neighbours and residual0.959 limit PSF precision; compact target-centered deficit and small-aperture eclipse support attribution.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R191. TIC 152086769 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.332203976073646

period_error_days: 7.904701005610135e-06

reference_epoch_btjd: 1356.809297220481

depth_ppt: 279.7427141291881

novelty_status: no identified prior EB

limitations: Deep flat-bottom repeated eclipses; sector96 aperture depth is lower than earlier sectors. No capable neighbour within60arcsec in approximate G flux budget.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R192. TIC 270260727 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.291649902528073

period_error_days: 1.9455996780702123e-05

reference_epoch_btjd: 3129.8498677040475

depth_ppt: 5.804677308979594

novelty_status: no identified prior EB

limitations: Shallow localized repeated eclipses are resolved in sectors67 and95; raw scatter and trapezoid approximation limit precision. No capable neighbour within60arcsec under G flux budget.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R193. TIC 112827464 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 3.4225952625012397

period_error_days: 4.65653931800416e-06

reference_epoch_btjd: 3129.3941410026505

depth_ppt: 33.4564194422391

novelty_status: no identified prior EB

limitations: Raw local baseline slopes are fitted explicitly. One capable neighbour is separated from the target-centered deficit; no calibrated centroid confidence ellipse. Gaia nominal phase offset0.063 and period drift remain explicit; full covariance and near-reference projection support association without arbitrary sigma veto.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R194. TIC 423657976 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.616416793242259

period_error_days: 1.647746040723537e-05

reference_epoch_btjd: 2310.599671098064

depth_ppt: 93.97262387817834

novelty_status: no identified prior EB

limitations: Repeated isolated eclipses with stable approximately94ppt QLP depth. No capable neighbour within60arcsec under approximate G flux budget.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R195. TIC 343926224 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.728977906284216

period_error_days: 0.00019858296949633555

reference_epoch_btjd: 1716.285636447413

depth_ppt: 42.29190632036057

novelty_status: no identified prior EB

limitations: Five clean primary repeats determine the ephemeris. S76 historical cycle 154 has structured excursions and an anomalously weak fitted dip; S84 cycle 176 has a broad distorted profile. Both remain in raw-windows.csv and plots but are excluded from the timing fit. The capable neighbour at 55.00 arcsec is separated from the compact target deficit. Aperture and local baseline variations limit depth precision. The retained timing fit has reduced chi-square 18.08; residual timing scatter materially inflates the stated period error and shows model/baseline limitations.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R196. TIC 53993311 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 14.542833409357298

period_error_days: 2.314386027834743e-05

reference_epoch_btjd: 3157.670659801766

depth_ppt: 97.18640828460472

novelty_status: no identified prior EB

limitations: Aperture-dependent primary depth changes from about 107 ppt in S68 to 85 ppt in S95; the same localized eclipse morphology repeats. Gaia conjunction uncertainty is substantial; a phase offset alone is not a veto.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R197. TIC 43785616 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 17.91112281610357

period_error_days: 0.0001892149302470355

reference_epoch_btjd: 1440.850824404922

depth_ppt: 23.115471716169672

novelty_status: no identified prior EB

limitations: Observed trapezoid duration exceeds a central estimate based on default primary mass and minimum companion mass. That approximate estimate is not an upper bound: stellar properties and inclination are not fitted. Repeated flat-bottomed events and matching period support association; detailed binary modeling remains optional.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R198. TIC 397178704 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 11.009685047241446

period_error_days: 1.7717221664631576e-05

reference_epoch_btjd: 1578.4727019498134

depth_ppt: 175.66990489468662

novelty_status: no identified prior EB

limitations: Unequal alternating primary and secondary eclipses establish twice the historical 5.50484 d event spacing. Only deeper odd cycles enter the primary ephemeris. The historical stack mixes both kinds. Large PSF residual 0.676 reflects surrounding light rather than an off-target core; the one capable neighbour lies 45.82 arcsec away from the compact deficit.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R199. TIC 77550184 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.214416259590474

period_error_days: 1.4190042358189235e-05

reference_epoch_btjd: 1527.7251647231262

depth_ppt: 60.64366018596226

novelty_status: no identified prior EB

limitations: Repeated isolated eclipses have aperture-dependent depths of approximately 56 to 63 ppt. No capable neighbour lies within the 60 arcsec screen.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R200. TIC 314041837 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 4.477995650988368

period_error_days: 1.534143438335759e-05

reference_epoch_btjd: 1770.0417342831945

depth_ppt: 44.521015368131536

novelty_status: no identified prior EB

limitations: The target is a large star (TIC radius 5.09 solar radii) with ellipsoidal modulation plus localized eclipses. A two-harmonic baseline fitted outside the primary core in S77 leaves a 30.31 ppt localized deficit; its double-wave modulation does not explain the eclipse. See harmonic-test.json and png. The 18.79 h local-linear trapezoid width includes smooth variability and is not a physical eclipse duration. The reported depth likewise describes that local model, not intrinsic eclipse depth. Capable neighbours at 27.40 and 53.17 arcsec are away from the target-centered deficit; PSF residual 0.595 limits morphology precision.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R201. TIC 7446763 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 5.91190666822017

period_error_days: 5.481789245650164e-06

reference_epoch_btjd: 1415.6444464852714

depth_ppt: 72.07551814166084

novelty_status: no identified prior EB

limitations: Primary eclipses are approximately twice secondary depth. Only deeper odd historical half-period cycles determine the ephemeris; the reported orbital period doubles the historical 2.95595 d spacing. Historical S98 aperture stack mixes both eclipse kinds.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R202. TIC 22705959 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 11.376810452243653

period_error_days: 5.702228089004493e-05

reference_epoch_btjd: 1843.6400352066814

depth_ppt: 140.88533223993306

novelty_status: no identified prior EB

limitations: A capable neighbour at 39.75 arcsec is separated from the target deficit. The target sits near a pixel boundary: the global deficit peak rounds to an adjacent pixel, while the centroid is only 3.64 arcsec from the actual target. This is not evidence for a different source.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R203. TIC 62940616 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 4.998587293904812

period_error_days: 1.3192060525370641e-05

reference_epoch_btjd: 2556.444026565471

depth_ppt: 52.59401180953602

novelty_status: no identified prior EB

limitations: One capable neighbour at 22.67 arcsec requires a 0.329 fractional eclipse to explain the signal, but the lost-light centroid lies 0.205 pixel from the target versus 1.33 pixels from that neighbour, displaced away from it. Its pixel fractional loss 0.0416 is a blended-pixel value and does not measure the neighbour intrinsic eclipse depth. The target-centered depth increases in the 1x1 aperture, consistent with target dilution in larger apertures. TESS resolution and G-to-TESS colour differences limit attribution precision.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R204. TIC 100911870 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 12.637002421838197

period_error_days: 1.9952345042033016e-05

reference_epoch_btjd: 2833.849725715909

depth_ppt: 84.03451128246843

novelty_status: no identified prior EB

limitations: Four clean primary events in S56 and S83 support the timing solution and approximately 84 ppt raw QLP depth. No capable neighbour is present in the 60 arcsec screen.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R205. TIC 77176486 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 13.753691913407923

period_error_days: 1.89931259791415e-05

reference_epoch_btjd: 2153.315120594371

depth_ppt: 12.943912782527418

novelty_status: no identified prior EB

limitations: The 13.75369 d signal lies near the TESS orbital timescale, but repeated localized raw SAP dips in S31 and S98 and the target-centered cutout deficit support astrophysical eclipses. Quality-nonzero cadences are excluded; no flagged or uncovered window is a clean null. Cycle188 is visibly present but its trapezoid covariance is singular, so it is excluded from timing. Bounded S98 pixel/background check (background-check.json) finds target-pixel losses11.99/22.43/16.35 electrons/s versus3.24/9.85/7.73 at the close capable neighbour and0.70/6.30/3.36 at the farther one. Outer-pixel median losses are -0.027/5.68/4.04: common-mode background affects later events, while cycle187 is localized with negligible outer deficit. Core quality flags are0/0/3; flagged cadences are excluded. FLUX_BKG is zero throughout and supplies no independent background measurement. These are descriptive diagnostics, not a calibrated artifact veto. Gaia conjunction offset is 0.04208 phase with 0.549 d propagated uncertainty; this is association support with appreciable orbital uncertainty, not a precise phase coincidence.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R206. TIC 155916287 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.053154397574601

period_error_days: 1.5431906036480347e-05

reference_epoch_btjd: 1741.6770525360378

depth_ppt: 31.52043029966451

novelty_status: no identified prior EB

limitations: Seven fitted repeats span S16,50,77. Sector-dependent depth variation and long-cadence integration limit intrinsic depth and duration precision.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R207. TIC 383603595 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 5.0148153411521825

period_error_days: 4.955091059655567e-06

reference_epoch_btjd: 1520.2020451552114

depth_ppt: 32.025684556497076

novelty_status: no identified prior EB

limitations: Eleven fitted repeated eclipses span S8,62,90. Modest aperture/baseline depth variation limits precision; no missing bounded check.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R208. TIC 436724968 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 2.732155078385491

period_error_days: 3.73397661927236e-06

reference_epoch_btjd: 2475.274686430569

depth_ppt: 48.52986342115804

novelty_status: no identified prior EB

limitations: QLP free-trapezoid depths show modest parity differences, especially in S71; local baseline and aperture dependence are material. Independent S43 3x3 cutout depths are 35.34-37.19 ppt for eight events including consecutive odd/even cycles, without a stable unequal alternating pattern. Gaia P=2.732167 d matches the measured event spacing. These data do not establish two distinct alternating eclipse kinds or require doubling; report the fitted period as the adopted orbit with this depth-model limitation, and the epoch as an observed reference eclipse. Capable neighbours at 35.19 and 35.89 arcsec are displaced from the compact target deficit.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R209. TIC 37948318 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 8.056727017738979

period_error_days: 1.2086497005613364e-05

reference_epoch_btjd: 1444.2226216138542

depth_ppt: 11.181622059581528

novelty_status: no identified prior EB

limitations: The historical absolute target-to-deficit offset is 15.1 arcsec (0.719 pixel), while light-to-deficit separation is 3.51 arcsec; these are distinct comparisons. The capable G13.73 neighbour is 53.65 arcsec away and opposite the deficit displacement. Independent per-event pixel checks find target-pixel losses 9.03-10.59 electrons/s against 0.185-0.437 at that neighbour in cycles317,319,321. This supports the target over the capable neighbour at TESS resolution, with an uncalibrated absolute-centroid systematic and residual crowding uncertainty. Historical cycles316/318 have common-mode excursions and are not individually decisive. QLP cycle93 has singular timing covariance and is excluded from timing.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R210. TIC 373034189 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 12.98870409783554

period_error_days: 2.8433415380927524e-05

reference_epoch_btjd: 1571.7245142398501

depth_ppt: 12.69611305405158

novelty_status: no identified prior EB

limitations: Four raw fitted eclipses in S10,38,90 support the period. The Gaia conjunction uncertainty is substantial (0.3105 d); local model widths and depths are descriptive.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R211. TIC 300969635 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 7.420727532798374

period_error_days: 7.815655460458437e-06

reference_epoch_btjd: 1328.2542487236803

depth_ppt: 217.5379472546418

novelty_status: no identified prior EB

limitations: Deep repeated eclipses persist in S1,37,98. Aperture depth changes from about210 to248 ppt; pooled parity depths confound sector coverage and do not establish stable unequal eclipse kinds. Gaia period matches the measured spacing. Intrinsic depth requires dilution modeling.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R212. TIC 317800184 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 3.689780944715975

period_error_days: 5.125646723563834e-06

reference_epoch_btjd: 1475.655235020276

depth_ppt: 50.47396076159312

novelty_status: no identified prior EB

limitations: Repeated eclipses span S6,87,98 with sector-dependent depths around45-62 ppt. This limits precision but does not erase the localized repeat pattern.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R213. TIC 219033622 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.9464765821983425

period_error_days: 0.00042454076935334877

reference_epoch_btjd: 1713.808706332515

depth_ppt: 132.32584082017803

novelty_status: no identified prior EB

limitations: Three constrained same-kind events in S15 and S23 determine the ephemeris; S23 cycle32 is retained as a raw repeat but excluded from timing because sparse sampling leaves its egress poorly constrained. S76 did not supply a usable selected event window and is not a clean non-detection. The independent S16 cutout contains four repeating deep target-centered deficits. The timing baseline is shorter than the available sector list suggests.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R214. TIC 312725755 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 11.304994219364756

period_error_days: 3.431475573613791e-05

reference_epoch_btjd: 1823.1102836630625

depth_ppt: 47.710924471257606

novelty_status: no identified prior EB

limitations: Five fitted repeating eclipses span S19,45,73. The measured widths are long but compatible with approximate Gaia/TIC geometry; intrinsic depths and stellar parameters need detailed modeling.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R215. TIC 167178300 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 12.215667671701095

period_error_days: 2.524519726874651e-05

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

reference_epoch_btjd: 1479.059382158827

epoch_btjd: 1479.059382158827

epoch_kind: A

depth_ppt: 70.54060949519207

novelty_status: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown

limitations: Capable Gaia neighbours remain at 58.583arcsec, 41.62arcsec, 12.984arcsec, 24.468arcsec, 18.174arcsec, 59.428arcsec. Deficit centroid favors target, but close blend is unresolved without a PRF/deblend uncertainty test.

next_check: Resolved eclipse photometry or multi-source PRF localization

Full report and illustrated deciding evidence

R216. TIC 325573958 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 13.08172273071557

period_error_days: 1.3997427571534246e-05

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

reference_epoch_btjd: 2772.4906957388416

epoch_btjd: 2772.4906957388416

epoch_kind: A

depth_ppt: 202.25348219264384

novelty_status: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown

limitations: Capable Gaia neighbours remain at 57.77arcsec, 7.958arcsec, 20.628arcsec. Deficit centroid favors target, but close blend is unresolved without a PRF/deblend uncertainty test.

next_check: Resolved eclipse photometry or multi-source PRF localization

Full report and illustrated deciding evidence

P024. TIC 155303130 - CONFIRMED

count_addition: false

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 20.524777015731278

period_error_days: 0.0003376256272186696

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

epoch_btjd: 3238.1960535727508

epoch_kind: B

depth_ppt: 104.05549881862092

novelty_status: known IJspeert2024 EB recovery; zero new-system addition

limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim.

next_check: Optional precision characterization; no missing bounded admission check

Archived source: sanitized report text.

> Supervisor-accepted physical review: CONFIRMED. Known recovery; excluded from new-system count. Earlier pending-admission wording below records the original handoff.

TIC 155303130: CONFIRMED

Scientific method/population: fresh Gaia-conjunction appointments G11.5-12 P20-60d. Physical review completion: complete bounded first pass, independently audited in helper/AUDIT.md; supervisor admission pending. Catalogue novelty is separate: known IJspeert2024 EB recovery; zero new-system addition.

1. Repeated raw eclipses: PASS. 4 fit-eligible raw SAP events in sectors [71, 72, 87]. Trapezoid depths median 104.06 ppt; duration median 13.10 h. All saved events, rejected fits and raw files are identified in timing-results.json; no claim that other sectors are clean nulls. QUALITY==0 used. Existing raw observations were remeasured, not merely called eclipses from BLS.
2. Eclipse timing: PASS. P=20.5247770157 d; B-eclipse reference epoch BTJD 3238.19605357; OLS scatter SE 0.000337625627 d; residual RMS 5.974 min. This SE excludes cadence/template/extraction systematics and is not calibrated coverage. Fits are representative, not all-event precision ephemerides.
3. Target pixels: PASS. Sector 71, 2 event stack, peak pixel [7, 8] equals target [7, 8]; centroid-target 1.51 arcsec. 1 distinct Gaia neighbours could supply the flux under the conservative total-flux bound. The marginally capable neighbour at32.394arcsec is spatially distinct from the1.51arcsec target-centered deficit; it is not discarded merely because it is faint. Catalogue incompleteness/unresolved companions are not excluded. Gaia position used consistently with target ID. The older diffstack n_possible field includes the target itself; this report explicitly removes it.

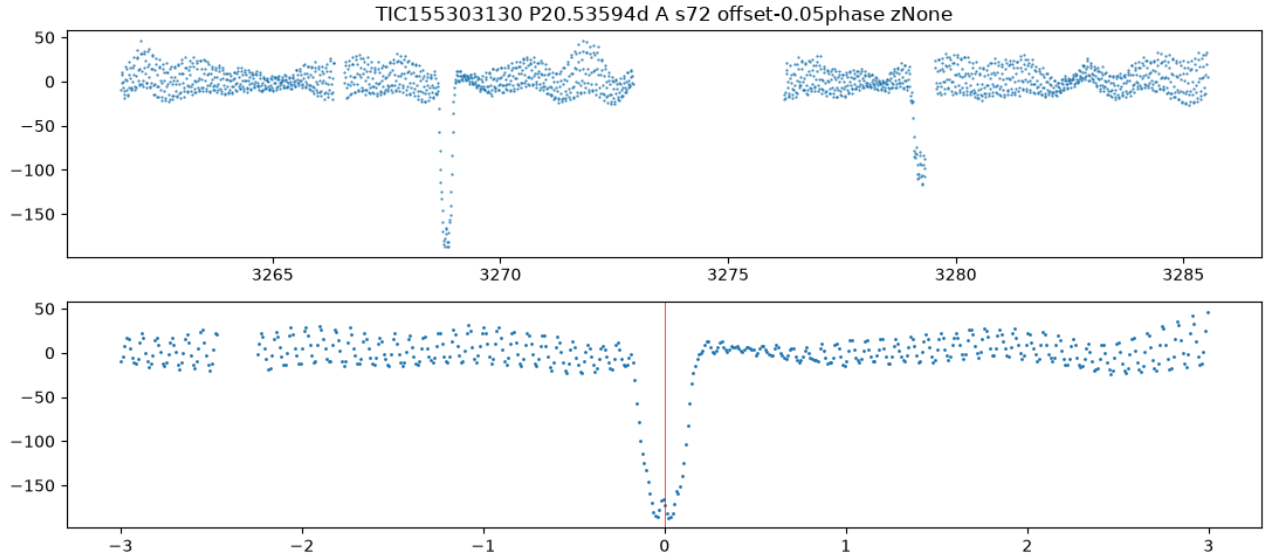
4. Alternative reduction: PASS as a processing check. Own TESScut extraction using median faint-pixel background and target 3x3 aperture reproduces deficit 98.38 ppt. The difference-stack receipt records per-event aperture depths. These are the same TESS photons, not an independent observing occasion; dilution/apertures explain differing QLP depths.
5. Gaia geometry: PASS for consistency, not a precision solution. Eclipse P differs from Gaia by -0.523 quoted Gaia period errors. B-conjunction epoch difference -0.522 propagated marginal-error sigma. Gaia mass function 0.0783 solar masses, minimum companion mass 0.736 solar masses for TIC primary mass 1.520. TIC R1=2.196 solar radii. Illustrative edge-on R2=1solar-radius duration at this conjunction 13.37 h, longer than observed; impact/inclination can shorten it. This is an existence/consistency check, not a fitted mass-radius claim. Correlated Gaia errors were not modeled; no arbitrary sigma veto.

Seven catalogue groups: fresh VSX60arcsec, ASAS-SN60arcsec, SIMBAD exactGaiaDR3, Gaia vari_eclipsing_binary exactID, Prsa2022 5arcsec, IJspeert2024 5arcsec, vetted Kostov2025 tables3+4 5arcsec all returned successful receipts. IJspeert contains exactTIC/Gaia. VSX DSCT and ASAS-SN ROT: classify the same star differently; its eclipses were already published. Full rows, UTC queries and response hashes are in catalogue coverage. The latter two Kostov tables form one group. No external messaging or public/paper edits.

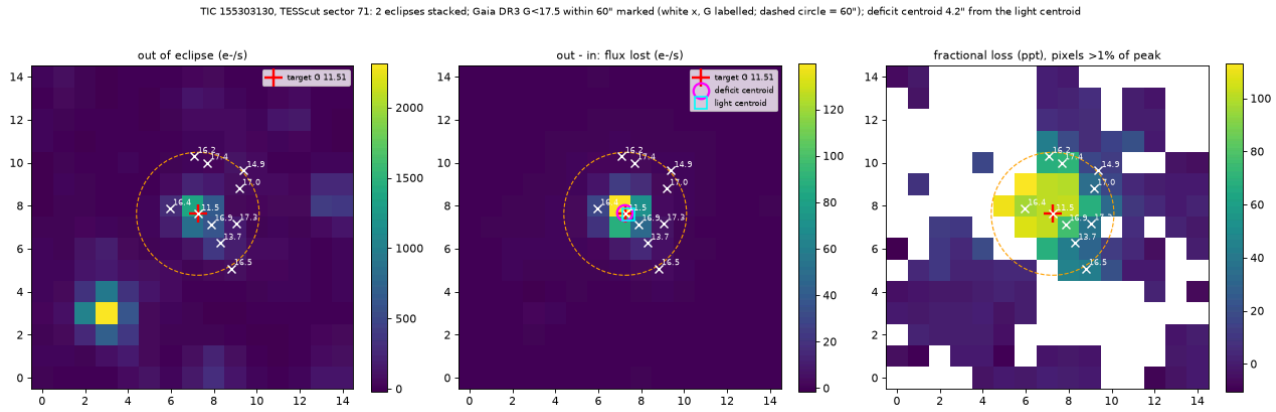
Limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim. Next check: Supervisor review before admission; independent helper audit completed.

Lead audit: scripts inspected and deciding numeric outputs reconciled with raw SAP plots, full Gaia NSS metadata, source IDs and real TESScut pixels. Apparent failure/non-detection is not called physical refutation unless contrary evidence is present. Independent helper audit uncertainty caveats are retained. The helper verified deciding artifacts, not an exhaustive literature search.

Accepted review key: P024. Pilot selection and limitations, independent audit, and catalogue receipts. Nearby files contain the deciding pixel measurements and hashes. This public archive omits raw FITS and sanitizes local paths; scripts are archival rather than turnkey.



P024; fresh-155303130.png. Exploratory selection and photometric evidence; selection and period qualifications remain as stated.



P024; diff_pilot.png. Target-pixel deficit evidence; the near-orbital-period cases remain probable because capable close neighbours are unresolved.

P025. TIC 382042754 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 25.89267238600749

period_error_days: 4.419206553916731e-05

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

reference_epoch_btjd: 3979.672005735797

epoch_btjd: 3979.672005735797

epoch_kind: A

depth_ppt: 87.77880028840502

novelty_status: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown

limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim.

next_check: Optional precision characterization; no missing bounded admission check

Archived source: sanitized report text.

> Supervisor-accepted physical review: CONFIRMED. Included under the paper's stated physical tier and bounded catalogue scope. Earlier pending-admission wording below records the original handoff.

TIC 382042754: CONFIRMED

Scientific method/population: fresh Gaia-conjunction appointments G11.5-12 P20-60d. Physical review completion: complete bounded first pass, independently audited in helper/AUDIT.md; supervisor admission pending. Catalogue novelty is separate: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown.

1. Repeated raw eclipses: PASS. 4 fit-eligible raw SAP events in sectors [96, 97, 98]. Trapezoid depths median 87.78 ppt; duration median 5.68 h. All saved events, rejected fits and raw files are identified in timing-results.json; no claim that other sectors are clean nulls. QUALITY==0 used. Existing raw observations were remeasured, not merely called eclipses from BLS.
2. Eclipse timing: PASS. P=25.8926723860 d; A-eclipse reference epoch BTJD 3979.67200574; OLS scatter SE 4.41920655e-05 d; residual RMS 0.133 min. This SE excludes cadence/template/extraction systematics and is not calibrated coverage. Fits are representative, not all-event precision ephemerides.
3. Target pixels: PASS. Sector 97, 2 event stack, peak pixel [7, 8] equals target [7, 8]; centroid-target 1.23 arcsec. 0 distinct Gaia neighbours could supply the flux under the conservative total-flux bound. All distinct Gaia neighbours within 60arcsec are too faint under the conservative bound. Catalogue incompleteness/unresolved companions are not excluded. Gaia position used consistently with

target ID. The older diffstack n_possible field includes the target itself; this report explicitly removes it.

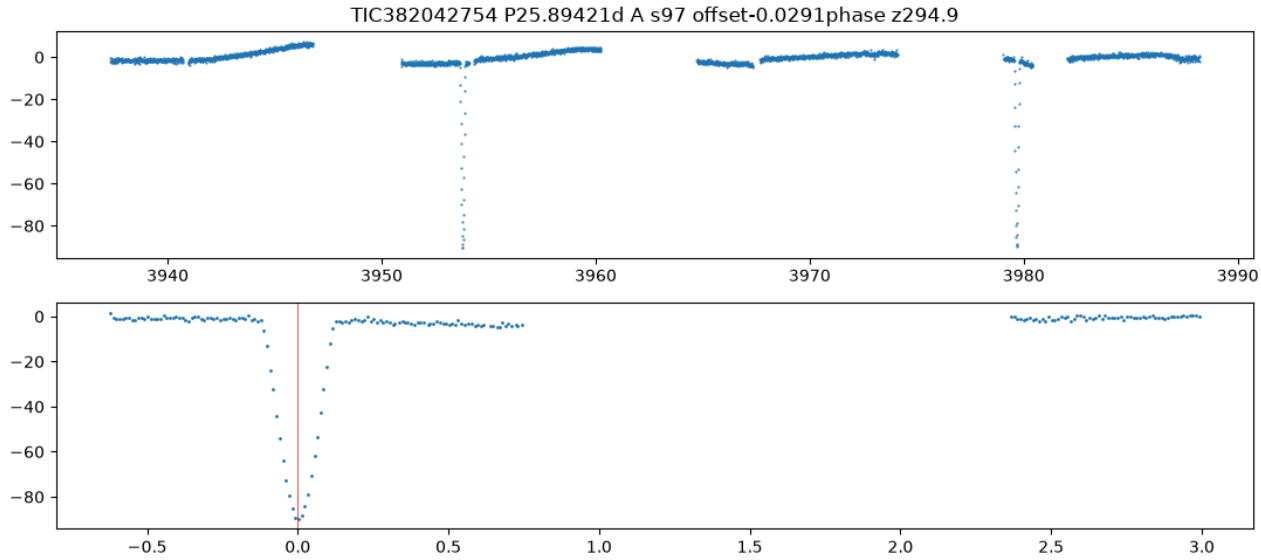
4. Alternative reduction: PASS as a processing check. Own TESScut extraction using median faint-pixel background and target 3x3 aperture reproduces deficit 90.79 ppt. The difference-stack receipt records per-event aperture depths. These are the same TESS photons, not an independent observing occasion; dilution/apertures explain differing QLP depths.
5. Gaia geometry: PASS for consistency, not a precision solution. Eclipse P differs from Gaia by -0.293 quoted Gaia period errors. A-conjunction epoch difference -0.911 propagated marginal-error sigma. Gaia mass function 0.1377 solar masses, minimum companion mass 0.835 solar masses for TIC primary mass 1.220. TIC R1=1.546 solar radii. Illustrative edge-on R2=1solar-radius duration at this conjunction 9.08 h, longer than observed; impact/inclination can shorten it. This is an existence/consistency check, not a fitted mass-radius claim. Correlated Gaia errors were not modeled; no arbitrary sigma veto.

Seven catalogue groups: fresh VSX60arcsec, ASAS-SN60arcsec, SIMBAD exactGaiaDR3, Gaia vari_eclipsing_binary exactID, Prsa2022 5arcsec, IJspeert2024 5arcsec, vetted Kostov2025 tables3+4 5arcsec all returned successful receipts. No on-target EB match in this bounded scope. Any VSX/ASAS-SN neighbour is retained as attribution evidence, not mistaken for target identity. Full rows, UTC queries and response hashes are in catalogue coverage. The latter two Kostov tables form one group. No external messaging or public/paper edits.

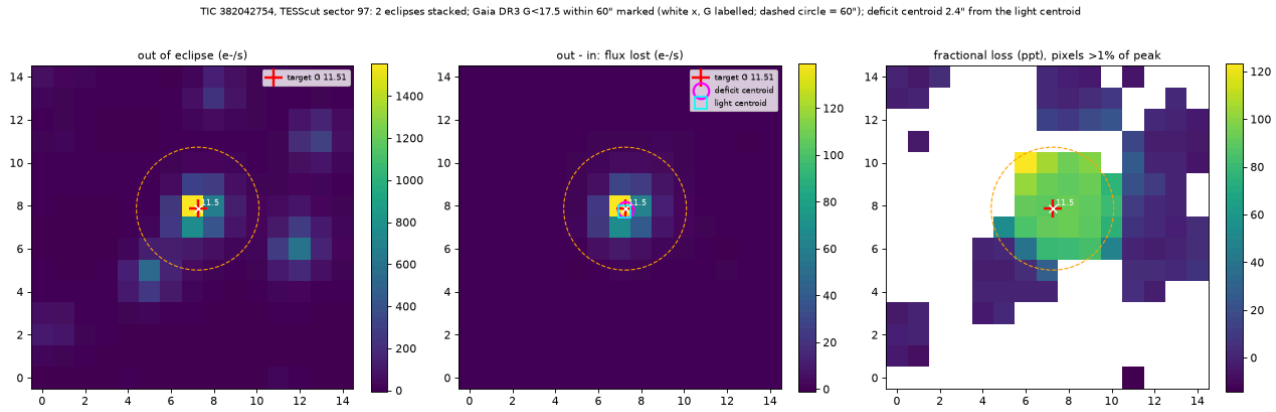
Limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim. Next check: Supervisor review before admission; independent helper audit completed.

Lead audit: scripts inspected and deciding numeric outputs reconciled with raw SAP plots, full Gaia NSS metadata, source IDs and real TESScut pixels. Apparent failure/non-detection is not called physical refutation unless contrary evidence is present. Independent helper audit uncertainty caveats are retained. The helper verified deciding artifacts, not an exhaustive literature search.

Accepted review key: P025. Pilot selection and limitations, independent audit, and catalogue receipts. Nearby files contain the deciding pixel measurements and hashes. This public archive omits raw FITS and sanitizes local paths; scripts are archival rather than turnkey.



P025; fresh-382042754.png. Exploratory selection and photometric evidence; selection and period qualifications remain as stated.



P025; diff_pilot.png. Target-pixel deficit evidence; the near-orbital-period cases remain probable because capable close neighbours are unresolved.

P026. TIC 389128842 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 22.666342834406446

period_error_days: 5.4096653855964156e-05

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

reference_epoch_btjd: 4134.517930923769

epoch_btjd: 4134.517930923769

epoch_kind: A

depth_ppt: 339.70241891594145

novelty_status: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown

limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim.

next_check: Optional precision characterization; no missing bounded admission check

Archived source: sanitized report text.

> Supervisor-accepted physical review: CONFIRMED. Included under the paper's stated physical tier and bounded catalogue scope. Earlier pending-admission wording below records the original handoff.

TIC 389128842: CONFIRMED

Scientific method/population: fresh Gaia-conjunction appointments G11.5-12 P20-60d. Physical review completion: complete bounded first pass, independently audited in helper/AUDIT.md; supervisor admission pending. Catalogue novelty is separate: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown.

1. Repeated raw eclipses: PASS. 4 fit-eligible raw SAP events in sectors [102, 103, 104]. Trapezoid depths median 339.70 ppt; duration median 8.39 h. All saved events, rejected fits and raw files are identified in timing-results.json; no claim that other sectors are clean nulls. QUALITY==0 used. Existing raw observations were remeasured, not merely called eclipses from BLS.
2. Eclipse timing: PASS. P=22.6663428344 d; A-eclipse reference epoch BTJD 4134.51793092; OLS scatter SE 5.40966539e-05 d; residual RMS 0.123 min. This SE excludes cadence/template/extraction systematics and is not calibrated coverage. Fits are representative, not all-event precision ephemerides.
3. Target pixels: PASS. Sector 102, 1 event stack, peak pixel [7, 8] equals target [7, 8]; centroid-target 0.54 arcsec. 0 distinct Gaia neighbours could supply the flux under the conservative total-flux bound. All distinct Gaia neighbours within 60arcsec are too faint under the conservative bound. Catalogue incompleteness/unresolved companions are not excluded. Gaia position used consistently with

target ID. The older diffstack n_possible field includes the target itself; this report explicitly removes it.

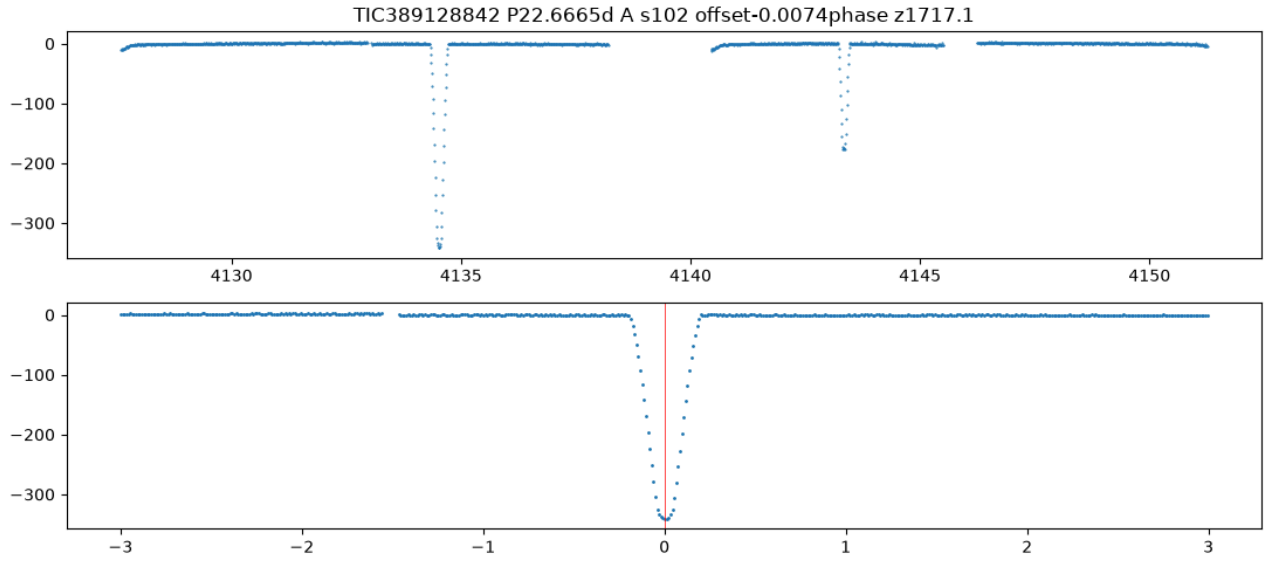
4. Alternative reduction: PASS as a processing check. Own TESScut extraction using median faint-pixel background and target 3x3 aperture reproduces deficit 371.64 ppt. The difference-stack receipt records per-event aperture depths. These are the same TESS photons, not an independent observing occasion; dilution/apertures explain differing QLP depths.
5. Gaia geometry: PASS for consistency, not a precision solution. Eclipse P differs from Gaia by -0.046 quoted Gaia period errors. A-conjunction epoch difference -0.235 propagated marginal-error sigma. Gaia mass function 0.1004 solar masses, minimum companion mass 0.649 solar masses for TIC primary mass 1.000. TIC R1=1.372 solar radii. Illustrative edge-on R2=1solar-radius duration at this conjunction 13.04 h, longer than observed; impact/inclination can shorten it. This is an existence/consistency check, not a fitted mass-radius claim. Correlated Gaia errors were not modeled; no arbitrary sigma veto.

Seven catalogue groups: fresh VSX60arcsec, ASAS-SN60arcsec, SIMBAD exactGaiaDR3, Gaia vari_eclipsing_binary exactID, Prsa2022 5arcsec, IJspeert2024 5arcsec, vetted Kostov2025 tables3+4 5arcsec all returned successful receipts. No on-target EB match in this bounded scope. Any VSX/ASAS-SN neighbour is retained as attribution evidence, not mistaken for target identity. Full rows, UTC queries and response hashes are in catalogue coverage. The latter two Kostov tables form one group. No external messaging or public/paper edits.

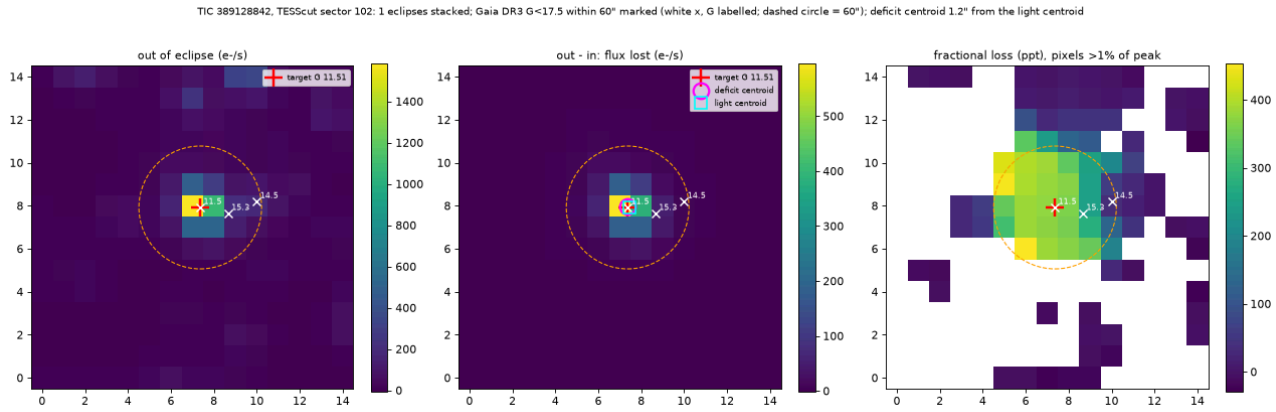
Limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim. Next check: Supervisor review before admission; independent helper audit completed.

Lead audit: scripts inspected and deciding numeric outputs reconciled with raw SAP plots, full Gaia NSS metadata, source IDs and real TESScut pixels. Apparent failure/non-detection is not called physical refutation unless contrary evidence is present. Independent helper audit uncertainty caveats are retained. The helper verified deciding artifacts, not an exhaustive literature search.

Accepted review key: P026. Pilot selection and limitations, independent audit, and catalogue receipts. Nearby files contain the deciding pixel measurements and hashes. This public archive omits raw FITS and sanitizes local paths; scripts are archival rather than turnkey.



P026; fresh-389128842.png. Exploratory selection and photometric evidence; selection and period qualifications remain as stated.



P026; diff_pilot.png. Target-pixel deficit evidence; the near-orbital-period cases remain probable because capable close neighbours are unresolved.